

QUANTUM MECHANICS II

QUANTUM MECHANICS II

A Second Course in Quantum Theory

SECOND EDITION

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To Jan and my parents

PREFACE

*Though a seeker since my birth,
Here is all I've learned on earth,
This is the gist of what I know:
Give advice and buy a foe.*

—Phyllis McGinley

Preface to the Second Edition

“How often in life do you get a chance to go back and do something right?” Such was the response of a valued colleague of mine upon hearing of my revision of *Quantum Mechanics II*. That observation, and memories of the thanks I have received from students over the last seven years, has been in my thoughts often as I labored to improve and correct an obsolete electronic draft of the first edition of *QMII*. Every page in this new edition has been edited with the aim of making the concepts clearer, the prose more fluid, and the equations more consistent, more readable and less erroneous. What has not changed is my view, confirmed during these last seven years, that many graduate students need the concrete footing of a text such as this before studying topics in modern field theory. My challenge, in addition to the usual physics ones, has been to make this author-prepared manuscript not look it.

In addition to a large number of small changes, in *Part I* I have added a discussion of the density matrix as part of Spin Phenomenology, and two new chapters on the Feynman path integral formulation, one on its theory and one on its application as a lattice computation. In *Part II* I have added a section on the momentum-space description of scattering from a combined short range plus Coulomb potential. The coverage of many-body theory has been increased and collected into *Part IV*, where a new chapter on fermion pairing has been added as well as a tutorial on Hartree–Fock theory.

A number of people have helped make this second edition an improved one. In particular, I wish to acknowledge the valuable comments I have received from students at

the University of Illinois, the University of Wisconsin, New Mexico State University, the University of Oregon, the University of Heidelberg, Rutgers University, and Oregon State University. I am also grateful to Albert Petschek, Kirk McVoy, Marvin Girardeau, Victor Madsen, Al Wasserman, Manuel Paez, and Al Stetz for helpful comments and contributions. I have been ably and cheerfully assisted by Dr. Bijan Shahir and Ms. Melanie Johnson; without their many hours of work, the painstaking conversions and revisions would not have been made. In spite of all these people's best efforts, I suspect that errors remain and that I am to blame. Finally, I wish to thank the many good people at John Wiley, especially Greg Franklin and Rosalyn Farkas, whose dedication to quality publishing has led to a second edition we are all proud of.

Preface to the First Edition

This book is a labor of love aimed at helping students see the beauty and unity of physics. It evolved from a second-year graduate course in quantum mechanics (QMII), in which I joked to my students that in one year we covered three or four books on different topics, each of which normally takes a year's course! While at times it seemed difficult to strike a satisfying balance between depth and breadth, at some point the material began to congeal into a collection of topics which I believed all graduate students in physical sciences wanted to study before taking research-oriented specialty courses. Whereas much space in the book is given to developing the needed tools, much is also given to developing new ideas about physics and nature.

Officially the purpose of this book is to help experimental and theoretical students in all disciplines learn intermediate, graduate-level physics. As such it is not the ultimate research tool or the last word on any subject but rather a vehicle to move a student from pupil to researcher. To achieve this, I have tried to emphasize physical understanding and intuition, starting with the concrete, progressing to the formal, and eventually facing the uncertain. Accordingly, Part I, *Scattering and Integral Quantum Mechanics*, moves slowly and systematically through the material, while the optional material (marked with an $\odot$) and some of Part III, *Quantum Fields* have a sketchiness similar to that found in research literature.

Scattered through all Parts of the text are over 200 exercises to ensure students contribute to their own education (and stay awake). At the end of the chapters are over 200 multi-part problems which assist in the understanding of the text, test the reader's understanding and recall, or extend the coverage to fascinating examples (which I had to restrain myself from covering in the text). In addition, some variety is provided by a number of "tutorials" within the chapters which lead the reader through programmed learning on special topics. I don't believe physics of this difficulty is understood without the student's commitment to work through these exercises and problems.

This book differs from others in its scope and its design to be accessible to a wide range of students. Whereas most effective when used in conjunction with regular lectures, I tried to keep in mind students throughout the world who need a physics book they can "read." I assumed a familiarity with quantum mechanics at the level of Merzbacher (up to scattering) and familiarity with special relativity and classical electromagnetism at the level of Jackson. There are, however, appendices on momentum states, representations,

special relativity, and the Dirac equation which can be consulted. As such, a good deal of the text should also be accessible for perusal by advanced undergraduates and practicing scientists.

The subject matter of this text is divided into three [now four] Parts. It is more than enough for a one-year, or one-semester course (it all depends on whether the instructor starts at Part I, *Scattering and Integral Quantum Mechanics*, or III, *Quantum Fields*, and how much of the optional material is included). The emphasis in all parts is toward developing understanding and intuition, which is tested in the world of experimental physics; I have not tried to lay a mathematical foundation through rigorous proofs and definitions. There is nonetheless a slant toward integral formulations, momentum-space techniques, and computational physics because I see this as the modern use of quantum mechanics.

Part I, *Scattering and Integral Quantum Mechanics*, is the most concrete and detailed, and most like first-year quantum mechanics. Part II, *Relativistic Quantum Mechanics*, while quite captivating, is new, different, and often confusing for the students; consequently, I here adopted a more operational point of view. After vigorous exercises burn off the haze of Dirac accounting from Part II, the physics should emerge. Part III, *Quantum Fields*, is difficult material rarely mastered in a first-time encounter. I emphasized the physics and connections with familiar quantum mechanics by using the (old-fashion) perturbation theory, avoiding proofs, “deriving” the Coulomb potential, and examining actual physical processes as proof that field theory works. To me this is the most effective introduction to the subject. Feynman diagrams and the now-standard metric and Dirac matrices of Bjorken and Drell are used.

A detailed description of the chapter-by-chapter contents is found in the Table of Contents and will not be repeated. I shall, however, give some suggestions regarding rearrangements and omissions.

It often fits in better with the specialty courses of second- and third-year graduate studies to start with the second quantization techniques of Chapter 19, *Second Quantization*, possibly supplemented by its applications in Chapter 20, *Quantized Electromagnetic Fields*. The course can then proceed with Part I, *Scattering and Integral Quantum Mechanics*, and in course pick up the rest of Part III, *Quantum Fields*. If the previous QMI course included an adequate study of scattering and its applications, Chapters 1–4 can be passed through quickly or only reviewed. Chapters 5–17 are probably new for most students, and with their emphasis on Green’s function techniques in quantum mechanics, are useful for much of modern physics. The description in Chapter 8, *The Angular Momentum Basis*, is valuable for practical problem-solving but is not essential to the logical development of the theory. It can be scanned on first reading and then used for reference. Part IV *Many-Body Theory* gives some basics and a survey of many-body physics. It fits in well with the preceding formal development, and while not needed for the logical development, should be covered somewhere in a graduate curriculum. The instructor may care to enlarge upon it, particularly if second quantization had been covered first.

Part II, *Relativistic Quantum Mechanics*, has optional chapters on integral forms of the Dirac equation (17) and on solving even relativistic integral equations (18). This material is not usually found in texts and gives an understanding of how the Dirac equation and computers are used in modern physics; it would be a shame if it were never read.

Part II covers the basic subjects in Chapters 19–22. Chapter 23, *The Breit–Pauli and Meson-Exchange Interactions*, is optional but mind-expanding as the reader sees how the Coulomb potential arises as an approximation to the real interaction between two electrons, and how similar it is to the nuclear force. I end Part II looking at the practical world, the

elegant, and the unknown. Chapter 24 surveys the weak interaction, and Chapter 25, *Wave Equations from Field Theory*, examines attempts at relativistic, 2-body wave equations. The use in Chapter 25 of so many of the theoretical tools in this book, along with the beauty, simplicity, and remaining mysteries of these equations, is the climax of the first three parts of the book.

As indicated by the quoted references, hardly any of this book describes my own contributions to physics. I am consequently deeply indebted to those who have put me in contact with this material and have tried to help me to understand it. It is my pleasure to acknowledge my teachers (who have no doubt been plagiarized by having their lectures and problems absorbed beyond recognition into “my” physics), my colleagues, in particular V. Madsen and A. Wasserman for their perceptive and caring comments, F. Tabakin for proselytizing momentum-space techniques, H. Jansen, J. Milana, D. Griffiths, M. Sagen, J. Schnick, G. He, and P. Fink for their input and assistance, and my students, who have helped me learn. Thanks also to the staff of John Wiley & Sons for their encouragement and assistance, in particular, Beatrice Shube, Maria Taylor, and Bob Hilbert.

Essentially all the writing of this text was undertaken at Oregon State University and my home starting in the summer of the 1986 and ending in the fall of 1988. Accordingly, I wish to acknowledge the support of the U.S. Department of Energy at Oregon State University and that of my family.

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*Your comedy I've read, my friend,
And like the half you pilfer'd best;
But sure the piece you yet may mend:
Take courage, man! and steal the rest.*

—Anonymous

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- “On His Books,” by Hilaire Belloc, *Complete Verse*, Gerald Duckworth & Co., Ltd.
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- “Upon Julia’s Clothes,” by Robert Herrick, *The Norton Anthology of English Literature*, Vol. 1, W. W. Norton & Co., Inc., New York.
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PART I

SCATTERING AND INTEGRAL QUANTUM MECHANICS

CHAPTER 1

SCATTERING

We study scattering theory because it is beautiful and it works. It works in the sense that scattering a beam of particles off a target is a successful technique to learn about a target which is too small to “see” in any other way (atoms, nuclei and particles). The beauty of scattering theory is that it works so well and that it is often an elegant application of quantum mechanics which becomes even more elegant with the use of modern computational techniques.

1.1 The Scattering Experiment

Those attributes of a scattering experiment making it such a powerful tool also limit what can be learned. To see this we examine Figure 1.1 which shows a schematic representation of a typical scattering experiment. On the left is an accelerator producing streams of particles which by passing through shielding, collimators, bending magnets and electromagnetic devices end up as a localized *beam* of definite energy and particle type. These *projectiles* are directed at a *target* which scatters some of them. We assume that the scattering is *gentle* enough for most of the beam to pass through the target and continue undisturbed (it may then be used in other experiments or discarded into the “beam dump”).

To describe Figure 1.1 quantitatively, we take the initial beam direction as the positive z axis and assign the projectile the momentum $\hbar\mathbf{k}_0$. The scattered particles are detected at a distance $\mathbf{r}$ from the target and at a scattering angle $\Omega \equiv (\theta, \phi)$.¹ The experimenter who does the detecting is free to vary the scattering angle Ω , and is obligated to use a variety of devices to determine the exact energy, position and nature of the scattered particle.

In the simplest scenario, a scattering experiment is conducted to determine the strength and functional dependence of the potential V acting between the projected particle P and the target particle T .² The potential, in turn, provides information on the structure and

¹It is common to use “natural” units in which $\hbar = 1$ and $c = 1$, that is, to measure angular momentum in units of $\hbar$ and velocities in units of c . With these units there is no distinction between momentum and wave vector. We will use natural units wherever possible, but sometimes convert the final equations to conventional units. See Appendix A, *Natural Units and Plane Waves*, and the Problems for exercises with natural units.

²See § 1.4 for a discussion of the particle concept.

interaction of the target and beam particles. In a theorist's eye, there is no distinction between the projectile and target particles. An experimentalist may view it differently because the target is usually fabricated from stable, workable, or abundant material, while the beam particle may be rare, unstable or even have an exceedingly short lifetime, in which case its lifetime is extended by relativistic time dilation.

Another assumption of scattering theory is that the projectile–target potential has a *finite range* R which is small compared with laboratory dimensions. While the exact value of R may be unclear for all but the square well, it is a measure of the distance beyond which the potential V becomes negligibly small relative to other energies in the problem.³ Because the range is of atomic size or smaller, on a scale set by R the observer and all the experimental equipment are effectively at infinity. And there is the rub, for with our equipment set at infinity we hope to learn about a submicroscopic potential!

Time-Dependent Approach

The preceding ideas are translated to quantum mechanics by describing Figure 1.1 with wave packets.⁴ The incident beam particle is described by a “free” wave packet ϕ_k :

$$\phi_k(\mathbf{r}, t) = \int d^3k e^{i\mathbf{k}\cdot\mathbf{r}} e^{-iE_k t} A_{k_0}(\mathbf{k}). \quad (1.1)$$

Here $A_{k_0}(\mathbf{k})$ is a sharply-peaked function about the beam momentum $\hbar\mathbf{k}_0 \equiv \mathbf{k}_o$, E_k is the energy of a free particle with momentum $\mathbf{k}$, and $k = |\mathbf{k}|$. For a *nonrelativistic* projectile of mass m_P incident upon an infinitely heavy target, the energy is:

$$E \equiv E_{k_0} = \frac{k_0^2}{2m_P}. \quad (1.2)$$

For a *relativistic* projectile,

$$E = E_{k_0} = \sqrt{\hbar^2 k_0^2 c^2 + m_P^2 c^4} \equiv \sqrt{k_0^2 + m_P^2}. \quad (1.3)$$

In either case the energy is positive, and because solutions of the Schrödinger equation exist for all positive energies, we are applying our quantum mechanics in the *continuum*. In § 1.2 and § 1.3 we discuss more general cases and kinematics.

The wave packet ψ for the entire system is the sum of the incident beam's wave packet ϕ_k plus another wave packet ψ^{sc} describing the scattered particle:

$$\psi(\mathbf{r}, t) = \phi_k(\mathbf{r}, t) + \psi^{\text{sc}}(\mathbf{r}, t), \quad (1.4)$$

$$\psi^{\text{sc}}(\mathbf{r}, t) = \int d^3k \frac{e^{i\mathbf{k}\cdot\mathbf{r}}}{r} e^{-iE_k t} A_{k_0}(\mathbf{k}) F(\mathbf{k}, \hat{\mathbf{r}}), \quad (1.5)$$

where $\hat{\mathbf{r}}$ is the unit vector in direction of the detected scattered wave. In (1.5), the function $A_{k_0}(\mathbf{k})$ is the same one used to describe the incident wave packet (1.1), $F(\mathbf{k}, \hat{\mathbf{r}})$ describes

³The Coulomb potential has infinite range and thus requires special treatment. See § 4.3.

⁴Most of our discussion of scattering will be done in a time-independent approach. The present discussion of wave packets is qualitative so the reader can appreciate the approximations we are making and the road we have not traveled. We will, however, use a time-dependent approach in our discussions of field theory, path integration, and fermion pairing. See Gottfried (1966), Goldberger and Watson (1963), or Rodberg and Thaler (1967) for a more complete discussion of wave packets.

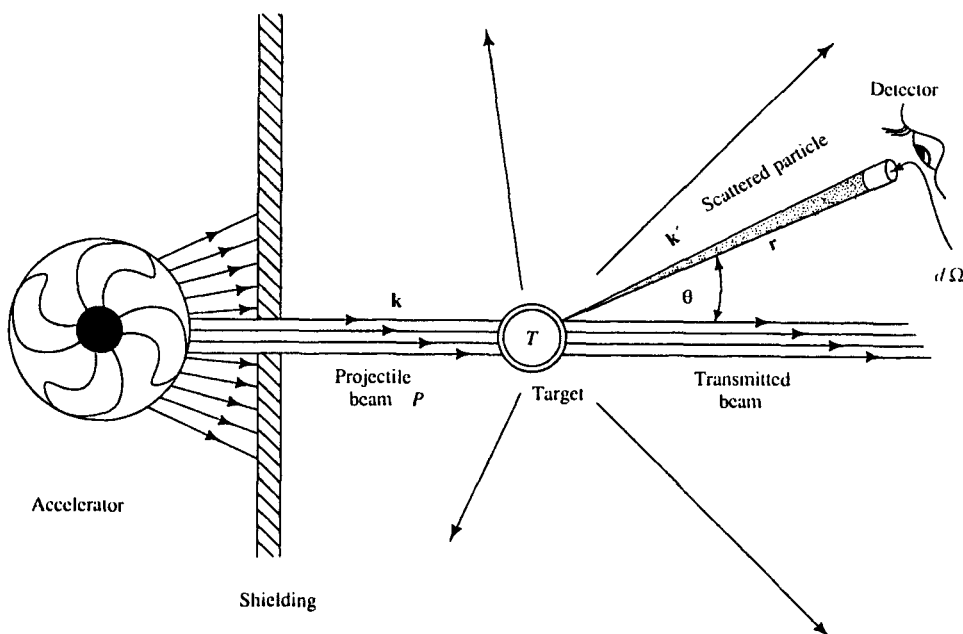


Figure 1.1: Some of the elements in an experiment to measure a differential cross section. The incident momentum of the projectile P is $\mathbf{k}$, the final momentum is $\mathbf{k}'$, and the location of the detector relative to the target is $\mathbf{r}$.

the strength and angular dependence of the scattering, and the outgoing spherical wave $\exp(i\mathbf{k}\mathbf{r})/r$ produces radiation of the scattered particle.

If the scattering is a gentle process, the outgoing wave packet will remain sharp and slowly spreading as it moves away from the target.⁵ The scattering-amplitude function $F(\mathbf{k}, \hat{\mathbf{r}})$ is then sharply peaked around the beam momentum $\mathbf{k}_0$, and so we evaluate it at $\mathbf{k}_0$ and factor it out of the integral:

$$\psi^{\text{sc}}(\mathbf{r}, t) \sim \frac{f_E(\theta, \phi)}{r} \phi_{\mathbf{k}}(r\hat{\mathbf{z}}, t), \quad (1.6)$$

$$f_E(\theta, \phi) = F(\mathbf{k}_0, \hat{\mathbf{r}}), \quad (1.7)$$

where we use the symbol $\sim$ for limit $r \rightarrow \infty$. Equation (1.6) says that the scattered wave packet has the same shape as the incident packet, that the intensity of the scattered beam falls off with an inverse square law, and that the amplitude of the scattered wave $f_E(\theta, \phi)$ is independent of the details of the experiment. This universal amplitude $f_E(\theta, \phi)$ is called the *scattering amplitude* and has the dimension of length. The goal of scattering theory is to determine as much about the potential V as possible from knowledge of $f_E(\theta, \phi)$ [or if life is being kind, to determine $f_E(\theta, \phi)$ given the potential V].

⁵If the scattering is too disruptive it may be impossible to separate the spreading from the scattering and the scattering theory we develop loses validity.

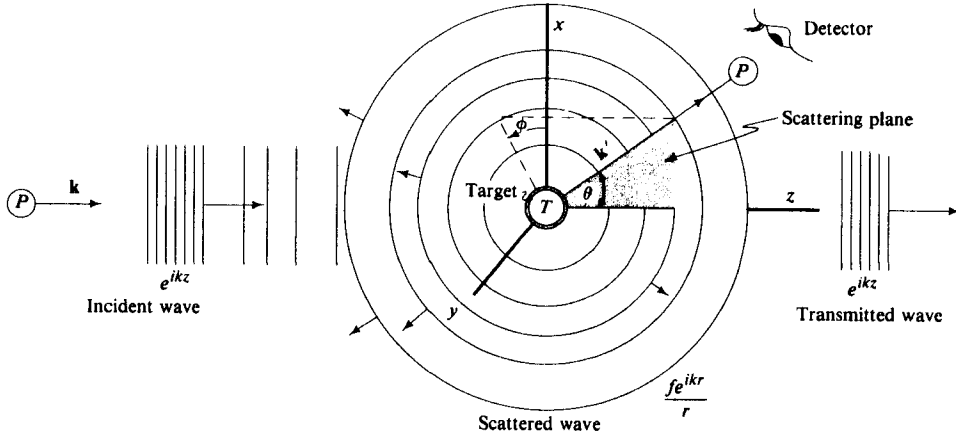


Figure 1.2: The incident, scattered, and transmitted waves. Note that the scattering plane is that formed by the vectors $\mathbf{k}$ and $\mathbf{k}'$, and the scattering angle θ and azimuth ϕ locate the detector relative to the beam direction.

The Time-Independent Trick

While wave packets are the most realistic description of a scattering experiment, they are so complicated that the less-realistic time-independent picture is almost always used. Because the time-independent approach deals with non-normalizable states, its mathematics is less rigorous and at times we will need our knowledge of wave packets to guide us. Nevertheless, the remainder of this book uses the time-independent view because it predicts the same observables in a less complicated way. The time-independent view is shown in Figure 1.2 where we imagine a plane wave continuously entering along the negative z axis. It is of course an idealization to have a wave of infinite extent in the xy plane which flows for all time.

In mathematical terms, the incident beam is represented by the plane wave:

$$\phi_{\mathbf{k}}(\mathbf{r}, t) = \phi_{\mathbf{k}}(\mathbf{r})e^{-iE_{\mathbf{k}_0}t}, \quad (1.8)$$

$$\phi_{\mathbf{k}}(\mathbf{r}) = Ne^{i\mathbf{k}\cdot\mathbf{r}} = Ne^{ikz}, \quad (1.9)$$

where the subscript $\mathbf{k}$ indicates the beam's momentum and thus energy $E_{\mathbf{k}}$ [since the plane wave describes free particles, its energy and momentum are related by (1.2) or (1.3)]. The constant N is a normalization factor which we leave arbitrary to emphasize the generality of the present concepts.⁶ The incident-plus-scattered-wave form of the wave packet (1.4) is also valid for time-dependent wave functions. When each term has the stationary-state time dependence $\exp[-iE_{\mathbf{k}_0}t]$ factored out, we obtain:

$$\psi_{\mathbf{k}}(\mathbf{r}) = \phi_{\mathbf{k}}(\mathbf{r}) + \psi_{\mathbf{k}}^{\text{sc}}(\mathbf{r}). \quad (1.10)$$

⁶In Chapter 3, *Partial-Wave Functions and Expansions*, we choose $N = 1/(2\pi)^{3/2}$.

Likewise, the radiation condition on the scattered wave (1.6) becomes

$$\psi_{\mathbf{k}}^{\text{sc}}(\mathbf{r}) \sim f_{\mathbf{E}}(\theta, \phi) \frac{e^{i\mathbf{k}\mathbf{r}}}{r}. \quad (1.11)$$

The full (*distorted*) wave function accordingly has the asymptotic form

$$\psi_{\mathbf{k}}(\mathbf{r}) \sim N \left[e^{i\mathbf{k}\mathbf{z}} + f_{\mathbf{E}}(\theta, \phi) \frac{e^{i\mathbf{k}\mathbf{r}}}{r} \right]. \quad (1.12)$$

1.2 The 1- and 2-Body Schrödinger Equations

The dynamics of projectile scattering is determined by the 2-particle Schrödinger equation,

$$\mathbf{H}_{\text{tot}} \Psi(\mathbf{r}_P, \mathbf{r}_T, t) = i \frac{\partial}{\partial t} \Psi(\mathbf{r}_P, \mathbf{r}_T, t). \quad (1.13)$$

Here $\mathbf{H}_{\text{tot}}$ is the 2-particle Hamiltonian operator⁷

$$\mathbf{H}_{\text{tot}} = \frac{\mathbf{p}_P^2}{2m_P} + \frac{\mathbf{p}_T^2}{2m_T} + V(\mathbf{r}_P - \mathbf{r}_T), \quad (1.14)$$

where $\mathbf{p}$ and $\mathbf{r}$ represent the momentum operator and the position, and P and T denote the projectile and target particles. For stationary states,

$$\Psi = \Psi(\mathbf{r}_P, \mathbf{r}_T) e^{-iE_{\text{tot}}t}, \quad (1.15)$$

the Schrödinger equation (1.13) takes the time-independent form

$$\mathbf{H} \Psi(\mathbf{r}_T, \mathbf{r}_P) = E_{\text{tot}} \Psi(\mathbf{r}_T, \mathbf{r}_P). \quad (1.16)$$

As it stands, (1.16) is a six-dimensional, partial differential equation, and is too difficult to solve directly. Fortunately a straightforward change of variables reduces this 2-body problem to two separate, effective 1-body problems.⁸ The transformation is treated in elementary textbooks and the Problems section. We let $\mathbf{r}$ be the relative position (separation) of the two particles, and $\mathbf{R}$ the position of their center of mass (CM):

$$\mathbf{r} = \mathbf{r}_P - \mathbf{r}_T, \quad (1.17)$$

$$\mathbf{R} = \frac{m_P \mathbf{r}_P + m_T \mathbf{r}_T}{m_T + m_P}. \quad (1.18)$$

The *conjugate* variables to $\mathbf{r}$ and $\mathbf{R}$ are then:⁹

$$\mathbf{p} \equiv \mathbf{k} = \frac{m_T \mathbf{p}_P - m_P \mathbf{p}_T}{m_T + m_P}, \quad (1.19)$$

$$\mathbf{P} = \mathbf{p}_T + \mathbf{p}_P. \quad (1.20)$$

⁷To avoid confusion or add emphasis, we sometimes denote coordinate-space operators by boldface print.

⁸The separation is not straight forward for the relativistic problem where each particle gets its own time. See Chapter 25, *Wave Equations from Field Theory*, (but maybe not until after you have read the whole book).

⁹See Messiah (1961), Newton (1966), and the Problems section for a discussion of canonical and conjugate variables, and Chapter 7, *Formal Quantum Mechanics*, for a discussion of operators and their representations.

Because these are *canonical* quantum-mechanical variables, they are represented by the coordinate-space operators

$$\mathbf{p} = -i\nabla_{\mathbf{r}} \equiv -i\nabla, \quad \mathbf{P} = -i\nabla_{\mathbf{R}}. \quad (1.21)$$

Conveniently, if the single-particle coordinates $(\mathbf{r}_P, \mathbf{p}_P)$ and $(\mathbf{r}_T, \mathbf{p}_T)$ are canonical, then the collective coordinates $(\mathbf{r}, \mathbf{p})$ and $(\mathbf{R}, \mathbf{P})$ are also, and in quantum mechanics this means they satisfy the commutation relations,

$$[r_j, p_j] = \hbar \equiv i, \quad [R_j, P_j] \equiv i, \quad [r_j, P_j] = 0, \quad \text{etc.} \quad (1.22)$$

Also conveniently, the Hamiltonian (1.14) separates into two independent parts:

$$\mathbf{H}_{\text{tot}} = \left[\frac{\mathbf{p}^2}{2\mu} + V(\mathbf{r}) \right] + \frac{\mathbf{P}^2}{2M}, \quad (1.23)$$

$$\mu = \frac{m_T m_P}{m_T + m_P}, \quad M = m_T + m_P. \quad (1.24)$$

We now guess that the total wave function is the product of a function of $\mathbf{r}$ times a function of $\mathbf{R}$:

$$\Psi(\mathbf{r}_T, \mathbf{r}_P) = \psi(\mathbf{r})\phi_{\text{CM}}(\mathbf{R}). \quad (1.25)$$

After substituting (1.25) into (1.16), we obtain one equation describing the relative motion

$$\left[\frac{\mathbf{p}^2}{2\mu} + V(\mathbf{r}) \right] \psi(\mathbf{r}) = E\psi(\mathbf{r}), \quad (1.26)$$

and another describing the CM's motion,

$$\frac{\mathbf{P}^2}{2M} \phi_{\text{CM}}(\mathbf{R}) = E_{\text{CM}} \phi_{\text{CM}}(\mathbf{R}), \quad (1.27)$$

$$E_{\text{tot}} = E + E_{\text{CM}}. \quad (1.28)$$

The solution of (1.26) will be our concern throughout much of Part I, *Scattering and Integral Quantum Mechanics*, of this book. In contrast, the solution of (1.27) is not much of a concern since it is the familiar plane wave:

$$\phi_{\text{CM}}(\mathbf{R}) = e^{i\mathbf{P} \cdot \mathbf{R}}, \quad (1.29)$$

$$E_{\text{CM}} = \frac{P^2}{2M}. \quad (1.30)$$

Equation (1.29) means that the system's CM moves as a free particle with momentum $\mathbf{P}$, regardless of how complicated the relative motion within the CM may be.¹⁰

¹⁰The center-of-mass frame has the position of the center of mass $\mathbf{R}$ (1.18) at the origin. In this frame, the total momentum (1.20) $\mathbf{P} = 0$. This sets $\mathbf{R}$ to rest so it may as well be at the origin. The center-of-momentum is more universal than the center of *mass* because it is well defined even for relativistic particles where the interparticle distances are distorted. In this book the two terms are used interchangeably with both abbreviated as "CM".

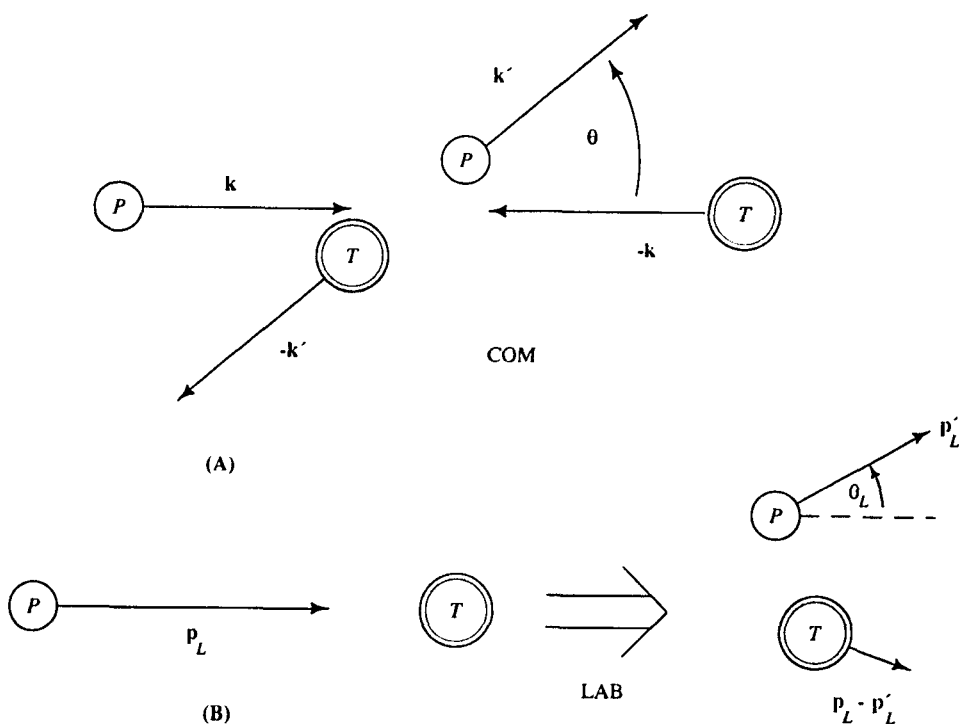


Figure 1.3: Kinematics for projectile–target scattering in (A), the center-of-momentum reference frame, and (B), in the laboratory.

1.3 Coordinate Systems and Elastic Scattering

Because $\mathbf{k}$ and E of the last section are the momentum and energy *in* (not of) the CM reference frame, when we speak of “solving the Schrödinger equation” in this book, we usually mean solving (1.26) for the relative wave function *in* the CM frame. For nonrelativistic problems the wave function describes an effective particle with reduced mass μ ; for relativistic problems the mass and equation may change, yet we will still do the solution in the CM. Consequently, in analyzing an actual experiment (or developing a multiparticle theory) we need to know how to transform to and from the CM frame. In this section we review the relation of variables in different frames.

As seen in Figure 1.3A, scattering in the CM is like a rotation in which the projectile and target make a head-on collision and then recoil back-to-back, the scattering angle (θ, ϕ) measuring the deflection of the projectile. Other than using the beam direction as the convention for positive z , there is no distinction in the CM between projectile and target particle. Our convention has the projectile receiving a momentum transfer $\mathbf{q}$:

$$\mathbf{q} = \mathbf{k}' - \mathbf{k}, \quad (1.31)$$

$$q^2 = (\mathbf{k}' - \mathbf{k})^2 = k'^2 + k^2 - 2kk' \cos \theta. \quad (1.32)$$

Because momentum is conserved, the target receives a momentum transfer of $-\mathbf{q}$. If the

scattering is *elastic*,¹¹ energy in the entrance channel is conserved, $k' = k$, and (1.32) reduces to

$$q^2 = 2k^2(1 - \cos\theta) = 4k^2 \sin^2 \frac{\theta}{2}. \quad (1.33)$$

The momentum transfer q clearly vanishes in the *forward* direction ($\theta = 0$) for elastic scattering, and is largest (twice the incident k) in the *backward* direction ($\theta = 180^\circ$).

While it is natural to formulate theories in the CM, it is usually more convenient to conduct experiments in the laboratory frame of reference (“lab”).¹² As shown in Figure 1.3B, in the lab frame the target is initially at rest, but recoils after the collision and thus carries off some of the energy of the incident projectile. The scattering angle in the lab θ_L refers to the deflection of the projectile relative to its initial direction and differs from θ in the CM.

Exercise¹³ Show that for nonrelativistic scattering the momentum transfer $\mathbf{q}$ is the same in the lab and CM frames:

$$\mathbf{q} = \mathbf{p}'_L - \mathbf{p}_L = \mathbf{k}' - \mathbf{k}. \quad \diamond \quad (1.34)$$

Exercise Show that for relativistic scattering the 3-momentum transfer $\mathbf{q}$ and the 4-momentum transfer q^μ differs in both frames, but that $q^\mu q_\mu$ is invariant.¹⁴ $\diamond$

When scattering is *elastic* the total kinetic energy remains unchanged in the collision process. In the CM this means $|\mathbf{k}'| = |\mathbf{k}|$, so the final (rotated) momenta lie on the shell formed in Figure 1.3A by using the initial momentum as a radius. If the kinetic energy (relativistically or nonrelativistically) changes in the collision, the scattering is *nonelastic*.¹⁵ In this case the collision will still look like a rotation in the CM, Figure 1.3A, yet because k' is generally less than k , it will not lie on the elastic shell of radius k . Note, energy is not “lost” in nonelastic collisions, but rather goes into internal excitation or rearrangement of the system, or creation of new particles.

In the lab frame it is less simple to tell if the collision is elastic because there is always a transfer of kinetic energy to the recoiling target particle (that is, $p'_L \leq p_L$). We know the scattering is elastic if the kinetic energy gained by the recoiling target equals the energy lost by the projectile,

$$E_T(\mathbf{p}'_L - \mathbf{p}_L) = m_T + E_P(\mathbf{p}_L) - E_P(\mathbf{p}'_L), \quad (1.35)$$

where we have included the target rest mass energy as part of E .

¹¹Definitions of elastic and channel to follow.

¹²Colliding beam experiments in which the target is also a beam of particles are exceptional because then the experiment is performed in the CM.

¹³We use the symbol $\diamond$ to denote the end of an exercise.

¹⁴Four vectors and Lorentz transformations are reviewed in Appendix C, *Four-Vectors and Lorentz Transformations*.

¹⁵We do not use the term “inelastic” because it is often reserved to mean a nonelastic collision in which a specific energy level is excited or deexcited. Nonelastic also includes reactions, breakup, particle creation, charge exchange, and so forth.

Elastic Kinematics

The invariance of the square of the total 4-momentum (the CM energy squared):¹⁶

$$s = (\mathbf{p}_P + \mathbf{p}_T)^2 = (E_P + E_T)^2 - (\mathbf{p}_P + \mathbf{p}_T)^2 \quad (1.36)$$

$$= m_P^2 + m_T^2 + 2E_P E_T - 2\mathbf{p}_T \cdot \mathbf{p}_P, \quad (1.37)$$

gives relations between the CM and lab momenta k and p_L (s can be evaluated in either frame).

Exercise Show that

$$s = [E_P(k) + E_T(k)]^2, \quad (1.38)$$

$$\Rightarrow k^2 = \frac{[s - (m_P + m_T)^2][s - (m_P - m_T)^2]}{4s}, \quad (1.39)$$

so knowing the CM energy gives the CM momentum k . $\diamond$

Exercise Show that

$$s = m_P^2 + m_T^2 + 2m_T E_P(p_L), \quad (1.40)$$

$$\Rightarrow p_L = \frac{k\sqrt{s}}{m_P}, \quad (1.41)$$

so knowing the lab momentum p_L gives the CM energy and thus CM momentum (and vice versa). $\diamond$

In elastic scattering the scattering angle and cross sections in the CM and lab frames are related by

$$\tan \theta_L = \frac{\sin \theta}{\cos \theta + m_P/m_T}, \quad (1.42)$$

$$\left. \frac{d\sigma}{d\Omega}(\theta, \phi) \right|_L = \left. \frac{d\sigma}{d\Omega}(\theta, \phi) \right|_{\text{CM}} \frac{d(\cos \theta)}{d(\cos \theta_L)}, \quad (1.43)$$

where (1.42) is valid only for nonrelativistic collisions. For relativistic collisions we again invoke Lorentz invariance to relate the scattering angles.

Exercise Show that the invariance of the magnitude of 4-momentum transfer,

$$t = (\mathbf{p}'_P - \mathbf{p}_P)^2 = (\mathbf{p}'_T - \mathbf{p}_T)^2, \quad (1.44)$$

leads to the relation of scattering angles

$$k^2 \cos \theta = E_P(k)^2 - E_P(p_L)E_P(p'_L) + p_L p'_L \cos \theta_L. \quad \diamond \quad (1.45)$$

¹⁶See Hagedorn (1963), Sard (1970), Jackson (1975), and Eden (1967).

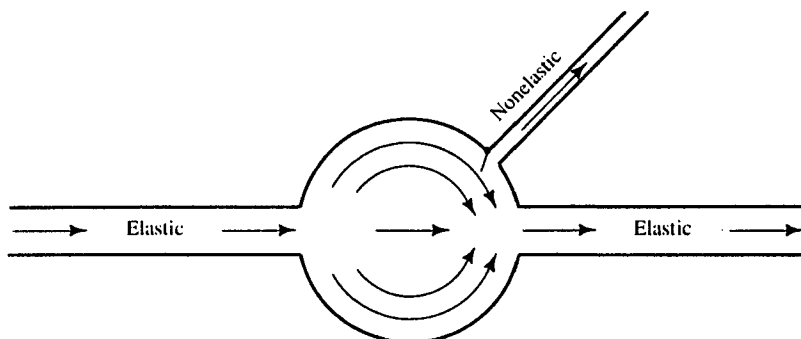


Figure 1.4: Flux flowing from the entrance elastic channel into the exit elastic channel and into a nonelastic channel. The gate in front of the nonelastic channel indicates that it may be closed for low energies.

Exercise Lorentz transformations do not affect momenta transverse to the transformation direction. Show that the invariance of the transverse momenta means

$$p'_L \sin \theta_L = k \sin \theta. \quad \diamond \quad (1.46)$$

As the target particle becomes heavier and heavier, the velocity difference between the CM and lab frames becomes smaller and smaller, as an inspection of these equations shows. For an extremely massive target, a solution of the 2-body Schrödinger equation becomes equivalent to scattering from a *fixed* (nonrecoiling) center of force. When the target and beam particles are of *equal* mass and nonrelativistic, we obtain the noteworthy constraint

$$\theta_L = \frac{\theta}{2}, \quad (m_P = m_T). \quad (1.47)$$

This means that 90° is the largest value possible for θ_L .

1.4 The Particle and Channel Concepts

We have been using the term “particle” often without saying what it means. When we speak of a “particle” we envision an elementary object occupying only a point in space and having no internal structure. This concept is obviously situation- or energy-dependent; if the beam particle does not have enough energy to excite the target or itself, then both appear elementary. This idealization is used throughout physics. In calculating astronomical orbits a rock is elementary, yet given more and more energy, the rock breaks into pieces, the pieces break into molecules, the molecules into atoms, the atoms into electrons and nuclei, the nuclei into nucleons, and so forth. (Electrons, however, appear to be elementary, no measurements have shown them to have excited states or spatial extent.) In quantum mechanics we account for the energy dependence of the particle concept by viewing each excited state of the system as a different particle. For example, hydrogen in an excited state is no longer the “hydrogen” it was in its ground state. This is not really so strange, for when relativistic effects are included, the excited states have different masses.

The term *channel* is an analogy which has crept into quantum mechanical jargon. We use it as an analogy to the flow of a fluid through a canal or channel. The incident channel is the elastic one and is always *open*. Each excited state of either the projectile or target is considered a separate channel, and the channel is *closed* (as if a flood gate is down) unless there is enough energy to reach that excited state (open the gate). When a channel opens, additional flux from the incident channel flows into that open channel, as pictured in Figure 1.4. When a virtual process occurs in quantum mechanics, it is analogous to fluid leaking through a flood gate—but with no net current flowing into the inelastic channel. Likewise, as the energy increases, we envision the gates on some energy-forbidden (closed) channels as gradually lifting open and permitting current to flow into them.

1.5 Problems

1. A projectile P scatters from a target particle T in an arbitrary reference frame.
 - (a) Under what conditions can a change of variables be made permitting the 2-particle Schrödinger equation to be separated into two, 1-particle equations?
 - (b) What are the two 1-particle equations and what are the “particles” in each?
 - (c) The new variables satisfy commutation relations of the form:

$$[r_j, p_j] \equiv i, \quad [R_j, P_j] \equiv i, \quad [r_j, R_j] = 0, \quad \text{etc.} \quad (1.48)$$

Derive or verify at least one of these commutation relations.

2. Examine the scattering of a projectile with mass m_P from a target of mass m_T as viewed in the laboratory and center-of-momentum reference frames.
 - (a) Relate the scattering angle in the two frames for nonrelativistic scattering.
 - (b) Relate the differential cross sections in the two frames for nonrelativistic scattering.
 - (c) Relate the CM momentum k to the lab momentum p_L for relativistic scattering.
 - (d) If the target and projectile particles have equal mass, what is the maximum possible scattering angle that can occur in the CM?
 - (e) Can a 1-eV electron scatter nonelastically from hydrogen? Explain.
 - (f) Can a 1-eV positron scatter nonelastically from hydrogen? Explain.
3. An interaction (event A) gives rise to an electron of speed 0.5. The electron travels 100m at a constant velocity and then hits an atom (event B). In the rest frame of the electron, what is
 - (a) The spatial separation of the two events.
 - (b) The distance between A and B.
 - (c) The temporal separation of the two events.
 - (d) The velocity of the laboratory.

4. A neutron with its spin aligned along its velocity and with a velocity $v = 0.99$ decays into an electron, proton, and antineutrino:

$$n \rightarrow p + e + \bar{\nu}. \quad (1.49)$$

Because the antineutrino is massless, it has speed $c = 1$. In the neutron's rest frame, the antineutrino has an angular distribution

$$\sigma(\theta) = \sigma_0(1 - \frac{1}{2} \cos \theta), \quad (1.50)$$

where θ is the angle between the neutron's spin and the antineutrino's momentum.

- (a) What is the angular distribution of the antineutrino as observed in the laboratory reference frame?
 - (b) Plot the distributions in the two frames and note if there is any "beacon effect" (a narrowing of emitted radiation).
5. Calculate in SI units the following natural-unit quantities for the electron (*Hint*: Use $mc^2 = 0.511$ MeV, $\hbar c = 197.32$ MeV fm, $c = 3 \times 10^{10}$ cm/sec):
- (a) The mass m .
 - (b) The radius $1/m$.
 - (c) The momentum m .
 - (d) The velocity $\frac{1}{2}$.
 - (e) The fine structure constant e^2 .
6. Consider scattering in which $m_T = m_P/6$, the beam has a lab kinetic energy ("energy") of m_P , and in which the projectile is observed at 30° in the lab.
- (a) For elastic scattering, how much lab energy does the projectile lose? (Either relativistic or nonrelativistic kinematics is acceptable.)
 - (b) If only the projectile is observed, how would an experimenter distinguish elastic from nonelastic scattering?
 - (c) What is θ_{CM} as calculated with the relativistic and nonrelativistic expressions?
 - (d) What is the relation between cross sections as calculated with the relativistic and nonrelativistic expressions?

CHAPTER 2

CURRENTS AND CROSS SECTIONS

In the last chapter we discussed how the physical arrangement of a scattering experiment determines the boundary conditions on the wave function we use to describe scattering. It would be pleasant to think that a physicist's work is done once he or she solves the Schrödinger equation for a wave function. Unfortunately, wave functions are not observable, and part of the value of scattering theory is its reformulation of quantum mechanics to make the dynamical quantities more directly related to experiment. In this section we review how quantum-mechanical currents are related to experimental cross sections and scattering amplitudes; later we will introduce a dynamical quantity equivalent to the scattering amplitude.

2.1 Elastic Scattering Currents

A classical current is a density (number per unit volume) times a velocity. In quantum mechanics, the density is $\psi^* \psi$, the “velocity” $\mathbf{v}$ is the momentum operator ∇/i divided by the mass, and so the current becomes the function

$$\mathbf{j}(\mathbf{r}) = \text{Re} [\psi^*(\mathbf{r}) \mathbf{v} \psi(\mathbf{r})] \equiv \text{Re} \left[\psi^*(\mathbf{r}) \frac{\nabla}{i\mu} \psi(\mathbf{r}) \right] \quad (2.1)$$

$$= \frac{1}{2\mu i} [\psi^* \nabla \psi - \psi \nabla \psi^*]. \quad (2.2)$$

If we choose an incident plane wave e^{ikz} for ψ , we obtain $\mathbf{j}_{\text{in}}$, the current incident upon the target:

$$\mathbf{j}_{\text{in}} = \text{Re} \left[\frac{N^* e^{-ikz} i k \hat{\mathbf{z}} N e^{ikz}}{i\mu} \right] = \frac{|N|^2}{\mu} \mathbf{k} = |N|^2 \mathbf{v}_{\text{in}} \quad (2.3)$$

$$\equiv j_{\text{in}} \hat{\mathbf{z}}, \quad (2.4)$$

where $\mathbf{v}_{\text{in}}$ is the incident beam velocity and $\hat{\mathbf{z}}$ is a unit vector in the positive z direction. If we choose a scattered wave,

$$\psi_{\mathbf{k}}^{\text{sc}} = N \frac{e^{i\mathbf{k}\mathbf{r}}}{r} f_{\mathbf{E}}(\theta, \phi), \quad (2.5)$$

we obtain the current scattered (radiating) from the target:

$$\mathbf{j}_{\text{sc}}(\mathbf{r}) = \frac{1}{\mu} |N|^2 |f_{\mathbf{E}}(\theta, \phi)|^2 \text{Re} \left[\frac{e^{-i\mathbf{k}\mathbf{r}}}{ir} \left(i\mathbf{k} \frac{e^{i\mathbf{k}\mathbf{r}}}{r} - \frac{e^{i\mathbf{k}\mathbf{r}}}{r^2} \right) \right] \hat{\mathbf{r}} \quad (2.6)$$

$$\sim |N|^2 \frac{|f_{\mathbf{E}}(\theta, \phi)|^2}{r^2} v_{\text{out}} \hat{\mathbf{r}}. \quad (2.7)$$

Exercise Verify (2.7) making sure to consider only the radial part of the gradient operator (the angular parts do not lead to radiation); drop terms which fall off faster than r^{-2} , and add the label “out” to v as a reminder that it is the velocity of the wave scattered “out” of the target. $\diamond$

Equations (2.4) and (2.7) can be combined to yield the proportionality between scattered and incident currents:

$$\mathbf{j}_{\text{sc}} = j_{\text{in}} \frac{|f_{\mathbf{E}}(\theta, \phi)|^2}{r^2} \frac{v_{\text{out}}}{v_{\text{in}}} \hat{\mathbf{r}}. \quad (2.8)$$

Equation (2.8) again illuminates why $f_{\mathbf{E}}(\theta, \phi)$ is called the scattering amplitude, and why it has the dimension of length.

2.2 Differential Cross Sections

The scattering cross section is an experimental observable which is independent of the equipment used to measure it; as such it is a convenient and universal measure of the strength of scattering. Classically it is the effective cross sectional area the target presents to the beam (indeed, it is πR^2 for a hard sphere of radius R). The quantum-mechanical cross section describes the scattering of waves, and much like the scattering of light waves, often varies rapidly with energy and angle. The analogy is so strong that the literature of scattering theory borrows optical terms such as diffraction, dispersion, absorption, interference, and index of refraction.

The differential cross section for elastic scattering $d\sigma/d\Omega$ is defined with reference to the scattering setup of Figure 1.2,

$$\frac{d\sigma}{d\Omega}(\theta, \phi) = \lim_{\Delta\Omega \rightarrow 0} \frac{N(\theta, \phi)/\Delta\Omega}{N_{\text{in}}/\Delta A_{\text{in}}}. \quad (2.9)$$

That is, to obtain a quantity independent of detector size, we divide the number of particles scattered into the detector N , by the solid angle subtended by the detector $\Delta\Omega$, and then take the limit for infinitesimally small detector size and infinitesimally thin targets. To obtain a quantity independent of beam current and target size, we divide by the number of incident particles per unit target area ΔA_{in} orthogonal to the beam. Because both N and

N_{in} increase with time, it is convenient to divide numerator and denominator by some time interval Δt , and take the limit as $\Delta t \rightarrow 0$ so as to get currents:¹

$$\frac{d\sigma}{d\Omega}(\theta, \phi) = \frac{(N/\Delta t) \div \Delta\Omega}{\text{Flux}} \quad (2.10)$$

$$= \frac{(j_{\text{sc}}\Delta A) \div (\Delta A/r^2)}{j_{\text{in}}}. \quad (2.11)$$

If we substitute (2.8) for j_{sc} and cancel the elementary area ΔA , we obtain the basic relation of the differential cross section to the scattering amplitude:

$$\frac{d\sigma}{d\Omega}(\theta, \phi) = |f_{\mathbf{E}}(\theta, \phi)|^2 \frac{v_{\text{out}}}{v_{\text{in}}}. \quad (2.12)$$

For elastic scattering the incident and outgoing velocities are equal, and so

$$\boxed{\frac{d\sigma}{d\Omega}(\theta, \phi) = |f_{\mathbf{E}}(\theta, \phi)|^2 \quad (\text{elastic scattering}).} \quad (2.13)$$

It is clear from (2.13) that $d\sigma/d\Omega$ has the dimension of area, or area per solid angle (per steradian). For atomic scattering, a natural unit for $d\sigma/d\Omega$ is angstrom squared/steradian or Bohr radius squared/steradian; for nuclear scattering a natural unit is Fermi squared/steradian or millibarn/steradian.² (The *Barn* is a convenient unit for cross sections arising in the scattering of low energy neutrons from nuclei; for most other nuclear processes it is so large that a target producing a cross section of a Barn would be as easy to hit as “the broad side of a barn.”)

The cross section (2.13) is the differential cross section [sometimes called $\sigma(\theta, \phi)$] and is a measure of the strength of scattering into a detector at angle (θ, ϕ) subtending a solid angle $d\Omega$. If the forces causing the scattering are *central*, the scattering must be *axially symmetric*, that is, $d\sigma/d\Omega$ will be independent of ϕ :

$$\frac{d\sigma}{d\Omega}(\theta, \phi) = \frac{d\sigma}{d\Omega}(\theta) \quad (\text{central forces}). \quad (2.14)$$

In general the differential cross section varies with scattering angle and energy.

Nonelastic Cross Sections (Absorption)

In a measurement of $d\sigma/d\Omega$ we would know if the scattering is elastic by determining the energy of the scattered particle E' and checking that it agrees with the energy expected (§ 1.3) for elastic scattering at this angle, that is, if $k' = k$ in the CM. When nonelastic scattering occurs there will be an energy loss even in the CM, and it is natural to make the differential cross section a doubly-differential one by including the differential energy or momentum loss:

$$\frac{d\sigma}{d\Omega}(\theta, \phi) \rightarrow \frac{d^2\sigma}{d\Omega dE'}(\theta, \phi, E'). \quad (2.15)$$

Because real detectors have finite energy resolution, a measured $d^2\sigma/(d\Omega dE')$ is finite, even for discrete final states. In the limit of infinitely high energy resolution, scattering to discrete final states (such as elastic scattering) would produce a doubly differential cross section proportional to $\delta(E - E')$.

¹ You will recall that the current density is a number per unit time per unit area.

² 1 Fermi = 10^{-13} cm $\equiv$ 1 fm, 1 barn = 10^{-24} cm², 1 millibarn $\equiv$ mb = fm²/10.

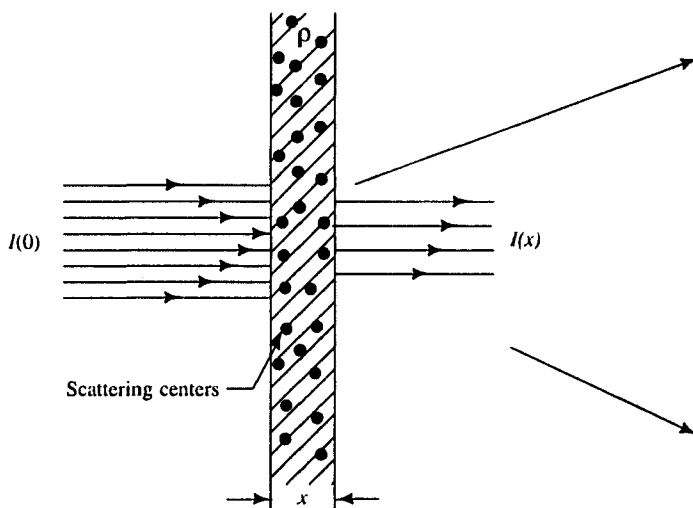


Figure 2.1: A setup for measuring a total cross section by observing the depletion of a beam's intensity from $I(0)$ before the target to $I(x)$ after the target.

2.3 Total Cross Section

The *total cross section*, or more precisely the *integrated elastic cross section*, is defined as:

$$\sigma_{\text{el}}(E) = \int d\Omega \frac{d\sigma}{d\Omega}(\theta, \phi) = \int_0^{2\pi} d\phi \int_{-1}^1 d(\cos \theta) \frac{d\sigma}{d\Omega}(\theta, \phi). \quad (2.16)$$

When $d\sigma/d\Omega$ is isotropic, $\sigma_{\text{el}} = 4\pi d\sigma/d\Omega$; otherwise an integration is necessary. Knowledge of the total cross sections for each final state provides a *phenomenological* picture of the strength and nature of the scattering as a function of energy. As such, total cross sections are valuable quantities.

If nonelastic scattering occurs, beam particles are said to be *absorbed* out of the elastic channel; there may still be some states of the beam or target particles around, but if they are not identical to the initial ones they are no longer the same “particles.”³ When nonelastic scattering occurs, the concept of total cross section is extended to include integrated scattering for both the elastic and nonelastic currents:

$$\sigma_{\text{t}} = \int d\Omega \left(\frac{d\sigma_{\text{el}}}{d\Omega} + \frac{d\sigma_{\text{ne}}}{d\Omega} \right) \quad (2.17)$$

$$= \sigma_{\text{el}} + \sigma_{\text{ne}}. \quad (2.18)$$

The importance of total cross sections in understanding a reaction is increased by the relative ease with which they are measured. In Figure 2.1 we present a schematic diagram of the setup of an experiment to measure total cross sections. Because we are interested in the total scattering out of the beam, it is easier to compare the beam intensity before and after the scattering than it is to measure all possible scattered particles. A beam of *intensity* (number of particles per unit area per unit time) $I(0)$ enters a slab of target material of

³The particle and channel concepts were discussed in § 1.4

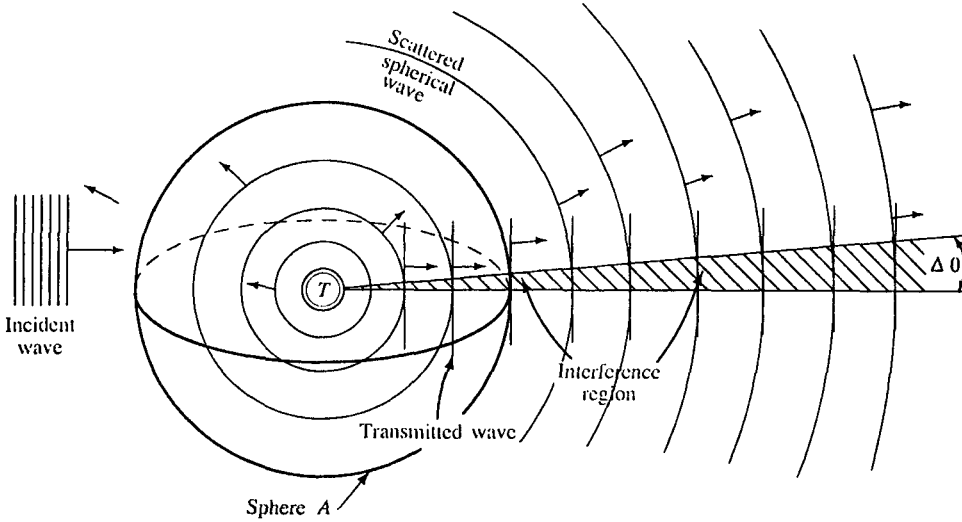


Figure 2.2: A sphere A surrounding the target T . The incident plane wave interferes with the forward-scattered scattered wave in the shaded region.

thickness x and number density (scattering centers per unit volume) ρ , and emerges with intensity $I(x)$. For a target of infinitesimal thickness dx , the total number of scatterings is proportional to the decrease in intensity dI . To obtain the total cross section, we divide dI by those quantities which depend on the “details” of the experiment: the beam intensity I and the number of target particles per unit area perpendicular to the beam, ρdx . This yields:

$$\sigma_t = \frac{-dI}{\rho dx I}. \quad (2.19)$$

The decrease in intensity in passing through dx is thus:

$$dI = -\sigma_t \rho dx I. \quad (2.20)$$

For a target of finite thickness, the intensity decreases continuously as the beam passes through the target, and so we integrate to obtain the final intensity,

$$\int_{I(0)}^{I(x)} \frac{dI}{I} = - \int_0^x dx \sigma_t \rho, \quad (2.21)$$

$$\ln I(x) - \ln I(0) = -\sigma_t \rho x, \quad (2.22)$$

$$I(x) = I(0)e^{-\sigma_t \rho x}, \quad (2.23)$$

where we have assumed the density is uniform and the beam’s energy is not changed as it passes through the target. The total cross section is thus determined from the simple relation

$$\sigma_t = \frac{\ln[I(0)/I(x)]}{\rho x}, \quad (2.24)$$

which only requires a measurement of the relative change in beam intensity.

Because an experimenter is free to change target materials and vary the beam energy, it is a simple matter to determine the dependences of the total cross section on target material

and beam energy. In practice there are important corrections to this experiment making it less simple. For example, while scattered particles should not be counted as part of the transmitted beam, finite-sized detectors must be used and it is very difficult to separate particles scattered at small angles from those of the beam. (The solution often lies in using detectors of varying sizes and then extrapolating to zero size.) Furthermore, the question of small-angle scattering becomes more of a problem if the beam and the target particles are charged, since then there are infinitely many small-angle scatterings of all beam particles by the Coulomb force. This leads to a serious problem in defining the total cross section for charged particles. While in a strict sense, Coulomb scattering should be ruled out anyway because it does not agree with our assumption of a finite-range interaction, it does become finite when all screening effects are included, and in the real world is always present for charged particles.⁴

2.4 The Optical Theorem

We know that the full wave function has the asymptotic form of a plane plus scattered spherical wave (Figures 1.2 and 2.2):

$$\psi_{\mathbf{k}}(\mathbf{r}) \sim N \left[e^{i\mathbf{k}z} + \frac{e^{i\mathbf{k}r}}{r} f_{\mathbf{B}}(\theta, \phi) \right], \quad (2.25)$$

and it is logical to assume that the full current also has this form, that is,

$$\mathbf{j} \sim \mathbf{j}_{\text{in}} + \mathbf{j}_{\text{sc}}. \quad (2.26)$$

However, this is wrong because the quantum-mechanical current (2.2) is not a linear operator on the wave function, and there is an interference term to include:

$$\mathbf{j} = \text{Re} \left[(\psi_{\text{sc}} + \psi_{\text{in}})^* \frac{\nabla}{i\mu} (\psi_{\text{sc}} + \psi_{\text{in}}) \right] \quad (2.27)$$

$$\sim \mathbf{j}_{\text{in}} + \mathbf{j}_{\text{sc}} + \mathbf{j}_{\text{int}}. \quad (2.28)$$

The insertion of the wave function (2.25) into the expression for the current (2.2) yields:

$$\begin{aligned} \mathbf{j} &\sim \frac{|N|^2}{\mu} \text{Re} \left[\left(e^{-i\mathbf{k}z} + f_{\mathbf{B}}(\theta, \phi)^* \frac{e^{-i\mathbf{k}r}}{r} \right) i^{-1} \right. \\ &\quad \times \left(i\mathbf{k}e^{i\mathbf{k}z} \hat{\mathbf{z}} + \frac{f_{\mathbf{B}}(\theta, \phi)}{r} i\mathbf{k}e^{i\mathbf{k}r} \hat{\mathbf{r}} - \frac{f_{\mathbf{B}}(\theta, \phi)}{r^2} e^{i\mathbf{k}r} \hat{\mathbf{r}} \right) \Big] \end{aligned} \quad (2.29)$$

$$\begin{aligned} &= \frac{|N|^2}{\mu} \text{Re} \left[\mathbf{k} \hat{\mathbf{z}} + \frac{f_{\mathbf{B}}}{r} \mathbf{k} e^{i\mathbf{k}(r-z)} \hat{\mathbf{r}} + i \frac{f_{\mathbf{B}}}{r^2} e^{i\mathbf{k}(r-z)} \hat{\mathbf{r}} + \frac{f_{\mathbf{B}}^*}{r} \mathbf{k} e^{i\mathbf{k}(z-r)} \hat{\mathbf{z}} \right. \\ &\quad \left. + \frac{|f_{\mathbf{B}}|^2}{r^2} \mathbf{k} \hat{\mathbf{r}} + i \frac{|f_{\mathbf{B}}|^2}{r^3} \hat{\mathbf{r}} \right]. \end{aligned} \quad (2.30)$$

We recognize in (2.30) the incident and scattered currents, (2.3) and (2.5). After dropping the r^{-3} and $e^{i\mathbf{k}(r-z)} r^{-2}$ terms because they do not radiate scattered particles from the target, we obtain the interference current [last term in (2.32)]:

$$\mathbf{j} \sim j_{\text{in}} \hat{\mathbf{z}} + j_{\text{sc}} \hat{\mathbf{r}} + j_{\text{in}} \frac{1}{2} \text{Re} [f_{\mathbf{B}} r^{-1} e^{i\mathbf{k}(r-z)} \hat{\mathbf{r}} + f_{\mathbf{B}}^* r^{-1} e^{-i\mathbf{k}(r-z)} \hat{\mathbf{z}}] \quad (2.31)$$

$$= j_{\text{in}} \hat{\mathbf{z}} + j_{\text{sc}} \hat{\mathbf{r}} + j_{\text{in}} \text{Re} [f_{\mathbf{B}}(\theta, \phi) r^{-1} e^{i\mathbf{k}(r-z)} (\hat{\mathbf{r}} + \hat{\mathbf{z}})]. \quad (2.32)$$

⁴See § 4.3 for a discussion of Coulomb scattering.

Exercise Verify (2.32). ◇

Although the physics of the interference current may not appear evident at first, we discover the need for it by examining the continuity equation. For wave functions described by the Schrödinger equation with Hermitian potentials,⁵ the continuity equation has the form:

$$\nabla \cdot \mathbf{j} + \frac{\partial \rho}{\partial t} = 0. \quad (2.33)$$

If we multiply both terms by d^3r , integrate over all space, interchange the order of integration and time differentiation, we obtain

$$\int d^3r \nabla \cdot \mathbf{j} + \frac{\partial}{\partial t} \int d^3r \rho = 0. \quad (2.34)$$

The second integral vanishes because it is the time derivative of the total probability (we assume no sinks or sources of particles). The first integral is converted to a surface integral by invoking the divergence theorem:

$$\int d^3r \nabla \cdot \mathbf{j} = \int d\mathbf{A} \cdot \mathbf{j} = 0, \quad (d\mathbf{A} \stackrel{\text{def}}{=} r^2 d\Omega). \quad (2.35)$$

Here we imagine a sphere A as in Figure 2.2 surrounding the target. When we substitute the total current, we obtain

$$\int d\mathbf{A} \cdot \mathbf{j}_{\text{in}} + \int d\mathbf{A} \cdot \mathbf{j}_{\text{sc}} + \int d\mathbf{A} \cdot \mathbf{j}_{\text{int}} = 0. \quad (2.36)$$

The first term in (2.36) must vanish because whatever incident current enters on the left must leave on the right, that is,

$$\int d\mathbf{A} \cdot \mathbf{j}_{\text{in}} = j_{\text{in}} \int_{-1}^1 d \cos \theta \int_0^{2\pi} d\phi \hat{\mathbf{z}} \cdot \hat{\mathbf{r}} = j_{\text{in}} \int_{-1}^1 d \cos \theta 2\pi \cos \theta = 0. \quad (2.37)$$

This leaves

$$\int d\mathbf{A} \cdot \mathbf{j}_{\text{sc}} + \int d\mathbf{A} \cdot \mathbf{j}_{\text{int}} = 0. \quad (2.38)$$

The first term in (2.38) measures the total scattering out of the sphere and is thus proportional to the integrated (total) elastic cross section:

$$\int d\mathbf{A} \cdot \mathbf{j}_{\text{sc}} = \frac{j_{\text{in}}}{r^2} \int r^2 d\Omega |f_{\mathbf{E}}(\theta, \phi)|^2 = j_{\text{in}} \int d\Omega \frac{d\sigma}{d\Omega} = j_{\text{in}} \sigma_{\text{el}}, \quad (2.39)$$

where we have used (2.16). The second term in (2.38) is tricky to evaluate because it contains a delicate interference between the scattered and the incident waves, an interference vanishing in all but the forward direction. We write it as

$$\int d\mathbf{A} \cdot \mathbf{j}_{\text{int}} = \frac{j_{\text{in}}}{r} 2\pi r^2 \text{Re} \int_{-1}^{+1} d \cos \theta e^{ikr(1-\cos \theta)} (1 + \cos \theta) f_{\mathbf{E}}(\theta), \quad (2.40)$$

⁵For most cases we can safely think of a Hermitian potential as a real potential. It is possible, however, to construct real, nonhermitian potentials. See the Problems section for a discussion of the modifications of the continuity equation for nonhermitian potentials

and evaluate the integral using the stationary phase approximation (Mathews and Walker, 1964). Because the scattering cross sections and amplitudes are defined in the asymptotic limit $r \sim \infty$, we normally expect this integral to vanish as the exponential develops infinitely many oscillations. The exception occurs when $\cos \theta \simeq 1$ (the shaded region in Figure 2.2) because here the exponential has a constant phase of zero. [These oscillations do average to zero, the limits do become well-defined, and the approximations do become exact when we integrate over k in the more rigorous wave packet (time-dependent) picture.] We therefore approximate the integral by evaluating it over a small range of angles $\Delta\theta$, and factor out $f_E(\theta)$ evaluated at $\theta = 0$:⁶

$$\int d\mathbf{A} \cdot \mathbf{j}_{\text{int}} \sim \frac{j_{\text{in}}}{r} 4\pi r^2 \text{Re} \left[f_E(0^\circ) \int_{1-\frac{1}{2}\Delta\theta^2}^{+1} d(\cos \theta) e^{ikr(1-\cos \theta)} \right] \quad (2.41)$$

$$= j_{\text{in}} 4\pi r \text{Re} \left[\frac{f_E(0^\circ)}{-ikr} (1 - e^{ikr\Delta\theta^2/2}) \right]. \quad (2.42)$$

We drop the exponential term because it too would vanish if we integrated over k . By combining these equations we obtain the optical theorem:

$$\sigma_{\text{el}} = \frac{4\pi}{k} \text{Im} f_E(0^\circ). \quad (2.43)$$

For elastic scattering the integrated elastic cross section σ_{el} is also the total cross section, and we rewrite (2.43) in its more general form,

$$\sigma_t = \frac{4\pi}{k} \text{Im} f_E(0^\circ). \quad (2.44)$$

To shed light on the importance of the optical theorem, we present these reflections:

- Those particles removed from the forward part of the beam feed all other scattering or reaction processes. The depletion of the incident beam arises from its interference with the scattered beam in the shaded area of Figure 2.2.
- The optical theorem is a statement of *unitarity*, that is, probability conservation, and is expected to be true under the most general of circumstances, even when the Schrödinger equation is not valid.
- Equation (2.44) holds even in the presence of absorption or reactions, as proven by extending the continuity equation to include sources or sinks of flux (see the Problems section).
- Because σ_t cannot vanish if there is any scattering at all, the scattering amplitude always has an imaginary part (even for real potentials).
- In order for the division by k in (2.44) not to yield an infinite cross section as k approaches zero, the scattering amplitude must have a linear dependence on momentum:

$$\lim_{k \rightarrow 0} \text{Im} f_E(0^\circ) = C k + O(k^2). \quad (2.45)$$

⁶This procedure fails if f is singular at 0° . For finite range potentials, f is usually finite; the infinite range Coulomb potential has a singular f at the 0° and the total cross section is not well defined.

- Because total cross sections are easier to measure than differential ones, (2.44) is often used to determine $f_E(0^\circ)$ over a wide range of energies. In addition, because differential cross sections provide only the magnitude of $f_E(\theta, \phi)$, total cross sections help determine the separate real and imaginary parts of f_E . By use of complex analysis it is also possible to relate $\text{Re} f_E$ to $\text{Im} f_E$ by dispersion relations. Introductory discussions of dispersion relations are found in Sakurai (1964), Newton (1966), and Goldberger and Watson (1963).

2.5 Problems

1. We derived the optical theorem for pure elastic scattering, in which case $\sigma_t = \sigma_{el}$. Nonelastic scattering and reactions are included in the single-channel formalism by viewing these processes as “absorbing” particles from the incident beam, with the absorption described by a complex potential,

$$V(r) = U(r) + iW(r). \quad (2.46)$$

- (a) Derive the continuity equation for a *time-dependent* Schrödinger equation including a complex potential.
- (b) Show that this leads to the relation

$$\frac{\partial}{\partial t} \int d^3r |\psi|^2 = 2 \int d^3r W(r) |\psi|^2 - \int dA \cdot \frac{\psi^* \nabla \psi - \psi \nabla \psi^*}{2\mu i}. \quad (2.47)$$

- (c) Show that $W(r)$ must be negative to be a “sink” rather than “source” of flux.
2. Prove that the optical theorem is valid with σ_t including nonelastic events.
 3. Consider the *index of refraction* n of classical optics.
 - (a) What is its definition?
 - (b) Indicate how a complex n is related to the wave vector k of a plane wave. In particular, what is the form of the wave inside an absorptive medium?
 - (c) How is the mean free path of a particle related to the index of refraction?
 4. Relate the index of refraction concept to the Schrödinger equation:
 - (a) Use the Schrödinger equation to determine the wave vector k of a plane wave traveling through a region in which there is a constant potential V .
 - (b) Show that for weak potentials the index of refraction is

$$n \simeq 1 - \frac{\mu V}{k^2}. \quad (2.48)$$

- (c) Under what conditions can this relation be used for nonhomogeneous media which also absorb waves?
- (d) Use the index of refraction concept to deduce the relation between the complex (optical) potential and the elementary scattering amplitude:

$$V(r) = -\frac{2\pi\rho(r)f_E(\theta = 0^\circ)}{\mu}. \quad (2.49)$$

5. Use the above relations between index of refraction and potentials to estimate the central well depths for both the real and imaginary parts of the optical potential.
 - (a) Calculate the potential's depths for a pi meson of mass 139 MeV and kinetic energy 100 MeV incident upon a ^{12}C nucleus of radius 2.5 fm. [*Hint:* $f(0^\circ) \simeq (0.25 + 0.5i)$ fm, and the nucleus can be considered a uniform sphere.]
 - (b) How does the mean free path of the pion compare to the size of the nucleus?
6. Construct a real, non-Hermitian potential, and investigate how this changes the continuity equation.

CHAPTER 3

PARTIAL-WAVE EXPANSIONS

3.1 Shifted Waves

Our efforts of Chapter 1, *Scattering*, in reducing the 2-body Schrödinger equation to a 1-body equation still leave us with a 3-dimensional differential equation to solve. Solving this equation and understanding the particle-particle interaction itself are made easier by replacing the 3-dimensional equation with a series (albeit infinite) of 1-dimensional equations. This is an advance because once the conceptual and computational techniques are developed which solve the 1-dimensional problem, we can let a computer do it as many times as needed. The approach is not unique to quantum mechanics and indeed is similar to the separation of variables techniques used in classical studies of the scattering of waves (Konopinski, 1969; Newton, 1966; Jackson, 1975).

In the first part of this chapter we introduce the general partial-wave functions for the Schrödinger equation and then look at several forms for the free partial waves, each with a characteristic asymptotic behavior appropriate to a specific boundary condition. The scattering phase shifts are then defined in terms of these partial waves and the expansions of the scattering amplitude and cross section in terms of these phase shifts are deduced. (Although strictly only a wave can be expanded in partial “waves,” the expansions of amplitudes and cross sections are also commonly referred to as partial-wave expansions). Finally, we end this chapter with a discussion of the technique for actually solving the differential Schrödinger equation. In Chapter 4, *Scattering Applications: Lengths, Resonances, Coulomb*, we apply that technique to several important problems.

Plane Waves

We have already stated that the incident part of the scattering wave function is the *plane wave* shown in Figure 1.2. We now normalize it:

$$\phi_{\mathbf{k}}(\mathbf{r}) \stackrel{\text{def}}{=} \frac{e^{i\mathbf{k}\cdot\mathbf{r}}}{(2\pi)^{3/2}} = \frac{e^{ikr \cos \theta}}{(2\pi)^{3/2}} = \frac{e^{ikz}}{(2\pi)^{3/2}}, \quad (3.1)$$

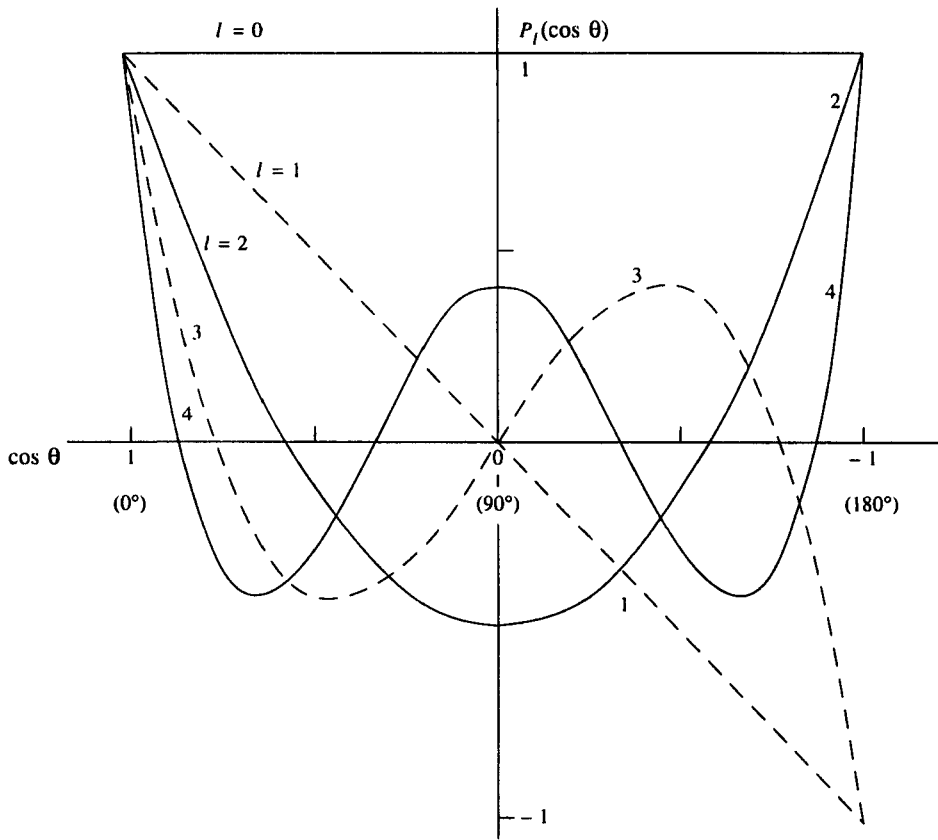


Figure 3.1: The Legendre functions $P_l(\cos \theta)$ for the first five l values. (Note the universal positive sign in the forward direction and the alternating signs in the backward direction.)

and decompose it into the sum of infinitely many *spherical* or *partial* waves (Gottfried, 1966; Bohm, 1951; Abramowitz and Stegun, 1964):

$$\phi_{\mathbf{k}}(\mathbf{r}) = \sum_{l=0}^{\infty} \frac{(2l+1)i^l j_l(kr) P_l(\cos \theta)}{(2\pi)^{3/2}} \quad (3.2)$$

$$= \sum_{l=0}^{\infty} \sum_{m=-l}^l \frac{i^l j_l(kr) 4\pi Y_{lm}^*(\Omega_{\mathbf{k}}) Y_{lm}(\Omega_{\mathbf{r}})}{(2\pi)^{3/2}}. \quad (3.3)$$

Here $P_l(\cos \theta)$ is the Legendre polynomial illustrated in Figure 3.1 and $j_l(kr)$ is the spherical Bessel function illustrated in Figure 3.2. Because the j_l 's have the limiting values:

$$j_l(kr) = \begin{cases} (kr)^l / [1 \cdot 3 \cdot 5 \cdots (2l+1)], & \text{for } kr \rightarrow 0, \\ \sin(kr - l\pi/2)/kr, & \text{for } kr \sim \infty, \end{cases} \quad (3.4)$$

each j_l is small¹ until $kr \simeq l$, and oscillatory for $kr \gg l$. This is visible in Figure 3.1.

¹Physically, the small kr behavior of j_l arises from the repulsion of the angular momentum barrier $l(l+1)/r^2$.

Consequently, if a plane wave is incident upon a potential $V(r)$, there will always be some partial waves which lie outside of the potential and are unaffected by it. This is simply a consequence of some j_l 's being essentially zero for the r region in which the potential exists. This also means that as we lower the energy E or wave vector k , we lower the number of partial waves getting scattered.

These properties of the plane wave are the basis of the *semiclassical* approximation in which a projectile of impact parameter b (a classical concept) is identified with the angular momentum quantum number l :

$$l + \frac{1}{2} \simeq kb. \quad (3.5)$$

Because a classical particle with an impact parameter b will not be scattered from a potential with range R if $b > R$, replacing b by R in (3.5) provides a handy estimate of the number of partial waves affected by the potential:

$$l_{\max} \geq kR. \quad (3.6)$$

Distorted Waves

We need to solve the 1-particle, stationary-state Schrödinger equation (1.26),

$$\left[\frac{-\nabla^2}{2\mu} + V(\mathbf{r}) \right] \psi(\mathbf{r}) = E\psi(\mathbf{r}), \quad (3.7)$$

or in *reduced* form:

$$[-\nabla^2 + k^2] \psi(\mathbf{r}) = U(\mathbf{r})\psi(\mathbf{r}), \quad (3.8)$$

$$[U(\mathbf{r}) = 2\mu V(\mathbf{r}), \quad k^2 = 2\mu E]. \quad (3.9)$$

We assume that the *scattering* (or *full*, or *distorted*) wave function $\psi_{\mathbf{k}}(\mathbf{r})$ has the same form for its partial-wave decomposition as does the plane wave (3.2):

$$\psi_{\mathbf{k}}(\mathbf{r}) = \sum_{l=0}^{\infty} (2l+1) i^l P_l(\cos \theta) \frac{u_l(kr)}{kr(2\pi)^{3/2}}, \quad (3.10)$$

where the subscript $\mathbf{k}$ indicates the beam momentum and thereby the energy $E_{\mathbf{k}}$. When (3.10) is substituted into (3.8), we find that the radial wave function $u_l(kr)$ satisfies the 1-dimensional radial Schrödinger equation with an angular momentum barrier added to the potential,

$$\left(\frac{d^2}{dr^2} + k^2 \right) u_l(kr) = \left[U(r) + \frac{l(l+1)}{r^2} \right] u_l(kr). \quad (3.11)$$

If the wave function u_l is known over all space for all l 's, we have completely solved the Schrödinger equation and should be able to obtain all experimental observables. Yet this is somewhat of an overkill for scattering in which all measurements are made in the region of space asymptotically far from the target. We build that constraint into our theory by introducing the phase-shift concept.

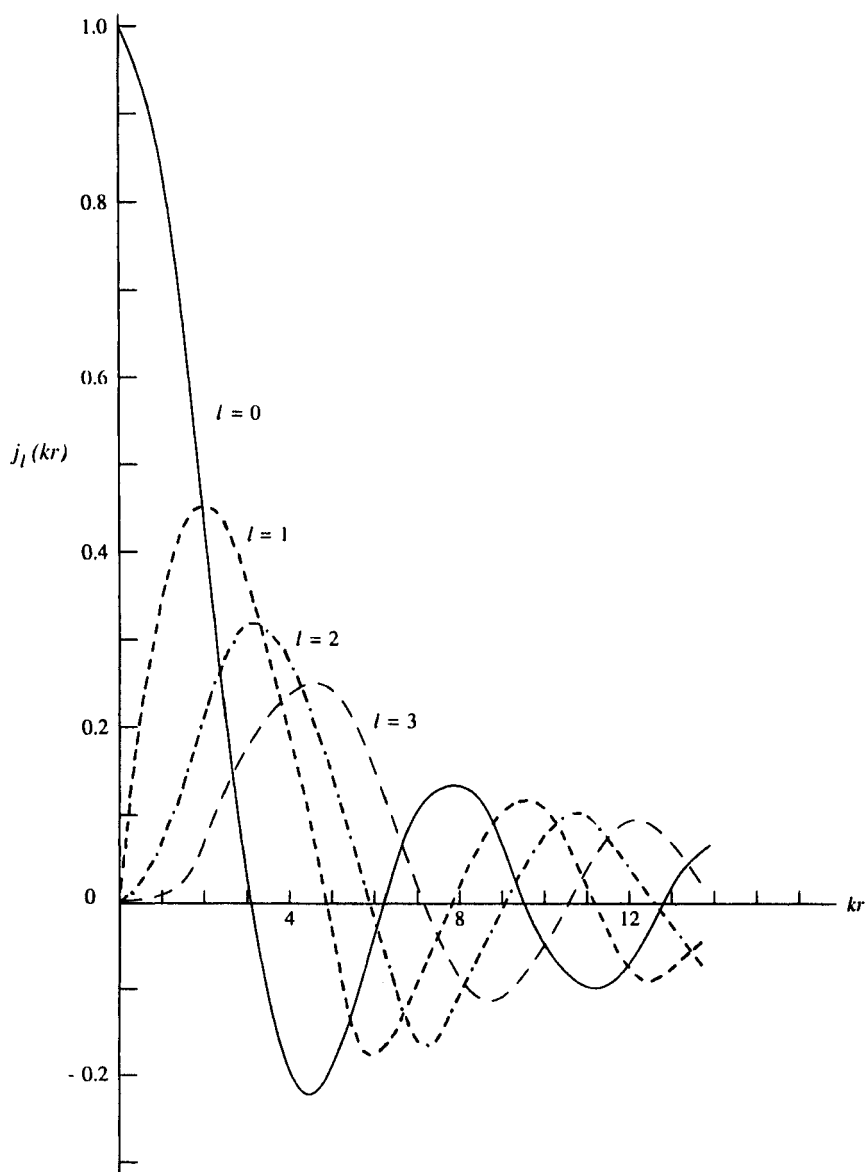


Figure 3.2: Spherical Bessel functions $j_l(kr)$. Note the asymptotic oscillatory behavior for $kr \geq l$.

Phase Shifts

The solutions of the radial Schrödinger equation with the potential V set to zero,

$$\left(\frac{d^2}{dr^2} + k^2\right) \begin{Bmatrix} F_l(kr) \\ G_l(kr) \end{Bmatrix} = \frac{l(l+1)}{r^2} \begin{Bmatrix} F_l(kr) \\ G_l(kr) \end{Bmatrix}, \quad (3.12)$$

are denoted by F_l and G_l , and are called the “free” solutions since they describe the motion of a free particle. Because (3.12) is a second-order differential equation there are two independent solutions. We choose F_l to be *regular* (nonsingular) at the origin and G_l to be *irregular* (divergent):

$$F_l(kr) \equiv kr j_l(kr) = \begin{cases} (kr)^{l+1}/[1 \cdot 3 \cdot 5 \cdots (2l+1)], & r \rightarrow 0, \\ \sin(kr - l\pi/2), & r \sim \infty, \end{cases} \quad (3.13)$$

$$G_l(kr) \equiv -kr n_l(kr) = \begin{cases} 1 \cdot 3 \cdot 5 \cdots (2l-1)/(kr)^l, & r \rightarrow 0, \\ \cos(kr - l\pi/2), & r \sim \infty. \end{cases} \quad (3.14)$$

We see that F_l is proportional to the spherical Bessel function and G_l is proportional to the spherical Neumann function.

If the potential V is finite-ranged and nonsingular, the radial wave function u_l must be regular at the origin but will not be equal to F_l because V is acting. In contrast, because F_l and G_l form a complete set wherever $V = 0$, the outer part of the full wave function, that is, u_l in the region outside of the potential's range, must be a linear combination of the two free solutions:

$$u_l(kr) = AG_l(kr) + BF_l(kr), \quad (r > R), \quad (3.15)$$

with A and B constants. Note that (3.15) is a true even for nonasymptotic r as long as r is in a region in which $V = 0$. Rather than deal with the constants A and B , we choose a normalization of u_l more consistent with the free-wave form (3.4):

$$u_l(kr) \stackrel{\text{def}}{=} e^{i\delta_l(E)} [\sin \delta_l G_l(kr) + \cos \delta_l F_l(kr)], \quad (r > R). \quad (3.16)$$

There is now only one energy-dependent constant $\delta_l(E)$ and it will be real as long as only the elastic channel is open. The origin of the name *scattering phase shift* is evident if we examine the asymptotic form of the wave function (3.16):

$$u_l(kr) \sim e^{i\delta_l(E)} [\sin \delta_l \cos(kr - l\pi/2) + \cos \delta_l \sin(kr - l\pi/2)] \quad (3.17)$$

$$= e^{i\delta_l} \sin(kr - l\pi/2 + \delta_l). \quad (3.18)$$

Equation (3.18) means that u_l has the same asymptotic form as the free wave (3.4), only with its “phase shifted” by δ_l (the exponential factor is needed to match normalizations). Furthermore, if δ_l equals zero, there is no irregular solution outside the potential ($u_l \equiv F_l$), which means no scattering! As discussed in § 3.3, this is closely related to the solution of the bound-state problem which has a decaying outer solution $\exp(-\kappa r)$.

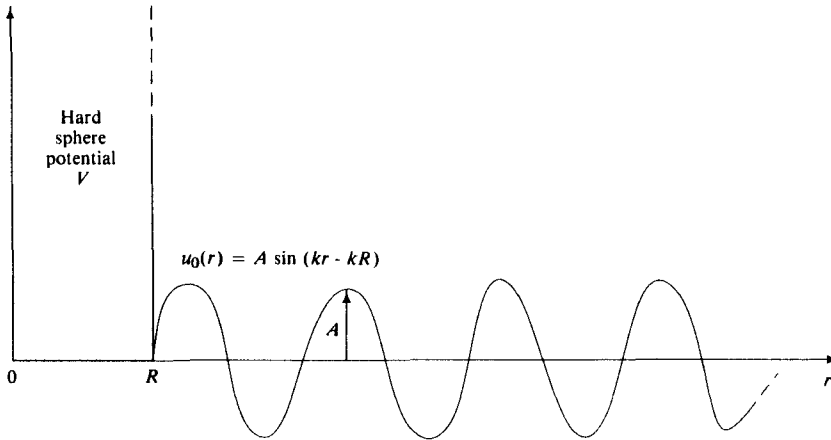


Figure 3.3: The $l = 0$ wave function of a particle experiencing a hard-sphere potential.

Example: Hard Sphere Scattering

Imagine a *hard sphere* of radius R , that is, a potential infinitely repulsive for $r < R$ and zero elsewhere. As we see in Figure 3.3, because the wave function does not penetrate the sphere, the inner solution ($r < R$) vanishes. The outer solution must match the inner one at the sphere's radius, and so

$$u_0(kr) = \begin{cases} 0, & r < R, \\ \sin(kr - kR), & r > R. \end{cases} \quad (3.19)$$

A comparison with (3.18) yields:

$$\delta_0(E) = -kR, \quad (\text{hard sphere}). \quad (3.20)$$

Equation (3.20) is useful as a reminder that *negative phase shifts usually arise from repulsive potentials* and that even “simple” potentials produce energy-dependent phase shifts.²

In Figure 3.4 we show some typical wave functions for attractive, repulsive and zero-strength potentials. We see that relative to the free wave function F_l (solid curve), the repulsive potential produces a wave function (dotted curve) which is pushed out or “repelled” from the origin (the region where the potential acts) and accordingly δ_l is negative. Likewise, a weakly attractive potential attracts the wave function back to the origin (dashed curve) and accordingly δ_l is positive. Of course interactions in the real world do not have to be simple: the nucleon–nucleon interaction (for spin-singlet S waves) is attractive at low energies and repulsive at higher energies (> 240 MeV).³ The usual understanding of this behavior is that the potential has a long-range attraction plus a short-range, but strong, *repulsive core*, and that high-energy projectiles can get close enough to be influenced by the core's repulsion.

Seeing that the phase shifts determine the outer ($r > R$) wave function, and that all experimental equipment is at $r \gg R$, it follows that all observables can be expressed in

²See § 4.1 for a discussion of some exceptional cases.

³See F. Tabakin (1968) for a discussion of some creative techniques to generate a potential producing both attraction and repulsion.

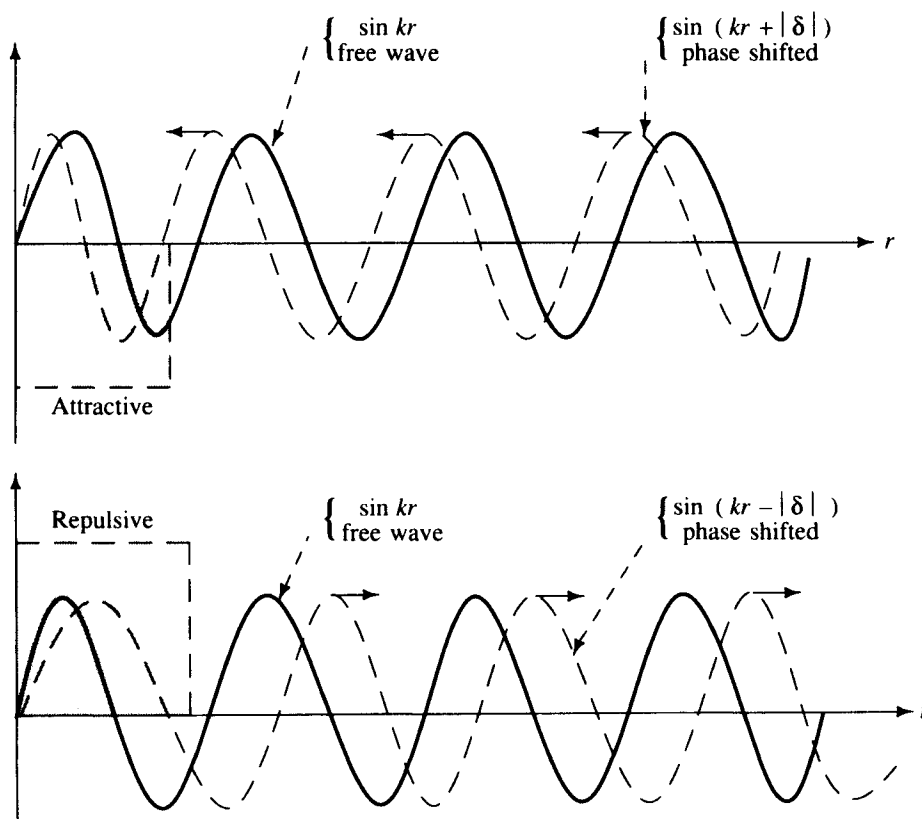


Figure 3.4: The shift in phase of a wave function caused by an attractive potential (upper dashed curve), and a repulsive potential (lower dashed curve) relative to the wave function for no potential (solid curve).

terms of the δ_l 's. For this reason the phase shifts are a convenient *model-independent* (that is, potential-independent) method to parameterize all possible scattering measurements. The only real assumptions are the existence of a wave function and *unitarity* of the S matrix, that is, continuity and conservation probability. Even if the potential concept were inapplicable, the phase shift parameterization would still be valid.

Incoming and Outgoing Waves

The wave functions F_l and G_l are called *standing wave* solutions of the free Schrödinger equation because they behave like sines and cosines. Although we defined the phase shift in terms of standing waves (3.18),

$$u_l(kr) \sim e^{i\delta_l} \sin(kr - l\pi/2 + \delta_l), \quad (3.21)$$

we can also define it in terms of *incoming* and *outgoing* waves. By decomposing the sine in (3.21) into exponentials we obtain

$$u_l(kr) \sim \frac{e^{2i\delta_l} e^{i(kr-l\pi/2)} - e^{-i(kr-l\pi/2)}}{2i}, \quad (3.22)$$

which says that a shift of the standing wave's phase by δ_l relative to the regular solution is equivalent to a shift of the outgoing wave's phase by $2\delta_l$ relative to the incoming wave. To make all this official, we explicitly define incoming and outgoing, free-particle solutions proportional to the spherical Hankel functions:

$$H_l^{(+)}(kr) = G_l(kr) + iF_l(kr), \quad (3.23)$$

$$H_l^{(-)}(kr) = G_l(kr) - iF_l(kr). \quad (3.24)$$

If we rewrite the outer solution (3.15) in terms of the H_l 's, the full outside solution is

$$\psi_{\mathbf{k}}(\mathbf{r}) = \sum_{l=0}^{\infty} (2l+1) i^l P_l(\cos \theta) \frac{e^{2i\delta_l} H_l^{(+)}(kr) - H_l^{(-)}(kr)}{2ikr(2\pi)^{3/2}}, \quad (r > R). \quad (3.25)$$

Elastic Waves with Absorption

The decomposition (3.25) is useful in extending the scattering formalism to include *absorption* (any process removing flux from the incident channel).

Exercise Show that the plane wave is the sum of incoming and outgoing waves,

$$\phi_{\mathbf{k}}(\mathbf{r}) \sim \sum_{l=0}^{\infty} (2l+1) i^l P_l(\cos \theta) \frac{e^{i(kr-l\pi/2)} - e^{-i(kr-l\pi/2)}}{2ikr(2\pi)^{3/2}}. \quad \diamond \quad (3.26)$$

Because absorption affects the wave only after it reaches the target, only the outgoing wave [first exponential in (3.26)] is affected. We build this physics into the mathematics by replacing the unit amplitude in front of the outgoing wave by an *absorption parameter* η_l :

$$\psi_{\mathbf{k}}(\mathbf{r}) \sim \sum_{l=0}^{\infty} (2l+1) i^l P_l(\cos \theta) \frac{\eta_l e^{2i\delta_l} e^{i(kr-l\pi/2)} - e^{-i(kr-l\pi/2)}}{2ikr(2\pi)^{3/2}}, \quad (3.27)$$

where the loss (in contrast to production) of flux leads to the restriction $\eta_l \leq 1$. The combination $\eta_l e^{2i\delta_l}$ appears often and is frequently called the S matrix element (see also Chapter 7, *Formal Quantum Mechanics*),

$$S_l(E) = \eta_l e^{2i\delta_l}. \quad (3.28)$$

The restriction on η_l is equivalent to the restriction on the S matrix, $|S_l(E)| \leq 1$. An equivalent inclusion of absorption is to retain the usual expressions involving phase shifts, but treat δ_l as a complex number:

$$\delta_l \rightarrow \delta_{l,R} + i\delta_{l,I}, \quad (3.29)$$

$$\Rightarrow \eta_l = e^{-2\delta_{l,I}}. \quad (3.30)$$

3.2 Partial-Wave Amplitudes

In Chapter 1, *Scattering*, we stated the basic assumption of scattering theory that the full wave function has the asymptotic form (1.12):

$$\psi_{\mathbf{k}}(\mathbf{r}) \sim \frac{e^{ikz}}{(2\pi)^{3/2}} + f_E(\theta, \phi) \frac{e^{ikr}}{r(2\pi)^{3/2}}. \quad (3.31)$$

Because we now can express $\psi_{\mathbf{k}}(\mathbf{r})$ in terms of the phase shifts, we can solve for the scattering amplitude in terms of the δ_l 's:

$$f_E(\theta, \phi) \frac{e^{ikr}}{r} = \lim_{r \rightarrow \infty} [\psi_{\mathbf{k}} - \phi_{\mathbf{k}}](2\pi)^{3/2}. \quad (3.32)$$

Exercise Show that substitution of the partial-wave expansions into (3.32) yields the scattering amplitude expansion:

$$f_E(\theta) = \sum_{l=0}^{\infty} (2l+1) P_l(\cos \theta) \frac{(\eta_l e^{2i\delta_l} - 1)}{2ik}. \quad \diamond \quad (3.33)$$

The scattering amplitude is the prime experimental observable in scattering and different versions of it often are used. For example, the S -matrix element S_l (3.28) can replace $\eta_l e^{2i\delta_l}$:

$$f_E(\theta) = \sum_{l=0}^{\infty} (2l+1) P_l(\cos \theta) \frac{S_l(E) - 1}{2ik}. \quad (3.34)$$

Or the partial-wave scattering amplitude,

$$f_l(E) = \frac{\eta_l e^{2i\delta_l} - 1}{2i}, \quad (3.35)$$

can be used:

$$f_E(\theta) = \sum_{l=0}^{\infty} (2l+1) P_l(\cos \theta) \frac{f_l(E)}{k}. \quad (3.36)$$

If there is no absorption (the energy E is below thresholds for excitation or reactions), the partial wave amplitude takes the simpler form:

$$f_l(E) = e^{i\delta_l} \sin \delta_l = \frac{1}{\cot \delta_l - i}, \quad (\eta_l \equiv 1). \quad (3.37)$$

The full amplitude is then

$$f_E(\theta) = \sum_{l=0}^{\infty} (2l+1) P_l(\cos \theta) \frac{e^{i\delta_l} \sin \delta_l}{k}, \quad (\eta_l \equiv 1). \quad (3.38)$$

Differential Cross Section

The elastic differential cross section of Chapter 2, *Currents and Cross Sections*, is simply the squared modulus of the scattering amplitude,

$$\frac{d\sigma}{d\Omega}(\theta, \phi) = |f_E(\theta, \phi)|^2. \quad (3.39)$$

Exercise Show by substitution of the partial-wave expansion for f_E that

$$\frac{d\sigma}{d\Omega}(\theta) = \frac{1}{k^2} \left| \sum_{l=0}^{\infty} (2l+1) P_l(\cos \theta) \frac{\eta_l e^{2i\delta_l} - 1}{2i} \right|^2. \quad \diamond \quad (3.40)$$

Equation (3.40) implies that for each l entering into the sum, a polynomial of degree $2l$ in $\cos \theta$ is added to the differential cross section. Since for a given beam energy there is a limit on the number of l 's contributing, there is also a constraint on the complexity of the angular dependence of the differential cross section.

S- and P-Wave Example

Imagine an experiment conducted to learn about the interaction between two particles. As the energy is made lower and lower, there will always be an energy at which only S-waves (and maybe a little bit of P) are scattered.⁴ Because the energy is low, we also assume $\eta_l \equiv 1$. If there are only S and P waves, the cross section in the CM must then have the form:

$$\frac{d\sigma}{d\Omega} = \frac{1}{k^2} |e^{i\delta_0} \sin \delta_0 + 3e^{i\delta_1} \sin \delta_1 \cos \theta|^2 \quad (3.41)$$

$$= \frac{\sin^2 \delta_0 + 9 \sin^2 \delta_1 \cos^2 \theta + 6 \sin \delta_0 \sin \delta_1 \cos(\delta_0 - \delta_1) \cos \theta}{k^2}. \quad (3.42)$$

In Figure 3.5 we show the cross section obtained from (3.42) where there are S and P waves (solid curve), only S wave (dot-dashed curve), and only P wave (dashed curve). We see that

⁴ A fuller discussion of low-energy scattering is found in § 4.1.

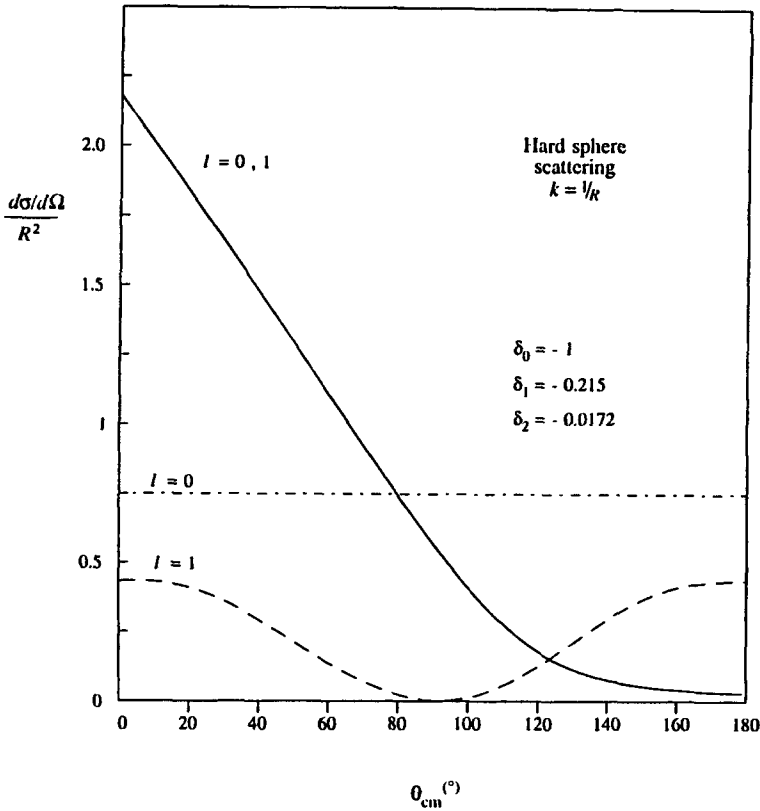


Figure 3.5: The differential cross section for scattering from a hard sphere of radius R for pure $l = 0$, pure $l = 1$, and combined $l = 0$ and $l = 1$ partial waves.

pure S-wave scattering is isotropic, pure P wave is symmetric about 90° (where $d\sigma/d\Omega$ vanishes), and the combination of S and P waves has a forward-backward asymmetry.

If an experimenter were to measure a cross section with the shape of the solid curve in Figure 3.5, he or she would need more than a visual inspection to determine which partial waves have entered. In practice, the experimenter might fit the cross section with a polynomial of degree n in $\cos \theta$,

$$\frac{d\sigma}{d\Omega}(\theta) = \frac{1}{k^2} (c_0 + c_1 \cos \theta + c_2 \cos^2 \theta + \cdots + c_n \cos^n \theta), \quad (3.43)$$

where the c 's are energy-dependent constants. The experimenter next finds the smallest value of n which reproduces the cross section (within experimental errors), and then know that partial waves up to $l = n/2$ are important. For our example this is

$$\frac{d\sigma}{d\Omega}(\theta) = \frac{1}{k^2} (a_0 + a_1 \cos \theta + a_2 \cos^2 \theta). \quad (3.44)$$

If the conditions (3.45) are not met by the measurement, the experimenter should conclude that either more than two l 's are entering, or that some error was made in the experiment (for example analyzing $d\sigma/d\Omega$ in the lab rather than CM). If the conditions (3.45) are met,

and a good enough experiment is performed, then it should be possible to deduce (modulo π) the phase shifts (δ_0, δ_1) at this energy.

Exercise Show that the partial-wave expansion (3.38) (that is, unitarity) restricts the expansion coefficients to the ranges:

$$0 \leq a_0 \leq 1, \quad -6 \leq a_1 \leq 6, \quad 0 \leq a_2 \leq 9. \quad \diamond \quad (3.45)$$

Total Cross Sections

In (2.16) we defined $\sigma_{\text{el}}(E)$ as the integrated differential cross section for elastic scattering:

$$\sigma_{\text{el}}(E) = \int d\Omega \frac{d\sigma}{d\Omega}(\theta). \quad (3.46)$$

Exercise Show that substitution of the partial-wave expansion for $d\sigma/d\Omega$ and the orthogonality of the Legendre polynomials,

$$\int_{-1}^1 dx P_l(x) P_{l'}(x) = \frac{2}{2l+1} \delta_{ll'}, \quad (3.47)$$

determine the expansion for the integrated cross section:

$$\sigma_{\text{el}}(E) = \frac{4\pi}{k^2} \sum_{l=0}^{\infty} (2l+1) \left| \frac{\eta_l e^{2i\delta_l} - 1}{2i} \right|^2 \quad (3.48)$$

$$= \frac{4\pi}{k^2} \sum_{l=0}^{\infty} (2l+1) \sin^2 \delta_l, \quad (\eta_l \equiv 1). \quad \diamond \quad (3.49)$$

For the preceding S- and P-wave example we have

$$\sigma_{\text{el}}(E) = \frac{4\pi}{k^2} (\sin^2 \delta_0 + 3 \sin^2 \delta_1). \quad (3.50)$$

The total *nonelastic* cross section introduced in § 2.3 also has a partial-wave expansion. To derive it we need to evaluate the net flow of current through the sphere in Figure 2.2 (that is, the difference between the elastic current in and out) and ascribe the loss to nonelastic events:

$$\sigma_{\text{ne}}(E) = \int r^2 d\Omega \cdot \frac{\mathbf{j}_t}{j_{\text{in}}}. \quad (3.51)$$

We leave it to the Problems to perform the integration and show that

$$\sigma_{\text{ne}}(E) = \frac{\pi}{k^2} \sum_{l=0}^{\infty} (2l+1) (1 - \eta_l^2). \quad (3.52)$$

Clearly, the nonelastic cross section vanishes unless there is absorption in some partial wave, $\eta_l \neq 1$. Yet the expansion for the elastic cross section (3.48) then implies that σ_{el}

cannot vanish, that is, the *shadow* of absorption appears as elastic scattering (the inverse is not true). The effect is well known in optics where an absorptive disk causes *diffraction*, that is, elastic scattering of an incident light wave. Because quantum mechanics is also ruled by a wave equation, it is no surprise that a similar effect occurs.

The total cross section σ_t is the sum of the integrated elastic and nonelastic cross sections:

$$\sigma_t = \sigma_{el} + \sigma_{ne}. \quad (3.53)$$

Exercise Show by substitution that

$$\sigma_t = \frac{2\pi}{k^2} \sum_{l=0}^{\infty} (2l+1) [1 - \eta_l \cos(2\delta_l)] \quad (3.54)$$

$$= \frac{4\pi}{k^2} \sum_{l=0}^{\infty} (2l+1) \sin^2 \delta_l, \quad (\eta_l = 1). \quad \diamond \quad (3.55)$$

Exercise Show that the total cross section in the presence of absorption, (3.54), is obtained by applying the optical theorem,

$$\sigma_t = \frac{4\pi}{k} \operatorname{Im} f_E(0^\circ), \quad (3.56)$$

to the partial-wave expansion of $f_E(\theta)$. ◇

The surprising validity of the optical theorem in the presence of absorption indicates that the depletion of the beam in the incident direction feeds all scattering and reactions.⁵

The $1/k^2$ factor in (3.48) and the energy dependences of the phase shifts produce total cross section with some energy-dependent structure. If the variation with energy is particularly rapid, this may indicate a *resonance* (see § 4.2); if the cross section diverges at zero energy this indicates an exothermic reaction in which $\delta_l(0) \neq 0$; if the cross section is rather flat this may indicate “diffractive” scattering.

3.3 Actually Solving Schrödinger's Equation

Most analytic and numerical solutions of the differential Schrödinger equation (3.11),

$$\left(\frac{d^2}{dr^2} + k^2 \right) u_l(kr) = \left[U(r) + \frac{l(l+1)}{r^2} \right] u_l(kr), \quad (3.57)$$

follow a multistage approach. First we solve for u_l in the inner region, that is, for r values for which $U = 2\mu V \neq 0$. Even though this is the region where the potential acts, as long as the potential is not too singular, the form of the wave function is universal because for

⁵For further illumination of this point see Schiff (1954, 1968).

$r \simeq 0$ the angular momentum barrier dominates and the Schrödinger equation becomes:⁶

$$\frac{d^2 u_l(kr)}{dr^2} \simeq \frac{l(l+1)}{r^2} u_l(kr). \quad (3.58)$$

Equation (3.58) has the power-law solution

$$u_l(kr) \rightarrow r^n \quad (r \rightarrow 0), \quad (3.59)$$

with $n = l + 1$ being *regular* at the origin and $n = -l$, *irregular*. Because the physical wave function is proportional to $u_l(kr)/r$ and cannot produce infinite probability, we choose $n = l + 1$:

$$u_l(kr) \rightarrow C r^{l+1}. \quad (3.60)$$

In numerical solutions we start with an initial value of zero for the wave function at the origin and an arbitrary value for its first derivative:

$$u_l(0) = C 0^{l+1} = 0, \quad (3.61)$$

$$u'_l(0) = C(l+1)0^l = \begin{cases} C, & l = 0, \\ 0, & l > 0. \end{cases} \quad (3.62)$$

We then choose a small step size h and invoke a numerical algorithm to determine $u_l(kh)$.⁷ We continue integrating outwards, finding $u_l(2kh)$, $u_l(3kh)$, etc. until a (*matching*) radius R_m is reached at which $U(R_m) \simeq 0$. In analytic approaches we determine the inside solution by solving the differential equation analytically.

The technique for solving the analytic or numeric equation is now the same. In the second stage we determine the outer solution and in the third stage we match the inner u_l and u'_l to the outer ones at R_m to ensure continuity of probability and current (see later). While any matching radius R_m greater than the range R of the potential should be acceptable, realistic potentials tend to become progressively smaller as r increases but never truly vanish. In this case we choose R_m such that $U(R_m)$ is small with respect to the precision of the problem at hand (say down to 10^{-7} of its value at the origin if we want wave functions with six to seven places of accuracy).

Because the overall normalization of the wave function is unimportant for determining the scattering phase shifts or the bound-state energies, it is simplest to match the logarithmic derivative $\mathcal{L}$,

$$\mathcal{L}(R_m) \stackrel{\text{def}}{=} \left. \frac{du_l(kr)/dr}{u_l(kr)} \right|_{r=R_m}, \quad (3.63)$$

of the inner and outer solutions at R_m . The logarithmic derivative $\mathcal{L}$ is a function of energy E_k and contains all the information we need to know about the wave function and potential to solve the Schrödinger equation.⁸ For the outer solution is either known analytically as part of the boundary conditions or determined by numerical integration inwards from an arbitrary constant (or zero) value for $u_l(\infty)$.

⁶The exceptions being $l = 0$ and the Coulomb potential, where the same results are obtained using probability conservation arguments.

⁷Because h sets the fineness of the grid upon which $u_l(kr)$ is determined, good precision demands that h be small with respect to variations of the wave function. Nevertheless, a compromise is needed since very small h 's lead to unacceptably long computing times and large round-off errors (Thompson, 1993). Note, this h does not belong to Planck.

⁸For example, a square well of reduced depth U_0 yields $\mathcal{L} = F'_l(k'\tau)/F_l(k'\tau)|_{r=R_m}$, where $k' = (k^2 - U_0)^{1/2}$.

Just for Scattering

For scattering, $u_l(kr)$ must have the form (3.16) and so $\mathcal{L}$ must have the form

$$\mathcal{L}(R_m) = \frac{\tan \delta_l G'_l(kR_m) + F'_l(kR_m)}{\tan \delta_l G_l(kR_m) + F_l(kR_m)}, \quad (3.64)$$

where $F'_l(kR_m) = dF_l(kr)/dr|_{r=R_m}$. Yet because all quantities except $\tan \delta_l$ in (3.64) are known, we can just solve it for the phase shift:

$$\tan \delta_l = -\frac{F'_l(kR_m) - F_l(kR_m)\mathcal{L}(R_m)}{G'_l(kR_m) - G_l(kR_m)\mathcal{L}(R_m)}. \quad (3.65)$$

Bound State Connection

This same technique is used to solve for bound states, that is, solutions of the Schrödinger equation confined to some region of space. The confinement requires that $u_l(kr)$ be normalizable, and this is possible only if $u_l(\infty) = 0$. Yet because the positive energy solutions behave like $\sin(kr - l\pi/2)$, $\cos(kr - l\pi/2)$, or $\exp[\pm i(kr - l\pi/2)]$, they are not normalizable and there are no positive-energy bound states. For negative energy the momentum k becomes a pure imaginary number,

$$E_k = \frac{k^2}{2\mu} = \frac{-\kappa^2}{2\mu} \Rightarrow k = \pm i\kappa, \quad (3.66)$$

and the outgoing (incoming) wave solutions analytically continue into the dying (growing) exponentials:

$$e^{\pm i(kr - l\pi/2)} \rightarrow e^{\mp \kappa r} i^l \quad (k \rightarrow \pm i\kappa). \quad (3.67)$$

While $k = \pm i\kappa$ both give negative energies, only $k = i\kappa$ produces a normalizable wave function. With this choice the outer solution is a Hankel function of imaginary argument:

$$u_l(kr) = H_l^{(+)}(i\kappa r) \sim e^{-\kappa r} i^l, \quad (3.68)$$

and has the logarithmic derivative

$$\mathcal{L}(R_m) = -\kappa. \quad (3.69)$$

The *eigenvalue* condition for bound states is that the inner value of $\mathcal{L}$ [obtained by integration of (3.1) with $k^2 \rightarrow -\kappa^2$] equals $-\kappa$. This extra condition makes the bound-state problem an eigenvalue problem with solutions for only certain energies; the phase shifts, in contrast, exist for all positive energies in the continuum.

Exercise Show that $\tan \delta_l$ becomes infinite at bound state energies. $\diamond$

3.4 Problems

1. To help test your understanding of the material in this chapter, try answering the following questions without resort to the text.

- (a) How is the phase shift δ_l defined in terms of solutions of the Schrödinger equation?
 - (b) If the *irregular* solution of the Schrödinger equation is not analytical for all r , how then can it enter into a physical solution for the wave function?
 - (c) What is the maximum allowed cross section (the unitarity limit) $\sigma_l^{\max}$ for each partial wave?
 - (d) What conclusions might you draw if an experiment gave $\sigma_l > \sigma_l^{\max}$?
2. If we assume stationary states in the *time-independent* view of scattering from a complex potential, the interpretation of the current must change.

- (a) Indicate the change in probability conservation,

$$\frac{\partial}{\partial t} \int d^3r |\psi(\mathbf{r})|^2 = 0. \quad (3.70)$$

- (b) Show that for complex phase shifts, the integral of the total flux through a sphere surrounding the scattering center yields

$$\sigma_{\text{ne}} = \frac{\pi}{k^2} \sum_{l=0}^{\infty} (2l+1) [1 - \eta_l^2]. \quad (3.71)$$

- (c) Comment on why this is the same expression as obtained with the optical theorem.
3. Consider projectile–target scattering in which $m_P = m_T/6$ and the energy is low enough for only S wave scattering (in the CM).
- (a) Estimate which partial waves make greater than 1% contribution to the scattering in the laboratory (that is, if an experimenter took his or her lab cross section and expanded it as a series of Legendre polynomials).
 - (b) Based on your analysis of the previous part, why would you say it is a bad idea to do a partial-wave decomposition in the lab (in particular, which physical arguments may lose validity)?
4. A particle of wave vector $k = 1/R$ scatters from an infinitely heavy and repulsive sphere of radius R .
- (a) What are the phase shifts for the first three partial waves?
 - (b) Show on a plot of $d\sigma/d\Omega$ versus $\cos \theta$, the contribution from each partial wave separately and the result for all three together.
 - (c) Evaluate the total cross section and estimate the error from truncating the number of l values. How good is the semiclassical estimate (3.5) of the number of partial waves contributing?
5. Two particles of mass m scatter. The potential between them is approximated by the attractive square well:

$$V(r) = \begin{cases} -V_0, & \text{for } r < b, \\ 0, & \text{for } r > b. \end{cases} \quad (3.72)$$

- (a) Solve for $k \cot \delta_0$, where δ_0 is the S wave phase shift.
- (b) Show explicitly that the condition for the scattering amplitude for this partial wave,

$$f_0 = \frac{e^{i\delta_0(k)} \sin \delta_0(k)}{k}, \quad (3.73)$$

to have a pole on the positive imaginary k axis, is also the condition for this potential to produce a bound state.

- (c) Based on the above expression for f_0 , verify that the scattering amplitude is an analytic function of the energy $E = k^2/2\mu$ with a branch cut from 0 to ∞ , and bound-state poles on the negative axis.
6. An experiment measures the differential cross section for the elastic scattering of two particles with wave vector k in the center of mass to have the form:

$$\frac{d\sigma}{d\Omega}(\theta) = \frac{1}{k^2} e^{-2(1-\cos\theta)}. \quad (3.74)$$

- (a) Make a crude sketch of $d\sigma/d\Omega$ versus the scattering angle θ for all allowed values of θ .
- (b) Without any detailed calculation, deduce the number of partial waves which contribute to the scattering and indicate if this is compatible with scattering from a finite-range potential.
- (c) What must be the modulus of the angle-dependent scattering amplitude, $|f_E(\theta)|$?
[Note: A complex number $z = x + iy = Re^{i\theta}$ has modulus $R = (x^2 + y^2)^{1/2}$ and phase θ .]

The experimentalist next measures the total cross section for the same particles and finds it to have the form

$$\sigma_t = \frac{4\pi}{k^2}. \quad (3.75)$$

- (d) What is the value of scattering amplitude in the forward direction, $f_E(0^\circ)$?
 - (e) Assuming that the scattering amplitude has constant phase, what is $f_E(\theta)$?
 - (f) What is the total elastic (integrated elastic) cross section for this reaction? Comment on why this is the same or different from the total cross section.
 - (g) Why must the phase shift $\delta_l(k)$ be complex for this interaction?
 - (h) Find the $l = 0$ phase shift for this interaction (you should be able to evaluate any integrals involved).
7. Consider the scattering of a neutron with 5.32 MeV kinetic energy from a very heavy, very black (that is, highly absorptive) nucleus of 4 fm in diameter.
- (a) What is the approximate relative size of the real and imaginary parts of the phase shift for elastic scattering from this nucleus?
 - (b) Approximately how many partial waves enter into the scattering?
 - (c) Calculate the scattering amplitude $f_E(\theta)$.
 - (d) Make a simple plot of the differential cross section $d\sigma/d\Omega$ versus θ .
 - (e) Calculate the integrated elastic, integrated reaction and total cross sections.

CHAPTER 4

SCATTERING APPLICATIONS: LENGTHS, RESONANCES, COULOMB

In this chapter we examine several applications of the preceding theory. This will demonstrate its power and method of application and help to develop physical understanding needed as we formalize the theory in the following chapters. We look at low-energy scattering, resonant and Gamow states, and scattering from the Coulomb potential.

4.1 The Low-Energy Limit

Let us again consider the radial Schrödinger equation (3.11) for the inner wave function:

$$\left(\frac{d^2}{dr^2} + k^2\right) u_l(kr) = \left[U(r) + \frac{l(l+1)}{r^2}\right] u_l(kr). \quad (4.1)$$

If in the interaction region, the energy is much less than the potential, $E \ll |V|$ or $k^2 \ll |U|$, then U dominates k^2 and the inner wave function is approximately independent of energy. Because the logarithmic derivative $\mathcal{L}$ (3.64) is also independent of energy, the energy dependence of $\tan \delta_l(E)$ (3.65) arises from that of the wave function. We use this observation to deduce properties of low-energy scattering which are independent of the nature of the potential, that is, model-independent.

Scattering Length

We know the free waves have the small argument limits (3.13)–(3.14):

$$F_l(kr) = (kr)^{l+1}/1 \cdot 3 \cdot 5 \cdots (2l+1), \quad (4.2)$$

$$G_l(kr) = 1 \cdot 3 \cdot 5 \cdots (2l-1)/(kr)^l. \quad (4.3)$$

Substitution of these limits into our expression for $\tan \delta_l$ (3.65) produces

$$\tan \delta_l \rightarrow \frac{R_m^{2l+1}[(l+1) - \mathcal{L}R_m]}{(2l+1)[1 \cdot 3 \cdot 5 \cdots (2l-1)]^2(l + \mathcal{L}R_m)} k^{2l+1}. \quad (4.4)$$

This has the potential dependence in the $\mathcal{L}$ of the first factor, and the k dependence in the second factor. Because the first factor is independent of k , we replace it with the constant a_l :

$$\tan \delta_l \rightarrow a_l k^{2l+1} \quad (k \rightarrow 0). \quad (4.5)$$

The $l = 0$ value a_0 is called the *scattering length* and the $l = 1$ value a_1 is called the *scattering volume*. An equivalent form of (4.5) is

$$k^{2l+1} \cot \delta_l \rightarrow \frac{1}{a_l}, \quad (k \cot \delta_0 \rightarrow \frac{1}{a_0}). \quad (4.6)$$

These equations mean that the phase shifts from any finite-ranged potential all vanish as $k \rightarrow 0$. The S-wave phase shift vanishes most slowly,

$$\delta_0 \simeq \tan \delta_0 \simeq a_0 k \rightarrow 0. \quad (4.7)$$

This is the proof that as the energy decreases there is always a value below which only S-waves contribute to scattering.

Exercise Show that the S-wave amplitude has the form

$$f_0(E) = e^{i\delta_0} \sin \delta_0 = \frac{\tan \delta_0}{1 - i \tan \delta_0}. \quad \diamond \quad (4.8)$$

Exercise Show that the scattering amplitude has the low energy limit

$$f_E(\theta, \phi) \rightarrow \frac{a_0}{1 - i a_0 k} \equiv \frac{a_0 + i(a_0)^2 k}{1 + (a_0)^2 k^2} \quad (k \rightarrow 0). \quad \diamond \quad (4.9)$$

We see that as $k \rightarrow 0$, the real part of the scattering amplitude equals the scattering length and the imaginary part approaches $(a_0)^2 k$, that is, zero. [The P-wave contributes a term of $O(k^2)$ to $\text{Re} f_E$ and $O(k^5)$ to $\text{Im} f$]. The cross sections likewise have the limits

$$\frac{d\sigma}{d\Omega} \rightarrow a_0^2, \quad \sigma_{\text{el}} = \int d\Omega \frac{d\sigma}{d\Omega} = 4\pi a_0^2. \quad (4.10)$$

We note that (4.10) is four times the classical cross section. This is consistent with the hard-sphere scattering length $a_0 = -R$ found in § 3.1.

Exercise Show that these low-energy limits preserve the analytic properties of the Schrödinger equation by showing that they satisfy the optical theorem,

$$\sigma_{\text{el}} = \frac{4\pi}{k} \text{Im} f_E(0^\circ). \quad \diamond \quad (4.11)$$

An important consequence of (4.9) and (4.10) is that a single real number a_0 completely parameterizes all of low-energy scattering. This is fortunate for those people using low-energy neutron scattering to study the structure of solids, but unfortunate for those using

scattering to deduce the properties of the projectile–target potential. In fact, this limit plagued the early days of nuclear physics when attempts were made to discover a unique potential acting between two nucleons, yet there always seemed to be many potentials which produced the same low energy scattering (Breit, et al., 1936; Bethe and Morrison, 1966). We recognize from our analysis that all these potentials must have produced the same a_0 and bound-state energy.

Low-Energy Wave Function

Although the scattering length produced by a potential does not directly measure the potential's range, it does tell us the low-energy wave function.

Exercise Show that the outer, S-wave wave function,

$$u_0(r > R) = e^{i\delta_0} [\sin \delta_0 G_0(kr) + \cos \delta_0 F_0(kr)] , \quad (4.12)$$

has the low-energy limit (ignoring normalization),

$$u_0(r > R) \rightarrow kr + \tan \delta_0 \propto r + a_0. \quad \diamond \quad (4.13)$$

We see that $u_0(r)$ outside of the potential is proportional to $r + a_0$. This means the outer wave function intercepts the r axis at $r = -a_0$. This result, combined with the realization that the curvature of the inner wave function equals reduced potential,

$$\frac{1}{u_0(r)} \frac{d^2 u_0(r)}{dr^2} = U(r), \quad (4.14)$$

gives meaning to the sign and magnitude of a_0 . Specifically:

- As illustrated in Figure 4.1A, if we have a repulsive potential $U > 0$, the inner wave function has positive curvature and therefore is expelled from the origin. If we extrapolate the outer wave function to smaller r , it intercepts the r axis at a positive r which implies $a_0 < 0$, that is, *repulsive* potentials have *negative* scattering lengths (and phase shifts). If the potential is made more repulsive, the curvature of the inner wave function increases and a_0 becomes more negative.
- As illustrated in Figure 4.1B, if we have an attractive potential $U < 0$, the inner wave function has negative curvature and is thus drawn in towards the origin. If we extrapolate the outer wave function to smaller r , it intercepts the r axis at a negative r which implies $a_0 > 0$, that is, weakly *attractive* potentials (generally) have *positive* scattering lengths (and phase shifts).
- As illustrated in Figure 4.1C, if an attractive potential is made more attractive the positive scattering length becomes larger and larger until the inner wave function matches to a horizontal outer one; at this point $a_0 \rightarrow \infty$. Because the inner wave function now matches onto a dying exponential, the solution is that of a confined particle, that is, a *zero-energy bound state* is accompanied by an *infinite* scattering length.

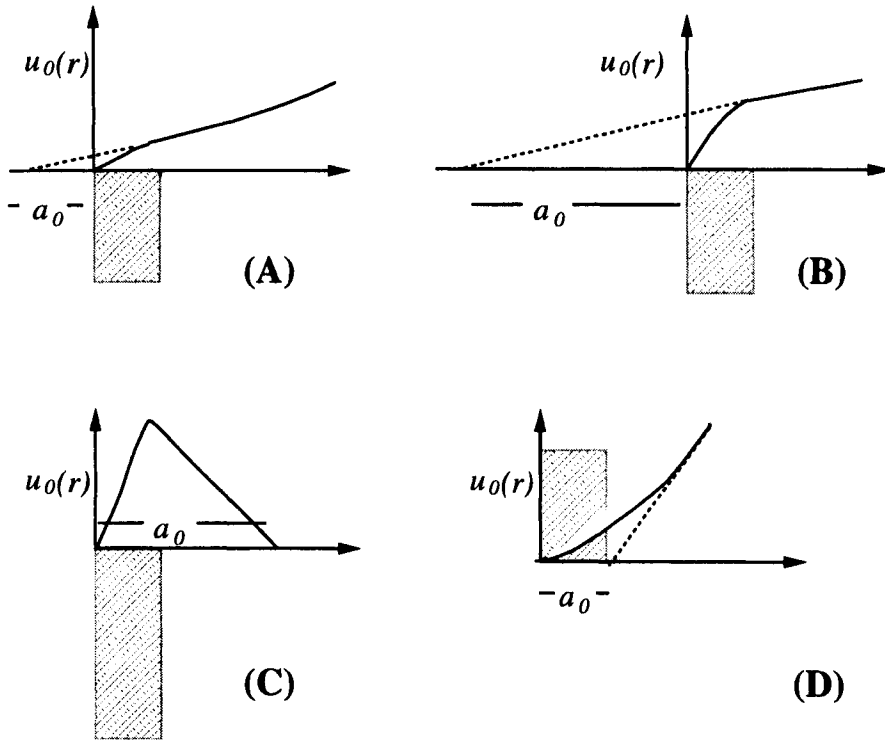


Figure 4.1: The relation between the scattering length and wave function for potentials of differing strengths and signs. The dashed curve is the extrapolation of the outer wave function to the point where it intersects the r axis, and the intercept is the negative scattering length: (A) Weak attraction, $a_0 > 0$; (B) Stronger attraction, $a_0 \gg 0$; (C) Attractive enough for a bound state, $a_0 < 0$; (D) Repulsive, $a_0 < 0$. (Computed by M. Paez.)

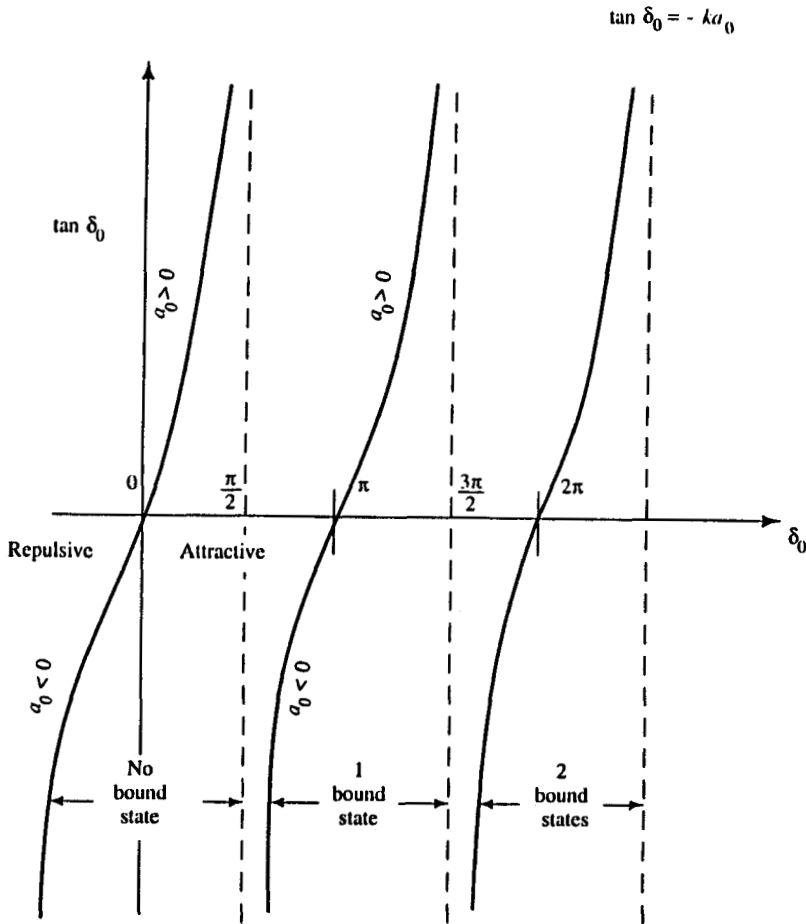


Figure 4.2: The function $\tan \delta_0$ versus δ_0 . Successive branches correspond to increasing numbers of bound states. Note how the sign of $\tan \delta_0$, and accordingly a_0 , changes as the potential is made more attractive.

- As illustrated in Figure 4.1D, if the potential is more attractive than that just needed to form a bound state, the inner wave function turns over within the potential well, and the matching outer wave function intercepts the positive r axis. This produces a large and negative scattering length. Further increase in attraction causes the scattering length to decrease in magnitude!

Levinson's Theorem ☉

It should be clear from Figure 4.1 that a_0 is not simply a measure of the potential size or strength. Moreover, as the potential strength continues to increase, even more bound states are supported and the entire cycle repeats. This is consistent with the relation $\tan \delta_0 = ka_0$; as Figure 4.2 shows, the tangent (or accordingly a_0) varies periodically and discontinuously from $+\infty$ to $-\infty$. If the potential is strong enough to have a bound state at $k = 0$, we must be on the second branch of the tangent function and the zero-energy phase shift is π rather

than 0. This result is actually quite general and is an example of Levinson's theorem:¹

$$\delta(k=0) - \delta(k=\infty) = n_B \pi, \quad (4.15)$$

which relates the phase shifts at zero and infinite energy to the number of bound states n_B .

Example: Two Nucleon Scattering

By keeping the lowest-order terms in the low-energy limit of the wave functions we have deduced that a_0 parameterizes low-energy scattering. If the next-higher-order term is included it is found that any finite-ranged force is parameterized with just two number (a_0, r_0), the scattering length and *effective range*:

$$k \cot \delta_0 \rightarrow \frac{1}{a_0} + \frac{1}{2} r_0 k^2 + O(k^4). \quad (4.16)$$

As derived in the references and the Problems section, the effective range r_0 is a measure of the distance over which the distorted wave u_0 differs from a free wave and is thus a better measure of the range of the potential than the scattering length.

Neutron-proton scattering in the spin 0 (singlet s) and spin 1 (triplet t) states is a good example of the preceding theory. The experimental values for the scattering lengths and effective ranges are (Brown and Jackson, 1976):

$$(a_0^s = 23.7 \text{ fm}, r_0^s = 2.76 \text{ fm}), \quad (a_0^t = -5.41 \text{ fm}, r_0^t = 1.75 \text{ fm}). \quad (4.17)$$

It is known that the triplet potential is attractive enough to bind the deuteron by 2.22 MeV; this is consistent with the negative sign for a_0^t . Yet it is also known that the potential has a depth $\simeq 70$ MeV and a range $\simeq 2$ fm; this range is consistent with r_0^t . The small binding energy relative to the depth reveals that the bound state lies near the top of the well, the large negative value of a_0^t (relative to r_0) indicates that the deuteron is weakly bound, that is, like the dot-dashed curve in Figure 4.1D. In contrast, the positive sign for the singlet scattering length a_0^s indicates that the potential for this spin state is attractive, but not enough to form a bound state. The very large magnitude of a_0^s indicates that the wave function is almost turned over, like Figure 4.1C; a slight increase in attraction would form a singlet bound state.

Relation to Bound States

We have just seen that a potential which forms a zero-energy bound state also produces an infinite scattering length. As proven in § 7.5, the result is more general: bound states occur at energies for which the scattering amplitude has poles. This is a profound thought. We are analytically continuing what was the amplitude of a scattered wave of positive energies to negative energies (complex wave numbers k) where scattering cannot be defined. The mathematical justification of the continuation arises from the analytic continuation of the Schrödinger equation into the complex energy plane.²

As an example, consider small k for which the scattering amplitude takes the approximate form:

$$f_B \simeq \frac{f_{l=0}}{k} = \frac{\tan \delta_0(k)/k}{1 - i \tan \delta_0(k)} \simeq \frac{a_0}{1 - i a_0 k}. \quad (4.18)$$

¹Newton (1966) and Tabakin (1968) give general proofs. Note too that $\delta(\infty) = 0$ for most cases.

²Park (1966), Courant and Hilbert (1962), Chew (1962), Newton (1966), Omnes and Froissart (1963).

Because the circle of convergence extends into the complex k plane, (4.18) should be a good approximation for small but complex k (we know it has the analytic behavior required by the optical theorem). That being so, the poles of f_E occur at the positive imaginary momenta for which $\tan \delta_0(k) = 1/i$, which in this case means $k = -i/a_0$.

Exercise Show that the location of these poles and our sign convention makes sense by showing that the plane wave continues into the dying exponential $e^{ikr} \rightarrow e^{r/a}$ with $a < 0$ for bound states. $\diamond$

The imaginary momentum of the pole positions correspond to negative energies,

$$E = \frac{k^2}{2\mu} = \frac{-1}{2\mu(a_0)^2} = -E_B, \quad (4.19)$$

as expected for bound states. Because knowledge of the binding energy is equivalent to knowledge of the scattering length, if there is a bound state, it determines the low energy scattering:

$$\sigma^{\text{el}} = |f_E|^2 \simeq \frac{1}{2\mu(E + E_B)}. \quad (4.20)$$

Conversely, knowledge of the scattering length determines the bound state energy. For example, from knowledge of the deuteron's binding energy we predict the spin-triplet, nucleon-nucleon scattering length to be

$$a_0^t \simeq \frac{1}{\sqrt{E_B 2\mu}} = \frac{\hbar c}{\sqrt{E_B 2(m_N/2)c^2}} = -4.32 \text{ fm}. \quad (4.21)$$

The difference of (4.21) from the observed value of -5.41 fm indicates that the bound state is not truly at zero energy and some higher-order terms are needed. (Note that while the scattering length approximation may not be true if the energy gets far from zero, the relation of the poles of the T matrix to bound states is general.)

4.2 Resonances

Many elementary text such as Baym (1969) and Merzbacher (1970) discuss transmission and potential-well resonances. We indicate here how the resonance idea fits into the discussion of scattering and bound states of this chapter.

Consider the bound-state levels of a typical potential with an angular momentum barrier (solid lines in Figure 4.3). Because solutions of the Schrödinger equation are continuous in all parameters, a bound state with nearly-zero negative energy should not change much if it were continued to a nearly-zero positive energy. In a negative-energy bound state the two-body system is trapped in the well forever; in a positive-energy scattering state the system eventually tunnels through the barrier, remaining confined for only a finite (but possibly long) time. These positive-energy nearly-confined states are called *resonances*, and, like bound states, occur at the poles of the scattering amplitude.

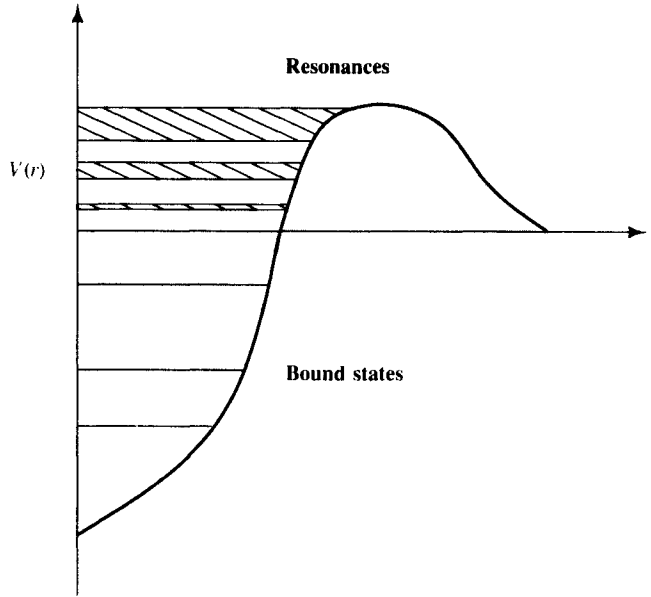


Figure 4.3: Bound state ($E < 0$) and resonance ($E > 0$) levels. Note how the resonance level's width increases (lifetime decreases) as there is less barrier through which to tunnel.

Breit–Wigner Resonances

We saw in § 4.1 that $\tan \delta_l$ becomes infinite when analytically continued to (negative) bound-state energies. Let us now examine the possibility of $\tan \delta_l$ becoming infinite at small positive energies. We start with (3.65):

$$\tan \delta_l = -\frac{F'_l(kR_m) - F_l(kR_m)\mathcal{L}(R_m)}{G'_l(kR_m) - G_l(kR_m)\mathcal{L}(R_m)} \quad (4.22)$$

$$\rightarrow \frac{k^{2l+1} R_m^{2l+1} [(l+1) - \mathcal{L}R_m]}{(2l+1)[1 \cdot 3 \cdot 5 \cdots (2l-1)]^2 (l + \mathcal{L}R_m)}. \quad (4.23)$$

We see that the denominator will vanish at an energy we call E_R for which $\mathcal{L}R_m = -l$, that is, $\tan \delta_l = \infty$, $\delta_l = \frac{1}{2}\pi + n\pi$. Accordingly, for energies near E_R , the logarithmic derivative must have the expansion

$$\mathcal{L}R_m \simeq -l + (E - E_R) \left. \frac{d(\mathcal{L}R_m)}{dE} \right|_{E=E_R}. \quad (4.24)$$

If we substitute this into (4.23), we find that for E 's close to E_R ,

$$\tan \delta_l \simeq \frac{\Gamma_l/2}{E_R - E}, \quad (4.25)$$

$$\Gamma_l = 2 \frac{-(kR_m)^{2l+1}}{[(2l-1)!!]^2 d(\mathcal{L}R_m)/dE}. \quad (4.26)$$

Such being the case, the not-so-transparent assumption (4.24) for $\mathcal{L}$ leads to the *Breit–Wigner* resonance form for the amplitude,

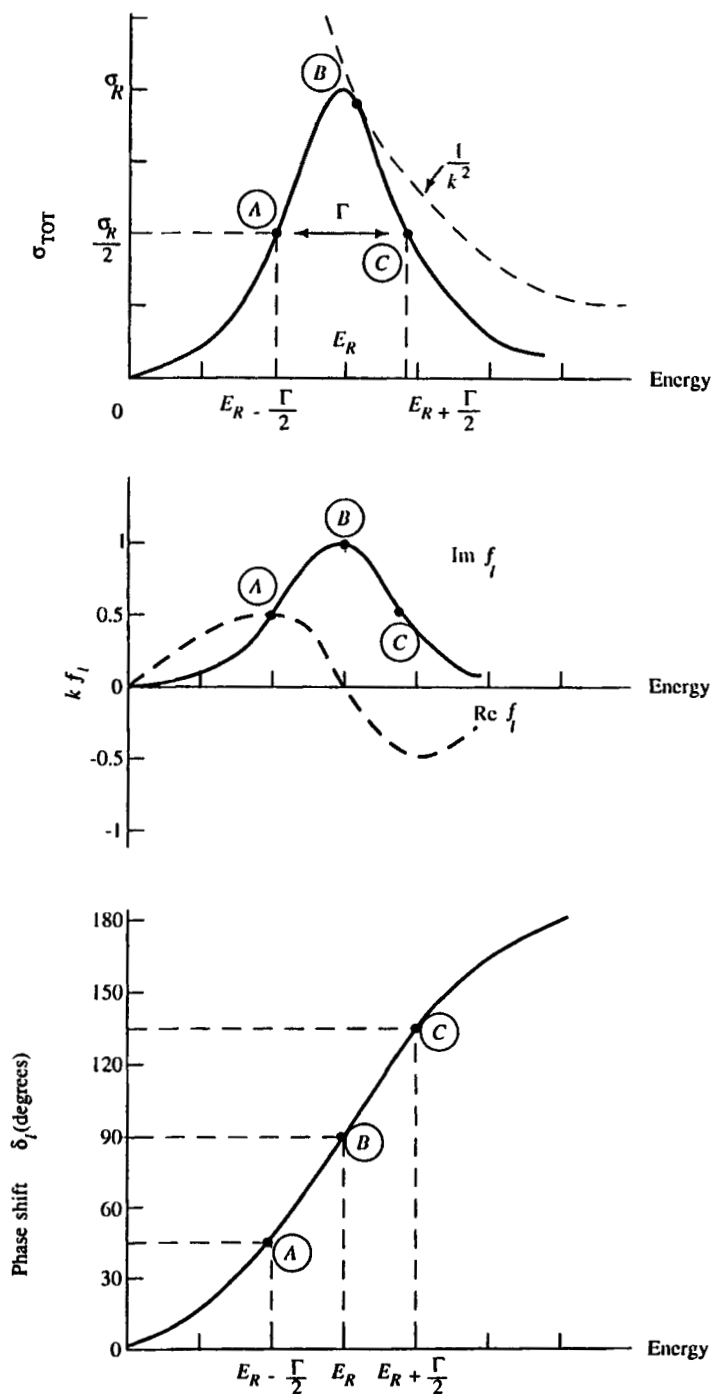


Figure 4.4: Resonance behavior in a total cross section (*top*); in the real and imaginary parts of a scattering amplitude (*middle*); and in the phase shift (*bottom*).

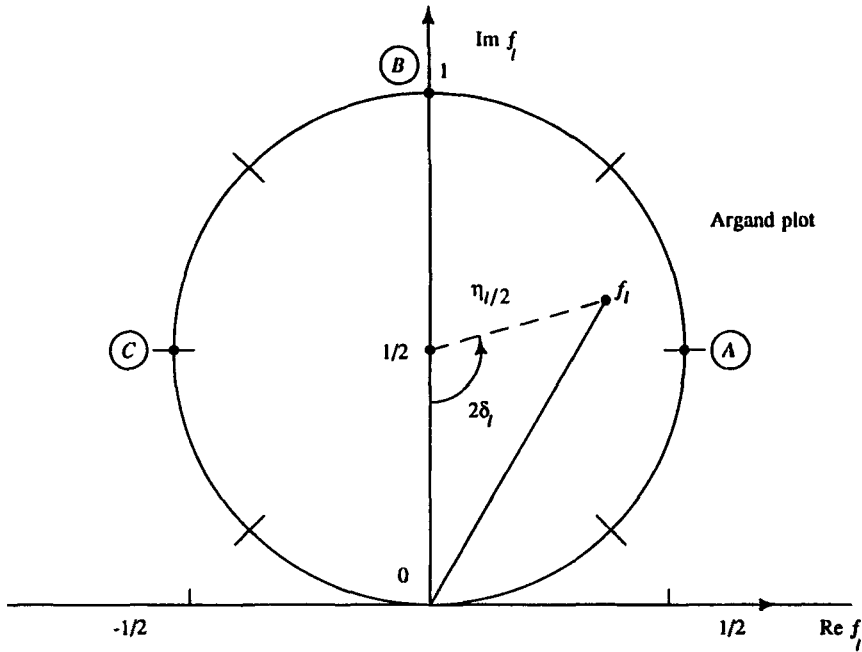


Figure 4.5: Argand plot ($\text{Im} f_l$ versus $\text{Re} f_l$) of the scattering amplitude. The complex amplitude f_l rotates counterclockwise around the circle as the energy passes through resonance. The points A, B, and C correspond to those in Figure 4.4.

$$f_l = e^{i\delta_l} \sin \delta_l = \frac{\Gamma_l/2}{E_R - E - i\Gamma_l/2}, \quad (4.27)$$

and to the famous resonance shape for the cross section,

$$\sigma_l^{\text{tot}} = \frac{4\pi(2l+1)}{k^2} \frac{\Gamma_l^2/4}{(E_R - E)^2 + \Gamma_l^2/4}. \quad (4.28)$$

Exercise Derive (4.27) from (4.25). ◇

Exercise Verify that (4.28) follows from (4.27) by use of the optical theorem. ◇

We see from (4.28) that if Γ_l is small, that is, $\mathcal{L}$ and δ_l vary rapidly near E_R , the resonance will be *sharp* and σ_l^{tot} will have a narrow peak as a function of energy, as indicated in Figure 4.4. If we look at the second factor in (4.28), we see that it equals half of its peak value when $E = E_R \pm \frac{1}{2}\Gamma_l$; for this reason Γ_l is called the *full width at half maxima* and is always a positive number (Baym, 1969). Note that unless the resonance is exceedingly narrow (very small Γ_l), the $1/k^2$ factor in σ_l^{tot} will distort the shape of the cross section, and shift its peak from E_R to a lower energy.³

³Consequently, to deduce a peak's width and energy it is best to remove the distorting kinematic factors before analyzing cross section data, or use an Argand plot.

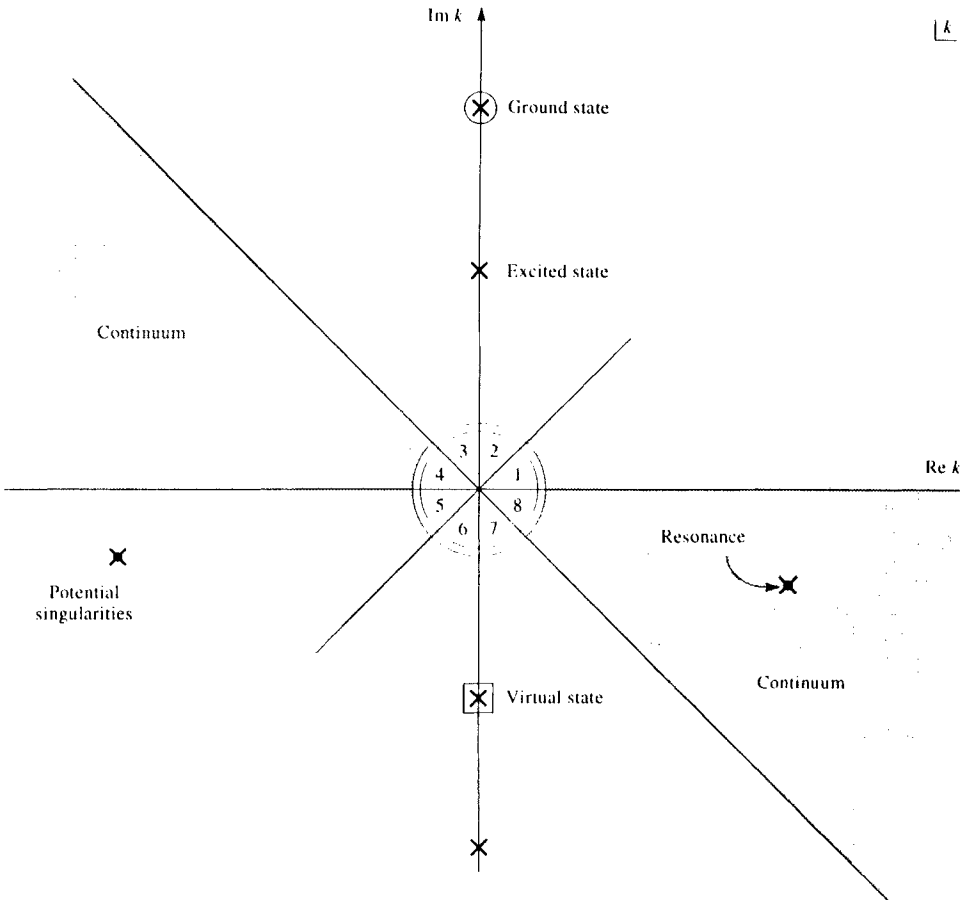


Figure 4.6: Some possible poles and cuts of the scattering amplitude in the complex momentum plane. The shading indicates the continuum.

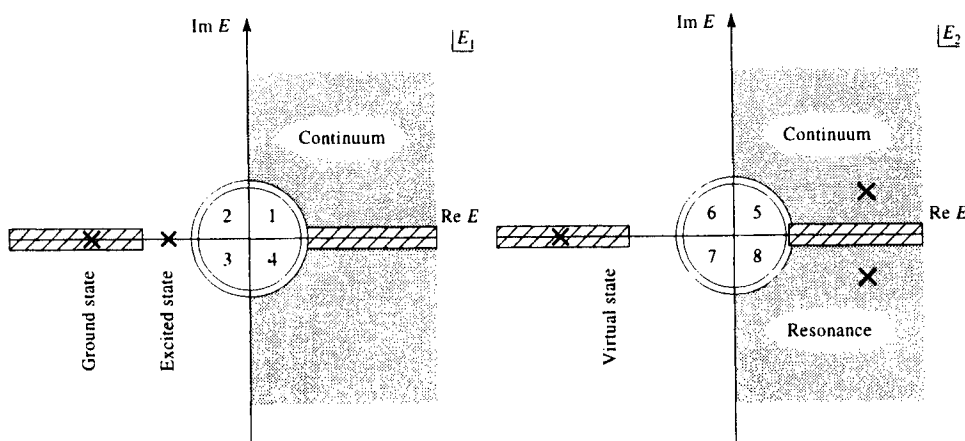


Figure 4.7: Some possible poles and cuts of the scattering amplitude in the complex energy plane. Only poles on first energy sheet correspond to bound states. The shading indicates the continuum. The right-hand cut, which separates the two energy planes, begins at the origin. The left-hand cuts arise from the potential itself.

As the energy increases through resonance, the imaginary part of the scattering amplitude $\text{Im} f_l$ increases and then decreases, the real part changes from positive (the sign expected for attraction) to negative (Figure 4.4), and the partial-wave total cross section obtains its largest possible value (the *unitarity limit*), $4\pi(2l + 1)/k^2$. A useful way of examining the energy dependence of f_l is with an Argand plot of $\text{Im} f_l$ versus $\text{Re} f_l$ (see Figure 4.5 and the Problems section).

Physically, a sharp peak in the energy dependence of a cross section indicates a dynamical happening such as a strong attraction at that energy. If the phase shift passes rapidly through $\pi/2$ (modulo π) as shown in Figure 4.4, this probably means a resonance, that is, the beam and target particles are temporarily binding themselves together and then breaking apart.⁴ These positive-energy resonant states are often included in energy-level diagrams with broad lines, as we have done in Figure 4.3, with the widths of these levels given by Γ_l , the uncertainty in the value of the resonant energy. If the well depth is increased sufficiently, the resonances would be drawn down into the bound-state spectrum and their widths would disappear (unless there is still some coupling to open channels, in which case they may become *unstable bound states*, in which case they may still be considered resonances). If the width of a resonance is small, it is similar to a bound state which decays to physical particles.⁵ We see part of this similarity in the analytic structure of the scattering amplitude f_l where both bound states and resonances occur as poles, but in different quadrants of the complex plane. The position of these poles, as well as some other analytic properties of the amplitude, are indicated in Figure 4.6 for the complex momentum plane, and in Figure 4.7 for the upper and lower Riemann sheets of the complex energy plane. We see there that the resonant amplitude (4.27) has poles at

$$E = E_R - i\Gamma_l/2, \quad (4.29)$$

⁴Often a number of indirect tests are used to isolate a resonance and even then the conclusion is rated in regards to reliability, see Particle Data Group (1994).

⁵Except for just a few, most of the tabulated elementary "particles" are actually resonances.

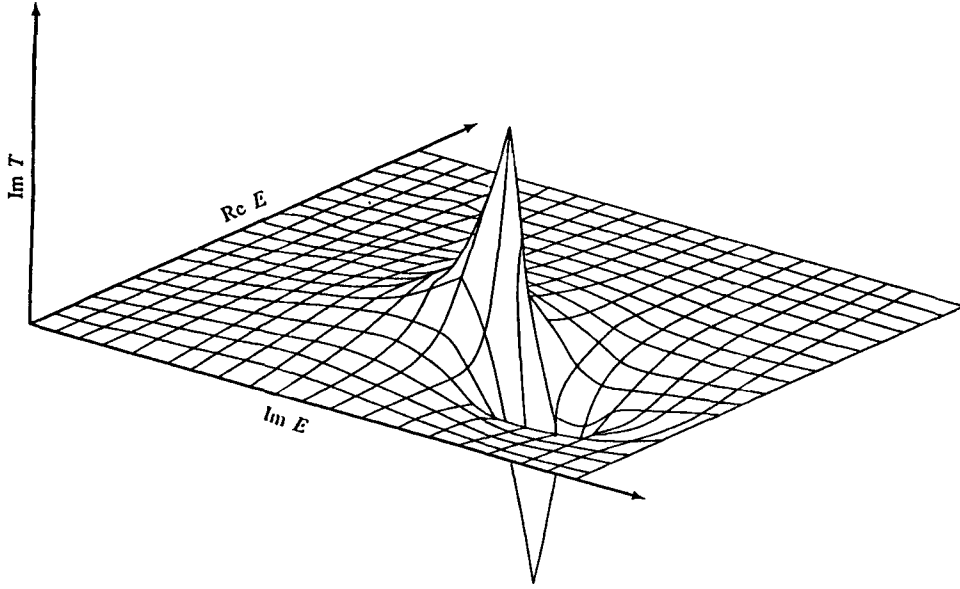


Figure 4.8: The imaginary part of the scattering amplitude describing a resonance as a function of complex energy [with the origin displaced from (0, 0)]. The interaction is that of an antikaon with a nucleon as generated by a quark bag model. (Courtesy of G. He and P. Fink.)

$$k = \sqrt{2\mu E} = \sqrt{2\mu E_R - i\mu\Gamma_I} = \sqrt{2\mu E_R} \sqrt{1 - i\Gamma_I/2E_R} \quad (4.30)$$

$$\simeq k_R - i \frac{\mu\Gamma_I/2}{k_R} \quad (\text{for small } \Gamma_I). \quad (4.31)$$

If Γ_I is small, this pole is right below the real positive k axis in Figure 4.6, that is, in the eighth octant of the complex k plane.⁶ When the corresponding complex energy $E = k^2/2m$ is formed, the bound-state poles map to the first Riemann energy sheet (Figure 4.7), while the resonance poles move into the lower half of the second (“unphysical”) Riemann energy sheet (Figure 4.7). The smaller is Γ_I , the sharper is σ ’s peak; the longer-lived is the resonance, the closer the poles are to the real axis, and the clearer it is that an actual “state” exists.

While it may be clear from looking at the analytic expression for the Breit–Wigner resonance amplitude (4.27) that it has a singularity in the complex energy plane at $E = E_R - i\Gamma_I/2$, actually visualizing a resonance as a function of complex energy may be less clear. We leave it as an exercise to verify that the three-dimensional plot in Figure 4.8 is that of the imaginary part of a resonance’s scattering amplitude as a function of complex energy. Note the characteristic pole behavior of a peak as a function of real energy and a sign change as a function of imaginary energy.

⁶We have chosen the positive square root in (4.30), yet there are cases where the negative root also leads to physical states. See for example Newton (1966) and Park (1964) for discussions of bound states in the continuum and negative energy resonances.

Complex Energy States and Exponential Decay ◉

*When in eternal lines to time thou grow'st:
So long as men can breathe, or eyes can see,
So long lives this, and this gives life to thee.*

—William Shakespeare

A fascinating application of the resonance concept concerns the time dependence of resonant states and its relation to *Gamow states* (Gamow, 1928; Blatt and Weisskopf, 1952; Kapur and Peirles, 1938; Kwon and Tabakin, 1978; Hernandez and Mondragon, 1984; Landau, 1983). We start with the assumed time dependence of stationary states $\psi \propto \exp(-iEt)$. A resonance with complex energy $E = E_R - i\Gamma/2$ has a wave function ψ and probability density $|\psi|^2$ with the time dependences:

$$\psi \propto e^{-iEt} = e^{-i(E_R - i\Gamma/2)t} = e^{-iE_R t} e^{-\Gamma t/2}, \quad (4.32)$$

$$|\psi|^2 \propto e^{-\Gamma t} = e^{-t/\tau}. \quad (4.33)$$

The probability density of a resonance accordingly decays exponentially in time with a time constant or *lifetime*

$$\tau = \frac{1}{\Gamma} \equiv \frac{\hbar}{\Gamma}. \quad (4.34)$$

Equation (4.34) is the uncertainty principle's relation of the time uncertainty τ to the energy uncertainty $\Delta E = \Gamma$.

Decaying states were introduced into quantum mechanics by Gamow in 1928 while developing a model for alpha-particle emission from nuclei. The concept of resonant states having a complex energy was developed by Kapur and Peirles in 1938. Both models envision a confined state which decays slowly in time by emitting particles, that is, a state with *pure outgoing waves*.

In mathematical terms, we are looking for a wave function whose logarithmic derivative $\mathcal{L}$ matches onto the logarithmic derivative of the pure, outgoing wave e^{ikr} :

$$\mathcal{L}(R_m) = \frac{1}{u_l(kr)} \left. \frac{du_l(kr)}{dr} \right|_{r=R_m} = ik. \quad (4.35)$$

For a given energy there are usually two solutions.⁷ We restrict our considerations to states which decay exponentially in time, that is, which are described by (4.32) with $\Gamma > 0$. This requires the imaginary part of the energy to be negative, which, in turn, requires the k of the outgoing wave e^{ikr} to be complex:

$$2\mu E = k^2 = (k_R^2 - k_I^2) + 2ik_R k_I. \quad (4.36)$$

Exercise Show that two states are possible with negative imaginary energy:

$$\psi = \begin{cases} \psi_{\text{res}} = e^{ik_R r} e^{|k_I| r}, & \text{with } k_R > 0, k_I < 0, \\ \psi_{\text{bs}} = e^{-i|k_R| r} e^{-k_I r}, & \text{with } k_R < 0, k_I > 0. \end{cases} \quad \diamond \quad (4.37)$$

⁷Since resonant states are in the continuum, physical solutions to the Schrödinger equation exist for all energies. The additional outgoing-wave constraint on the energy leads to discrete energies.

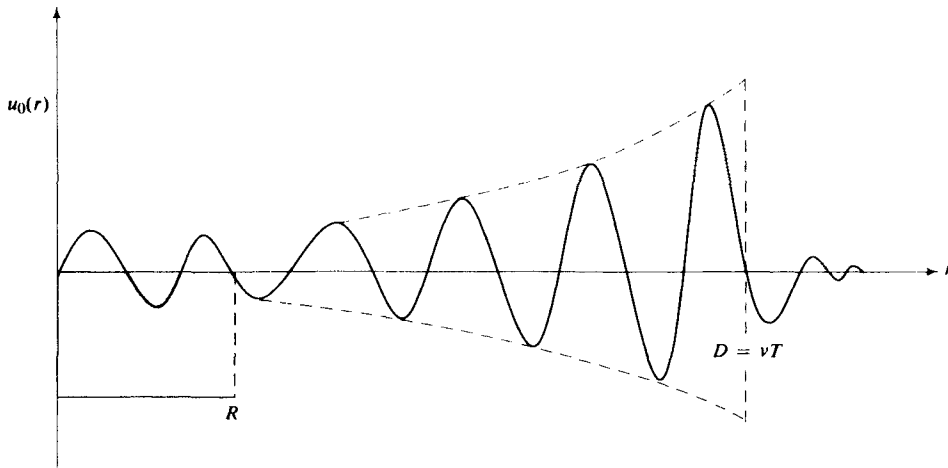


Figure 4.9: A model of the wave function of a resonant (decaying) state. The time since the state was formed is T , the propagation velocity is v , the distance the wave has traveled is D , and the range of the interaction is R .

We identify the outgoing wave state ψ_{res} as a *resonant* state if its momentum is in the eighth octant of the complex-momentum plane and close to the axis (right below the real energy axis on the second, “physical” energy sheet in Figure 4.7). We identify the incoming wave state ψ_{bs} containing a decaying exponential in space, as an *unstable bound state* if its momentum is in the third octant of the complex-momentum plane and close to the imaginary axis. The exponentially increasing amplitude of the resonant state (4.37) is related to the exponentially decreasing time dependence; as we go further out in r we examine the wave emitted at earlier times when there was more of a state! The exponentially decreasing space dependence in (4.37) indicates a bound state, while its incoming plane-wave amplitude indicates an incoming flux needed to feed a state which keeps decaying in time.

If these complex energy states seem peculiar, in part that is because we are using a time-independent formalism to describe what is truly a time-dependent situation (resonance formation via scattering and subsequent decay into the particle–target system). A more realistic picture of the wave function may be that in Figure 4.9 in which there is a confined state within the potential’s regime ($r < R$), an increasing-exponential space dependence outside of the potential’s regime, and a decrease of the wave function to zero at some large distance $D \geq vT$. Here T is the time since the state was formed, and v is the velocity at which the waves radiate outward. This resonant state is manifestly finite and thus normalizable; it is the limit of this state as $T \rightarrow \infty$ that we are attempting to model with our time-independent formalism. Nonetheless, resonances have proven to be useful models of nature and, as discussed in the references and the Problems section, has required us to extend the rules of quantum mechanics to handle complex energies. Although describing decaying states may appear as just an academic exercise, they are probably more realistic than conventional quantum mechanics; if we wait long enough many (if not all) particles, nuclei, and atoms do decay in time and thus must be described as a type of Gamow state.

4.3 Coulomb Scattering: A Bad Example

The theory and applications developed so far have assumed potentials with a finite range R because only then does the angular momentum barrier $l(l+1)/2\mu r^2$ determine the inner wave function. This assumption is violated by the Coulomb potential,

$$V(r) = \frac{Z_T Z_P e^2}{r}, \quad (4.38)$$

acting between a projectile with charge $Z_P e$ and a target with charge $Z_T e$.⁸ The extension of the theory requires either the inclusion of Coulomb distortion into what was previously our free waves, F_l and G_l , or the inclusion of electronic shielding into the theory (the universe is electrically neutral so a realistic V has finite range). In the next two sections we outline the changes.⁹

Pure Coulomb Scattering

To see the “problem” with the Coulomb potential, we write the 1D radial Schrödinger equation (3.11) in dimensionless form:¹⁰

$$\frac{d^2 u_l(\rho)}{d\rho^2} + \left(1 - \frac{2\eta}{\rho} - \frac{l(l+1)}{\rho^2}\right) u_l(\rho) = 0, \quad (\rho = k r), \quad (4.39)$$

$$\eta = \frac{\mu Z_T Z_P e^2}{k} = \frac{Z_T Z_P e^2}{v} = \frac{1}{a_B k}, \quad a_B = \frac{1}{\mu Z_T Z_P e^2}. \quad (4.40)$$

Here η is the *Coulomb parameter* (positive for a repulsive interaction), v is the target-beam relative velocity, and a_B is the Bohr radius. The “problem” is that

$$u_l(\rho) \sim e^{\pm i\rho} e^{\mp i\eta \ln(2\rho)} \quad (\rho \rightarrow \infty). \quad (4.41)$$

This means $u_l(\rho)$ behaves like it has a phase shift which depends on r and becomes infinite for large r ! Aside from this (quite significant) change, the solution is much like before, only now with an $F_l(\eta, \rho)$ which is regular at the origin, and a $G_l(\eta, \rho)$ which is irregular. The most important properties for us are the $r \rightarrow 0$ and $r \rightarrow \infty$ behaviors:

$$F_l(\eta, \rho) = \begin{cases} \rho^{l+1}, & \text{for } \rho \rightarrow 0, \\ \sin[\rho - \eta \ln(2\rho) - l\pi/2 + \sigma_l], & \text{for } \rho \rightarrow \infty, \end{cases} \quad (4.42)$$

$$G_l(\eta, \rho) = \begin{cases} \rho^{-l}, & \text{for } \rho \rightarrow 0, \\ \cos[\rho - \eta \ln(2\rho) - l\pi/2 + \sigma_l], & \text{for } \rho \rightarrow \infty, \end{cases} \quad (4.43)$$

$$\sigma_l = \arg \Gamma(l+1+i\eta) = -\frac{1}{2}i \ln \frac{\Gamma(l+1+i\eta)}{\Gamma(l+1-i\eta)}. \quad (4.44)$$

If $\eta = 0$, the Coulomb waves reduce to our previous expressions:

$$F_l(\eta = 0, \rho) = F_l(\rho) = \rho j_l(\rho), \quad (4.45)$$

$$G_l(\eta = 0, \rho) = G_l(\rho) = -\rho n_l(\rho). \quad (4.46)$$

⁸Our convention has a repulsive Coulomb potential. Reverse the sign of e^2 for the attractive interaction.

⁹A complete analysis of Coulomb scattering is rather involved and we refer the reader to Rodberg and Thaler (1967), Gottfried (1966), Schiff (1968), Abramowitz and Stegun (1964), and Yost et al. (1936), and the Problems section.

¹⁰The treatment of Coulomb scattering is very similar to that of the hydrogen atom, except for the sign of the energy.

Before we proceed with the phase-shift analysis based on these new functions, it is helpful to examine the exact solution of the 3D Schrödinger equation for the Coulomb potential. For our purposes we only need to know that the asymptotic limit of the exact wave function is

$$\psi(\mathbf{r}) \sim e^{i[kz + \eta \ln(k[r-z])]} + \frac{f_C(\theta)}{r} e^{i[kr - \eta \ln(2kr)]}. \quad (4.47)$$

In spite of the logarithmic-phase problem, ψ still decomposes into incident plus scattered waves, only now these waves are distorted by the long range of the Coulomb potential. Because the scattering observables are defined in terms of currents, and since currents are unaffected by phase factors, the theory is still applicable (except possibly at $\theta \simeq 0$). Specifically, the amplitude f_C of the outgoing, scattered spherical wave is

$$f_C(\theta) = \frac{-\eta e^{-i\eta \ln(\sin^2 \theta/2)} e^{2i\sigma_0}}{2k \sin^2 \theta/2}. \quad (4.48)$$

This we recognize as the amplitude for scattering from a point charge. Because $j_{sc} \sim (k/\mu)(1/r^2)|f_C|^2$, we deduce that the quantum-mechanical cross section for scattering of two point charges is

$$\boxed{\frac{d\sigma}{d\Omega}(\theta) = |f_C(\theta)|^2 = \frac{\eta^2}{4k^2 \sin^4 \theta/2} = \left(\frac{Z_T Z_P e^2}{4E \sin^2 \theta/2} \right)^2}. \quad (4.49)$$

In spite of the perplexing nature of the infinite phase of $f_C(0^\circ)$, we recognize equation (4.49) as the classical Rutherford cross section.¹¹

The connection of this exact analysis with the phase-shift analysis is obtained by expanding the analytic expression (4.48) in Legendre polynomials:

$$f_C(\theta) = \sum_{l=0}^{\infty} (2l+1) \frac{e^{2i\sigma_l(k)} - 1}{2ik} P_l(\cos \theta). \quad (4.50)$$

In analogy with (3.38), the σ_l 's are called *point Coulomb phase shifts* because they measure how much $F_l(\eta, \rho)$'s and $G_l(\eta, \rho)$'s phases are shifted from free waves [σ_l has already appeared in (4.42) and (4.43)].

Hidden Hydrogen Atoms

Inasmuch as the rigor of our analysis may be less than convincing, it is challenging to compare the results against the conclusions drawn earlier in this chapter regarding bound states and poles of the scattering amplitude. If we examine our expression for the Coulomb scattering amplitude (4.50), we see that poles enter through the $\exp[2i\sigma_l(k)]$ factor (we ignore those at $k = 0$ where there is also an infinite number of hydrogen bound states). We write this exponential factor as

$$e^{2i\sigma_l(k)} = \frac{\Gamma(l+1+i\eta)}{\Gamma(l+1-i\eta)}, \quad (4.51)$$

¹¹We will see in the next section that screening removes the $\theta \rightarrow 0$ singularity. However, a satisfying explanation of why classical and quantum mechanics give the same cross section is not known to this author.

where we have used the definition of σ_1 (4.44). Because the gamma function $\Gamma(z)$ has poles at $z = 0, -1, -2, \dots$, but no zeros, and because η must be negative for attraction (repulsive potential do not form bound states), the poles of (4.51) occur at:

$$l + 1 + i\eta = 0, -1, -2, \dots, \quad (4.52)$$

$$\Rightarrow i\eta \equiv \frac{i}{-a_B k} = -(l + 1), -(l + 2), \dots, \quad (4.53)$$

$$\Rightarrow E = \frac{k^2}{2\mu} = \frac{-1}{a_B^2 2\mu n^2}, \quad n = 1, 2, \dots \quad (4.54)$$

These are the bound-state energies of the hydrogen atom. Our deduction that the bound-state energies are determined by analytically continuing the positive-energy scattering amplitude to negative energies and looking for its poles, makes sense here too.

Regge Poles

Buoyed by our success in analytically continuing the Schrödinger equation to complex energies, we should feel confident to explore continuations in other parameters. In particular, we immediately deduce the angular momentum at which the scattering amplitude (4.50) has poles by solving (4.52) for l :

$$l = -1 - n - i\eta = -1 - n + \frac{i}{a_B k}, \quad n = 0, 1, 2, \dots \quad (4.55)$$

This means that Schrödinger's equation supports infinitely many bound states of energy $\kappa^2/2\mu$ for these noninteger values of l (recall that $k = +i\kappa$ and $\eta < 0$ for bound states). We identify those values of k for which l is an integer as physical bound states. The other *Regge poles* form a *Regge trajectory* in the complex k plane and reveal much about the dynamics.

An application of Regge poles arises in elementary particle physics where the Hamiltonian is quite different from the Coulomb potential. When the masses of particles are plotted as a function of their angular momenta ("spin"), they are found to fall on a continuous curve. Apparently, the particles on any one trajectory have the same internal quantum numbers n but different l 's, and are thus related in a way similar to the analogous states in hydrogen. More thorough discussions of Regge poles can be found in Collins and Squires (1968), Eden (1967), and Omnes and Froissart (1963).

Shielded Coulomb Potential

Because the universe appears to be electrically neutral, if you take a broad enough point of view there never are interactions with isolated point charges. We indicate this possibility schematically in Figure 4.10 which shows the electric charge $Z_T e$ at the origin and a thin shell of radius R_s and thickness Δr upon which a charge $-Z_T e$ resides. A detector is at a distance X . In a typical experiment $Z_T e$ resides on the nucleus of size R , and $-Z_T e$ resides on the atomic shells of size R_s ; thus $R \sim 10^{-13}$ cm, $X \sim 1$ cm, $R_s \sim 10^{-8}$ cm, and $R_s \ll X$.

The shielded charge is a finite-ranged potential:

$$V(r) = \begin{cases} Z_T Z_P e^2 (1/r - 1/R_s), & \text{for } r \leq R_s, \\ 0, & \text{for } r > R_s. \end{cases} \quad (4.56)$$

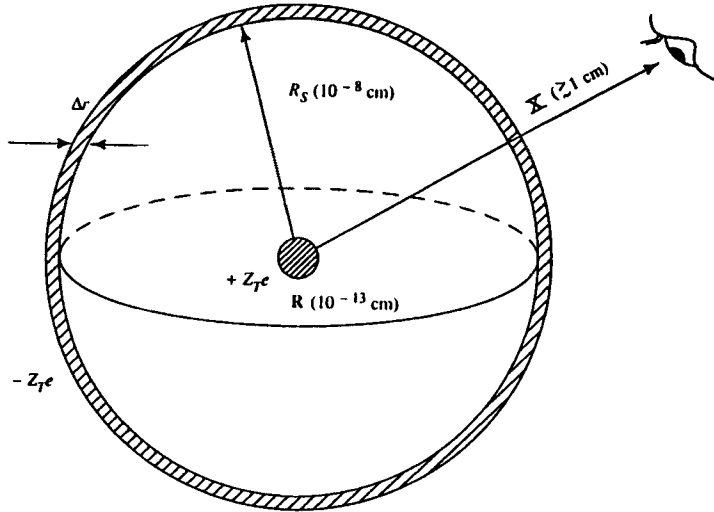


Figure 4.10: A positively charged nucleus of radius R at the origin and its electronic shell of radius R_s and thickness Δr .

Our conventional phase shift analysis is consequently valid, and there is a maximum (and very large) number of partial waves $l_{\max} \simeq kR_s$ which contribute [recall the impact parameter relation (3.5)]. Inside R_s the electrons do not contribute, so V is the Coulomb potential from $+Z_T e$ and the wave function is the regular Coulomb wave $F_l(\eta, kr)$. Exterior to the shielding, all potentials vanish so the wave function must be a combination of F_l and G_l , or equivalently, the phase-shifted regular solution $F_l(kr)$:

$$u_l(kr) \sim \begin{cases} \sin[kr - l\pi/2 + \sigma_l - \eta \ln(2kr)], & \text{for } R < r < R_s, \\ \sin[kr - l\pi/2 + \delta_l], & \text{for } R_s < r. \end{cases} \quad (4.57)$$

Exercise Show that for high enough energies the large kr form is accurate, and that matching the inner and outer solutions determines the phase shift

$$\delta_l = \begin{cases} \sigma_l - \eta \ln(2kR_s), & \text{for } l < l_{\max}, \\ 0, & \text{for } l > l_{\max}. \end{cases} \quad \diamond \quad (4.58)$$

The nonvanishing phase shifts are just the point-Coulomb phases σ_l plus a term which depends on the shielding.

Exercise Show by substitution that the shielded Coulomb scattering amplitude is

$$f_{\text{sc}}(\theta) = \sum_{l=0}^{l_{\max}} \frac{e^{2i\sigma_l} e^{-2i\eta \ln(2kR_s)} - 1}{2ik} (2l+1) P_l(\cos \theta). \quad \diamond \quad (4.59)$$

While in some sense we have now solved the problem, it is illuminating to rearrange (4.59) into a form which separates out the point Coulomb amplitude (4.48). We leave the analysis to the references and quote the reasonable result:

$$f_{\text{sc}}(\theta) \simeq e^{-2i\eta \ln(2kR_s)} [f_c(\theta) + F(\theta \leq 2\eta/kR_s)]. \quad (4.60)$$

We see an overall phase factor $\exp[-2i\eta \ln(2k\tau)]$ which depends on the screening and its details but is unobservable when a cross section is formed. The first term in brackets is the familiar point Coulomb scattering amplitude $f_c(\theta)$ (4.48); the second term F depends on the screening and is nonnegligible only if $\theta < 2\eta/kR_s$. Consequently, for high energy experiments only f_c enters for all but the most forward scattering, and it is possible to examine scattering from only the nucleus (and ignore the complications of this section). Likewise, for experiments at low k , only F will be important and it is possible to examine scattering from only the atomic charge.

Coulomb Plus Short-Range Potentials $\odot$

Often the Coulomb potential acts in conjunction with a potential V' which has a short range in space (e.g., the correction due to the finite size of a charge distribution, or the nuclear potential):

$$V(r) = \begin{cases} Z_T Z_P e^2/r + V'(r), & \text{for } r < R, \\ Z_T Z_P e^2/r, & \text{for } r > R. \end{cases} \quad (4.61)$$

For this case we define a phase shift δ'_l caused by the potential $V'(r)$ as the shift in phase of u_l relative to the Coulomb distorted waves $F_l(\eta, kr)$ and $G_l(\eta, kr)$:

$$u_l(kr > kR) = e^{i\delta'_l} [F_l(\eta, kr) \cos \delta'_l + G_l(\eta, kr) \sin \delta'_l] \quad (4.62)$$

$$\sim \sin(kr - l\pi/2 - \eta \ln 2kr + \sigma_l + \delta'_l). \quad (4.63)$$

If we ignore the Coulomb distortion of the phase, the total phase shift is just the sum of the point Coulomb phase σ_l plus that due to the short range potential V' :

$$\delta_l = \sigma_l + \delta'_l. \quad (4.64)$$

An examination of the expression for the current based on the wave function (4.63) yields¹² the same expression for the scattering amplitude as would be obtained by direct substitution of the phase shift $\delta = \sigma + \delta'$ into our standard expression (3.38):

$$f_{c'}(\theta) = \sum_{l=0}^{\infty} (2l+1) \frac{e^{2i\sigma_l} e^{2i\delta'_l} - 1}{2ik} P_l(\cos \theta). \quad (4.65)$$

Exercise Show by simple algebraic manipulation that the amplitude separates as:

$$f_{c'}(\theta) = f_c(\theta) + f'(\theta), \quad (4.66)$$

$$f'(\theta) = \sum_{l=0}^{\infty} (2l+1) \frac{e^{2i\sigma_l} (e^{2i\delta'_l} - 1)}{2ik} P_l(\cos \theta). \quad \diamond \quad (4.67)$$

This result states that the scattering amplitude for scattering from the combined short-range plus Coulomb potential (4.61) is the sum of the amplitude for pure Coulomb scattering f_c plus an amplitude for scattering from V' , the latter amplitude containing a correction factor

¹²These same results are derived more rigorously once the two-potential formula is derived. This is described in § 7.8.

for the Coulomb distortion of the free wave upon which V' acts. Because of this distortion, the phase shift for the scattering from the sum of two potentials does not equal the sum of phase shifts for the scattering from the individual potentials. Likewise, the separate scattering amplitudes do not add.

The addition of the two amplitudes in (4.66) illustrates a basic result of quantum mechanics: When there is no experimental way to distinguish the internal steps for some process (that is, scattering from V' or V_c), one adds together all the possible probability amplitudes to obtain the complete amplitude. The probability (in our case cross section) is the amplitude squared:

$$\frac{d\sigma}{d\Omega}(\theta) = |f_c + f'|^2 = |f_c|^2 + |f'|^2 + 2 \operatorname{Re} [f_c^*(\theta) f'(\theta)] . \quad (4.68)$$

The last term is the quantum-mechanical interference term and provides an experimental means to determine the relative sign (that is, attraction or repulsion) of f' . For example, if there is *destructive interference* in the scattering of two particles with the same sign of electric charge, we would know that V' has *opposite* sign to that of V_c , that is, it is attractive. We give an example in Problem 6.

4.4 Problems

1. To help test your understanding of the material in this chapter, try answering the following questions without resort to the text.
 - (a) What happens to the magnitude and phase of a scattering amplitude as the energy of the scattered particle passes through a resonance?
 - (b) The Coulomb potential $V_c = e^2/r$ produces a scattering amplitude $f_c \propto 1/\sin^2(\theta/2)$. How would this amplitude change if
 - i. V_c were *not* singular at the origin.
 - ii. V_c had a finite range.
2. Two particles each of mass m scatter. The potential between them is approximated by the attractive square well:

$$V(r) = \begin{cases} -V_0, & \text{for } r < b, \\ 0, & \text{for } r > b. \end{cases} \quad (4.69)$$

- (a) Determine the *scattering* length a_0 for this potential.
- (b) Sketch a_0 versus the potential strength V_0 with V_0 starting at zero and getting large enough to produce a bound state.
- (c) Expand a_0 in powers of V_0 and thereby generate the *Born series*.
- (d) Show that this Born series diverges when the potential is strong enough to form a bound state.
- (e) The same analysis can be used for repulsive potentials by changing the sign of V_0 . Plot a_0 for a repulsive potential as a function of potential strength and deduce if the Born series can fail for a repulsive potential.

3. The *deuteron* is a bound state of the neutron and proton with a binding energy of $E = -2.22$ MeV. We model the deuteron as an S-wave bound state in a square well of radius $R = 1.2$ fm.

- (a) What must be the depth V_0 of this well be so that it produces just one bound state at 2.22 MeV? (*Hint*: The binding energy is small compared with the depth of the well and so it is possible to expand some of the functions appearing in the transcendental equation; or you can just compute the answer.)
- (b) How does the scattering length for your well compare to the experimental value $a_T = -5.41$ fm?

4. A particle with mass m scatters elastically from a fixed potential and produces the S matrix:

$$e^{2i\delta_0(k)} = \frac{(k + ib)(k + ic)}{(k - ib)(k - ic)}. \quad (4.70)$$

- (a) Show that the scattering length a_0 is:

$$a_0 = \frac{b + c}{bc}. \quad (4.71)$$

- (b) Investigate the three cases: $(b > 0, c > 0)$, $(b > 0, c < 0)$, and $(b < 0, c < 0)$.
 - i. Use the relation of bound states to poles in the scattering amplitude to deduce the bound-state energies for each of these cases.
 - ii. For each case give a qualitative plot of the low-energy radial wave function $u_0(r)$.
 - (c) What conditions on b and c might lead to the potential having a resonance? (There are a number of conditions required for a truly realistic resonance, and this simple a model does not meet them all.)
 - (d) What conditions on b and c might lead to the potential absorbing flux?
 - (e) What conditions on b and c might lead to the potential producing flux?
5. Resonances often occur in interactions with attractive potentials having a repulsive “lip” or barrier above $E = 0$ that helps confine the particle. Since the lip is often the angular momentum barrier $l(l + 1)/2\mu r^2$ (which vanishes for $l = 0$), S-wave ($l = 0$) resonances are unlikely; here is an exception.

- (a) Show that a very deep square well of depth V_0 and radius R can produce S-wave scattering resonance levels near the top of the well with:

$$E_n = \frac{1}{2\mu} \left(\frac{(2n + 1)\pi}{2R} \right)^2. \quad (4.72)$$

- (b) Calculate the width Γ and verify that for a very deep well the resonances do not overlap, that is, $\Gamma \ll E_n - E_{n-1}$.
- (c) What would you expect for the behavior of $\tan \delta_0(k)$ versus k for a very shallow well?

6. Alpha particles of 8 MeV energy are scattered from a gold foil target. The back-angle cross section, $d\sigma/d\Omega(180^\circ)$, is found to be reduced by 5% from that given by the Rutherford cross section. Assume that the reduction is caused by a short range modification of the Coulomb potential which affects only S-waves.
- Is the short range potential attractive or repulsive?
 - Deduce the modification in the S-wave phase shift caused by the short range potential.
 - What change would there be to the Rutherford result at 10° ?
 - At what angles would you expect corrections due to the atomic shielding to be important (that is, for the theory to break down)?
7. We wish to derive the *Eikonal* or *semiclassical* approximation for the scattering amplitude in which the sum over partial waves l , is replaced by an integral over the impact parameters b .

- Show that if many partial waves enter the scattering, and if the scattering angle is very small, that the scattering amplitude can be written as

$$f_E(\theta) \simeq \frac{1}{2ik} \int_0^\infty db k(2kb) P_{kb}(\cos \theta) (e^{2i\delta(b)} - 1). \quad (4.73)$$

- Show further that for very large k ,

$$f_E(\theta) \simeq -ik \int_0^\infty db b (e^{2i\delta(b)} - 1) J_0[2kb \sin(\theta/2)]. \quad (4.74)$$

- Use this expression to calculate the scattering amplitude for scattering from a black disc. Sketch $d\sigma/d\Omega(\theta)$ for this case.

8. Derive the effective range expansion,

$$k \cot \delta_0 = \frac{1}{a_0} + \frac{1}{2} r_0 k^2 + O(k^4), \quad (4.75)$$

$$r_0 = 2 \int_0^R dr [w_0^2(r) - u_0^2(r)], \quad (4.76)$$

where w_0 is the $k = 0$ limit of the potential-free wave function. Accordingly, the effective range r_0 is a measure of the distance over which the distorted wave u_0 differs from the free wave w_0 , and is thus a better measure of the range of the potential than the scattering length.

9. Two particles each of mass m are in a quasi-bound or Gamow state. Assume their wave function is the same as a bound state inside their interaction potential and a pure outgoing wave outside of the potential. The potential between them is approximated by the attractive square well:

$$V(r) = \begin{cases} V_0, & \text{for } r < b, \\ 0, & \text{for } r > b. \end{cases} \quad (4.77)$$

- (a) If κ is the inner wave number and k the outer one, determine the relation between κ and k .
 - (b) Show that the above boundary conditions can only be satisfied for certain values of k , and that these values are complex.
 - (c) Show that the energy for this system is also complex and that the imaginary part has the sign expected for a state which decays exponentially over time.
10. A quantum state has an exponentially decaying time dependence,

$$\psi(x, t) = e^{-iE_R t} e^{-\Gamma t/2} \psi(x). \quad (4.78)$$

Determine the energy uncertainty in this state by calculating its Fourier transform:

$$\tilde{\psi}(E) = \int_{-\infty}^{\infty} dt \psi(x, t) e^{-iEt}. \quad (4.79)$$

Show that if we approximate the lower limit on this integral by $-T$ (the time since creation), the energy spectrum has the Breit-Wigner form

$$\tilde{\psi}(E) \propto \frac{1}{E - E_R + i\Gamma/2} \Rightarrow |\tilde{\psi}(E)|^2 \propto \frac{1}{(E - E_R)^2 + \Gamma^2/4}. \quad (4.80)$$

11. Deduce the appropriate normalization for Gamow states and show that ψ^2 and *not* $|\psi|^2$ should be normalized.
12. **The Ramsauer-Townsend Effect** Consider scattering from a strongly attractive potential at a low enough energy for only S-waves to contribute.
 - (a) What condition on the wave function leads to an S-wave phase shift $\delta_0 = \pi$?
 - (b) What would the scattering amplitude be in this case and what would you predict as its energy dependence?
 - (c) Sketch the expected energy dependence of the total cross section.
 - (d) What must be the relation between the depth and range of a 3D square well for the scattering amplitude to vanish at some energy?
 - (e) Indicate the physical arguments against this Ramsauer-Townsend effect occurring for repulsive potentials.
13. Plot $\text{Im} f_l$ and $\text{Re} f_l$ on an Argand plot for a Breit-Wigner resonance. Take $\Gamma = E_R/10$ and plot your points for enough energies to complete a full loop. Mark the (equally spaced) energy values on your diagram.
14. Derive (4.60) from (4.59).
15. Verify that Figure 4.9 shows the behavior expected for a Breit-Wigner resonance as a function of complex energy.

CHAPTER 5

GREEN'S FUNCTIONS AND INTEGRAL QUANTUM MECHANICS

In this chapter we discuss the techniques used to convert the Schrödinger equation we have been using into an integral equation. Although this introduces no new physics, the integral formulation is important for the following reasons:

- It is able to incorporate boundary conditions into the dynamical equation rather than imposing them at the end of solution.
- It easily deals with nonlocal interactions or coupled channels.
- It is easily extended to include relativity, many-particle effects, field theory, and so on.
- It leads to *abstract* or *formal quantum mechanics*.
- It leads to equations which are solved easily with modern computational techniques, thereby permitting exact solutions of problems which otherwise could only be approximated.
- It is elegant and illuminating.

In the first part of this chapter we define and determine the Green's functions for the three-dimensional problem and derive the integral form of the Schrödinger equation known as the *Lippmann–Schwinger equation*. In the course of the solution we utilize plane waves, orthogonal expansions and Cauchy's theorem. (Appendices A, *Natural Units and Plane Waves*, and B, *Dirac Notation and Representations*, review plane waves in finite and infinite boxes and some basics of orthogonal expansions.) In the latter part of this chapter we apply these techniques to develop the Born approximation for the wave function and scattering amplitude.

Chapter 8, *The Angular Momentum Basis*, follows a similar development for the one-dimensional Schrödinger equation. On first reading it may be best to learn the formalism in its 3D form in Chapter 5, *Green's Functions: Integral Quantum Mechanics*, and Chapter 6, *The Transition and Potential Matrices*, in its abstract form in Chapter 7, *Formal Quantum*

Mechanics, and then use Chapter 8, *The Angular Momentum Basis*, as a review of a specific representation. On further reading and for problem solving, Chapter 8 is valuable.

Much of the complex analysis and Green's functions techniques also appear in the classical physics references. There is particularly strong overlap with Jackson (1975), Mathews and Walker (1964), and Wyld (1976).

5.1 Definition of Green's Function

Because the Green's function technique is general, we rewrite the Schrödinger wave equation in a general form,

$$\mathcal{D}_k \psi(\mathbf{r}) = \mathcal{V}(\mathbf{r}), \quad (5.1)$$

$$\mathcal{D}_k(\mathbf{r}) = \frac{\nabla^2}{2\mu} + E_k, \quad (5.2)$$

$$\mathcal{V}(\mathbf{r}) = V(\mathbf{r})\psi(\mathbf{r}), \quad (5.3)$$

where we note that the driving term $\mathcal{V}$ contains the solution ψ . A special case of (5.1) is the *free wave* ϕ which is a solution of the *homogeneous equation*,

$$\mathcal{D}_k \phi_k(\mathbf{r}) = 0. \quad (5.4)$$

Unless indicated to the contrary, the homogeneous solutions are usually plane or standing waves.

The Green's function $G_E(\mathbf{r}, \mathbf{r}')$ is a specific solution of this wave equation for a unit driving term (a system's response to a unit impulse); in particular, the wave at $\mathbf{r}$ due to a delta function source at $\mathbf{r}'$:

$$\mathcal{D}_k G_E(\mathbf{r}, \mathbf{r}') = \delta(\mathbf{r} - \mathbf{r}'). \quad (5.5)$$

Because $G_E(\mathbf{r}, \mathbf{r}')$ is a solution of the wave equation it must be well behaved, with the exception of the point $\mathbf{r} = \mathbf{r}'$ where (5.5) says its second derivative is infinite. Whereas solving for $G_E(\mathbf{r}, \mathbf{r}')$ may not be easy, once it is known, obtaining all other solutions is straight forward because they must have the form:¹

$$\psi_k(\mathbf{r}) = \phi_k(\mathbf{r}) + \int d^3r' G_E(\mathbf{r}, \mathbf{r}') \mathcal{V}(\mathbf{r}'). \quad (5.6)$$

The exact amount of homogeneous solution ϕ added in is dependent on the boundary conditions which must be satisfied. If $\mathcal{V}$ were a known function, (5.6) would be the full solution. For the Schrödinger equation, $\mathcal{V}$ contains the (unknown) wave function, and this leads to an integral equation:

$$\psi_k(\mathbf{r}) = \phi_k(\mathbf{r}) + \int d^3r' G_E(\mathbf{r}, \mathbf{r}') V(\mathbf{r}') \psi_k(\mathbf{r}'). \quad (5.7)$$

Exercise Verify (5.7) by acting on both sides of the equation with $\mathcal{D}_k$. (You need assume

¹The question of generality and uniqueness of our solution should be familiar from the study of classical electrodynamics as given by Jackson (1975). The present problem corresponds to a Neumann boundary condition because we include a homogeneous solution. The reader should be aware that there are problems, such as Gamow states, in which the appropriate boundary conditions must be reformulated (see § 4.2).

the functions are sufficiently well behaved to permit interchange of the order of integration and differentiation.) $\diamond$

We now look at two ways to solve for $G_E(\mathbf{r}, \mathbf{r}')$.

5.2 Solution via Eigenfunction Expansion

Because the Green's function is a solution of the Schrödinger equation, it has the expansion

$$G_E(\mathbf{r}, \mathbf{r}') = \sum_i c_i \phi_i(\mathbf{r}). \quad (5.8)$$

We take ϕ_i to be the free solutions

$$\mathcal{D}_k \phi_i(\mathbf{r}) = 0, \quad (5.9)$$

$$\left[\frac{\nabla^2}{2\mu} \phi_i(\mathbf{r}) + E_i \right] \phi_i(\mathbf{r}) = 0. \quad (5.10)$$

To solve for c_i , we require the definition (5.5) of $G_{\mathbf{k}}$ be satisfied:

$$\delta(\mathbf{r} - \mathbf{r}') = \mathcal{D}_k G_E(\mathbf{r}, \mathbf{r}') \quad (5.11)$$

$$= \sum_i c_i \mathcal{D}_k \phi_i = \sum_i c_i \frac{-k_i^2 + k^2}{2\mu} \phi_i(\mathbf{r}). \quad (5.12)$$

Yet we know (Appendices A, *Natural Units and Plane Waves*, and B, *Dirac Notation and Representations*) that the delta function on the RHS of (5.5) has the expansion

$$\delta(\mathbf{r} - \mathbf{r}') = \sum_i \phi_i^*(\mathbf{r}') \phi_i(\mathbf{r}). \quad (5.13)$$

A comparison of (5.12) and (5.13) tells us that

$$G_E(\mathbf{r}, \mathbf{r}') = \sum_i \frac{\phi_i(\mathbf{r}) \phi_i^*(\mathbf{r}')}{E - E_i} \equiv \sum_i \frac{\phi_i(\mathbf{r}) \phi_i^*(\mathbf{r}')}{(k^2 - k_i^2)/2\mu}. \quad (5.14)$$

As we shall verify, (5.14) is often a convenient form for $G_E(\mathbf{r}, \mathbf{r}')$.

5.3 Solution via Spectral Representation

We obtain the spectral or Fourier-integral representation of $G_E(\mathbf{r}, \mathbf{r}')$ by taking the eigenfunction expansion and going to the limit where the ϕ_i 's are plane waves in a box (see Appendix B, *Dirac Notation and Representations*):

$$G_E(\mathbf{r}, \mathbf{r}') = \int \frac{d^3 k'}{(2\pi)^3} \frac{e^{i\mathbf{k}' \cdot (\mathbf{r} - \mathbf{r}')}}{(k^2 - k'^2)/2\mu}. \quad (5.15)$$

Here $\mathbf{k}'$ is a dummy integration variable and k is equivalent to the energy parameter $E = k^2/2\mu$. Equation (5.15) displays some general properties of $G_E(\mathbf{r}, \mathbf{r}')$:

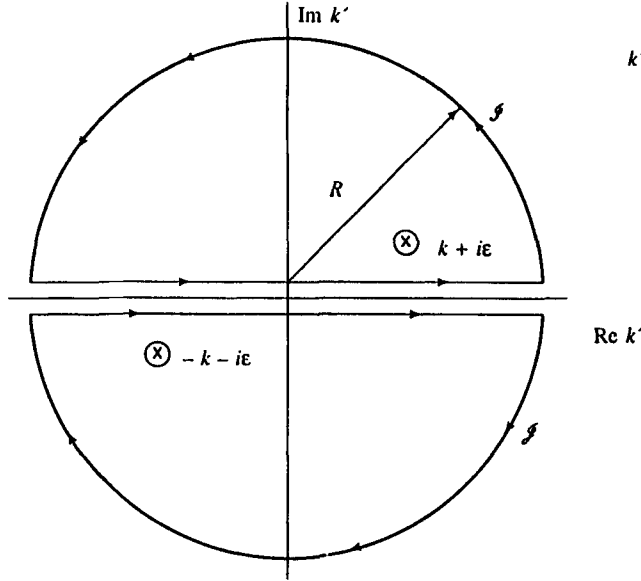


Figure 5.1: Contours of integration in the complex k' plane used to evaluate the Green's function. Because the poles at $\pm(k + i\epsilon)$ are displaced off the real axis, the integration paths may be placed along the real axis.

- It is symmetric in $\mathbf{r}$ and $\mathbf{r}'$ and only depends on the magnitude of their difference:

$$G_E(\mathbf{r}, \mathbf{r}') = G_E(\mathbf{r}', \mathbf{r}) = G_E(|\mathbf{r} - \mathbf{r}'|). \quad (5.16)$$

This is reasonable because waves propagate symmetrically in an isotropic space.

- It depends *parametrically* on the energy $E = k^2/2\mu$ at which we are solving the Schrödinger equation.
- The character of G_E changes at $\mathbf{r} = \mathbf{r}'$ (its second derivative does not exist).
- There is a singular point in the integrand at $k = k'$, and the choice of treatments of this singularity is equivalent to imposing specific boundary conditions on the wave function.
- Much like energy denominators in second-order perturbation theory, integration over the dummy variable k' is equivalent to including transitions to virtual intermediate states with energies $k'^2/2\mu \neq k^2/2\mu$.

Evaluation of G with Residues

Evaluation of (5.15), the integral expression for G_E , is an exercise in applying *Cauchy's theorem*. Those readers familiar with this technique may choose to proceed to (5.27). We let $\mathbf{x} = |\mathbf{r} - \mathbf{r}'|$ and evaluate the integration over the direction of $\mathbf{k}'$:

$$G_E(\mathbf{r}, \mathbf{r}') = \int_0^\infty \frac{dk' 2\pi k'^2}{(2\pi)^3} \int_{-1}^{+1} \frac{d(\cos \theta) e^{i\mathbf{k}' \cdot \mathbf{x} \cos \theta}}{(k^2 - k'^2)/2\mu} = \frac{-2i\mu (\mathcal{I} - \mathcal{J})}{(2\pi)^2 \mathbf{x}}, \quad (5.17)$$

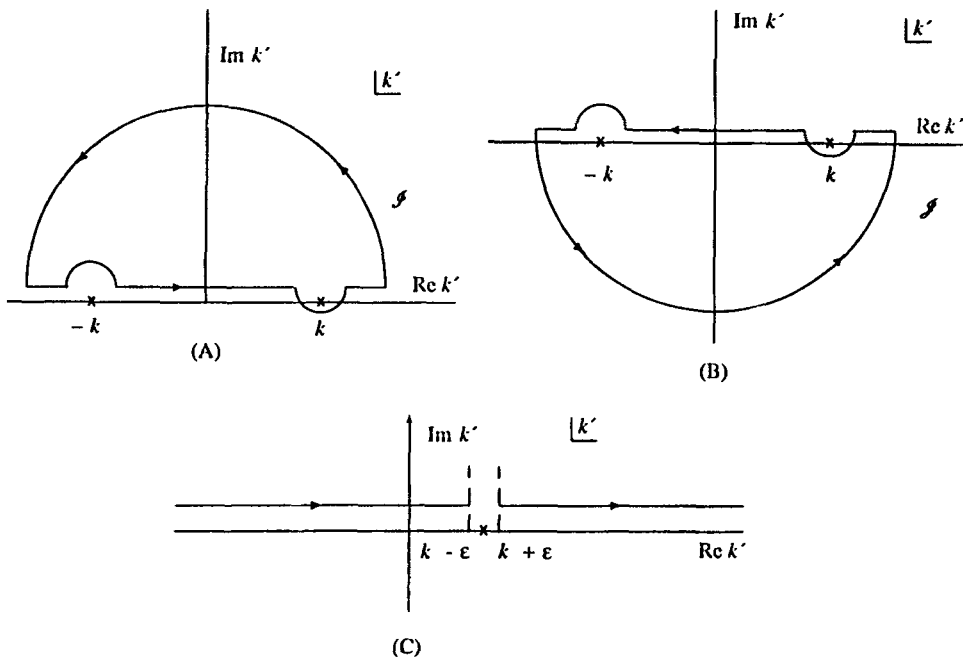


Figure 5.2: Three routes in the k' plane for avoiding the singularities of the Green's function at $k' = \pm k$.

$$\mathcal{I}, \mathcal{J} = \frac{1}{2} \int_{-\infty}^{\infty} \frac{dk' k' e^{\pm i k' x}}{k^2 - k'^2}. \quad (5.18)$$

Exercise Verify (5.18). (*Hint:* Note the symmetry of the integrand about 0.) ◇

We first evaluate the integral $\mathcal{I}$ by factorizing the denominator so the poles are evident:

$$\mathcal{I} = \frac{1}{2} \int_{-\infty}^{+\infty} \frac{dk' k' e^{i k' x}}{(k - k')(k + k')}. \quad (5.19)$$

Given a single-valued function $f(z)$ whose only singularities are poles at $z = a_i$, Cauchy's theorem expresses the (counterclockwise) line integral of f around a closed contour C as the sum over the residues of f within C :

$$\oint_C dz f(z) = 2\pi i \sum_i \text{Residues } f(a_i), \quad (5.20)$$

$$\text{Residue } f(a_i) \stackrel{\text{def}}{=} \lim_{z \rightarrow a_i} [(z - a_i) f(z)]. \quad (5.21)$$

The definition (5.21) assumes the singularities are simple poles at $z = a_i$. For $\mathcal{I}$ we choose the contour shown in the upper half k' plane, Figure 5.1 or Figure 5.2A. The function $e^{i k' x} = e^{i k'_R x} e^{-k'_I x}$ vanishes in the limit of $R \rightarrow \infty$, and so the contour integral equals the

line integral:

$$\oint_C f(z) dz \rightarrow \int_{-\infty}^{+\infty} f(x) dx. \quad (5.22)$$

To evaluate the residues we must know if the poles of f occurring on the real axis at $k' = \pm k$ should lie within or without C . We remove this ambiguity by assuming the energy E has a *small, positive imaginary part* $+i\epsilon$. We then evaluate the integral, take the limit as $\epsilon \rightarrow 0$, and affix a $(+)$ to G_E as a reminder of our choice. Because the resulting $G_E^{(+)}(\mathbf{r}, \mathbf{r}')$ is an outgoing wave, this procedure is equivalent to the boundary condition that the *scattered wave radiates outwards*. As shown in Figure 5.1, the poles in the k' plane occur at

$$k'^2 = k^2 + i\epsilon, \quad (5.23)$$

$$\Rightarrow k' = k + i\epsilon, \quad -k - i\epsilon. \quad (5.24)$$

Because only the pole at positive k resides within the contour, only its residue contributes to $\mathcal{I}$ and we obtain

$$\mathcal{I} = -\frac{\pi i}{2} e^{ikx}. \quad (5.25)$$

Exercise Show that the integral $\mathcal{J}$ must be evaluated with the lower contour in Figure 5.2, and when evaluated,

$$\mathcal{J} = +\frac{\pi i}{2} e^{ikx}. \quad \diamond \quad (5.26)$$

Accordingly, the Green's function with *outgoing wave* boundary conditions is

$$G_E^{(+)}(\mathbf{r}, \mathbf{r}') \equiv G_{E+i\epsilon}(\mathbf{r}, \mathbf{r}') = \frac{-\mu e^{ik|\mathbf{r}-\mathbf{r}'|}}{2\pi |\mathbf{r}-\mathbf{r}'|}, \quad (5.27)$$

which certainly looks like the wave at $\mathbf{r}$ due to a unit source at $\mathbf{r}'$!

Other Boundary Conditions

Because integrations over energy denominators occur frequently in physics, it is worth examining the consequences of using other prescriptions for handling the singularity. Had we chosen a negative sign for ϵ (that is, added a small negative imaginary part to the energy), only the pole at $k' = -k + i|\epsilon|$ would lie within the contour for $\mathcal{I}$ and we would have obtained

$$G_E^{(-)}(\mathbf{r}, \mathbf{r}') \equiv G_{E-i\epsilon}(\mathbf{r}, \mathbf{r}') = \frac{-\mu e^{-ik|\mathbf{r}-\mathbf{r}'|}}{2\pi |\mathbf{r}-\mathbf{r}'|}. \quad (5.28)$$

We see that a small negative imaginary part $-i\epsilon$ in the energy is equivalent to choosing an *incoming* scattered wave.

In contrast to moving the singularities up and down in the complex plane, the $i\epsilon$ procedure is equivalent to distorting the contour around the singularities without disturbing the singularities (see Figure 5.2A and B). The distortion is such that it has no effect on the residue theorem. A different approach would be to leave the singularities on the real axis at

$k' = \pm k$, and integrate up to—but not through—them. This is the *Cauchy principal-value prescription*² indicated in Figure 5.2C and is denoted by $\mathcal{P}$:

$$\mathcal{P} \int_{-\infty}^{+\infty} dk' f(k') = \lim_{\epsilon \rightarrow 0} \left[\int_{-\infty}^{k-\epsilon} dk' f(k') + \int_{k+\epsilon}^{+\infty} dk' f(k') \right]. \quad (5.29)$$

By use of Cauchy's theorem and some contour distortions, these different prescriptions are related:

$$\int_{-\infty}^{+\infty} \frac{dk' f(k')}{k' - k \pm i\epsilon} = \mathcal{P} \int_{-\infty}^{+\infty} \frac{dk' f(k')}{k' - k} \mp i\pi f(k). \quad (5.30)$$

Exercise Prove (5.30). ◇

A short hand version of the relation (5.30) which shows its independence of $f(k')$ is

$$\frac{1}{k' - k \mp i\epsilon} = \frac{\mathcal{P}}{k' - k} \pm i\pi \delta(k' - k) \quad (dk' \text{ integral}), \quad (5.31)$$

where the equality holds only when integrated over dk' .³

Because a second-order differential equation has only two independent solutions, the incoming and outgoing wave Green's functions form a complete set. Indeed, the principal value G ,

$$G_E^{(P)}(\mathbf{r}, \mathbf{r}') = \mathcal{P} \int \frac{d^3k}{(2\pi)^3} \frac{e^{i\mathbf{k}' \cdot (\mathbf{r} - \mathbf{r}')}}{(k^2 - k'^2)/2\mu} = \frac{-2\mu \cos k|\mathbf{r} - \mathbf{r}'|}{4\pi|\mathbf{r} - \mathbf{r}'|}, \quad (5.32)$$

is clearly the Green's function for *standing-waves* boundary conditions and is clearly a linear combination of $G_E^{(+)}$ and $G_E^{(-)}$.

Exercise Use (5.31) to prove (5.32). ◇

5.4 Lippmann–Schwinger Wave Equation

We return to the solution (5.7) of the Schrödinger equation in terms of the Green's function. In order to have $\psi_{\mathbf{k}}(\mathbf{r})$ describe the conventional scattering setup of Chapter 1, *Scattering*, (incident beam of momentum $\mathbf{k}$ and outgoing, spherical scattered wave), we substitute the plane wave states $\phi_{\mathbf{k}}$ for ϕ , and use the outgoing Green's function $G_E^{(+)}$:

$$\psi_{\mathbf{k}}^{(+)}(\mathbf{r}) = \phi_{\mathbf{k}}(\mathbf{r}) + \int d^3r' G_E^{(+)}(\mathbf{r}, \mathbf{r}') V(\mathbf{r}') \psi_{\mathbf{k}}^{(+)}(\mathbf{r}'). \quad (5.33)$$

In explicit form this is

$$\psi_{\mathbf{k}}^{(+)}(\mathbf{r}) = \frac{e^{i\mathbf{k} \cdot \mathbf{r}}}{(2\pi)^{3/2}} - \frac{\mu}{2\pi} \int d^3r' \frac{e^{ik|\mathbf{r} - \mathbf{r}'|}}{|\mathbf{r} - \mathbf{r}'|} V(\mathbf{r}') \psi_{\mathbf{k}}^{(+)}(\mathbf{r}). \quad (5.34)$$

²As discussed in Chapter 18, *Solving Even Relativistic Integral Equations*, the $\mathcal{P}$ prescription can easily be evaluated on a computer.

³Note that if the integration is over k with k' as a dummy variable, we would have $1/(k - k' + i\epsilon) = \mathcal{P}/(k - k') - i\pi\delta(k - k')$ (dk integral), which is not simply the negative of (5.31).

We note again that the superscript (+) indicates the outgoing wave boundary condition imposed on $\psi_{\mathbf{k}}$, and the subscript $\mathbf{k}$ indicates the physical scattering configuration with a beam of momentum $\mathbf{k}$ and energy $E_k = k^2/2\mu$.⁴ The $\psi_{\mathbf{k}}$'s and $\phi_{\mathbf{k}}$'s have the normalization

$$\langle \phi_{\mathbf{k}} | \phi_{\mathbf{k}'} \rangle = \langle \psi_{\mathbf{k}} | \psi_{\mathbf{k}'}^{(+)} \rangle = \delta(\mathbf{k}' - \mathbf{k}), \quad (5.35)$$

which is evident for the $\phi_{\mathbf{k}}$'s, but will not be proved until Chapter 7, *Formal Quantum Mechanics*, for the $\psi_{\mathbf{k}}$'s.

Integral Expression for f

The integral equation (5.33) is equivalent to the differential Schrödinger equation with outgoing scattered-wave boundary conditions. It is also called the *Lippmann–Schwinger (LS) equation* and is the basic integral equation of scattering theory. To help make that point we note that because the elastic scattering amplitude $f_E(\theta, \phi)$ of Chapter 1, *Scattering*, and Chapter 2, *Currents and Cross Sections*, was defined in terms of the asymptotic wave function ($\psi \sim e^{ikz} + f e^{ikr}/r$), it must be contained (somewhere) in the second term of (5.33). To extract it we look at that integral for asymptotically large r and finite r' (r' is restricted to be less than the range R of the potential).

Exercise The scattering amplitude is defined in the $r \gg r'$ limit. Show that in this limit,

$$|\mathbf{r} - \mathbf{r}'| = \sqrt{r^2 + r'^2 - 2\mathbf{r}' \cdot \mathbf{r}} \simeq r - \frac{\mathbf{r} \cdot \mathbf{r}'}{r}, \quad (5.36)$$

and so the second term of (5.33) tells us that

$$f_E(\theta, \phi) = -4\pi^2\mu \int d^3r' \frac{e^{-i\mathbf{k}' \cdot \mathbf{r}'}}{(2\pi)^{3/2}} V(\mathbf{r}') \psi_{\mathbf{k}}^{(+)}(\mathbf{r}'), \quad (5.37)$$

or in terms of free and distorted waves,

$$f_E(\theta, \phi) = -4\pi^2\mu \langle \phi_{\mathbf{k}'} | V | \psi_{\mathbf{k}}^{(+)} \rangle. \quad \diamond \quad (5.38)$$

We see that $f_E(\theta, \phi)$ is “nothing more” than the matrix element of the potential between an initial distorted wave function $\psi_{\mathbf{k}}^{(+)}$, and a final plane wave function $\phi_{\mathbf{k}'}$. While $k = k'$ for elastic scattering, (5.37) is valid even for nonelastic reactions.

Exercise Explain how it is compatible for the scattering amplitude $f_E(\theta, \phi)$ to be defined in terms of the asymptotic wave function and simultaneously be equal to the integral (5.37) over the potential and *inner* wave function. $\diamond$

⁴It is often true that unless the context demands it the superscripts and subscripts are left off; at some point choices have to be made between completeness with opaqueness versus simplicity with clarity.

5.5 Born Approximation: The Neumann Series

An integral equation such as (5.7) may appear useless because it looks like you need to know the answer in order to solve for the answer. Nevertheless, this equation is used to develop approximation schemes, to derive the formal theory, and is solved exactly (see the Problems section) or numerically (see Chapter 18, *Solving Even Relativistic Integral Equations*).

One of the most famous approximation schemes in physics is the Born approximation, which is actually two schemes, one for the wave function ψ and one for the scattering amplitude f . Each scheme produces a series of approximations, with the lowest order approximation (the *first Born*) often called *the* Born approximation.

Consider the Lippmann–Schwinger equation (5.33):

$$\psi_{\mathbf{k}}^{(+)}(\mathbf{r}) = \frac{e^{i\mathbf{k}' \cdot \mathbf{r}}}{(2\pi)^{3/2}} - \frac{\mu}{2\pi} \int d^3r' \frac{e^{i\mathbf{k} \cdot |\mathbf{r} - \mathbf{r}'|}}{|\mathbf{r} - \mathbf{r}'|} V(\mathbf{r}') \psi_{\mathbf{k}}^{(+)}(\mathbf{r}'). \quad (5.39)$$

If the potential is weak enough we should be able to ignore the distortion (scattering) of the incident beam caused by the potential and make the approximation:

$$\psi_{\mathbf{k}}^{(+)}(\mathbf{r}) \simeq \psi_{\mathbf{k}}^{(+)} \Big|_1 = \phi_{\mathbf{k}}(\mathbf{r}) = \frac{e^{i\mathbf{k} \cdot \mathbf{r}}}{(2\pi)^{3/2}}. \quad (5.40)$$

This is the *first Born approximation* for the wave function. We (probably) improve it with an *iteration* technique in which we generate the next higher order approximation by using a lower order approximation for $\psi_{\mathbf{k}}$ in the distortion (second) term of (5.39):⁵

$$\psi_{\mathbf{k}}^{(+)} \Big|_2 = \frac{e^{i\mathbf{k} \cdot \mathbf{r}}}{(2\pi)^{3/2}} - \frac{\mu}{2\pi} \int d^3r' \frac{e^{i\mathbf{k} \cdot \mathbf{x}}}{x} V(\mathbf{r}') \psi_{\mathbf{k}}^{(+)}(\mathbf{r}') \Big|_1, \quad (5.41)$$

$$\psi_{\mathbf{k}}^{(+)} \Big|_3 = \frac{e^{i\mathbf{k} \cdot \mathbf{r}}}{(2\pi)^{3/2}} - \frac{\mu}{2\pi} \int d^3r' \frac{e^{i\mathbf{k} \cdot \mathbf{x}}}{x} V(\mathbf{r}') \psi_{\mathbf{k}}^{(+)}(\mathbf{r}') \Big|_2, \quad (5.42)$$

⋮

where $x = |\mathbf{r} - \mathbf{r}'|$. Note that each successive term in this series involves an additional three-dimensional integration; accordingly, it is not too surprising that the first term is usually the only one used.

We now apply this iterative technique to determine a series expansion for the scattering amplitude. Starting with (5.37) we obtain,

$$f_E(\theta, \phi) = -4\pi^2\mu \int d^3r \frac{e^{-i\mathbf{k}' \cdot \mathbf{r}}}{(2\pi)^{3/2}} V(\mathbf{r}) \psi_{\mathbf{k}}^{(+)}(\mathbf{r}), \quad (5.43)$$

$$f_E(\theta, \phi) \Big|_B^1 = -4\pi^2\mu \int d^3r \frac{e^{-i\mathbf{k}' \cdot \mathbf{r}}}{(2\pi)^{3/2}} V(\mathbf{r}) \psi_{\mathbf{k}}^{(+)}(\mathbf{r}) \Big|_1. \quad (5.44)$$

In a form which may be easier to remember, the first Born approximation is

⁵This is the an application of the Neumann technique for the solution of integral equations, and in our case is known to be convergent if the potential is *not* strong enough to form a bound state.

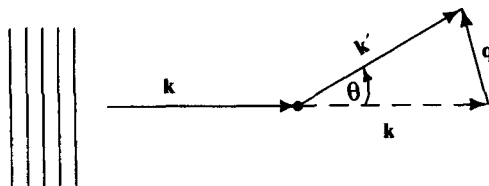


Figure 5.3: The initial and scattered beam momenta, $\mathbf{k}$ and $\mathbf{k}'$, and the momentum transfer $\mathbf{q}$. In elastic scattering, $k = k'$ and the triangle is isosceles.

$$f_B(\theta, \phi) = \frac{-\mu}{2\pi} \int d^3r e^{-i\mathbf{q} \cdot \mathbf{r}} V(\mathbf{r}) = \frac{-\mu}{2\pi} \tilde{V}(q) = -4\pi^2 \mu \langle \phi_{\mathbf{k}'} | V | \phi_{\mathbf{k}} \rangle, \quad (5.45)$$

where $\mathbf{q} = \mathbf{k}' - \mathbf{k}$ is the *momentum transferred* to the target by the beam (Figure 5.3).

Exercise Show that iteration produces the *second Born approximation*:

$$f_B^2 = f_B^1 + \frac{m^2}{4\pi^2} \int d^3r d^3x e^{-i\mathbf{q} \cdot \mathbf{r}} V(\mathbf{r}) \frac{e^{i\mathbf{k} \cdot \mathbf{x}}}{x} V(\mathbf{r} - \mathbf{x}). \quad \diamond \quad (5.46)$$

Note that in the first Born approximation the scattering amplitude is real, is proportional to $\tilde{V}(q)$ (the Fourier transform of the potential), and is a function of just q . Its reality is not such a good thing because it means the optical theorem (2.44) is manifestly violated. This is not a shattering of some basic principle since, after all, the Born approximation is just an approximation (it does get better as higher-order terms are included; see, for example, the R matrix in § 8.6).

Exercise Show that the integrated, first Born cross section $\sigma_t^1|_B = 4\pi \text{Im}f(0^\circ)|_B^2/k$. This means there is an approximate optical theorem relating different orders of approximation. $\diamond$

As is known from the Neumann theory of integral equations, iteration of the Born series does not always work. In fact, the series diverges for potentials which strongly distort the incident plane wave (for example, those strong enough to form a bound state if they were attractive). For further discussion we refer the reader to Gottfried (1966) who develops some convenient rules regarding the applicability of the Born approximation.

Yukawa and Coulomb Potentials

Consider the Yukawa potential,

$$V(r) = \frac{V_0 e^{-r/R}}{r}, \quad (5.47)$$

with the range⁶ R . The Born approximation for scattering of CM momentum $\mathbf{k} \rightarrow \mathbf{k}'$ from this potential is the Fourier transform

$$f|_B = \frac{-\mu}{2\pi} \int d^3r e^{-i\mathbf{q}\cdot\mathbf{r}} V(\mathbf{r}) = \frac{-\mu}{2\pi} \tilde{V}(q) \quad (5.48)$$

$$= -\frac{\mu 4\pi V_0}{2\pi(q^2 + 1/R^2)} = -\frac{2\mu V_0}{(q^2 + 1/R^2)}. \quad (5.49)$$

Here $q = |\mathbf{k}' - \mathbf{k}| = \sqrt{2k^2(1 - \cos\theta)} = 2k \sin\theta/2$ is the momentum transferred to the target (see Figure 5.3), and for small q is approximately perpendicular to $\mathbf{k}$. The differential cross section is

$$\left. \frac{d\sigma}{d\Omega} \right|_B = |f|_B|^2 = \left(\frac{2\mu V_0}{q^2 + 1/R^2} \right)^2, \quad (5.50)$$

and has the angular dependence shown in Figure 5.4. We see that the cross section appears isotropic and independent of R until the momentum transfer is comparable to $1/R$. Unless the experiment is capable of reaching this large a value of q it will not be possible to learn about the range of the potential.

In spite of our better judgment that the Coulomb potential with its many bound states and high distortion is too strong for the Born approximation to be good, let us examine the first Born approximation for the Coulomb scattering. To obtain Coulomb scattering we take the limit of (5.49) and (5.50) for $R \rightarrow \infty$ and $V_0 \rightarrow Z_T Z_P e^2$:

$$f_{\text{coul}}|_B = \frac{-\mu Z_T Z_P e^2}{2k^2 \sin^2 \frac{1}{2}\theta} \Rightarrow \left. \frac{d\sigma}{d\Omega} \right|_B = \frac{(\mu Z_T Z_P e^2)^2}{4 (k \sin \frac{1}{2}\theta)^4}. \quad (5.51)$$

Although the phase of the amplitude differs from the exact result given in § 4.3, miraculously, its magnitude (and thus $d\sigma/d\Omega$) agree with the exact answer.

5.6 Scattering from Bound Systems ⊙

The Born scattering amplitude (5.37) is an integral of a potential between the initial and final nondistorted wave functions Φ_i and Φ_f^* describing the system. When scattering a particle α from a bound particle β , Figure 5.5, we need to generalize that expression. We assume we know the wave functions ϕ_n which are the solutions of the Schrödinger equation describing the binding of particle β by the potential $W(r_\beta)$,

$$\left[\frac{-\nabla_{r_\beta}^2}{2\mu} + W(r_\beta) \right] \phi_n(r_\beta) = E_n \phi_n(r_\beta). \quad (5.52)$$

We also assume that the target particle β is initially in its ground state $n = 0$, but that after the scattering [interaction through the potential $V(\mathbf{r}_\beta - \mathbf{r}_\alpha)$], β may be left in an arbitrary

⁶ Yukawa (1935) postulated that the exchange of mesons between nucleons generates a potential of this form. Calling R a “range” does not strictly agree with the convention in Chapter 1, *Scattering*, and Chapter 2, *Currents and Cross Sections*, because this potential never quite vanishes; for example, for $r > 14R$ there is still one part in a million left. Yet for a given precision, there is always some r beyond which the potential is zero.

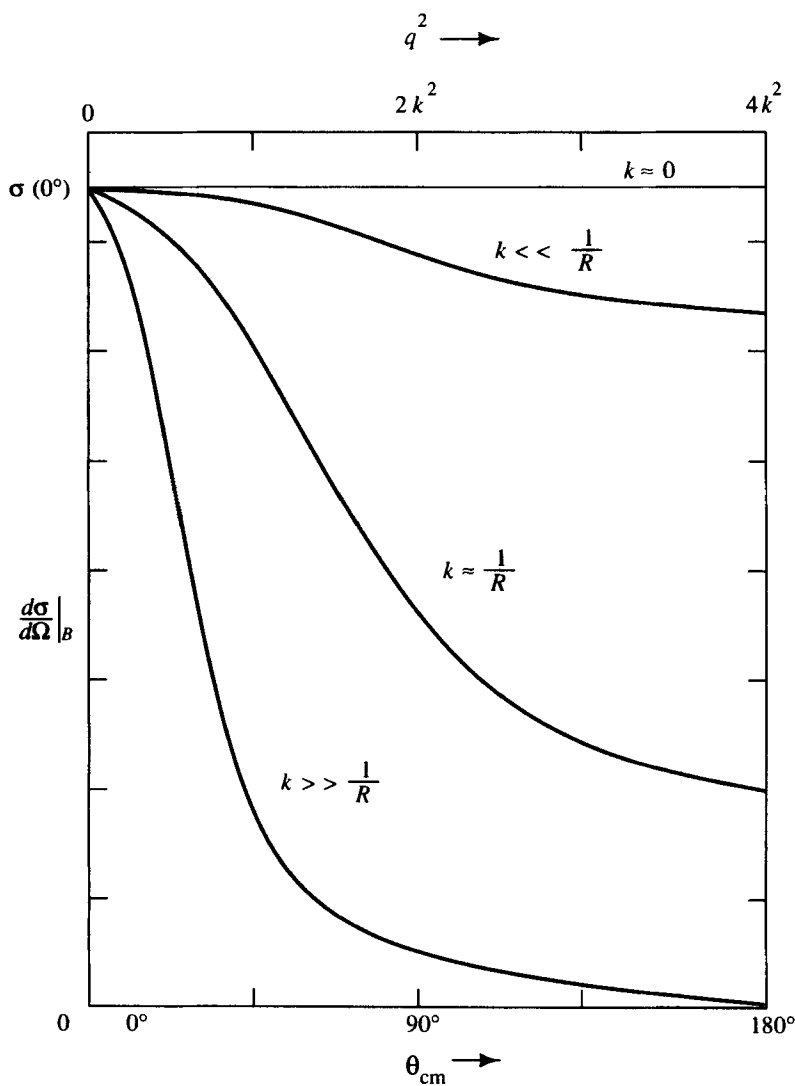


Figure 5.4: The differential cross section in Born approximation for scattering from a Yukawa potential at different energies. (Adapted from Gottfried, 1966.)

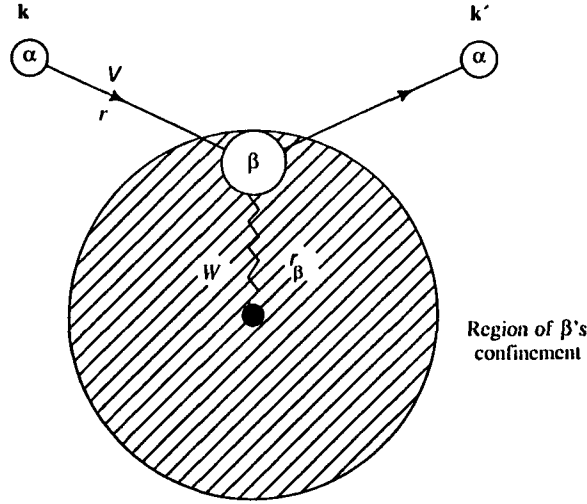


Figure 5.5: Scattering of a particle α from a particle β bound by the potential W . The α - β potential V causes the scattering.

state n . The generalization of (5.37) replaces the unperturbed, initial and final free wave functions by the direct product of unperturbed projectile and target wave functions:

$$\Phi_f \rightarrow e^{i\mathbf{k}' \cdot \mathbf{r}_\alpha} \phi_n(\mathbf{r}_\beta), \quad (5.53)$$

$$\Phi_i \rightarrow e^{i\mathbf{k} \cdot \mathbf{r}_\alpha} \phi_0(\mathbf{r}_\beta). \quad (5.54)$$

The Born approximation for the scattering amplitude is thereby:

$$f|_B = \frac{-\mu}{2\pi} \int d^3\mathbf{r}_\alpha d^3\mathbf{r}_\beta e^{-i\mathbf{k}' \cdot \mathbf{r}_\alpha} \phi_n^*(\mathbf{r}_\beta) V(\mathbf{r}_{\alpha\beta}) e^{i\mathbf{k} \cdot \mathbf{r}_\alpha} \phi_0(\mathbf{r}_\beta), \quad (5.55)$$

where $\mathbf{r}_{\alpha\beta} = |\mathbf{r}_\alpha - \mathbf{r}_\beta|$ and the scattering potential V is assumed spherically symmetric. The relation (5.55) says that the effective potential experienced by the projectile α is the 2-body interaction with β averaged over all possible positions of β :

$$V(\mathbf{r}_\alpha) = \int d^3\mathbf{r}_\beta |\phi_0(\mathbf{r}_\beta)|^2 V(\mathbf{r}_\alpha - \mathbf{r}_\beta). \quad (5.56)$$

This viewpoint also arises in Hartree's view of interactions in many-electron systems (Chapter 26, *Many-Body Problems*).

The integral in (5.55) is evaluated with a change to the relative variables $\mathbf{r} = \mathbf{r}_\alpha - \mathbf{r}_\beta$:

$$f(q)|_B = -4\pi^2 \mu \tilde{V}(q) F_{n0}(q), \quad (q = |\mathbf{k}' - \mathbf{k}|), \quad (5.57)$$

where $\tilde{V}(q)$ is the Fourier transform of the 2-body potential,

$$\tilde{V}(q) = \int d\mathbf{r} e^{i\mathbf{q} \cdot \mathbf{r}} V(r). \quad (5.58)$$

The function F_{n0} is the *form factor* or *structure function*:

$$F_{n0}(q) = \int d^3\mathbf{r}_\beta e^{i\mathbf{q} \cdot \mathbf{r}_\beta} \phi_n^*(\mathbf{r}_\beta) \phi_0(\mathbf{r}_\beta), \quad (5.59)$$

and is seen to be the Fourier transform of a bound-state transition density. The Born amplitude for scattering from a bound system (5.57), is just our previous Born approximation result for scattering of α from a free β times a form factor to account for the confinement (or motion) of β within the potential W .

Exercise Derive (5.57) from (5.55). ◇

The α - β cross section now becomes

$$\left. \frac{d\sigma}{d\Omega} \right|_B = (-4\pi^2\mu)^2 |\tilde{V}(q)|^2 F_{n0}^2(q) = \left. \frac{d\sigma}{d\Omega} \right|_{\alpha\beta}^{\text{free}} F_{n0}^2(q). \quad (5.60)$$

That is, the Born approximation cross section for scattering from a bound system is the Born approximation cross section for free scattering times a form factor to account for the finite size of the region in which β is bound.⁷

In *elastic scattering*, the target remains in its ground state $n \equiv 0$, and the form factor becomes the Fourier transform of the ground state density:

$$F_{00}(q) = \int d^3r e^{i\mathbf{q}\cdot\mathbf{r}} \rho(r). \quad (5.61)$$

Exercise Show by expanding the exponential in (5.61), that knowledge of the form factor for small momentum transfers q is equivalent to knowledge of the root-mean-square radius of β 's distribution, that is, show that

$$F_{00}(q) \equiv F(q) \simeq 1 - \frac{1}{6}q^2 R_{\text{rms}}^2 + O(q^4). \quad \diamond \quad (5.62)$$

As an example, the form factor for the electronic distribution in a hydrogen-like atom is

$$F_{00}(q) = \frac{1}{[1 + (a_B q/2\alpha)^2]^2}, \quad (5.63)$$

where a_B is the Bohr radius and α is the effective charge (1 for H, 1.69 for He, see Chapter 26, *Many-Body Problems*). We see that $F(0) = 1$, which just means that ρ is normalized to 1, and the leading power in q^2 measures the gross size of the system. Because the ratio of bound to free cross sections is the square of this form factor, one way to learn about $F(q)$, or indirectly $\rho(r)$, is to measure this ratio over a wide range of q values. The utility of this technique is shown in Figure 5.6 where the deviation from the point charge curve of the small q -value measurements determines the root-mean-square radius R_{rms} , and the larger q measurements determine higher Fourier components of the density.

After the ground state-density has been determined, the next step is to measure nonelastic scatterings that leave the target in an excited state. This determines F_{n0} and consequently the overlap of the ground-state and excited-state wave functions.

⁷It is a better approximation to use the exact $\alpha - \beta$ cross section rather than the Born approximation to it. This approximation, the single-scattering or *impulse* approximation, is derived in multiple scattering theory; see Goldberger and Watson (1963).

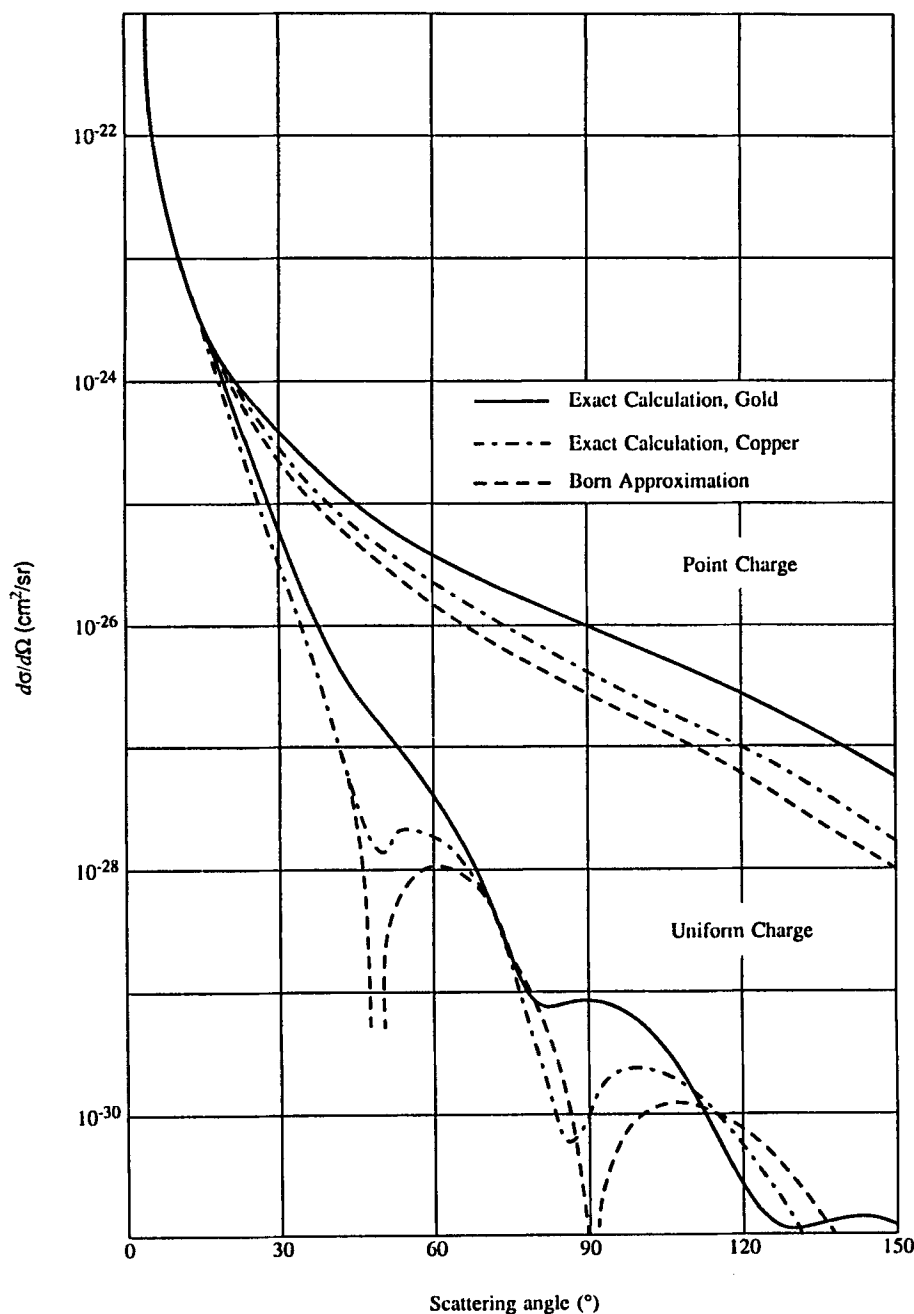


Figure 5.6: The differential cross section for the scattering of 150-MeV electrons from the nuclei of gold ($Z = 79$) and copper ($Z = 29$). The cross section for copper has been multiplied by a factor to make it have the same first Born approximation as gold. The Born approximation is seen to be more accurate for copper. From Yennie, Ravenhall, and Wilson (1954).

Exercise Show that $F_{00}(0) = 1$ and that $F_{n0}(0) = 0$. $\diamond$

If the scattering ejects or *knocks out* a bound particle, a different organization of (5.55) is useful. Specifically, if in some cases the ejected β is observed with a momentum $\mathbf{k}_\beta$ close to that expected for free α - β scattering (that is, $\mathbf{q}_\beta \simeq -\mathbf{q}_\alpha$), it is probably a good assumption that no other bound particles have interfered with the scattering. In these *quasielastic events* we would approximate the final state wave function as just a free α particle and a free β particle:

$$\Phi_f \simeq e^{i\mathbf{k}' \cdot \mathbf{r}_\alpha} e^{i\mathbf{k}_\beta \cdot \mathbf{r}_\beta}. \quad (5.64)$$

Exercise Show that substitution of (5.64) into (5.55) yields the quasielastic cross section:

$$\frac{d\sigma}{d\Omega}(\text{quasielastic}) = \frac{d\sigma}{d\Omega} \bigg|_{\alpha\beta}^{\text{free}} |\phi_0(\mathbf{k}' - \mathbf{k} - \mathbf{k}_\beta)|^2. \quad (5.65)$$

Here the momentum of α changes from $\mathbf{k}$ to $\mathbf{k}'$ and β is ejected with momentum $\mathbf{k}_\beta$. $\diamond$

Quasielastic or quasifree scattering thereby provides information on $|\phi_0(\mathbf{k}' - \mathbf{k} - \mathbf{k}_\beta)|^2$, the *distribution of momenta* within the target's ground state, in contrast to elastic scattering which tells about β 's matter distribution within the target. Yet because the r - and p -space wave functions are transforms of each other, these distributions are related, as shown in the Problems section.

5.7 Problems

1. Singularities and boundary conditions.

- Define the principal-value prescription for defining singular integrals.
- Show that

$$\frac{1}{x - x' \pm i\epsilon} = \mathcal{P} \frac{1}{x - x'} \mp i\pi\delta(x - x'). \quad (5.66)$$

- Obtain the Green's function for standing-wave boundary conditions from those for ingoing and outgoing waves.
- How are the phase shifts for the Schrödinger equation with standing wave boundary conditions related to those obtained with outgoing wave boundary conditions? Explain.
- Show that the partial-wave decomposition of the 3D Green's function,

$$G_E(\mathbf{r}, \mathbf{r}') = \frac{1}{4\pi r r'} \sum (2l + 1) g_l(r, r') P_l(\cos \theta), \quad (5.67)$$

produces the same $g_l(r, r')$ as obtained with the eigenfunction expansion (8.13).

2. The Schrödinger equation with a *nonlocal* potential has the form:

$$\frac{1}{2\mu} \nabla^2 \psi(\mathbf{r}) + \int d^3r' V(\mathbf{r}, \mathbf{r}') \psi(\mathbf{r}') = E\psi(\mathbf{r}). \quad (5.68)$$

A special case of a nonlocal potential (especially easy to use) is the *separable potential*:

$$V(\mathbf{r}, \mathbf{r}') = \frac{-1}{2\mu} \lambda v(\mathbf{r})v(\mathbf{r}'). \quad (5.69)$$

- (a) Show that only S-waves ($l = 0$) are affected by (5.69).
- (b) What form must the separable potential have to affect all l 's?
- (c) Establish the integral equation equivalent to (5.68) including outgoing, scattered wave boundary conditions.
- (d) Because of the special form of (5.69), this integral equation can be reduced to an algebraic equation which can be solved directly. Show that the scattering amplitude for momentum k is:

$$f_E = \frac{4\pi\lambda|g(k)|^2}{1 - 2\lambda J(k)/\pi}, \quad J(k) = \int d^3q \frac{g(q)^2}{k^2 - q^2 + i\epsilon}, \quad (5.70)$$

$$g(k) = \int_0^\infty r^2 dr \frac{\sin kr}{kr} v(r). \quad (5.71)$$

- (e) Show that the expansion of (5.70) as a power series in λ produces the Born series.
 - (f) Show that the condition on λ for the preceding Born series to converge is related to the condition for the existence of a bound state.
3. Slow neutrons of momentum $\mathbf{k}$ are scattered by a diatomic molecule aligned along the y axis with one atom at $y = -b$ and the other at $y = b$. The neutrons are directed along the z axis as usual. Assume the atoms to be infinitely heavy so that they remain fixed during the scattering process. In this case, the potential seen by the neutrons can be approximated as

$$V(\mathbf{r}) = a\delta(y - b)\delta(x)\delta(z) + a\delta(y + b)\delta(x)\delta(z). \quad (5.72)$$

- (a) Calculate the scattering amplitude and differential cross section in the first Born approximation.
 - (b) Sketch $d\sigma/d\Omega$ versus $\cos\theta$ for $0 \leq \theta \leq \pi$ if $kb = 2\pi$, and if $kb < \pi/2$ (θ is the angle with the z axis, and ϕ is the angle with the x axis in the xy plane).
 - (c) In what way does the quantum result differ from the classical result?
 - (d) How would the answer to (a) change if the molecule was oriented along the z axis?
4. The diamagnetic susceptibility of an atom is proportional to the mean-square radius $\langle r^2 \rangle$ of the electronic distribution. What is the connection, in Born approximation, of this quantity to small-angle elastic scattering from atoms?

5. Consider N static, spherically symmetric scatterers placed on a straight line such that the n^{th} scatterer is at the point $(n - 1)\mathbf{a}$.

(a) Show that for weak scattering, the elastic differential cross section is:

$$\frac{d\sigma}{d\Omega}(\theta, \phi) \simeq \left. \frac{d\sigma}{d\Omega}(\theta, \phi) \right|_B F^2(q), \quad (5.73)$$

where $d\sigma/d\Omega|_B$ is the Born approximation for a single scatterer.

(b) Find the form factor $F(q)$.

6. The time-independent Green's function with outgoing-wave boundary conditions for an incident wave of momentum $\mathbf{k}$ and energy $E = E_k$ is:

$$G_E^{(+)}(\mathbf{r}, \mathbf{r}') = \frac{-2\mu}{4\pi} \frac{e^{ik|\mathbf{r}-\mathbf{r}'|}}{|\mathbf{r}-\mathbf{r}'|}. \quad (5.74)$$

- (a) Write down a representation of $G_E^{(+)}$ as an integral over momentum $\mathbf{p}$.
 (b) Recall the Eikonal (or diffraction) limit of scattering examined in the Problems section of Chapter 4, *Scattering Applications: Lengths, Resonances, Coulomb*. Show that in this limit in which many partial waves contribute to small angle scattering,

$$G_E^{(+)}(\mathbf{r}, \mathbf{r}') \simeq \frac{2\mu i}{2k} e^{i\mathbf{k} \cdot (\mathbf{r}-\mathbf{r}')} \delta^{(2)}(\mathbf{b}-\mathbf{b}') \theta(z-z'), \quad (5.75)$$

$$\mathbf{r} = \mathbf{b} + \mathbf{z}, \quad \mathbf{r}' = \mathbf{b}' + \mathbf{z}', \quad (5.76)$$

and that the impact parameter vectors $\mathbf{b}$ and $\mathbf{b}'$ are perpendicular to the beam direction $\mathbf{k}$. (*Hint*: Assume the intermediate (virtual) momentum $\mathbf{p}$ in part (a) is nearly equal to $\mathbf{k}$, the usual type of Eikonal approximation):

$$\mathbf{p} \simeq \mathbf{k} + \boldsymbol{\eta}, \quad \eta^2 \ll k^2. \quad (5.77)$$

You may find useful the representation of the unit step function:

$$\theta(z) = \begin{cases} 1, & \text{for } z > 0, \\ 0, & \text{for } z < 0, \end{cases} = \frac{1}{2\pi i} \lim_{\epsilon \rightarrow 0^+} \int_{-\infty}^{+\infty} dx \frac{e^{ixz}}{x - i\epsilon}. \quad (5.78)$$

- (c) Give a physical explanation of the meaning of the terms in the approximate expression for $G_E^{(+)}$.
7. Consider the scattering of a fast electron from a hydrogen atom.
- (a) What is $d\sigma/d\Omega$ in Born approximation for
- elastic scattering at 0° ?
 - inelastic scattering at 0° ?
- (b) How would the above two results change if the calculation was not done in Born approximation?
8. Prove that the form factor $F(q)$ can be written as the overlap of a displaced momentum-space wave function $\psi(\mathbf{k})$ with itself:

$$F(q) = \int d^3k \psi^*(\mathbf{k} + \mathbf{q}) \psi(\mathbf{k}). \quad (5.79)$$

CHAPTER 6

TRANSITION AND POTENTIAL MATRICES

In some ways our procedures and equations are unnecessarily complicated. We have been solving for wave functions and then extracting scattering amplitudes from them, when it would be more direct to just solve for scattering amplitudes. Furthermore, our equations have indices and variables which hide their basic, simple structure. In this chapter we alleviate these shortcomings by introducing the T and V matrices (operators). While the reader will conclude if this uncomplicates things, it is needed to develop the abstract theory. The theoretical developments in this chapter are not lengthy, yet they are fundamental and lend themselves to the many applications, as found in the Problems section at the end of the chapter. We recommend working on the Problems as part of mastering the theory.

6.1 T - and V -Matrix Elements

We start with the amplitude for the scattering of a particle with momentum $\mathbf{k}$ to $\mathbf{k}'$, (5.37):

$$f_E(\mathbf{k}' \leftarrow \mathbf{k}) = -4\pi^2\mu \int d^3r \frac{e^{-i\mathbf{k}' \cdot \mathbf{r}}}{(2\pi)^{3/2}} V(\mathbf{r}) \psi_{\mathbf{k}}^{(+)}(\mathbf{r}) \quad (6.1)$$

$$= -4\pi^2\mu \langle \phi_{\mathbf{k}'} | V | \psi_{\mathbf{k}}^{(+)} \rangle, \quad (6.2)$$

where the subscripts on the scattering amplitudes and wave functions give the energy and momentum respectively. For elastic scattering $k = k' = k_0$, so only the directions change. We define the T matrix as the matrix element in (6.1):

$T_E(\mathbf{k}', \mathbf{k}) \stackrel{\text{def}}{=} \langle \phi_{\mathbf{k}'} | T | \phi_{\mathbf{k}} \rangle = \langle \phi_{\mathbf{k}'} | V | \psi_{\mathbf{k}}^{(+)} \rangle.$

(6.3)

In explicit form, this is

$$T_E(\mathbf{k}', \mathbf{k}) = \int d^3r \frac{e^{-i\mathbf{k}' \cdot \mathbf{r}}}{(2\pi)^{3/2}} V(\mathbf{r}) \psi_{\mathbf{k}}^{(+)}(\mathbf{r}). \quad (6.4)$$

Note that (6.4) denotes the momentum representation of the T operator in several ways.¹ Here T_E looks very similar to f_E except that $\mathbf{k}'$ is now arbitrary. The magnitude of $\mathbf{k}$ is fixed at $k = k_0$ because $\psi_{\mathbf{k}}$ is a solution of the Schrödinger equation for energy E_{k_0} . The amplitude f_E and $T_E(\mathbf{k}', \mathbf{k})$ are proportional when $k' = k$:

$$f_E(\theta, \phi) = -4\pi^2 \mu T_E(\mathbf{k}', \mathbf{k}) \Big|_{k'=k}. \quad (6.5)$$

Because the matrix element of T between plane waves $\phi_{\mathbf{k}}$ is proportional to the scattering amplitude f_E , T is sometimes called a *pseudopotential* (“potential” because its behavior is similar to that of the potential; “pseudo” because in Born approximation T gives the exact scattering amplitude). As we shall see, knowledge of T is equivalent to knowledge of the wave function throughout all of space.

The potentials considered up to now have all been *local*, that is, in solving the Schrödinger equation,

$$\frac{-\nabla^2}{2\mu} \psi(\mathbf{r}) + V(\mathbf{r})\psi(\mathbf{r}) = E\psi(\mathbf{r}), \quad (6.6)$$

$V(\mathbf{r})$ is evaluated at the same point $\mathbf{r}$ as the wave function. When a many-body problem is reduced to an effective 1-body problem (see Chapter 23, *The Breit–Pauli and Meson-Exchange Interactions*, Chapter 26, *Many-Body Problems*, and Chapter 27, *Statistical Help with Many-Body Problems*, for examples) we encounter a *nonlocal potential* $V(\mathbf{r}, \mathbf{r}')$, which requires knowledge of the wave function throughout all of space to determine the interaction at $\mathbf{r}$:

$$\frac{-\nabla^2}{2\mu} \psi(\mathbf{r}) + \int d^3\mathbf{r}' V(\mathbf{r}, \mathbf{r}')\psi(\mathbf{r}') = E\psi(\mathbf{r}). \quad (6.7)$$

Here we use the same symbol V for local and nonlocal potentials, but with a different number of arguments. Because a local potential is a special case of a nonlocal potential,

$$V(\mathbf{r}, \mathbf{r}')|_{\text{local}} = \delta(\mathbf{r} - \mathbf{r}')V(\mathbf{r}), \quad (6.8)$$

we develop the formalism assuming nonlocal potentials.

Exercise Generalize the integral form of the Schrödinger equation (5.7) for nonlocal potentials. $\diamond$

The momentum-space representations of the T and V operator (or matrices) for nonlocal potentials are

$$T_E(\mathbf{k}', \mathbf{k}) = \int d^3\mathbf{r} d^3\mathbf{r}' \frac{e^{-i\mathbf{k}' \cdot \mathbf{r}'}}{(2\pi)^{3/2}} V(\mathbf{r}', \mathbf{r}) \psi^{(+)}(\mathbf{r}) \quad (6.9)$$

$$= \langle \phi_{\mathbf{k}'} | V | \psi_{\mathbf{k}}^{(+)} \rangle \quad (6.10)$$

$$\equiv \langle \mathbf{k}' | T | \mathbf{k} \rangle, \quad (6.11)$$

$$V(\mathbf{k}', \mathbf{k}) = \int d^3\mathbf{r} d^3\mathbf{r}' \frac{e^{-i\mathbf{k}' \cdot \mathbf{r}'}}{(2\pi)^{3/2}} V(\mathbf{r}', \mathbf{r}) \frac{e^{i\mathbf{k} \cdot \mathbf{r}}}{(2\pi)^{3/2}} \quad (6.12)$$

$$= \langle \phi_{\mathbf{k}'} | V | \phi_{\mathbf{k}} \rangle \quad (6.13)$$

$$\equiv \langle \mathbf{k}' | V | \mathbf{k} \rangle, \quad (6.14)$$

¹ See Appendix B, *Dirac Notation and Representations*, for a summary of Dirac notation and representations.

where in (6.10) and (6.13) we use Dirac notation. These expressions can be inverted to coordinate space, for example,

$$V(\mathbf{r}, \mathbf{r}') = \int d^3k d^3k' \frac{e^{i\mathbf{k}' \cdot \mathbf{r}}}{(2\pi)^{3/2}} V(\mathbf{k}', \mathbf{k}) \frac{e^{-i\mathbf{k} \cdot \mathbf{r}'}}{(2\pi)^{3/2}} \quad (6.15)$$

$$\equiv \langle \mathbf{r} | V | \mathbf{r}' \rangle. \quad (6.16)$$

With our choice of plane waves, (6.13) and (6.15) are seen to be highly symmetric.

Exercise Verify (6.15) by direct substitution. $\diamond$

6.2 Lippmann–Schwinger Equation for T

We deduce an integral equation for T by starting with the integral form of the Schrödinger equation (5.7):

$$\psi_{\mathbf{k}}^{(+)}(\mathbf{r}) = \phi_{\mathbf{k}}(\mathbf{r}) + \int d^3r' G_E(\mathbf{r}, \mathbf{r}') V(\mathbf{r}') \psi_{\mathbf{k}}^{(+)}(\mathbf{r}'). \quad (6.17)$$

Multiplying all terms by $\phi_{\mathbf{k}'}^*(\mathbf{r}) V(\mathbf{r})$ and integrating over d^3r yields

$$\begin{aligned} \int d^3r \phi_{\mathbf{k}'}^*(\mathbf{r}) V(\mathbf{r}) \psi_{\mathbf{k}}^{(+)}(\mathbf{r}) &= \int d^3r \phi_{\mathbf{k}'}^*(\mathbf{r}) V(\mathbf{r}) \phi_{\mathbf{k}}(\mathbf{r}) \\ &+ \int d^3r d^3r' \phi_{\mathbf{k}'}^*(\mathbf{r}) V(\mathbf{r}) G_E(\mathbf{r}, \mathbf{r}') V(\mathbf{r}') \psi_{\mathbf{k}}^{(+)}(\mathbf{r}'). \end{aligned} \quad (6.18)$$

We now substitute the integral representation (5.15) for $G_E(\mathbf{r}, \mathbf{r}')$ and replace the first two integrals by the T - and V -matrix elements,

$$T_E(\mathbf{k}', \mathbf{k}) = V(\mathbf{k}', \mathbf{k}) + \int d^3p V(\mathbf{k}', \mathbf{p}) \frac{1}{E - E_p + i\epsilon} T_E(\mathbf{p}, \mathbf{k}), \quad (6.19)$$

where $E_p = p^2/2\mu$ for the nonrelativistic problem. Equation (6.19) is called the Lippmann–Schwinger integral equation for the T matrix. It is a form of the Schrödinger equation with boundary conditions built into the energy denominator. In fact, we recognize this energy denominator as the momentum-space *representation* of the Green's function (as we will see in Chapter 7, *Formal Quantum Mechanics*),

$$G_E^{(+)}(p) = \frac{1}{E - E_p + i\epsilon}. \quad (6.20)$$

Equation (6.19) now takes the neat form

$$T_E(\mathbf{k}', \mathbf{k}) = V(\mathbf{k}', \mathbf{k}) + \int d^3p V(\mathbf{k}', \mathbf{p}) G_E^{(+)}(p) T_E(\mathbf{p}, \mathbf{k}). \quad (6.21)$$

Exercise Show that (6.21) is also valid for nonlocal potentials. $\diamond$

Easy Derivation of Born Series

We have seen in Chapter 5, *Green's Functions: Integral Quantum Mechanics*, that the Born approximation for the scattering amplitude is a power series in matrix elements of the potential V . This same series is obtained by the simple iteration of (6.21):

$$T_E(\mathbf{k}', \mathbf{k})|_B^1 = V(\mathbf{k}', \mathbf{k}), \quad (6.22)$$

$$T_E(\mathbf{k}', \mathbf{k})|_B^2 = V(\mathbf{k}', \mathbf{k}) + \int d^3p V(\mathbf{k}', \mathbf{p}) G_E^{(+)}(p) V(\mathbf{p}, \mathbf{k}), \quad (6.23)$$

$$T_E(\mathbf{k}', \mathbf{k})|_B^3 = T_E(\mathbf{k}', \mathbf{k})|_B^2 + \int d^3p d^3p' V(\mathbf{k}', \mathbf{p}) G_E^{(+)}(p) G_E^{(+)}(p') V(\mathbf{p}', \mathbf{k}),$$

⋮

This embarrassingly easy derivation of the Born series demonstrates the power and convenience of these integral formulations of quantum mechanics.

6.3 Off the Energy Shell

*As I was going up the stair
I met a man who wasn't there!
He wasn't there again to-day!
I wish, I wish he'd stay away!*

—Hughes Mearns

The Lippmann–Schwinger (LS) equation (6.15) is an integral equation for $T_E(\mathbf{k}', \mathbf{k})$. It defines the functional dependence of T on the momenta $\mathbf{k}$ and $\mathbf{k}'$, and the parametric dependence on the energy E . Because the energy E enters as a parameter (it does not get integrated over), $T_E(\mathbf{k}', \mathbf{k})$ can be determined for any E independently of $\mathbf{k}$ and $\mathbf{k}'$. Even though the physical scattering amplitude $f_E(\theta, \phi)$ is related to the T -matrix element for $k = k' = k_0$, equation (6.19) determines $T_E(\mathbf{k}', \mathbf{k})$ for all values of $\mathbf{k}$ and $\mathbf{k}'$, that is, for virtual transitions as well as physical scattering. In contrast, the original definition $\langle T \rangle = \langle \phi_{\mathbf{k}'} | V | \psi_{\mathbf{k}} \rangle$ has the constraint $E = E_k$.

These kinematics are made somewhat clearer if we examine Figure 6.1 which shows the scattering process in momentum space. Physical elastic scattering is characterized by the energy conservation condition:

$$E \equiv E_{k_0} = \frac{k^2}{2\mu} = \frac{k'^2}{2\mu} = \frac{k_0^2}{2\mu}. \quad (6.25)$$

If a shell of radius k is drawn in momentum space, physical scatterings have both $\mathbf{k}$ and $\mathbf{k}'$ lying on this *energy shell*. Physical scatterings is therefore also called *on-energy-shell* or *on-shell* scattering.² When solving the integral equation (6.19), $T_E(\mathbf{k}', \mathbf{k})$ enters with

²In the noncovariant wave equations in this book, all particles are on the mass shell; that is, we always conserve mass but not necessarily kinetic energy. In relativistic treatments the same idea is incorporated by requiring the energy to be conserved but permitting the mass of an interacting particle to differ from that of a free particle. In either case one is simply saying that the energy, momentum, and mass have a different relation for free and interacting particles.

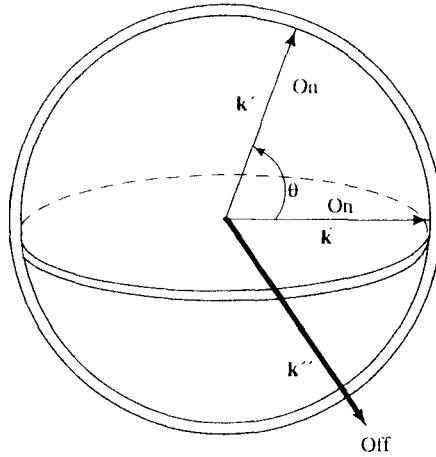


Figure 6.1: The energy shell (shaded). The momenta $\mathbf{k}'$ and $\mathbf{k}$ are on the energy shell, while $\mathbf{k}''$ is off-shell.

$\mathbf{k} \neq \mathbf{k}'$. This is *off-energy-shell* scattering and occurs when kinetic energy is not conserved in a scattering process, that is, whenever a potential acts. The off-shell scattering present from the $\mathbf{p}$ integration in the LS equation is analogous to the energy-nonconserving *virtual transitions* familiar in atomic perturbation theory.

This discussion of the physical content of the off-shell T matrix is made more rigorous by rearranging the LS equation as

$$\begin{aligned}\psi_{\mathbf{k}}^{(+)}(\mathbf{r}) &= \phi_{\mathbf{k}}(\mathbf{r}) + \int d^3r' G_E(\mathbf{r}, \mathbf{r}') V(\mathbf{r}') \psi_{\mathbf{k}}(\mathbf{r}') \\ &= \phi_{\mathbf{k}}(\mathbf{r}) + \int d^3r' d^3p \frac{e^{i\mathbf{p}\cdot\mathbf{r}}/(2\pi)^{3/2} e^{-i\mathbf{p}\cdot\mathbf{r}'} / (2\pi)^{3/2}}{E_{\mathbf{k}} - E_{\mathbf{p}} + i\epsilon} V(\mathbf{r}') \psi_{\mathbf{k}}(\mathbf{r}') \\ &= \phi_{\mathbf{k}}(\mathbf{r}) + \int d^3p \frac{e^{i\mathbf{p}\cdot\mathbf{r}}/(2\pi)^{3/2}}{E_{\mathbf{k}} - E_{\mathbf{p}} + i\epsilon} T_E(\mathbf{p}, \mathbf{k}),\end{aligned}\quad (6.26)$$

$$\psi_{\mathbf{k}}^{(+)}(\mathbf{r}) = \phi_{\mathbf{k}}(\mathbf{r}) + \int d^3p \frac{e^{i\mathbf{p}\cdot\mathbf{r}}}{(2\pi)^{3/2}} G_E^{(+)}(p) T_E(\mathbf{p}, \mathbf{k}). \quad (6.27)$$

This shows that a knowledge of $T_E(\mathbf{p}, \mathbf{k})$ for all values of $\mathbf{p}$ permits calculation of the wave function for all values of $\mathbf{r}$; knowledge at the on-shell point $T_E(\mathbf{k}', \mathbf{k})_{\mathbf{k}'=\mathbf{k}}$ determines the scattering amplitude and thus the asymptotic ($r \rightarrow \infty$) wave function only.

Example of Off-Shell T

Consider the *nonlocal, separable potential*:

$$V(\mathbf{r}', \mathbf{r}) = \lambda \bar{g}(\mathbf{r}') \bar{g}(\mathbf{r}). \quad (6.28)$$

Exercise Show that the momentum-space representation of this potential is

$$V(\mathbf{k}', \mathbf{k}) = \lambda g(\mathbf{k}')g(\mathbf{k}) \quad (6.29)$$

where $g(\mathbf{k})$ is the Fourier transform of $\tilde{g}(\mathbf{r})$. $\diamond$

For this potential, the LS equation (6.21) is

$$T_E(\mathbf{k}', \mathbf{k}) = \lambda g(\mathbf{k}')g(\mathbf{k}) + \lambda g(\mathbf{k}') \int d^3p g(\mathbf{p}) G_E^{(+)}(\mathbf{p}) T_E(\mathbf{p}, \mathbf{k}). \quad (6.30)$$

This shows that $T_E(\mathbf{k}', \mathbf{k})$ is proportional to $g(\mathbf{k}')$. Yet because the equation is symmetric in $\mathbf{k}$ and $\mathbf{k}'$, T must also be proportional to $g(\mathbf{k})$, and so we guess

$$T_E(\mathbf{k}', \mathbf{k}) = C \lambda g(\mathbf{k}')g(\mathbf{k}), \quad (6.31)$$

with C a constant that may be a function of energy. If we substitute this guess into (6.30), we obtain an algebraic equation for C :

$$C \lambda g(\mathbf{k}')g(\mathbf{k}) = \lambda g(\mathbf{k}')g(\mathbf{k}) + C \lambda^2 g(\mathbf{k}')g(\mathbf{k}) \int d^3p g(\mathbf{p})^2 G_E^{(+)}(\mathbf{p}). \quad (6.32)$$

This equation has the simple solution

$$C = \frac{1}{1 - \lambda J(E)}, \quad (6.33)$$

$$J(E) = \int d^3p g(\mathbf{p})^2 G_E^{(+)}(\mathbf{p}) = \int d^3p \frac{g(\mathbf{p})^2}{E - E_p + i\epsilon}. \quad (6.34)$$

The off-energy-shell T matrix is thus

$$T_E(\mathbf{k}', \mathbf{k}) = \frac{\lambda g(\mathbf{k}')g(\mathbf{k})}{1 - \lambda J(E)}. \quad (6.35)$$

This equation shows that $\mathbf{k}$, $\mathbf{k}'$ and E are independent variables, and that for this potential, the numerator has the momenta dependences and the denominator has the energy dependence. Note that this is the same solution obtained in the Problems section of Chapter 5, *Green's Functions: Integral Quantum Mechanics*, after considerably more effort.

Exercise Show that the ratio of off-energy-shell to on-energy-shell T matrix elements for the potential (6.29) is

$$\frac{T_E(\mathbf{k}', \mathbf{k})}{T_E(\mathbf{k}_0', \mathbf{k}_0)} = \frac{g(\mathbf{k}')g(\mathbf{k})}{g^2(\mathbf{k}_0)}. \quad \diamond \quad (6.36)$$

We can use the analytic expression for the T matrix (6.35) to deduce the requirements on the potential for it to support a bound state. The bound states occur when there are poles in the T matrix, and the poles occurs when

$$1 - \lambda J(E) = 0 \Rightarrow J(E) = \frac{1}{\lambda}. \quad (6.37)$$

Yet if E is negative (standard for a bound state), there is no contribution from the $i\epsilon$ in (6.34) and J is negative. The equality in (6.37) then also requires λ to be negative, that is, an attractive potential is needed for a bound state.

6.4 Problems

- To help test your understanding of the material in this chapter, try answering the following questions without resort to the text.
 - Present an argument that knowing the off-energy-shell T matrix $\langle \mathbf{k}' | T | \mathbf{k} \rangle$ is equivalent to knowing the entire wave function $\psi(\mathbf{r})$.
 - What is the momentum-space representation of the Schrödinger equation with outgoing-wave boundary conditions?
- In the relativistic Schrödinger equation the kinetic energy is $E_p = \sqrt{p^2 + m^2}$.
 - What is the coordinate-space representation of the kinetic energy operator $\langle \mathbf{r}' | K | \mathbf{r} \rangle$?
 - Show that this relativistic kinetic energy operator is nonlocal with a range of nonlocality given by $R_{nl} \simeq 1/m$.
 - Give an explanation for why the nonrelativistic K is local.
 - What effect does the use of a relativistic kinetic energy operator have on the uncertainty principle?
- In close analogy to our study of scattering from a bound system in Born approximation in Chapter 5, *Green's Functions and Integral Quantum Mechanics*, the more complete multiple scattering theory gives the momentum-space potential as:

$$\langle \mathbf{k}' | V | \mathbf{k} \rangle = \langle \mathbf{k}' | t | \mathbf{k} \rangle F(\mathbf{k}' - \mathbf{k}), \quad (6.38)$$

where $F(\mathbf{q})$ is the form factor for a bound system of density $\rho(\mathbf{r})$, and $\langle \mathbf{k}' | t | \mathbf{k} \rangle$ is the elementary (2-body) T matrix.

- Show that if $\langle t \rangle$ has S- and P-wave parts a and b :

$$\langle \mathbf{k}' | t | \mathbf{k} \rangle = a + b \mathbf{k}' \cdot \mathbf{k}, \quad (6.39)$$

then the potential is the operator

$$\langle \mathbf{r}' | V | \mathbf{r} \rangle = \delta(\mathbf{r}' - \mathbf{r}) \{ a \rho(\mathbf{r}) - b \nabla \cdot [\rho(\mathbf{r}) \nabla] \}. \quad (6.40)$$

- Show that this form of the potential has the $\mathbf{r}$ representation

$$\langle \mathbf{r} | V | \psi \rangle = a \rho(\mathbf{r}) \psi(\mathbf{r}) - b \nabla \cdot [\rho(\mathbf{r}) \nabla \psi(\mathbf{r})]. \quad (6.41)$$

- Why is this type of potential called *velocity-dependent*?
 - Write down the Schrödinger equation for this potential and deduce its effective mass m^* .
 - If the density were proportional to a step function, $\rho(\mathbf{r}) = \rho_o \Theta(\mathbf{R} - \mathbf{r})$, what would be the functional form of the potential? (Yes, this is an unphysical, extreme case, but it shows the effect well.)
- A potential scatters only in S wave when located at the origin. It is now displaced from the origin a distance $a = \pi/k$ along the positive z axis and scatters a particle of momentum $\mathbf{k} = k \hat{\mathbf{e}}_z$ into $\mathbf{k}'$.

- (a) Show that if the T matrix element describing scattering from this force center located at the origin is $\langle \mathbf{k}' | t | \mathbf{k} \rangle$, the T matrix for the displaced center t_1 is

$$\langle \mathbf{k}' | t_1 | \mathbf{k} \rangle = e^{-i\mathbf{q} \cdot \mathbf{a}} \langle \mathbf{k}' | t | \mathbf{k} \rangle, \quad (6.42)$$

where $\mathbf{q}$ is the momentum transfer. (*Hint:* First consider $\langle \mathbf{r}' | t | \mathbf{r} \rangle$.)

- (b) Is the cross section with a displaced center isotropic?
- (c) We place one potential at $z = a$ and a second at $z = -a$. Approximately³ how does the experimental differential cross section compare to that of just one potential at the origin? Make a simple sketch.
5. Generalize the preceding problem by considering two fixed centers, one at $\mathbf{a}$ and the other at $-\mathbf{a}$. Let v_1 and v_2 be the potential interaction of each fixed center with the projectile.

- (a) What is the Hamiltonian, Green's function, and Lippmann–Schwinger equation for this problem?
- (b) If t_1 and t_2 are the T matrices for scattering from each individual atom (at the displaced position),

$$t_1 = v_1 + v_1 G t_1, \quad t_2 = v_2 + v_2 G t_2, \quad (6.43)$$

show that the T matrix for scattering from both centers is;

$$T = (t_1 + t_2) + (t_1 G t_2 + t_2 G t_1) + (t_1 G t_2 G t_1 + t_2 G t_1 G t_2) + \cdots \quad (6.44)$$

[*Hint:* Show that the LS equation for T is a series in v_1 and v_2 which when summed yields (6.44).]

- (c) Evaluate $\langle \mathbf{k}' | T | \mathbf{k} \rangle$ up to third order in the t 's (integral expressions are fine).
- (d) Show that a full evaluation of the double scattering term requires knowledge of $\langle \mathbf{k}' | t_i(E) | \mathbf{k} \rangle$ for $\mathbf{k}' \neq \mathbf{k}$.
6. Consider the previous two-center problem in which each center scatters only in S wave:

$$\langle \mathbf{k}' | t_i(E) | \mathbf{k} \rangle = F_i(E). \quad (6.45)$$

- (a) Show that the double-scattering term is

$$\langle \mathbf{k}' | t^{\text{DS}}(E) | \mathbf{k} \rangle = F_1(E) F_2(E) 2 \cos[(\mathbf{k}_1 + \mathbf{k}_2) \cdot \mathbf{a}] K(a), \quad (6.46)$$

where $K(a)$ is an integral independent of the F 's.

- (b) Show that with this simple function for t , the multiple scattering series can be summed explicitly to:

$$\langle \mathbf{k}' | T^{\text{MS}} | \mathbf{k} \rangle = \frac{\langle \mathbf{k}' | T^{\text{SS}} + T^{\text{DS}} | \mathbf{k} \rangle}{1 - F_1(E) F_2(E) K(a)^2}, \quad (6.47)$$

where T^{SS} and T^{DS} represent the single-scattering and double-scattering terms in the multiple-scattering (MS) series. (*Hint:* Prove the result for triple and quadruple scattering and then apply induction.)

³One way of answering this “approximately” is to treat the problem in Born approximation; another way is to ignore multiple scattering, that is, to solve the problem in *impulse approximation*. In any case, you should make some approximation which will provide you with the qualitative behavior of the scattering.

7. Given an experiment which can determine the functional form of a scattering amplitude and T matrix, we would like to deduce information about the potential which produced the scattering. If the (off-energy-shell) T matrix were a constant,

$$\langle \mathbf{k}' | t(E) | \mathbf{k} \rangle = C, \quad (6.48)$$

what could you deduce about the range, functional form, and energy dependence of the potential?

8. Consider the separable potential

$$V(\mathbf{r}', \mathbf{r}) = -4\pi\lambda\delta(\mathbf{r}' - \mathbf{a})\delta(\mathbf{r} - \mathbf{a}). \quad (6.49)$$

- What is $V_l(k', k)$ for this potential for all l 's?
- Solve for the off-energy-shell T matrix $T_l(k', k; E)$ for this potential. (This can be done algebraically.)
- Indicate why this solution is valid for *any* value of k' , k and E .
- Show that the T matrix, considered as a function of the complex energy E , is discontinuous along the real positive energy axis. (This is called the *unitarity* or *right-hand* cut.)

9. Consider the separable potential

$$V(\mathbf{r}', \mathbf{r}) = \lambda v(\mathbf{r})v(\mathbf{r}'). \quad (6.50)$$

- Derive (or verify) that the off-energy-shell T matrix for this potential has the form

$$\langle \mathbf{k}' | t(E) | \mathbf{k} \rangle = \frac{\lambda g(k')g(k)}{1 - \lambda J(E)}, \quad (6.51)$$

where $E = k_0^2/2m$ is the energy.

- Determine the integral expressions for $g(k)$ and $J(E)$.
- Relate the value of the zero-momentum *form factor* $g(k = 0)$ to the scattering length.
- Express the bound-state condition for this potential in terms of the integral $J(E)$.
- Deduce the number of bound states that occurs with this potential.
- Express the condition for this potential to produce a Breit-Wigner resonance at energy $E_r = k_r^2/2m$.
- Show that the width of this resonances is

$$\Gamma = Bg^2(k_r), \quad (6.52)$$

where B is a constant you should determine.

10. Prove that a $\mathbf{p}$ -space potential which depends on only the momentum transfer $\mathbf{k}' - \mathbf{k}$ must be local in $\mathbf{r}$ space.

11. Show that the $\mathbf{r}$ -space representation of the potential

$$\langle \mathbf{k}' | V | \mathbf{k} \rangle = [a + b(\mathbf{k}' - \mathbf{k})^2] \tilde{\rho}(\mathbf{k}' - \mathbf{k}) \quad (6.53)$$

is the local potential

$$\langle \mathbf{r}' | V | \mathbf{r} \rangle = C \delta(\mathbf{r}' - \mathbf{r}) [a \rho(\mathbf{r}) - b \nabla^2 \rho(\mathbf{r})], \quad (6.54)$$

where ρ is the Fourier transform of $\tilde{\rho}$ and C is a constant.

CHAPTER 7

FORMAL QUANTUM MECHANICS

In this chapter we develop abstract forms of the Schrödinger equation. The notation is intended to be abstract and simple. In Chapter 8, *The Angular Momentum Basis*, most of these same relations are expressed in the one-dimensional energy-angular momentum basis.

7.1 Operator Schrödinger's Equation

Some of the complexity of our equations arise from the specifics of a particular representation and the mixing of different representations. To better display the fundamental structure of the theory we now use the Dirac notation (reviewed in Appendix B, *Dirac Notation and Representations*) to express Schrödinger theory in *abstract* (operator, matrix, or representation-independent) form. We start with the differential Schrödinger equation for a general potential:

$$-\frac{\nabla^2}{2\mu}\psi(\mathbf{r}) + \int d^3\mathbf{r}' V(\mathbf{r}, \mathbf{r}')\psi(\mathbf{r}') = E\psi(\mathbf{r}), \quad (7.1)$$

$$-\frac{\nabla^2}{2\mu}\psi(\mathbf{r}) + \int d^3\mathbf{r}' \langle \mathbf{r} | V | \mathbf{r}' \rangle \langle \mathbf{r}' | \psi \rangle = E\psi(\mathbf{r}). \quad (7.2)$$

We recognize the potential term as the $\mathbf{r}$ representation of $V |\psi\rangle$:

$$\langle \mathbf{r} | V |\psi\rangle = \int d^3\mathbf{r}' \langle \mathbf{r} | V | \mathbf{r}' \rangle \langle \mathbf{r}' | \psi \rangle = \int d^3\mathbf{r}' V(\mathbf{r}, \mathbf{r}')\psi(\mathbf{r}'). \quad (7.3)$$

After defining the $\mathbf{r}$ representation of the kinetic energy operator, we identify the $\mathbf{r}$ representation of $K |\psi\rangle$:

$$\langle \mathbf{r}' | K | \mathbf{r} \rangle = \delta(\mathbf{r} - \mathbf{r}') \left(-\frac{\nabla_{\mathbf{r}}^2}{2\mu} \right), \quad (7.4)$$

$$\Rightarrow \langle \mathbf{r} | K |\psi\rangle = \int d^3\mathbf{r}' \langle \mathbf{r} | K | \mathbf{r}' \rangle \langle \mathbf{r}' | \psi \rangle = -\frac{\nabla_{\mathbf{r}}^2\psi(\mathbf{r})}{2\mu}. \quad (7.5)$$

The Schrödinger equation consequently reduces to the abstract form:

$$\langle \mathbf{r} | K | \psi \rangle + \langle \mathbf{r} | V | \psi \rangle = E \langle \mathbf{r} | \psi \rangle, \quad (7.6)$$

$$\Rightarrow (K + V) | \psi \rangle = E | \psi \rangle, \quad (7.7)$$

$$H | \psi \rangle = E | \psi \rangle, \quad (7.8)$$

where H is the Hamiltonian operator,

$$H = K + V. \quad (7.9)$$

Equation (7.8) should *not* be thought of as a short-hand form of the differential Schrödinger equation. Rather, it is a relation among abstract operators and Hilbert-space vectors which has (7.1) as an explicit representation.

We now look at the Schrödinger equation in the momentum representation. The kinetic energy operator's representation is fixed from our previous definitions:

$$\begin{aligned} \langle \mathbf{k}' | K | \mathbf{k} \rangle &= \int d^3r d^3r' \langle \mathbf{k}' | \mathbf{r}' \rangle \langle \mathbf{r}' | K | \mathbf{r} \rangle \langle \mathbf{r} | \mathbf{k} \rangle \\ &= \int d^3r d^3r' \frac{e^{-i\mathbf{k}' \cdot \mathbf{r}'}}{(2\pi)^{3/2}} \delta(\mathbf{r} - \mathbf{r}') \left(\frac{-\nabla_{\mathbf{r}}^2}{2\mu} \right) \frac{e^{i\mathbf{k} \cdot \mathbf{r}}}{(2\pi)^{3/2}} \\ &= \int d^3r \frac{e^{-i\mathbf{k}' \cdot \mathbf{r}}}{(2\pi)^{3/2}} \frac{k^2}{2\mu} \frac{e^{i\mathbf{k} \cdot \mathbf{r}}}{(2\pi)^{3/2}}, \end{aligned} \quad (7.10)$$

$$\Rightarrow \langle \mathbf{k}' | K | \mathbf{k} \rangle = \delta(\mathbf{k}' - \mathbf{k}) \frac{k^2}{2\mu}. \quad (7.11)$$

The Schrödinger equation is thus

$$\langle \mathbf{k} | K | \psi \rangle + \langle \mathbf{k} | V | \psi \rangle = E \langle \mathbf{k} | \psi \rangle, \quad (7.12)$$

$$\frac{k^2}{2\mu} \psi(\mathbf{k}) + \int d^3k' \langle \mathbf{k} | V | \mathbf{k}' \rangle \psi(\mathbf{k}') = E \psi(\mathbf{k}), \quad (7.13)$$

where only the arguments are used to distinguish the function $\psi(\mathbf{r})$ from its Fourier transform $\psi(\mathbf{k})$.

Exercise Fill in the steps leading to (7.13). ◇

Exercise Relate $\langle \mathbf{k} | \psi \rangle$ to $\langle \mathbf{k} | \psi \rangle$ using representation theory, and deduce the conditions on ψ for this relation to exist. ◇

7.2 Operator Lippmann–Schwinger Equations

The discussion in Chapter 5, *Green's Functions and Integral Quantum Mechanics*, of the spectral representation (eigenfunction expansion) of the Green's function has already given us the $\mathbf{k}$ and $\mathbf{r}$ representations of G :

$$\langle \mathbf{r}' | G_E^{(+)} | \mathbf{r} \rangle \stackrel{\text{def}}{=} G_E(\mathbf{r}, \mathbf{r}') = \frac{-\mu}{2\pi} \frac{e^{i\mathbf{k}|\mathbf{r}-\mathbf{r}'|}}{|\mathbf{r}-\mathbf{r}'|} \quad (7.14)$$

$$= \int d^3 p' \frac{[e^{i\mathbf{p}' \cdot \mathbf{r}'} / (2\pi)^{3/2} e^{-i\mathbf{p}' \cdot \mathbf{r}} / (2\pi)^{3/2}]}{E_k - E_{p'} + i\epsilon} \quad (7.15)$$

$$= \int d^3 p' d^3 p \langle \mathbf{r}' | \mathbf{p}' \rangle \left\langle \mathbf{p}' \left| \frac{1}{E_k - K + i\epsilon} \right| \mathbf{p} \right\rangle \langle \mathbf{p} | \mathbf{r} \rangle. \quad (7.16)$$

Here K is the kinetic energy operator and k is the on-shell momentum directly related to the energy (we drop the zero subscript):

$$E \equiv E_k = E_{k_0}. \quad (7.17)$$

We have also made the identification of the Green's function:

$$\langle \mathbf{p}' | G_E | \mathbf{p} \rangle = \left\langle \mathbf{p}' \left| \frac{1}{E - K} \right| \mathbf{p} \right\rangle = \frac{\delta(\mathbf{p} - \mathbf{p}')}{E - p^2/2\mu}. \quad (7.18)$$

The *abstract Green's operator* for outgoing waves (the (+) superscript) is thus

$$G_E^{(+)} = \frac{1}{E - K + i\epsilon} \equiv \frac{1}{E - H_0 + i\epsilon} \quad (7.19)$$

$$= \frac{\mathcal{P}}{E - H_0} - i\pi\delta(E - H_0), \quad (7.20)$$

where $H_0 \equiv K$ is the unperturbed Hamiltonian operator. Equation (7.20) is the operator version of (5.30).

We write the LS equation for the wave function (5.33) in abstract form by removing the explicit representation dependences:

$$\begin{aligned} \langle \mathbf{r} | \psi_{\mathbf{k}}^{(+)} \rangle &= \langle \mathbf{r} | \phi_{\mathbf{k}} \rangle + \int d^3 r' d^3 r'' \langle \mathbf{r} | G_E^{(+)} | \mathbf{r}' \rangle \langle \mathbf{r}' | V | \mathbf{r}'' \rangle \langle \mathbf{r}'' | \psi_{\mathbf{k}}^{(+)} \rangle \\ &= \langle \mathbf{r} | \phi_{\mathbf{k}} \rangle + \langle \mathbf{r} | G_E^{(+)} V | \psi_{\mathbf{k}}^{(+)} \rangle. \end{aligned} \quad (7.21)$$

We now remove the representation dependence to obtain

$$\boxed{|\psi_{\mathbf{k}}^{(+)}\rangle = |\phi_{\mathbf{k}}\rangle + G_E^{(+)} V |\psi_{\mathbf{k}}^{(+)}\rangle,} \quad (7.22)$$

where $G_E^{(+)}$ is given by (7.19).

The fractions in (7.19) and (7.20) with operators in the denominators are inverse operators. For example, if we start with the Schrödinger equation in operator form,

$$(E - H_0) |\psi\rangle = V |\psi\rangle, \quad (7.23)$$

we obtain its solution by acting upon both sides with the operator $(E - H_0)^{-1}$:

$$(E - H_0)^{-1} (E - H_0) |\psi\rangle = (E - H_0)^{-1} V |\psi\rangle, \quad (7.24)$$

$$|\psi\rangle = (E - H_0)^{-1} V |\psi\rangle. \quad (7.25)$$

To make this solution meaningful we must define the inverse when E is an eigenvalue of H_0 (since scattering is in the continuous-energy regime, this is bound to happen). We avoid

this singularity, and build in the outgoing scattered-wave boundary conditions, by adding a small positive imaginary part $i\epsilon$ to the energy E :

$$|\psi_{\mathbf{k}}^{(+)}\rangle = (E - H_0 + i\epsilon)^{-1} V |\psi_{\mathbf{k}}^{(+)}\rangle. \quad (7.26)$$

For completeness and to match the scattering-experiment boundary conditions, the full solution must also include $\phi_{\mathbf{k}}$, the solution to the homogeneous equation,

$$|\psi_{\mathbf{k}}^{(+)}\rangle = |\phi_{\mathbf{k}}\rangle + (E - H_0 + i\epsilon)^{-1} V |\psi_{\mathbf{k}}^{(+)}\rangle. \quad (7.27)$$

We recognize (7.27) as our previous solution and G as the inverse of $(E - H_0)$.

Momentum Space LS Wave Equation

Inasmuch as momentum and coordinate spaces just provide different representations of the same equation, we should be able to solve the Schrödinger equation equally well in either. The mathematics, however, is quite different for scattering states compared with bound states because scattering states are in the *continuum of energy* and consequently are not normalizable. For this reason the Fourier transform of the coordinate-space wave function may cease to exist, and we need to solve separately for the $\mathbf{r}$ - and $\mathbf{p}$ -space wave functions. An example of this difference is seen by looking at the momentum representation of the ψ LS equation (7.22):

$$|\psi_{\mathbf{k}}^{(+)}\rangle = |\phi_{\mathbf{k}}\rangle + (E_{\mathbf{k}} - H_0 + i\epsilon)^{-1} V |\psi_{\mathbf{k}}^{(+)}\rangle, \quad (7.28)$$

$$\langle \mathbf{k}' | \psi_{\mathbf{k}}^{(+)} \rangle = \langle \mathbf{k}' | \phi_{\mathbf{k}} \rangle + \langle \mathbf{k}' | (E_{\mathbf{k}} - H_0 + i\epsilon)^{-1} V | \psi_{\mathbf{k}}^{(+)} \rangle, \quad (7.29)$$

$$\Rightarrow \psi_{\mathbf{k}}^{(+)}(\mathbf{k}') = \delta(\mathbf{k}' - \mathbf{k}) + \frac{1}{E_{\mathbf{k}} - E_{\mathbf{k}'}} \int d^3p \langle \mathbf{k}' | V | \mathbf{p} \rangle \psi_{\mathbf{k}}^{(+)}(\mathbf{p}). \quad (7.30)$$

We see that there is a delta function in $\psi_{\mathbf{k}}(\mathbf{k}')$ arising from the incident plane wave which can make a solution for $\psi_{\mathbf{k}}(\mathbf{k}')$ difficult.

Other Operator Forms

As long as we respect the noncommuting properties of operators, we can manipulate them algebraically. We manipulate in order to explore new forms of the Schrödinger equation, and start by iterating the LS equation (7.22):

$$|\psi\rangle = |\phi\rangle + (E - H_0)^{-1} V |\psi\rangle \equiv |\phi\rangle + GV |\psi\rangle \quad (7.31)$$

$$= |\phi\rangle + GV |\phi\rangle + GVG V |\phi\rangle + GVG VGV |\phi\rangle + \dots \quad (7.32)$$

$$\Rightarrow |\psi\rangle = \frac{1}{1 - GV} |\phi\rangle. \quad (7.33)$$

We recognize (7.32) as the operator form of the Born series for the wave function (5.40), and (7.33) as the formal summation of that series (the summation remains valid even for bound states where the Born series diverges).

Exercise Show that the full wave function also has the formal solution

$$|\psi\rangle = |\phi\rangle + G(1 - VG)^{-1}V|\phi\rangle. \quad \diamond \quad (7.34)$$

The operator $(1 - GV)^{-1}$ in (7.33) and (7.34) is given meaning by the series (7.32), and is seen to convert the free wave ϕ into a full, or distorted, wave ψ . It occurs often and is called the Moller or *wave operator* Ω :

$$|\psi_{\mathbf{k}}\rangle = \Omega^{(+)}|\phi_{\mathbf{k}}\rangle, \quad (7.35)$$

$$\Omega^{(+)} = \frac{1}{1 - G^{(+)}V}. \quad (7.36)$$

Here we include the subscript $\mathbf{k}$ in (7.35) as a reminder that the free and distorted waves must correspond to the same beam momentum, and add a plus superscript in (7.36) to indicate the outgoing-wave boundary conditions (obtained by adding $+i\epsilon$ to the energy).

Exercise Show that the wave operator satisfies the LS equation

$$\Omega^{(+)} = 1 + G^{(+)}V\Omega^{(+)}. \quad \diamond \quad (7.37)$$

The Green's operator $G_E^{(+)} = (E - H_0 + i\epsilon)^{-1}$ we have been using is often called the *free particles' Green's operator*. A related operator is the *full* or *interacting* Green's function $G|_{\text{full}}$ which has the full Hamiltonian H in its denominator:

$$G_E^{(+)}|_{\text{full}} = \frac{1}{E - H + i\epsilon} = \frac{1}{E - H_0 - V + i\epsilon}. \quad (7.38)$$

Exercise Use the operator relation

$$\frac{1}{A^{-1}B^{-1}} = \frac{1}{(BA)^{-1}} = BA \quad (7.39)$$

to show that

$$G|_{\text{full}} = \frac{1}{1 - GV}G = G\frac{1}{1 - VG}. \quad \diamond \quad (7.40)$$

Exercise Use (7.40) to show that the full Green's operator satisfies the LS-like equation:

$$G|_{\text{full}} = G + GV G|_{\text{full}}. \quad \diamond \quad (7.41)$$

Another useful form of the ψ LS equation utilizes the full Green's operator:

$$|\psi_{\mathbf{k}}^{(+)}\rangle = |\phi_{\mathbf{k}}\rangle + G_{\text{full}}^{(+)} V |\phi_{\mathbf{k}}\rangle \quad (7.42)$$

$$= |\phi_{\mathbf{k}}\rangle + (E - H + i\epsilon)^{-1} V |\phi_{\mathbf{k}}\rangle \quad (7.43)$$

$$= |\phi_{\mathbf{k}}\rangle + (E - K - V + i\epsilon)^{-1} V |\phi_{\mathbf{k}}\rangle. \quad (7.44)$$

Exercise Prove (7.42) by expanding the denominator in (7.44) in powers of V , thus obtaining the familiar Born series for $|\psi\rangle$. $\diamond$

Return of the Schrödinger Equation $\odot$

If our manipulations makes sense, we should be able to convert (7.34) back to the usual form of the Schrödinger equation.¹ Consider the inverse of the Green's function:

$$G^{-1} = E - H_0 + i\epsilon. \quad (7.45)$$

If we let G^{-1} act on both sides of (7.34) we obtain:

$$G^{-1} |\psi\rangle = (E - H_0 + i\epsilon) |\phi\rangle + G^{-1} G (1 - VG)^{-1} V |\phi\rangle, \quad (7.46)$$

$$(E - H_0 + i\epsilon) |\psi\rangle = 0 + \frac{1}{1 - VG} V |\phi\rangle. \quad (7.47)$$

Equation (7.47) is almost the Schrödinger equation.

Exercise Show that the series expansion of the denominator on the RHS of (7.47) leads to the identity,

$$(1 - VG)^{-1} V |\phi\rangle = V |\psi\rangle, \quad (7.48)$$

which shows that (7.47) is the familiar Schrödinger equation. $\diamond$

An alternate and illuminating approach is to multiply both sides of (7.46) by $(1 - VG)$:

$$(1 - VG)G^{-1} |\psi\rangle = 0 + (1 - VG)(1 - VG)^{-1} V |\phi\rangle, \quad (7.49)$$

$$(G^{-1} - V) |\psi\rangle = V |\phi\rangle, \quad (7.50)$$

$$(E - H_0) |\psi\rangle = V |\psi\rangle + V |\phi\rangle = V(|\psi\rangle + |\phi\rangle), \quad (7.51)$$

$$(E - H_0)(|\psi\rangle + |\phi\rangle) = V(|\psi\rangle + |\phi\rangle). \quad (7.52)$$

Equation (7.52) shows that $|\psi\rangle$ plus any amount of $|\phi\rangle$ satisfies the Schrödinger equation with scattering boundary conditions.

7.3 Proof of Orthogonality $\odot$

An example of the power and simplicity of the operator formalism is afforded by the proof of orthogonality of the scattering wave functions $|\psi\rangle$'s for different $\mathbf{k}$'s, a proof of significant complication if an explicit representation is assumed. We start by noting that the

¹We thank Kirk McVoy for some interesting discussions on this point.

free waves satisfy the orthogonality relation (see Appendices A, *Natural Units and Plane Waves*, and B, *Dirac Notation and Representations*),

$$\langle \phi_i | \phi_j \rangle = \delta_{ij}, \quad (7.53)$$

where i and j can be vectors, and the delta function is Kronecker's for box normalization and Dirac's for the infinite domain. We want to prove that the distorted waves of different momentum i and j ,

$$|\psi_i^{(+)}\rangle = |\phi_i\rangle + (E_i - H_0 + i\epsilon)^{-1} V |\psi_i^{(+)}\rangle, \quad (7.54)$$

$$|\psi_j^{(+)}\rangle = |\phi_j\rangle + (E_j - H_0 + i\epsilon)^{-1} V |\psi_j^{(+)}\rangle, \quad (7.55)$$

are orthogonal:

$$\langle \psi_j^{(+)} | \psi_i^{(+)} \rangle = \delta_{ji}. \quad (7.56)$$

To prove (7.56) we first determine the dual-space bra $\langle \psi_j^{(+)} |$ using the full Green's operator form (7.42) of the ψ LS equation:

$$|\psi_j^{(+)}\rangle = |\phi_j\rangle + (E_j - H + i\epsilon)^{-1} V |\phi_j\rangle \quad (7.57)$$

$$\Rightarrow \langle \psi_j^{(+)} | = \langle \phi_j | + \langle \phi_j | V (E_j - H - i\epsilon)^{-1}, \quad (7.58)$$

where we have used $(AB)^\dagger = B^\dagger A^\dagger$, reversed the sign of ϵ , and assumed V and H are Hermitian. We now form the inner product, with care to distinguish the full and free Green's operators (and, for simplicity, leave off the $(+)$ superscript):

$$\langle \psi_j | \psi_i \rangle = \langle \phi_j | \phi_i \rangle + \left\langle \phi_j \left| \frac{1}{E_i - H_0 + i\epsilon} V \right| \psi_i \right\rangle \left\langle \phi_j \left| V \frac{1}{E_j - H - i\epsilon} \right| \psi_i \right\rangle. \quad (7.59)$$

We let the Green's operator in the second term act to the left (we know H_0 's effect on a free state ϕ_j), and the full Green's operator act to the right (we know H 's effect on ψ):

$$\begin{aligned} \langle \psi_j | \psi_i \rangle &= \delta_{ij} + \frac{1}{E_i - E_j + i\epsilon} \langle \phi_j | V | \psi_i \rangle + \frac{1}{E_j - E_i - i\epsilon} \langle \phi_j | V | \psi_i \rangle \\ &= \delta_{ij}. \end{aligned} \quad (7.60)$$

7.4 Operator Equation for T

We now reduce the LS equation for the T matrix (6.21),

$$T_E(\mathbf{k}', \mathbf{k}) = V(\mathbf{k}', \mathbf{k}) + \int d^3p V(\mathbf{k}', \mathbf{p}) \frac{1}{E - E_p + i\epsilon} T_E(\mathbf{p}, \mathbf{k}), \quad (7.61)$$

to its abstract form.² We use Dirac notation (Appendix B, *Dirac Notation and Representations*) and the momentum representation of the completeness relation to write

$$\langle \mathbf{k}' | T_E | \mathbf{k} \rangle = \langle \mathbf{k}' | V | \mathbf{k} \rangle + \int d^3p d^3p' \langle \mathbf{k}' | V | \mathbf{p}' \rangle \langle \mathbf{p}' | G_E^{(+)} | \mathbf{p} \rangle \langle \mathbf{p} | T_E | \mathbf{k} \rangle \quad (7.62)$$

$$= \langle \mathbf{k}' | V | \mathbf{k} \rangle + \langle \mathbf{k}' | V G_E^{(+)} T_E | \mathbf{k} \rangle. \quad (7.63)$$

²The symbol T clearly represents the T matrix whereas $\langle \mathbf{k}' | T_{\mathbf{k}} | \mathbf{k} \rangle$ is a T -matrix element. In practice, the elements are often referred to as *the* T matrix.

In abstract form this is

$$T = V + VG_E^{(+)}T_E. \quad (7.64)$$

Equation (7.64) can also be written with the T before the G ,

$$T_E = V + T_E G_E^{(+)}V. \quad (7.65)$$

Equations (7.64) and (7.65) are equivalent since they both have the same Born series:

$$T = V + VGV + VGVGV + \cdots. \quad (7.66)$$

We note again that because T and V are equal in the Born approximation, they have similar behaviors in $\mathbf{k}$ or $\mathbf{r}$ space. This is explored further in the Problems section of Chapter 6, *Transition and Potential Matrices*, and in what follows.

Exercise Show that a constant T matrix,

$$T_E(\mathbf{k}', \mathbf{k}) = C(E), \quad (7.67)$$

corresponds to a local, zero-range pseudopotential, that is,

$$\langle \mathbf{r}' | T | \mathbf{r} \rangle \propto C(E) \delta(\mathbf{r} - \mathbf{r}') \delta(\mathbf{r}). \quad \diamond \quad (7.68)$$

These operator equations provide a formal solution of (7.64) for the T matrix:

$$T_E = V + VG_E^{(+)}T_E, \quad (7.69)$$

$$(1 - VG_E^{(+)})T_E = V, \quad (7.70)$$

$$\Rightarrow T_E = (1 - VG_E^{(+)})^{-1}V = \Omega^{(+)}V. \quad (7.71)$$

Or if (7.65) is used,

$$T_E = V(1 - G_E^{(+)}V)^{-1} = V\Omega^{(+)}, \quad (7.72)$$

where $\Omega^{(+)}$ is the wave operator (7.36). Thus again, (7.72) show that knowledge of all elements of T is equivalent to knowledge of $\Omega^{(+)}$ and accordingly of the wave function with outgoing wave boundary conditions.

Exercise Prove (7.71) and (7.72) by use of the Born series. $\diamond$

7.5 The Bound-State Connection

The observation in Chapter 4, *Scattering Applications: Lengths, Resonances, Coulomb*, that bound states of a potential occur at the complex-energy poles of the scattering amplitude (or T matrix) is transparent in the operator formalism. The formal solutions for the T matrix,

$$T_E = \frac{1}{1 - VG_E^{(+)}}V = V \frac{1}{1 - G_E^{(+)}V}, \quad (7.73)$$

is singular wherever the denominators has zeros, that is, where

$$1 - VG_E = 1 - GV_E = 0. \quad (7.74)$$

If these are simple zeros, T has simple poles. In an explicit representation, the conditions (7.74) lead to the requirement that the determinants of the matrices that represent these operators vanish,

$$\det(1 - VG_E) = \det(1 - G_E V) = 0. \quad (7.75)$$

This condition is met only by specific value of E , some of which are bound-state poles.³ The T matrix also will have poles wherever the numerator V has poles, but these *potential poles* are less interesting because they reflect some mathematical property of the potential rather than some dynamical happening like a bound state.

The bound states of a potential are probably more familiar as the solutions of the eigenvalue problem (7.23):

$$(H_0 + V)\psi = E\psi. \quad (7.76)$$

Formally, we have been using the Green's function technique to solve the scattering problem $\psi = \phi + GV\psi$, where ϕ , the solution of the homogeneous equation $H_0\phi = E\phi$, was included for completeness. For the eigenvalue or bound-state problem, ϕ should not be included because we have different boundary conditions; being confined, bound states must vanish at infinity and thus cannot contain any plane waves. Likewise, the Green's operator for true bound states has no singularities in its integral representation (7.16) and thus vanishes at infinity.⁴ Accordingly we solve

$$\psi = G_E V \psi = \frac{1}{E - H_0} V \psi, \quad (7.77)$$

or

$$(1 - G_E V)\psi = 0, \quad (7.78)$$

$$\Rightarrow \det(1 - G_E V) = 0. \quad (7.79)$$

This is the same condition obtained above for the poles of the T matrix, yet is now obtained from the solution for the bound-state wave function.

It is satisfying to note that the momentum-space representation of the bound state Schrödinger equation (7.77) has no delta function singularities (as occurred for scattering):

$$\langle \mathbf{k} | \psi \rangle = \langle \mathbf{k} | G_E V | \psi \rangle = \left\langle \mathbf{k} \left| \frac{1}{E - H_0} V \right| \psi \right\rangle, \quad (7.80)$$

$$\Rightarrow \psi(\mathbf{k}) = \frac{1}{E - E_k} \int d^3p V(\mathbf{k}, \mathbf{p}) \psi(\mathbf{p}). \quad (7.81)$$

This means that for bound states, the Fourier transform of $\psi(\mathbf{r})$ exists, and that the bound-state wave function is normalizable in both $\mathbf{p}$ and $\mathbf{r}$ spaces.

³While all bound states occur at poles of T , not all poles of T are bound states. As we have seen in Chapter 4, *Scattering Applications: Lengths, Resonances, Coulomb*, true bound states are those poles located along the positive imaginary axis $k = +i\kappa$; other poles may correspond to resonances, virtual states, or kinematic effects, see, e.g., Park (1966), Newton (1966), Gasirowicz (1966).

⁴These different conditions on the Green's function and the solution of the wave equation should be familiar from classical electrodynamics (e.g. Jackson, 1975, Chapter 1), where the application of Green's theorem is subject to either Dirichlet or Neumann boundary conditions. The scattering problem, which has a homogeneous solution included, is a Neumann problem, while the bound-state problem, in which the solution must vanish at infinity to be confined, is a Dirichlet problem. We thank A. Wasserman for a discussion on this point.

7.6 Unitarity of T and the Optical Theorem

In classical mechanics it is useful when trying to understand phenomena to separate kinematic and dynamics effects. In quantum mechanics we do something similar by separating kinematic effects, such as phase space factors and Clebsch-Gordan coefficients, from the more dynamic matrix elements. The separation is continued even further by examining which properties of the matrix elements are general, say, following from some symmetry principle, and which are specific to the potential being used.

An example is *unitarity*. In Chapter 2, *Currents and Cross Sections*, our study of current and probability conservation culminated in the (rather complicated) derivation of the *optical theorem*. We now start by asking where probability conservation is contained within the LS equation (7.64):

$$T = V + VGT. \quad (7.82)$$

The Hermiticity of V and the $i\epsilon$ part of the energy should guarantee that particles scatter out from the target. This follows from (7.82) and its Hermitian adjoint:

$$T^\dagger = V^\dagger + T^\dagger G^* V^\dagger = V + T^\dagger G^* V, \quad (7.83)$$

where $G^\dagger = G^*$ because the Green's operator is diagonal. We “solve” (7.82) and (7.83) for V :

$$V = T - VGT, \quad V = T^\dagger - T^\dagger G^* V. \quad (7.84)$$

We combine these solutions with (7.82) and (7.83) to obtain

$$T = V + T^\dagger GT - T^\dagger G^* VGT, \quad (7.85)$$

$$T^\dagger = V + T^\dagger G^* T - T^\dagger G^* VGT, \quad (7.86)$$

$$\Rightarrow T - T^\dagger = T^\dagger (G - G^*) T. \quad (7.87)$$

Exercise Use (7.87) to show that

$$G - G^* = -2\pi i \delta(E - H_0), \quad (7.88)$$

which leads to the *unitarity relation*:

$$T - T^\dagger = -2\pi i T^\dagger \delta(E - H_0) T. \quad \diamond \quad (7.89)$$

(Calling (7.89) a *unitarity relation* is somewhat of a misnomer because a unitary T would satisfy $T^\dagger = T$.)

To derive the optical theorem from (7.89), we look at its momentum representation:

$$\langle \mathbf{k}' | T | \mathbf{k} \rangle - \langle \mathbf{k}' | T^\dagger | \mathbf{k} \rangle = -2\pi i \int d^3 p d^3 p' \langle \mathbf{k}' | T | \mathbf{p} \rangle \langle \mathbf{p} | \delta(E - H_0) | \mathbf{p}' \rangle \langle \mathbf{p}' | T | \mathbf{k} \rangle.$$

$$\text{Yet } \langle \mathbf{p} | \delta(E - H_0) | \mathbf{p}' \rangle = \delta(\mathbf{p} - \mathbf{p}') \delta(E - E_p), \quad \text{so} \quad (7.90)$$

$$\langle \mathbf{k}' | T | \mathbf{k} \rangle - \langle \mathbf{k}' | T^\dagger | \mathbf{k} \rangle = -2\pi i \int d^3 p \langle \mathbf{k}' | T | \mathbf{p} \rangle \delta[E - E_p] \langle \mathbf{p} | T | \mathbf{k} \rangle. \quad (7.91)$$

This last equation is known as the *off-shell unitarity relation* since there is no restriction to $k' = k$. If we limit ourselves to on-energy-shell forward scattering, $k' = k = k_0$, and (7.91) reduces to:⁵

$$\begin{aligned}
 2i \operatorname{Im} \langle \mathbf{k} | T | \mathbf{k} \rangle &= -2\pi i \int d^3p |\langle \mathbf{k} | T | \mathbf{p} \rangle|^2 \delta[E_k - E_p] \\
 &= -2\pi i \int d\Omega \int_0^\infty dp p^2 |\langle \mathbf{k} | T | \mathbf{p} \rangle|^2 \delta[E_k - E_p] \\
 &= -2\pi i \int d\Omega \int_0^\infty dE_p \mu p |\langle \mathbf{k} | T | \mathbf{p} \rangle|^2 \delta[E_k - E_p] \\
 &= -2\pi i k \mu \int d\Omega |\langle \mathbf{k} | T | \mathbf{p} \rangle|^2.
 \end{aligned} \tag{7.92}$$

We relate these on-shell T -matrix elements to the scattering amplitude via (??):

$$2i \operatorname{Im} f_E(\mathbf{k} \leftarrow \mathbf{k}) \frac{-1}{4\pi^2 \mu} = -2\pi i k \mu \left(\frac{-1}{4\pi^2 \mu} \right)^2 \int d\Omega |f_E(\mathbf{k} \leftarrow \mathbf{p})|^2. \tag{7.93}$$

This implies the optical theorem of Chapter 2, *Currents and Cross Sections*:

$$\frac{4\pi}{k} \operatorname{Im} f_E(\theta = 0^\circ) = \sigma_t. \tag{7.94}$$

7.7 Reaction and Scattering Matrices

The transition matrix T is an extension of the scattering amplitude that contains all information about the elastic-scattering wave function. It is also useful to define the reaction matrix R and the scattering matrix S . The R matrix elements satisfies an integral equation similar to that of T , only with the standing-wave or principal-value Green's function (??):

$$R_E(\mathbf{k}', \mathbf{k}) = V(\mathbf{k}', \mathbf{k}) + \mathcal{P} \int d^3p V(\mathbf{k}', \mathbf{p}) \frac{1}{E - E_p} R_E(\mathbf{p}, \mathbf{k}). \tag{7.95}$$

This equation has the operator form:

$$R_E = V + V G^{(P)} R_E = \frac{1}{1 - V G^{(P)}} V. \tag{7.96}$$

Exercise Show by examining the integral equations for T and R that

$$R_E = T_E + i\pi T_E \delta(E - H_0) R_E. \quad \diamond \tag{7.97}$$

Exercise Solve for the R operator in terms of the T operator. \(\diamond\)

Because R is determined once T is known, and vice versa, (7.95) is an alternative form of the Schrödinger equation. Solving (7.95) for R has some advantages:

⁵Rather than change variables to match those of the delta function, it is often convenient to use the identity $\delta[f(x)] = f'(x)\delta(x)$.

- The Green's operator is real, and therefore R will be real (if the potential is real).
- It is possible to follow the principal value prescription on a computer and convert (7.95) to a matrix equation which can be solved directly (see Chapter 18, *Solving Even Relativistic Integral Equations*).
- The unitarity relation (7.91) for T will be satisfied automatically if the R matrix elements are real (so even an approximate R can produce a unitary T).
- We shall see that if R is Hermitian, T will be unitarity.

The S matrix arises naturally in the time-dependent view of scattering as an operator which transforms the $t = -\infty$ physical state into a $t = +\infty$ one.⁶ Because we are using a time-independent approach, we cannot use that definition; instead, we use the equivalent one of the overlap of $\psi^{(+)}$ (the distorted state containing an outgoing scattered wave) with ψ^- (the distorted state containing an incoming scattered wave, that is, with opposite sign for $i\epsilon$):

$$\langle \phi_f | S | \phi_i \rangle \stackrel{\text{def}}{=} \langle \psi_f^{(-)} | \psi_i^{(+)} \rangle \quad (7.98)$$

$$| \psi_i^{(+)} \rangle = | \phi_i \rangle + (E_i - H_0 + i\epsilon)^{-1} V | \psi_i^{(+)} \rangle \quad (7.99)$$

$$| \psi_f^{(-)} \rangle = | \phi_f \rangle + (E_f - H_0 - i\epsilon)^{-1} V | \psi_f^{(-)} \rangle. \quad (7.100)$$

Because the S operator relates physical states, S -matrix elements are defined only for initial and final states having the same energy, $E_i = E_f$. We leave it to the Problems section to show that

$$\langle \phi_f | S | \phi_i \rangle = \delta(\mathbf{k}_f - \mathbf{k}_i) - 2\pi i \delta(E_f - E_i) \langle \phi_f | V | \psi_i^{(+)} \rangle \quad (7.101)$$

$$= \delta(\mathbf{k}_f - \mathbf{k}_i) - 2\pi i \delta(E_f - E_i) \langle \phi_f | T_i | \phi_i \rangle. \quad (7.102)$$

Exercise Use (7.102) to deduce the relation between S and T :

$$S = 1 - 2\pi i \delta(E - H_0) T. \quad \diamond \quad (7.103)$$

The delta function in (7.103) asserts that only on-shell matrix elements are used (the off-shell elements of T correspond to the non-asymptotic parts of the wave function which are not affected by S).

Because the S operator actually maps one state into another, conservation of probability requires it to be a unitary operator. This also follow directly from (7.103) and the unitarity relation for T :

$$\begin{aligned} S^\dagger S &= [1 + 2\pi i T^\dagger \delta(E - H_0)] [1 - 2\pi i \delta(E - H_0) T] \\ &= 1 - 2\pi i \delta(E - H_0) T + 2\pi i T^\dagger \delta(E - H_0) + 4\pi^2 T^\dagger \delta(E - H_0) \delta(E - H_0) T \\ &= 1 - 2\pi i \delta(E - H_0) [T - T^\dagger + 2\pi i T^\dagger \delta(E - H_0) T] \\ &= 1. \end{aligned} \quad (7.104)$$

⁶Our treatment of the S matrix is sketchy. For fuller treatment, see Newton (1966), Goldberger and Watson (1963), Chew (1962), and Taylor (1972).

7.8 The Two-Potential Formula, Tutorial

Assume you know the solutions of the Schrödinger equation $|\chi_i\rangle$ for a given potential V_0 :

$$(H_0 + V_0) |\chi_i\rangle = E_i |\chi_i\rangle. \quad (7.105)$$

We want to use the *distorted* waves χ_i and the plane waves ϕ_i to obtain solutions of the Schrödinger equation

$$(H_0 + V) |\psi\rangle = E_i |\psi\rangle, \quad (7.106)$$

for a potential V which is the sum of two potentials

$$V = V_0 + V_1. \quad (7.107)$$

1. What are the LS wave equations for χ and ψ ? (Operator form is fine.)
2. Express your equation for ψ in the momentum representation, being sure all functions can be explicitly evaluated.
3. Show that a redefinition of the unperturbed Hamiltonian H_0 to include V_0 leads to the new LS equation

$$|\psi_i^{(+)}\rangle = |\chi_i^{(+)}\rangle + (E - H_0 - V_0 + i\epsilon)^{-1} V_1 |\psi_i^{(+)}\rangle. \quad (7.108)$$

4. Explain in physical terms what is meant by the subscript i and the $(+)$ superscript.
5. Take your expression for χ and use it to show that

$$\langle\phi_f| = \langle\chi_f^{(\pm)}| - \langle\phi_f| V_0^\dagger (E - H_0^\dagger - V_0^\dagger \mp i\epsilon)^{-1}. \quad (7.109)$$

6. Let T_{fi} be the T matrix for the total potential V . Give the definition of T_{fi} in terms of ϕ , ψ and V .
7. Because we do not want to solve for the full wave function ψ , we need to convert this expression for T into one with only ϕ 's and χ 's. Show (or verify) that for Hermitian potentials, T_{fi} can be written as:

$$T_{fi} = \langle\phi_f| V_0 |\chi_i^{(+)}\rangle + \langle\chi_f^{(-)}| V_1 |\psi_i^{(+)}\rangle \stackrel{\text{def}}{=} T_0 + T_1. \quad (7.110)$$

This is the *two-potential formula*.

8. Show that if we ignore the distortion in ψ caused by V_1 , an approximation (the *Distorted Waved Born Approximation*) to the scattering produced by V_1 is

$$T_1 \simeq \langle\chi_f^{(-)}| V_1 |\chi_i^{(+)}\rangle. \quad (7.111)$$

7.9 Problems

1. To help test your understanding of the material in this chapter, try answering the following question without resort to the text.

Given the momentum-space representation of some operator

$$\langle \mathbf{p}' | X | \mathbf{p} \rangle = (2\pi)^3 X(\mathbf{p}) \delta(\mathbf{p}' - \mathbf{p}), \quad (7.112)$$

determine $\langle \mathbf{r}' | X | \mathbf{r} \rangle$. (*Hint:* It may be less confusing to first evaluate $\langle \mathbf{r} | X | \psi \rangle$.)

2. In Chapter 4, *Scattering Applications: Lengths, Resonances, Coulomb*, we gave the form for the scattering amplitude (4.66) when a short range potential acts in addition to the Coulomb one. Use the two-potential formula to derive this separation, making sure to identify the point Coulomb amplitude.
3. Try to solve this problem without resort to the text. Start with the operator form of the Schrödinger equation,

$$(K + V)\psi = E\psi. \quad (7.113)$$

- (a) Derive the Lippmann–Schwinger equation for the wave function,

$$\psi = \phi + GV\psi, \quad (7.114)$$

making sure to define all symbols and explain each step.

- (b) Check that (7.114) is an actual solution of (7.113).
- (c) What is the definition of the scattering T matrix and its elements?
- (d) Prove the following operator relation

$$V\psi = T\phi. \quad (7.115)$$

- (e) Derive, using formal scattering theory, the Lippmann–Schwinger equations for the T matrix

$$T = V + VGT. \quad (7.116)$$

- (f) How would (7.114) change if we wanted to solve the bound state Schrödinger equation at an energy $E = -E_B$? (*Hint:* Explain why we would or would not include a solution to the homogeneous problem.)
- (g) Show that the bound-state condition can be stated in terms of these operators as

$$\det(1 - GV) = 0, \quad (7.117)$$

where you may think of the determinant as the usual linear algebra determinant of the matrix representation of the operator. [*Hint:* Make sure to explain the relation of this eigenvalue problem to the existence of non-trivial solutions of (7.114).]

4. We want to model a physical system which can exist in either its ground state 1 or an excited state 2. We take the ground state energy as zero and the excited state energy as e_2 . We now scatter from this system with the system initially in its ground state, but the possibility of the state 2 getting excited.

- (a) Explain why the full wave function can be written as a sum of products of scattering parts and internal parts (similar to the generalization of the Born approximation to bound systems).
- (b) Justify that this system is described by the coupled Schrödinger equations

$$K_1\psi_1 + V_{11}\psi_1 + V_{12}\psi_2 = E\psi_1, \quad (7.118)$$

$$(K_2 + e_2)\psi_2 + V_{21}\psi_1 = E\psi_2. \quad (7.119)$$

- (c) For this problem, what would be the boundary conditions on the coordinate-space wave functions $\psi_1(\mathbf{r})$ and $\psi_2(\mathbf{r})$?
- (d) Show that the algebraic elimination of ψ_2 from the above equations produces an equation for ψ_1 which can be viewed as the scattering from an effective (*optical*) potential. Show (or argue) that this optical potential must in general be energy dependent, nonlocal, and complex.
- (e) Show that if E is less than e_2 , the optical potential must be real.
5. Consider again the preceding, coupled-channels problem. Show that for this problem the Lippmann–Schwinger equation has the form

$$\psi_1 = \phi_1 + G_1 V_{11}\psi_1 + G_1 V_{12}\psi_2, \quad (7.120)$$

$$\psi_2 = G_2 V_{21}\psi_1 + G_2 V_{22}\psi_2. \quad (7.121)$$

6. Prove from their definitions, that S and T are related by

$$\langle \phi_f | S | \phi_i \rangle = \delta(\mathbf{k}_f - \mathbf{k}_i) - 2\pi i \delta(E_f - E_i) \langle \phi_f | T_i | \phi_i \rangle. \quad (7.122)$$

(The proof is very similar to those of orthogonality and unitarity given in the text.)

CHAPTER 8

THE ANGULAR MOMENTUM BASIS

In this chapter we apply the Green's function and operator techniques to the 1D Schrödinger equation. First we develop the Lippmann–Schwinger equation for the wave function in the partial-wave basis and then the Born series for the partial-wave amplitudes. The approach is quite similar to Chapter 5, *Green's Functions: Integral Quantum Mechanics*, except that we use the eigenfunction expansion as opposed to the Fourier integral to obtain the Green's function. In the second part of this chapter we develop the energy-angular momentum basis $|klm\rangle$ or l , and project the formal equations onto this basis. These results, and other representations, are summarized in Appendix B, *Dirac Notation and Representations*. There are lots of equations in this chapter, and those readers interested in the logical development of the theory may care to skip this chapter and use it and Appendix B for reference and problem solving (the Problems section do have practical importance). The related numerical techniques for solving the integral equations are discussed in Chapter 18, *Solving Even Relativistic Integral Equations*.

8.1 Partial-Wave Green's Function

To use Green's function techniques with the one-dimensional Schrödinger equation, we start with some basic ingredients. There is the 1D differential operator $d_l(r)$, the radial wave function $u_l(kr)$, the radial free wave $w_l(r)$, and the 1D Green's function $g_l(r, r'; E)$. They have the properties:

$$d_l(r) = \frac{1}{2\mu} \frac{d^2}{dr^2} + E - \frac{l(l+1)}{2\mu r^2}, \quad (8.1)$$

$$d_l(r)u_l(kr) = V(r)u_l(kr), \quad (8.2)$$

$$d_l(r)w_l(r) = 0, \quad (8.3)$$

$$d_l(r)g_l(r, r'; E) = \delta(r - r'). \quad (8.4)$$

The 1D LS equation for a local potential is, accordingly:

$$u_l(kr) = w_l(r) + \int_0^\infty dr' g_l(r, r'; E) V(r') u_l(kr'). \quad (8.5)$$

There are several ways to determine the Green's function g_l . One is to expand G_B in partial waves (see the Problems section), another is to use a Fourier integral, and a third (and the one we use) is to deduce g_l directly from the definition

$$d_l(r)g_l(r, r'; E) = \delta(r - r'). \quad (8.6)$$

Because the RHS of (8.6) equals zero at all points except $r = r'$, at all other points g_l must be a linear combination of two independent solutions of the free wave equation, and we take them to be any of the four free solutions $F_l(kr)$, $G_l(kr)$, $H_l^{(+)}(kr)$, and $H_l^{(-)}(kr)$. Unless $r' = 0$, the function $g_l(r, r')$ must be regular at $r = 0$, namely, proportional to $F_l(kr)$. If $r > r'$ we are in the outer region and g_l should be an outgoing wave, namely $H_l^{(+)}$. Putting these facts together means

$$g_l(r, r') = \begin{cases} C F_l(kr), & \text{for } r < r', \\ D H_l^{(+)}(kr), & \text{for } r > r'. \end{cases} \quad (8.7)$$

Yet because free space is isotropic, a wave propagates in the same way from r to r' as does one from r' to r , and so we must have $g_l(r, r') = g_l(r', r)$. This means

$$g_l(r, r') = \begin{cases} A H_l^{(+)}(kr') F_l(kr), & \text{for } r < r', \\ A H_l^{(+)}(kr) F_l(kr'), & \text{for } r > r'. \end{cases} \quad (8.8)$$

To determine the constant A , we integrate (8.4) over the $r = r'$ singular point,

$$\frac{1}{2\mu} \int_{r'-\epsilon}^{r'+\epsilon} \left(\frac{d^2 g_l}{dr^2} + k^2 g_l - \frac{l(l+1)}{r^2} g_l \right) dr = 1, \quad (8.9)$$

$$\Rightarrow \frac{1}{2\mu} \frac{dg_l}{dr} \Big|_{r'-\epsilon}^{r'+\epsilon} = 1. \quad (8.10)$$

Upon substituting (8.8) into (8.10) we find

$$\frac{A}{2\mu} \left[\frac{dH_l^{(+)}(kr')}{dr} F_l(kr') - H_l^{(+)}(kr') \frac{dF_l(kr)}{dr} \right]_{r=r'} = 1. \quad (8.11)$$

The term in brackets is the *Wronskian* and is known from the theory of differential equations to be a constant, independent of r .

Exercise Verify that the value of the Wronskian is:

$$\frac{dH_l^{(+)}(kr)}{dr} F_l(kr) - H_l^{(+)}(kr) \frac{dF_l(kr)}{dr} = -k. \quad (8.12)$$

(Hint: Evaluate it at $r \simeq \infty$ where you know the asymptotic forms.) $\diamond$

We thereby know that $A = -2\mu/k$, and so we write (8.8) in the compact form

$$g_l^{(+)}(r, r'; E) = \frac{-2\mu}{k} F_l(kr_{<}) H_l^{(+)}(kr_{>}), \quad (8.13)$$

where $r_{>}$ ($r_{<}$) is the larger (smaller) of r and r' .

8.2 The Radial Wave Function

The integral form of the Schrödinger equation with outgoing-wave boundary conditions is obtained by substituting the outgoing Green's function into the general solution (8.5):

$$u_l(kr) = w_l(r) + \int_0^\infty dr' g_l(r, r'; E) V(r') u_l(r') \quad (8.14)$$

$$= F_l(kr) - \frac{2\mu}{k} \int_0^\infty dr' F_l(kr_<) H_l^{(+)}(kr_>) V(r') u_l(kr'). \quad (8.15)$$

This equation makes it clear that $u_l(kr)$ is regular at the origin and behaves like the outgoing wave $H_l^{(+)}$ as $r \rightarrow \infty$. Yet we also know from (3.16) that

$$u_l(r > R) = e^{i\delta_l} [\sin \delta_l G_l(kr) + \cos \delta_l F_l(kr)]. \quad (8.16)$$

Exercise Substitute $H_l^{(+)} = G_l + iF_l$ into (8.15), compare factors multiplying G_l , and thereby derive the integral expression for the partial-wave scattering amplitude:

$$f_l(E) = e^{i\delta_l} \sin \delta_l = \frac{-2\mu}{k} \int_0^\infty dr' F_l(kr') V(r') u_l(kr'). \quad \diamond \quad (8.17)$$

With this expression it is easy to derive the scattering length approximation for low-energy scattering.

Exercise Show that (8.17) has the limit $f_l(E \rightarrow 0) \propto k^{2l+1}$. ◇

8.3 The T Matrix

We determine the integral equation for the T matrix in the partial-wave basis either by starting with the 3D LS equation (6.21) for T :

$$T_E(\mathbf{k}', \mathbf{k}) = V(\mathbf{k}', \mathbf{k}) + \int d^3p V(\mathbf{k}', \mathbf{p}) G_E^{(+)}(p) T_E(\mathbf{p}, \mathbf{k}), \quad (8.18)$$

and expanding all matrix elements in series of Legendre polynomials,¹ or by starting with the formal operator equation,

$$T_E = V + V G_E^{(+)} T_E, \quad (8.19)$$

and projecting it onto energy-angular momentum basis. We will follow the first path.

We assume we know the double Fourier transform,

$$\langle \mathbf{k}' | V | \mathbf{k} \rangle = \int d^3r' d^3r \langle \mathbf{k} | \mathbf{r}' \rangle \langle \mathbf{r}' | V | \mathbf{r} \rangle \langle \mathbf{r} | \mathbf{k} \rangle, \quad (8.20)$$

¹Differences in normalizations for partial-wave T matrices abound. Ours is related to that of Goldberger and Watson (1963) via $T_l = (\pi/2)T_l(G\&W)$, and is the same as LPOTT, Landau (1982). For other references it is best to search for the expression for T_l in terms of phase shifts and work back from there.

and define identical forms for the partial-wave expansions of T and V :

$$V(\mathbf{k}', \mathbf{k}) = \frac{1}{2\pi^2} \sum_{l=0}^{\infty} (2l+1) V_l(k', k) P_l(\cos \theta_{\mathbf{k}\mathbf{k}'}) \quad (8.21)$$

$$= \frac{2}{\pi} \sum_{l=0}^{\infty} \sum_{m=-l}^l V_l(k', k) Y_{lm}^*(\Omega_{\mathbf{k}'}) Y_{lm}(\Omega_{\mathbf{k}}), \quad (8.22)$$

$$T_E(\mathbf{k}', \mathbf{k}) = \frac{1}{2\pi^2} \sum_{l=0}^{\infty} (2l+1) T_l(k', k; E) P_l(\cos \theta_{\mathbf{k}\mathbf{k}'}) \quad (8.23)$$

$$= \frac{2}{\pi} \sum_{l=0}^{\infty} \sum_{m=-l}^l T_l(k', k; E) Y_{lm}^*(\Omega_{\mathbf{k}'}) Y_{lm}(\Omega_{\mathbf{k}}). \quad (8.24)$$

For a spherically symmetric potential we have axial symmetry and so $T_E(\mathbf{k}', \mathbf{k})$ and $V(\mathbf{k}', \mathbf{k})$ will be independent of ϕ . In this case the expansion in terms of Legendre polynomials can be used. For a non-central potential there will be a dependence on ϕ and the spherical-harmonic expansion is required. Note that because $\cos \theta_{\mathbf{k}\mathbf{k}'}$ depends only on the angle between $\mathbf{k}$ and $\mathbf{k}'$, the arguments the Y_{lm} 's can be reversed:

$$T_E(\mathbf{k}', \mathbf{k}) = \frac{2}{\pi} \sum_{l=0}^{\infty} \sum_{m=-l}^l T_l(k', k; E) Y_{lm}^*(\Omega_{\mathbf{k}}) Y_{lm}(\Omega_{\mathbf{k}'}). \quad (8.25)$$

Because the on-energy-shell T matrix is proportional to the scattering amplitude (6.5):

$$f_E(\theta, \phi) = -4\pi^2 \mu T_E(\mathbf{k}', \mathbf{k})|_{\mathbf{k}'=\mathbf{k}} = -4\pi^2 \mu \langle \phi_{\mathbf{k}'} | V | \psi_{\mathbf{k}}^{(+)} \rangle, \quad (8.26)$$

and because we know the expansion of f_E , (3.33), we can make the identification:

$$T_l(k, k; E_k) = \frac{f_l(E_k)}{-\rho_T} = \frac{e^{i\delta_l} \sin \delta_l}{-\rho_T}, \quad (8.27)$$

$$\rho_T = 2\mu k. \quad (8.28)$$

Here ρ_T is a density-of-states factor which depends on our choice for the wave function normalizations and μ is the reduced mass (or its relativistic generalization given in Chapter 18, *Solving Even Relativistic Integral Equations*). Now we place these expansions for T and V into the 3D Lippmann–Schwinger equation (8.18) and use the orthogonality conditions:

$$\int_{-1}^1 dx P_{l'}(x) P_l(x) = \frac{2}{2l+1} \delta_{l'l}, \quad (8.29)$$

$$\int d\Omega Y_{lm}^*(\Omega) Y_{l'm'}(\Omega) = \delta_{ll'} \delta_{mm'}, \quad (8.30)$$

to obtain the desired 1D form of the LS equation:

$$T_l(k', k; E) = V_l(k', k) + \frac{2}{\pi} \int_0^{\infty} \frac{dp p^2 V_l(k', p) T_l(p, k; E)}{E - E_p + i\epsilon}. \quad (8.31)$$

Exercise Verify the steps leading to (8.31). ◇

8.4 Energy-Angular Momentum Basis

The preceding manipulations have taken the abstract form of the Schrödinger equation $T = V + VGT$ and projected it onto the energy-angular momentum basis $|klm\rangle$. We now examine the use of this basis in more detail.

Completeness Relation

A comparison of (8.31) with its abstract form (8.19) infers the completeness relation:

$$\tilde{1} = \frac{2}{\pi} \sum_{l=0}^{\infty} \sum_{m=-l}^{+l} \int_0^{\infty} dk k^2 |klm\rangle \langle klm| \quad (8.32)$$

$$= \frac{2\mu}{\pi} \sum_{l,m=-l}^{\infty,+l} \int_0^{\infty} dE k |klm\rangle \langle klm|. \quad (8.33)$$

Exercise Derive (8.33) and its generalization for relativistic energy $E_k = \sqrt{m^2 + k^2}$. $\diamond$

The $|\mathbf{k}\rangle$ Expansion

Because the $|klm\rangle$ basis is often used to reduce the Schrödinger equation, we already know much about it. For example, we know the plane wave expansion:

$$\langle \mathbf{r} | \mathbf{k} \rangle \stackrel{\text{def}}{=} \langle \mathbf{r} | \phi_{\mathbf{k}} \rangle \stackrel{\text{def}}{=} \frac{e^{i\mathbf{k}\cdot\mathbf{r}}}{(2\pi)^{3/2}} \quad (8.34)$$

$$= \sum_{l=0}^{\infty} \frac{(2l+1)i^l [F_l(kr)/kr] P_l(\cos \theta_{kr})}{(2\pi)^{3/2}} \quad (8.35)$$

$$= \sum_{l,m} \frac{4\pi i^l [F_l(kr)/kr] Y_{lm}^*(\Omega_{\mathbf{k}}) Y_{lm}(\Omega_{\mathbf{r}})}{(2\pi)^{3/2}}. \quad (8.36)$$

From this we may deduce:

$$Y_{lm}(\Omega_{\mathbf{r}}) \propto \langle \mathbf{r} | klm \rangle, \quad (8.37)$$

$$Y_{lm}^*(\Omega_{\mathbf{k}}) \propto \langle \mathbf{k} | klm \rangle. \quad (8.38)$$

We eliminate the bra $\langle \mathbf{r} |$ and deduce that $|\mathbf{k}\rangle$ is a sum of Y_{lm} 's for its angular part and $|klm\rangle$'s for its energy-angular momentum part,

$$|\mathbf{k}\rangle = \sum_{lm} C Y_{lm}^*(\Omega_{\mathbf{k}}) |klm\rangle. \quad (8.39)$$

The normalization constant C is determined by comparing the unit operator in terms of momentum states,

$$\tilde{1} = \int d^3k |\mathbf{k}\rangle \langle \mathbf{k}| \quad (8.40)$$

$$= \int_0^{\infty} dk k^2 \sum_{l,m} C^2 |klm\rangle \langle klm|, \quad (8.41)$$

to the expansion (8.32) of $\tilde{I}$ in $|klm\rangle$ states

$$\Rightarrow |\mathbf{k}\rangle = \sqrt{\frac{2}{\pi}} \sum_{l,m} Y_{lm}^*(\Omega_{\mathbf{k}}) |klm\rangle. \quad (8.42)$$

Thus the momentum state $|\mathbf{k}\rangle$ is a sum of infinitely many angular momentum states, $|klm\rangle$'s.

Normalization

The normalization of the momentum state $|\mathbf{k}\rangle$ determines the normalization of $|klm\rangle$:

$$\langle \mathbf{k}' | \mathbf{k} \rangle = \delta(\mathbf{k}' - \mathbf{k}) = \frac{\delta(k' - k)}{k^2} \sum_{l,m} Y_{lm}^*(\Omega_{\mathbf{k}'}) Y_{lm}(\Omega_{\mathbf{k}}), \quad (8.43)$$

$$\langle \mathbf{k}' | \mathbf{k} \rangle = \frac{2}{\pi} \sum_{l,m,l',m'} Y_{lm}^*(\Omega_{\mathbf{k}}) Y_{l'm'}(\Omega_{\mathbf{k}'}) \langle k'l'm' | klm \rangle, \quad (8.44)$$

$$\Rightarrow \langle k'l'm' | klm \rangle = \frac{\pi \delta(k' - k)}{2 k^2} \delta_{ll'} \delta_{mm'}. \quad (8.45)$$

Momentum Space Wave Function

Because the $\mathbf{r}$ representation of $|\mathbf{k}\rangle$ must be the plane wave expansion (8.35), we also obtain the $\mathbf{r}$ representation of the energy-angular momentum states:

$$\langle \mathbf{r} | klm \rangle \equiv \langle \mathbf{r} | \phi_{klm} \rangle = i^l \frac{F_l(kr)}{kr} Y_{lm}(\Omega_{\mathbf{r}}), \quad (8.46)$$

which is what we started with in Chapter 3, *Partial-Wave Functions and Expansions*! Analogously, the distorted wave has the same expansion form,

$$\langle \mathbf{r} | \psi_{\mathbf{k}} \rangle \equiv \psi_{\mathbf{k}}(\mathbf{r}) = \sum_{l=0}^{\infty} \frac{(2l+1)i^l [u_l(r, E_k)/kr] P_l(\cos \theta_{\mathbf{k}\mathbf{r}})}{(2\pi)^{3/2}} \quad (8.47)$$

$$= \sum_{l,m} \frac{4\pi i^l [u_l(r, E_k)/kr] Y_{lm}(\Omega_{\mathbf{r}}) Y_{lm}^*(\Omega_{\mathbf{k}})}{(2\pi)^{3/2}}. \quad (8.48)$$

The $\mathbf{r}$ representation of the distorted waves is consequently

$$\langle \mathbf{r} | \psi_{klm} \rangle = i^l \frac{u_l(r, E_k)}{kr} Y_{lm}(\Omega_{\mathbf{r}}). \quad (8.49)$$

The p -Space Wave Functions

It is intuitively reasonable that the vector nature of angular momentum is described by the spherical harmonic $Y_{lm}(\Omega_{\mathbf{k}})$. To get the factors right, we examine the expansion of the full wave $\psi_{\mathbf{k}}$:

$$\langle \mathbf{r} | \psi_{\mathbf{k}} \rangle = \sum_{l,m} 4\pi i^l \frac{u_l(r, E_k)}{kr} Y_{lm}^*(\Omega_{\mathbf{r}}) Y_{lm}(\Omega_{\mathbf{k}}). \quad (8.50)$$

To deduce $\langle k'lm | \psi_{\mathbf{k}} \rangle$, we expand $\langle \mathbf{r} | \psi_{\mathbf{k}} \rangle$ in the $|k'lm\rangle$ basis [insert the completeness relation (8.32) between the $\langle \mathbf{r} |$ and $|\psi\rangle$]:

$$\langle \mathbf{r} | 1 | \psi_{\mathbf{k}} \rangle = \langle \mathbf{r} | \frac{2}{\pi} \sum_{l,m} \int_0^\infty dp p^2 |plm\rangle \langle plm | \psi_{\mathbf{k}} \rangle \quad (8.51)$$

$$\begin{aligned} &= \frac{2}{\pi} \sum_{l,m} \int_0^\infty dp p^2 \langle \mathbf{r} | plm \rangle \langle plm | \psi_{\mathbf{k}} \rangle \\ &= \frac{2}{\pi} \sum_{l,m} \int_0^\infty dp p^2 i^l \frac{F_l(pr)}{pr} Y_{lm}(\Omega_{\mathbf{r}}) \langle plm | \psi_{\mathbf{k}} \rangle, \end{aligned} \quad (8.52)$$

where we have also substituted (8.49) for the $\mathbf{r}$ representation of a $|k'lm\rangle$. Compared with (8.52), the distorted wave's representation (8.48) permits the identification

$$\int_0^\infty dp p^2 \frac{F_l(pr)}{pr} \langle plm | \psi_{\mathbf{k}} \rangle = \sqrt{\frac{\pi}{2}} \frac{u_l(r; E_{\mathbf{k}})}{kr} Y_{lm}^*(\Omega_{\mathbf{k}}). \quad (8.53)$$

Exercise Extract the $\langle plm | \psi_{\mathbf{k}} \rangle$ from under the integral by multiplying both sides of (8.53) by $F_l(k'r)$, integrating over r , and using orthogonality to obtain

$$\langle k'lm | \psi_{\mathbf{k}} \rangle = \sqrt{\frac{\pi}{2}} \frac{Y_{lm}^*(\Omega_{\mathbf{k}})}{kk'} u_l(k'; E_{\mathbf{k}}), \quad (8.54)$$

where $u_l(k'; E_{\mathbf{k}})$ is the momentum-space wave function,

$$u_l(k'; E_{\mathbf{k}}) = \int_0^\infty dr \frac{F_l(k'r)}{\pi/2} u_l(kr). \quad \diamond \quad (8.55)$$

Because we are dealing with continuum wave functions, special care is needed in defining their normalizations and Fourier transforms. We see this if we use the preceding formulation to deduce the $|k'lm\rangle$ representation of a momentum eigenstate. We replace the distorted wave $u_l(kr)$ with the free wave $F_l(kr)$ in (8.54) and (8.55):

$$\bar{F}_l(k') = \delta(k' - k), \quad (8.56)$$

$$\langle k'lm | \phi_{\mathbf{k}} \rangle \equiv \langle k'lm | \mathbf{k} \rangle \quad (8.57)$$

$$= \sqrt{\frac{\pi}{2}} \frac{Y_{lm}^*(\Omega_{\mathbf{k}})}{kk'} \delta(k' - k). \quad (8.58)$$

As expected, unless the plane wave $|\phi_{\mathbf{k}}\rangle$ has the same energy as the basis vector $|k'lm\rangle$, the projection unto the basis vector yields zero.

Schrödinger Equation

To obtain the Schrödinger equation in the $|k'lm\rangle$ basis, we start with the abstract form

$$(K + V) |\psi\rangle = E |\psi\rangle, \quad (8.59)$$

and operate on the left with the bra $\langle klm|$. If we insert the completeness relations (8.32) between V and $|\psi\rangle$, we obtain the desired result:

$$\frac{k^2}{2\mu} \langle klm|\psi\rangle + \frac{2}{\pi} \sum_{l',m'} \int_0^\infty dk' k'^2 \langle klm|V|k'l'm'\rangle \langle k'l'm'|\psi\rangle = E_k \langle klm|\psi\rangle. \quad (8.60)$$

Momentum Space Wave Function

Because the plane waves are proportional to delta functions in the $|klm\rangle$ basis, the distorted waves, which contain a plane-wave piece, must also contain delta functions in this basis. We see this by taking the LS equation for the wave function and projecting it unto the $|klm\rangle$ basis:

$$|\psi_k^{(+)}\rangle = |\phi_k\rangle + G_E^{(+)} V |\psi_k^{(+)}\rangle, \quad (8.61)$$

$$\langle k'lm|\psi_k^{(+)}\rangle = \langle k'lm|\phi_k\rangle + \langle k'lm|G_E^{(+)} V |\psi_k^{(+)}\rangle. \quad (8.62)$$

Exercise Use the completeness relation and the Green's function representation

$$\langle k'lm|G_E^{(+)}|k'l'm'\rangle = \frac{\pi}{2} \frac{\delta(k-k')}{k^2} \frac{\delta_{ll'}\delta_{mm'}}{E-E_{k'}+i\epsilon}, \quad (8.63)$$

to derive

$$u_l(k'; E_k) = \delta(k'-k) + \frac{2k'}{\pi(E_k - E_{k'} + i\epsilon)} \int_0^\infty dp p V_l(k', p) u_l(p; E_k). \quad \diamond \quad (8.64)$$

Accordingly, solving for the continuum wave functions in momentum space requires some care since $u_l(k'; E_k)$ contain singularities. The bound-state wave functions contain no momentum states (solutions of the homogeneous equation) and consequently are well behaved.

Exercise Deduce the LS equation for a bound state wave function $u_l(k)$. $\diamond$

V and T Matrix Elements

Given a potential $V(\mathbf{r})$, we must determine its matrix elements $V_l(k', k)$ in order to solve (or approximate) the LS equation. One way is to evaluate the double Fourier transform $V(k', k)$ and then expand this as a series of Legendre polynomials. We derive a direct integral expression starting with the $\mathbf{r}$ -space expansion,

$$\langle \mathbf{r}'|V|\mathbf{r}\rangle = \sum_{l,m} V_l(\mathbf{r}', \mathbf{r}) Y_{lm}^*(\Omega_{\mathbf{r}'}) Y_{lm}(\Omega_{\mathbf{r}}) \quad (8.65)$$

$$= \sum_{l=0}^{\infty} V_l(r', r) \frac{2l+1}{4\pi} P_l(\cos \theta_{r'r}). \quad (8.66)$$

Note, for a *local* potential $V(r)$, the matrix elements for all l 's are equal:

$$V_l(r', r)|_{\text{local}} = \frac{\delta(r' - r)}{r^2} V(r). \quad (8.67)$$

Exercise Verify (8.67). $\diamond$

Exercise Show that the expansion of the local potential is equivalent to the expansion of the delta function

$$\delta(\mathbf{r}' - \mathbf{r}) = \frac{\delta(r' - r)}{r^2} \sum_{l=0}^{\infty} \sum_{m=-l}^{+l} Y_{lm}^*(\Omega_{\mathbf{r}}) Y_{lm}(\Omega_{\mathbf{r}}). \quad \diamond \quad (8.68)$$

To finally deduce the expression for V_l , we evaluate the six-dimensional, double Fourier transform (8.20):

$$\langle \mathbf{k}' | V | \mathbf{k} \rangle = \int d^3 r' d^3 r \langle \mathbf{k}' | \mathbf{r}' \rangle \langle \mathbf{r}' | V | \mathbf{r} \rangle \langle \mathbf{r} | \mathbf{k} \rangle. \quad (8.69)$$

We expand the plane waves, expand the potential, do the integrals over the angle arguments of the spherical harmonics (they are just the orthogonality integrals), compare the resulting coefficient of $Y_{lm}^*(\Omega_{\mathbf{k}'}) Y_{lm}(\Omega_{\mathbf{k}})$ to the expansion (8.22), and thus deduce:

$$\langle \mathbf{k}' l' m' | V | \mathbf{k} l m \rangle = V_l(k', k) \delta_{ll'} \delta_{mm'}, \quad (8.70)$$

$$V_l(k', k) = \frac{1}{kk'} \int_0^{\infty} dr' r' \int_0^{\infty} dr r F_l(k' r') V_l(r', r) F_l(kr). \quad (8.71)$$

The delta functions in (8.70) ensure angular momentum conservation and arise from the rotational invariance of the potential.

Exercise Fill in the steps leading to (8.71). (*Hint:* Note the interchangeability of the arguments of the Y_{lm} 's.) $\diamond$

Exercise Show that for a local potential $V(r)$,

$$V_l(k', k) = \frac{(2\pi)^3}{kk'} \int_0^{\infty} dr F_l(k' r) V(r) F_l(kr). \quad \diamond \quad (8.72)$$

The definition of the T matrix tells us that

$$\langle \phi_{\mathbf{k}'} | T | \phi_{\mathbf{k}} \rangle \stackrel{\text{def}}{=} \langle \phi_{\mathbf{k}'} | V | \psi_{\mathbf{k}}^{(+)} \rangle. \quad (8.73)$$

It is simplest to assume that T and V have the same expansion *form*, and this leads to a slight modification of (8.71) in changing from V to T :

$$\langle \mathbf{k}' l' m' | T | \mathbf{k} l m \rangle = T_l(k', k) \delta_{ll'} \delta_{mm'}, \quad (8.74)$$

$$\langle \mathbf{k}' l' m' | T | \mathbf{k} l m \rangle = \frac{1}{kk'} \int_0^{\infty} dr' r' \int_0^{\infty} dr r F_l(k' r') V_l(r', r) u_l(kr). \quad (8.75)$$

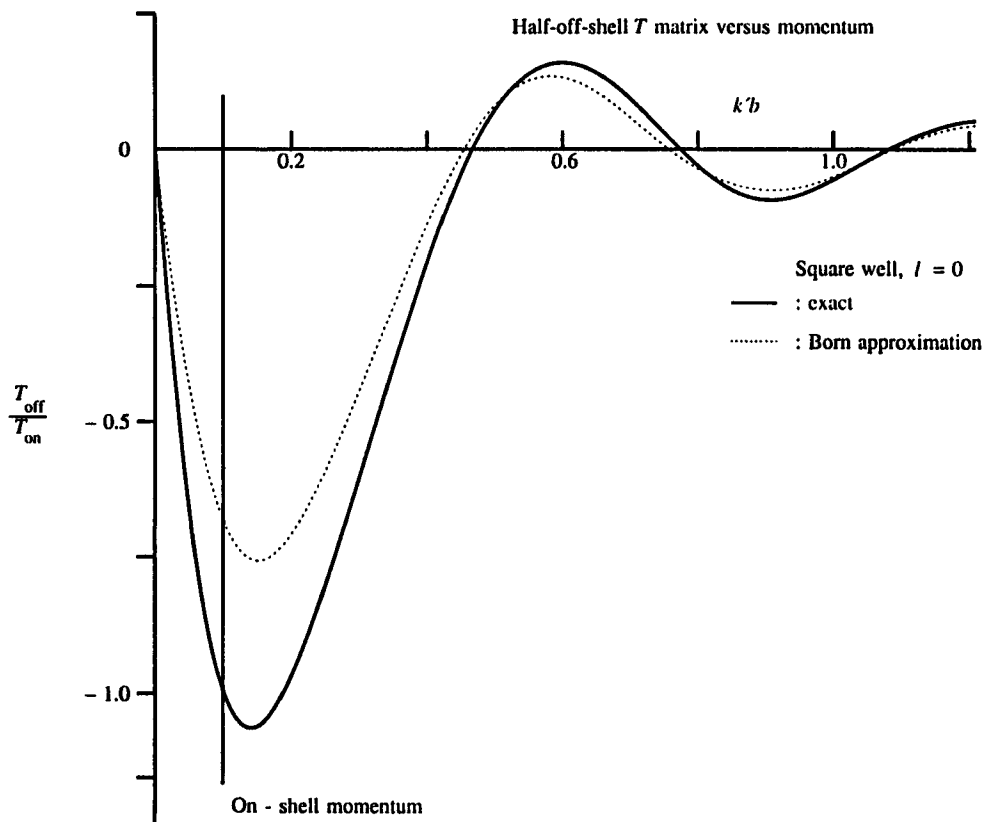


Figure 8.1: The ratio of half-off to on-shell T matrices for a square well of radius b . The solid curve is the exact result, the dashed curve is the first Born approximation, and the vertical line is where k' equals the on-shell momentum k .

Because the $\mathbf{p}$ - and $\mathbf{r}$ -space descriptions of quantum mechanics are equally valid, it is convenient to transform between the two.² We are aided in these transformations by the relations

$$\int_0^\infty d\mathbf{r} F_l(k\mathbf{r}) F_l(k'\mathbf{r}) = \frac{\pi}{2} \delta(k' - k), \quad (8.76)$$

$$\int_0^\infty dk F_l(k\mathbf{r}) F_l(k'\mathbf{r}) = \frac{\pi}{2} \delta(\mathbf{r}' - \mathbf{r}), \quad (F_l = k\mathbf{r} j_l). \quad (8.77)$$

It is easy to show that $V_l(\mathbf{r}', \mathbf{r})$ is the double transform

$$V_l(\mathbf{r}', \mathbf{r}) = \frac{4}{\pi^2 r r'} \int_0^\infty dk' k' \int_0^\infty dk k F_l(k' \mathbf{r}') V_l(k', k) F_l(k \mathbf{r}). \quad (8.78)$$

Example: Off-Shell T for Square Well

A complete solution of the integral equation (8.31) determines T_l for arbitrary values of k , k' , and E . If we have a potential whose wave function is known, then it is possible to determine the T matrix directly from (8.75). The $T_l(k', k; E_k)$ is only half-off-shell because the k dependence is that of the wave function $u_l(r; E_k)$ and consequently k must correspond to the energy. For example, in the Problems section the $l = 0$ wave function for a square well is used to deduce the ratio of off-shell to on-shell elements:

$$\frac{T_0(k', k; E_k)}{T_0(k, k; E_k)} = \frac{k'(k'^2 - \kappa^2)(\kappa \sin k'b \cos \kappa b - k' \cos k'b \sin \kappa b)}{k(k^2 - \kappa^2)(\kappa \sin kb \cos \kappa b - k \cos kb \sin \kappa b)}, \quad (8.79)$$

where b is the well radius and $\kappa^2 = k^2 - U$ is the wave vector within the well. We plot this ratio in Figure 8.1, which shows that as you go off shell, the T matrix oscillates and simultaneously decays to smaller and smaller values. The large number of oscillations is a characteristic of the sharp corners of a square well, whereas the decay is characteristic of the finite range of the interaction.

Born Approximation for T_l

Iteration of the LS equation for T (8.31) produces the Born series,

$$T_l(k', k; E) = V_l(k', k) + \frac{2}{\pi} \int_0^\infty \frac{dp p^2 V(k', p) V_l(p, k)}{E - E_p + i\epsilon} + \dots \quad (8.80)$$

Applying this series is quite simple because the matrix elements of V are given by the Fourier transforms (8.70) and (8.71). However, if the series is slow to converge (or if there is a bound state), a direct solution of the LS equation may be just as easy.

8.5 Optical Theorem

In the last chapter we derived the abstract *unitarity relation* (7.89):

$$T_E - T_E^\dagger = -2\pi i T_E^\dagger \delta[E_k - H_0] T_E. \quad (8.81)$$

²This does not mean, however, that both are equally convenient for a given problem; for example, the bound-state boundary conditions are easier to handle in $\mathbf{r}$ space, while nonlocal potentials are easier to handle in $\mathbf{p}$ space.

Evaluated between on-shell ($k' = k$) angular momentum energy states, this is:

$$\langle klm | T_E | klm \rangle - \langle klm | T_E^\dagger | klm \rangle = -2\pi i \langle klm | T_E^\dagger \delta[E_k - H_0] T_E | klm \rangle, \quad (8.82)$$

$$2i \operatorname{Im} \langle klm | T_E | klm \rangle = -2\pi i \langle klm | T_E^\dagger \delta[E_k - H_0] T_E | klm \rangle. \quad (8.83)$$

We use the completeness relations (8.32) and (8.33) between operators, integrate over E to evaluate the delta function, and obtain:

$$\begin{aligned} \langle klm | T_E^\dagger \delta(E_k - H_0) T_E | klm \rangle &= \\ \sum_{l'm'l''m''} \int_0^\infty \frac{2\mu dE_{p'} p'}{\pi} \frac{2}{\pi} dp p^2 \langle klm | T_E^\dagger | p'l'm' \rangle \langle p'l'm' | \delta(E_k - H_0) | p'l''m'' \rangle \\ &\quad \times \langle p'l''m'' | T_E | klm \rangle \\ &= \int_0^\infty \frac{2\mu dE_{p'} p'}{\pi} \int_0^\infty \frac{2}{\pi} dp p^2 \langle klm | T_E^\dagger | p'l'm \rangle \delta(E_k - E_{p'}) \langle p'l'm | p'l'm \rangle \\ &\quad \times \langle p'l'm | T_E | klm \rangle \\ &= \int_0^\infty \frac{2\mu dE_{p'} p'}{\pi} \frac{2}{\pi} \int_0^\infty dp p^2 \langle klm | T_E^\dagger | p'l'm \rangle \frac{\delta(E_k - E_{p'})}{2p^2} \pi \delta(p' - p) \\ &\quad \times \langle p'l'm | T_E | klm \rangle \end{aligned} \quad (8.84)$$

$$= \frac{-2\mu k |T_l(k, k; E)|^2}{\pi}, \quad (8.85)$$

$$\Rightarrow \operatorname{Im} T_l(k, k; E) = -2\mu k |T_l(k, k; E)|^2 = -|T_l(k, k; E)|^2 \rho_T, \quad (8.86)$$

where ρ_T is defined in (8.28). Equation (8.86) is the on-energy-shell unitarity relation in the partial-wave basis, that is, the constraint on T arising from probability conservation. The only way to satisfy this constraint is for T_l to have the familiar form (8.27) in terms of phase shifts:

$$T_l(k, k; E_k) = \frac{e^{i\delta_l} \sin \delta_l}{-2\mu k}. \quad (8.87)$$

8.6 On-Shell R_l and T_l

In the last chapter we introduced the reaction matrix R as the operator which satisfies the LS equation with the principal-value prescription (T employs the $E + i\epsilon$ prescription), that is, standing-wave boundary conditions in contrast to outgoing scattered waves. The mathematical manipulations of this chapter remain unchanged when treating the R matrix, and the corresponding LS equation is

$$R_l(k', k; E) = V_l(k', k) + \frac{2}{\pi} \mathcal{P} \int_0^\infty \frac{dp p^2 V_l(k', p) R_l(p, k; E)}{E - E_p}. \quad (8.88)$$

Solving for R_l is as good as knowing T_l because

$$R_l[k, k; E_k] = -\frac{\tan \delta_l}{\rho_T}. \quad (8.89)$$

Exercise Prove (8.89) by starting with (7.97),

$$R_E = T_E + i\pi T_E \delta(E - H_0) R_E, \quad (8.90)$$

and following similar steps as used in deducing the partial-wave unitarity relation (8.86) to deduce

$$R_l(k, k; E) = T_l(k, k; E) + i\rho_T R_l(k, k; E) T_l(k, k; E), \quad (8.91)$$

$$T_l(k, k; E) = \frac{R_l(k, k; E)}{1 + i\rho_T R_l(k, k; E)}, \quad (8.92)$$

$$R_l(k, k; E) = \frac{T_l(k, k; E)}{1 - i\rho_T R_l(k, k; E)}. \quad \diamond \quad (8.93)$$

Exercise Show that (8.93) implies (8.89). $\diamond$

Exercise Verify that a real R_l produces an Hermitian T_l . $\diamond$

Finally, we deduce the partial-wave matrix elements of the the S matrix (7.103):

$$S = 1 - 2\pi i \delta(E - H_0), \quad (8.94)$$

$$\Rightarrow \langle k'l'm' | S | klm \rangle = \frac{\pi \delta(k' - k) \delta_{ll'} \delta_{mm'}}{2k^2} S_l(k), \quad (8.95)$$

$$S_l(k) = e^{2i\delta_l}. \quad (8.96)$$

This means that S_l is defined only on the energy shell, that S_l is unitary ($|S_l| = 1$), and that S_l is related to the on-shell T_l element via

$$S_l(k) = 1 - 2i\rho_T T_l(k, k; E_k) = e^{2i\delta_l}. \quad (8.97)$$

8.7 Born Series for Wave Function

We obtain the Born series for the wave function $u_l(kr)$ simply by iterating (8.15):

$$u_l(kr)|_B^1 = F_l(kr), \quad (8.98)$$

$$\begin{aligned} u_l(kr)|_B^2 &= F_l(kr) - \frac{2\mu}{k} \int_0^\infty dr' F_l(kr_{<}) H_l^{(+)}(kr_{>}) V(r') u_l(kr')|_B^1 \\ &= F_l(kr) - \frac{2\mu}{k} \int_0^\infty dr' F_l(kr_{<}) H_l^{(+)}(kr_{>}) V(r') F_l(kr'), \end{aligned} \quad (8.99)$$

$$\begin{aligned} u_l(kr)|_B^3 &= F_l(kr) - \int_0^\infty dr' g_l(r, r'; E) V(r') u_l(kr')|_B^2 \\ &= F_l(kr) - \int_0^\infty dr' g_l(r, r'; E) V(r') F_l(kr') \\ &\quad + \int_0^\infty dr' \int_0^\infty dr'' g_l(r, r''; E) V(r'') g_l(r'', r'; E) V(r') F_l(kr') + \dots \end{aligned} \quad (8.100)$$

where we keep the expressions compact by considering a local potential. Likewise, by substituting these expansions for u_l into the integral expression for f_l , (8.17), we obtain the Born series for the partial-wave scattering amplitude:

$$\begin{aligned} f_l(E)|_B^1 &= \frac{-2\mu}{k} \int_0^\infty dr' F_l(kr') V(r') u_l(kr')|_B^1 \\ &= \frac{-2\mu}{k} \int_0^\infty dr' F_l(kr') V(r') F_l(kr'), \end{aligned} \quad (8.101)$$

$$f_l(E)|_B^2 = \frac{-2\mu}{k} \int_0^\infty dr' F_l(kr') V(r') u_l(kr')|_B^2 \quad (8.102)$$

$$\begin{aligned} &= f_l(E)|_B^1 - \frac{2\mu}{k} \int_0^\infty dr' F_l(kr') V(r') \int_0^\infty dr g_l(r', r'') V(r'') F_l(kr) + \dots \\ &\quad (8.103) \end{aligned}$$

Although the higher-order terms look very complicated here, they are just a coordinate-space representation of the abstract series $T = V + VGV + \dots$, and $\psi = \phi + GV\phi + \dots$.

8.8 Problems

- To help test your understanding of the material in this chapter, try answering the following questions without resort to the text.
 - What is the form of the Green's function $g_l(r', r)$ for the radial Schrödinger equation? (You can ignore proportionality constants.)
 - What boundary conditions are contained in this $g_l(r', r)$?
 - What must be the $r \rightarrow 0$ limit of the Green's function?
- A square well of range R and depth V_0 scatters particles of energy $E = k_0^2/2m$.
 - What is the on-energy-shell S-wave T matrix $T_{l=0}(k_0, k_0)$?
 - What is the half off-energy-shell T matrix $T_0(k_0, k)$?
 - What are the Born approximation results for parts a and b?
 - Do you expect the oscillatory behavior of T to be a general feature of off-shell T matrices? Explain.
- If an interaction is confined to a very small region of space, its scattering can be approximated by that of a *delta-shell potential*:

$$V(r) = \frac{1}{2\mu} \lambda \delta(r - b). \quad (8.104)$$

- Is this potential of zero or finite range? Explain.
- Deduce which partial waves you expect to be scattered.
- If $b = 0$ how would the previous two answers change?
- What is the scattering amplitude in first Born approximation?
- What is the radial wave function?

- (f) Use the radial wave function and the definition of the T matrix to solve for the (on-energy-shell) scattering amplitude and thus show that

$$e^{i\delta_l} \sin \delta_l = \frac{-CF_l^2(kb)}{1 + iCF_l(kb)H_l^{(+)}(kb)}, \quad C = -\lambda/k. \quad (8.105)$$

4. Again consider the delta shell potential. Use the partial-wave LS equation to determine the off-shell $T_0(k', k)$.
5. Use the results just derived for the delta-shell potential.
 - (a) Determine the $l = 1$ and $l = 2$ phase shifts δ_1 and δ_2 .
 - (b) Plot δ_1 and δ_2 as functions of kb for the two cases:
 - i. $\lambda b = 2\pi$.
 - ii. $\lambda b = \pi/2$.

In both cases choose the modulus of δ_l such that it goes to zero at infinite energy.

6. Consider again the delta-shell potential.
 - (a) Determine the necessary conditions for this potential to have an $l = 0$ bound state.
 - (b) What is the scattering length a_0 for this potential?
 - (c) Determine the conditions for an $l = 0$ resonance to occur.
7. Use the r representation $u_l(kr)$ to develop the Born series.
8. Evaluate the LS equation, $T = V + VGT$, in the angular momentum basis.
9. Consider the delta-function separable potential:

$$V(\mathbf{r}', \mathbf{r}) = -4\pi\lambda\delta(r' - a)\delta(r - a). \quad (8.106)$$

- (a) What is $V_l(k', k)$ for this potential?
 - (b) What is $T_l(k', k; E)$ for this potential?
 - (c) Why is the preceding solution valid for all values of k' , k , and E ?
 - (d) Show that the T matrix, considered as a function of the complex energy E , is discontinuous along the real positive energy axis (the unitarity or right-hand cut).
10. Prove that the 3D and 1D Green's functions are related by the partial-wave decomposition:

$$G_E(\mathbf{r}, \mathbf{r}') = \frac{1}{4\pi r r'} \sum_{l=0}^{\infty} (2l+1) g_l(r, r'; E) P_l(\cos \theta). \quad (8.107)$$

CHAPTER 9

SPIN THEORY

Extending quantum mechanics to include spin is sometimes subtle and sometimes hard. From a fundamental viewpoint it is important to see the inclusion of spin into quantum mechanics in order to understand relativity. From a practical viewpoint it is important to understand how to solve the Schrödinger equation for spin-dependent potentials and to understand the *phenomenology* of how spin affects experimental observables. In this chapter we discuss the meaning of spin and how it is included in the Schrödinger equation. In the next chapter we discuss the phenomenology of spin and the relation to experimental observables. In Chapter 16, *Interactions in Dirac Theory*, we apply identical mathematics for the Dirac equation, and in Chapter 17, *Integral Forms of the Dirac Equation, Scattering*, identical phenomenology for Dirac equation.

To remain within the scope of this book we limit ourselves to the spin $0 \times \frac{1}{2}$ interaction, leaving the treatment of general spins to references such as Goldberger and Watson (1963), Gottfried (1966), Rodberg and Thaler (1967), McGregor et al. (1960), and Moravscik (1963). Also left to sources such as Thompson (1994), Edmonds (1957), Gottfried (1966), Lipkin (1966), and Rose (1957) are the group-theoretic treatment of rotations and its connection to spin.

9.1 Basics

Definitions

When we say a particle or system of particles has “spin,” we mean it has a quantized internal structure which can be described by an angular-momentum quantum number. The extra quantum number needed to describe the internal structure implies extra states which ultimately show up as multiplets in energy-level diagrams.¹ It is intuitively appealing to think that the internal structure we call spin arises from a physical extension in space, as occurs if constituents orbit within a *composite* particle. While true for some situations, that reasoning is classical yet spin is quantum-mechanical; *point* particles like electrons and

¹ Doublets correspond to spin $\frac{1}{2}$; triplets to spin 1, quadruplets to spin $\frac{3}{2}$, etc..

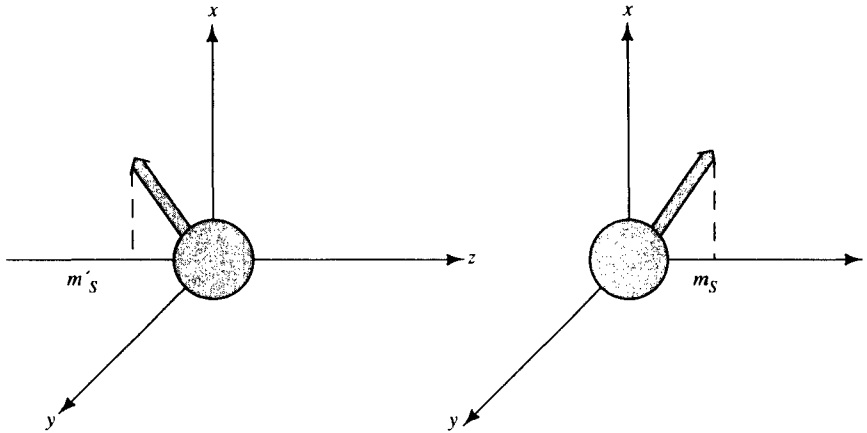


Figure 9.1: Rotating the internal space of the particle on the left changes the spin's projection along the z axis. An additional rotation of the z axis is required to obtain a configuration physically equivalent to the initial one.

neutrinos have spin but no size and thus cannot be “spinning”. Even in the *rest* frame of a particle with mass, orbital motion may cease but there is no way to “stop” the particle’s spin. Spin can be reoriented such that its projection on some axis changes, but if we “change” the spin of a particle we change the internal structure of the particle and thereby have a different particle.

Even though a particle may not be “spinning,” there is a basic connection between its spin and how one rotates the particle. To have our theory preserve the isotropy of space, when we rotate a particle we must rotate both its external and internal spaces (as suggested in Figure 9.1). Because the orbital angular momentum $\mathbf{l}$ is the generator of infinitesimal rotations for the external space, the need for a complete rotation means that we must add the spin $\mathbf{s}$ to $\mathbf{l}$ to obtain the generator $\mathbf{j} = \mathbf{l} + \mathbf{s}$ of total rotations.²

Mathematical Description

We now state these concepts mathematically. Because spin generates rotations (albeit internal ones), it obeys angular momentum commutation relations,

$$[s_j, s_k] = i\epsilon_{jkl}s_l, \quad (9.1)$$

where ϵ_{jkl} is the totally antisymmetric tensor. Because a total rotation rotates both internal plus external spaces, the generator of total rotations $\mathbf{j}$ must be the sum:

$$\mathbf{j} = \mathbf{l} + \mathbf{s}. \quad (9.2)$$

Because there is no preferred direction in space (isotropy), the Hamiltonian H is invariant under rotations, and the generator of rotations is a *symmetry operator*, that is,

$$[H, \mathbf{j}] = 0. \quad (9.3)$$

²Gottfried (1966), Merzbacher (1970), Thompson (1994).

Equation (9.3) implies that j is a conserved quantity and so the potentials and transition matrices must conserve it:

$$\left\langle j\alpha \left| \begin{pmatrix} V \\ T \end{pmatrix} \right| j'\beta \right\rangle = \delta_{jj'} \left\langle j\alpha \left| \begin{pmatrix} V \\ T \end{pmatrix} \right| j\beta \right\rangle. \quad (9.4)$$

Spin Space

To include spin in our theory, we extend the Hilbert space in which we work to a *direct product* of an external space with an internal one:

$$|\psi\rangle = |\psi_{\text{space}}\rangle \times |\chi_{\text{spin}}\rangle \equiv |\psi_{\text{space}}\rangle |\chi_{\text{spin}}\rangle \equiv |\psi\chi\rangle. \quad (9.5)$$

The independence of the two spaces requires

$$[\mathbf{s}, \mathbf{l}] = 0, \quad (9.6)$$

and this means that spin operators do not act on the external space (and vice versa).

The ket $|\chi\rangle$ in (9.5) is the spin-space part of the state vector. It cannot be represented as a continuous function, but instead by $2s + 1$ discrete components.³ To span a spin space of dimension $2s + 1$ requires $2s + 1$ independent basis vectors. We denote these basis vectors by $|sm_s\rangle$, with m_s having all values from s to $-s$ in steps of 1: $m_s = \{s, s-1, \dots, -s\}$. These vectors are eigenstates of the spin operators:

$$\mathbf{s} \cdot \mathbf{s} |sm_s\rangle = s(s+1) |sm_s\rangle \neq s^2 |sm_s\rangle, \quad (9.7)$$

$$s_z |sm_s\rangle = m_s |sm_s\rangle. \quad (9.8)$$

The preceding kets are in an abstract space. An *explicit representation* (in the group-theory sense) is given by the set of $2s + 1$ numbers (each 0 or 1):

$$\langle \sigma | sm_s \rangle \stackrel{\text{def}}{=} \chi_{sm_s}(\sigma) = \begin{cases} 1, & \text{for } \sigma = m_s, \\ 0, & \text{for } \sigma \neq m_s. \end{cases} \quad (9.9)$$

Once we have basis vectors, we can expand an arbitrary spin-state vector $|\chi_s\rangle$ as a linear combination of them:⁴

$$|\chi_s\rangle = \sum_{m_s=-s}^s \alpha_{m_s} |sm_s\rangle, \quad (9.10)$$

where α_{m_s} is a number which give the projection of $|\chi_s\rangle$ onto (overlap with) the basis vector $|sm_s\rangle$. Whereas the number of components $2s + 1$ is finite, the number of $|\chi_s\rangle$'s which can be formed by (9.10) is infinite. The common *spin vector* is a representation of the state vector (9.10) in column vector form:

$$\langle \sigma | \chi_s \rangle : \begin{pmatrix} \alpha_s \\ \alpha_{s-1} \\ \vdots \\ \alpha_{-s} \end{pmatrix}. \quad (9.11)$$

³Although not strictly correct, it is common to ignore the differences among the abstract state vector $|\chi\rangle$, its explicit representation $\langle \sigma | \chi \rangle$, and its matrix representation, e.g., to write $|\chi\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$. We try to keep this distinction clear in this chapter; but not in the next one.

⁴This spin state is arbitrary but still a *coherent* state. In § 9.2 we describe *incoherent* spin states.

In this representation, the basis vectors have the form

$$|s, s\rangle = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \quad |s, s-1\rangle = \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}, \quad \dots, \quad |s, -s\rangle = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix}. \quad (9.12)$$

The operators are now matrices, for example, the unit operator

$$\tilde{1}_s = \sum_{m_s=-s}^s |sm_s\rangle \langle sm_s| = \begin{pmatrix} 1 & 0 & 0 & \cdots \\ 0 & 1 & 0 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}. \quad (9.13)$$

The spin states $|sm_s\rangle$ combine with the external (orbital angular momentum) states $|lm_l\rangle$ to form states $|lsjm_j\rangle$ of total angular momentum $\mathbf{j} = \mathbf{l} + \mathbf{s}$. The direct product of the two spaces,

$$|lm_l\rangle |sm_s\rangle \equiv |ls; m_l m_s\rangle, \quad (9.14)$$

can be expanded as a direct sum of spaces spanned by basis vectors $|lsjm_j\rangle$. To find the expansion coefficients we take the unit operator (completeness relation) in the ls space:

$$\tilde{1}_{l,s} = \sum_{m_l=-l}^l \sum_{m_s=-s}^s |ls; m_l m_s\rangle \langle ls; m_l m_s|, \quad (9.15)$$

and have it act on the full state $|lsjm_j\rangle$:

$$|lsjm_j\rangle \equiv \tilde{1} |lsjm_j\rangle = \sum_{m_l, m_s} |ls; m_l m_s\rangle \langle ls; m_l m_s | lsjm_j \rangle. \quad (9.16)$$

We recognize the matrix element $\langle ls; m_l m_s | lsjm_j \rangle$ as a Clebsch-Gordan coefficient $C(jmls; m_l m_s)$. We obtain an explicit representation of these $|lsjm_j\rangle$'s by representing them in the spin-angle basis $|\theta\phi\sigma\rangle$. Not surprisingly, these explicit functions $\mathcal{Y}(\theta\phi\sigma)$ are called the spin-angle functions:

$$\mathcal{Y}_{jm}^{ls}(\theta\phi\sigma) \equiv \langle \theta\phi\sigma | ls; jm \rangle \quad (9.17)$$

$$= \sum_{m_l, m_s} Y_{lm_l}(\theta\phi) \chi_{sm_s}(\sigma) \langle ls; m_l m_s | lsjm_j \rangle, \quad (9.18)$$

where we have used the explicit representation

$$\langle \theta\phi\sigma | ls; m_l m_s \rangle = Y_{lm_l}(\theta\phi) \chi_{sm_s}(\sigma). \quad (9.19)$$

The spin-angle functions are useful because they are simultaneous eigenfunctions of j , l , and s .

Just for Spin One Half

Systems with total spin $\frac{1}{2}$ are two-dimensional, and this is at least twice as complicated to describe as a system with spin 0. The two basis vectors for spin $\frac{1}{2}$ have the representation:

$$|\frac{1}{2} \frac{1}{2}\rangle \equiv |\uparrow\rangle \equiv |+\rangle, \quad |\frac{1}{2} -\frac{1}{2}\rangle \equiv |\downarrow\rangle \equiv |-\rangle, \quad (9.20)$$

$$\langle \sigma | \chi_{\frac{1}{2}\frac{1}{2}} \rangle : \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} \chi_{\frac{1}{2}\frac{1}{2}}(\frac{1}{2}) \\ \chi_{\frac{1}{2}\frac{1}{2}}(-\frac{1}{2}) \end{pmatrix}, \quad (9.21)$$

$$\langle \sigma | \chi_{\frac{1}{2}-\frac{1}{2}} \rangle : \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} \chi_{\frac{1}{2}-\frac{1}{2}}(\frac{1}{2}) \\ \chi_{\frac{1}{2}-\frac{1}{2}}(-\frac{1}{2}) \end{pmatrix}. \quad (9.22)$$

Accordingly, an arbitrary spin- $\frac{1}{2}$ state is represented as

$$|\chi\rangle = \alpha_+ |\uparrow\rangle + \alpha_- |\downarrow\rangle : \begin{pmatrix} \alpha_+ \\ \alpha_- \end{pmatrix}. \quad (9.23)$$

The spin operator $\mathbf{s}$ is proportional to the Pauli matrices:

$$\mathbf{s} = \frac{1}{2}\boldsymbol{\sigma} = \frac{1}{2}[\sigma_x \hat{\mathbf{e}}_x + \sigma_y \hat{\mathbf{e}}_y + \sigma_z \hat{\mathbf{e}}_z], \quad (9.24)$$

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (9.25)$$

The 2×2 unit operator $\tilde{I}_s (\equiv I)$ plus the three Pauli matrices form a complete set of 2×2 matrices in which any general 2×2 matrix may be expanded.

9.2 Polarized Beams

The fact that much has been written about beams and polarizations should serve as a warning that there is some subtlety here. For a *coherent state* $|\chi\rangle$ describing a single particle (9.23), the meaning of polarization is quite clear, it is the normalized expectation value of the spin,

$$\mathbf{P}_i \stackrel{\text{def}}{=} \frac{\langle \chi | \mathbf{s} | \chi \rangle}{s \langle \chi | \chi \rangle} = \langle \chi | \boldsymbol{\sigma} | \chi \rangle, \quad (9.26)$$

where the last equality holds for normalized states. An arbitrary coherent state $|\chi_i\rangle$ can always be expressed as a linear combination of states with spins up and down:

$$|\chi_i\rangle = \alpha_+ |\uparrow\rangle + \alpha_- |\downarrow\rangle, \quad (9.27)$$

where different i values correspond to different values of the complex constants α_+ and α_- . These constants are arbitrary except for the normalization constraint:

$$\langle \chi_i | \chi_i \rangle = 1 \quad \Rightarrow \quad |\alpha_+|^2 + |\alpha_-|^2 = 1. \quad (9.28)$$

According to its definition (9.26), the polarization $\mathbf{P}_i$ in state χ_i is

$$\mathbf{P}_i = \langle \chi_i | \boldsymbol{\sigma} | \chi_i \rangle \quad (9.29)$$

$$= \langle \chi_i | \sigma_x | \chi_i \rangle \hat{\mathbf{e}}_x + \langle \chi_i | \sigma_y | \chi_i \rangle \hat{\mathbf{e}}_y + \langle \chi_i | \sigma_z | \chi_i \rangle \hat{\mathbf{e}}_z \quad (9.30)$$

$$= 2 \operatorname{Re}(\alpha_+^* \alpha_-) \hat{\mathbf{e}}_x + 2 \operatorname{Im}(\alpha_+^* \alpha_-) \hat{\mathbf{e}}_y + (|\alpha_+|^2 - |\alpha_-|^2) \hat{\mathbf{e}}_z. \quad (9.31)$$

Note that a choice of the α 's makes the polarization $\mathbf{P}_i$ point in a particular direction but that regardless of the direction, the magnitude of the polarization is always 100%,

$$\mathbf{P}_i \cdot \mathbf{P}_i \equiv 1. \quad (9.32)$$

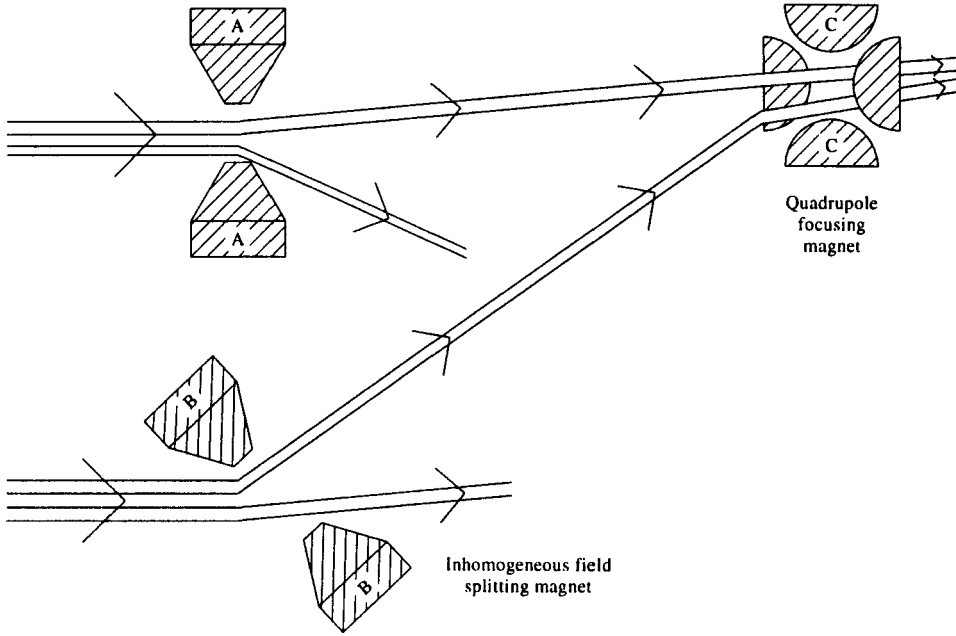


Figure 9.2: The formation of an incoherent beam. The focusing magnets at C mix two coherent beams produced by the Stern-Gerlach apparatuses at A and B.

It is only when we deal with states of partial coherence [that is, ones which cannot be expressed in the form (9.27)] that a polarization less than 100% is possible (see what follows).

Exercise Verify (9.31) and (9.32) for a general state and for a pure up or down state. ◇

Exercise What state vector has spin pointing in the x direction? Verify your answer by having σ_x act on it. ◇

Due to the nature of experimental equipment, in actual experiments it is rare to have a beam of 100% polarization or coherence. Laboratory beams are represented as *incoherent sums* of a number of coherent states or, alternatively, many-particle states formed by adding together a number of single-particle states. An example of this is shown in Figure 9.2 where inhomogeneous magnetic fields in the Stern-Gerlach apparatuses A and B act as polarizers to produce coherent beams of definite spin-states, while the collimator C passes a particular mixture of the individual beams to produce a new one of partial coherence.

Let us say we have coherent beams $|\chi_i\rangle$'s with polarizations $\mathbf{P}_i$, each of magnitude 1 but aligned in an arbitrary direction. The polarization $\mathbf{P}$ of a beam formed by mixing these coherent beams is

$$\mathbf{P} = \sum_i w_i \mathbf{P}_i = \sum_i w_i \left[2 \operatorname{Re}(\alpha_+^{i*} \alpha_-^i) \hat{\mathbf{e}}_x + 2 \operatorname{Im}(\alpha_+^{i*} \alpha_-^i) \hat{\mathbf{e}}_y + (|\alpha_+^i|^2 - |\alpha_-^i|^2) \hat{\mathbf{e}}_z \right], \quad (9.33)$$

where w_i is the probability of state i being in the beam and we have substituted (9.31) for the individual polarizations. We see that an incoherent beam is described by an *ensemble average* such as (9.33) in which we add together the observables $\mathbf{P}_i$'s appropriate to each coherent state which gets mixed. Each observable in the sum is weighted by the probability w_i of state $|\chi_i\rangle$ occurring. The exact values for the w_i 's and α^i 's depends on the details of the experimental apparatus. In practice, $\mathbf{P}$ could be determined empirically using the methods discussed in Chapter 10, *Spin Phenomenology and Identical Particles*, or from our analytic expressions. For example, if we look at the expression for the polarization (9.33), we see that a beam of zero polarization requires

$$\sum_i w_i \operatorname{Re}(\alpha_+^{i*} \alpha_-^i) = \sum_i w_i \operatorname{Im}(\alpha_+^{i*} \alpha_-^i) = \sum_i w_i (|\alpha_+^i|^2 - |\alpha_-^i|^2) = 0. \quad (9.34)$$

Equation (9.34) can be satisfied by summing over a large number of states with random phases which average out to zero, or by a judicious choice of α 's. For example, $\alpha^1 = (1, 0)$, $\alpha^2 = (0, 1)$, $w_1 = w_2 = \frac{1}{2}$, which is the mixture of two beams, one with spin up and one with spin down, is unpolarized.

The description of beams with partial coherence and the calculation of experimental observables are made simpler and more transparent by use of the *density matrix* formalism. We discuss it in § 10.3.

9.3 Wave Equation with a Spin-Orbit Potential

We apply the general formalism to the interaction of a spin $\frac{1}{2}$ particle with a spin-0 potential. Because the analysis is in the CM system, the target and projectile are equivalent and either may have the spin. We quantize the spin along the *beam direction* (our z axis) as indicated in Figure 9.3A. An equally valid choice would be to quantize the spin along an axis perpendicular to the beam, as indicated in Figure 9.3B. Although the results with one choice of axis can effectively be rotated into the other, the language or phenomenology appears different in the two cases.

Since spin-dependent potentials produce non-central forces, the scattering may depend on the azimuthal angle ϕ . Yet when the spin is oriented along the beam direction as in Figure 9.3A, the interaction cannot be ϕ dependent because the physical arrangement is cylindrically symmetric. If the spin is oriented in some other direction as in Figure 9.3B, there will be ϕ dependence to the scattering.

What is the most general spin dependence of the potential V for the scattering of a particle with spin $\mathbf{s}$ from the spinless force center in Figure 9.3? Because V is invariant under rotations (isotropy of space), it may contain a central term V_c (which can be a function of r , k , and k') plus scalar products formed by combining the vector $\mathbf{s}$ with $\mathbf{k}$, $\mathbf{k}'$, and $\mathbf{r}$:

$$V = V_c + V_a \mathbf{s} \cdot \mathbf{k} + V_b \mathbf{s} \cdot \mathbf{k}' + V_c \mathbf{s} \cdot (\mathbf{k} \times \mathbf{k}') + \dots \quad (9.35)$$

Sometimes it is convenient to replace $\mathbf{k} \times \mathbf{k}'$ by $\hat{\mathbf{n}}$, the unit normal to the scattering plane,

$$\hat{\mathbf{n}} = \frac{\mathbf{k} \times \mathbf{k}'}{|\mathbf{k} \times \mathbf{k}'|}. \quad (9.36)$$

Note that higher powers of $\mathbf{s}$ are redundant because they can be reduced to $\mathbf{s}$ and $\hat{\mathbf{I}}$, (which form a complete set in spin space). If we demand that V be invariant under time reversal

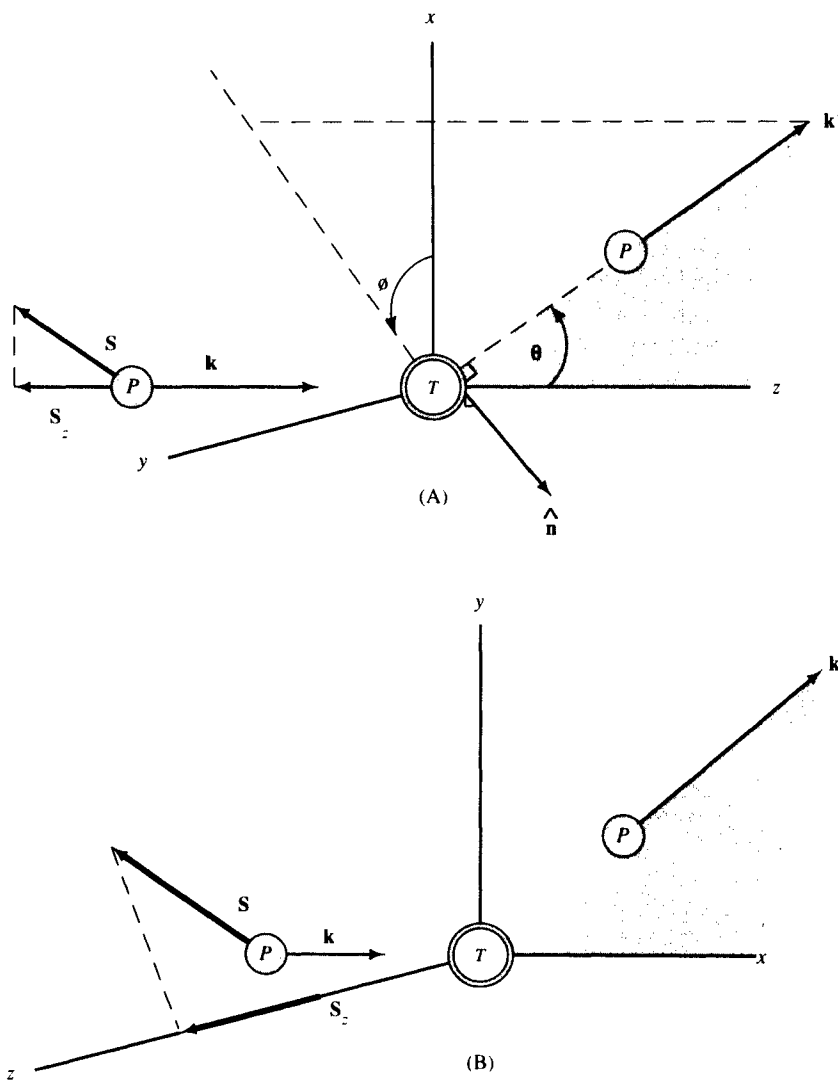


Figure 9.3: Scattering of a projectile P from a target T with the spin quantized along the z axis. The shaded scattering plane is defined by $\mathbf{k}$ and $\mathbf{k}'$. In (A) the spin is along the beam direction, while in (B) the spin is perpendicular to beam.

($\mathbf{k} \rightarrow -\mathbf{k}$, $\mathbf{k}' \rightarrow -\mathbf{k}'$, $\mathbf{s} \rightarrow -\mathbf{s}$) and the parity transformation ($\mathbf{k} \rightarrow -\mathbf{k}$, $\mathbf{k}' \rightarrow -\mathbf{k}'$, $\mathbf{s} \rightarrow \mathbf{s}$), the only scalar product of $\mathbf{s}$ (an axial vector) which survives is that with the only other axial vector $\hat{\mathbf{n}}$:

$$V = V_c + V_s \mathbf{s} \cdot \hat{\mathbf{n}}. \quad (9.37)$$

Equation (9.37) is a convenient form for the potential in momentum space with V_c and V_s functions of k and k' . For coordinate-space calculations it is more conveniently written as

$$V(\mathbf{r}) = V_c(\mathbf{r}) + V_s(\mathbf{r}) \mathbf{s} \cdot \mathbf{l}, \quad (9.38)$$

where $\mathbf{l}$ is the orbital angular momentum of the system, and we have assumed a local potential to minimize notation. The spin-dependent term should be familiar from the hydrogen atom as the *spin-orbit potential*.

Exercise Prove the equivalence of (9.37) and (9.38). $\diamond$

Expansion of Wave Function

The Schrödinger equation with the spin-orbit potential (9.38) is

$$\left[\frac{-\nabla^2}{2\mu} \tilde{\mathbf{l}}_s + V_c(\mathbf{r}) \tilde{\mathbf{l}}_s + V_s(\mathbf{r}) \mathbf{s} \cdot \mathbf{l} \right] \psi(\mathbf{r}\sigma) = E\psi(\mathbf{r}\sigma), \quad (9.39)$$

where we have indicated the implied spin dependence of the first two terms by the unit operator in spin space $\tilde{\mathbf{l}}_s$. At first it may seem that we could solve (9.39) by taking our previous partial-wave expansion of the wave function (8.50),

$$\psi_k(\mathbf{r}) = \sum_{l=0}^{\infty} \frac{(2l+1)i^l [u_l(kr)/kr] P_l(x)}{(2\pi)^{3/2}}, \quad x \stackrel{\text{def}}{=} \cos \theta_{k\mathbf{r}}, \quad (9.40)$$

multiply it by a spin wave function $|\uparrow\rangle$ or $|\downarrow\rangle$ for the initial spin up or down, and substitute it into (9.39). The difficulty is that the $\mathbf{l} \cdot \mathbf{s}$ term flips the spin:

$$\mathbf{l} \cdot \mathbf{s} Y_{l,0}(\theta\phi) \chi_{\pm} = \frac{1}{2} \sqrt{l(l+1)} Y_{l,\pm 1}(\theta\phi) \chi_{\mp}. \quad (9.41)$$

This causes the individual terms in (9.40) to no longer be eigenstates of the Hamiltonian and thereby couples together the partial waves for different l values.

Exercise Prove (9.41) by expressing $\mathbf{l} \cdot \mathbf{s}$ in terms of of raising and lowering operators. $\diamond$

We obtain a solution of (9.39) if, instead of expanding the wave function in eigenstates of l , we expand in eigenstates of j and $\mathbf{s} \cdot \mathbf{l}$. This is not as difficult as it may seem because we have just derived the spin-angle functions (9.17) and (9.18) which are these eigenstates. For spin $s = \frac{1}{2}$, we use the somewhat condensed notation:

$$y_l^{(+)}(\theta\phi\sigma) \stackrel{\text{def}}{=} y_{j=l+1/2, m=1/2}^{ls=1/2}(\theta\phi\sigma) \quad (9.42)$$

$$= \sqrt{\frac{l+1}{2l+1}} Y_{l0}(\theta\phi) |\uparrow\rangle + \sqrt{\frac{l}{2l+1}} Y_{l1}(\theta\phi) |\downarrow\rangle, \quad (9.43)$$

$$\mathcal{Y}_l^{(-)}(\theta\phi\sigma) \stackrel{\text{def}}{=} \mathcal{Y}_{j=l-1/2, m=1/2}(\theta\phi\sigma) \quad (9.44)$$

$$= \sqrt{\frac{l}{2l+1}} Y_{l0}(\theta\phi) |\uparrow\rangle - \sqrt{\frac{l+1}{2l+1}} Y_{l1}(\theta\phi) |\downarrow\rangle, \quad (9.45)$$

$$Y_{lm}(\theta\phi) = \sqrt{\frac{(2l+1)(l-m)!}{4\pi(l+m)!}} (-1)^m e^{im\phi} P_l^m(\cos\theta), \quad (9.46)$$

$$P_l^m(x) = (x^2 - 1)^{m/2} \frac{d^m P_l}{dx^m}. \quad (9.47)$$

With this notation, the plus and minus superscripts indicate the j value:

$$(\pm) \Rightarrow j = l \pm \frac{1}{2}. \quad (9.48)$$

Exercise Derive (9.43) and (9.45) from (9.17) by substitution of the Clebsch-Gordan coefficients. $\diamond$

Exercise Express the $\mathcal{Y}$'s in matrix form. $\diamond$

Exercise Verify that the spin-angle functions are eigenstates of $\mathbf{l} \cdot \mathbf{s}$ and j^2 with the specific eigenvalues:

$$\mathbf{l} \cdot \mathbf{s} \mathcal{Y}_l^{(\pm)} = \left(-\frac{1}{2}(l+1) \right) \mathcal{Y}_l^{(\pm)}, \quad (9.49)$$

$$j^2 \mathcal{Y}_l^{(\pm)} = (l \pm \frac{1}{2})(l \pm \frac{1}{2} + 1) \mathcal{Y}_l^{(\pm)}. \quad \diamond \quad (9.50)$$

To expand the wave function in spin-angle functions, we examine the incident plane wave of a spin-up particle:

$$\phi_{\uparrow}(\mathbf{r}) = \frac{e^{i\mathbf{k} \cdot \mathbf{r}}}{(2\pi)^{3/2}} |\uparrow\rangle = \sum_{l=0}^{\infty} \frac{(2l+1)i^l F_l(kr)}{(2\pi)^{3/2} kr} P_l(x) |\uparrow\rangle \quad (9.51)$$

$$= \sum_{l=0}^{\infty} i^l \frac{F_l(kr)}{kr} \sqrt{4\pi(2l+1)} \frac{Y_{l0}(\theta\phi)}{(2\pi)^{3/2}} |\uparrow\rangle, \quad (9.52)$$

where we have right juxtaposed a spin-up ket and expressed the Legendre function P_l in terms of the spherical harmonic Y_{l0} . We next express $Y_{l0} |\uparrow\rangle$ in spin-angle functions:

$$Y_{l0}(\theta\phi) |\uparrow\rangle = \sqrt{\frac{l+1}{2l+1}} \mathcal{Y}_l^{(+)} + \sqrt{\frac{l}{2l+1}} \mathcal{Y}_l^{(-)}, \quad (9.53)$$

$$\Rightarrow \phi_{\uparrow}(\mathbf{r}) = \sum_{l=0}^{\infty} i^l \frac{F_l(kr)}{\sqrt{2\pi} kr} \left[\sqrt{l+1} \mathcal{Y}_l^{(+)}(\theta\phi\sigma) + \sqrt{l} \mathcal{Y}_l^{(-)}(\theta\phi\sigma) \right]. \quad (9.54)$$

Equation (9.54) is a *plane wave* with spin up expanded in terms of spherical waves of definite $j = l \pm \frac{1}{2}$. As in our previous discussions, we obtain the *distorted-wave expansion*

by replacing the free radial waves F_l by the distorted radial waves u_l :

$$\psi_{\uparrow}(\mathbf{r}) = \sum_{l=0}^{\infty} \frac{i^l}{\sqrt{2\pi}} \left[\sqrt{l+1} \frac{u_l^{(+)}(kr)}{kr} \mathcal{Y}_l^{(+)}(\theta\phi\sigma) + \sqrt{l} \frac{u_l^{(-)}(kr)}{kr} \mathcal{Y}_l^{(-)}(\theta\phi\sigma) \right]. \quad (9.55)$$

Note that the up arrow indicates the spin direction of the initial beam and the plus and minus that $j = l \pm \frac{1}{2}$.⁵ In contradistinction to the plane-wave expansion (9.54), when there is a spin-dependent potential the wave functions $u_l^{(+)}$ and $u_l^{(-)}$ will not be equal and so the terms in bracket in (9.55) will *not* add back up to a pure spin-up state. Accordingly, for spin-dependent potentials the scattered wave contains a piece whose spin has been *flipped*.

Solution

To check if our deduced partial-wave expansion works, we only need to substitute it back into the Schrödinger equation (9.39) and see if it leads to 1D differential equations.

Exercise Show that the substitution of (9.55) into (9.39) leads to the two, *uncoupled* one-dimensional, differential equations, each for a state of different j :

$$\left[\frac{-1}{2\mu} \frac{d^2}{dr^2} + \frac{l(l+1)}{2\mu r^2} + V_c(r) + \frac{l}{2} V_s(r) \right] u_l^{(+)}(kr) = E u_l^{(+)}(kr), \quad (9.56)$$

$$\left[\frac{-1}{2\mu} \frac{d^2}{dr^2} + \frac{l(l+1)}{2\mu r^2} + V_c(r) - \frac{(l+1)}{2} V_s(r) \right] u_l^{(-)}(kr) = E u_l^{(-)}(kr). \quad (9.57)$$

◇

Note the *spin dependence* of the scattering arises solely from the different coefficients multiplying the V_s terms in (9.56) and (9.57). If V_s equals zero, the equations are identical, the scattered wave has the same spin as the incident wave, and the spin dependence vanishes.

The equations for each j can now be solved in the usual way with the wave functions having the usual scattering normalization and boundary conditions:

$$u_l^{(\pm)}(kr) \sim \sin(kr - l\pi/2) + e^{i\delta_l^{\pm}} \sin \delta_l^{\pm} e^{i(kr - l\pi/2)}. \quad (9.58)$$

If we substitute (9.58) into the partial-wave expansion of the distorted wave (9.55), we find that the scattered wave has a part for scattering of $j = l + \frac{1}{2}$ and a different part for $j = l - \frac{1}{2}$:

$$\psi_{\uparrow}(\mathbf{r}) \sim \phi_{\uparrow}(\mathbf{r}) + \frac{e^{ikr}}{kr\sqrt{2\pi}} \sum_l \left[\sqrt{l+1} e^{i\delta_l^{+}} \sin \delta_l^{+} \mathcal{Y}_l^{(+)} + \sqrt{l} e^{i\delta_l^{-}} \sin \delta_l^{-} \mathcal{Y}_l^{(-)} \right]. \quad (9.59)$$

If we decompose the spin-angle functions $\mathcal{Y}^{(\pm)}$ back into spinors and Legendre functions, we see that the scattered wave has a part with spin up and a part in which the spin has been *flipped* to down:

$$\psi_{\uparrow}(\mathbf{r}) \sim \phi_{\uparrow}(\mathbf{r}) + \frac{e^{ikr}}{r} \frac{f_{++}(\theta) |\uparrow\rangle + f_{-+}(\theta, \phi) |\downarrow\rangle}{(2\pi)^{3/2} r}. \quad (9.60)$$

⁵We leave off our earlier (+) indicating outgoing wave boundary conditions to avoid confusion with the + from $j = l + \frac{1}{2}$.

Here f_{++} has the clumsy, but descriptive, name of *spin-nonflip amplitude*; we denote it by $f(\theta)$ because it is related to the scattering amplitude for spinless scattering.⁶ The expansion of $f(\theta)$ in phase shifts and spherical harmonics is read off after comparing (9.59) and (9.60):

$$f_{++}(\theta) = \sum_{l=0}^{\infty} \frac{\sqrt{4\pi(2l+1)}}{k} \left[(l+1)e^{i\delta_l^+} \sin \delta_l^+ + le^{i\delta_l^-} \sin \delta_l^- \right] Y_{l0}(\theta). \quad (9.61)$$

This is usually written in the simpler form

$$f(\theta) \stackrel{\text{def}}{=} f_{++} = \frac{1}{k} \sum_{l=0}^{\infty} \left[(l+1)e^{i\delta_l^+} \sin \delta_l^+ + le^{i\delta_l^-} \sin \delta_l^- \right] P_l(x). \quad (9.62)$$

This spin-nonflip amplitude is independent of the azimuthal angle ϕ . If $\delta_l^+ = \delta_l^-$, that is, if there were no spin dependence in the scattering, $f(\theta)$ would equal the familiar expression for the scattering amplitude.

The amplitude for scattering from spin up to down, f_{-+} , is also deduced from comparing (9.59) with (9.60):

$$f_{-+}(\theta, \phi) = -\frac{1}{k} \sum_l \sqrt{\frac{4\pi l(l+1)}{2l+1}} \left[e^{i\delta_l^+} \sin \delta_l^+ - e^{i\delta_l^-} \sin \delta_l^- \right] Y_{l-1}(\theta, \phi). \quad (9.63)$$

We remove the ϕ and $\sin \theta$ dependences and define⁷

$$g(\theta) \stackrel{\text{def}}{=} -\frac{f_{-+}(\theta, \phi)e^{-i\phi}}{\sin \theta}. \quad (9.64)$$

This leads to the simpler form for the *spin-flip* amplitude:

$$g(\theta) = \frac{1}{k} \sum_{l=1}^{\infty} \left[e^{i\delta_l^+} \sin \delta_l^+ - e^{i\delta_l^-} \sin \delta_l^- \right] \frac{dP_l(x)}{d(x)}. \quad (9.65)$$

Here $g(\theta)$ is the sum of derivatives of Legendre polynomials. Its S-wave contribution vanishes and it has an angle dependence different from $f(\theta)$'s. The full spin-flip amplitude f_{-+} in (9.64) is proportional to $\sin \theta$, so it vanishes for 0° and 180° scattering. We see an example of this in Figure 9.4 where the spin-flip scattering fills the minima in the nonflip cross section (we discuss cross-section calculations in Chapter 10, *Spin Phenomenology and Identical Particles*).

The preceding equations describe a spin-up beam scattering into spin-up and spin-down waves. Essentially the same analysis follows if the incident beam were spin down. As expected from symmetry considerations, the down-to-down (nonflip) amplitude equals the up-to-up one:

$$f_{--}(\theta) = f_{++}(\theta). \quad (9.66)$$

⁶We follow here the spin conventions of Goldberger and Watson (1963) which are the same as used by Landau (1982) in the computer codes LPOTT.

⁷Note that some authors do not remove $\sin \theta$ from $g(\theta)$, some authors have $\hat{n}$ in the opposite direction, and some authors call our f their g , and our g their h .

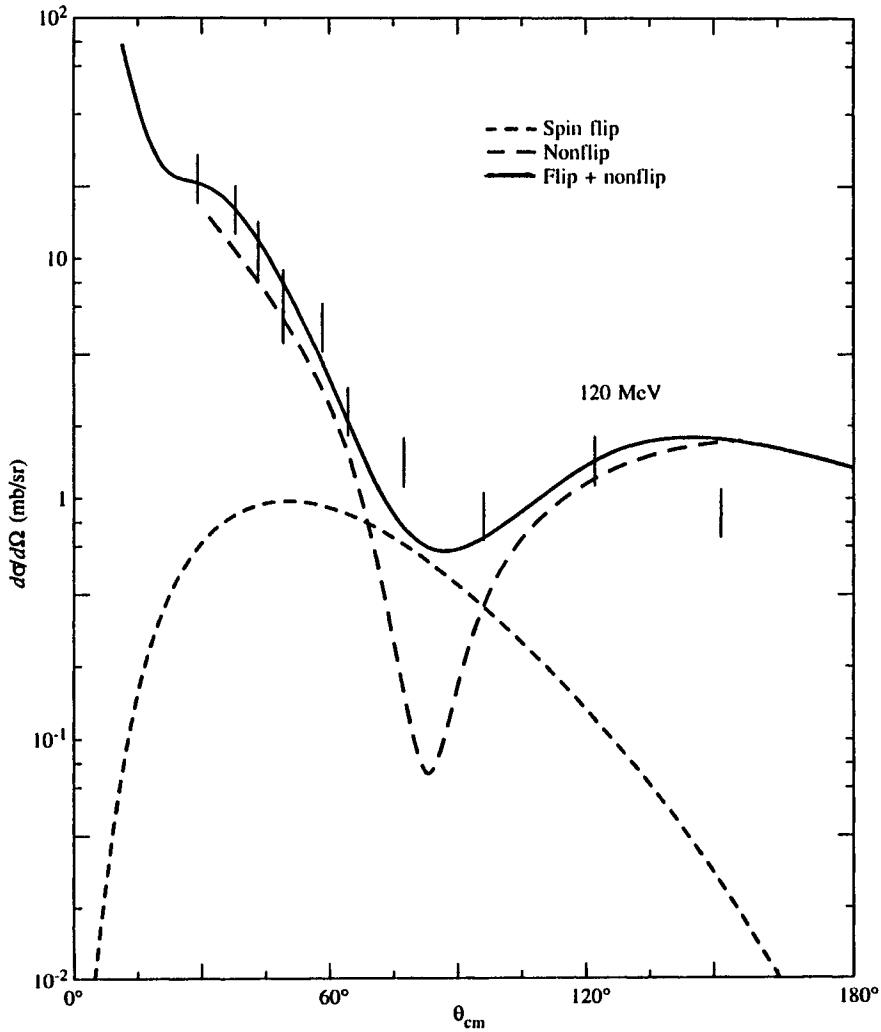


Figure 9.4: The effect of spin flip on the scattering of 120-MeV pi mesons from an unpolarized ^3He nucleus. The long-dashed curve is nonflip scattering, the short dashed curve is spin flip scattering, and the solid curve is the sum. Spin flip is needed to explain the experimental data. Adapted from Landau (1977).

The spin-flip ones differ only in phase:

$$f_{+-}(\theta, \phi) \stackrel{\text{def}}{=} e^{-i\phi} \sin \theta g(\theta) = -e^{-2i\phi} f_{-+}(\theta, \phi). \quad (9.67)$$

Exercise Verify that (9.67) follow from substitution into (9.59) and (9.60). (Be careful with the signs.) $\diamond$

We now have the formalism which permits us to start with a potential containing a spin-orbit term (9.38), solve the differential equations (9.56)–(9.57), and thereby deduce the phase shifts (δ_l^+ , δ_l^-) for $j = l \pm \frac{1}{2}$. The cited references and the next section give an equivalent formulation in momentum space, and we describe some of it in the context of Dirac theory in Chapter 17, *Integral Forms of the Dirac Equation, Scattering*. In the next chapter we indicate how experimental observables are calculated for spin-dependent interactions.

Momentum Space Spin Zero $\times$ Spin One Half

The hard work needed to solve the momentum-space spin $0 \times \frac{1}{2}$ problem has all been done in our study of coordinate-space potentials. We need to only proceed by analogy.

- We know the potential has the general form (9.37):

$$V = V_c + V_s \mathbf{s} \cdot \hat{\mathbf{n}}. \quad (9.68)$$

- We also know the scattering matrix has the spin-space structure:⁸

$$\mathcal{F} = f(\theta) \tilde{1}_s + i\sigma \cdot \hat{\mathbf{n}} \sin \theta g(\theta). \quad (9.69)$$

- Because the V and T operators have the same structure in momentum space, and because T is proportional to $\mathcal{F}$, we can write both as having a nonflip (NF , or central) term like the first one in (9.35), plus a flip (F , or spin orbit) term like the second one:

$$\begin{pmatrix} V(\mathbf{k}', \mathbf{k}) \\ T(\mathbf{k}', \mathbf{k}) \end{pmatrix} = \begin{pmatrix} V(\mathbf{k}', \mathbf{k}) \\ T(\mathbf{k}', \mathbf{k}) \end{pmatrix}_{NF} + i\sigma \cdot \hat{\mathbf{n}} \sin \theta_{kk'} \begin{pmatrix} V(\mathbf{k}', \mathbf{k}) \\ T(\mathbf{k}', \mathbf{k}) \end{pmatrix}_F. \quad (9.70)$$

- We know that nonflip and flip amplitudes f and g [(9.62) and (9.65), respectively] have the expansions:

$$f(\theta) = \frac{1}{k} \sum_{l=0}^{\infty} \left[(l+1) e^{i\delta_l^+} \sin \delta_l^+ + l e^{i\delta_l^-} \sin \delta_l^- \right] P_l(x), \quad (9.71)$$

$$g(\theta) = \frac{1}{k} \sum_{l=1}^{\infty} \left[e^{i\delta_l^+} \sin \delta_l^+ - e^{i\delta_l^-} \sin \delta_l^- \right] \frac{dP_l(x)}{dx}. \quad (9.72)$$

⁸We will study the consequences of this structure in Chapter 10, *Spin Phenomenology and Identical Particles*.

- Likewise, the nonflip and flip parts of V and T must have the expansions:

$$\begin{pmatrix} V \\ T \end{pmatrix}_{NF} = \sum_{l=0}^{\infty} \left[(l+1) \begin{pmatrix} V_{l+}(k', k) \\ T_{l+}(k', k) \end{pmatrix} + l \begin{pmatrix} V_{l-}(k', k) \\ T_{l-}(k', k) \end{pmatrix} \right] \frac{P_l(x)}{2\pi^2}, \quad (9.73)$$

$$\begin{pmatrix} V \\ T \end{pmatrix}_F = \sum_{l=0}^{\infty} \left[\begin{pmatrix} V_{l+}(k', k) \\ T_{l+}(k', k) \end{pmatrix} - \begin{pmatrix} V_{l-}(k', k) \\ T_{l-}(k', k) \end{pmatrix} \right] \frac{1}{2\pi^2} \frac{dP_l(x)}{d(x)}, \quad (9.74)$$

where again, the plus and minus indicate $j = l \pm \frac{1}{2}$.

- Substitution of these expansions reduces the 3D coordinate-space Schrödinger equation to two, uncoupled 1D radial equations for $j = l \pm \frac{1}{2}$. They must also reduce the 3D, Lippmann–Schwinger equation to uncoupled 1D integral equations for each j value:

$$T_{l\pm}(k', k) = V_{l\pm}(k', k) + \frac{2}{\pi} \int_0^{\infty} \frac{dp p^2 V_{l\pm}(k', p) T_{l\pm}(p, k)}{E - E_p + i\epsilon}. \quad (9.75)$$

- Solution of this equation yields the scattering amplitude (or T matrix) for this (j, l) state with energy E_k :

$$T_{l\pm}(k, k) = \frac{e^{i\delta_l^{\pm}} \sin \delta_l^{\pm}}{-2\mu k}. \quad (9.76)$$

Once the phase shifts are known, the flip and nonflip amplitudes (and from them, all experimental observables) can be computed.

9.4 Problems

1. As seen in the text, solving the Schrödinger equation for a spin 0 particle interacting with a $\frac{1}{2}$ one via a spin–orbit interaction is simplified considerably once appropriate spin-angles eigenfunctions are known.

- (a) Derive the spin-angle functions $\mathcal{Y}_{j=l\pm 1/2, m=1/2}^{l, s=1/2} \stackrel{\text{def}}{=} \mathcal{Y}^{(l\pm)}$.
- (b) Show that these functions have the property:

$$\mathbf{l} \cdot \mathbf{s} \mathcal{Y}^{(l\pm)} = \left(-\frac{1}{2}(l+1) \right) \mathcal{Y}^{(l\pm)}, \quad (9.77)$$

$$\mathbf{j} \cdot \mathbf{j} \mathcal{Y}^{(l\pm)} = (l \pm \frac{1}{2})(l \pm \frac{1}{2} + 1) \mathcal{Y}^{(l\pm)}. \quad (9.78)$$

2. Use the properties of the spin-angle functions to show that the Schrödinger equation with a spin–orbit potential reduces to two uncoupled equations, both with the *same* l but *different* j 's. Derive these equations.
3. When solving the Dirac equation for the hydrogen atom, we will need to know the effect of $\boldsymbol{\sigma} \cdot \mathbf{r}$ on the spin-angle functions. Prove the following (and in the process determine the constant C):

$$\boldsymbol{\sigma} \cdot \mathbf{r} \mathcal{Y}^{(l=j-1/2)} = C \mathcal{Y}^{j+1/2}, \quad (9.79)$$

$$\boldsymbol{\sigma} \cdot \mathbf{r} \mathcal{Y}^{(l=j+1/2)} = C \mathcal{Y}^{(j-1/2)}. \quad (9.80)$$

4. Two non-identical spin $\frac{1}{2}$ particles interact via the potential

$$V(\mathbf{r}) = V_c(r) + V_s(r)\boldsymbol{\sigma}_1 \cdot \boldsymbol{\sigma}_2. \quad (9.81)$$

- (a) Indicate the possible values for the total spin $\mathbf{s} = \mathbf{s}_1 + \mathbf{s}_2$ and the form of the potential in each spin state.
 - (b) Is this a central potential?
 - (c) For each allowed value of s , how many values for m_s are possible? These quantum numbers enumerate the possible states and thus scattering amplitudes for this problem.
 - (d) Write down the partial-wave expansion for a general state Ψ_m^s in terms of spin functions, spherical harmonics, and radial functions.
 - (e) Deduce the one-dimensional form of the Schrödinger equation for each value of l .
 - (f) Assume that you could solve the preceding Schrödinger equations for the scattering amplitude $f_{s,m}$ in each state. What is the differential cross section for an unpolarized (random) beam of spin $\frac{1}{2}$ particles interacting via the preceding potential with an unpolarized spin $\frac{1}{2}$ target when the final polarization is undetected?
5. To test your understanding of the material in this chapter, try answering the following questions with minimal resort to the text. Consider the interaction of two, non-identical spin $\frac{1}{2}$ particles. A simple interaction between these two particles may include central, spin–spin, and spin–orbit forces:

$$V(\mathbf{r}) = V_c(\mathbf{r}) + V_s(\mathbf{r})\mathbf{s}_1 \cdot \mathbf{s}_2 + V_{12}(\mathbf{s}_1 + \mathbf{s}_2) \cdot \mathbf{l}, \quad (9.82)$$

where $\mathbf{r}$ is the interparticle distance $\mathbf{r}_1 - \mathbf{r}_2$, $\mathbf{s}_i$ is the spin operator for particle i , and $\mathbf{l}$ is the orbital angular momentum of the relative motion of the particles.

- (a) Indicate the appropriate form of the Schrödinger equation if there were no interaction between these two particles.
- (b) Is this potential “central”?
- (c) What are the allowed values of the total spin s , and the possible $s_z = m$ values for each s ?
- (d) Use the preceding result to deduce the multiplicity (degeneracy) of each spin state.
- (e) Deduce and interpret the 1D Schrödinger equations appropriate for this potential.
- (f) What quantum numbers are “good” (conserved) for the Hamiltonian containing this potential?
- (g) For a given value of l , what quantum numbers are good, and what is the potential in the channel defined by these quantum numbers?
- (h) Make use of the fact that V is diagonal in some basis to write down the partial-wave expansion of the wave function for the Schrödinger equation with this potential.

- (i) What are all the Schrödinger equations for each l value?
- (j) List all the different scattering amplitudes which can occur for this problem.
- (k) In terms of the “triplet” and “singlet” scattering amplitudes, what is the differential cross section for an unpolarized beam of spin $\frac{1}{2}$ particles scattering from an unpolarized spin $\frac{1}{2}$ target when the final polarization is undetected?

CHAPTER 10

SPIN PHENOMENOLOGY AND IDENTICAL PARTICLES

In the preceding chapter we developed the formalism appropriate for the scattering of a spin 0 and a spin $\frac{1}{2}$ particle. We know how to start with a spin-dependent potential, write down the proper 1D Schrödinger equations incorporating the spin's degrees of freedom, solve these equations, and deduce the phase shifts for each j value. With these phase shifts we can directly calculate the amplitudes for spin-flip and non-flip scattering which, as we see in this chapter, permits us to calculate the experimental observables. For many purposes it is easier to perform calculations and visualize the phenomenology of spin by considering the scattering amplitude as an operator or matrix in spin space. This view is developed in this chapter. We thereby make the connection between spin as an additional angular momentum, the introductory view, and as an extra degree of freedom in an internal space, the view in Chapter 9, *Spin Theory*.

10.1 $\mathcal{F}$ as a Matrix in Spin Space

A convenient and compact way to express the spin dependence of the scattering amplitude is to write it as an operator in spin space [that is, a 2×2 matrix in the representation (9.20)–(9.22)]:

$$\mathcal{F} = f(\theta) \tilde{1}_s + i\sigma \cdot \hat{n} \sin \theta g(\theta), \quad (10.1)$$

where $\hat{n}$, shown in Figure 9.3A, is the unit normal to the scattering plane,

$$\hat{n} = \frac{\mathbf{k} \times \mathbf{k}'}{\sin \theta}. \quad (10.2)$$

Even if $\mathcal{F}$ is not computed from phase shifts, knowledge that it has the form (10.1) permits the phenomenological analysis of experiments.

The proof and use of (10.1) is straightforward. Consider spin *non-flip* of an up beam scattered into an up final state. We take the appropriate matrix element of $\mathcal{F}$:

$$\langle \uparrow | \mathcal{F} | \uparrow \rangle = f(\theta) \langle \uparrow | \uparrow \rangle + ig(\theta) \langle \uparrow | \sigma | \uparrow \rangle \cdot \hat{n} \sin \theta. \quad (10.3)$$

Because $\hat{\mathbf{n}}$ must be normal to the beam direction and because we quantize σ along the beam direction,

$$\langle \uparrow | \sigma | \uparrow \rangle \cdot \hat{\mathbf{n}} = \langle \downarrow | \sigma | \downarrow \rangle \cdot \hat{\mathbf{n}} = 0. \quad (10.4)$$

The non-flip scattering amplitudes are correspondingly

$$\langle \uparrow | \mathcal{F} | \uparrow \rangle = f_{++} = f(\theta) = f_{--} = \langle \downarrow | \mathcal{F} | \downarrow \rangle. \quad (10.5)$$

To check (10.1) for *spin-flip* scattering, we note that for the geometry of Figure 9.3A [$\mathbf{k}$ along the z (spin quantization) axis]:

$$\hat{\mathbf{n}} = (-\sin \phi, \cos \phi, 0), \quad (10.6)$$

$$\Rightarrow \sigma \cdot \hat{\mathbf{n}} = -\sin \phi \sigma_x + \cos \phi \sigma_y, \quad (10.7)$$

$$\langle \uparrow | \sigma \cdot \hat{\mathbf{n}} | \downarrow \rangle = -\sin \phi \langle \uparrow | \sigma_x | \downarrow \rangle + \cos \phi \langle \uparrow | \sigma_y | \downarrow \rangle \quad (10.8)$$

$$= -\sin \phi - i \cos \phi = -ie^{-i\phi}. \quad (10.9)$$

So the spin-flip amplitude

$$f_{+-} = \langle \uparrow | \mathcal{F} | \downarrow \rangle = e^{-i\phi} \sin \theta g(\theta). \quad (10.10)$$

Exercise Evaluate the matrix element of $\mathcal{F}$ to prove that

$$f_{-+} = -e^{i\phi} \sin \theta g(\theta). \quad \diamond \quad (10.11)$$

Exercise Show that there is no spin-flip for a particle whose spin is normal to the scattering plane. $\diamond$

10.2 Observables

Calculating spin observables in quantum mechanics is challenging, in part because it is necessary to account for the spin state of the beam and scattered particle and the spin sensitivity of the detector. We shall keep things simple by considering scattering only for spin 0 on $\frac{1}{2}$. Systematic calculations of spin observables is best done with the density matrix formalism after working through the results of this chapter and developing some physical intuition.

Cross Sections

As in the spinless case, the differential cross section equals the square of the scattered wave's amplitude. The challenge with spin is deducing the appropriate amplitude or combination of amplitudes a particular experiment measures. Let us look at the possibilities.

- If the incident beam has spin *up* and the scattered beam is *observed* to be up, the cross section for this non spin-flip scattering is

$$\begin{aligned} \frac{d\sigma}{d\Omega}(\uparrow \leftarrow \uparrow) &\stackrel{\text{def}}{=} \left. \frac{d\sigma}{d\Omega} \right|_{\text{non-flip}} \\ &= |\langle \uparrow | \mathcal{F} | \uparrow \rangle|^2 = |f_{++}|^2 = |f(\theta)|^2. \end{aligned} \quad (10.12)$$

Obviously (that is, by symmetry), we get the same answer for down-to-down scattering.

- If the incident particle has spin up and the observed particle has spin *down*, the cross section for this spin-flip scattering is

$$\begin{aligned} \frac{d\sigma}{d\Omega}(\downarrow \leftarrow \uparrow) &\stackrel{\text{def}}{=} \left. \frac{d\sigma}{d\Omega} \right|_{\text{flip}} \\ &= |\langle \downarrow | \mathcal{F} | \uparrow \rangle|^2 = |f_{-+}|^2 = |-e^{i\phi} \sin \theta g(\theta)|^2 = \sin^2 \theta |g(\theta)|^2. \end{aligned} \quad (10.13)$$

Somewhat less obviously, we would get the same answer for down-to-up scattering because the phase factor cancels.

- If the incident beam has its spin quantized along the z axis (say up) and the spin of the scattered beam is not observed (either intentionally or because the measuring device does not have that capability), we add the *probabilities* for scattering to spin up and to spin down:

$$\frac{d\sigma}{d\Omega}(0 \leftarrow \uparrow) = \frac{d\sigma}{d\Omega}(\uparrow \leftarrow \uparrow) + \frac{d\sigma}{d\Omega}(\downarrow \leftarrow \uparrow) \quad (10.14)$$

$$= |f(\theta)|^2 + \sin^2 \theta |g(\theta)|^2. \quad (10.15)$$

We recognize (10.14) as the sum of cross sections for spin-flip and non-flip scattering. This addition of probabilities (cross sections) is a basic premise of quantum mechanics: If it is possible for a transition to occur in more than one way and the ways could have been distinguished experimentally, we add together the probability for each way even if the different ways were not distinguished. In contrast, if it is not possible in principle to distinguish the ways, we add the probability *amplitudes* and then square the modulus to obtain the physical probability for the transition. It is of course only in this latter case that the amplitudes interfere with each other.

- If the incident beam is in the spin state χ_i , it must be polarized with a *polarization* $\mathbf{P}_i$ given by (9.31). In this case the cross section for scattering to spin up is:

$$\frac{d\sigma}{d\Omega}(\uparrow \leftarrow i) = |\langle \uparrow | \mathcal{F} | \chi_i \rangle|^2 = |\alpha_+ f_{++} + \alpha_- f_{+-}|^2 \quad (10.16)$$

$$= |\alpha_+ f(\theta) + \alpha_- e^{-i\phi} \sin \theta g(\theta)|^2 \quad (10.17)$$

$$\begin{aligned} &= |f(\theta)|^2 |\alpha_+|^2 + |g(\theta)|^2 \sin^2 \theta |\alpha_-|^2 \\ &\quad + 2 \sin \theta \operatorname{Re}[\alpha_-^* \alpha_+ e^{i\phi} g^*(\theta) f(\theta)]. \end{aligned} \quad (10.18)$$

We note the interference of the spin-flip and non-flip amplitudes produces a ϕ dependence, which, in turn, produces a right-left asymmetry in the scattering (see Figure 9.3). This is reasonable because the normal to the scattering plane combined with the polarization vector, orients a plane in space and thus breaks the axial symmetry.

- If the incident beam has the polarization $\mathbf{P}_i$ (9.31), the cross section for scattering to spin down is:

$$\frac{d\sigma}{d\Omega}(\downarrow \leftarrow i) = |\langle \downarrow | \mathcal{F} | \chi_i \rangle|^2 = |\alpha_+ f_{-+} + \alpha_- f_{--}|^2 \quad (10.19)$$

$$= |-\alpha_+ e^{i\phi} \sin \theta g(\theta) + \alpha_- f(\theta)|^2 \quad (10.20)$$

$$= |f(\theta)|^2 |\alpha_-|^2 + |g(\theta)|^2 \sin^2 \theta |\alpha_+|^2 - 2 \sin \theta \operatorname{Re}[\alpha_+^* \alpha_- e^{-i\phi} g(\theta)^* f(\theta)]. \quad (10.21)$$

Note, in general this is different from (10.18).

- If the incident beam has polarization $\mathbf{P}_i$ and the spin of the scattered particle is not detected, the cross section is the sum of the preceding two (addition of probabilities):

$$\frac{d\sigma}{d\Omega}(0 \Leftarrow i) = \frac{d\sigma}{d\Omega}(\uparrow \Leftarrow i) + \frac{d\sigma}{d\Omega}(\downarrow \Leftarrow i) \quad (10.22)$$

$$= |f(\theta)|^2 + |g(\theta)|^2 \sin^2 \theta - 2 \operatorname{Im}[g(\theta)^* f(\theta)] \operatorname{Im}[\alpha_-^* \alpha_+ e^{i\phi}], \quad (10.23)$$

where we have assumed the initial state is normalized. Note that even if the final spin is undetected, the scattering exhibits a right-left asymmetry (ϕ dependence) arising from the initial polarization, and contains interference terms which make $d\sigma/d\Omega$ differ from the sum of spin-flip and non-flip cross sections.

Exercise Show that for a state in which $\alpha_+^* \alpha_- = 0$, the spin is aligned along the beam, the right-left asymmetry vanishes, and spin flip vanishes. $\diamond$

Exercise Explain in physical terms why these consequences should be related. $\diamond$

- There is an important connection to beam polarization in (10.23) that is somewhat obfuscated by all the complex algebra. If the final spin is not observed, the only way the cross section depends on the polarization is through a scalar product of $\mathbf{P}_i$ with another pseudovector (this preserves rotational invariance and parity invariance).¹ Because the only other pseudovector available is the normal to the scattering plane $\hat{\mathbf{n}} = \mathbf{k} \times \mathbf{k}' / \sin \theta$ in Figure 9.1, the polarization's angle dependence must be through

$$\mathbf{P}_i \cdot \hat{\mathbf{n}} = \mathbf{P}_i \cdot (-\sin \phi, \cos \phi, 0). \quad (10.24)$$

Exercise Verify that for beam polarization $\mathbf{P}_i$ given by (9.31),

$$\mathbf{P}_i \cdot \hat{\mathbf{n}} = -\sin \phi [2 \operatorname{Re}(\alpha_+^* \alpha_-)] - \cos \phi [2 \operatorname{Im}(\alpha_-^* \alpha_+)], \quad (10.25)$$

$$\begin{aligned} \frac{d\sigma}{d\Omega}(0 \Leftarrow i) &= |f(\theta)|^2 + |g(\theta)|^2 \sin^2 \theta \\ &\quad + 2 \mathbf{P}_i \cdot \hat{\mathbf{n}} \sin \theta \operatorname{Im}[g(\theta)^* f(\theta)]. \end{aligned} \quad \diamond \quad (10.26)$$

We have arrived at this expression for the cross section by explicitly enumerating the possible initial and final states and then summing them appropriately. We can also obtain this last result, and express it in a more transparent form, by directly utilizing the spin

¹Because the polarization is proportional to the expectation value of the spin, and because the spin being an angular momentum, is a pseudovector, the polarization $\mathbf{P}$ must be a pseudo- or axial vector.

structure of $\mathcal{F}$ (10.1). If an experiment does not detect the final state spin, this is equivalent to summing over all possible final-spin states, and so:

$$\frac{d\sigma}{d\Omega}(0 \leftarrow i) = \sum_{\chi_f} |\langle \chi_f | \mathcal{F} | \chi_i \rangle|^2 \quad (10.27)$$

$$= \sum_{\chi_f} \langle \chi_i | \mathcal{F}^\dagger | \chi_f \rangle \langle \chi_f | \mathcal{F} | \chi_i \rangle = \langle \chi_i | \mathcal{F}^\dagger \mathcal{F} | \chi_i \rangle \quad (10.28)$$

$$= \langle \chi_i | \left\{ |f(\theta)|^2 + (\boldsymbol{\sigma} \cdot \hat{\mathbf{n}})^\dagger \boldsymbol{\sigma} \cdot \hat{\mathbf{n}} \sin^2 \theta |g(\theta)|^2 + 2\boldsymbol{\sigma} \cdot \hat{\mathbf{n}} \sin \theta \operatorname{Im}[f(\theta)g(\theta)^*] \right\} | \chi_i \rangle, \Rightarrow \quad (10.29)$$

$$\frac{d\sigma}{d\Omega}(0 \leftarrow i) = |f(\theta)|^2 + |g(\theta)|^2 \sin^2 \theta + \mathbf{P}_i \cdot \hat{\mathbf{n}} \sin \theta \operatorname{Im}[g(\theta)^* f(\theta)], \quad (10.30)$$

where $\mathbf{P}_i$ is the polarization of the initial spin state.

Equation (10.30) is truly illuminating. If there is no initial polarization, that is, $\mathbf{P}_i = 0$, we have the incoherent sum of spin-flip and non-flip scattering:

$$\frac{d\sigma}{d\Omega}(0 \leftarrow 0) = |f(\theta)|^2 + |g(\theta)|^2 \sin^2 \theta. \quad (10.31)$$

If there is an initial polarization, there is a preferred direction to space and the ϕ dependence enters through $\mathbf{P}_i \cdot \hat{\mathbf{n}}$, (10.25).

Exercise Use the completeness relation (9.13), the identity $\boldsymbol{\sigma} \cdot \mathbf{A} \boldsymbol{\sigma} \cdot \mathbf{B} = \mathbf{A} \cdot \mathbf{B} + i\boldsymbol{\sigma} \cdot (\mathbf{A} \times \mathbf{B})$, and the definition of polarization to fill in the steps leading to (10.30). $\diamond$

Polarizations

We have seen the usefulness of treating $\mathcal{F}$, a matrix in the spin space, as a scattering “amplitude” whose matrix elements between spinors must be evaluated before obtaining scattering cross sections. We now use this shortcut to calculate the polarization of the scattered particle.

- We start by assuming the initial beam is in the coherent state χ_i with polarization $\mathbf{P}_i$. Because the scattering amplitude is the amplitude of the scattered wave, the scattered wave has a spin state,

$$|\chi_f\rangle = N\mathcal{F}|\chi_i\rangle, \quad (10.32)$$

where N is a normalization constant (which eventually cancels out). Note that $|\chi_f\rangle$ does *not* represent the full, final state, which also contains the unscattered part of the incident beam. By using $|\chi_f\rangle$ in the definition of polarization (9.26), we obtain the scattered particle’s polarization:

$$\mathbf{P}_f = \frac{\langle \chi_f | \boldsymbol{\sigma} | \chi_f \rangle}{\langle \chi_f | \chi_f \rangle} = \frac{\langle \chi_i | \mathcal{F}^\dagger \boldsymbol{\sigma} \mathcal{F} | \chi_i \rangle}{\langle \chi_i | \mathcal{F}^\dagger \mathcal{F} | \chi_i \rangle}, \quad (10.33)$$

where $\mathcal{F} = f + i\boldsymbol{\sigma} \cdot \hat{\mathbf{n}} \sin \theta g$. We have already evaluated the denominator and found it to be the differential cross section (10.30) for undetected final spin $d\sigma/d\Omega(0 \leftarrow i)$.

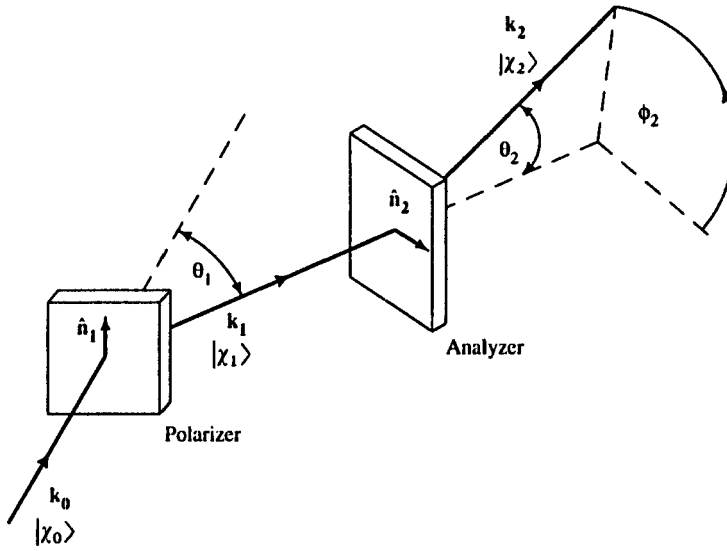


Figure 10.1: A double scattering experiment. The first scattering polarizes an unpolarized beam and the second scattering analyzes the polarization by measuring the azimuthal-angle ϕ_2 dependence of the scattering.

Evaluation of the numerator involves similar spin algebra; there results:

$$\mathbf{P}_f = \left[\frac{d\sigma(0 \leftarrow i)}{d\Omega} \right]^{-1} \left\{ \hat{\mathbf{n}} \left(2 \operatorname{Im}[g(\theta)^* f(\theta)] \sin \theta + 2 |g(\theta)|^2 \sin^2 \theta \hat{\mathbf{n}} \cdot \mathbf{P}_i \right) + \mathbf{P}_i \left(|f(\theta)|^2 - |g(\theta)|^2 \sin^2 \theta \right) + \mathbf{P}_i \times \hat{\mathbf{n}} \left(2 \operatorname{Re}[g^* h] \right) \right\}. \quad (10.34)$$

We see that $\mathbf{P}_f$ has components along all three possible pseudovectors.

- We generalize (10.33) to an initial state of partial polarization (and accordingly partial coherence) by letting $\mathbf{P}_i$ have a magnitude less than 1.
- For an unpolarized initial beam $\mathbf{P}_i = 0$, the polarization of the scattered beam is simply

$$\mathbf{P}_f(f \leftarrow 0) = \hat{\mathbf{n}} \frac{2 \operatorname{Im}[g(\theta)^* f(\theta)] \sin \theta}{|f(\theta)|^2 + |g(\theta)|^2 \sin^2 \theta}. \quad (10.35)$$

As could have been anticipated, the pseudovector $\mathbf{P}_f$ is proportional to the only other surviving pseudovector, $\hat{\mathbf{n}}$.

Polarization Analysis

In Chapter 1, *Scattering*, we indicated that differential cross sections are measured by counting the number of particles scattered into a detector. Analyzing the polarization of a beam is also possible by counting particles, as shown by applying the results of this chapter. Consider the experimental setup described in Figure 10.1. An unpolarized beam in state

$|\chi_0\rangle$ having momentum $\mathbf{k}_0$ is incident on target 1 (*the polarizer*) and gets scattered by the polar angle θ_1 into momentum $\mathbf{k}_1$ and spin state $|\chi_1\rangle$. The once-scattered beam, in turn, is scattered by target 2 (*the analyzer*) into an angle θ_2 and momentum $\mathbf{k}_2$. We assume the initial polarization $\mathbf{P}_0 = 0$, while the polarizations $\mathbf{P}_1$ and $\mathbf{P}_2$ of the scattered beams are what we wish to measure. The normals to the first and second scattering planes,

$$\hat{\mathbf{n}}_1 = \frac{\mathbf{k}_0 \times \mathbf{k}_1}{\sin \theta_1}, \quad \hat{\mathbf{n}}_2 = \frac{\mathbf{k}_1 \times \mathbf{k}_2}{\sin \theta_2}, \quad (10.36)$$

are known. The scattered state $|\chi_1\rangle$ is related to the initial state by

$$|\chi_1\rangle = N \mathcal{F}_1(\theta_1) |\chi_0\rangle, \quad (10.37)$$

where $\mathcal{F}_1(\theta_1)$ is the spin-space scattering amplitude (10.1) and N is a normalization constant.

The cross section for scattering at center 1 is that describing a nonpolarized beam (10.31):

$$\frac{d\sigma}{d\Omega}(1 \Leftarrow 0) = |f(\theta_1)|^2 + |g(\theta_1)|^2 \sin^2 \theta_1. \quad (10.38)$$

We know from (10.35) that the first scattering polarizes the beam, so the cross section for scattering at center 2 is that describing a polarized beam (10.30):

$$\begin{aligned} \langle \chi_1 | \chi_1 \rangle \frac{d\sigma}{d\Omega}(2 \Leftarrow 1) = & \langle \chi_1 | \left\{ |f(\theta_2)|^2 + |g(\theta_2)|^2 \sin^2(\theta_2) \right. \\ & \left. + 2\sigma \cdot \hat{\mathbf{n}}_2 \sin \theta_2 \operatorname{Im}[f(\theta_2)g^*(\theta_2)] \right\} | \chi_1 \rangle. \end{aligned} \quad (10.39)$$

In (10.39) the matrix element of $d\sigma/d\Omega$ between the bra $\langle \chi_2 |$ and the ket $|\chi_1\rangle$ equals the polarization $\mathbf{P}_1$, while the bracket on the LHS handles the possibility of χ_1 not being normalized. If we substitute for χ_1 and $\mathcal{F}_1$, we obtain

$$\begin{aligned} \frac{d\sigma}{d\Omega}(2 \Leftarrow 1) = & \frac{|N|^2 \langle \chi_0 | \mathcal{F}_1^\dagger \{ \cdots \} \mathcal{F}_1 | \chi_0 \rangle}{|N|^2 \langle \chi_0 | \mathcal{F}_1^\dagger \mathcal{F}_1 | \chi_0 \rangle} \\ = & [|f(\theta_2)|^2 + |g(\theta_2)|^2 \sin^2(\theta_2)] [1 + P_2(\theta_2)P_1(\theta_1)\hat{\mathbf{n}}_2 \cdot \hat{\mathbf{n}}_1]. \end{aligned} \quad (10.40)$$

This rather simple expression tells us that the scattering at target 2 has a linear dependence on the cosine of the angles between the normal to the scattering planes in Figure 10.1 (the familiar $\cos \theta$ dependence of crossed polarizers).

A simple experiment that applies this result is sketched in Figure 10.2 where we examine only those double scatterings having parallel scattering planes, that is, for which $\hat{\mathbf{n}}_1 \cdot \hat{\mathbf{n}}_2 = \pm 1$. In Figure 10.2A we look at scatterings to the left of target 2 and in Figure 10.2B to the right. With this configuration it is possible to measure the *asymmetry* parameter e by the difference in right (R) and left (L) scattering:

$$e = \frac{\sigma_{21}(L) - \sigma_{21}(R)}{\sigma_{21}(L) + \sigma_{21}(R)}, \quad (10.41)$$

where σ_{21} is the cross section for fixed values of θ_1 and θ_2 , and the R and L values correspond to different ϕ values. Because e is a ratio, this measurement does not require

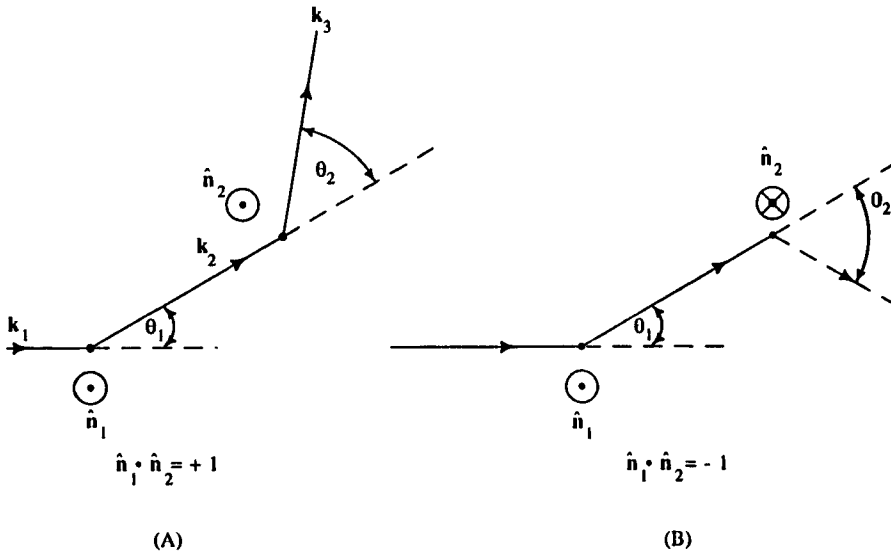


Figure 10.2: The normals to the scattering planes in double scattering. In (A) both $\hat{n}_1$ and $\hat{n}_2$ are out of the page (left scattering by analyzer), while in (B) $\hat{n}_1$ is out and $\hat{n}_2$ is in (right scattering by the analyzer).

the knowledge of beam normalization and detector acceptance angles required in absolute, cross-section measurements [overall factors cancel out of the ratio (10.41)]. When we substitute (10.40) for the cross sections into (10.41), we obtain an asymmetry

$$e = P_2(\theta_2)P_1(\theta_1), \quad (10.42)$$

which is the product of the polarizations of the intermediate and final beams. In practice, a target of known polarizability $P_1(\theta_1)$ could be used at target 1 (*the polarizer*) and then the polarization at target 2 (*the analyzer*) measured. Alternatively, the geometries and equipment could be arranged to ensure that the two targets are equivalent so $P_1(\theta_1) = P_2(\theta_2)$. In this case $P_2(\theta_2) = \sqrt{e}$ and P_2 is determined directly, or the polarizer is calibrated.

10.3 The Density Matrix

Those readers who have worked through the preceding derivations of spin observables for various states of polarizations may well be wondering “Is there not a more mechanical way to determine spin observables which avoids detailed analysis for each individual case?” Well there is and it is called the density matrix formalism. This formalism also addresses the classical “problem” with quantum mechanics that we must go through lots of work to calculate wave functions, but since wave functions are not observables, we must go through yet more work to reduce the information in the wave function to obtain an observable. It clearly would be more direct to deal with observables than with nonobservables. The density matrix formalism does this too.

The density matrix is a particularly valuable tool for dealing with the complications of mixed states, that is, with statistical ensembles in which probabilities and not amplitudes

add. You will recall from the last chapter that a pure or coherent spin state is one which can be expressed in the explicit form

$$|\chi\rangle = \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_N \end{pmatrix}. \quad (10.43)$$

Here $\alpha_j = \langle j | \chi \rangle$ is the probability amplitude for state $|j\rangle$ to be present in $|\chi\rangle$, and $N = (2s_1 + 1)(2s_2 + 1)$ is the total dimension of the spin space for a particle of spin s_1 interacting with one of spin s_2 . Other than being a complete set of states in spin space, there is no restriction on what the states $|j\rangle$ must be. For the spin $0 \times \frac{1}{2}$ examples considered in this chapter, the dimension $N = 2$ and the spin part of the state vector has the form

$$|\chi\rangle = \begin{pmatrix} \alpha_+ \\ \alpha_- \end{pmatrix} = \begin{pmatrix} |\alpha_+| e^{i\phi_+} \\ |\alpha_-| e^{i\phi_-} \end{pmatrix}, \quad (10.44)$$

where the plus and minus indicate $\pm \frac{1}{2}$ as the total spin's z component.

For pure states, the expectation value of the polarization vector always equals unity, $\langle \mathbf{P} \rangle = \langle \chi | \mathbf{P} | \chi \rangle = 1$. This means the direction of polarization can be modified but not its magnitude. For a mixed state, the polarization is defined to be a weighted sum of the polarizations of the coherent states mixed into the beam. For the $0 \times \frac{1}{2}$ system this is given by (9.33):

$$\mathbf{P} = \sum_i w_i \mathbf{P}_i = \sum_i w_i \left[2 \operatorname{Re}(\alpha_+^{i*} \alpha_-^i) \hat{\mathbf{e}}_x + 2 \operatorname{Im}(\alpha_+^{i*} \alpha_-^i) \hat{\mathbf{e}}_y + (|\alpha_+^i|^2 - |\alpha_-^i|^2) \hat{\mathbf{e}}_z \right]. \quad (10.45)$$

The magnitude of the $\mathbf{P}$ defined by (10.45) can have any value between 0 and 1.

We start our discussion of the density matrix for the simpler case of pure states, but we shall see that the key relation is also true for mixed states. For each “state” of a beam, be it pure or mixed, there is a density matrix. If anything modifies the state, such as a scattering event, the density matrix changes. This means there is a different density matrix for the initial and final states. For a coherent state expressed as (10.43), the density matrix is defined to be

$$\rho = [\rho_{jk}] \stackrel{\text{def}}{=} [\alpha_j \alpha_k^*]. \quad (10.46)$$

So ρ is a matrix in which the diagonal elements are probability *densities*, and in which the off-diagonal elements contain information about the *phases* within the beam. We see from this definition that the density matrix is generally a complex but Hermitian ($\rho_{jk}^* = \rho_{kj}$) matrix. For the spin $0 \times \frac{1}{2}$ system, the density matrix is

$$\rho = \begin{pmatrix} |\alpha_+|^2 & \alpha_+ \alpha_-^* \\ \alpha_- \alpha_+^* & |\alpha_-|^2 \end{pmatrix}. \quad (10.47)$$

For spin up and down states, ρ is diagonal with the values

$$\rho(\uparrow) = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad \rho(\downarrow) = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}. \quad (10.48)$$

An examination of (10.44) again shows us that “problem” of quantum mechanics requiring four real numbers to define the spin state χ of our system, yet experimental measurements yielding only two real numbers (e.g., the relative probability $|\alpha_+|^2/|\alpha_-|^2$ and the relative phase $\phi_+ - \phi_-$). We can make sense of this, and see how the density matrix improves the situation, by noting that one real number is always determined by the normalization of the state:

$$\langle \chi | \chi \rangle = 1 \Rightarrow |\alpha_+|^2 + |\alpha_-|^2 = 1. \quad (10.49)$$

This, in turn, means that for normalized states the density matrix is constrained to have a unit trace,

$$\text{Tr } \rho = |\alpha_+|^2 + |\alpha_-|^2 = 1. \quad (10.50)$$

If the condition (10.50) is not met, it can be imposed afterward as one does with wave function normalization. Note, however, that the absolute phase ambiguity of quantum mechanics is handled automatically by the density matrix (10.47); setting $\alpha_i = e^{i\phi} \alpha_i$ does not affect the density matrix or observables. Thus there are only two independent elements in the density matrix (10.47), and if we can find a way to express all observables in terms of ρ , then some of our hardships will be ended.

Relation to Observables

Let us say we have an operator $\tilde{O}$ with some arbitrary spin dependence in it and we want to evaluate its expectation value in the general spin state (10.44). In the conventional approach we do it by sandwiching $\tilde{O}$ between $|\chi\rangle$ and $\langle\chi|$:

$$\begin{aligned} \langle \tilde{O} \rangle &= \langle \chi | \tilde{O} | \chi \rangle = \begin{pmatrix} \alpha_+^* & \alpha_-^* \end{pmatrix} \begin{pmatrix} O_{++} & O_{+-} \\ O_{-+} & O_{--} \end{pmatrix} \begin{pmatrix} \alpha_+ \\ \alpha_- \end{pmatrix} \\ &= |\alpha_+|^2 O_{++} + \alpha_+ \alpha_-^* O_{-+} + \alpha_- \alpha_+^* O_{+-} + |\alpha_-|^2 O_{--}. \end{aligned} \quad (10.51)$$

Yet we get the same result by evaluating the trace of the matrix product:

$$\begin{aligned} &|\alpha_+|^2 O_{++} + \alpha_+ \alpha_-^* O_{-+} + \alpha_- \alpha_+^* O_{+-} + |\alpha_-|^2 O_{--} \\ &= \text{Tr} \left\{ \begin{pmatrix} |\alpha_+|^2 & \alpha_+ \alpha_-^* \\ \alpha_- \alpha_+^* & |\alpha_-|^2 \end{pmatrix} \begin{pmatrix} O_{++} & O_{+-} \\ O_{-+} & O_{--} \end{pmatrix} \right\}. \end{aligned} \quad (10.52)$$

This result is true for all spins and can be written as

$$\boxed{\langle \chi | \tilde{O} | \chi \rangle = \text{Tr} \{ \rho \tilde{O} \} \equiv \text{Tr} \{ \tilde{O} \rho \},} \quad (10.53)$$

where ρ clearly have some functional dependence on the state $|\chi\rangle$.

Equation (10.53) is just what we have been searching for. It is a recipe for calculating the expectation value of a general spin-dependent operator between a general spin state; we need only multiply the matrix of $\tilde{O}$'s matrix elements by the density matrix and evaluate a trace. Its generality follows from the general relation of a spin state to the $(2s_1 + 1)(2s_2 + 1)$ orthonormal basis vectors which span the spin space:

$$|\chi\rangle = \sum_j |j\rangle \langle j | \chi \rangle = \sum_j \alpha_j |j\rangle, \quad (10.54)$$

$$\begin{aligned}\Rightarrow \langle \tilde{O} \rangle &= \sum_{j,k} \langle \chi | k \rangle \langle k | \tilde{O} | j \rangle \langle j | \chi \rangle = \sum_{j,k} \alpha_k^* \alpha_j O_{jk} = \sum_{j,k} \rho_{kj} O_{jk}, \\ \langle \tilde{O} \rangle &= \text{Tr } \rho \tilde{O}.\end{aligned}\quad (10.55)$$

We see that even for this general spin case, the diagonal elements of ρ are probability densities, and this means $\text{Tr } \rho = 1$ still holds. We further note, that for a pure state the density matrix is *idempotent*, that is, $\rho^2 = \rho$:

$$(\rho^2)_{jk} = \sum_m \rho_{jm} \rho_{mk} = \sum_m \alpha_j^* \alpha_m \alpha_m^* \alpha_k = \alpha_j^* \alpha_k = \rho_{jk}, \quad (10.56)$$

$$\Rightarrow \rho^2 = \rho \quad (\text{pure state}). \quad (10.57)$$

The basic observable for a polarized beam is the polarization $\mathbf{P} = \langle \boldsymbol{\sigma} \rangle$. It has a simple relation to the density matrix ρ and the unit matrix I :

$$\rho = \frac{1}{2} (I + \mathbf{P} \cdot \boldsymbol{\sigma}). \quad (10.58)$$

Exercise Verify (10.58) by noting that

$$\rho = \frac{1}{2} (I + \mathbf{P} \cdot \boldsymbol{\sigma}) = \frac{1}{2} \begin{pmatrix} 1 + P_z & P_x - iP_y \\ P_x + iP_y & 1 - P_z \end{pmatrix}, \quad (10.59)$$

and that

$$\mathbf{P} = \text{Tr } \rho \boldsymbol{\sigma} = [2\text{Re}(\alpha_+^* \alpha_-), 2\text{Im}(\alpha_+^* \alpha_-), (|\alpha_+|^2 - |\alpha_-|^2)]. \quad \diamond \quad (10.60)$$

The relation (10.58) makes it clear that the density matrix is the sum of one term which depends linearly on the polarization of the beam, and another term which is proportional to the unit matrix. In summary, the density matrix is diagonal if the polarization is along the z axis, contains nondiagonal elements if there are x or y components of polarization, and is proportional to the unit matrix if there is no polarization.

Mixed State Density Matrix

The real utility of the density matrix is in evaluating expectation values for mixed states, that is, for incoherent states in which $|\langle \mathbf{P} \rangle| \neq 1$. The basis of the description is the definition of the expectation value for such a state as an ensemble average, which we denote by a bar,

$$\overline{\langle \tilde{O} \rangle} \stackrel{\text{def}}{=} \sum_i w_i \langle \chi(i) | \tilde{O} | \chi(i) \rangle. \quad (10.61)$$

Here the mixed state is composed of the individual coherent states, the $\chi(i)$'s, each occurring with relative probability w_i . The individual $\chi(i)$'s need not be orthogonal, but they do need be normalized, and the relative probabilities must add to 1, $\sum_i w_i = 1$. By substituting the general relation (10.43) for χ , we obtain a relation to the density matrix elements in each

$\chi(i)$:

$$\begin{aligned}\overline{\langle \tilde{O} \rangle} &= \sum_{i,n,m} w_i \alpha_n^*(i) \alpha_m(i) \langle n | \tilde{O} | m \rangle = \sum_{i,n,m} w_i \rho_{nm}(i) \langle n | \tilde{O} | m \rangle \\ &= \sum_{n,m} \left[\sum_i w_i \rho_{nm}(i) \right] \langle n | \tilde{O} | m \rangle.\end{aligned}\quad (10.62)$$

An examination of (10.62) reveals that each ρ_{nm} is being averaged independently over all the states comprising the beam, and so we have an *ensemble average* of the density matrix:

$$\bar{\rho}_{nm} \stackrel{\text{def}}{=} \sum_i w_i \rho_{nm}(i) \equiv \sum_i w_i \alpha_n^*(i) \alpha_m(i). \quad (10.63)$$

With this ensemble-average notation, the expectation value of a general operator $\tilde{O}$ takes the same form as we had before:

$$\overline{\langle \tilde{O} \rangle} = \sum_{n,m} \bar{\rho}_{nm} \langle n | \tilde{O} | m \rangle = \text{Tr } \bar{\rho} \tilde{O}. \quad (10.64)$$

As an example, we investigate the form of the density matrix for the scattered beam. If we do not worry about normalization for the moment, then knowledge that the scattering amplitude f_{jm} is the amplitude of the final state $|j\rangle$ when the initial state is $|m\rangle$, means that the states are related by

$$\alpha_j = \sum_m f_{jm}(\theta, \phi) \alpha_m. \quad (10.65)$$

We now insert this into the definition (10.46) to obtain the density matrix for the final state:

$$\rho_{jk}^{(f)} = \sum_m (f_{jm} \alpha_m) \sum_l (f_{kl} \alpha_l)^* = \sum_{l,m} f_{jm} \rho_{ml}^{(i)} f_{kl}, \quad (10.66)$$

$$\Rightarrow \rho^{(f)} = \tilde{f}(\theta, \phi) \rho^{(i)} \tilde{f}^\dagger(\theta, \phi). \quad (10.67)$$

Equation (10.67) indicates that the initial and final density matrices are related by an orthogonal transformation with the scattering matrix generating the transformation. Since, in general, $\tilde{f}$ is not a unitary matrix, expectation values calculated with $\rho^{(f)}$ need be divided by $\text{Tr } \rho^{(f)}$ to ensure normalization. In fact, if we evaluate the trace we find that it equals the differential cross section:

$$\text{Tr } \rho^{(f)} = \text{Tr } [\tilde{f} \rho^{(i)} \tilde{f}^\dagger] = \sum_j \left| \sum_i f_{ij} \alpha_j \right|^2 = \sum_j \frac{d\sigma_j}{d\Omega} = \frac{d\sigma}{d\Omega}(\theta, \phi). \quad (10.68)$$

10.4 Extensions for Identical Particles

If the particles in the target and incident beam are identical, it is impossible to tell if we are observing a scattered beam particle or a recoiling target particle, as we indicate in Figure 10.3. For classical scattering, the additional particles increase the cross section relative to the nonidentical result by making σ symmetric about $\theta = \pi/2$:

$$\sigma(\theta)_{\text{class}} = \sigma(\theta) + \sigma(\pi - \theta). \quad (10.69)$$

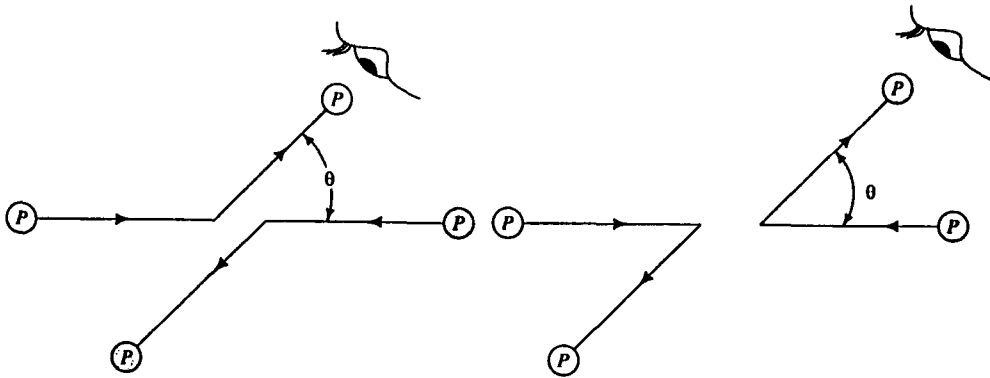


Figure 10.3: Two indistinguishable scatterings of identical particles P (the shading is invisible to experimentalists).

For quantum scattering, interference occurs at the amplitude level and can result in dramatic changes.

In Chapter 1, *Scattering*, we assumed that when the projectile and target particles interact, their spatial wave function has the form [(1.17), (1.18), and (1.25)]:

$$\Psi(\mathbf{r}_T, \mathbf{r}_P) = \psi(\mathbf{r})\phi_{\text{CM}}(\mathbf{R}), \quad (10.70)$$

$$\mathbf{r} = \mathbf{r}_P - \mathbf{r}_T, \quad \mathbf{R} = \frac{m_P \mathbf{r}_P + m_T \mathbf{r}_T}{m_T + m_P}. \quad (10.71)$$

So far we have been solving for the relative wave function $\psi(\mathbf{r})$ with ϕ_{CM} , the overall CM plane wave, not affecting ψ . In Chapter 9, *Spin Theory*, we extended Ψ by including the spin-space wave function:

$$\Psi(T, P) = \psi(\mathbf{r}) \phi_{\text{CM}}(\mathbf{R}) \chi(T, P). \quad (10.72)$$

The exclusion principle requires that the interchange of two identical particles change the overall sign of the wave function,

$$\Psi(T, P) = \begin{cases} +\Psi(P, T), & \text{for bosons,} \\ -\Psi(P, T), & \text{for fermions,} \end{cases} \quad (10.73)$$

where the interchange includes *all* coordinates: space, spin, isospin, color, and so on.

Spin Zero—Spin Zero

It is an experimental fact that particles with integer spins obey Bose statistics. Because the spin wave function of two spin-zero particles is always even under interchange, the required plus sign in (10.73) means that the spatial wave function must also be even, that is,

$$\Psi(T, p) = \phi_{\text{CM}}(\mathbf{R}) [\psi(\mathbf{r}) + \psi(-\mathbf{r})] \chi_{\text{even}}(T, P), \quad (10.74)$$

where $\psi(\mathbf{r})$ is the (unsymmetrized) wave function we calculate with the Schrödinger equation. The symmetry in (10.74) carries over to the scattered wave, hence to the

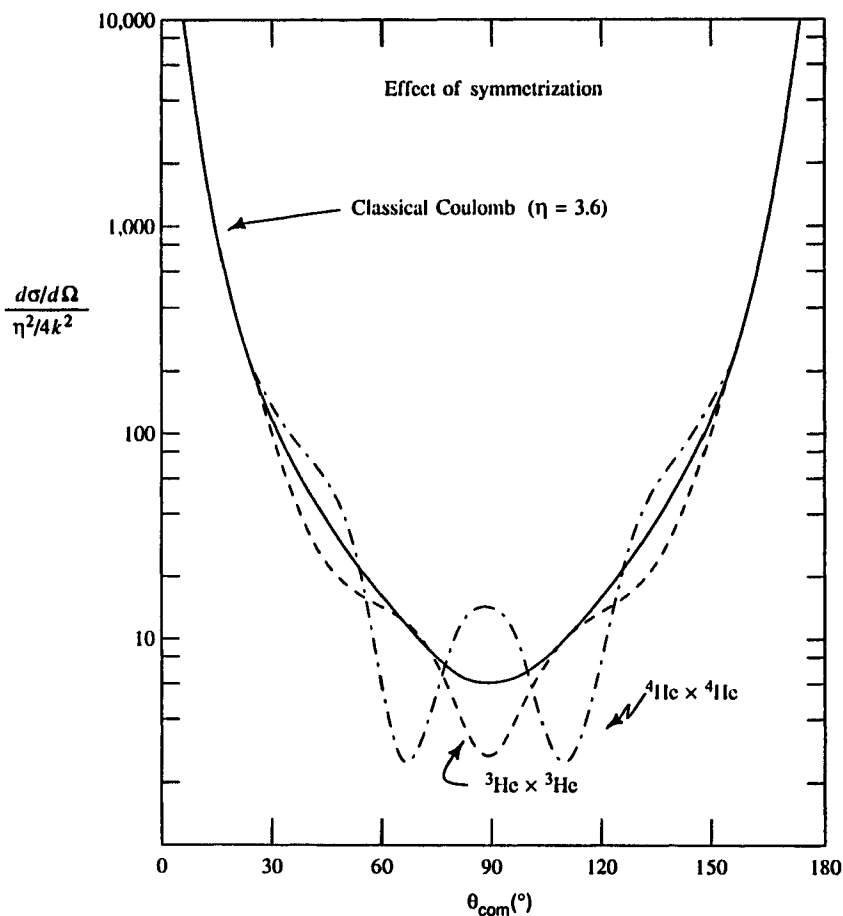


Figure 10.4: The effect of symmetrization on the Coulomb scattering of two $Z = 2$ particles. The solid curve is the classical, symmetrized result (10.69), the long-dashed curve is the quantum result for two bosons (10.77), and the short-dashed curve is the quantum result for two fermions (10.82). Note the large differences at 90° .

scattering amplitude $f_E(\theta, \phi) \equiv f_E(\mathbf{k}', \mathbf{k})$, and finally to the cross section:

$$\frac{d\sigma}{d\Omega}(\theta, \phi) = |f_E(\mathbf{k}', \mathbf{k}) + f_E(-\mathbf{k}', \mathbf{k})|^2. \quad (10.75)$$

Exercise Show that with the geometry of Figure 10.3, this symmetrization condition is

$$\begin{aligned} \frac{d\sigma}{d\Omega} &= |f_E(\theta, \phi) + f_E(\pi - \theta, \pi + \phi)|^2 \\ &= |f_E(\theta, \phi)|^2 + |f_E(\pi - \theta, \pi + \phi)|^2 + 2 \operatorname{Re}[f_E(\theta, \phi)f_E^*(\pi - \theta, \pi + \phi)] \\ &= \sigma(\theta, \phi) + \sigma(\pi - \theta, \pi + \phi) + 2 \operatorname{Re}[f_E(\theta, \phi)f_E^*(\pi - \theta, \pi + \phi)]. \quad \diamond \end{aligned} \quad (10.76)$$

The dashed curve in Figure 10.4 shows an example of the symmetrized Coulomb scattering of two ^4He nuclei for which (10.77) modifies the Rutherford cross section (4.49) to

$$\sigma = \frac{\eta^2}{4k^2} \left(\frac{1}{\sin^4(\theta/2)} + \frac{1}{\cos^4(\theta/2)} + \frac{8 \cos[2\eta \ln(\tan \theta/2)]}{\sin^2 \theta} \right). \quad (10.77)$$

Because the nuclei are spin 0, they are bosons and have a cross section which is symmetric about 90° . Note the differential cross section at 90° equals four times the unsymmetrized σ , and this is twice what is expected classically (solid curve) from (10.69).

Exercise Show that the total cross section agrees with the classical result. $\diamond$

If we undertook a partial-wave analysis of the cross section for boson–boson scattering (10.75) (the dashed curve in Figure 10.4), we would find that only even l values enter:

$$\begin{aligned} \frac{d\sigma}{d\Omega}(\theta, \phi) &= \left| \frac{1}{k} \sum_{l=0}^{\infty} (2l+1) e^{i\delta_l} \sin \delta_l [P_l(\cos \theta) + P_l(-\cos \theta)] \right|^2 \\ &= \left| \frac{1}{k} \sum_{l=0}^{\infty} (2l+1) e^{i\delta_l} \sin \delta_l [P_l(\cos \theta) + (-1)^l P_l(\cos \theta)] \right|^2 \\ &= \left| \frac{2}{k} \sum_{l=0,2,4,\dots}^{\infty} (2l+1) e^{i\delta_l} \sin \delta_l P_l(\cos \theta) \right|^2. \quad (10.78) \end{aligned}$$

This means that it is impossible to measure or even define odd- l phase shifts for boson–boson scattering.

Spin One Half–Spin One Half

It is a fact of nature that all half-integer particles obey Fermi statistics. In the Problems section of Chapter 9, *Spin Theory*, we studied the scattering of two spin $\frac{1}{2}$ particles (fermions) and found that the Schrödinger equation separates into equations for total spin $S = 1$ (the “triplet” state) and an equation for total spin $S = 0$ (the “singlet”).

Exercise Show by adding two spin $\frac{1}{2}$'s, that the symmetry of the singlet-state spin wave function is odd, while that of the triplet state is even under interchange of the fermions. $\diamond$

For the complete wave function to be antisymmetric, the spatial wave function for the singlet spin state must be even, while that for the triplet spin-state must be odd. Fermions thus appear to “attract” each other (symmetric wave function) one-fourth of the time and “repel” each other three-fourths of the time.

Exercise Follow the reasoning of the boson–boson case to show that the symmetry of the fermion–fermion wave function translates into the corresponding symmetry for the singlet and triplet scattering amplitudes (matrices) $\mathcal{F}_s$ and $\mathcal{F}_t$:

$$\mathcal{F} = [\mathcal{F}_s(\mathbf{k}', \mathbf{k}) + \mathcal{F}_s(-\mathbf{k}', \mathbf{k})] + [\mathcal{F}_t(\mathbf{k}', \mathbf{k}) - \mathcal{F}_t(-\mathbf{k}', \mathbf{k})]. \quad \diamond \quad (10.79)$$

In the simplest case of an unpolarized beam, unpolarized target, and unobserved final spins, the cross section is the statistically-weighted sum of singlet and triplet scattering:

$$\frac{d\sigma}{d\Omega}(\theta, \phi) = \frac{1}{4} |f_s(\theta, \phi) + f_s(\pi - \theta, \pi + \phi)|^2 + \frac{3}{4} |f_t(\theta, \phi) - f_t(\pi - \theta, \pi + \phi)|^2. \quad (10.80)$$

Analogous to the boson case, the singlet term in (10.80) is a sum over only even l 's, while the triplet term is a sum over only odd l 's. The resulting cross section contains mixed symmetry with respect to θ .

Exercise Show that for spin $\frac{1}{2} \times \frac{1}{2}$ scattering with spin-independent forces, $f_s = f_t = f$, and therefore,

$$\frac{d\sigma}{d\Omega}(\theta, \phi) = \sigma(\theta, \phi) + \sigma(\pi - \theta, \pi + \phi) - \text{Re}[f(\theta, \phi)f^*(\pi - \theta, \pi + \phi)]. \quad \diamond \quad (10.81)$$

This differs from the boson–boson case by the sign of the interference term, and in this case produces a minimum at 90° . For example, the dot-dashed curve in Figure 10.4 shows the Coulomb scattering of two ^3He nuclei, where (10.81) modifies the Rutherford result to

$$\sigma = \frac{\eta^2}{4k^2} \left[\frac{1}{\sin^4(\theta/2)} + \frac{1}{\cos^4(\theta/2)} - \frac{4 \cos[2\eta \ln(\tan \theta/2)]}{\sin^2 \theta} \right]. \quad (10.82)$$

Inasmuch as the ^3He – ^3He Coulomb interaction is identical to that in ^4He – ^4He , the difference in cross sections between (10.77) and (10.82) is a striking example of quantum statistics.

10.5 Problems

1. We have seen that the scattering amplitude for spin $0 \times \frac{1}{2}$ can be considered as the operator in spin space,

$$\mathcal{F} = f(\theta) + i\boldsymbol{\sigma} \cdot \hat{\mathbf{n}} \sin \theta g(\theta), \quad (10.83)$$

where $\hat{n}$ is a unit vector in the $\mathbf{k} \times \mathbf{k}'$ direction, $\mathbf{k}'$ is in the xz plane, and $\mathbf{k}$ in the z direction (Figure 9.3A).

We now wish to consider the spin quantized along the z axis (like Figure 9.3B), say as a consequence of an experimental spin rotation. Show the following by calculating $d\sigma/d\Omega$:

- (a) There is now no “spin-flip” scattering.
 - (b) There is a spin dependence of the scattering; that is, up and down spins lead to different cross sections.
 - (c) There is a “right-left asymmetry;” that is, $d\sigma/d\Omega$ depends on whether $\mathbf{k}'$ has a positive or negative projection along the x axis.
2. The ^{12}C nucleus has spin 0 while the ^{13}C nucleus, with an extra neutron, has spin $\frac{1}{2}$. (You may assume for this problem, that the nuclear binding forces are *not* spin dependent).
- (a) What is the numerical value for the differential cross section for pure Coulomb (Rutherford) scattering of ^{12}C from ^{13}C ? Evaluate it at 90 degrees.
 - (b) What is the differential cross section for pure Coulomb scattering of ^{12}C from ^{12}C ? Evaluate it at 90 degrees.
 - (c) What is the differential cross section for pure Coulomb scattering of ^{13}C from ^{13}C ? Evaluate it at 90 degrees.
3. We have studied the scattering of spin 0 and $\frac{1}{2}$ particles on each other and based much phenomenology on the expression for the scattering amplitude as a matrix in spin space:

$$\mathcal{F} = f(\theta) + i\boldsymbol{\sigma} \cdot \hat{n} g(\theta) \sin \theta. \quad (10.84)$$

- (a) Is this the most general form of the scattering amplitude possible? Specifically, why not include terms like $\boldsymbol{\sigma} \cdot \mathbf{k}$ and $(\boldsymbol{\sigma} \cdot \hat{n})^2$?
 - (b) Let the spin be quantized along the z axis and the incident beam be along the z axis. What are the components of $\hat{n}$ for this case?
 - (c) Use the matrix $\mathcal{F}$ to calculate the probability amplitude for the scattering of a spin-down system to one with spin up.
 - (d) Use this matrix to calculate the probability amplitude for the scattering of a spin-up system to one with spin down.
 - (e) Use this matrix to calculate the cross section for the scattering of a spin-up system to a final state in which the spin is not observed.
 - (f) Explain in physical terms why there should not be a right-left asymmetry in the scattering of (3e).
4. Determine the polarization for a spin $0 \times \frac{1}{2}$ system when it is described by the (unnormalized) density matrices:

$$\rho = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}. \quad (10.85)$$

5. Use the density matrix formalism to derive the differential cross section for the scattering of a polarized beam to a spin up state.
6. Use the density matrix formalism to derive the differential cross section for the scattering of a polarized beam to an unpolarized state.
7. Use the density matrix formalism to derive the final polarization arising from the scattering of a polarized beam.
8. When light scatters from an atom or molecule, it can lose or gain energy if the bound system changes state. This *Raman effect* is used to investigate the vibrational and rotational level structures of molecules. For rotational transitions in diatomic molecules, there is the selection rule $\Delta K = 2$, where $K(K + 1)$ is the eigenvalue of the square of the nuclear orbital angular momentum. Show that the energy shift for the observed lines should be given by

$$\Delta E = \pm B(2K + 3) \quad (10.86)$$

where B is a constant and K is the rotational quantum number of the lower state involved.

9. Consider again Raman scattering of light (preceding problem). If the two nuclei of a diatomic atom are identical, the required symmetry of the molecular wave function restricts the values possible for the nuclear orbital angular momentum and for the nuclear spin.

- (a) The rotational Raman spectrum of the O_2 molecule is observed to fit the formula:

$$E = \pm B(4n + 5), \quad (10.87)$$

where B is a constant and n is not an integer. If the electronic wave function is an odd function of $\mathbf{R}_{12}$ (the vector connecting the two nuclei), deduce the type of statistics that the spin-zero ^{16}O nucleus must obey.

- (b) In the H_2 molecule, the nuclei are protons of spin $\frac{1}{2}$ and the electronic wave function is even. Show that if a gas of molecules is formed from protons with random spin orientations, there will be an alternation in intensity of the rotational Raman lines of about 3:1.
 - (c) Which lines will be more intense?
 - (d) The rotational Raman spectrum of N_2 is observed to have an alternation in intensity of about 2:1, the line with the least shift being more intense than its neighbor. Find the spin and statistics of the nitrogen nucleus.
10. Is there an “exclusion principle” for bosons? Explain.

CHAPTER 11

PATH INTEGRALS AND LATTICE QUANTUM MECHANICS

*"How does it work? What is the machinery behind the law?"
No one has found any machinery behind the law.
No one can "explain" any more than we have just "explained."
No one will give you any deeper representation of the situation.
We have no ideas about a more basic mechanism
from which these results can be deduced.*

—Richard P. Feynman

11.1 A New View

We assume throughout this book that the reader is familiar with the basic philosophical and mathematical postulates of quantum mechanics and that he or she will benefit from a simple and direct path to the physics of any topic. For those reasons we have followed the time-independent approach to quantum mechanics for scattering and will follow the historical perturbation-theory approach to field theory. In the present chapter we make some exceptions. First, we explicitly include time as a dynamical element in quantum mechanics, and second, we study Feynman's formulation of quantum mechanics in terms of path integrals (Feynman, 1948; Feynman et al., 1965; Feynman and Hibbs, 1965). Part of that formulation envisions solving the dynamical equations on a grid of points throughout space and time, as one might do in solving an integral equation on a computer. In recent times the latter is exactly what has been happening, and so in the next chapter we examine a computing project dealing with an explicit evaluation of a path-integral as a Monte Carlo calculation on a lattice.

While many physicists have grown up with Feynman diagrams and their unique view of particle motion forwards and backwards in spacetime, it is valuable to understand how this picture evolves from the familiar Schrödinger equation. We study this path integral formulation not because it is an easy way to do calculations, for actually, if the situation permits the use of operators and the Schrödinger equation then that is probably easier.

in presenting an alternative point of view, more general than the Schrödinger equation, and may be our only alternative when all else fails. By more general we mean that the same path integral formulation can be used for nonrelativistic quantum mechanics and relativistic quantum mechanics, as well as second quantization (field theory), with or without perturbation expansions being made. Yet the path-integral formulation is more than just a nice presentation, it makes renormalization easier than with conventional theories and is presently the only way to solve non-Abelian gauge field theories.

The theory and calculations of this chapter will be simple; we start by taking the formalism used elsewhere in this book and show how it leads to Feynman's postulates. In formulating the theory, we limit ourselves to a single, nonrelativistic, spinless particle in one space dimension. Even if the reader is not planning to implement a numerical computation of a path integral sum, he or she may benefit from passing through the following chapter with its somewhat more applied viewpoint. Its concrete representation may make clearer some of the abstract ideas of this chapter. Once the reader understands the physics and techniques of the 1D quantum mechanics problem, it is basically just "accounting" to increase the number of dimensions or the complexity of the potential. We leave to the references the extensions to field theory and to fermions.

11.2 Propagation in Spacetime

"How does the Schrödinger equation actually advance a particle from time t_a to time t_b ?" In its time-dependent form,

$$i \frac{\partial \psi(\mathbf{x}, t)}{\partial t} = \left[-\frac{\nabla^2}{2\mu} + V(\mathbf{x}, t) \right] \psi(\mathbf{x}, t), \quad (11.1)$$

it clearly relates the space and time derivatives of ψ and the effect of the potential V on ψ , and so this must be the answer for infinitesimal advances. In its operator form,

$$i \frac{\partial \psi(\mathbf{x}, t)}{\partial t} = \hat{H} \psi(\mathbf{x}, t), \quad (11.2)$$

we would say that the Schrödinger equation relates ψ 's time derivative to the action of the Hamiltonian. Even though the $\hat{H}$ in (11.2) is an operator, one way to understand the nature of the time dependence in ψ is to integrate (11.2) while treating $\hat{H}$ as a constant:

$$\psi(\mathbf{x}, t) = e^{-i\hat{H}t} \psi(\mathbf{x}, t_0 = 0) \stackrel{\text{def}}{=} \left[1 - i\hat{H}t + \frac{[-i\hat{H}t]^2}{2!} + \dots \right] \psi(\mathbf{x}, 0), \quad (11.3)$$

where the series expansion defines what is meant by an operator in the exponent. This provides one answer to our original question: The Schrödinger equation starts with the wave function at time $t_0 = 0$ and propagates it forward with the *operator* $\exp(-i\hat{H}t)$.

Another way of gaining insight into how quantum mechanics advances a free particle from one time to another is to start way back, at the very beginning of this book, where we discussed wave packets. That discussion gives us the integral relation between the wave function at different times,

$$\psi(\mathbf{r}_b, t_b) = \int d\mathbf{r}_a G(\mathbf{r}_b, t_b; \mathbf{r}_a, t_a) \psi(\mathbf{r}_a, t_a), \quad (11.4)$$

where G is the time-dependent Green's function,

$$G(\mathbf{r}_b, t_b; \mathbf{r}_a, t_a) \stackrel{\text{def}}{=} \int d\mathbf{k} \langle \mathbf{r}_b | \mathbf{k} \rangle e^{-iE_{\mathbf{k}}(t_b - t_a)} \langle \mathbf{k} | \mathbf{r}_a \rangle. \quad (11.5)$$

To prove (11.4), we start with equation (1.1) in Chapter 1 which indicates that the motion of a localized particle of momentum $\mathbf{k}_0$ is described most realistically as a wave function (*wave packet*). This wave packet is a function of both space and time, and can be expressed as an integral over momentum $\mathbf{k}$:

$$\psi(\mathbf{r}, t) = \int d\mathbf{k} e^{i\mathbf{k} \cdot \mathbf{r}} e^{-iE_{\mathbf{k}} t} A_{\mathbf{k}_0}(\mathbf{k}), \quad (11.6)$$

where for nonrelativistic problems the energy is $E_{\mathbf{k}} = k^2/2\mu$. As shown in basic quantum mechanics texts, if $A_{\mathbf{k}_0}(\mathbf{k})$ is peaked about $\mathbf{k}_0$, this wave packet propagates through space and disperses with time. We do not wish to repeat those arguments here, but rather to better understand the time dependence of (11.6) by relating it to the time-independent Dirac notation and the abstract form of the Schrödinger equation already studied in Chapter 5, *Green's Functions: Integral Quantum Mechanics*, Chapter 7, *Formal Quantum Mechanics*, and Appendix B, *Dirac Notation and Representations*. Equation (B.23) tells us that the general $\mathbf{r}$ -space representation of the wave function at time $t = 0$ can also be written as an integral (Fourier transform) over momentum:

$$\langle \mathbf{r} | \psi(t=0) \rangle = \int d\mathbf{k} \langle \mathbf{r} | \mathbf{k} \rangle \psi(\mathbf{k}, t=0) \equiv \int d\mathbf{k} \frac{e^{i\mathbf{k} \cdot \mathbf{r}}}{(2\pi)^{3/2}} \langle \mathbf{k} | \psi(t=0) \rangle. \quad (11.7)$$

By comparing this with (11.6) we can identify the momentum-space representation (wave function) of a wave packet as

$$\psi(\mathbf{k}, t=0) \equiv \langle \mathbf{k} | \psi(t=0) \rangle = (2\pi)^{3/2} A_{\mathbf{k}_0}(\mathbf{k}). \quad (11.8)$$

Whereas in the past we have applied the Dirac notation only to time-independent formulations, we now generalize the Fourier transform (B.23) and the identity operator to include the time dependence:

$$\langle \mathbf{r} | \psi(t) \rangle = \int d\mathbf{k} \langle \mathbf{r} | \mathbf{k} \rangle e^{-iE_{\mathbf{k}}(t-t')} \langle \mathbf{k} | \psi(t') \rangle, \quad (11.9)$$

$$\tilde{1} = \int d\mathbf{k} |\mathbf{k}\rangle e^{-iE_{\mathbf{k}}(t)} \langle \mathbf{k}|. \quad (11.10)$$

What we called time zero in (11.6) is arbitrary, and so in writing (11.9) we use a general time difference.

We now can see that by sandwiching the operator between plane waves, (11.9) gives an explicit representation of (11.3)'s description of how the Schrödinger equation propagates the wave function forward in time. If we now go back to (11.7) and substitute the momentum representation of the wave function,

$$\langle \mathbf{k} | \psi(t') \rangle = \int d\mathbf{r}' \langle \mathbf{k} | \mathbf{r}' \rangle \langle \mathbf{r}' | \psi(t') \rangle, \quad (11.11)$$

we obtain the desired rule for advancing the time in a wave function:

$$\langle \mathbf{r} | \psi(t) \rangle = \int d\mathbf{r}' \int d\mathbf{k} \langle \mathbf{r} | \mathbf{k} \rangle e^{-iE_{\mathbf{k}}(t-t')} \langle \mathbf{k} | \mathbf{r}' \rangle \langle \mathbf{r}' | \psi(t') \rangle. \quad (11.12)$$

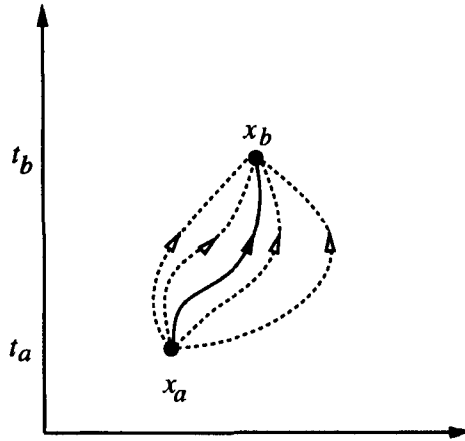


Figure 11.1: A collection of paths connecting the initial and final spacetime points. The solid line represents the classical trajectory $\bar{x}(t)$. The transition amplitude from a to b includes all paths.

We make it clear that this is the quantum-mechanical rule for propagating a particle from time t_a to t_b by rewriting (11.12) as

$$\psi(\mathbf{r}_b, t_b) = \int d\mathbf{r}_a G(\mathbf{r}_b, t_b; \mathbf{r}_a, t_a) \psi(\mathbf{r}_a, t_a). \quad (11.13)$$

The function,

$$G(\mathbf{r}_b, t_b; \mathbf{r}_a, t_a) \equiv G(b, a) \stackrel{\text{def}}{=} \int d\mathbf{k} \langle \mathbf{r}_b | \mathbf{k} \rangle e^{-iE_{\mathbf{k}}(t_b - t_a)} \langle \mathbf{k} | \mathbf{r}_a \rangle, \quad (11.14)$$

goes by various names including *kernel*, *propagator*, *transition amplitude*, *Green's function*, or just plain “amplitude”. These are the relations (11.4) and (11.5) we wanted to prove, and they tell us that to propagate a free particle from the time t_a to the time t_b , and from the position $\mathbf{r}_a$ to the position $\mathbf{r}_b$, we take the wave function $\psi(\mathbf{r}_a, t_a)$ at time t_a , multiply it by $G(b, a)$, and integrate over all values of $\mathbf{r}_a$ (the time variable does not get integrated over). Accordingly, we recognize $G(b, a)$ as the amplitude for a wave to be at b if it was at a at an earlier time. With this in mind, we recognize (11.4) as a statement of *Huygen's principle* that each point on a wavefront emits a new spherical wavelet which propagates forward in space and time. By interfering with all the other wavelets a new wavefront is created.¹ As we shall show later, interactions modify the expression we use for $G(b, a)$, but not the method of wave function propagation.

One way of interpreting (11.4) is illustrated in Figure 11.1 for one space dimension. We visualize the probability amplitude (wave function) for the particle to be at b as the

¹The traditional path integral formulation postulates (11.4) and (11.5) and then derives the wave function relations as an application. We reverse that treatment and use our wave function formulation to motivate the path integral formulation.

sum over all *paths* through spacetime originating at a . In this view, the statistical nature of quantum mechanics is implicit in the existence of an infinite number of paths. Furthermore, the philosophical difficulties of how a single particle can interfere with itself is avoided because the path summation interpretation of (11.4) means that a particle always travels along an infinite number of paths connecting the initial and final times. Likewise, if an obstacle is placed in Figure 11.1 such that any of the paths are obstructed, they no longer get included in the sum, the interference among the different terms changes, and we have *diffraction* with just one particle.

11.3 The Free Particle Propagator

We have used our 3D description of free-particle wave functions to derive Huygen's principle (11.4) and thereby motivate the path-integral view of quantum mechanics. For simplicity, and without forsaking any physics, we now switch to the 1D formulation (because computational approaches to the 3D problem often use multiple 1D approaches, this simpler formulation is also useful). The 1D version of the free particle propagator (11.5) is:

$$G(x_b, t_b; x_a, t_a) = \int_{-\infty}^{+\infty} dk \langle x_b | k \rangle e^{-iE_k(t_b-t_a)} \langle k | x_a \rangle \quad (11.15)$$

$$= \int_{-\infty}^{+\infty} dk \frac{e^{ix_b k}}{\sqrt{2\pi}} e^{-i(t_b-t_a)k^2/2\mu} \frac{e^{-ix_a k}}{\sqrt{2\pi}}, \quad (11.16)$$

where we have used the nonrelativistic expression for the energy. Equation (11.16) is called a *Gaussian integral* because it has an exponential with k^2 in it. These types of integrals are particularly nice to work with analytically because they often can be evaluated just by completing the square in the exponent of the exponential, and using the relation

$$\int_0^\infty dk k^n e^{-r^2 k^2} = \frac{\Gamma(\frac{n+1}{2})}{2r^{n+1}} \quad \left(= \frac{\sqrt{\pi}}{2r}, \text{ for } n = 0 \right). \quad (11.17)$$

For example, we let $\epsilon = t_b - t_a$ and $\delta = x_b - x_a$, and write the combined exponentials in (11.16) as

$$\exp\left(-i\left[-k\delta + \frac{k^2\epsilon}{2\mu}\right]\right) = \exp\left(-\frac{i\epsilon}{2\mu}\left[k^2 - k\frac{2\mu\delta}{\epsilon} + \left(\frac{\mu\delta}{\epsilon}\right)^2\right]\right) \exp\left[i\frac{\epsilon\mu^2\delta^2}{2\mu\epsilon^2}\right]. \quad (11.18)$$

The k -independent term comes outside the integral and a change of variable converts the resulting integral into a tabulated one:

$$\begin{aligned} G(x_b, t_b; x_a, t_a) &= \frac{\exp\left[\frac{i\epsilon\mu^2\delta^2}{2\mu\epsilon^2}\right]}{2\pi} \int_{-\infty}^{+\infty} dk \exp\left[\frac{-i\epsilon}{2\mu}(k - \mu\delta/\epsilon)^2\right] \\ &= \frac{\exp\left[\frac{i\epsilon\mu^2\delta^2}{2\mu\epsilon^2}\right]}{2\pi} \int_{-\infty}^{+\infty} dp \exp\left[\frac{-i\epsilon}{2\mu}p^2\right]. \end{aligned} \quad (11.19)$$

The free propagator is thus

$$G(x_b, t_b; x_a, t_a) = \frac{\theta(t_b - t_a)}{\sqrt{2\pi i(t_b - t_a)/\mu}} \exp\left[\frac{i\mu(x_b - x_a)^2}{2(t_b - t_a)}\right]. \quad (11.20)$$

We leave it as a problem to show that the 3D propagator obtained by explicit evaluation of (11.5) is just the “cube” of the 1D propagator (11.20):

$$G(\mathbf{r}_b, t_b; \mathbf{r}_a, t_a) = \frac{\theta(t_b - t_a)}{[2\pi i(t_b - t_a)/\mu]^{3/2}} \exp \left[\frac{i\mu(\mathbf{x}_b - \mathbf{x}_a)^2}{2(t_b - t_a)} \right]. \quad (11.21)$$

Relation to Free Green’s Function

The form (11.20) of the free propagator is familiar looking. We see that $G(b, a)$ is just a function of the time separation $t_b - t_a$ and of the space separation $\mathbf{x}_b - \mathbf{x}_a$, as expected for the amplitude of the wavelet in Huygen’s principle (11.4). In addition, because a delta function is the limit of a Gaussian whose width approaches zero,

$$\delta(\mathbf{x}_b - \mathbf{x}_a) = \lim_{\gamma \rightarrow 0} \frac{1}{\sqrt{2\pi}} \exp \left[-\frac{(\mathbf{x}_b - \mathbf{x}_a)^2}{2\gamma^2} \right], \quad (11.22)$$

the free propagator G approaches a delta function for zero time separations:

$$\lim_{t_b \rightarrow t_a} G(\mathbf{x}_b, t_b; \mathbf{x}_a, t_a) = \delta(\mathbf{x}_b - \mathbf{x}_a). \quad (11.23)$$

This delta function is reminiscent of the Green’s function of Chapter 5, *Green’s Functions: Integral Quantum Mechanics*. There we defined the Green’s function as the solution of the time-independent Schrödinger equation for a delta function source at the origin, and so the solution to the Schrödinger equation is the integral of the Green’s function times the driving term, as in (5.6). If we inspect the Huygen relation (11.4), we see that $G(\mathbf{x}_b, t_b; \mathbf{x}_a, t_a)$ must be some type of Green’s function for the time-dependent, free particle Schrödinger equation with $\psi(\mathbf{x}_a, t_a)$ considered as the driving term.

Exercise Show that $G(b, a)$ is the solution of the time-dependent, free particle Schrödinger equation with a delta function source at $(\mathbf{x}_a, t_a)$:

$$\left(i \frac{\partial}{\partial t_b} - \frac{1}{2\mu} \frac{\partial^2}{\partial \mathbf{x}_b^2} \right) G(\mathbf{x}_b, t_b; \mathbf{x}_a, t_a) = \delta(\mathbf{x} - \mathbf{x}_a) \delta(t - t_a). \quad \diamond \quad (11.24)$$

11.4 Feynman’s Variation on a Theme by Hamilton

Path integrals arose originally in Feynman’s quest for a least-action principle of quantum mechanics in which classical mechanics occurs as the special case of quantum mechanics for vanishingly small values of $\hbar$. In developing the theory, Feynman followed a suggestion of Dirac that the exponential of $i \int dt$ times the Lagrangian might be a time translation operator for the wave function.² We do the same here in order to generalize Huygen’s principle (11.4) for a particle under the influence of a potential.

Hamilton’s Principle

Our discussion of a particle sampling various paths through spacetime with their endpoints fixed, as in Figure 11.1, is reminiscent of *Hamilton’s principle of least action* in classical

²Feynman and Hibbs (1965).

mechanics. This principle can be used to derive Newton's laws of motion as well as many other, otherwise disparate, physical laws such as Einstein's equations of general relativity (Scheck, 1994; Fetter and Walecka, 1980; Landau and Lifschift, 1975). Hamilton's principle states that:

The most general motion of a physical system moving along the classical trajectory $\bar{\mathbf{x}}(t)$ through configuration space from time t_a to t_b , is along a path such that the action $S[\bar{\mathbf{x}}(t)]$ is an extremum:

$$\delta S[\bar{\mathbf{x}}(t)] \stackrel{\text{def}}{=} S[\bar{\mathbf{x}}(t) + \delta \mathbf{x}(t)] - S[\bar{\mathbf{x}}(t)] = 0, \quad (11.25)$$

where the paths are constrained to pass through the endpoints:

$$\delta(\mathbf{x}_a) = \delta(\mathbf{x}_b) = 0. \quad (11.26)$$

Here we imagine many paths such as those in Figure 11.1 with the *generalized coordinate* $\mathbf{x}$ subject to whatever constraints are imposed on the system. Along each path we calculate the classical action S as a line integral of the classical Lagrangian:

$$S[\bar{\mathbf{x}}(t)] = \int_{t_a}^{t_b} dt L(\mathbf{x}, \dot{\mathbf{x}}; t), \quad (11.27)$$

$$L(\mathbf{x}, \dot{\mathbf{x}}; t) = T(\mathbf{x}, \dot{\mathbf{x}}; t) - V(\mathbf{x}; t), \quad (11.28)$$

where T is the kinetic energy expressed in terms of $\mathbf{x}$, $\dot{\mathbf{x}} \equiv d\mathbf{x}/dt$ and the time t , and V is a conservative potential function. The square brackets around $\bar{\mathbf{x}}(t)$ on the LHS of (11.27) indicates that S is a *functional*³ of the function $\mathbf{x}(t)$.

Hamilton's principle states that the physical, classical trajectory (the dark one in Figure 11.1) is embedded in a set of trajectories, all with their endpoints fixed. Furthermore, a classical particle somehow "knows" ahead of time to travel along just that trajectory for which S is an extremum, that is, for which S does not change (to lowest order in α) when we vary the paths as

$$\bar{\mathbf{x}}(t) \rightarrow \bar{\mathbf{x}}(t) + \alpha \eta(t), \quad \eta(t_a) = \eta(t_b) = 0. \quad (11.29)$$

The boundary condition requiring the vanishing of $\eta(t)$ is just the condition in Hamilton's principle that the endpoints of the paths remain fixed, $\delta \mathbf{x}(t_a) = \delta \mathbf{x}(t_b) = 0$, as we vary the paths.

The Postulates of Quantum Mechanics

Now that we know what we are looking for, we can go back to our expression (11.20) for the free particle's propagator and try to relate it to Hamilton's principle of least action. First we observe that for a free particle the classical Lagrangian is

$$L = T - V = T = \frac{\mu}{2} \left(\frac{d\mathbf{x}}{dt} \right)^2 = \frac{\mu}{2} \left(\frac{\mathbf{x}_b - \mathbf{x}_a}{t_b - t_a} \right)^2, \quad (11.30)$$

³A functional is a number whose value depends upon the complete behavior of some function and not just its behavior at one point. For example, the derivative $f'(x)$ depends on just the value of f at x , yet the integral $\int_a^b dx f(x)$ depends upon the entire function and is therefore a functional of f .

where we have used the fact that for a free particle the velocity $d\mathbf{x}/dt$ is constant and equal to $\Delta\mathbf{x}/\Delta t$. The classical action for a free particle traveling along one of our paths is

$$S[\mathbf{x}(t)] = \int_{t_a}^{t_b} dt L(\mathbf{x}, \dot{\mathbf{x}}; t) = \int_{t_a}^{t_b} dt \frac{\mu}{2} \left(\frac{d\mathbf{x}}{dt} \right)^2 \quad (11.31)$$

$$= \frac{\mu}{2} \left(\frac{d\mathbf{x}}{dt} \right)^2 (t_b - t_a) = \frac{\mu}{2} \frac{(\mathbf{x}_b - \mathbf{x}_a)^2}{t_b - t_a}. \quad (11.32)$$

As expected, while S depends upon the functional dependence of $\mathbf{x}$ on t , its explicit value depends on just the endpoints.

We see that the free propagator (11.20) is related to the classical action in a simple way:

$$G(\mathbf{x}_b, t_b; \mathbf{x}_a, t_a) = \sqrt{\frac{\mu}{2\pi i(t_b - t_a)}} \exp \left\{ \frac{i\mu(\mathbf{x}_b - \mathbf{x}_a)^2}{2(t_b - t_a)} \right\} \quad (11.33)$$

$$= \sqrt{\frac{\mu}{2\pi i(t_b - t_a)}} \exp \{ iS[\mathbf{x}(t)]/\hbar \}. \quad (11.34)$$

Note, in (11.34) we have divided S (which has dimension of an angular momentum) by $\hbar$ (which we usually set equal to one) in order to help us identify the classical limit. If we again examine our expression of Huygen's principle (11.4) (you will recall, $G(b, a)$ is the amplitude of wavelets being integrated over), it becomes easier to understand how Feynman could base quantum mechanics on three postulates:

Postulate I The probability density for a transition from $a \equiv (\mathbf{x}_a, t_a)$ to $b \equiv (\mathbf{x}_b, t_b)$ equals the squared modulus of an amplitude:

$$P(b, a) = |G(b, a)|^2. \quad (11.35)$$

Postulate II The transition amplitude $G(b, a)$ for a particle making a transition from a to b is a sum over all paths connecting a to b . Each path makes a contribution of the same magnitude to the sum, but with a phase equal to the classical action evaluated along that path:

$$G(b, a) = \sum_{\text{paths}} A e^{iS[b, a]/\hbar}. \quad (11.36)$$

Here A is a constant determined, eventually, by the wave function normalization (which is a good thing because we will see that A is usually infinite). This sum over paths is also called a *path integral* because the classical action $S[b, a]$ is a line integral from time t_a to time t_b , and, in part, because when we approximate the line integral as a finite sum over discrete time steps, the sums over paths and time steps appear as one multidimensional sum over small elements in space and time. The sum over paths thus enter just like integrals over paths.

Postulate III If, as illustrated in Figure 11.2, an experiment is capable of determining which alternative paths (say 1 and 2) were traveled by the particle, then the probabilities instead of the paths are summed:

$$P(b, a) = P_1(b, a) + P_2(b, a). \quad (11.37)$$

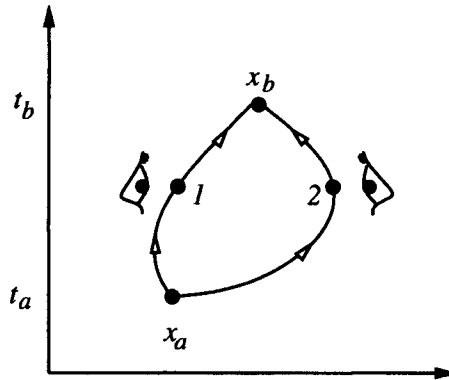


Figure 11.2: An experiment which observes if path 1 or 2 is traveled by a particle. Here the probabilities and not the paths are summed.

Exercise Show that for a free particle

$$P(b, a) \propto |G(b, a)|^2 = \frac{\mu}{2\pi\hbar(t_b - t_a)}. \quad \diamond \quad (11.38)$$

Exercise Explain why this last exercise indicates that all momenta are equally likely, and that the probability to be in any region of space decreases with time. $\diamond$

It is valuable to mull over the implication of Postulate II. You will recall that for classical systems, Hamilton's principle tells us that a particle is smart enough to know to take only that one path for which the action $S[b, a]$ is an extremum, the classical trajectory $\bar{x}(t)$ in Figure 11.1. The key to understanding how this classical limit arises from Postulate II is the realization that in units of $\hbar$, a classical action is a very large number, $S_{cl}/\hbar \simeq \infty$. This means that while all paths enter into the sum in (11.36) with a weight of equal magnitude, because S is a constant to first order in the variation of paths, (11.25), those paths adjacent to the classical trajectory $\bar{x}$ have phases which vary smoothly and relatively slowly. In contrast, those paths far from the classical trajectory enter with rapidly varying phases, and when many are included they tend to cancel each other out. In the classical limit, $\hbar \rightarrow \infty$, only the classical trajectory contributes and Postulate II becomes Hamilton's principle of least action!⁴

We see that a quantum particle has the freedom (and obligation) to sample the surrounding paths in which the classical trajectory is embedded, but because of the increasing variation of its phase, if it strays too far it will not make much of a contribution to the sum in (11.36). The path integral formulation implicitly includes quantum corrections to

⁴The alert reader may be wondering "How can the classical trajectory be determined when only the position x_a at time t_a is fixed and not the initial velocity?" The answer is that the position x_b at time t_b is also fixed and this is equivalent to the second initial condition.

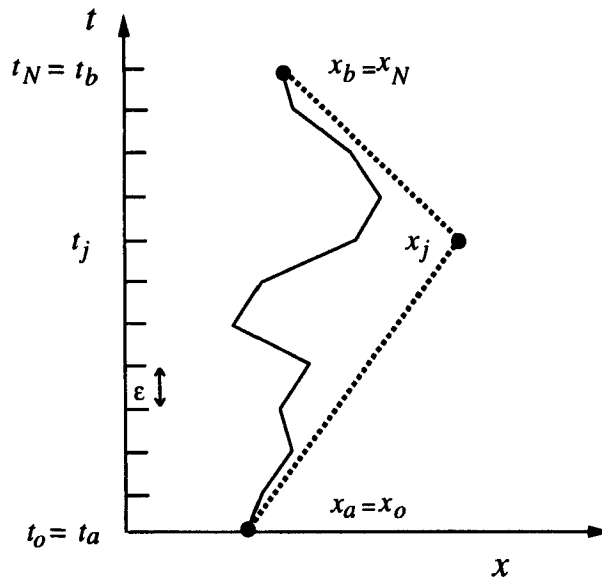


Figure 11.3: Spacetime with time as a discrete variable. The dashed path joins the initial and final times in two equal time steps, the solid curve uses N steps each of size ϵ . The position of the curve at time t_j defines the position x_j .

the classical motion via its statistical summation over paths. Also within the constraint of Postulate III, the interference of alternative paths provides a natural explanation of how a quantum particle travels through a screen with multiple slits (you sum over the paths passing through every slit), and how diffraction occurs (a diffracting object blocks some paths and thereby changes the sum). Because these conceptual questions related to the statistical aspects of quantum mechanics can be tricky in other formulations, having them woven into the very fabric of the path integral formulation is attractive.

Although some philosophical questions about quantum mechanics are inherently easier to address with the path integral formulation, there are still fundamental difficulties with it. For example, a trajectory and the integral along a trajectory appear as more classical than quantum concepts, and it is not clear if this trajectory formulation is well defined in the Minkowski space of special relativity. In practise, the theory becomes difficult to work with analytically when nongaussian integrals are encountered or when the Hamiltonian is complicated, and if the Hamiltonian contains nonlinear terms in the momentum, the noncommuting aspects of position and momentum can result in difficulties.

11.5 Path Integration on a Lattice

Although we have shown that Postulate II's description of path summation illuminates the conceptual foundation of quantum mechanics, the practical physicist may well be asking "How do I really perform such a path summation, let alone use one to determine an

observable?" In some cases the integrals can be evaluated analytically, and in some cases they can be approximated numerically (we show how in the next chapter). In either case, we start by deriving the *composition law* for propagators. Consider Huygen's principle (11.4) for a 1D system:

$$\psi(x_b, t_b) = \int dx_a G(x_b, t_b; x_a, t_a) \psi(x_a, t_a). \quad (11.39)$$

We must somehow evaluate this integral for all paths connecting a to b . As it stands, (11.39) is an integral equation for the wave function ψ and so it is reasonable to base our evaluation on the Fredholm theory of integral equations (Hilderbrant, 1963; Mathews and Walker, 1964). Fredholm theory is based on discretizing the variables of the equation and thereby converting an integral equation into a set of simultaneous linear equations. The linear equations are then solved with standard techniques, and when the limit is taken such that the discrete variables become continuous, the solution of the simultaneous linear equations becomes the solution of the integral equation.

As a first step, consider evaluating (11.39) for the dashed path in Figure 11.3 which joins the initial and final times in two time steps through the intermediate point (x_j, t_j) . Because Huygen's principle (11.39) must hold for each step, we know how to propagate from a to j and then from j to b :

$$\psi(x_j, t_j) = \int dx_a G(x_j, t_j; x_a, t_a) \psi(x_a, t_a), \quad (11.40)$$

$$\psi(x_b, t_b) = \int dx_j G(x_b, t_b; x_j, t_j) \psi(x_j, t_j). \quad (11.41)$$

If we substitute the expression for $\psi(x_j, t_j)$ into the expression for $\psi(x_b, t_b)$, we obtain the double integral

$$\psi(x_b, t_b) = \int dx_j \int dx_a G(x_b, t_b; x_j, t_j) G(x_j, t_j; x_a, t_a) \psi(x_a, t_a). \quad (11.42)$$

For this to be equivalent to our original expression (11.39) for $\psi(x_b, t_b)$, the propagators must combine as

$$G(x_b, t_b; x_a, t_a) = \int dx_j G(x_b, t_b; x_j, t_j) G(x_j, t_j; x_a, t_a), \quad (t_a < t_j, t_j < t_b). \quad (11.43)$$

The rule (11.43) indicates that:

Paths always move forward in time.

Amplitudes (propagators) multiply for events occurring at successive times.

We obtain some further insight by substituting the expression (11.20) for the free propagator into (11.43) to obtain

$$\begin{aligned} G(x_b, t_b; x_a, t_a) &= \sqrt{\frac{\mu}{2\pi i(t_b - t_j)}} \sqrt{\frac{\mu}{2\pi i(t_j - t_a)}} \\ &\times \int dx_j \exp \left[\frac{i\mu(x_b - x_j)^2}{2(t_b - t_j)} + \frac{i\mu(x_j - x_a)^2}{2(t_j - t_a)} \right] \end{aligned}$$

$$\begin{aligned}
&= \sqrt{\frac{\mu}{2\pi i(t_b - t_j)}} \sqrt{\frac{\mu}{2\pi i(t_j - t_a)}} \int dx_j e^{i(S[b,j] + S[j,a])/\hbar} \\
&= A' \int dx_j e^{iS[b,a]/\hbar}.
\end{aligned} \tag{11.44}$$

Here we replace the function in the exponents by the action, add the actions together since line integrals combine as

$$S[b, j] + S[j, a] = S[b, a], \tag{11.45}$$

and replace the square root factors by the x -independent constant A' . Equation (11.44) tells us that as we combine propagators by multiplying and integrating, we only need keep a running sum of the total action along the path, and that the x -independent normalization factors just accumulate outside of the integral.

We now take the big step (well, actually the many little steps illustrated in Figure 11.3) and “discretize” the time interval between a and b into N equal steps of size ϵ :

$$\epsilon \stackrel{\text{def}}{=} \frac{t_b - t_a}{N}. \tag{11.46}$$

Because we made time discrete, we go ahead and label it with the index j :

$$t_j = t_a + j\epsilon, \quad (j = 0, N) \tag{11.47}$$

$$t_0 \equiv t_a, \quad t_N \equiv t_b. \tag{11.48}$$

In (11.48) we give the fixed endpoints a discrete index even though they will not vary as we change paths. Next we draw the solid curve in Figure 11.3 as a representation of a single path through spacetime with the position of the curve at time t_j defining the position x_j as

$$x_j \stackrel{\text{def}}{=} x(t_j). \tag{11.49}$$

Although only a crude numerical approximation, in drawing paths we assume that *successive x_j 's are connected by straight lines*. By considering all curves which connect points a and b , the associated x_j 's and t_j 's map out a *lattice* in spacetime upon which time and space are discrete. Presumably, the solution obtained on the lattice approaches the true solution in the limit of infinitesimal lattice spacing ϵ .

The lattice construct is convenient. We can replace time differentials dt by $\Delta t = t_{j+1} - t_j \equiv \epsilon$, and space differentials dx by $\Delta x = x_{j+1} - x_j = \delta$. Thus, if we use Euler's rule for the first derivative,⁵ the velocity and action take the simple lattice forms

$$\dot{x}_j \stackrel{\text{def}}{=} \frac{dx}{dt} \simeq \frac{x_j - x_{j-1}}{t_j - t_{j-1}} = \frac{x_j - x_{j-1}}{\epsilon}, \tag{11.50}$$

$$S_j \stackrel{\text{def}}{=} S[j, j-1] \simeq L_j \Delta t \simeq \frac{1}{2} \mu \frac{(x_j - x_{j-1})^2}{\epsilon} - V(x_j) \epsilon, \tag{11.51}$$

where we assume that the Lagrangian is constant for each path segment on the lattice.

⁵A number of rules for numerical derivatives can be found in Thompson (1992). Even though the Euler rule for the derivative yields the largest approximation error, it is usually used in lattice calculations because it makes the entire calculation simpler. Note, however, if the Lagrangian involves a second derivative, to avoid infinities you may have to use the more precise central difference method, $\ddot{x}_j \simeq (\ddot{x}_{j+\frac{1}{2}} - \ddot{x}_{j-\frac{1}{2}})/\epsilon \simeq ([x_{j+1} - x_j] - [x_j - x_{j-1}])/\epsilon^2$.

Next we generalize (11.43) or (11.44) for the N -segmented path of Figure 11.3,

$$G(x_b, t_b; x_a, t_a) = A^N \int dx_1 \dots dx_{N-1} e^{i \sum_{i=1}^N L_i \epsilon / \hbar} \quad (11.52)$$

$$= A^N \int dx_1 \dots dx_{N-1} e^{i S[b, a] / \hbar}, \quad (11.53)$$

$$S[b, a] = \sum_{j=1}^N S_j, \quad A = \sqrt{\frac{\mu}{2\pi i \epsilon}}. \quad (11.54)$$

We use here the expression (11.51) for the value of the action for each segment, and the normalization factor appropriate to a free propagator. Consequently we see that even without a path variation, the integral over the *single* path in Figure 11.3 is an N -dimensional integral which becomes an infinite-dimensional integral as the time step ϵ approaches zero.

Postulate II says that to obtain the transition amplitude $G(b, a)$, we should “sum over all paths connecting a to b ”. This means we must not only integrate over the links in one path, but *also* sum over all paths in order to produce the variation in paths which is at the heart of Hamilton’s principle. The constraint in the sum is that the paths must pass through a and b and cannot double back on themselves [because G is nonzero only for forward steps in time (11.20)]. This is the essence of *path integration* and is given the special symbol $\mathcal{D}$ meaning:

$$\int \mathcal{D}x \equiv \sum_{\text{paths}} \stackrel{\text{def}}{=} \lim_{N \rightarrow \infty} \prod_{i=1}^N \delta x_j, \quad (11.55)$$

where the symbol δx indicates a path variation. The path integration symbol $\int \mathcal{D}x$ represents an infinite product of line integrals (variations of paths) with each line integral approximated as an infinite number of integrals over the links comprising the path. Because we are integrating over functions as well as along paths, the technique is also known as *functional integration*.

In terms of the path integration symbol, Postulate II’s prescription for the propagator is

$$G(b, a) = \mathcal{A} \int_a^b \mathcal{D}x e^{i S[b, a] / \hbar}, \quad (11.56)$$

where the endpoints a and b are held fixed, and for free particle $\mathcal{A}$ is just the normalization constant $\lim_{N \rightarrow \infty} (\frac{\mu}{2\pi i \epsilon})^{N/2}$. The fact that $\epsilon \rightarrow 0$ and $\mathcal{A} \rightarrow \infty$ in this limit may appear troublesome, but in fact this ϵ dependence is needed for the total integral defined by (11.55) to exist (this is analogous to the convergence of the numerical evaluation of a Riemann integral as $1/N$ times the sum over N values of the integrand). In practical calculations, the normalization factor is often ignored because it is determined when the wave function is normalized. Alternatively, $\mathcal{A}$ can be evaluated by using the functional form of the combination rule (11.43),

$$G(b, a) = \int_a^b \mathcal{D}x_j G(b, i) G(i, a), \quad (11.57)$$

where the integration is over all paths connecting a and b through the intermediate x_j .

Exercise Consider a free particle for which $L = \mu \dot{x}^2/2$. Work backwards and prove that the path integral expression for G equals the expression from which we started:

$$\begin{aligned} G(b, a) &= \lim_{\epsilon \rightarrow 0} \left(\frac{\mu}{2\pi\hbar\epsilon} \right)^{N/2} \int dx_1 dx_2 \cdots dx_{N-1} \exp \left[\frac{i\mu}{2\hbar\epsilon} \sum_{i=1}^N (x_i - x_{i-1})^2 \right] \\ &= \sqrt{\frac{2\pi\hbar(t_b - t_a)}{\mu}} \exp \left[\frac{i\mu(x_b - x_a)^2}{2\hbar(t_b - t_a)} \right]. \end{aligned} \quad (11.58)$$

(Hint: All of the integrals you need to evaluate are Gaussian, so you only need to do the first and then proceed by induction.) $\diamond$

11.6 Paths Through Electromagnetic Gauge Fields

There is a connection between the equations describing the electromagnetic field and those describing quantum mechanics. Specifically, many of the unique aspects of quantum mechanics arise from the phases of transition amplitudes, and yet phases are also related to the gauge symmetry of the electromagnetic field. There is a further connection in that the path integral formulation of quantum mechanics has the dynamics entering through the classical action, which in turn also modifies the phase of the amplitude. In this section we examine these connections. First we review the inclusion of the electromagnetic force into classical Hamiltonian dynamics and its related gauge symmetry. Then we extend the classical treatment to the Schrödinger formulation and finally to the path integral formulation. We end with a look at the *Aharonov–Bohm effect* as a direct application of how a magnetic field changes the phase of amplitudes.

Classical Dynamics and the EM Force

Classical Hamiltonian dynamics is usually formulated for conservative forces which do not depend on velocity. But because the basic experimental fact of electromagnetism is the velocity-dependent *Lorentz force*,

$$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}) = \frac{d\mathbf{p}}{dt}, \quad (11.59)$$

the Hamiltonian formulation must be extended to include nonconservative forces. The theory is simpler with the electromagnetic vector and “scalar” potentials $\mathbf{A}$ and Φ , even though the $\mathbf{E}$ and $\mathbf{B}$ fields are more closely related to experiment. These fields are obtained through differentiation:

$$\mathbf{E} = -\nabla\Phi(\mathbf{r}, t) - \frac{\partial\mathbf{A}(\mathbf{r}, t)}{\partial t}, \quad \mathbf{B} = \nabla \times \mathbf{A}(\mathbf{r}, t). \quad (11.60)$$

The Lorentz force is introduced through the effective potential (Scheck, 1994),

$$V_{\text{eff}} = q\Phi(\mathbf{r}, t) - q\mathbf{v} \cdot \mathbf{A}(\mathbf{r}, t). \quad (11.61)$$

The Lorentz force is produced by the addition of this potential to the Lagrangian $L = T - V$, and its use in Lagrange's equations:

$$L \rightarrow L - q\Phi(\mathbf{r}, t) + q\mathbf{v} \cdot \mathbf{A}(\mathbf{r}, t), \quad (11.62)$$

$$\frac{\delta L}{\delta \mathbf{x}_j} = \frac{\partial L}{\partial \mathbf{x}_j} - \frac{d}{dt} \frac{\partial L}{\partial \dot{\mathbf{x}}_j} = 0. \quad (11.63)$$

Once we have made this change in L , the canonical theory leads us to the corresponding changes in the generalized momenta and Hamiltonian:

$$p_j \stackrel{\text{def}}{=} \frac{\partial L}{\partial \dot{\mathbf{x}}_j} \rightarrow p_j - qA_j, \quad (11.64)$$

$$H(\mathbf{p}, \mathbf{x}) \stackrel{\text{def}}{=} \sum_j p_j \dot{\mathbf{x}}_j - L \rightarrow H(\mathbf{p} - q\mathbf{A}, \mathbf{x}) + q\Phi. \quad (11.65)$$

The procedure (11.65) is known as *minimal coupling* of the electromagnetic field. It is “minimal” because it couples the field to the elementary charge at one point, which is appropriate for a *point particle* with no charge or magnetic moments. As we shall see next, it also leaves the gauge symmetry of the equations intact.

A gauge transformation is

$$\mathbf{A} \rightarrow \mathbf{A} + \nabla\chi(\mathbf{r}, t), \quad (11.66)$$

$$\Phi \rightarrow \Phi - \frac{\partial\chi(\mathbf{r}, t)}{\partial t}, \quad (11.67)$$

where $\chi(\mathbf{r}, t)$ is a scalar function of space and time. This gauge transformation does not affect $\mathbf{E}$ and $\mathbf{B}$, and thus does not affect Maxwell's equations or the experimental observable, the Lorentz force. The invariance of Maxwell's equations under this transformation is called *gauge invariance*, and the invariance is a consequence of a symmetry in the equations which is called *gauge symmetry*. Because χ changes with $\mathbf{r}$ and t , the transformation (11.66)–(11.67) is sometimes called a *local gauge transformation*.

Because the gauge transformation (11.66)–(11.67) modifies $\mathbf{A}$ and Φ , it follows from (11.62) that it also modifies the Lagrangian:

$$L \rightarrow L + q \left[\frac{\partial\chi(\mathbf{r}, t)}{\partial t} + \mathbf{v} \cdot \nabla\chi(\mathbf{r}, t) \right] \equiv L + \frac{d}{dt} [q\chi(\mathbf{r}, t)]. \quad (11.68)$$

This means the classical Lagrangian cannot be defined uniquely in the presence of the electromagnetic field.

Schrödinger Dynamics and the EM Force

A benefit of the Schrödinger equation having a Hamiltonian structure is that we know the minimal coupling prescriptions (11.64) and (11.65) is the way to include the electromagnetic field (see too § 13.5 and § 20.3):

$$i\frac{\partial}{\partial t} \rightarrow i\frac{\partial}{\partial t} - q\Phi, \quad \frac{\nabla}{i} \rightarrow \frac{\nabla}{i} - q\mathbf{A}. \quad (11.69)$$

This produces the Schrödinger equation with minimal electromagnetic coupling:

$$i \frac{\partial \psi(\mathbf{r}, t)}{\partial t} = \frac{1}{2\mu} [\nabla/i - q\mathbf{A}]^2 \psi(\mathbf{r}, t) + q\Phi \psi(\mathbf{r}, t). \quad (11.70)$$

We know that classical Hamiltonian dynamics with minimal electromagnetic coupling is gauge invariant, yet it is not obvious how gauge invariance fits into the quantum world.

Exercise Show that the gauge transformation (11.66)–(11.67) applied to the Schrödinger equation with minimal electromagnetic coupling (11.70) induces a phase change in the wave function:

$$\psi(\mathbf{r}, t) \rightarrow e^{i\chi(\mathbf{r}, t)} \psi(\mathbf{r}, t). \quad \diamond \quad (11.71)$$

In general, the phase of a complex wave function may be changed by an arbitrary amount with no observable consequences. If the χ in (11.71) were a constant, this phase change would be called a *global* gauge transformation. Since χ is not constant, we have a *local* gauge transformation which changes the phase of the wave function by an amount which varies from spacetime point to spacetime point. This transformation will have no affect on observables *if and only if* the simultaneous transformations (11.66)–(11.67) are made on the fields. In this way, minimal coupling (11.69) is also minimal because it is the simplest prescription that is consistent with gauge invariance.

In summary, the electromagnetic field's change of the wave function is included implicitly when we solve the Schrödinger equation (11.70). However, when a gauge transformation is made on the fields, only the phase of the wave function changes. In the next section we show how the entire effect of the fields can be interpreted as yet another type of phase change, this time of the wave function along each path.

Quantum Path Integrals and the EM Force

In (11.55) and (11.56) we expressed the amplitude for a transition from a to b as the sum over paths,

$$G(b, a) = \mathcal{A} \sum_{\text{paths}} e^{iS[b, a]/\hbar}. \quad (11.72)$$

While we “derived” this relation by observing the propagation of a free particle, we postulated that it should be valid even for a particle under the influence of a force. To see how the electromagnetic field affects S we look at equation (11.62). It shows that the field modifies the Lagrangian:

$$L \rightarrow L - q\Phi(\mathbf{r}, t) + q\mathbf{v} \cdot \mathbf{A}(\mathbf{r}, t), \quad (11.73)$$

and correspondingly changes S :

$$S \rightarrow S - q \int dt \Phi(\mathbf{r}, t) + q \int dt \mathbf{v} \cdot \mathbf{A}(\mathbf{r}, t) \quad (11.74)$$

$$= S - q \int dt \Phi(\mathbf{r}, t) + q \int ds \cdot \mathbf{A}(\mathbf{r}, t), \quad (11.75)$$

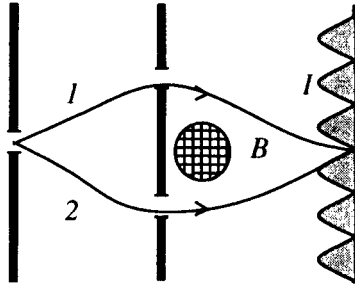


Figure 11.4: A solenoid B produces a confined magnetic field which cannot be entered by the particles passing along paths 1 or 2. The two slit interference pattern I shifts as the magnetic field changes.

where in (11.75) we have substituted ds/dt for $\mathbf{v}$, where $\mathbf{s}$ is the path variable. We see that the effect of the electromagnetic field on the propagator (11.72) is

$$G(b, a) \rightarrow \mathcal{A} \sum_{\text{paths}} e^{iS[b, a]/\hbar} \exp \left[\frac{iq}{\hbar} \int ds \cdot \mathbf{A}(\mathbf{r}, t) \right] \exp \left[\frac{-iq}{\hbar} \int dt \Phi(t) \right], \quad (11.76)$$

where S does not include the electromagnetic field, and where we have assumed no space dependence in the “scalar” potential Φ .

Equation (11.76) says inclusion of the Lorentz force into the path integral formalism is equivalent to a modification of the phase of the amplitude *along each path*. This seems similar to local gauge invariance which also modifies the phase of the wave function, yet here a force is actually acting and this is implemented by there being a different phase along each path. This phase modification is such that the sum over all paths produces the same results as the Schrödinger equation (11.70).

Aharonov–Bohm Effect

In Figure 11.4 we illustrate how beams of particles emanating from a source on the left will form an interference pattern I on the right after passing through two slits. We assume there is no external electric fields present, but that a solenoid B produces a confined magnetic field which (at least classically) cannot be entered by the particles passing along paths 1 or 2. We know that there will be an interference pattern when no magnetic field is present. We want to determine if the magnetic field affects the interference. The answer is given by (11.76) which tells us that the phase difference between paths 1 and 2 arising from the presence of the magnetic field is

$$\Delta \text{phase} = \frac{e}{\hbar} \left[\int_1 ds \cdot \mathbf{A} - \int_2 ds \cdot \mathbf{A} \right] = \frac{e}{\hbar} \left[\oint_{\text{cw}} ds \cdot \mathbf{A} \right] = \frac{e}{\hbar} \phi_B. \quad (11.77)$$

Here the difference in path integrals results in a clockwise loop integral, which, after application of Stoke’s theorem, we recognize as ϕ_B , the *magnetic flux* threading the loop.

The result (11.77), while simple to derive via the path integral formulation, is not trivial. It expresses an experimental observable directly in terms of the vector potential $\mathbf{A}$, and therefore says that the vector potential is more fundamental in quantum mechanics than is the magnetic field $\mathbf{B}$. If we look at Figure 11.4 we see that the mere existence of the magnetic field at point B is enough to shift the interference pattern I even if the beams of particles do not pass through the magnetic field (as evidenced, for example, by the particles not being deflected by the magnetic field).

How strong a magnetic flux is necessary to cause such a shift in the interference pattern? Since we wish to observe a phase change which causes a shift, we can state that there will be a displacement of the interference pattern and a return to the original pattern every time the phase shift changes by 2π , that is, each time

$$\frac{q}{\hbar}\phi_B = 2\pi n, \Rightarrow \phi_B = n \left(\frac{2\pi\hbar}{q} \right) \equiv n \left(\frac{2\pi\hbar c}{e} \right). \quad (11.78)$$

Apparently, the term in parenthesis in (11.78) must be a fundamental flux unit, and we predict that the interference pattern will shift and return to its original pattern periodically each time the magnetic field is increased by one such flux unit. This shifting was predicted by Aharonov and Bohm (1959), and later observed by Chamber (1960) and Tonomura et al. (1986).

11.7 Problems

1. Consider the 3D form for the propagator $G(\mathbf{r}_b, t_b; \mathbf{r}_a, t_a)$ given by (11.5). Evaluate it explicitly as done in the text for the 1D form, and thus prove that

$$G(\mathbf{r}_b, t_b; \mathbf{r}_a, t_a) = \left(\frac{\mu}{2\pi i(t_b - t_a)} \right)^{3/2} \exp \left[\frac{i\mu(\mathbf{x}_b - \mathbf{x}_a)^2}{2(t_b - t_a)} \right] \theta(t_b - t_a). \quad (11.79)$$

2. Show that the time-independent Schrödinger equation,

$$i \frac{\partial \psi(\mathbf{x}, t)}{\partial t} = \frac{-1}{2m} \frac{\partial^2 \psi}{\partial \mathbf{x}^2}, \quad (11.80)$$

follows from the 1D version of Huygen's principle (11.39):

$$\psi(\mathbf{x}_b, t_b) = \int d\mathbf{x}_a G(\mathbf{x}_b, t_b; \mathbf{x}_a, t_a) \psi(\mathbf{x}_a, t_a). \quad (11.81)$$

- (a) Consider an infinitesimal time separation $t_b = t + \epsilon$.
 - (b) Consider the infinitesimal space variation $\mathbf{x}_b = \mathbf{x} + \delta$ with δ a variable.
 - (c) Expand the time-dependence of the LHS of (11.81) as a Taylor series in ϵ , and space-dependence of the wave function on the RHS as a Taylor series in δ .
 - (d) Show that the Schrödinger equation is obtained as the $\delta \rightarrow 0$ and $\epsilon \rightarrow 0$ limit of these expansions.
3. In the classical, periodic motion of a particle with generalized coordinate $\mathbf{x}$ and generalized momentum $\mathbf{p}$, the energy $E(\mathbf{p}, \mathbf{x})$ is a constant (integral) of the motion. The surface in phase space enclosed by the periodic orbit is

$$F(E) = \oint \mathbf{p} d\mathbf{x} = 2 \int_{\mathbf{x}_{\min}}^{\mathbf{x}_{\max}} \mathbf{p} d\mathbf{x}.$$

- (a) Explain why the orbit in phase space (the “phase portrait”) must be symmetric with respect to the x -axis?
- (b) Show that the classical period of the motion must be the derivative of F with respect to the energy

$$T = \frac{dF(E)}{dE}.$$

(Hint: Deduce an integral expression for dF/dE and show that it is proportional to the integral expression for the period T in classical mechanics.)

- (c) Calculate F and T for the harmonic oscillator,

$$E(p, x) = \frac{p^2}{2m} + \frac{m\omega^2 x^2}{2},$$

and draw the analogy to the quantum-mechanical energy $E = (n + \frac{1}{2})\hbar\omega$.

4. **Hamilton’s Principle** We are given the experimental fact that in classical mechanics a particle falls a known distance y_0 in a known time $t_0 = \sqrt{2y_0/g}$, and we guess that the functional dependence of distance on time is

$$y = at + bt^2.$$

Show that for reasonable conditions, the integral

$$S = \int_0^{t_0} L dt$$

is an extremum only when $a = 0$ and $b = g/2$.

5. Consider a *classical* particle in the harmonic oscillator potential

$$V(x) = \frac{-\mu\omega^2}{2}x^2. \quad (11.82)$$

Show that the classical action for this particle is

$$S[b, a] = \frac{\mu\omega}{2 \sin \omega T} [(x_a^2 + x_b^2) \cos \omega T - 2x_a x_b], \quad (11.83)$$

where $T = t_b - t_a$.

Hint: First show that the classical trajectory through (x_a, t_a) and (x_b, t_b) is

$$x(t) = \frac{1}{\sin \omega T} [x_b \sin \omega(t - t_a) + x_a \sin \omega(t_b - t)], \quad (11.84)$$

and then evaluate L and from that S .

6. Consider a *quantum* particle in the potential well of a harmonic oscillator

$$V(x) = \frac{-\mu\omega^2}{2}x^2. \quad (11.85)$$

By performing a path integration, show that the quantum propagator for this particle has the form

$$G(b, a) = A(T) \exp \left[\frac{i\mu\omega}{2\hbar \sin \omega T} \{ (x_a^2 + x_b^2) \cos \omega T - 2x_a x_b \} \right], \quad (11.86)$$

where $T = t_b - t_a$. The multiplicative function of time can be determined by use of the combination rule (11.57)

$$A(T) = \sqrt{\frac{\mu\omega}{2\pi\hbar \sin \omega T}}. \quad (11.87)$$

7. (Based on Feynman and Hibbs, 3-12) Given the wave function for a harmonic oscillator at time $t = 0$,

$$\psi(x, 0) = \exp \left[-\frac{\mu\omega}{2\hbar} (x - a)^2 \right]. \quad (11.88)$$

- (a) Use Huygen's principle (11.39) and the preceding expression for the propagator to show that

$$\psi(x, t) = \exp \left[-\frac{i\omega t}{2} - \frac{\mu\omega}{2\hbar} \{ x^2 - 2ax e^{-i\omega t} + \frac{a^2}{2}(1 + e^{-2i\omega t}) \} \right]. \quad (11.89)$$

- (b) Find the probability distribution $|\psi|^2$ and comment on its x and t dependences.

8. Consider a *quantum* particle that is constrained to move along the circumference of a unit circle. Let the distance along the circumference be measured by the angle ϕ .

- (a) Show that if the initial wave function of the particle is $\psi(\phi_a, 0)$, then at a later time

$$\psi(\phi, t) = A(t) \sum_{m=-\infty}^{\infty} \int_0^{2\pi} d\phi_a \exp \left[\frac{i\mu}{2\hbar t} (\phi + 2\pi m - \phi_a)^2 \right] \psi(\phi_a, 0), \quad (11.90)$$

where the sum is over paths making m revolutions from the time they leave ϕ_a to the time they arrive at ϕ .

- (b) Express (11.90) as a single integral and show that it is single valued.
 (c) Determine the function $A(t)$.

CHAPTER 12

APPLIED PATH INTEGRALS

Part of the art of computational physics is deriving theoretical equations which are amenable to computation and which yield reliable and useful information. Part of the value of computational physics is that it permits us to look at physics in new ways and permits us to solve problems with theories which previously appeared to be of only “academic” interest. Accordingly, rather than forge ahead and try to directly compute the equations of Chapter 11, *Path Integrals and Lattice Quantum Mechanics*, we start this chapter by deriving a rather ingenious relation between the Feynman propagator and the ground state wave function.¹ We then develop a project which computes path-integrals on a lattice (Mannheim, 1983; MacKeown, 1987). The developments in this chapter should make the abstract path integration of the last chapter more concrete. We advise all readers (even those with no interest in computation) to read this chapter. Those with interest in the computation will find a general description of a realistic program and an actual source listing at the end of the chapter.

The assumption in lattice calculations is that for small enough lattice spacings and large enough lattices, these discrete and finite calculations become a good approximation to continuous reality. In recent times path integral formulations have been used to solve the field theoretic equations of quantum chromodynamics on a lattice (Wilson, 1974; Kogut, 1983), although not with fine enough lattices to be certain about extrapolations to the continuum limit. These calculations are not meant to be a demonstration of an alternative formulation of quantum mechanics, but rather are the last resort at solving the equations of quantum chromodynamics.

12.1 The Propagator–Bound State Connection

As we have seen in (11.4) of the last chapter, the propagation of a wave function forward in time can be described in terms of the propagator G :

$$\psi(x, t) = \int_{-\infty}^{+\infty} G(x, t; x_0, t_0) \psi(x_0, t_0) dx_0, \quad (12.1)$$

¹ In this chapter we also eliminate the $\hbar$'s we included for pedagogical purposes in Chapter 11.

where we have gone to one dimension and have labeled the point a as (x_0, t_0) , and b as (x, t) . The integral equation (12.1) for the wave function does not treat time and space symmetrically—Schrödinger theory is after all not covariant. Yet (12.1) is equivalent to the usual Schrödinger theory, and we use that equivalence to solve for the ground state wave function in terms of G . We assume the Hamiltonian $\hat{H}$ supports a spectrum of eigenfunctions,

$$\hat{H}\psi_n = E_n\psi_n, \quad (n = 0, 1, 2, \dots, \infty). \quad (12.2)$$

With a Hermitian Hamiltonian $\hat{H}$, these solutions form a complete orthonormal set, and so we expand the general time-independent solution as a series in these eigenfunctions, and then generalize it to obtain the time dependence:

$$\psi(x, t = 0) = \sum_{n=0}^{\infty} c_n \psi_n(x), \quad (12.3)$$

$$\Rightarrow \psi(x, t) = e^{-i\hat{H}t} \psi(x, t = 0) = \sum_{n=0}^{\infty} c_n e^{-iE_n t} \psi_n(x). \quad (12.4)$$

The exponential time dependence on the extreme RHS of (12.4) can be considered as the usual time dependence associated with a stationary eigenstate, or as the effect of $\exp(-i\hat{H}t)$ on each eigenstate. The time-independent expansion coefficient c_n is determined by multiplying both sides of (12.3) by $\psi_n^*(x)$, integrating over x , and invoking the orthonormality of the ψ_n 's to obtain:

$$c_n = \int_{-\infty}^{+\infty} dx_0 \psi_n^*(x_0) \psi(x_0, t = 0), \quad (12.5)$$

where we have used x_0 as a dummy integration variable. If we substitute this c_n back into the expansion (12.4), we obtain the identity:

$$\psi(x, t) = \int_{-\infty}^{+\infty} dx_0 \sum_n \psi_n^*(x_0) \psi_n(x) e^{-iE_n t} \psi(x_0, t = 0). \quad (12.6)$$

The preceding steps should appear familiar because they are a spectral representation of the rule (11.4) we derived in the last chapter. A comparison of (12.6) with (12.1) yields an expression for the propagator:

$$G(x, t; x_0, t_0 = 0) = \sum_n \psi_n^*(x_0) \psi_n(x) e^{-iE_n t}. \quad (12.7)$$

This expression is another representation of the definition (11.5) we deduced in the last chapter.

Wick's Time Rotation

The time t in the time-dependent Schrödinger equation is just a variable; there is nothing sacrosanct about it. In fact, it is a standard procedure in solving elliptic partial differential equations to change to imaginary variables, and we will be so bold as to evaluate the Schrödinger equation at a negative *imaginary time*:

$$\psi(x, t = -i\tau) = e^{-\hat{H}\tau} \psi(x, t = 0). \quad (12.8)$$

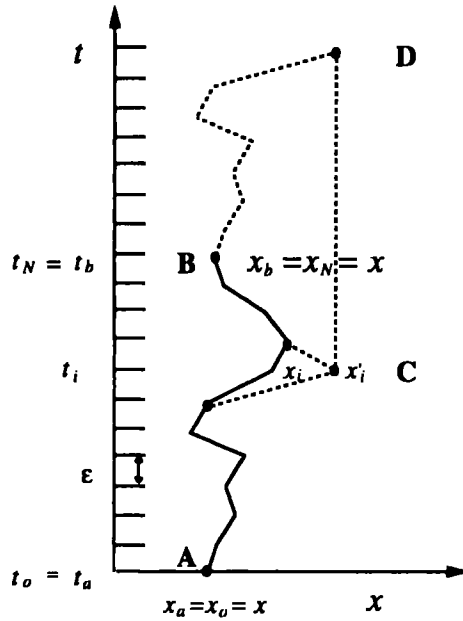


Figure 12.1: A path through the spacetime lattice which starts and ends at $x = x_a = x_b$. The action is an integral over this path, while the “path integral” is a sum of integrals over all paths. The dotted path BD is a transposed replica of the path AC.

Here τ is a time variable measured along an imaginary axis. Alternatively, one can think of (12.8) as an analytic continuation of the Schrödinger equation in which time has been rotated into the imaginary axis, a procedure known as a *Wick rotation*.

A general $\psi(x, t)$ we compute approximately need not be an eigenfunction of $\tilde{H}$ (that is, a stationary state of definite energy). This being the case, (12.8) indicates that if we wait a long enough imaginary time (on a scale set by $\hbar/\Delta E_n$), the parts of ψ with higher energies will decay away most quickly and leave only the ground state ψ_0 :

$$\lim_{\tau \rightarrow \infty} \psi(x, t = -i\tau) = e^{-E_0\tau} \psi_0(x), \quad (12.9)$$

where the 0 subscript on the RHS, but not the LHS, indicates the ground state.

If we play this same imaginary time trick on the eigenfunction expansion of the propagator (12.7), we obtain

$$G(x, t = -i\tau; x_0, 0) = \sum_n \psi_n^*(x_0) \psi_n(x) e^{-E_n\tau} \quad (12.10)$$

$$\longrightarrow \psi_0^*(x_0) \psi_0(x) e^{-E_0\tau} \quad (\tau \rightarrow \infty). \quad (12.11)$$

If, as shown in Figure 12.1, we set $x_0 = x$, we obtain for the wave function:

$$|\psi_0(x)|^2 = \lim_{\tau \rightarrow \infty} e^{E_0\tau} G(x, t = -i\tau; x_0 = x, 0). \quad (12.12)$$

Although in the last chapter we assumed $x_a \neq x_b$, assuming them equal just means that the particle is at the same place in space at times t_a and t_b .² Equation (12.12) provides a closed-form solution for the ground state wave function directly in terms of the propagator G . If we evaluate the propagator using Feynman's postulates, we obtain a closed-form solution for the ground-state wave function.

You may also recall from the last chapter that there are difficulties in determining the normalization factors for G and that these normalization factors are truly determined only after the limit to infinitesimal times are taken (something we do not have the time to do with a simulation on a finite computer). A solution to this difficulty is to build the normalization constraint into the solution for the wave function. We do this by first determining the normalization $\mathcal{N}$ of the wave function,

$$\mathcal{N} = \int_{-\infty}^{+\infty} dx |\psi_0(x)|^2 = \lim_{\tau \rightarrow \infty} e^{E_0 \tau} \int_{-\infty}^{+\infty} dx G(x, -i\tau; x, 0). \quad (12.13)$$

We then divide the solution for $|\psi_0|^2$ by $\mathcal{N}$ to obtain

$$|\psi_0(x)|^2 = \frac{\lim_{\tau \rightarrow \infty} e^{E_0 \tau} G(x, t = -i\tau; x, 0)}{\lim_{\tau \rightarrow \infty} e^{E_0 \tau} \int_{-\infty}^{+\infty} dx G(x, -i\tau; x, 0)} = \lim_{\tau \rightarrow \infty} \frac{G(x, t = -i\tau; x, 0)}{\int_{-\infty}^{+\infty} dx G(x, -i\tau; x, 0)}. \quad (12.14)$$

The $x = x_0 \equiv x_a$ appearing in (12.14) is the initial and final positions of the system at the time $t = t_0 \equiv 0$. While the general paths shown in Figure 11.1 do not have to start and end at the same space position, this expression for $|\psi_0(x)|^2$ requires that the paths return to the same x value. This is as shown in Figure 12.1.

12.2 Computation of the Propagator

Recall Figure 11.1 and Feynman's postulate that the propagator $G(b, a)$ is a sum over all paths connecting a to b , with each path weighted by the exponential of the action evaluated along that path:

$$G(b, a) = \mathcal{A} \sum_{\text{paths}} e^{iS[b, a]} \stackrel{\text{def}}{=} \mathcal{A} \int_a^b \mathcal{D}x e^{iS[b, a]}. \quad (12.15)$$

Equation (12.15) gives a form appropriate to evaluation on a lattice such as that in Figure 12.1. The lattice is made up of N equal time intervals of length

$$\epsilon = t_j - t_{j-1} = \frac{t_N - t_0}{N} \quad (12.16)$$

connecting the initial time $t_a = t_0$ to the final time $t_b = t_N$, and some appropriate region in space with a comparable number of space intervals of length δ . (The actual values for ϵ and δ are chosen and adjusted empirically when the computation is carried out.) Once we

²We have already shown in (11.23) that a delta function occurs in G if we set $t_a = t_b$, yet there are no singularities in ψ or G if we set $x_a = x_b$. However, (11.24) shows that the second derivative of ψ will be singular in this case.

have our lattice, each path connecting a to b is approximated as a sum of N straight-line links through time and space. Since the sum of any function over discrete $x_j \equiv x(t_j)$ points on the lattice is equivalent to the integral of that function over all the x values of the trajectories, the sum over paths in (12.15) is equivalent to integrations over x and is often written as an integration rather than a sum. That is, the continuous variation of paths δx_j , which is at the heart of Hamilton's principle, is approximated by a sum over x_j values on the lattice. While this may appear to be a crude approximation to the infinite number of paths which should be included, most paths lie far from the classical trajectory with its minimum action, and so contribute very little to the sum. We thus approximate the infinite number of paths by a relatively small number of paths lying close to the classical trajectory, all of which have approximately the minimum action.

Our computation of the propagator is based on (12.15) which we write as:

$$G(x, t; x_0, t_0) = A^N \oint dx_1 dx_2 \dots dx_{N-1} e^{iS[x, x_0]}, \quad A = \sqrt{\frac{\mu}{2\pi i \epsilon}}, \quad (12.17)$$

where the integration (summation) is over all paths, and the paths will be picked with a Monte Carlo technique. The exponential of the total action along a path weights that path, and we take the total action to be the sum of the action for N links,

$$S[x, x_0] = \sum_{j=1}^{N-1} S[x_{j+1}, x_j] \simeq \sum_{j=1}^{N-1} L(x_j, \dot{x}_j; t_j) \epsilon, \quad (12.18)$$

where $L(x_j, \dot{x}_j; t_j)$ is the average value of the Lagrangian on link j which corresponds to time $t = j\epsilon$.

We now need to specify how the action is to be computed. To keep the project as simple as possible, we assume a local and velocity-independent potential $V(x)$. Next we observe that for the calculation of $|\psi_0|^2$ with the expression (12.14), G is evaluated with a negative imaginary time, which in turn means the time variable t in L should also change to $-i\tau$:

$$L\left(x, \frac{dx}{dt}; t\right) = T - V(x) = +\frac{1}{2}m\left(\frac{dx}{dt}\right)^2 - V(x), \quad (12.19)$$

$$L\left(x, \frac{dx}{-id\tau}; t = -i\tau\right) = -\frac{1}{2}m\left(\frac{dx}{d\tau}\right)^2 - V(x). \quad (12.20)$$

We see that the imaginary time leads to a reversal in the sign of the kinetic energy term in L , and this leads to the Lagrangian evaluated at a negative time $t = -i\tau$ being equal to the negative of the Hamiltonian evaluated at a real positive time $t = \tau$:

$$H\left(x, \frac{dx}{d\tau}; t = \tau\right) = \frac{1}{2}m\left(\frac{dx}{d\tau}\right)^2 + V(x) = E, \quad (12.21)$$

$$\Rightarrow L\left(x, \frac{dx}{-id\tau}; t = -i\tau\right) = -H\left(x, \frac{dx}{d\tau}; t = \tau\right). \quad (12.22)$$

We thus rewrite the t -path integral of L as a τ -path integral of H along a trajectory and thereby express the propagator completely in terms of the Hamiltonian,

$$S[x_{j+1}, x_j] = \int_{t_j}^{t_{j+1}} L(x, t) dt = -i \int_{\tau_j}^{\tau_{j+1}} H(x, \tau) d\tau. \quad (12.23)$$

This then gives us the solution for G :

$$G(x, t = -i\tau; x_0, 0) = A^n \int dx_1 \dots dx_{N-1} e^{-\int_0^\tau H(x, \tau') d\tau'}, \quad (12.24)$$

where this line integral of H is over an entire trajectory. To make the expressions simpler we express this path integral in terms of an average energy of the particle on each link, $E_j = T_j + V_j$, and then sum over links to obtain the summed energy $\mathcal{E}$,

$$\int_0^\tau H(x, \tau') d\tau' \simeq \sum_j \epsilon E_j = \epsilon \mathcal{E}(\{x_j\}), \quad (12.25)$$

$$\mathcal{E}(\{x_j\}) \approx \sum_{j=1}^N \left[\frac{m}{2} \left(\frac{x_j - x_{j-1}}{\epsilon} \right)^2 + V \left(\frac{x_j + x_{j-1}}{2} \right) \right]. \quad (12.26)$$

In (12.26) we have approximated each path link as a *straight line*, have used Euler's derivative rule $dx/dt \simeq \Delta x / \Delta t$ for the velocity on that link, and have evaluated the potential at the midpoint of each link.³

We now substitute this expression for G into our solution (12.14) for the ground state wave function, which you will recall requires that the initial and final points in space be the same, $x_0 = x$:

$$|\psi_0(x)|^2 = \lim_{\tau \rightarrow \infty} \frac{G(x, t = -i\tau, x_0 = x, t = 0)}{\int_{-\infty}^{\infty} dx G(x, t = -i\tau, x_0 = x, t = 0)} \quad (12.27)$$

$$= \frac{A^N \int dx_1 \dots dx_{N-1} \exp \left[-\int_0^\tau H d\tau' \right]}{A^N \int dx dx_1 \dots dx_{N-1} \exp \left[-\int_0^\tau H d\tau' \right]} \quad (12.28)$$

$$= \frac{1}{Z} \lim_{\tau \rightarrow \infty} \int dx_1 \dots dx_{N-1} e^{-\epsilon \mathcal{E}}, \quad (12.29)$$

$$Z = \lim_{\tau \rightarrow \infty} \int dx dx_1 \dots dx_{N-1} e^{-\epsilon \mathcal{E}}. \quad (12.30)$$

The similarity to thermodynamics, even with a partition function Z , is no accident; by making the time parameter of quantum mechanics imaginary we have converted the time-dependent Schrödinger equation to the heat diffusion equation. This similarity leads to the integrand in (12.29) being called f , as in statistical mechanics, and the identification of f as a Boltzmann distribution function with a temperature proportional to the inverse of the time step:

$$f = e^{-\epsilon \mathcal{E}} = e^{-\mathcal{E}/k_B T}, \quad (12.31)$$

$$\Rightarrow k_B T = 1/\epsilon \equiv \hbar/\epsilon. \quad (12.32)$$

Accordingly, the limit $\epsilon \rightarrow 0$ which makes time continuous is a “high temperature” limit. The limit $\tau \rightarrow \infty$, which is required to project out the ground state wave function, means we must integrate over a path which is long in imaginary time (i.e. long compared with

³We could also have used the linear average, $V \simeq [V(x_j) + V(x_{j-1})]/2$. For more precision we could use the Simpson rule for the potential, $V \simeq [V(x_j) + 4V(\frac{x_j + x_{j-1}}{2}) + V(x_{j-1})]/6$, and the central difference rule for the derivative, $\dot{x}_j \simeq (x_{j+1} - x_{j-1})/(2\epsilon)$.

$\hbar/\Delta E$). In addition we wait around a long real time while the algorithm repeats in order for our system to equilibrate (find the classical trajectory). Once the system equilibrates the remaining thermal fluctuations will simulate Feynman's quantum fluctuations about the classical trajectory in spacetime. In principle this is a closed form solution of the Schrödinger's equation for the wave function on a discrete lattice in spacetime which becomes exact as the lattice becomes infinitely large and dense.

The evaluation of path integrals like (12.29) requires the integration of the Hamiltonian along each trajectory and the summation over all trajectories. We evaluate this so-called "path integration" as the sum of 1 over all space and time points in our lattice, with the points chosen with a Monte Carlo technique so that they are random but weighted by the Boltzmann factor. We generate this weighted distribution with the Metropolis (heat-bath) algorithm (which we describe in the next section). In general, Monte Carlo Green's function techniques work best if we start off with a good guess at the correct answer and have the algorithm calculate corrections to our guess. For the present problem this means that if we start off with a path close to the classical trajectory in spacetime, the algorithm may be expected to do a good job at simulating the quantum fluctuations about our initial guess.

As we have formulated our computation, we would pick a value of $\mathbf{x}$ and perform a rather lengthy computation of line integrals over all space and time to obtain the modulus of the wave function at this $\mathbf{x}$. To obtain the wave function at another value of $\mathbf{x}$, the entire simulation would have to be repeated from scratch. Rather than go through all that trouble again and again, there is a trick which permits us to compute the entire $\mathbf{x}$ dependence of the wave function in a single run. The trick is to insert in the probability integral (12.29) a delta function which fixes the initial position $\mathbf{x}_0$, and then to integrate over all initial positions:

$$|\psi_0(\mathbf{x})|^2 = \lim_{\tau \rightarrow \infty} \frac{1}{Z} \int d\mathbf{x}_1 \dots d\mathbf{x}_{N-1} e^{-\epsilon \mathcal{E}(\mathbf{x}, \mathbf{x}_1, \dots, \mathbf{x}_N)} \quad (12.33)$$

$$= \lim_{\tau \rightarrow \infty} \frac{1}{Z} \int d\mathbf{x}_0 d\mathbf{x}_1 \dots d\mathbf{x}_{N-1} \delta(\mathbf{x} - \mathbf{x}_0) e^{-\epsilon \mathcal{E}(\mathbf{x}_0, \mathbf{x}_1, \dots, \mathbf{x}_N)}. \quad (12.34)$$

Equation (12.34) expresses the wave function as an average of a delta function over all paths, a procedure which might appear totally inappropriate for numeric computation since there is tremendous error in representing a singular function on a finite-word-length computer. Yet when we simulate the sum over all paths with (12.34), there will always be some $\mathbf{x}$ value for which the integral is nonzero, and we need only accumulate the solution for various $\mathbf{x}$ values to determine $|\psi_0(\mathbf{x})|^2$ for all $\mathbf{x}$. To understand how this works, consider path AB in Figure 12.1 for which we have just calculated the summed energy. We next let one point on the chain jump to point C (which changes two links) to form a new path. If we replicate section AC and use it as the extension AD to form the top path, we see that the path CBD has the same summed energy (action) as the path ACB and thus can be used to determine $|\psi(\mathbf{x}'_j)|^2$. Accordingly, once the system is equilibrated, we determine new values of the wave function at new locations $\mathbf{x}'_j$ by flipping links to new values and calculating new actions. The more often some $\mathbf{x}'_j$ occurs (is accepted) in the simulation, the greater is the wave function at that point.

12.3 Metropolis Algorithm ☉

In their simulation of neutron transmission through materials, Metropolis, Rosenbluth, Teller, and Teller (1953) found a Monte Carlo algorithm which has proven to be well suited

for simulating the fluctuations which occur when a system is in thermal equilibrium. With this *Metropolis algorithm* for path generation, individual links which comprise the path are varied randomly in such a way that on the average, the probability of any one path occurring follows a Boltzmann distribution based on the value of the action for that path. This means that for a given value of the time step $\epsilon \propto 1/T$, the algorithm is repeated until the classical trajectory (thermal equilibrium) is reached, and then the statistical fluctuations about equilibrium are used to calculate the average values of the quantities of interest. Since there are N time steps, each trajectory is composed of N links. But since there is no limit to the number of trajectories which may contribute, even though N is finite the number of paths can be made arbitrarily large. In any case, the number of configurations of N links can be very, very large, on the order of e^N , and so the amount of computer time needed to run this algorithm can be very, very long. Our hope is that after waiting for thermal equilibrium so that the paths summed over all lie close to the classical trajectory, good answers are obtained with a relatively small number of paths, $M \simeq 10N$. Since the statistical relative error is proportional to $1/\sqrt{M}$, we are bound to get better and better answers if we sample more and more paths (larger M). Yet since the average roundoff error arising from the finite word length on the computer increases as more terms are summed, at some point increasing M will only makes the answer worse.

We define an individual path by an array of N numbers $\alpha = \{x_1, x_2, \dots, x_N\}$ which gives the values of the endpoints of the individual links comprising the path. The Metropolis algorithm is used to generate a nonuniform, random distribution of α_j 's, each α_j with probability

$$\mathcal{P}(\alpha_j) = \frac{1}{Z} e^{-\epsilon \mathcal{E}(\alpha_j)}. \quad (12.35)$$

This algorithm is a variation of the von Neumann rejection technique (Koonin, 1986) in which we start with an initial path, and then vary it in a random way to obtain a *trial* path. If ΔE is the change in the summed energy of the trial path relative to the previous path, then the Boltzmann factor tells us that the relative probability of this trial path is proportional to $\Delta \mathcal{P} = \exp(-\epsilon \Delta E)$. If the trial path has a lower energy, ΔE will be negative, the relative probability will be greater than one, and we accept the trial path as the new path with no further ado. If on the other hand the trial path has a higher energy, we do not reject it out of hand, but instead accept it with the relative probability $\Delta \mathcal{P} = \exp(-\epsilon \Delta E) < 1$. To accept with a probability, we pick a uniform random number between 0 and 1, and if the relative probability is greater than this number, we accept the trial path; if the Boltzmann factor is smaller than the chosen random number, we reject it. When the trial path is not accepted, the next path is identical to the preceding one.

The key aspects of the Metropolis algorithm is that the weight given to a trial path depends upon how far it is from the classical trajectory. Those paths which stray far from the classical trajectory are deemphasized but not completely discarded. By permitting the system to deviate away from the classical trajectory (go “uphill” for a while), this technique is successful at finding a global extremum for situations in which other techniques are successful at finding only local ones. Its success relies on it not being too quick in jumping (cooling) to the classical trajectory, and for this reason the algorithm is sometimes called *simulated annealing*. For path integral evaluations, we would end up with the classical action if the system always went downhill. The algorithm's fluctuations in its search for the classical trajectory simulate Feynman's postulate that a quantum system fluctuates about the classical trajectory. And while the algorithm may not to quick to find the classical trajectory from scratch, it is expected to be good at finding the fluctuations about the

classical trajectory.

The explicit rules for the Metropolis algorithm which changes one link in a path at a time are:

1. Start with some arbitrary, initial path $\alpha_k = \{x_1, x_2, \dots, x_N\}$.
2. To generate the new path α_{k+1} :
 - (a) Pick a uniform random integer between 1 and N which determines the link i .
 - (b) Create a trial path α_{tr} by having a link jump to a nearby lattice site.
 - (c) Calculate the energy $\mathcal{E}(\alpha_{tr})$ of the trial path.
 - (d) Make the decision:
 - i. If $\mathcal{E}(\alpha_{tr}) \leq \mathcal{E}(\alpha_k)$, accept the trial: $\alpha_{k+1} = \alpha_{tr}$.
 - ii. If $\mathcal{E}(\alpha_{tr}) > \mathcal{E}(\alpha_k)$, calculate $\Delta\mathcal{P} = \exp(-\epsilon\Delta E)$ and accept the trial with probability $\Delta\mathcal{P}$ by:
 - A. Choosing a uniform random number: $0 \leq r \leq 1$.
 - B. Let $\alpha_{k+1} = \begin{cases} \alpha_{tr}, & \text{if } \Delta\mathcal{P} \geq r \text{ (accept),} \\ \alpha_k, & \text{if } \Delta\mathcal{P} < r \text{ (reject).} \end{cases}$

The paths change slowly with the Metropolis algorithm, analogously to thermal fluctuations in a physical system placed in contact with a heat bath. Therefore, some computational physicists wait for a number $\simeq N$ link rearrangements before calculating quantities in order to ensure that the initial path has had a good chance to get close to the classical trajectory (to equilibrate). Our goal is to find the quantum average path after running a relatively small number of simulations which sample the most important, but not all, possible paths. The statistics will be improved by repeating the calculation for different starting paths and taking the average of results.

12.4 Computational Project

Figure 12.2 shows some results from an application of the Metropolis algorithm using the program given at the end of this section. In this computation we start off with an initial path close to the classical trajectory and then examine one half million variations about this path. All paths are constrained to begin and end at $x = 1$ (which turns out to be somewhat less than the amplitude of the classical oscillation). In the subroutine *Pathenergy* we defined the energy such that $\mu = k = 1$ and correspondingly the oscillator has a period $T = 2\pi$. When we run the program with a time difference $t_b - t_a$ equals the short time $2T$, the system does not have enough time to decay to its ground state and, as we see in the top of Figure 12.2, the wave function looks like the probability distribution of an excited state (nearly classical in fact with the probability highest for the particle to be near its turning points where its velocity vanishes). However when the time difference $t_b - t_a$ equals the long time $20T$, the system has enough time to decay to its ground state and the wave function looks like the Gaussian probability distribution of the ground state. In either case we see in the bottom part of Figure 12.2 that the trajectory through spacetime fluctuates about the classical trajectory. This fluctuation is a consequence of the Metropolis algorithm occasionally going uphill in its search; if you modify the program so that searches only go downhill, the spacetime trajectory will be a very smooth trigonometric function (the

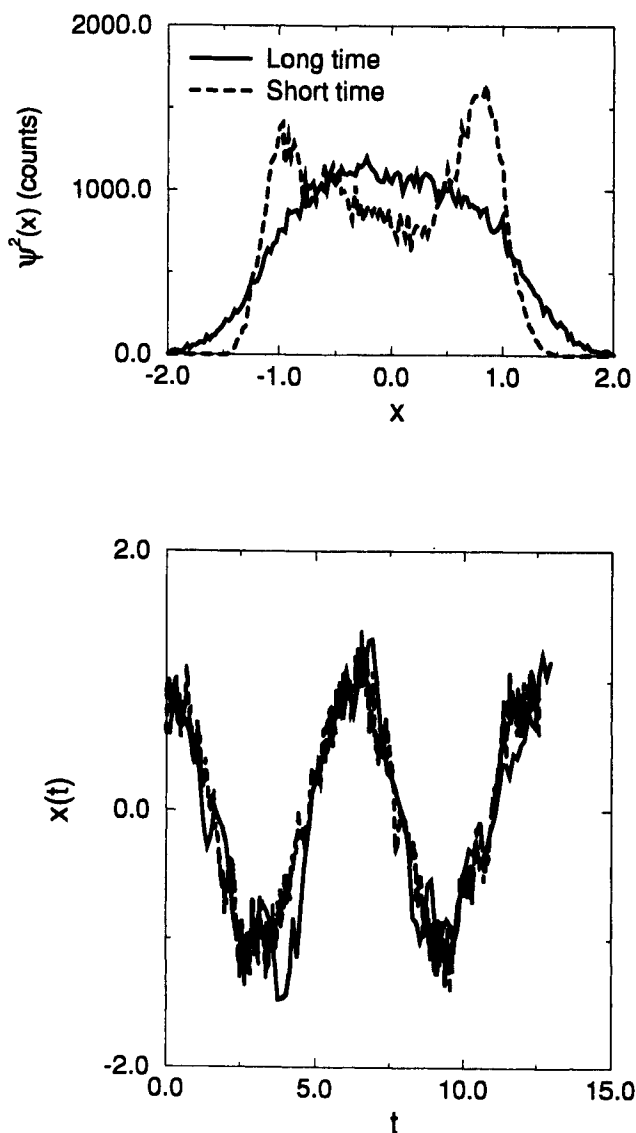


Figure 12.2: The ground state wave function of the harmonic oscillator as determined with the Metropolis algorithm. *Upper:* The dashed curve is the wave function for a value of $t_b - t_a$ equal to twice the classical period and the solid curve for 20 times the classical period. The long time yields a wave function closer to the Gaussian form expected for the ground state. *Lower:* The trajectories in space time used last in the solutions for the wave function. The oscillator has initial and final amplitudes of $x = 1$, $m = k = 1$, and accordingly a period of $T = 2\pi$.

classical trajectory), but the wave function, which is a measure of the fluctuations about the classical trajectory, will vanish!

The program evaluates the integral (12.15) by finding the average of the integrand $\delta(\mathbf{x}_0 - \mathbf{x})$ with paths distributed according to the weighting function $\exp[-\epsilon \mathcal{E}(\mathbf{x}_0, \mathbf{x}_1, \dots, \mathbf{x}_N)]$. The physics enters via (12.37), the calculation of the summed energy $\mathcal{E}(\mathbf{x}_0, \mathbf{x}_1, \dots, \mathbf{x}_N)$. For this program we evaluate the action integral for the harmonic oscillator potential

$$V(x) = \frac{1}{2}x^2, \quad (12.36)$$

and for a particle of mass $\mu = 1$. A convenient set of natural units is to measure lengths in $\sqrt{1/\mu\omega} \equiv \sqrt{\hbar/\mu\omega} = 1$, and times in $1/\omega = 1$. Since the period is 2π , long times would be $\tau \simeq 20\pi$, and the wave function's range is approximately $-2 \leq x \leq 2$. The total time should be long enough for the system to decay to its ground state, and you can determine this empirically by producing a wave function which "looks good".

The steps of the calculation are:

1. Construct a time grid of N time steps of length ϵ as in Figure 12.1. Start at $t = 0$ and extend to time $\tau = N\epsilon$ [this means N time intervals and $(N + 1)$ lattice points in time]. Note that time always increases monotonically along a path.
2. Construct a space grid of M points separated by steps of size δ . Use a range of x values several times larger than the characteristic size or range of the potential being used, and start with $M \simeq N$.
3. When calculating the wave function, any x or t value falling between lattice points should be assigned to the closest lattice point.
4. Associate a position x_j with each time τ_j subject to the boundary conditions that the initial and final positions always remain the same, $x_N = x_0 = x$.
5. Choose an *arbitrary path* of straight-line links connecting the lattice points. For the most realistic simulation it may be best to start with something close to the classical trajectory since otherwise the simple numerical procedures may not converge rapidly. Note that the x values for the links of the path may have values which increase, decrease, or remain unchanged (in contrast to time which always increases).
6. Evaluate the energy $\mathcal{E}(\mathbf{x}_0, \mathbf{x}_1, \dots, \mathbf{x}_N)$ by summing the kinetic and potential energies for each link of the path starting at $j = 0$,

$$\mathcal{E}(\mathbf{x}_0, \mathbf{x}_1, \dots, \mathbf{x}_N) \simeq \sum_{j=1}^N \left[\frac{m}{2} \left(\frac{x_j - x_{j-1}}{\epsilon} \right)^2 + V \left(\frac{x_j + x_{j-1}}{2} \right) \right]. \quad (12.37)$$

7. Begin the first of a sequence of repetitive steps in which a random position x_j associated with time t_j is changed to the position x'_j . As shown by point C in Figure 12.1, this changes *two* links in the path.
8. For the coordinate which gets changed, weigh the change with the Boltzmann distribution (12.31) by using the *Metropolis algorithm*.
9. For each lattice point establish a running sum to represent the value of the wave function squared at that point.

10. After each single-link change (or decision not to change), increase the running sum for the new x value by 1. After a sufficiently long running time, the sum divided by the number of steps is the simulated value for $|\psi(x_j)|^2$ at each lattice point x_j .
11. Repeat the entire link-changing simulation using a different seed for the Metropolis algorithm. The average wave function from a number of intermediate-length runs should be better than that from one very long run.
12. For a more continuous picture of the wave function, make the x lattice spacing smaller; for a more precise value of the wave function at any particular lattice site, sample more points (run longer) and use a smaller time step ϵ .
13. Because there are no nodes in a ground state wave function, we can ignore the phase and assume

$$\psi(x) = \sqrt{\psi^2(x)}. \quad (12.38)$$

We then can estimate the energy via

$$E = \frac{\omega}{2} \int_{-\infty}^{+\infty} \psi^*(x) \left(-\frac{d^2}{dx^2} + x^2 \right) \psi(x) dx, \quad (12.39)$$

where we evaluate the space derivative numerically.

Sample Programs

Here is a sample program to evaluate a path integration using a quantum Monte Carlo method to solve for the ground state wave function of a one-dimensional potential. These programs are based on those written by Sean Fox, and we thank him for permission to use them.

qmc.f is the main program which computes $|\psi_0(x)|^2$.

pathenergy.f calculates the summed energy over a trajectory.

drand48, **srand48** are for random number generation. They may have to be replaced by local functions.

qmc.f

```

c      Main qmc computes ground state wave function**2 using
c      Monte Carlo (Metropolis) evaluation of path integrals
c      for 1D, local, quantum mechanical systems.
c      The mass value and potential constants are used in
c      subroutine Pathenergy.
c      Time interval 'tb_ta' should be large so excited states
c      decay relative to ground state.
c      PROGRAM qmc
c      External PathEnergy

c
c      drand48 generates random numbers [0,1];
c      srand48 or seed48 used to seed random number generator.
c      arrays are excessively large to allow later manipulations.
c
c      Double Precision xvalues(1000), drand48
c      Integer t, count, paths, ticks, intrange, ix, psi(1005)
c      Integer*4 seed
c      Double Precision xb4, xnew, Eb4, Enew, deltaE, deltaX, xrange

```

```

      Double Precision tb_ta, xstep, xchange
c
c      Input: number of time subdivisions, the range of x values,
c      the number of 'bins' into which x-values are subdivided,
c      total time path covers, and max xstep size between x's.
      ticks = 1000
      xrange = 4.
      deltaX = 1./40.
c      need run long tb_ta to converge to ground state
      tb_ta = 4.*3.141593*10.
      xstep = 0.25
      epsilon = tb_ta/ticks
c      intrange = number "bins" psi gets divided into
      intrange = xrange/deltaX + 1
      call seed48(seed)
c      set initial xvalues to 0, 1st & last xvalues equal
      Do 10 count = 2, ticks
         xvalues(count) = cos((count-1.)*epsilon)
10      Continue
      xvalues(1) = 1.
      xvalues(ticks) = xvalues(1)
c      set psi to zero
      Do 15 count = 1, intrange
         psi(count) = 0.0
15      Continue
c      Calculate Eb4 = epsilon * summed path energy
      Eb4 = PathEnergy(xvalues, ticks, tb_ta)
c
c      Generate a randomly evolving configuration of x values with
c      changes in x coordinates weighted by the relative energies of
c      new and old paths. The first 10,000 won't be
c      recorded as they just allow the system to relax
c
      paths=500000
      paths = paths + 10000
c      Choose a t randomly and record its x-value.
      Do 30 count = 1, paths
         t = (ticks - 1) * drand48() + 1.5
         xb4 = xvalues(t)
         xchange = xstep * (2.*drand48() - 1.)
c      If x outside of range, ignore the change.
         If (abs(xb4 + xchange).GT.(xrange/2.)) Then
            xnew = xb4
         Else
            xnew = xb4 + xchange
         Endif
         xvalues(t) = xnew
c      Initial and final x-positions equal and unchanged.
         If ((t .eq. ticks) .or. (t .eq. 1)) Then
            xvalues(t) = xb4
         Endif
         Enew = PathEnergy(xvalues, ticks, tb_ta)
c
c      calculate change in energy between old and new paths
c      Compare a random # (0, 1) to exp(delta E).
c      Either the new path is rejected and the x-coordinate is
c      reset to its old value, or the path is accepted and its
c      energy is stored in Eb4 so that it will be available next time.
c      Take care of probabilities greater than one.
c
         deltaE = (Enew - Eb4)
         If (exp(-deltaE) .GT. 1) Then
            deltaE = 0.
         Endif
c      Can reverse test here so only goes downhill in search
c      If (exp(-deltaE) .LT. 1. ) Then
         If (exp(-deltaE) .LT. drand48()) Then
            xvalues(t) = xb4
         Else
            Eb4 = Enew
         Endif

```

```

c      Increase wave function's x bin by 1.
c      Wait 10000 iterations for equilibrium
c      If (count .GT. 10000) Then
          ix = nint((xvalues(t) + (xrange/2.))/deltaX + 1)
          psi(ix) = psi(ix) + 1
      Endif
30    Continue
c      Output wave function
      print *, 'paths, ticks, intrange'
      print *, paths, ticks, intrange
      print *, 'x, psi**2(x)'
      Do 40 count=1, intrange
          x = (count-1.)*deltaX - xrange/2.
          print *, x, psi(count)
40    Continue
      print *
      print *, 'final x vs t'
      print*
          Do 51 count=1, ticks
              print *, count*tb_ta/ticks, xvalues(count)
51    Continue
      Stop
      End

```

pathenergy.f

```

c      PathEnergy computes epsilon*(sum KE + PE) for the
c      straight-line links of a path with N time intervals.
c      'path' is the array with dimension 'ticks'.
c      'path' contains x-values for the various time coordinates.
c      'tb_ta' is the time covered in 'path', i.e. the path is
c      'tb_ta' long and has been divided into 'ticks' number of
c      intervals, each 'epsilon' long.
c
      Double Precision Function PathEnergy(path, ticks, tb_ta)
      Double Precision path(1000)
      Double Precision Esegment, epsilon, tb_ta
      Integer time, ticks
      PathEnergy = 0.
      epsilon = tb_ta/ticks

c
c      Add KE and PE contributions for each segment.

      Do 10 time = 2, ticks
          Esegment = ((path(time) - path(time - 1))/(epsilon))**2
c      Can change sign of V so Ldt not H dtau; works either way
          Esegment = Esegment + 0.25*(path(time)+path(time - 1))**2
          PathEnergy = PathEnergy + (epsilon/2) * Esegment
10    Continue
      End

```

12.5 Problems

1. Modify the computer code so that the search always goes downhill, that is, to a state of lower summed energy. Verify that this leads to just the classical trajectory and no wave function. (There is a comment card in the code which can be included to make this change.)
2. Verify that the wave function obtained using time integral of the Lagrangian is the same as that obtained with the tau integral of the Hamiltonian.

3. Test the wave function computation for the gravitational potential

$$V(x) = mgx, \quad (12.40)$$

$$x(t) = x_0 + v_0 t + \frac{1}{2}gt^2. \quad (12.41)$$

You may want to set the initial positions to be close to the classical trajectory to ensure convergence.

PART II

RELATIVISTIC QUANTUM MECHANICS

CHAPTER 13

RELATIVISTIC WAVE EQUATIONS FOR SPINLESS PARTICLES

Principle of Relativity

The laws by which the states of physical systems alter are independent of the alternative, to which of two systems of coordinates, in uniform motion of parallel translations relatively to each other, these alterations of state are referred.

—Albert Einstein

We wish to extend quantum mechanics to incorporate the *postulates of special relativity* given by Einstein (Einstein et al., 1952):

1. The laws of physics have the same form in different, inertial reference frames. This *principle of special relativity* is also described as the “covariance of the laws of nature”.
2. Light is always propagated in empty space with the velocity c which is independent of the state of motion of the emitting body.

There are two approaches to this extension of quantum mechanics. The first makes intuitive extensions and investigates what must be done to make sense of them. This is the approach we prefer since it is historical and clearly displays the limits of the relativistic equations (even if they are more valid than our beloved Schrödinger equation). The other approach is more systematic and based on studying those operators which generate the group of Lorentz transformations. This is a natural way to develop wave equations for higher and higher spins (representations of higher and higher dimensionality) and makes the relativistic nature of spin an inherent part of the theory. We leave that approach for nonintroductory treatments.

In the next few chapters we concentrate on the one-particle, relativistic wave equations known as the Klein–Gordon and Dirac equations. Two-particle, relativistic wave equations are still very much an area of research as are many-body relativistic equations. Approximate two-particle equations such as the Bethe–Salpeter equation, the Blankenbecler–Sugar

equation, and the Breit equation are discussed in Chapter 23, *The Breit–Pauli and Meson-Exchange Interactions*, and Chapter 25, *Wave Equations from Field Theory*. It becomes clear there that they are neither complete nor without problems.

13.1 Canons

One “derives” the nonrelativistic Schrödinger equation by considering the Hamiltonian for a nonrelativistic, free particle of mass m ,¹

$$\tilde{H}_0 = \frac{\tilde{\mathbf{p}}^2}{2m}, \quad (13.1)$$

as acting on a wave function with an energy eigenvalue of E :

$$\tilde{H}_0\psi(\mathbf{x}, t) = \frac{\tilde{\mathbf{p}}^2}{2m}\psi(\mathbf{x}, t). \quad (13.2)$$

If we make the canonical substitutions,

$$\boxed{\tilde{H}_0 \rightarrow i\frac{\partial}{\partial t}, \quad \tilde{\mathbf{p}} \rightarrow \frac{\nabla}{i}}, \quad (13.3)$$

we obtain the time-dependent Schrödinger equation:

$$i\frac{\partial\psi(\mathbf{x}, t)}{\partial t} = -\frac{\nabla^2}{2m}\psi(\mathbf{x}, t). \quad (13.4)$$

Stationary-states solutions are obtained by guessing an exponential time dependence:

$$\psi(\mathbf{x}, t) = e^{-iEt}\psi(\mathbf{x}), \quad (13.5)$$

which produces the time-independent Schrödinger equation (for free particles):

$$E\psi(\mathbf{x}) = -\frac{\nabla^2}{2m}\psi(\mathbf{x}). \quad (13.6)$$

After substituting the relation between energy and on-shell momentum,

$$E = \frac{k^2}{2m}, \quad (13.7)$$

into (13.6), we obtain the time independent Schrödinger wave equation,

$$(\nabla^2 + k^2)\psi(\mathbf{x}) = 0. \quad (13.8)$$

Note, these procedures do not satisfy Einstein’s first postulate because they are *not* covariant. The LHS of (13.6) has E , the time component of a 4-vector, multiplying ψ , while the RHS has p^2 , the square of the spacelike component of a 4-vector; the two sides thereby transform differently under Lorentz transformations and consequently the equation is not covariant.²

¹To facilitate comparison with relativistic wave equations, we examine only the one-particle theory and consequently do not use reduced masses. Also, to make the operator nature of the symbols more apparent in these derivations, we use a tilde.

²We review relativistic notation and 4-vectors in Appendix C, *Four-Vectors and Lorentz Transformations*.

13.2 Relativistic Schrödinger Equation

The Schrödinger equation is clearly incompatible with relativity because it is based on (13.1), a nonrelativistic relation between energy and momentum. If we try to correct that by using

$$\tilde{H}_0\psi(\mathbf{x}, t) = \sqrt{\mathbf{p}^2 + m^2}\psi(\mathbf{x}, t), \quad (13.9)$$

and making the canonical substitutions (13.3), we obtain the *relativistic Schrödinger equation* in coordinate space:

$$\boxed{i\frac{\partial\psi(\mathbf{x}, t)}{\partial t} = \sqrt{-\nabla^2 + m^2}\psi(\mathbf{x}, t).} \quad (13.10)$$

While both sides of this equation have similar transformation properties, the mathematical operations implied by the radical are not well-defined. Specifically, the relativistic kinetic energy operator is *nonlocal*, which we see by expanding the square root in (13.10) to obtain:

$$i\frac{\partial\psi(\mathbf{x}, t)}{\partial t} = \left(m - \frac{\nabla^2}{2m} - \frac{\nabla^4}{8m^3} + \cdots\right)\psi(\mathbf{x}, t). \quad (13.11)$$

The infinite number of derivatives indicates the nonlocality of the energy operator, that is, the need to know the wave function over all of space to solve for it in one place (as happens with nonlocal potentials).

A proper way to handle the relativistic energy operator in (13.10) is to treat $\sqrt{\mathbf{p}^2 + m^2}$ as the momentum representation of the kinetic energy operator, and then form the coordinate representation of that operator using the techniques of Chapter 6, *Transition and Potential Matrices*, and Chapter 7, *Formal Quantum Mechanics*, and Appendix B, *Dirac Notation and Representations*. This yields:

$$K(\mathbf{x}, \mathbf{x}') = \int d^3p \frac{e^{i\mathbf{x}\cdot\mathbf{p}}}{(2\pi)^{3/2}} \sqrt{\mathbf{p}^2 + m^2} \frac{e^{-i\mathbf{x}'\cdot\mathbf{p}}}{(2\pi)^{3/2}}. \quad (13.12)$$

Because we no longer have square roots of Laplacians, (13.12) is well defined and the relativistic Schrödinger equation can be written as

$$i\frac{\partial\psi(\mathbf{x}, t)}{\partial t} = \int d^3x' K(\mathbf{x}, \mathbf{x}')\psi(\mathbf{x}', t). \quad (13.13)$$

Exercise Show that for very large m (the nonrelativistic limit), $K(\mathbf{x}, \mathbf{x}')$ approaches a local operator, that is, becomes proportional to $\delta(\mathbf{x} - \mathbf{x}')$. $\diamond$

For finite mass m , the K operator is nonlocal. The scale of nonlocality is set by the only parameter in the expression $1/m = \lambda_c$, the Compton wavelength of the particle [this is also evident from the expansion (13.11)]. Because in atomic physics the mass of the electron is much greater than typical momenta encountered, for distances much greater than $\lambda_c = 3.86 \times 10^{-13}\text{m}$, the kinetic energy is almost a local operator, and we do not expect this nonlocal effect to be important: it would be, however, if we tried to confine a high-energy electron.

13.3 Relativistic Lippmann–Schwinger Equation

While it is fairly rare to find solutions of the relativistic Schrödinger equation (13.10) in the literature, it is not unusual to find solutions of the equivalent generalization of the Lippmann–Schwinger (LS) equation of § 6.2:

$$T_I(k', k; E) = V_I(k', k) + \frac{2}{\pi} \int_0^\infty \frac{dp p^2 V_I(k', p) T_I(p, k; E)}{E - E_p + i\epsilon}, \quad (13.14)$$

$$E_p = \sqrt{p^2 + m_T^2} + \sqrt{p^2 + m_P^2}. \quad (13.15)$$

Equation (13.14) includes relativistic kinematics but not complete, covariant dynamics.³ It is solved using our previous formalism and the techniques described in Chapter 18, *Solving Even Relativistic Integral Equations*.

13.4 The Klein–Gordon Equation

One way to avoid the problem with the square root in the kinetic energy operator (13.10) is to apply the canonical quantization rule (13.3) to E^2 rather than to E :

$$E^2 = p^2 + m^2, \quad (13.16)$$

$$\Rightarrow \tilde{H}_0^2 \psi(\mathbf{x}, t) = [\tilde{\mathbf{p}}^2 + m^2] \psi(\mathbf{x}, t). \quad (13.17)$$

Or in a form which really does look like a wave equation,

$$0 = \left[\frac{\partial^2}{\partial t^2} - \nabla^2 + m^2 \right] \psi(\mathbf{x}, t). \quad (13.18)$$

This *Klein–Gordon equation* provides relativistic corrections to the bound-state energies of atoms and has antiparticle degrees of freedom but no spin degrees of freedom. For these reasons it is considered the appropriate wave equation for spinless particles like π mesons.

Properties

• **Lorentz Covariance** We need to show that the form of the equation (13.18) is the same in all reference frames. First we note that the canonical quantum rules (13.3) convert the momentum 4-vector p^μ into a 4-vector operator $i\partial^\mu$:

$$p^\mu = (E, \mathbf{p}) \rightarrow i \left(\frac{\partial}{\partial t}, -\nabla \right) = i \frac{\partial}{\partial x_\mu}. \quad (13.19)$$

The Klein–Gordon equation can then be written as

$$(\square + m^2) \psi(\mathbf{x}, t) = 0, \quad (13.20)$$

where $\square$ is the D'Alembertian, that is, the familiar wave operator:

$$\square = \partial_t^2 - \nabla^2. \quad (13.21)$$

³This is a two-body equation as written and, to have the correct nonrelativistic limit, it should be solved in the CM system.

Because the RHS of (13.20) is a Lorentz scalar (the same number in all reference frames), we have covariance if the LHS is also a Lorentz scalar. Because the term in parentheses on the LHS looks like the scalar product of a 4-vector with itself, we have covariance if the Klein–Gordon wave function ψ were a scalar,⁴ that is, if

$$\psi'(\mathbf{x}', t') = \psi(\mathbf{x}, t). \quad (13.22)$$

We can actually prove that the term in parentheses in (13.20) is a scalar by examining the Lorentz transformation of 4-vectors (as reviewed in Appendix C, *Four-Vectors and Lorentz Transformations*):

$$x'^{\nu} = a^{\nu}_{\mu} x^{\mu}. \quad (13.23)$$

Exercise Apply the chain rule to (13.23) to deduce the Lorentz transformation of derivatives:

$$\frac{\partial}{\partial x^{\mu}} = a^{\nu}_{\mu} \frac{\partial}{\partial x'^{\nu}}, \quad (13.24)$$

$$\partial_{\mu} = a^{\nu}_{\mu} \partial'_{\nu}. \quad \diamond \quad (13.25)$$

Exercise Show that these relations imply that

$$\partial^{\mu} \partial_{\mu} = \partial'^{\nu} \partial'_{\nu}, \quad (13.26)$$

that is, that the D'Alembertian $\square = \partial^{\mu} \partial_{\mu}$ is a scalar. $\diamond$

• **Definition of Probability** A definition of *probability* ρ different from the Schrödinger theory is needed in order for $\int d^3x \rho(\mathbf{x})$ to be a scalar. With the old definition, $\rho = \psi^* \psi$, the density ρ would be a scalar, in which case its volume integral would not be.

Because the Klein–Gordon equation is based on the relation among squares, $E^2 = p^2 + m^2$, there is a sign ambiguity when this is solved for the energy:

$$E = \pm E_p = \pm \sqrt{p^2 + m^2} = \pm \gamma_v m, \quad (13.27)$$

$$\gamma_v = (1 - v^2)^{-1/2}, \quad (13.28)$$

where we always define E_p to be positive. Although we can discard as unphysical the negative-energy solutions, we shall see that they return as *antiparticles*.

• **For Particles at Rest** For particles at rest, ($\nabla \psi = 0$), it is easy to solve the Klein–Gordon equation:

$$0 = \left[\frac{\partial^2}{\partial t^2} - 0 + m^2 \right] \psi(\mathbf{x}, t), \quad (13.29)$$

$$\Rightarrow \psi^{(\pm)}(\mathbf{x} = 0, t) = e^{\mp i m t} \psi^{(\pm)}(0, 0). \quad (13.30)$$

⁴The term “scalar” is used with two meanings. One is the scalar product which is an invariant under rotations and velocity boosts. The other is a scalar under transformations, that is, a one-component function like temperature or pressure which can change but does not mix with other functions (vectors have 3-components).

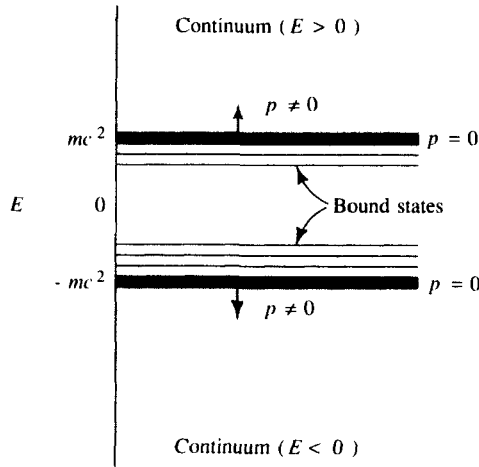


Figure 13.1: The energy spectrum for the Klein–Gordon equation. The upper continuum (shaded) is for positive-energy particles, the lower one is for negative-energy particles. The discrete bound states are in the gap.

The two solutions $\psi^{(\pm)}$ are independent and clearly have opposite signs for the rest energy, $E_0 = \pm mc^2$ (the eigenvalue of $i\partial/\partial t$).

• **Free Particles** For freely moving particles, the wave functions are covariant generalizations of the at-rest solutions (13.30). Because mt is the rest frame value of the scalar product,

$$p^\mu x_\mu = E_p t - \mathbf{p} \cdot \mathbf{x}. \quad (13.31)$$

We use Einstein's first postulate to generalize (13.30) for a moving particle:

$$\psi^{(+)}(\mathbf{x}, t) = e^{+i[\mathbf{p} \cdot \mathbf{x} - E_p t]} = e^{-ip^\mu x_\mu}, \quad (13.32)$$

$$\psi^{(-)}(\mathbf{x}, t) = e^{-i[\mathbf{p} \cdot \mathbf{x} - E_p t]} = e^{+ip^\mu x_\mu}, \quad (13.33)$$

We recognize the expression in brackets as the scalar product $p^\mu x_\mu$.

After examining (13.32) and (13.33), we see that the Klein–Gordon equation has two classes of solutions, with spectra which appear to be like those symbolized in Figure 13.1: positive energy solutions with energy beginning at $E = mc^2$ and rising to $+\infty$ as p increases, and negative-energy ones beginning at $E = -mc^2$ and falling to $-\infty$ as p increases. The existence of these continua of states is a bit of a problem. In classical physics the initial conditions can be set up so that there would be no interaction between these classes and no mixing. In quantum mechanics, however, transitions would occur from the positive- to negative-energy states and there would be a massive cascade, unless something blocked it. There is a similar problem with the Dirac equation which Dirac got around by arguing that the negative-energy levels are all filled with fermions so the exclusion principle blocks the transition. Because the Klein–Gordon equation deals with spinless particles (scalar wave functions), we cannot use that argument and we have a problem.⁵ It was this problem that led

⁵Note, however, that in § 13.5 we show these negative frequency solutions behave like positive energy antiparticles.

to the early discard of the Klein–Gordon equation as a valid, relativistic wave equation, and to the development of the Dirac equation. However, the modern interpretation, originally advanced by Pauli and Weisskopf, is that it is more appropriate to use a field theory than a wave equation for a “problem” in which particles are created or destroyed, and in that case the Klein–Gordon equation is the proper equation for spinless (scalar) fields, while the Dirac equation is the proper one for spin $\frac{1}{2}$ fields.

Probability and Current

If a wave equation is to make sense, it must provide a description of the probability of finding a particle in space and of how this probability changes from one region of space to another. In addition, if the new theory is to obey the *correspondence principle*,⁶ it should be possible to cast this description in the form of the familiar *continuity equation*. The procedure used to convert the Schrödinger equation into a continuity equation is repeated with relativistic equations as well. We take (13.18), multiply on the left by ψ^* , and then subtract ψ times (13.18) from this:

$$\psi^* \left[\frac{\partial^2}{\partial t^2} - \nabla^2 + m^2 \right] \psi(\mathbf{x}, t) - \psi \left[\frac{\partial^2}{\partial t^2} - \nabla^2 + m^2 \right] \psi^*(\mathbf{x}, t) = 0. \quad (13.34)$$

Exercise Show that (13.34) becomes the continuity equation,

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{j} = 0, \quad (13.35)$$

if you make the identification of the density,

$$\rho = \frac{i}{2m} \left(\psi^* \frac{\partial \psi}{\partial t} - \psi \frac{\partial \psi^*}{\partial t} \right), \quad (13.36)$$

and for the current,

$$\mathbf{j} = \frac{1}{2} \left(\psi^* \frac{\nabla \psi}{im} - \psi \frac{\nabla \psi^*}{im} \right). \quad \diamond \quad (13.37)$$

In spite of the familiarity of (13.35) from nonrelativistic physics, it is also quite acceptable relativistically since it is a covariant equation; namely,

$$\partial_\mu j^\mu = 0, \quad j^\mu = (\rho, \mathbf{j}). \quad (13.38)$$

We leave it to the Problems section to show explicitly that $(\rho, \mathbf{j})$ transforms as a 4-vector.

Although the density ρ is different from the familiar Schrödinger form, in the nonrelativistic limit:

$$\psi(\mathbf{x}, t) \rightarrow e^{-imt} \psi(\mathbf{x}), \quad (13.39)$$

the density ρ approaches the familiar form:

$$\rho(\mathbf{x}, t) \rightarrow \frac{i}{2m} [\psi^*(-im\psi) - \psi(im\psi^*)] = \psi^* \psi. \quad (13.40)$$

⁶A new theory must agree with the old theory in the old one's regime of validity.

An intriguing aspect of the density (13.36) is that there is no guarantee of it being positive, or of it remaining positive for all time if it were at an earlier time. This leads to the interpretation of ρ as a *charge density* rather than probability density, in which case its negative values become sensible. To see this, we start with the positive- and negative-energy solutions (13.32) and (13.33):

$$\psi^{(+)}(\mathbf{x}, t) = \psi^{(+)}(\mathbf{x})e^{-iE_p t}, \quad \psi^{(-)}(\mathbf{x}, t) = \psi^{(-)}(\mathbf{x})e^{+iE_p t}. \quad (13.41)$$

The density for each solution is

$$\rho^{(+)} = \frac{E_p}{m} |\psi^{(+)}(\mathbf{x})|^2, \quad \rho^{(-)} = -\frac{E_p}{m} |\psi^{(-)}(\mathbf{x})|^2. \quad (13.42)$$

Because we have defined E_p to be positive, $\rho^{(+)}$ is always positive, but $\rho^{(-)}$ is always negative. Aside from this mystery, the transformation properties make sense: Because E_p is proportional to γ_v (13.28), and ψ is a Lorentz-scalar, ρ is proportional to γ_v ; so ρ 's increase by γ_v under a Lorentz transformation exactly cancels the contraction of the volume element, $d^3\mathbf{x}' = d^3\mathbf{x}/\gamma_v$, and the total probability $\int d^3\mathbf{x} \rho(\mathbf{x})$ is a Lorentz invariant.

Exercise Uncover some of the physics of the negative-energy solutions by showing that the current densities for the states (13.32) and (13.33) are:

$$\mathbf{j}^{(+)} = \frac{\mathbf{p}}{m} = \frac{\mathbf{p}}{E_p} \rho^{(+)}, \quad (13.43)$$

$$\mathbf{j}^{(-)} = -\frac{\mathbf{p}}{m} = \frac{\mathbf{p}}{E_p} \rho^{(-)}, \quad (13.44)$$

where we have used the expression $\mathbf{p} = \gamma_v m \mathbf{v}$ for a relativistic particle. $\diamond$

We see that the negative-energy solutions have energy $-E_p$ and $\rho^{(-)} < 0$, yet move like real particles with the same relation between current and density as the positive-energy solutions. These are *antiparticles* and will be discussed further when we examine their interactions.

13.5 Interactions and the KGE

Minimal Electromagnetic Coupling

Inasmuch as classical electrodynamics is the origin of special relativity, we follow it as a guide for including interactions with *external fields*. Moreover, because both the classical and quantum theories are in Hamiltonian forms, we also assume the classical, canonical procedures (Goldstein, 1977; Jackson, 1975). For a free *point particle* of charge q described by the Hamiltonian H_0 , “turning-on” an electromagnetic field produces an interaction term H' in the Hamiltonian:

$$H_0 \rightarrow H = H_0 + H', \quad (13.45)$$

$$H' = -q\mathbf{A} \cdot \mathbf{v} + q\Phi, \quad (13.46)$$

where $\mathbf{v}$ is the velocity and the electric and magnetic fields are related to the electromagnetic potentials via

$$\mathbf{E} = -\nabla\Phi - \frac{\partial \mathbf{A}}{\partial t}, \quad \mathbf{B} = \nabla \times \mathbf{A}. \quad (13.47)$$

Likewise, the kinetic momentum $\boldsymbol{\pi} = m\mathbf{v}$ of a particle (which for a free particle equals its canonical momentum $\mathbf{p}$) changes its relation to

$$\mathbf{p} \rightarrow \boldsymbol{\pi} = \mathbf{p} - q\mathbf{A}. \quad (13.48)$$

Equation (13.48) means that the new canonical momentum $\mathbf{p}$ (which in quantum mechanics becomes an operator) includes the kinetic momentum $\boldsymbol{\pi}$ plus the momentum associated with the electromagnetic field:

$$\mathbf{p} = \boldsymbol{\pi} + q\mathbf{A}. \quad (13.49)$$

Consequently, we include the effect of an external electromagnetic field on a point particle and derive an H' (*minimal electromagnetic coupling*) by taking the unperturbed Hamiltonian and making the replacement

$$\mathbf{p} \rightarrow \mathbf{p} - q\mathbf{A}. \quad (13.50)$$

We now extend our relativistic theory with this prescription. While at first this may seem incongruent since relativity involves spacetime, we know that the “scalar”⁷ and vector potentials Φ and $\mathbf{A}$ form the 4-vector

$$A^\mu = (\Phi, \mathbf{A}). \quad (13.51)$$

Accordingly, the prescription (13.45)–(13.46) has the relativistic generalization

$$E \rightarrow E - q\Phi, \quad \mathbf{p} \rightarrow \mathbf{p} - q\mathbf{A}, \quad (13.52)$$

$$p^\mu \rightarrow p^\mu - qA^\mu. \quad (13.53)$$

We combine the recipe (13.3) for quantization with the recipe for electromagnetic coupling to obtain

$$\tilde{H}_0 = i\frac{\partial}{\partial t} \rightarrow i\frac{\partial}{\partial t} - q\Phi, \quad (13.54)$$

$$\tilde{\mathbf{p}} = \nabla/i \rightarrow \nabla/i - q\mathbf{A}. \quad (13.55)$$

This is clearly a covariant procedure, and in fact it is sometimes (Guidry, 1991) written in terms of the *covariant derivative* D^μ :

$$\frac{\partial}{\partial t} \rightarrow D_0 \stackrel{\text{def}}{=} \frac{\partial}{\partial t} + iq\Phi, \quad (13.56)$$

$$\nabla \rightarrow \mathbf{D} \stackrel{\text{def}}{=} \nabla - iq\mathbf{A}, \quad (13.57)$$

$$\partial^\mu \rightarrow D^\mu \stackrel{\text{def}}{=} \partial^\mu + iqA^\mu. \quad (13.58)$$

The *covariant derivative* is an essential ingredient in developing gauge invariant field theories.

⁷There is an unfortunate misnomer here. As is evident from (13.51), the electromagnetic “scalar” potential transforms as the timelike component of a 4-vector and *not* as a Lorentz scalar. It is, however, a “scalar” under rotations in space.

If we use these prescriptions in the Schrödinger equation, we obtain

$$i \frac{\partial \psi(\mathbf{x}, t)}{\partial t} - q\Phi \psi(\mathbf{x}, t) = \frac{[\nabla/i - q\mathbf{A}]^2}{2m} \psi(\mathbf{x}, t), \quad (13.59)$$

or in terms of the covariant derivative,

$$iD^0 \psi(\mathbf{x}, t) = \frac{-i\mathbf{D}}{2m} \psi(\mathbf{x}, t). \quad (13.60)$$

Here $q\mathbf{A}$ may be replaced by any vector potential and $q\Phi$ by any potential that behaves like the time component of a 4-vector.

Likewise, if we use these prescriptions in the Klein–Gordon equation, we obtain

$$[(i\partial_\mu - qA_\mu)^2 + m^2] \psi(x) = 0, \quad (13.61)$$

$$(D^\mu D_\mu + m^2) \psi(x) = 0. \quad (13.62)$$

In a more explicit form, this equation is:

$$\left(i \frac{\partial}{\partial t} - q\Phi \right)^2 \psi(\mathbf{x}, t) = \left[\left(\frac{\nabla}{i} - q\mathbf{A} \right)^2 + m^2 \right] \psi(\mathbf{x}, t). \quad (13.63)$$

The procedure (13.54)–(13.55) is called “minimal” in classical physics because it couples only to the point aspect of the particle and not to its spatial extension. Aside from being elegant, it is particularly significant that it preserves *gauge invariance*, a basic symmetry of the electromagnetic field (see the Problems section). This means the wave equation in the presence of the electromagnetic field is invariant under the full gauge transformation of the fields *and* of the wave function:

$$A^\mu \rightarrow A^\mu - \partial^\mu \chi(x), \quad \psi(x) \rightarrow e^{iq\chi(x)} \psi(x), \quad (13.64)$$

where $\chi(x)$ is an arbitrary function of spacetime. For this reason, the procedure (13.54)–(13.55) is also called minimal because it is the simplest prescription which contains this invariance.

Before we proceed to solve (13.63), we note that the canonical 4-current $j^\mu = (\rho, \mathbf{j})$, (13.36)–(13.37), changes in the presence of an electromagnetic field:

$$\rho \rightarrow \frac{1}{2m} \left[\psi^* \left(i \frac{\partial}{\partial t} - q\Phi \right) \psi - \psi \left(i \frac{\partial}{\partial t} + q\Phi \right) \psi^* \right], \quad (13.65)$$

$$\mathbf{j} \rightarrow \frac{1}{2m} \left[\psi^* \left(\frac{1}{i} \nabla - q\mathbf{A} \right) \psi - \psi \left(\frac{1}{i} \nabla + q\mathbf{A} \right) \psi^* \right]. \quad (13.66)$$

Exercise Derive these expressions from the Klein–Gordon equation and thereby show that the continuity equation is still satisfied. $\diamond$

Exercise Do the terms in j^μ depending on A^μ describe the particle’s or external field’s current? Why are they present? $\diamond$

Positive- and Negative-Energy Degrees of Freedom

Substituting the positive- and negative-energy solutions (13.32) and (13.33) into the Klein–Gordon equation (13.63) yields

$$(E_p \mp q\Phi)^2 \psi^{(\pm)}(\mathbf{x}, t) = [(\mathbf{p} \mp q\mathbf{A})^2 + m^2] \psi^{(\pm)}(\mathbf{x}, t). \quad (13.67)$$

We see that the positive- and negative-energy solutions correspond to a particle and antiparticle with the same mass m , same positive energy E_p , opposite charge, opposite current, and no spin (that is, no magnetic moment interaction). The π^+ and π^- mesons are such particles with opposite quantum numbers, as proved by placing them in contact and having them *annihilate* into just photons. Alternatively, rather than speak of positive- and negative-energy or frequency solutions, we could speak of any solution ψ and its complex conjugate ψ^* and ascribe different physical interpretations to each.

Exercise Show that if ψ describes a particle, then the Klein–Gordon equation satisfied by ψ^* describes an antiparticle. $\diamond$

Relation to Schrödinger Equation

There is an illuminating relation between the Klein–Gordon and Schrödinger equations which has found use in a number of applications. If we take the Klein–Gordon equation with a fourth-component potential $q\Phi = V$,

$$(E^2 + V^2 - 2EV) \psi = (\mathbf{p}^2 + m^2) \psi, \quad (13.68)$$

and substitute the relation $E^2 = k^2 + m^2$ between energy and wave vector, we obtain

$$(\nabla^2 + k^2) \psi = (2EV - V^2) \psi. \quad (13.69)$$

This looks like a Schrödinger equation with the equivalent energy-dependent potential

$$V^{\text{SE}} = \frac{2EV - V^2}{2m}. \quad (13.70)$$

As such, (13.70) is soluble as an eigenvalue problem with an energy-dependent potential. If the potential is weak enough to ignore the V^2 term, the relativistic formalism becomes equivalent to the Schrödinger equation with the replacement $m \rightarrow E$.

Another type of potential in relativity is the *Lorentz scalar* which adds into the mass (the mass is scalar because $p_\mu p^\mu = m^2$). The Klein–Gordon equation with coupling to the scalar potential S is:

$$E^2 \psi = [p^2 + (m + S)^2] \psi. \quad (13.71)$$

Exercise Compare (13.71) with the Schrödinger equation and show that the equivalent potential is

$$V^{\text{SE}} = \frac{S^2 + 2mS}{2m}. \quad \diamond \quad (13.72)$$

In this case the equivalent potential is energy independent, yet it has the same nonrelativistic ($m \rightarrow \infty$) limit as the vector potential (13.70).

An Atomic Solution

An *exotic* atom is formed by placing a negatively charged elementary particle, such as the π^- meson, within a regular atom. The system forms an excited atom that deexcites to its ground state by emitting X-rays. (Since the pion's mass is ~ 280 times an electron's mass, the emitted photons have ~ 280 times the energy of hydrogen photons, and are X-rays. Relativistic effects are proportionally more important.⁸) Because the π^- has zero spin, the Klein–Gordon equation appears appropriate. We solve it in the partial-wave basis,

$$\psi(\mathbf{x}, t) = e^{-iEt} \frac{u_l(kr)}{r} Y_{lm}(\theta, \phi), \quad (13.73)$$

with an external electric field generated by the nuclear charge Ze :

$$q\Phi = -\frac{Ze^2}{r}. \quad (13.74)$$

Exercise Show that the resulting radial Klein–Gordon equation is

$$\frac{d^2 u_l(kr)}{dr^2} + \left[\frac{2EZ\alpha}{r} - (m^2 - E^2) - \frac{l(l+1) - (Z\alpha)^2}{r^2} \right] u_l(kr) = 0, \quad (13.75)$$

where α is the fine-structure constant

$$\alpha = e^2 \equiv \frac{e^2}{\hbar c} \simeq \frac{1}{137}. \quad \diamond \quad (13.76)$$

The solution of (13.75) is very similar to that of Schrödinger hydrogen atom, and we lead the reader through it in the Problems section. Already evident in (13.75) is a relativistic (V^2) correction to the angular momentum barrier, $l(l+1) \rightarrow l(l+1) - (Z\alpha)^2$. This makes the orbits precess, makes the $l = 0$ wave function (but not probability) singular at the origin, and removes the l degeneracy in the energy.⁹ The eigenenergies are

$$E_{n,l} = \frac{m}{\left(1 + (Z\alpha)^2 / \left[n - l - \frac{1}{2} + \sqrt{\left(l + \frac{1}{2} \right)^2 - (Z\alpha)^2} \right]^2 \right)^{1/2}}, \quad (13.77)$$

where n is the principal quantum number. To reach the nonrelativistic limit (and improve the numerical precision of the energy computation) we expand E in powers of $Z\alpha$:

$$E_{n,l} \simeq m - \frac{m(Z\alpha)^2}{2n^2} - \frac{m(Z\alpha)^4}{2n^4} \left(\frac{n}{l + \frac{1}{2}} - \frac{3}{4} \right) + O(\alpha^6). \quad (13.78)$$

⁸We ignore here any effect of nuclear forces on the pion. Because the Bohr radius for this atom is 100–1000 times the range of the strong and weak forces, these nuclear interactions may be treated in perturbation theory. See the Problems section for a practical example.

⁹Note that for $Z\alpha > l + \frac{1}{2}$, the repulsive, angular-momentum barrier becomes attractive, and the solution becomes pathological. This is unnatural since for such large Z it is unrealistic to treat the nucleus as a point charge and the assumption of a $1/r$ potential is unrealistic. See Bethe and Jackiw (1968) for more details.

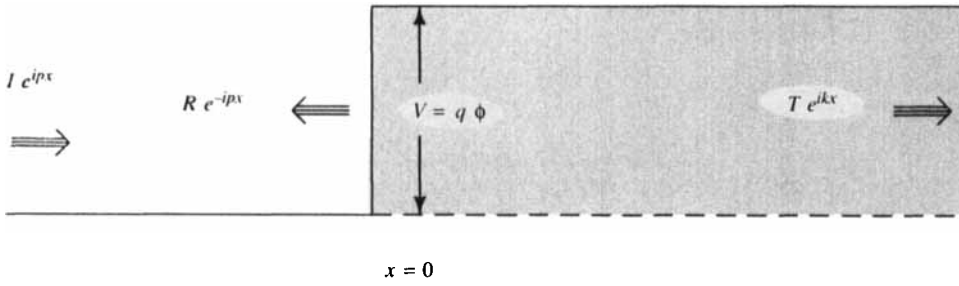


Figure 13.2: A timelike potential barrier. There are incident and reflected waves on the left but only a transmitted wave on the right.

We see that the total energy consists of the electron's rest energy plus Bohr's result for the binding energy with a relativistic correction that produces a fine structure and removes the l degeneracy.¹⁰ The difficulty generally is that relativistic two-body problems do not separate into relative and CM coordinates as cleanly as in the nonrelativistic problem. The correction terms provide agreement with experiment and this proves that the Klein–Gordon equation is more accurate than the Schrödinger equation.

A Paradoxical Solution

The preceding atomic solution, while successful at extending quantum mechanics to include relativistic kinematics and the covariant interaction with an external field, did uncover some possible problems for very strong potentials (large $Z\alpha$). We now study a related “problem” possessed by relativistic wave equations known as *Klein's paradox*. Consider the Klein–Gordon equation,

$$(E - V)^2 \psi(x) = -\frac{d^2 \psi(x)}{dx^2} + m^2 \psi(x), \quad (13.79)$$

describing a particle scattering from the timelike potential barrier V pictured in Figure 13.2. The wave function consists of incident, transmitted and reflected waves, with values on the left and right of

$$\psi_L = I e^{ipx} + R e^{-ipx}, \quad (13.80)$$

$$\psi_R = T e^{ikx}. \quad (13.81)$$

Exercise Show that substitution of ψ_L and ψ_R into the wave equation determines the relations of the two, local wave numbers to the energy:

$$E^2 = p^2 + m^2 \quad (\text{left}), \quad p = \pm \sqrt{E^2 - m^2}, \quad (13.82)$$

$$(E - V)^2 = k^2 + m^2 \quad (\text{right}), \quad k = \pm \sqrt{(E - V)^2 - m^2}. \quad \diamond (13.83)$$

¹⁰Some flexibility in interpretation is useful because we are applying a one-particle Klein–Gordon theory to a two-particle Schrödinger theory; to get the Bohr result we should use a reduced mass in the second term of (13.78), but an actual mass in the first term.

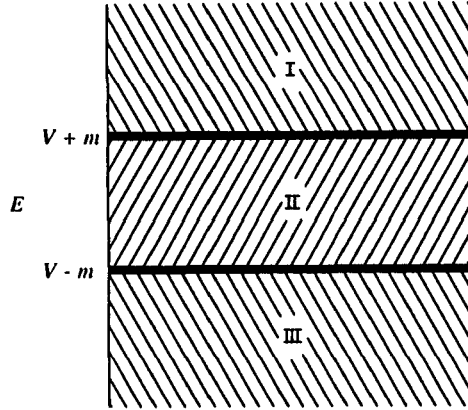


Figure 13.3: Three energy regions for the Klein–Gordon equation describing the scattering from the barrier in Figure 13.2. The solution has a different character in each region.

We choose the plus signs in (13.82) and (13.83) to obtain the directions of the arrows shown in Figure 13.2. Relation (13.83) is illuminating when written as

$$E^2 - k^2 = (m^*)^2 \quad (13.84)$$

with m^* the “effective” or off-shell mass given by

$$(m^*)^2 = m^2 + 2EV - V^2. \quad (13.85)$$

This shows how an interaction drives a particle’s mass off the mass shell. In other words, within the potential of Figure 13.2, the particle has the same energy–momentum relation as a free particle but with the effective mass m^* instead of m . An example is the A nucleons bound within a nucleus by a very large and negative V . The average bound nucleon mass m^* (the nucleus’s mass divided by A) is about 1% less than m [the decrease expected from (13.85)], yet returns to m if the nucleon is removed from the nucleus.

To solve the barrier problem, we demand that probability and current be continuous across the barrier:

Exercise Show that the continuity of ψ and $d\psi/dx$ requires

$$\begin{aligned} I + R &= T, & pI - pR &= kT, \\ \Rightarrow T &= \frac{2p}{p+k} I, & R &= \frac{p-k}{p+k} I. \end{aligned} \quad \diamond \quad (13.86)$$

The currents on the right and left are

$$j_L = \frac{p}{m} (|I|^2 - |R|^2), \quad (13.87)$$

$$j_R = \frac{k}{m} |T|^2. \quad (13.88)$$

The amplitude relations (13.86) translate into $R < I$ with currents moving along the positive x axis in both the right and left regions (as expected, there is less reflected wave than

incident wave). We look at these solutions for the three different energy regimes shown in Figure 13.3 (or, equivalently, for different strengths of V):

Above Barrier ($E > V + m$): In this regime $k = \sqrt{(E - V)^2 - m^2}$ is less than p as follows from

$$k^2 = p^2 - 2EV + V^2. \quad (13.89)$$

The probability density (13.36) is positive both to the right and left,

$$\rho_L = \frac{E}{m} |\Psi_L|^2, \quad \rho_R = \frac{E - V}{m} |\Psi_R|^2, \quad (13.90)$$

and we have the expected incident, reflected, and transmitted waves.

Within Barrier ($V - m < E < V + m$): In this regime k becomes pure imaginary,

$$k = i\sqrt{m^2 - (E - V)^2} = i\kappa, \quad (13.91)$$

and we have total internal reflection (equal magnitudes for reflected and incident currents and $j_L = 0$). This follows from

$$R = \frac{p - i\kappa}{p + i\kappa} I \Rightarrow |R| = |I|. \quad (13.92)$$

As also occurs with light, the particle's probability density decays exponentially as it tunnels through the barrier:

$$\rho_R = \frac{E - V}{m} |\psi_R|^2 = \frac{E - V}{m} e^{-2\kappa x}. \quad (13.93)$$

Yet, maybe not as expected, ρ can be positive or negative:

$$E > V \Rightarrow \rho_R > 0, \quad (13.94)$$

$$E < V \Rightarrow \rho_R < 0. \quad (13.95)$$

Apparently, when the potential is made strong enough it will create a particle-antiparticle pair. The created particle is repelled out of the barrier and the created antiparticle, feeling V with the opposite sign, is attracted into the barrier. While this *pair creation* is virtual (there is no flow of antiparticle current), it is typical of what happens near a potential that changes abruptly over a very short distance.

Ultrastrong Potential ($V > E + m$): In this regime k becomes pure real,

$$k = \pm\sqrt{(E - V)^2 - m^2}, \quad (13.96)$$

with $k^2 > p^2$. Again the density within the barrier is negative,

$$\rho_R = \frac{E - V}{m} |T|^2 < 0, \quad (13.97)$$

only now with a real current within the barrier:

$$j_R = \frac{k}{m} |T|^2. \quad (13.98)$$

To deduce if anything physical is entering the barrier, we pull out a theoretical tool from classical waves (Jackson, 1975) and evaluate the group velocity $v_g = \partial E / \partial k$, where k is the local wave number within the barrier:

$$k^2 = (E - V)^2 - m^2 \Rightarrow v_g = \frac{\partial E}{\partial k} = \frac{k}{E - V}. \quad (13.99)$$

If particles move to the right within the barrier, that is, $v_g > 0$, it must be that $k < 0$; yet if particles move to the right, we would expect a diminished reflected current.

Exercise Evaluate the reflected current and show that if $k < 0$ the reflected current is *larger* than the incident one. This is *Klein's paradox*. $\diamond$

The resolution of this paradox is to realize that there are two different kinds of particles present, particles and antiparticles. For case III, the very strong external potential provides the energy needed to produce these particle-antiparticle pairs. The current and density within the barrier are negative because the antiparticles are attracted to the barrier; they are moving to the right, however, because $v_g > 0$. The produced particles, on the other hand, are repelled by the barrier and thus increase the reflected current to a value greater than the incident one.¹¹

There is a theoretical problem with the preceding example aside from coping with the logical intricacies of a paradox. We have started with a theory of a single particle interacting with an external potential, and found that to make sense of it we must have antiparticle and particles created. Yet we have neither a many-particle theory nor a mathematical framework to describe the creation of particles, and so our theory appears to bear the seeds of its own downfall. Alternatively, it is probably prudent to conclude that the changing sign for ρ and j signals the need to treat relativistic problems within a *field theory* designed to handle the creation and destruction of particles.

13.6 Problems

1. Consider a free relativistic particle of rest mass m with wave function Φ satisfying the Klein–Gordon equation.
 - (a) Show that $\phi(\mathbf{x}, t)$ is a Lorentz invariant, that is, $\phi(\mathbf{x}, t) = \phi'(\mathbf{x}', t')$ for observers 0 and $0'$ related by a *homogeneous* Lorentz transformation (that is, a velocity boost or rotation, but not displacement).
 - (b) Prove that $\partial \rho / \partial t + \nabla \cdot \mathbf{j} = 0$.
 - (c) Show that $(\rho, \mathbf{j})$ transforms as a 4-vector under Lorentz transformations.
 - (d) Prove that $\int d^3x \rho$ is a Lorentz scalar and is time independent.
2. Prove that the minimal electromagnetic coupling procedure for a particle with charge q is gauge invariant:
 - (a) Define the classical gauge transformation and its effect on A^μ .

¹¹This may all seem far from reality, yet Hawkins (1977) has postulated a related effect as the basis for radiation from black holes.

- (b) Indicate how this transformation changes the Klein–Gordon equation containing minimal coupling.
- (c) Show that the Klein–Gordon wave function for the transformed equation differs from the original wave function by an $\mathbf{r}$ -dependent phase:

$$\psi'(\mathbf{x}, t) = e^{i q \chi(\mathbf{x}, t)} \psi(\mathbf{x}, t). \quad (13.100)$$

- (d) How is χ related to the functions in the Gauge transformation?

3. A particle of mass m is bound in a 1D square well of radius b and depth $V_0 = -M_0$. Determine the conditions necessary for there to be a Klein–Gordon bound state of total energy $E < m$. Let the potential be the fourth component of a 4-vector, and consider even-parity solutions only.

4. **Klein–Gordon Hydrogen Atom:** A spinless electron is bound by the Coulomb potential:

$$V(r) = -e\phi(r) = -\frac{Ze^2}{r} \quad (13.101)$$

in a stationary state of *total* energy $E \leq m$.

- (a) What is the time-independent Klein–Gordon equation for this potential?
- (b) Assume the radial and angular parts of the wave function $\psi(\mathbf{r})$ separate, and verify that this yields equation (13.75) for the radial wave function.
- (c) Show that this equation can be written in the dimensionless form

$$\frac{d^2 u_l(\rho)}{d\rho^2} + \left[\frac{2ZE\alpha}{\gamma\rho} - \frac{1}{4} - \frac{l(l+1) - (Z\alpha)^2}{\rho^2} \right] u_l(\rho) = 0, \quad (13.102)$$

$$\alpha = e^2, \quad \gamma^2 = 4(m^2 - E^2), \quad \rho = \gamma r.$$

- (d) Assume that this equation has a solution of the usual form of a power series times the $\rho \rightarrow \infty$ and $\rho \rightarrow 0$ solutions:

$$u_l(\rho) = \rho^k (1 + c_1 \rho + c_2 \rho^2 + c_3 \rho^3 + \dots) e^{-\rho/2}. \quad (13.103)$$

Show that

$$k = k_{\pm} = \frac{1}{2} \pm \sqrt{(l + \frac{1}{2})^2 - (Z\alpha)^2}. \quad (13.104)$$

- (e) Show that for both k_+ and k_- , the wave function is divergent at the origin yet normalizable.
- (f) Show that only for k_+ is the expectation value of the kinetic energy finite:

$$\int dr r^2 \left[\frac{d(u_l/r)}{dr} \right]^2 < \infty. \quad (13.105)$$

- (g) Show that the k_+ solution has a nonrelativistic limit which agrees with the solution found for the Schrödinger equation.
- (h) Determine the recurrence relation among the c_i 's for this to be a solution of the Klein–Gordon equation.

- (i) Show that unless the power series (13.103) terminates, the wave function will have an incorrect asymptotic form.
- (j) Show that the termination condition determines the eigenenergy for the k_+ solution to be

$$E = \frac{m}{\left(1 + (Z\alpha)^2 \left[n - l - \frac{1}{2} + \sqrt{\left(l + \frac{1}{2}\right)^2 - (Z\alpha)^2}\right]^{-2}\right)^{1/2}}, \quad (13.106)$$

where n is the principal quantum number.

- (k) Expand E in powers of α^2 and show that the α^2 term yields the Bohr formula, and that the higher-order terms can be identified with relativistic corrections.
- (l) Is the l degeneracy present in the nonrelativistic theory now removed? (And, if so, to what order in α ?)
5. Jenkins and Kunselman (1966) observed that the $3d \rightarrow 2p$ transition for π mesons in ^{59}Co emits a photon of energy 384.6 ± 1.0 KeV.
- (a) What is the result expected for this energy from Klein–Gordon theory? (*Hint:* Unless you calculate the preceding exact formula with high precision, the series in α^2 may be more accurate.)
- (b) Indicate three effects which may explain the discrepancy between theory and experiment.
6. Consider the Klein–Gordon equation with a potential coupling like the fourth component of a 4-vector:

$$(E - V)^2 \psi(\mathbf{x}) = (-\nabla^2 + m^2) \psi(\mathbf{x}). \quad (13.107)$$

- (a) If the potential is much weaker than the energy $V/E \ll 1$, show that this equation reduces to the Schrödinger equation with the reduced potential

$$U = 2EV. \quad (13.108)$$

(This means that people can use their nonrelativistic theories and computer codes to approximate the Klein–Gordon equation.)

- (b) Deduce, with only a line or two of analysis, the Lippmann–Schwinger equation that is equivalent to this approximate Klein–Gordon equation.
- (c) How would the scattering differ if there were a scalar potential S ,

$$E^2 \psi(\mathbf{x}) = [-\nabla^2 + (m + S)^2] \psi(\mathbf{x})? \quad (13.109)$$

(*Hint:* Examine the high- and low-energy limits.)

CHAPTER 14

DIRAC EQUATION

It appears that the simplest Hamiltonian for a point-charge electron satisfying the requirements of both relativity and the general transformation theory leads to an explanation of all the duplexity phenomena without further assumption. All the same there is a great deal of truth in the spinning electron model, at least as a first approximation.

—P. A. M. Dirac

The Dirac equation is one way to extend the Schrödinger equation to include relativity. It differs from the Klein–Gordon equation in having more intrinsic degrees of freedom and thus the capability to describe an electron’s spin. It consequently is considered the “proper” spin $\frac{1}{2}$ equation.¹ Historically, the Dirac equation is important because it led to the prediction of antiparticles, to a fundamental understanding of the electron’s magnetic moment, and to more precise predictions of the spectra in atoms. For these reasons we study it while keeping in mind that it too suffers from similar illnesses as does the Klein–Gordon equation, illnesses that can be cured only with a quantum field theory and a relativistic many-body theory.

For assistance there is a summary of 4-vectors and Lorentz transformations in Appendix C and a summary of the equations of Dirac theory in Appendix D.

14.1 Derivation

We follow Dirac’s search (Dirac, 1928), for a theory of the electron with the following properties:

- (i) There is a wave function that can be determined by an equation that is first order in the time derivative, so probability will be independent of time.
- (ii) The theory is canonical and contains a Hermitian Hamiltonian H_0 so the wave equation and quantization rules are conventional.

¹ See however our discussion in Chapter 25, *Wave Equations from Field Theory*, before accepting this.

- (iii) Probability is constructed to be positive definite by being postulated as proportional to the modulus of the wave function $\Psi^\dagger \Psi$.
- (iv) Total probability is conserved, that is,

$$\frac{\partial}{\partial t} \int d^3x \Psi^\dagger \Psi = 0. \quad (14.1)$$

- (v) A continuity equation $\partial^\mu j_\mu = 0$ exists.
- (vi) Relativistic (Lorentz) covariance holds.
- (vii) $E^2 = p^2 + m^2$ for a free particle.

In making our search, Maxwell's equations for the electrodynamic potential A^μ are a good guide: They are covariant and first order in space and time derivatives. They describe a field with spinlike internal degrees of freedom (the photon polarizations; see § 20.2), yet relate to observables (the $\mathbf{E}$ and $\mathbf{B}$ fields) that satisfy second-order differential equations. Accordingly, it makes sense that Dirac postulated (i) a linear equation with (ii) a standard Hamiltonian form:

$$i \frac{\partial \Psi}{\partial t}(\mathbf{x}, t) = \mathbf{H}_0 \Psi(\mathbf{x}, t). \quad (14.2)$$

To also satisfy (vi), the most general $\mathbf{H}_0$ must also be first order in the space derivatives:

$$\mathbf{H}_0 = \frac{1}{i} \left[\alpha_1 \frac{\partial}{\partial x^1} + \alpha_2 \frac{\partial}{\partial x^2} + \alpha_3 \frac{\partial}{\partial x^3} \right] + \beta m. \quad (14.3)$$

Here α_i and β are assumed to be dimensionless constants independent of space and time that commute with $\mathbf{r}$ and $\mathbf{p}$ (this guarantees translational invariance). By placing the three α 's together as a vector,

$$\boldsymbol{\alpha} \stackrel{\text{def}}{=} (\alpha_1, \alpha_2, \alpha_3), \quad (14.4)$$

and relating the space derivatives to the momentum operator

$$\mathbf{p} = \frac{\nabla}{i}, \quad (14.5)$$

we obtain the Hamiltonian form of the Dirac equation:

$$i \frac{\partial \Psi}{\partial t} = \mathbf{H}_0 \Psi(\mathbf{x}, t). \quad (14.6)$$

This looks like the Schrödinger equations, yet we shall see that the Dirac Hamiltonian,

$$\mathbf{H}_0 = \boldsymbol{\alpha} \cdot \mathbf{p} + \beta m, \quad (14.7)$$

has a very different structure. Because this is a canonical theory (ii), Hamiltonian dynamics (Scheck, 1994; Goldstein, 1977) determines the generalized velocity v_i to be α_i :

$$v_i = \frac{\partial H}{\partial p_i} = \alpha_i. \quad (14.8)$$

The relation (14.8) is a bit of a mystery for now because it seems to imply a constant velocity of $c \equiv 1$.

The major constraints on α and β arise from (vii)'s requiring the operator form of $E^2 = p^2 + m^2$ (that is, the Klein–Gordon equation) still to be valid when acting on Ψ . This is met by postulating Ψ to have a number N of components,

$$\Psi = \begin{pmatrix} \psi_1 \\ \psi_2 \\ \psi_3 \\ \vdots \\ \psi_N \end{pmatrix}, \quad (14.9)$$

and then requiring each component to satisfy the Klein–Gordon equation. Yet if Ψ has multiple components, β and each α_i must be matrices.

To systematically impose requirement (vii) on each component, we take the time derivative of the Dirac equation (14.6):

$$\frac{\partial}{\partial t} \left(i \frac{\partial \Psi}{\partial t} \right) = (\alpha \cdot \mathbf{p} + \beta m) \frac{\partial \Psi}{\partial t}, \quad (14.10)$$

$$\Rightarrow i \frac{\partial^2 \Psi}{\partial t^2} = (\alpha \cdot \mathbf{p} + \beta m) \left(\alpha \cdot \frac{\mathbf{p}}{i} \Psi + \frac{\beta}{i} m \Psi \right). \quad (14.11)$$

Because the LHS of (14.11) is the LHS of the Klein–Gordon equation operating on the vector Ψ , that is, each component of Ψ , we require likewise for the RHS:

$$\begin{aligned} i \frac{\partial^2 \Psi}{\partial t^2} &= \left(\sum_k \frac{\alpha_k}{i} \frac{\partial}{\partial x^k} + \beta m \right) \left(\sum_j \frac{\alpha_j}{i^2} \frac{\partial}{\partial x^j} + \frac{\beta m}{i} \right) \Psi \\ &= i \sum_{k,j} \frac{\alpha_j \alpha_k + \alpha_k \alpha_j}{2} \frac{\partial^2 \Psi}{\partial x^k \partial x^j} - m \sum_k (\alpha_k \beta + \beta \alpha_k) \frac{\partial \Psi}{\partial x^k} - i \beta^2 m^2 \Psi. \end{aligned} \quad (14.12)$$

If the RHS of (14.12) is to be equivalent to the Klein–Gordon equation (13.18) for each component,² we must have the anticommutation relations:

$$\alpha_k \alpha_j + \alpha_j \alpha_k \stackrel{\text{def}}{=} \{\alpha_k, \alpha_j\} = 2\delta_{kj}, \quad (14.13)$$

$$\alpha_k \beta + \beta \alpha_k \stackrel{\text{def}}{=} \{\alpha_k, \beta\} = 0, \quad (14.14)$$

$$\Rightarrow \alpha_i^2 = \beta^2 = 1. \quad (14.15)$$

Matrices that satisfy (14.13)–(14.15) form a representation of a *Clifford algebra*. Determining them is analogous to finding higher-dimension representations of the Pauli matrices.

Further Properties of α and β Matrices

• Hermiticity.

$$\alpha_i^\dagger = \alpha_i, \quad \beta^\dagger = \beta. \quad (14.16)$$

²Sakurai (1967) shows that these relations also follow from the requirement of there being a continuity equation. Note, care is needed when referring to other treatments, such as Sakurai (1967), Mandl (1966), Rose (1961), and Pauli (1921), because they use a different metric or different Dirac matrices than we.

This follows from (ii), Hermiticity of the Hamiltonian $\alpha \cdot \mathbf{p} + \beta m$. [With (14.15) this implies that the eigenvalues of α_i and β must be ± 1 , which is a little surprising if α is to behave like a velocity (14.8).]

- **Trace $\alpha_i = \text{Trace } \beta = 0$. (Proof)**

$$\begin{aligned}\alpha_i \beta &= -\beta \alpha_i \Rightarrow \alpha_i = -\beta \alpha_i \beta, \\ \Rightarrow \text{Tr}(\alpha_i) &= -\text{Tr}(\beta \alpha_i \beta) = -\text{Tr}(\alpha_i \beta^2) = -\text{Tr} \alpha_i, \\ &\Rightarrow \text{Tr}(\alpha_i) = 0.\end{aligned}$$

- **Dimensionality (N) Must be Even. (Proof):**

$$\begin{aligned}\alpha_i \beta &= -\beta \alpha_i, \\ \Rightarrow \det(\alpha_i \beta) &= \det(-\beta \alpha_i) = \det(-\beta) \det(\alpha_i) = (-1)^N \det \alpha_i \det \beta, \\ \Rightarrow N &= \text{even}.\end{aligned}$$

- **$N \geq 4$.** It is not possible to have $N = 2$ because the Pauli matrices (σ, I) form a complete set of 2×2 matrices; yet because the unit matrix I always commutes, (14.13)–(14.15) will not be satisfied. This leaves $N = 4$ as lowest possibility (and the one we use):

$$\Psi = \begin{pmatrix} \psi_1 \\ \psi_2 \\ \psi_3 \\ \psi_4 \end{pmatrix}, \quad \Psi^\dagger = (\psi_1^* \psi_2^* \psi_3^* \psi_4^*). \quad (14.17)$$

- **Standard α and β .** The α and β matrices used by Schweber (1959), Bjorken and Drell (1964), and Messiah (1961) have become the modern standard.³ They are:

$$\alpha_i = \begin{pmatrix} 0 & \sigma_i \\ \sigma_i & 0 \end{pmatrix}, \quad \beta = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad (14.18)$$

where the σ_i , 1, and 0 in (14.18) are the 2×2 Pauli matrices (14.34), for example:

$$\alpha_3 = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix}, \quad \beta = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}. \quad (14.19)$$

14.2 Electrons At Rest

Although we have yet to check that all requirements on the theory have been met, it may help our motivation to check that there is reasonable physics in the Dirac equation. Consider the Dirac equation for a free electron at rest, that is, $\mathbf{p} = 0$ so $\nabla \Psi = 0$ in (14.6),

$$i \frac{\partial \Phi}{\partial t}(\mathbf{x}, t) = \beta m \Phi(\mathbf{x}, t), \quad (14.20)$$

³ Although the old standards have some advantages for computer algebra.

where we use the conventional notation of Φ for a plane wave. In terms of Φ 's four components, the Dirac equation is

$$i \begin{pmatrix} \partial\phi_1/\partial t \\ \partial\phi_2/\partial t \\ \partial\phi_3/\partial t \\ \partial\phi_4/\partial t \end{pmatrix} = m \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} \phi_1 \\ \phi_2 \\ \phi_3 \\ \phi_4 \end{pmatrix} = m \begin{pmatrix} \phi_1 \\ \phi_2 \\ -\phi_3 \\ -\phi_4 \end{pmatrix}, \quad (14.21)$$

that is, four first-order differential equations (normally coupled by the $\alpha \cdot \mathbf{p}$ term, which, however, vanishes for an electron at rest). Four independent-solutions of (14.21) are

$$\Phi_1 = e^{-imt} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad \Phi_2 = e^{-imt} \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \quad (14.22)$$

$$\Phi_3 = e^{+imt} \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \quad \Phi_4 = e^{+imt} \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}. \quad (14.23)$$

From the t dependence of (14.22) and our familiarity with spin in Schrödinger theory, we identify Φ_1 and Φ_2 as positive-energy solutions for spin up and down, respectively, and Φ_3 and Φ_4 as negative-energy (negative-frequency), antiparticle solutions with spin up and spin down; whence the need for four components.⁴

14.3 Covariant Form, γ Matrices

The α and β matrices are historical and convenient for the Hamiltonian form of the Dirac equation. For other purposes it is convenient to have the equation in a form that at least “looks” covariant, that is, in which the space and time derivatives are treated similarly. We obtain this *covariant form* by multiplying the Dirac equation (14.6) by β , and replacing β^2 by 1,

$$\left(i\beta \frac{\partial}{\partial t} - \beta \alpha \cdot \frac{\nabla}{i} \right) \Psi(\mathbf{x}, t) = m\Psi(\mathbf{x}, t). \quad (14.24)$$

We next define new matrices,

$$\gamma^0 \stackrel{\text{def}}{=} \beta, \quad \gamma^i \stackrel{\text{def}}{=} \beta \alpha_i, \quad \gamma \stackrel{\text{def}}{=} \beta \alpha, \quad \gamma^\mu \stackrel{\text{def}}{=} (\gamma^0, \gamma), \quad (14.25)$$

so that the Dirac equation for a free-particle can be written as

$$\left(i\gamma^0 \frac{\partial}{\partial t} - \gamma \cdot \frac{\nabla}{i} \right) \Psi(\mathbf{x}, t) = m\Psi(\mathbf{x}, t), \quad (14.26)$$

$$(\gamma^0 p_0 - \gamma \cdot \mathbf{p}) \Psi(\mathbf{x}, t) = m\Psi(\mathbf{x}, t). \quad (14.27)$$

⁴Because we are in the rest frame of the particle, it is easy to discuss spin; it is more difficult for a moving particle as discussed in Chapters 15–17. Yet even in classical relativistic mechanics it is difficult to define the spin of a moving particle because its angular momentum precesses when viewed in different frames (Hagedorn, 1963).

In 4-vector form it obtains its maximum compactness:

$$\gamma^\mu p_\mu \Psi(\mathbf{x}, t) = m \Psi(\mathbf{x}, t), \quad (14.28)$$

where we have used $p_\mu = i\partial_\mu$ (see Appendix C, *Four-Vectors and Lorentz Transformations*).

If γ^μ were actually a 4-vector (that is, with components which transform like $\mathbf{x}$ and t), then $\gamma \cdot p \equiv \gamma^\mu p_\mu$ would be invariant, and we could make the Dirac equation (14.28) covariant (same form in all reference frames) by choosing $\Psi'(\mathbf{x}', t') = \Psi(\mathbf{x}, t)$. While we use the 4-vector notation γ^μ to denote the four γ matrices, we follow the conventional view that all frames agree on their γ matrices, as they do for their Pauli matrices, in which case the γ 's are constants and only the wave functions change under Lorentz transformation.⁵ We study the transformation in § 14.5.

Exercise Show that the properties of the α and β matrices lead to the following properties for the γ matrices:

$$\gamma^\mu \gamma^\nu + \gamma^\nu \gamma^\mu \equiv \{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu}, \quad (14.29)$$

$$(\gamma^i)^2 = -1, \quad (i = 1, 2, 3), \quad (\gamma^0)^2 = 1, \quad (14.30)$$

$$\gamma^\dagger = -\gamma, \quad \gamma_0^\dagger = \gamma_0. \quad \diamond \quad (14.31)$$

Standard γ Representation

The standard representation of the α and β matrices (14.18) leads to the standard γ matrices (also in Appendix C, *Four-Vectors and Lorentz Transformations*):

$$\gamma^i = \begin{pmatrix} 0 & \sigma_i \\ -\sigma_i & 0 \end{pmatrix}, \quad \gamma^0 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \gamma^5 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \quad (14.32)$$

It is also useful to define a σ matrix as a generalization of spin:

$$\sigma^{ij} = \begin{pmatrix} \sigma_k & 0 \\ 0 & \sigma_k \end{pmatrix}, \quad \sigma^{0i} = i\alpha_i = i \begin{pmatrix} 0 & \sigma_i \\ \sigma_i & 0 \end{pmatrix}. \quad (14.33)$$

Here (i, j, k) can take on any of the values 1, 2, and 3 (or equivalently x, y, z), and may be permuted cyclically. The σ 's are the Pauli matrices:

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (14.34)$$

Exercise Verify that these γ 's follow from the standard α 's and β . $\diamond$

Exercise Verify that these matrices satisfy (14.29)–(14.31). $\diamond$

⁵Good (1955) shows that all observers have gamma matrices that are related by an equivalence transformation $\gamma' = U^{-1}\gamma U$. Since they are “equivalent,” it is consistent but not necessary to have all frames with the same γ .

Table 14.1: Internal Parts of Bilinear Covariants

$\Gamma_{4 \times 4}$	Definition	Transforms	Number
γ	$\beta\alpha$	spacelike vector	3
γ_0	β	timelike vector	1
$1 \equiv I$	Identity	scalar	1
$\sigma^{\mu\nu}$	$\frac{i}{2} [\gamma^\mu, \gamma^\nu]$	traceless tensor	6
γ_5	$i\gamma^0\gamma^1\gamma^2\gamma^3$	pseudoscalar	1
$\gamma_5\gamma^\mu$	$\gamma_5\gamma^\mu$	pseudovector	4

All Possible 4D Square Matrices

Whereas it may be confusing to first introduce the α and β matrices and then the equivalent γ matrices, it may be comforting to know that this must end somewhere because there are only 16 independent 4×4 matrices! We call the set of all 16 matrices Γ_i and give a complete set in Table 14.3. Also indicated in the table are the transformation properties of $\bar{\Psi}\Gamma\Psi$ (we define $\bar{\Psi}$ in the next section where we discuss these transformations). We shall see that the spacelike parts σ^{ij} ($i, j = 1, 2, 3$) are spinlike, while the mixed spacetime parts $\sigma^{0\mu}$, $\sigma^{\mu 0}$ are velocitylike (they are proportional to α). To emphasize this identity we write

$$\Sigma_k \stackrel{\text{def}}{=} \sigma_{4 \times 4}^{ij}, \quad (14.35)$$

where Σ is to be viewed as a 4D generalization of the Pauli matrix σ . We shall see that Σ appears in Dirac Hamiltonians much the way σ does in nonrelativistic Hamiltonians.

14.4 Probability and Current

Now that we know there is interesting physics present, we return to verify that requirements (i), (ii), (iv), and (v) are met. As usual, probability and current are deduced by first assuming the validity of the continuity equation,

$$\partial^\mu j_\mu = \frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{j} = 0. \quad (14.36)$$

We then manipulate the Dirac equation until it has a time derivative and divergence like (14.36), and then read off ρ and $\mathbf{j}$ as the quantities whose time derivative and divergence are being taken. We multiply the Hamiltonian form of the Dirac equation (14.6) by $\Psi^\dagger$ to obtain

$$\Psi^\dagger i \frac{\partial \Psi}{\partial t} = \Psi^\dagger \mathbf{H}_0 \Psi(\mathbf{x}, t). \quad (14.37)$$

We subtract the Hermitian adjoint of this equation from itself to obtain:

$$\frac{\partial(\Psi^\dagger \Psi)}{\partial t} + \nabla \cdot (\Psi^\dagger \alpha \Psi) = 0. \quad (14.38)$$

Exercise Fill in the steps leading to (14.38). $\diamond$

Accordingly, probability is

$$\rho = \Psi^\dagger \Psi = |\psi_1|^2 + |\psi_2|^2 + |\psi_3|^2 + |\psi_4|^2, \quad (14.39)$$

and current is

$$\mathbf{j} = \Psi^\dagger \boldsymbol{\alpha} \Psi \equiv \Psi^\dagger \beta \boldsymbol{\alpha} \Psi \equiv \bar{\Psi} \boldsymbol{\gamma} \Psi. \quad (14.40)$$

Note, we have expressed $\mathbf{j}$ in terms of the *Dirac adjoint* $\bar{\Psi}$ which we have defined as:

$$\bar{\Psi} \stackrel{\text{def}}{=} \Psi^\dagger \beta \equiv \Psi^\dagger \gamma_0. \quad (14.41)$$

Exercise Verify that $j_3 = \text{Re}(\psi_1^* \psi_3) - \text{Re}(\psi_2^* \psi_4)$ $\diamond$

While we have manifestly succeeded in making the electron's probability ρ positive definite (iii) and subject to the continuity equation (v), our use of four components apparently requires us to include both positive- and negative-energy particles in the expressions for probability and current. Yet now one of our mysteries is unraveled: we see from (14.40) how $\boldsymbol{\alpha}$ plays the role of a generalized velocity [as expected from (14.8)].

We next employ the transformation properties of the Dirac equation to prove that the density and current form a 4-vector. Indeed, this is almost manifest when they are written in terms of $\boldsymbol{\gamma}$ matrices:

$$j^\mu = (\rho, \mathbf{j}) = \bar{\Psi} \boldsymbol{\gamma}^\mu \Psi. \quad (14.42)$$

Although (14.42) means that ρ , being the timelike component of a 4-vector, is not a scalar, we shall see that the Dirac scalar product,

$$\bar{\Psi} \Psi = \Psi^\dagger \gamma_0 \Psi = |\psi_1|^2 + |\psi_2|^2 - |\psi_3|^2 - |\psi_4|^2, \quad (14.43)$$

does produce an invariant.

14.5 Lorentz Transformation of Wave Functions

In Appendix C, *Four-Vectors and Lorentz Transformations*, we indicate that the Lorentz transformation of a 4-vector such as the coordinate x^μ is described by a matrix⁶ of numbers $a^\nu{}_\mu$:

$$x'^\nu = a^\nu{}_\mu x^\mu. \quad (14.44)$$

As illustrated in Figure 14.1, x' is the displacement vector in a reference frame O' moving with a constant velocity $\mathbf{u}$ relative to the reference frame O in which x is the displacement vector. An observer in O' measures a wave function $\Psi'(x')$ that, of course, is a function of x' . We define an operator $L_\nu(a)$ that transforms Ψ to Ψ' as

$$\Psi'(x') = L_\nu(a) \Psi(x), \quad (14.45)$$

where a in (14.45) represents the transformation parameters [the elements of the a matrix in (14.44)]. Accordingly, L_ν is represented by a matrix that operates on Dirac spinors (internal space), but *not* 4-vectors like x^μ (external space).

⁶This can be thought of as a matrix multiplication, but it is safer to think of it as a summation.

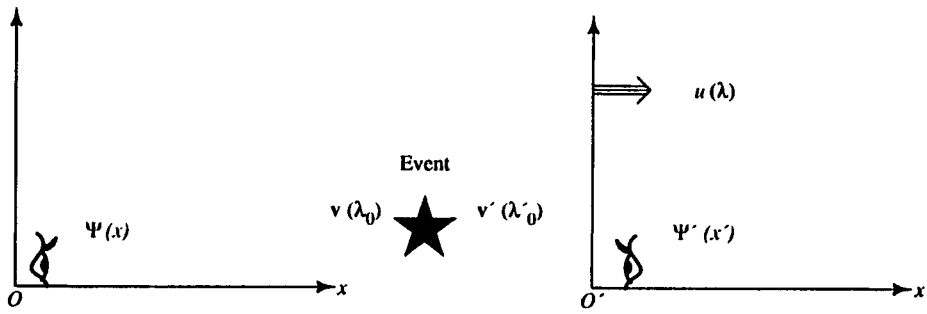


Figure 14.1: Two observers O and O' related by a Lorentz transformation of velocity u viewing the same event (the star). Observer O sees it with velocity v , observer O' sees it with velocity v' .

It is indubitably confusing to have four components for both Dirac spinors and 4-vectors. We try to decrease the confusion arising from this coincidence by using matrix multiplication for the spinor degrees of freedom, but *not* for 4-vector transformations like (14.44).

Requirements on Dirac Equation

The requirement of covariance is that observers in O and O' (who, by agreement, share the same γ matrices) describe nature with similar Dirac equations:⁷

$$i\gamma^\mu \frac{\partial \Psi(x)}{\partial x^\mu} = m\Psi(x), \quad (14.46)$$

$$i\gamma^\mu \frac{\partial \Psi'(x')}{\partial x'^\mu} = m\Psi'(x'). \quad (14.47)$$

To deduce the requirement on the γ matrices for (14.46) to imply (14.47), we express Ψ and the x derivatives in terms of quantities known by observer O' ,

$$\Psi(x) = L_v^{-1}(a)\Psi'(x'), \quad \frac{\partial}{\partial x^\mu} = a^\nu{}_\mu \frac{\partial}{\partial x'^\nu}. \quad (14.48)$$

Upon substitution into (14.46), this produces

$$i\gamma^\mu a^\nu{}_\mu \frac{\partial}{\partial x'^\nu} L_v^{-1}(a)\Psi'(x') = m L_v^{-1}(a)\Psi'(x'). \quad (14.49)$$

We make (14.49) have the same RHS as (14.47) by left-multiplying with $L_v(a)$. On the LHS, L_v commutes with the derivative (L_v has no x dependence), and with the $a^\nu{}_\mu$ (they are just numbers):

$$[a^\nu{}_\mu L_v(a)\gamma^\mu L_v^{-1}(a)] i \frac{\partial}{\partial x'^\nu} \Psi'(x') = m\Psi'(x'). \quad (14.50)$$

⁷Although important, the development of this section gets somewhat involved. The first-time reader may therefore decide to jump to the result (14.62) and then to the next section.

For this to be the same as (14.47) requires

$$[a^\nu{}_\mu L_\nu(a) \gamma^\mu L_\nu^{-1}(a)] = \gamma^\nu, \quad (14.51)$$

$$\Rightarrow a^\nu{}_\mu \gamma^\mu = L_\nu^{-1}(a) \gamma^\nu L_\nu(a). \quad (14.52)$$

Here the sum over μ is implied, the a 's are just numbers, and the γ 's and L 's are 4×4 matrices.

Exercise Show that for velocity boost directions $\nu \neq 0$, or k , (14.52) reduces to

$$I\gamma^\nu = \gamma^\nu \quad (\nu \neq 0, k). \quad \diamond \quad (14.53)$$

The Transformation Operator

We guess an L_ν to satisfy (14.52) by generalizing the analogous operator for 3D rotations to include velocity transformations (which are after all just rotations in spacetime.⁸) Because we can verify if our guess works, it does not matter that we just guessed it. We know that the operator U_R induces 3D rotations of nonrelativistic spin $\frac{1}{2}$ wave functions:

$$\Psi'(\mathbf{x}') = U_R(\theta)\Psi(\mathbf{x}), \quad (14.54)$$

$$U_R(\theta) = e^{-i\mathbf{J}\cdot\boldsymbol{\theta}} = e^{-i\boldsymbol{\theta}\cdot\boldsymbol{\sigma}/2}. \quad (14.55)$$

Here $\boldsymbol{\sigma}/2$ is the generator of infinitesimal rotations for spin $\frac{1}{2}$, $\boldsymbol{\theta}$ points along the axis of rotation, and the minus sign in the exponent indicates an *active* rotation (the physical system described by Ψ is transformed.⁹) Because the relativistic generalization of the Pauli σ_k is the 4×4 matrix $\sigma^{ij} \equiv \Sigma_k$ of Table 14.3, we guess that the relativistic operator for active 3D rotations along the k axis is

$$U_R(\theta) = e^{-i\Sigma_k \theta_k/2}. \quad (14.56)$$

To generalize (14.56) to velocity boosts, we imagine the Lorentz transformation (14.44) as a rotation in spacetime and replace σ^{ij} , the generator of infinitesimal rotations about the k axis, by σ^{0k} , the generator of velocity boosts along the k axis. We shall see that the “angle” of rotation is imaginary. For transformations along one axis, there is a rapidity parameter λ that determines the velocity $u = u(\lambda)$ of a Lorentz transformation and is additive for successive transformations:

$$\cosh \lambda = \gamma_u = (1 - \beta_u^2)^{-1/2} = (1 - u^2)^{-1/2}, \quad \sinh \lambda = \gamma_u \beta_u. \quad (14.57)$$

Exercise Show that Lorentz transformation along the x axis in Figure 14.1 can be expressed as a rotation by an imaginary angle,

$$x' = x \cosh \lambda - t \sinh \lambda, \quad (14.58)$$

$$t' = t \cosh \lambda - x \sinh \lambda. \quad \diamond \quad (14.59)$$

⁸The integration of space and time into a 4D space with Lorentz transformations as rotations was introduced by Minkowski (see Einstein et al., 1952).

⁹For active Lorentz transformations, we imagine boosting a particle at rest to some final velocity u . For passive transformations, we imagine moving to a new reference frame moving with velocity u (in which case a particle at rest in the initial frame would now be seen with velocity $-u$, just the negative of that from an active transformation).

When observer O sees a particle moving with velocity v along the x axis in Figure 14.1, he or she uses (14.57) to determine the particle's *rapidity* and calls it λ_0 . Likewise, observer O' sees the same particle moving with velocity v' and uses (14.57) to determine its rapidity and calls it λ'_0 .

Exercise Show that the velocity $v'(\lambda'_0)$ is related to the velocity $v(\lambda_0)$ (and their associated rapidity parameters) by:

$$v'(\lambda'_0) = \frac{u(\lambda) + v(\lambda_0)}{1 + uv'}, \quad (14.60)$$

$$\lambda'_0 = \lambda_0 + \lambda. \quad \diamond \quad (14.61)$$

As (14.61) shows, the parameter λ is additive for transformations along one axis (like angles of rotations about one axis), and so our analogy with rotations appears to be working. Furthermore, since this transformation boosts the particle's velocity, it is active, and since the conventional Lorentz transformation (14.58)–(14.59) is passive, we have a minus sign to watch out for. We guess that the generalization of (14.56) to rotations in spacetime is

$$L_v(a) \equiv L_v(\lambda) = e^{\mp \frac{1}{2} i \lambda_k \sigma^{0k}}, \quad (14.62)$$

where k is the axis of the velocity boost (1 or x in Figure 14.1), and the upper (lower) sign is for active (passive) transformations. To check if our guess works, that is, satisfies (14.52), we convert (14.62) to an explicit matrix by expanding the exponential:

$$\begin{aligned} e^{\mp \frac{1}{2} i \lambda_k \sigma^{0k}} &= 1 \mp i \frac{\lambda}{2} \frac{\sigma^{0k}}{1!} - \left(\frac{\lambda}{2}\right)^2 \frac{(\sigma^{0k})^2}{2!} \mp \dots \\ &= 1 + \frac{(\frac{\lambda}{2})^2}{2!} + \frac{(\frac{\lambda}{2})^4}{4!} + \dots + i \sigma^{0k} \left(\frac{\lambda}{2} + \frac{(\frac{\lambda}{2})^3}{3!} + \dots \right). \end{aligned} \quad (14.63)$$

Yet if we use identity

$$(\sigma^{0k})^2 = (i\alpha_k)^2 = -I_{4 \times 4}, \quad (14.64)$$

we obtain the simple relation

$$L_v = e^{\mp \frac{1}{2} i \lambda_k \sigma^{0k}} = I \cosh \frac{\lambda_k}{2} \mp i \sigma^{0k} \sinh \frac{\lambda_k}{2}. \quad (14.65)$$

Exercise Show that (14.52) is satisfied for $\nu = 0, k$, by showing that

$$\{\gamma^0, \sigma^{0k}\} = \{\gamma^k, \sigma^{0k}\} = 0, \quad (14.66)$$

$$\begin{aligned} L_v^{-1} \gamma^0 L_v &= e^{-i \lambda_k \sigma^{0k}} \gamma^0 = \cosh \lambda_k \gamma^0 - \sinh \lambda_k \gamma^k \\ &= a^0_\nu \gamma^\nu = \gamma_\nu \gamma^0 - \gamma_\nu \beta_\nu \gamma^k. \quad \diamond \end{aligned} \quad (14.67)$$

Exercise Similarly, show that

$$L_v^{-1} \gamma^k L_v = \gamma_v \gamma^k - \gamma_v \beta_v \gamma^0. \quad \diamond \quad (14.68)$$

One of the first things to note about L_v (14.65) is that it is *not unitary*, $L_v \neq L_v^{-1}$. This means Ψ 's normalization changes under transformation.

Exercise Prove that

$$L_v^\dagger = L_v \neq L_v^{-1}. \quad \diamond \quad (14.69)$$

Because the volume element also changes under transformation, this change in Ψ is needed to keep total probability constant (iv). Although $L_v^\dagger$ is not L_v^{-1} , there is a very handy relation between the two.

Exercise Verify that with the standard representation

$$L_v^{-1} = \gamma_0 L_v^\dagger \gamma_0. \quad \diamond \quad (14.70)$$

Bilinear Covariants

In Table 14.3 we gave 16 matrices that, when sandwiched between a Dirac spinor Ψ and its adjoint $\bar{\Psi}$, transform as indicated. For example, the current,

$$j^\mu(x) = \bar{\Psi}(x) \gamma^\mu \Psi(x) = \Psi^\dagger \gamma^0 \gamma^\mu \Psi, \quad (14.71)$$

is a 4-vector. We prove this by relating $j'^\mu(x')$ to $j^\nu(x)$:

$$\begin{aligned} j'^\mu(x') &= \Psi'^\dagger(x') \gamma^0 \gamma^\mu \Psi'(x') = \Psi^\dagger(x) L_v^\dagger \gamma^0 \gamma^\mu L_v \Psi(x) \\ &= \Psi^\dagger(x) \gamma^0 (\gamma^0 L_v^\dagger \gamma^0) \gamma^\mu L_v \Psi(x) = \Psi^\dagger \gamma^0 [(L_v^{-1})^\dagger \gamma^\mu L_v] \Psi \\ &= \Psi^\dagger(x) \gamma^0 [\alpha_\nu^\mu \gamma^\nu] \Psi(x) = \alpha_\nu^\mu \bar{\Psi}(x) \gamma^\nu \Psi(x). \end{aligned} \quad (14.72)$$

We recognize the RHS as the current again, and so

$$j'^\mu(x') = \alpha_\nu^\mu j^\nu(x). \quad (14.73)$$

Likewise, $\bar{\Psi} I \Psi$ is a scalar,

$$\bar{\Psi} I \Psi = \bar{\Psi}' I \Psi', \quad (14.74)$$

and $\bar{\Psi} \sigma^{\mu\nu} \Psi$ is a second-rank tensor,

$$\bar{\Psi}'(x') \sigma^{\mu\nu} \Psi'(x') = \alpha^\mu_\alpha \alpha^\nu_\beta \bar{\Psi}(x) \sigma^{\alpha\beta} \Psi(x). \quad (14.75)$$

Because the spin operator $\Sigma_k \equiv \sigma^{ij}$ and the velocity operator σ^{0k} mix under Lorentz transformations, a particle's spin (magnetic) and velocity (electrical) properties also mix. We cannot separate the two for a moving particle, and so analysis in the rest frame is usually simpler (see too the Gordon decomposition in Chapter 15, *Components of Dirac Wave Functions*).

Including Parity

The parity operation on a wave function is defined to have the effect

$$P\Psi(\mathbf{x}) \stackrel{\text{def}}{=} \Psi'(\mathbf{x}') = \Psi'(-\mathbf{x}, t). \quad (14.76)$$

To include it as part of the Lorentz group of operators L_v , we need to find a 4×4 matrix P with the property

$$P^{-1}\gamma^\nu P = a(P)^\nu{}_\mu \gamma^\mu, \quad (14.77)$$

$$a(P) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}. \quad (14.78)$$

Note this is an improper Lorentz transformation because $\det(a) = -1$, while our previous transformations were proper with $\det(a) = 1$. When the matrix P that satisfies (14.76) acts on a spinor Ψ , the new $P\Psi$ has the opposite *intrinsic parity*, that is, P changes the internal parts of the wave functions. The parity operator on the external part is distinct from the internal part and may have its own eigenvalues. For example, for external angular-momentum eigenstates, the external parity eigenvalues are the familiar $(-1)^l$.

Exercise Verify that the choice $P = \gamma^0 = \beta$ for the parity operator has the desired effect:

$$P \begin{pmatrix} \psi_1 \\ \psi_2 \\ \psi_3 \\ \psi_4 \end{pmatrix} = \beta \begin{pmatrix} \psi_1 \\ \psi_2 \\ \psi_3 \\ \psi_4 \end{pmatrix} = \begin{pmatrix} \psi_1 \\ \psi_2 \\ -\psi_3 \\ -\psi_4 \end{pmatrix}. \quad \diamond \quad (14.79)$$

This last relation is the microscopic explanation of why fermions and antifermions have opposite *intrinsic parity* in Dirac theory.

The *pseudoscalar* object γ^5 and *pseudovector* (or *axial vector*) object $\gamma^5\gamma^\mu$ of Table 14.3 behave like a scalar and a vector respectively for proper Lorentz transformation [$\det(a) = 1$] but pick up minus signs for the parity transformation:

$$P^{-1}\gamma^5 P = -\gamma^5, \quad P^{-1}\gamma^5\gamma^\mu P = \gamma^5\gamma^\mu. \quad (14.80)$$

In Part III, *Quantum Fields*, we shall find these objects and their transformation properties essential in constructing the elementary interactions of physics.

14.6 Problems

1. Verify that

- (a) $\{\alpha_k, \alpha_j\} = 2\delta_{kj}$.
- (b) $\{\alpha_k, \beta\} = 0$.
- (c) $\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu}$.

- (d) $\gamma^\dagger = -\gamma$.
 (e) $\gamma_0^\dagger = \gamma^0$.

2. Which of the relations 1a–1e are independent of representation?
3. Examine $\sigma^{\mu\nu}$ and discuss why it is a good relativistic generalization for the spin operator. In particular, show that its space components look like spin and its mixed components look like “velocity”. Why is this mixing needed in a relativistic theory?
4. The Lorentz transformation is defined such that a 4-vector B in frame O appears as a different 4-vector B' in frame O' ,

$$B'^\nu = a^\nu_\mu B^\mu. \quad (14.81)$$

- (a) Write down a^ν_μ for a transformation that “boosts” a particle at rest to one with velocity $\mathbf{v} = (v_x, v_y, v_z)$.
 (b) Show that the electromagnetic field tensor

$$F^{\mu\nu} = \frac{\partial A^\mu}{\partial x_\nu} - \frac{\partial A^\nu}{\partial x_\mu} \quad (14.82)$$

is a Lorentz tensor.

- (c) How are the electric and magnetic fields $\mathbf{E}$ and $\mathbf{B}$ related to $F^{\mu\nu}$?
 (d) Indicate how the electromagnetic fields $\mathbf{E}$ and $\mathbf{B}$ transform.
5. Consider the Lorentz transformation (including parity) on Dirac wave functions. Verify (for at least one space component and the time component) that the “bilinear covariants” transform as indicated:
- (a) $\bar{\Psi} I \Psi = \bar{\Psi} \Psi$, scalar.
 (b) $\bar{\Psi} \gamma_5 \Psi$, pseudoscalar.
 (c) $\bar{\Psi} \gamma^\mu \Psi$, 4-vector.
 (d) $\bar{\Psi} \gamma_5 \gamma^\mu \Psi$, pseudo (axial) vector.
 (e) $\bar{\Psi} \sigma^{\mu\nu} \Psi$, tensor.
6. View a Lorentz transformation as a rotation in space-time.
- (a) Show that a Lorentz transformation along one axis can be described as a rotation by an imaginary angle λ . Determine λ in terms of the transformation velocity.
 (b) Prove that the velocity parameter λ is additive for two successive Lorentz transformation along the same axis.
 (c) To complement the text’s derivation of the Dirac plane-wave spinors, prove that

$$\tanh \frac{\lambda}{2} = \frac{p}{E_p + m}, \quad \cosh \frac{\lambda}{2} = \frac{\sqrt{E_p^2 + m^2}}{2m}. \quad (14.83)$$

7. Verify or prove that a Lorentz transformation is proper, that is, that $\det[a] = 1$, unless a parity operation is involved.
8. Verify that for the matrix a^ν_μ of Problem 4,

$$L_\nu^{-1} \gamma^\nu L_\nu = a^\nu_\mu \gamma^\mu. \quad (14.84)$$

CHAPTER 15

COMPONENTS OF DIRAC WAVE FUNCTIONS

Now that we have mastered some Dirac accounting, we can uncover more of the physics of relativistic quantum mechanics. We start with the hole interpretation of antifermions, continue by decomposing the current into internal and external pieces, extend the theory by introducing interactions, and conclude that the Dirac equation contains much physics that previously had to be introduced phenomenologically. Klein's paradox, although present with the Dirac equation, will not be repeated (there are some exercises on it). We end with a tutorial on Dirac theory for mass-zero particles like neutrinos.

15.1 Holes in the Sea

In the last chapter we gave the following solutions for a Dirac particle at rest:

$$\Phi_1 \equiv \Phi_{0\uparrow}^{(+)} = e^{-imt} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad \Phi_2 \equiv \Phi_{0\downarrow}^{(+)} = e^{-imt} \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \quad (15.1)$$

$$\Phi_3 \equiv \Phi_{0\uparrow}^{(-)} = e^{+imt} \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \quad \Phi_4 \equiv \Phi_{0\downarrow}^{(-)} = e^{+imt} \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}, \quad (15.2)$$

where the first of the double subscripts refers to the momentum and the second to the spin.

Exercise Act on these Φ 's with the 4D spin operator σ^{12} (Σ_3) and show that Φ_1 and Φ_3 correspond to spin up (eigenvalues 1 for Σ_3), while Φ_2 and Φ_4 correspond to spin down. $\diamond$

Exercise Act with the energy operator $i\partial_t$ and show that Φ_1 and Φ_2 are positive-energy solutions, while Φ_3 and Φ_4 are negative-energy ones. $\diamond$

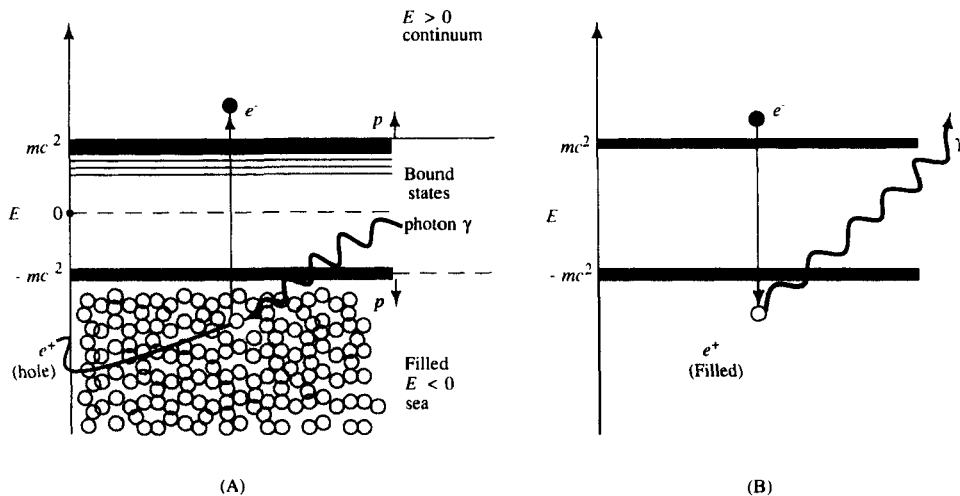


Figure 15.1: The energy spectrum for the Dirac equation. (A) A photon (wavy line) elevating a negative-energy electron from the filled sea into the positive-energy continuum, leaving a hole (positron). (B) An electron filling a negative-energy hole and emitting a photon (electron-positron annihilation).

As discussed for the Klein–Gordon equation, the existence of negative-energy solutions is a problem since the positive-energy states should make rapid, radiative transitions to them, in which case atoms as we know them would not exist. Dirac’s ingenious escape from this catastrophic collapse was to hypothesize that in the world, as we know it (Figure 15.1A), all the negative-energy levels are filled and thus forbidden by the exclusion principle from accepting more electrons. The *vacuum*, as we know it, is a sea of negative-energy electrons beginning at $E = -m$ and extending to $-\infty$ and an (empty) positive-energy continuum beginning at $E = +m$ and extending to $+\infty$. Bound states (atoms) are right below m (that is, with negative kinetic energy but positive total energy). Although the vacuum with its infinite sea of negative-energy electrons is anything but empty, it is what we have been brought up to think of as “nothing”.

Observable indications of the negative-energy sea occur when we modify it; for example, in Figure 15.1A we show by a wavy line a photon colliding with one of the sea’s electrons of energy $-E_p$ and charge $q = -e$, and knocking it into the positive-energy continuum. We observe the *hole* in the sea as the absence of $-E_p$ and $-e$, that is, as a particle of energy E_p and charge $+e$, the *positron*. The process just described is the creation of a particle-antiparticle pair (electron-positron) by a photon:¹

$$\gamma \rightarrow e^- + e^+. \quad (15.3)$$

As follows from conserving energy for the levels in Figure 15.1A, in order for the photon

¹ The one-photon process actually requires the presence of a heavy object like a nucleus to conserve momentum, $\gamma + (Z, A) \rightarrow e^- + e^+ + (Z, A)$. A free electron-positron pair must annihilate into two (or more) photons.

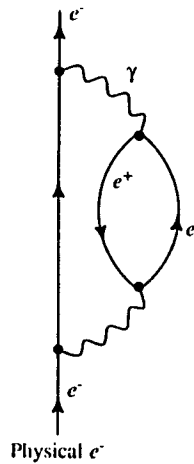


Figure 15.2: A bare electron “dressed” by a photon and an e^+e^- pair.

to lift the electron out of the sea and into the positive energy continuum, the photon must have an energy $E_\gamma > 2m$. Inversely, in Figure 15.1B we show a positive-energy electron falling into a hole in the negative-energy sea and emitting a gamma ray (photon) in the process; this process is *particle-antiparticle annihilation*:

$$e^- + e^+ \rightarrow \gamma. \quad (15.4)$$

Dirac’s prediction of positrons was confirmed by Anderson (1933) who observed positively charged electrons in cosmic rays.

Another consequence of this sea of fermions is *vacuum polarization*, an effect contributing to the Lamb shift in hydrogen (§ 21.3) and to the anomalous magnetic moment of fermions. We picture this diagrammatically in Figure 15.2 where a bare fermion (with the charge density of Figure 15.3A) “polarizes” the vacuum by emitting a virtual photon that in moving through space knocks an e^- from the sea, leaving an e^+ hole behind. The virtual positron is attracted to the physical electron, while the virtual electron is repelled from it; this leads to the polarization density of Figure 15.3B. Eventually the pair recombines and leaves the original fermion. This polarization gives the fermion an effective charge density as shown schematically in Figure 15.3C.²

While Dirac’s prediction of the positron e^+ is a supreme achievement, it is also the theory’s death knoll. As discussed with the Klein–Gordon equation, relativity’s incorporation of negative-energy degrees of freedom makes a one-particle theory inconsistent. Nonetheless, as long as the kinetic energy is not too high (or too negative), the Dirac equation does a very good (though not perfect) job; for example, it describes the hydrogen atom better than Schrödinger theory and is used within field theory as the correct description of noninteracting fermions.

Another view of antiparticles is the *Feynman–Stueckelberg picture* (Feynman, 1949, Stueckelberg, 1941), in which the negative energies or frequencies appearing in solutions such as (15.2) are viewed as describing particles running *backward* in time. In this case

²This is only symbolic of the physical effect. Electrons are actually point particles.

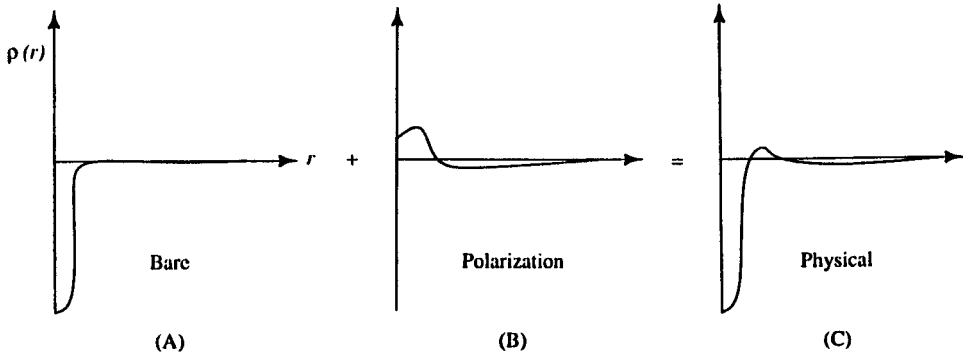


Figure 15.3: The effective charge density (A) of a bare particle, (B) of the polarized vacuum, and (C) of the physical particle.

the particle creation and annihilation events of Figure 15.1A are pictured in Figure 15.4 as photons scattering from electrons, sometimes knocking them forward in time, sometimes backward.

15.2 Plane Waves

Because the absence of a negative-energy particle is identified with a positive-energy antiparticle, it is conventional to reverse the spin eigenvalues on the negative-energy solutions and make them refer to physical antiparticles. We thus rewrite the particle-at-rest solutions (15.2) as

$$\Phi_1 = \Phi_{0\uparrow}^{(+)}(0) = e^{-imt} u_{\uparrow}^{(+)}(0), \quad \Phi_2 = \Phi_{0\downarrow}^{(+)}(0) = e^{-imt} u_{\downarrow}^{(+)}(0), \quad (15.5)$$

$$\Phi_3 = \Phi_{0\downarrow}^{(-)}(0) = e^{+imt} u_{\downarrow}^{(-)}(0), \quad \Phi_4 = \Phi_{0\uparrow}^{(-)}(0) = e^{+imt} u_{\uparrow}^{(-)}(0), \quad (15.6)$$

$$u_{\uparrow}^{(+)}(0) = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad u_{\downarrow}^{(+)}(0) = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \quad u_{\downarrow}^{(-)}(0) = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \quad u_{\uparrow}^{(-)}(0) = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}.$$

Here the first of the double subscripts (0) refers to the momentum, the second (m_s) to the spin, and the argument (0) to the position $\mathbf{x}$. It is important to remember that these particle-at-rest spinors are the *basis vectors* in which we expand a general 4×1 Dirac spinor.

We now take these at-rest solutions and deduce from them solutions of the Dirac equation for a free particle of momentum $\mathbf{p}$ and energy $E_p = \sqrt{p^2 + m^2}$:

$$\left(i\gamma^0 \frac{\partial}{\partial t} + i\boldsymbol{\gamma} \cdot \boldsymbol{\nabla} \right) \Phi(\mathbf{x}, t) = m\Phi(\mathbf{x}, t). \quad (15.7)$$

We recognize mt in the exponents in (15.5) and (15.6) as the scalar product $p^\mu x_\mu$ evaluated in the rest frame of the particle. We thus invoke covariance and write

$$\Phi_{p_s}^{(\pm)}(\mathbf{x}, t) = e^{\mp i p^\mu x_\mu} u_s^{(\pm)}(p). \quad (15.8)$$

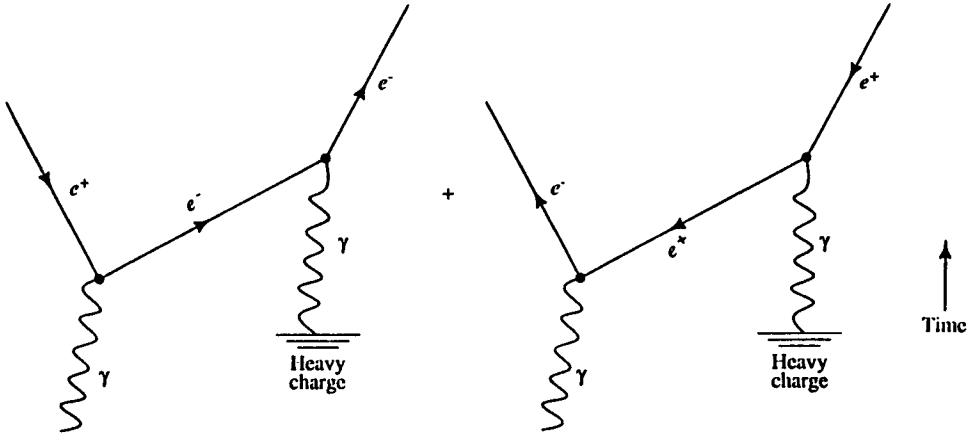


Figure 15.4: A spacetime diagram showing the creation of an electron-positron pair by a photon in the field of a heavy (nonrecoiling) charge. Because there is no way to separate off the different time orderings, the amplitudes corresponding to these diagrams must be added.

Next we determine the plane-wave spinors $u_s^{(\pm)}(p)$ by actively boosting the at-rest solutions along the k axis with the Lorentz transformation operator L_v (14.65):

$$u_s^{(\pm)}(p) = L_v(\lambda) u_s^{(\pm)}(0), \quad (15.9)$$

$$L_v(\lambda) = e^{\mp i \lambda \sigma^{0k}/2} = I \cosh \frac{\lambda}{2} \mp i \sigma^{0k} \sinh \frac{\lambda}{2}. \quad (15.10)$$

Exercise Verify that for a particle with momentum $\mathbf{p}$,

$$\tanh \frac{\lambda}{2} = \frac{p}{E_p + m}, \quad \cosh \frac{\lambda}{2} = \sqrt{\frac{E_p + m}{2m}}. \quad \diamond \quad (15.11)$$

Exercise Use (15.11) to rewrite $L_v(\lambda)$ as

$$L_v(\lambda) = \sqrt{\frac{E_p + m}{2m}} \left(1 + \frac{\boldsymbol{\alpha} \cdot \mathbf{p}}{E_p + m} \right). \quad \diamond \quad (15.12)$$

Exercise Substitute for $\boldsymbol{\alpha}$ into (15.12), perform the matrix multiplication, and thus derive

$$u_{\uparrow}^{(+)}(p) = N \begin{pmatrix} 1 \\ 0 \\ \frac{p_x}{E_p + m} \\ \frac{p_y}{E_p + m} \end{pmatrix}, \quad u_{\downarrow}^{(+)}(p) = N \begin{pmatrix} 0 \\ 1 \\ \frac{p_x}{E_p + m} \\ -\frac{p_y}{E_p + m} \end{pmatrix}, \quad (15.13)$$

$$u_{\uparrow}^{(-)}(p) = N \begin{pmatrix} \frac{p_x}{E_p + m} \\ \frac{p_y}{E_p + m} \\ 1 \\ 0 \end{pmatrix}, \quad u_{\downarrow}^{(-)}(p) = N \begin{pmatrix} \frac{p_x}{E_p + m} \\ -\frac{p_y}{E_p + m} \\ 0 \\ 1 \end{pmatrix}, \quad (15.14)$$

where

$$p_{\pm} = p_x \pm ip_y, \quad N = \sqrt{\frac{E_p + m}{2m}}. \quad \diamond \quad (15.15)$$

These Dirac spinors $u_s^{(\pm)}(p)$ describe free electrons of spin $s = \pm \frac{1}{2}$ (in their rest frame), energy $\pm E_p$, and 4-momentum p . (Note: The negative-energy solutions have the signs of their 3-momenta reversed, in accord with the hole interpretation of positron solutions.) The $u(p)$'s are the building blocks of all relativistic descriptions of spin- $\frac{1}{2}$ particles because they contain the proper transformation properties of spin, and will be used throughout the rest of this book.

Properties of Plane-Wave Spinors

It is worthwhile to verify, or at least contemplate, the following properties:

- The normalized, coordinate-space plane wave is

$$\Phi_{p,s}^{(\pm)}(x) = e^{\mp i p^\mu x_\mu} u_s^{(\pm)}(p). \quad (15.16)$$

The spacetime dependence is in the exponential, the 4D spin dependence is in the spinor, and the p is *not* an operator.

- Spin is a good quantum number only in the electron's rest frame or for motion along the z axis. If $p_x = p_y = 0$ while $p_z \neq 0$, the u 's will be eigenstates of Σ_z , but, interestingly enough, with the lower components contributing.
- *Helicity* $\stackrel{\text{def}}{=} \boldsymbol{\sigma} \cdot \hat{\mathbf{p}}$ is a good quantum number for these u 's.
- When the full wave function (15.16) is substituted into the coordinate-space Dirac equation (15.7), we obtain the *momentum-space Dirac equation* for the free particle spinors $u^{(\pm)}(p)$:

$$(\gamma^\mu p_\mu \mp m) u^{(\pm)}(p) = (\not{p} \mp m) u^{(\pm)}(p) = 0. \quad (15.17)$$

We again note that because p is *not* an operator, this is just a matrix equation.

- The explicit, 4D Dirac equations for positive- and negative-energy spinors are

$$[\gamma^0 p_0 - \boldsymbol{\gamma} \cdot \mathbf{p} - m] u^{(+)}(p) = 0, \quad (15.18)$$

$$[-\gamma^0 p_0 - \boldsymbol{\gamma} \cdot (-\mathbf{p}) - m] u^{(-)}(p) = 0. \quad (15.19)$$

From these equations we see that the negative-energy spinors have energy and momentum opposite to the positive-energy spinors.

- The Dirac adjoint spinor $\bar{u}(p) = u^\dagger(p) \gamma^0$ satisfies a transposed Dirac equation:

$$\bar{u}^{(\pm)}(p) (\not{p} \mp m) = 0. \quad (15.20)$$

- The u 's satisfy Lorentz-invariant *orthogonality relations*:

$$\bar{u}_s^{(b)}(p) u_{s'}^{(b')}(p) = b \delta_{bb'} \delta_{ss'}, \quad (15.21)$$

where $b = \pm$ for the positive (negative)-energy solutions.

- The probability density ρ for plane waves is

$$\rho = \Phi^\dagger \Phi = u_s^{(b)}(p)^\dagger u_{s'}^{(b')}(p) = \frac{E_p}{m} \delta_{bb'} \delta_{ss'}. \quad (15.22)$$

Because ρ is proportional to an energy, it is not Lorentz invariant.

- The *completeness relation* is

$$\sum_{b=\pm, s=\uparrow\downarrow} b u_s^{(b)}(p) \bar{u}_s^{(b)}(p) = I_{4 \times 4}. \quad (15.23)$$

Accordingly, mathematical completeness requires negative-energy degrees of freedom, even for low-energy processes.

- The *current*,

$$\mathbf{j} = \bar{u}_s^{(b)}(p) \boldsymbol{\gamma} u_{s'}^{(b')}(p) = \frac{\mathbf{p}}{m} \delta_{ss'} \delta_{bb'} = \frac{\mathbf{p}}{E_p} \rho \delta_{bb'} \delta_{ss'}, \quad (15.24)$$

has the form expected for plane waves, that is, $\mathbf{j} = \mathbf{v}_g \rho$, a group velocity times a density.

Projection Operators

When manipulating solutions to the Dirac equation within calculations, it is often useful to combine groups of them into simple operators. To do this, we note that because the spinors obey the momentum-space Dirac equations (15.17), the operators

$$\Lambda_\pm(p) = \frac{\pm \not{p} + m}{2m} \quad (15.25)$$

have the properties:

$$\begin{aligned} \Lambda_+ u(p) &= u(p), & \Lambda_- u(p) &= 0, \\ \Lambda_- v(p) &= v(p), & \Lambda_+ v(p) &= 0, \\ \Lambda_\pm^2 &= \Lambda_\pm, & \Lambda_+ \Lambda_- &= \Lambda_- \Lambda_+ = 0, \\ \Lambda_+ + \Lambda_- &= I, \end{aligned} \quad (15.26)$$

where we have defined electron and positron spinors as

$$u_s(p) = u_s^{(+)}(p), \quad v_s(p) = u_s^{(-)}(p). \quad (15.27)$$

Exercise Verify properties (15.26)–(15.27). $\diamond$

If we view the u 's and v 's as basis vectors, we see that Λ_+ is a *projection operator* that eliminates the positron part of a wave function by projecting out the electron part, and, in turn, Λ_- projects out the positron part. We obtain a particularly useful form for these projection operators by replacing the unit operator with the one from the completeness relation for spinors:

$$\sum_i |i\rangle \langle i| = I \Rightarrow \sum_s [u_s(p)\bar{u}_s(p) - v_s(p)\bar{v}_s(p)] = I. \quad (15.28)$$

Exercise Verify (15.28) by explicit substitution of the spinors (15.13). $\diamond$

By relating the individual terms in (15.28) to the operators in (15.26), we obtain alternate forms for the projection operators:

$$\Lambda_+(p) = \sum_{s=1}^2 u_s(p)\bar{u}_s(p) = \frac{\not{p} + m}{2m}, \quad (15.29)$$

$$\Lambda_-(p) = \sum_{s=1}^2 v_s(p)\bar{v}_s(p) = \frac{-\not{p} + m}{2m}. \quad (15.30)$$

15.3 Expansions in Plane Waves

The plane wave solutions to the Dirac equation given by (15.16) form a complete set of states which can be used as a basis for the expansion of more general solutions.

Exercise Verify that $\sqrt{m/(VE_p)} \Phi_s^{(b)}$ is normalized to 1 by showing that the plane wave solutions satisfy the orthogonality condition:

$$\int d^3x [\Phi_s^{(b)}(x)]^\dagger \Phi_{s'}^{(b')}(x) = \delta_{bb'} \delta_{ss'} \frac{E_p V}{m}. \quad \diamond \quad (15.31)$$

Hence, a general solution of the Dirac equation has the expansion

$$\begin{aligned} \Psi(\mathbf{x}, t) &= \sum_{ps} \sqrt{\frac{m}{VE_p}} \left[e^{i(\mathbf{p} \cdot \mathbf{x} - p_0 t)} u_s^{(+)}(p) b_{ps}^{(+)}(\mathbf{p}) + e^{-i(\mathbf{p} \cdot \mathbf{x} - p_0 t)} u_s^{(-)}(p) b_{ps}^{(-)}(\mathbf{p}) \right] \\ &= \sum_{ps} \sqrt{\frac{m}{VE_p}} \left[e^{-ip \cdot x} u_s^{(+)}(p) b_{ps}^{(+)}(\mathbf{p}) + e^{ip \cdot x} u_s^{(-)}(p) b_{ps}^{(-)}(\mathbf{p}) \right], \end{aligned} \quad (15.32)$$

where the b 's are the expansion coefficients determined by a projection of the p -space Dirac equation, and the u 's are 4×1 Dirac spinors.³ The internal space part of Ψ is thereby expanded in spinors with the functional dependence given by the exponentials; in other words, an eigenstate of the *full Hamiltonian* is expanded in eigenstates of the *free Hamiltonian*.

A natural question at this point concerns the need for the negative-energy states, particularly for solutions describing particles of nonrelativistic velocity. We repeat, the lower

³See also the Problems section and discussion in Chapter 17, *Integral Forms of the Dirac Equation, Scattering*.

components are needed for mathematical completeness. The Dirac equation is a 4D matrix equation that will always have solutions with some lower components; for the free Hamiltonian these lower components are negative-energy eigenstates, while, for an interacting particle, the wave functions can have lower components and still have positive-energy, for example, our solution for the hydrogen atom in Chapter 16, *Interactions in Dirac Theory*, has four components yet is a positive-energy eigenstate.

15.4 Gordon Decomposition of Current: Tutorial

The internal (spin) and external (charge) nature of the electron is also present in the expressions deduced for the density and current. To see this connection, work through this tutorial.

1. Consider first the probability density ρ .

(a) Show that ρ can be written

$$\rho = \frac{1}{2} \left[(\Psi^\dagger \beta) \beta \Psi + \Psi^\dagger \beta (\beta \Psi) \right]. \quad (15.33)$$

(b) Convert the Hamiltonian form of the Dirac equation into an equation for $\beta \Psi$:

$$\beta \Psi = \frac{1}{m} \left[\frac{i \partial \Psi}{\partial t} + i \alpha \cdot \nabla \Psi \right]. \quad (15.34)$$

(c) Convert the Dirac equation into an equation for $\Psi^\dagger$, with operators acting to the right:

$$\Psi^\dagger \beta = \frac{1}{m} \left[-i \frac{\partial \Psi^\dagger}{\partial t} - (i \nabla \Psi^\dagger) \cdot \alpha \right]. \quad (15.35)$$

(d) Show that substitution for $\beta \Psi$ and $\Psi^\dagger \beta$ yields

$$\rho = \frac{i}{2m} \left[\bar{\Psi} \frac{\partial \Psi}{\partial t} - \frac{\partial \bar{\Psi}}{\partial t} \Psi \right] + \frac{i}{2m} \left[\bar{\Psi} \alpha \cdot (\nabla \Psi) + (\nabla \bar{\Psi}) \cdot \alpha \Psi \right]. \quad (15.36)$$

(Hint: Be careful to indicate on what the ∇ 's operate.)

(e) Finally, show that for stationary states the probability density separates into the internal and convection parts:

$$\rho = \rho_{\text{conv}} + \rho_{\text{int}} \quad (15.37)$$

$$\rho_{\text{conv}} = \frac{E}{m} \bar{\Psi} \Psi, \quad \rho_{\text{int}} = \frac{i}{2m} \nabla \cdot (\bar{\Psi} \alpha \Psi). \quad (15.38)$$

2. Make the analogous decomposition of the 3-current (with constants!).

(a) Show that $\mathbf{j}$ separates as:

$$\mathbf{j} = \mathbf{j}_{\text{conv}} + \mathbf{j}_{\text{int}}, \quad (15.39)$$

$$\mathbf{j}_{\text{conv}} = \frac{1}{2im} \left[\bar{\Psi} \nabla \Psi - (\nabla \bar{\Psi}) \Psi \right] - \frac{e \mathbf{A}}{m} \bar{\Psi} \Psi, \quad (15.40)$$

$$\mathbf{j}_{\text{int}} = \nabla \times \mathbf{M} + \frac{\partial \mathbf{P}}{\partial t}, \quad (15.41)$$

$$\mathbf{M} = \frac{1}{2m} \bar{\Psi} \boldsymbol{\sigma} \Psi, \quad \mathbf{P} = \frac{1}{2m} \bar{\Psi} (-i \alpha) \Psi. \quad (15.42)$$

- (b) Hypothesize why it makes sense to identify $\mathbf{M}$ as a magnetization arising from the internal spin and $\mathbf{P}$ as an electric polarization.
3. Verify that the above decomposition is equivalent to the covariant form

$$\bar{\Psi}_1 \gamma^\mu \Psi_2 = \frac{1}{2m} [\bar{\Psi}_1 p^\mu \Psi_2 - (p^\mu \bar{\Psi}_1) \Psi_2 - i p_\nu (\bar{\Psi}_1 \sigma^{\mu\nu} \Psi_2)] . \quad (15.43)$$

(Hint: Either use the fact that Ψ satisfies the Dirac equation or explicitly verify (15.43) by substituting a plane wave.)

4. Show that for plane waves the current has the decomposition

$$\bar{u}(p') \gamma^\mu u(p) = \bar{u}(p') \left[\frac{(p + p')^\mu}{2m} + i \sigma^{\mu\nu} \frac{(p' - p)_\nu}{2m} \right] u(p) . \quad (15.44)$$

Here the internal and external or *convection* parts are separated.

15.5 Interactions and the Dirac Equation

Before accepting the Dirac equation, we need to test if it satisfies the correspondence principle, that is, if it reduces to the Schrödinger equation in the nonrelativistic limit even when interactions are included. We include interactions by invoking *minimal electromagnetic coupling*. As discussed in Chapter 13, *Relativistic Wave Equations for Spinless Particles*, this amounts to replacing the derivatives in the wave equation by the covariant derivatives,

$$\partial^\mu \rightarrow D^\mu \stackrel{\text{def}}{=} \partial^\mu + iqA^\mu, \quad (15.45)$$

or, equivalently, the 4-momentum operator p^μ by $p^\mu - qA^\mu$. There results the Dirac equations in Hamiltonian or covariant forms:

$$\left(i \frac{\partial}{\partial t} - q\phi \right) \Psi(\mathbf{x}, t) = [\boldsymbol{\alpha} \cdot (\mathbf{p} - q\mathbf{A}) + \beta m] \Psi(\mathbf{x}, t), \quad (15.46)$$

$$(i\gamma_\mu D^\mu - m)\Psi(x) \equiv (i\mathcal{D} - m)\Psi(x) = 0. \quad (15.47)$$

This equation incorporates the coupling of the electron's charge ($q = -e$) but no higher moments with the external field, and is gauge invariant under the set of transformations:

$$A^\mu \rightarrow A^\mu - \partial^\mu \chi(x), \quad \Psi(x) \rightarrow e^{iq\chi(x)} \Psi(x). \quad (15.48)$$

The Upper and Lower Split

Rather than solve (15.46) as four coupled differential equations, we illuminate its connection with Schrödinger theory by removing the time dependence associated with the rest energy m , and splitting Ψ into lower and upper components:

$$\Psi(\mathbf{x}, t) = e^{-imt} \begin{pmatrix} \psi_U(\mathbf{x}, t) \\ \psi_L(\mathbf{x}, t) \end{pmatrix} .$$

(15.49)

Here ψ_U and ψ_L are 2D (that is, 2×1 , or Pauli) spinors with the time dependence of the nonrest energy still remaining in them. If we substitute (15.49) into the Dirac equation

(15.46), we find that the α term couples ψ_U to ψ_L , and this results in the coupled, Pauli spinor equations,

$$i\frac{\partial\psi_U}{\partial t} = \boldsymbol{\sigma} \cdot (\mathbf{p} - q\mathbf{A}) \psi_L + q\phi\psi_U, \quad (15.50)$$

$$i\frac{\partial\psi_L}{\partial t} = \boldsymbol{\sigma} \cdot (\mathbf{p} - q\mathbf{A}) \psi_U + q\phi\psi_L - 2m\psi_L. \quad (15.51)$$

Nonrelativistic Limit, the Electron's Structure

The assumption of the time dependence (15.49) produces the asymmetrical $-2m$ term in (15.51). This has two important consequences. First, in the nonrelativistic limit $m \rightarrow \infty$, $\partial\psi_L/\partial t$ becomes exceptionally large; this leads to the image of an electron rapidly jumping back and forth between its positive- and negative-energy components (the so-called *Zitterbewegung*, or shaking motion). Second, if the kinetic energy is small relative to the rest mass, the lower component ψ_L is also small relative to the upper one. We see this by formally “solving” (15.51) for ψ_L :

$$\psi_L = \frac{\boldsymbol{\sigma} \cdot (\mathbf{p} - q\mathbf{A})}{2m} \psi_U - \frac{i(\partial/\partial t) - q\phi}{2m} \psi_L. \quad (15.52)$$

Exercise Show that solving (15.52) for ψ_L and then substituting that solution into the equation for ψ_U , yields the Klein–Gordon equation. $\diamond$

An approximation for the lower component is obtained by expanding it as a power series in the ratio of momentum to rest energy m :

$$\psi_L \simeq \psi_{L0}[O(v)^0] + \psi_{L1}[O(v)] + \cdots \quad (15.53)$$

$$\psi_{L0} \simeq 0, \quad \psi_{L1} \simeq \frac{\boldsymbol{\sigma} \cdot (\mathbf{p} - q\mathbf{A})}{2m} \psi_U. \quad (15.54)$$

Pauli equation (the e 's g)

We approximately uncouple ψ_U and ψ_L by inserting ψ_{L1} into (15.50):

$$i\frac{\partial\psi_{U_0}}{\partial t} \simeq q\phi\psi_{U_0} + \frac{[\boldsymbol{\sigma} \cdot (\mathbf{p} - q\mathbf{A})][\boldsymbol{\sigma} \cdot (\mathbf{p} - q\mathbf{A})]}{2m} \psi_{U_0}. \quad (15.55)$$

We use the identity

$$\boldsymbol{\sigma} \cdot \mathbf{A} \boldsymbol{\sigma} \cdot \mathbf{B} = \mathbf{A} \cdot \mathbf{B} + i\boldsymbol{\sigma} \cdot (\mathbf{A} \times \mathbf{B}) \quad (15.56)$$

to rewrite (15.55) as

$$i\frac{\partial\psi_{U_0}}{\partial t} \simeq \frac{(\mathbf{p} - q\mathbf{A})^2}{2m} \psi_{U_0} - \frac{q}{2m} \boldsymbol{\sigma} \cdot (\nabla \times \mathbf{A} + \mathbf{A} \times \nabla) \psi_{U_0} + q\phi\psi_{U_0}. \quad (15.57)$$

Since $\mathbf{p}$ is the operator $-i\nabla$ acting to the right, the $\nabla \times \mathbf{A}$ and $\mathbf{A} \times \nabla$ terms do *not* cancel. We evaluate the gradient terms using a vector identity and the relation of $\mathbf{A}$ to magnetic field:

$$\begin{aligned} (\nabla \times \mathbf{A} + \mathbf{A} \times \nabla) \psi_U &= \nabla \times (\mathbf{A} \psi_U) + \mathbf{A} \times \nabla \psi_U \\ &= \psi_U \nabla \times \mathbf{A} + (\nabla \psi_U) \times \mathbf{A} + \mathbf{A} \times \nabla \psi_U \\ &= \psi_U (\nabla \times \mathbf{A}) = \psi_U \mathbf{B}. \end{aligned} \quad (15.58)$$

We thereby obtain the *Pauli equation*,

$$i\frac{\partial\psi_{U_0}}{\partial t} \simeq H_P\psi_{U_0}, \quad (15.59)$$

which has the form of the Schrödinger equation but with the Pauli Hamiltonian:

$$H_P \stackrel{\text{def}}{=} \frac{(\mathbf{p} - q\mathbf{A})^2}{2m} - \frac{q}{2m} \boldsymbol{\sigma} \cdot \mathbf{B} + q\phi. \quad (15.60)$$

The Pauli equation was originally derived as the extension of the Schrödinger equation to include spin; to us it is the nonrelativistic limit of the Dirac equation with coupling to an external field. Lest we think the Dirac equation is really the “right” answer, we note that it too is only an approximation to the full interaction of two particles; in Chapter 23, *The Breit–Pauli and Meson-Exchange Interactions*, we discuss extensions to the Dirac equation known as the *Breit* interaction between two slow-moving electrons, and in Chapter 25, *Wave Equations from Field Theory*, the Bethe–Salpeter and Blankenbecler–Sugar equations for two relativistic electrons with retarded interactions.

While a solution of the Dirac equation automatically includes higher-order terms than those in the Pauli equation, we see that even in lowest order, the Dirac theory predicts the gyromagnetic ratio, or *g* factor, of the electron to be exactly 2. Specifically, knowing that a magnetic dipole’s interaction with an external magnetic field is

$$H' = -\boldsymbol{\mu} \cdot \mathbf{B}, \quad (15.61)$$

we identify this with the second term in the Pauli Hamiltonian (15.60), and deduce the electron’s magnetic dipole moment μ and *g* factor for its spin as

$$\boldsymbol{\mu} = -\mu_B g \mathbf{S} \equiv -\mu_B \boldsymbol{\sigma}, \quad (\mu_B = \frac{e}{2m}), \quad (15.62)$$

$$\Rightarrow g = g_D = 2, \quad (15.63)$$

where μ_B is the Bohr magneton for the electron. (The spin’s *g* should not be confused with the *g* factor for a charged particle’s orbital motion $g_l = 1$.) Equation (15.62) is a truly amazing result. We have not built in any magnetic moment into the equation (minimal coupling is for point particles) yet the equation predicts that every spin $\frac{1}{2}$ particle has $g = 2$. Apparently, the magnetic moment (usually associated with the finite size of a system) arises naturally from relativity or from the use of nonscalar wave functions.⁴

The prediction (15.63) agrees quite well with experiment for electrons and muons:⁵

$$\frac{g_e}{2} = 1.001159652193 \pm 1 \times 10^{-11}, \quad \frac{g_\mu}{2} = 1.001165923 \pm 8 \times 10^{-9}, \quad (15.64)$$

where the magnetic moment of a particle is measured in units of its own Bohr magneton [m is the mass of the particle in (15.62), for example, $m_\mu \simeq 205m_e$]. The deviation from 2 is due to quantum field-theory corrections, such as those in Figure 15.2. In contrast, the magnetic moments of the strongly interacting proton and neutron are quite different from the Dirac predictions of 2 and 0,

$$\frac{g_p}{2} = 2.7928474, \quad \frac{g_n}{2} = -1.9130427. \quad (15.65)$$

⁴Sakurai (1967).

⁵Particle Data Group (1994).

The deviations of magnetic moments from the Dirac value μ_D are characterized by an *anomalous moment* κ ,

$$\mu = \mu_D + \kappa \mu_B. \quad (15.66)$$

For nucleons, the anomalous moments arise from the strong interaction giving these hadrons a finite size. This anomalous magnetic interaction is included phenomenologically in Dirac theory by the addition of an explicit $-\kappa F_{\mu\nu} \sigma^{\mu\nu}/2$ term to the Hamiltonian. The moments of the nucleons can be explained, at least in part, with the quark model.

Spin-Orbit and Darwin Forces

We obtain corrections beyond the Pauli Hamiltonian by iteratively solving (15.52) [a systematic approach known as the Foldy–Wouthuysen transformation is described in Bjorken and Drell (1964)]. If we substitute ψ_{L_1} for ψ_L in (15.52), we obtain

$$\psi_{L_1} \simeq \frac{\boldsymbol{\sigma} \cdot [\mathbf{p} - q\mathbf{A}]}{2m} \psi_U - \frac{[i(\partial/\partial t) - q\phi][\boldsymbol{\sigma} \cdot (\mathbf{p} - q\mathbf{A})]}{2m} \psi_U. \quad (15.67)$$

We next substitute (15.67) into the equation for ψ_U , (15.50), and identify the RHS as $H\psi_U$. Before actually doing that, the wave function must be renormalized to make the Hamiltonian Hermitian and thereby produce the correct nonrelativistic limit. This complication is detailed in Bethe and Salpeter (1977) and Baym (1969), who find

$$\begin{aligned} H_0 \simeq & \left[\frac{(\mathbf{p} - q\mathbf{A})^2}{2m} - \frac{p^4}{8m^3} \right] + q\phi - \frac{q}{2m} \boldsymbol{\sigma} \cdot \mathbf{B} - \frac{iq}{8m^2} \mathbf{p} \cdot \mathbf{E} \\ & - \left[\frac{iq}{8m^2} \boldsymbol{\sigma} \cdot (\nabla \times \mathbf{E}) + \frac{q}{4m^2} \boldsymbol{\sigma} \cdot (\mathbf{E} \times \mathbf{p}) \right]. \end{aligned} \quad (15.68)$$

We recognize the second term in the first bracket as a relativistic correction to the kinetic energy

$$\sqrt{p^2 + m^2} - m \simeq \frac{p^2}{2m} - \frac{p^4}{8m^3} + \cdots. \quad (15.69)$$

This is familiar from atomic physics, where it lowers the levels slightly relative to their nonrelativistic values.

Exercise Show that the $\boldsymbol{\sigma} \cdot (\nabla \times \mathbf{E})$ term in (15.68) vanishes for a spherically symmetric potential. $\diamond$

The second bracket contains the *spin–orbit force* responsible for the fine structure of levels in atoms. To see that, we rewrite it as

$$\mathbf{H}_{\text{so}} = \frac{-q}{4m^2} \boldsymbol{\sigma} \cdot (\mathbf{E} \times \mathbf{p}) = \frac{q}{4m^2} \boldsymbol{\sigma} \cdot \frac{\partial V(r)}{\partial \mathbf{r}} \frac{\mathbf{r}}{r} \times \mathbf{p} \quad (15.70)$$

$$= \frac{q}{4m^2} \frac{1}{r} \frac{\partial V(r)}{\partial r} \boldsymbol{\sigma} \cdot \mathbf{l}. \quad (15.71)$$

Amazingly, (15.71) is the correct spin–orbit force including Thomas precession, that is, it is one half the naive result obtained by considering an electron at rest with its magnetic

moment interacting with the magnetic field of an orbiting proton.⁶ This precession, and more, is inherent in the Dirac equation.

The $\mathbf{p} \cdot \mathbf{E}$ term in the Hamiltonian (15.68) is known as the *Darwin term* and is related to the Laplacian of the central potential:

$$V_{\text{dar}} = \frac{-iq}{8m^2} \mathbf{p} \cdot \mathbf{E} = \frac{1}{8m^2} \nabla \cdot \nabla V(r) = \frac{1}{8m^2} \nabla^2 V(r). \quad (15.72)$$

Because $\nabla^2(1/r) \propto \delta(r)$, this is a contact interaction, and so mainly affects the S states in atoms. A $\nabla^2 V(r)$ term is suggestive of the variation in V as might be seen by an electron whose confinement to a bound orbit causes it to oscillate between positive- and negative-energy states. To carry this suggestion further, we imagine that this *Zitterbewegung* causes the electron to sample a region of space on the order of its Compton wavelength $\Delta r \simeq 1/m$ about the point r . Consequently, the Hamiltonian contains an extra term to account for this fluctuation:

$$H' \simeq \langle V(r + \Delta r) \rangle - \langle V(r) \rangle \quad (15.73)$$

$$\simeq \left\langle V(r) + \frac{\partial V}{\partial r} \Delta r + \frac{1}{2} \sum_{i,j} \Delta r_i \Delta r_j \frac{\partial^2 V}{\partial r_i \partial r_j} \right\rangle - \langle V(r) \rangle \quad (15.74)$$

$$\simeq \frac{1}{6} (\Delta r)^2 \nabla^2 V \simeq \frac{1}{6m^2} \nabla^2 V. \quad (15.75)$$

This in fact looks like (15.72).

15.6 Mass-Zero Dirac Equation: Tutorial

1. Recall Dirac's reasoning for his electron equation and proceed in a similar fashion to deduce a relativistic wave equation for a spin $\frac{1}{2}$ *massless* ($m = 0$) particle.
2. Write down your choice for the equation of motion stating clearly the conditions on any of the matrices that occur in it.
3. Show that either of the two equations

$$i \frac{\partial \Psi(\mathbf{x}, t)}{\partial t} = \pm i \boldsymbol{\sigma} \cdot \nabla \Psi(\mathbf{x}, t) \quad (15.76)$$

is a satisfactory equation with $\boldsymbol{\sigma}$ the 2×2 Pauli matrices.

4. Why was it not possible to use the Pauli matrices for $m \neq 0$?
5. Determine the plane-wave solutions of (15.76), and then use your Dirac equation to deduce the relation among energy, momentum, and velocity for them. Does it make sense?
6. Show that the positive- and negative-energy solutions for this equation are orthogonal, and that they have opposite eigenvalues for the helicity operator $\boldsymbol{\sigma} \cdot \hat{\mathbf{p}}$. Keep it simple by considering motion along only the z axis.

⁶A painfully (but necessarily) detailed, classical calculation is described by Jackson (1975) and Hagedorn (1963).

7. Use the hole interpretation of the negative-energy states and the fact that $\boldsymbol{\sigma} \cdot \hat{\mathbf{p}}$ is a pseudoscalar to show that the parity operator acting on these solutions produces new states that are no longer (consistent) solutions to the wave equation.
8. What then is the physical difference between positive- and negative-energy solutions for this case?
9. Which equation does the physical neutrino satisfy?

15.7 Problems

1. Verify or prove that the momentum-space Dirac spinors satisfy:
 - (a) $(\not{p} \mp m)u_s^\pm(\mathbf{p}) = 0$, (Dirac equation),
 - (b) $\bar{u}_s^\pm(\mathbf{p})(\not{p} \mp m) = 0$, (dual space Dirac equation),
 - (c) $\bar{u}_s^{(b)}(\mathbf{p})u_{s'}^{(b)'}(\mathbf{p}) = b\delta_{bb'}\delta_{ss'}$, (invariant normalization),
 - (d) $\bar{u}(\mathbf{p})\gamma u(\mathbf{p}) = \mathbf{p}\rho/E_p$, (current).
2. In relating field-theoretic amplitudes to nonrelativistic potentials, it will be necessary to know nonrelativistic (sometimes even "static") limits of the bilinear covariants. Determine the leading term in v for
 - (a) $\bar{u}_{s'}(\mathbf{p}')\gamma_0 u_s(\mathbf{p})$
 - (b) $\bar{u}_{s'}(\mathbf{p}')u_s(\mathbf{p})$
 - (c) $\bar{u}_{s'}(\mathbf{p} + \mathbf{q})\gamma_5 u_s(\mathbf{p})$
 - (d) $\bar{u}_{s'}(\mathbf{p}')\sigma^{\mu\nu} u_s(\mathbf{p})$
3. Try working through this problem on positive- and negative-frequency components without reference to the text.
 - (a) Write down the time-dependent Dirac equation for an electron *at rest* (make sure to explicitly indicate any derivatives).
 - (b) Solve for the two "positive frequency" solutions.
 - (c) Solve for the two "negative frequency" solutions.
 - (d) As indicated in the text, $\sigma^{12} \equiv \Sigma_3$ is the Dirac version of the operator for the 3 or z component of spin. Deduce which of the solutions of parts (b) and (c) are spin "up" and which are spin "down".
 - (e) An observer in a moving reference frame sees the electron of part (a) as having energy $E_p = \sqrt{p^2 + m^2}$. Use relativistic covariance to deduce what *must* be the time dependences that this observer observes for the four solutions of parts (b) and (c)?
4. An electron at time $t = 0$ is described by the normalized Dirac wave function

$$\Psi(\mathbf{x}, 0) = \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix} e^{i\mathbf{p}\cdot\mathbf{z}}, \quad (15.77)$$

where a , b , c , and d are independent of the spacetime coordinates. Calculate the probability of measuring this electron with

- (a) $E > 0$, spin up
- (b) $E > 0$, spin down
- (c) $E < 0$, spin up
- (d) $E < 0$, spin down

(Hint: This may not be as trivial as it seems at first.)

5. A time-independent scattering theory can be developed for a Dirac particle in a fixed potential much like we did for the Schrödinger equation.⁷

- (a) Verify that positive-energy spinors $u_s(\mathbf{p})$ satisfy

$$\sum_s u_s(\mathbf{p}) \bar{u}_s(\mathbf{p}) = \frac{\not{p} + m}{2m}. \quad (15.78)$$

- (b) We define free plane wave solutions

$$\Phi_{p,s}(\mathbf{x}) = C u_s(\mathbf{p}) e^{i\mathbf{p} \cdot \mathbf{x}}. \quad (15.79)$$

Determine C such that these solutions are normalized to $\delta(\mathbf{p} - \mathbf{p}')$.

- (c) A positive-energy 4-spinor $\Psi(\mathbf{x})$ is expanded in plane waves:

$$\Psi(\mathbf{x}) = \sum_s \int d^3p \Phi_{p,s}(\mathbf{x}) \hat{\Phi}_s(\mathbf{p}). \quad (15.80)$$

Invert this expansion to find the momentum-space wave function $\hat{\Phi}(\mathbf{p})$.

- (d) As is conventional for the Schrödinger equation, the full wave function is defined to be the sum of a plane plus outgoing, spherical wave:

$$\Psi_{p_0,s} = \Phi_{p_0,s}(\mathbf{x}) + \Psi'(\mathbf{x}). \quad (15.81)$$

Show that:

$$[H_0 - E(p_0)] \Psi'(\mathbf{x}) = -V \Psi_{p_0,s}(\mathbf{x}), \quad (15.82)$$

where H_0 is the free Dirac Hamiltonian, and V is the potential.

- (e) Show that in Born approximation the positive-energy part of Ψ' is

$$\begin{aligned} (2\pi)^{3/2} \Psi'(\mathbf{x}) &= - \sum_{s'} \int d^3p d^3x' e^{i\mathbf{p} \cdot (\mathbf{x} - \mathbf{x}')} e^{i\mathbf{p}_0 \cdot \mathbf{x}'} \\ &\times \left\{ \frac{m u_{s'}(\mathbf{p}) u_s^\dagger(\mathbf{p}) V(\mathbf{x}) u_s(\mathbf{p}_0) \sqrt{m}}{(2\pi)^3 E_p [E_p - E(p_0) - i\epsilon] \sqrt{E(p_0)}} \right\}. \end{aligned} \quad (15.83)$$

- (f) Does the negative-energy part contribute to the asymptotic wave function? If not, where does it contribute?

⁷ Adapted from class notes of J. Stack.

- (g) Explain why it makes sense to identify the Green's function as

$$G(\mathbf{x}, \mathbf{x}') = \sum_s \int d^3p \frac{m u_s(\mathbf{p}) u_s^\dagger(\mathbf{p})}{(2\pi)^3 E_p [E_p - E(\mathbf{p}_0) - i\epsilon]} e^{i\mathbf{p} \cdot (\mathbf{x} - \mathbf{x}')} \quad (15.84)$$

- (h) Show that for large $|\mathbf{x} - \mathbf{x}'|$,

$$G(\mathbf{x}, \mathbf{x}') \simeq \frac{[E(\mathbf{p}_0) - i\nabla \cdot \boldsymbol{\alpha} + m\beta] e^{ip_0|\mathbf{x} - \mathbf{x}'|}}{4\pi |\mathbf{x} - \mathbf{x}'|} \quad (15.85)$$

- (i) Finally, show that if the differential cross section is $|f_{s's}|^2$, the Born approximation for the scattering amplitude is:

$$f_{s's}(\mathbf{p}' - \mathbf{p}) = \frac{1}{4\pi} \int d^3x u_{s'}^\dagger(\mathbf{p}') [-2mV(\mathbf{x})] u_s(\mathbf{p}) e^{i(\mathbf{p} - \mathbf{p}') \cdot \mathbf{x}} \quad (15.86)$$

6. Consider the scattering of a Dirac electron from the Coulomb field of a very heavy nucleus.

- (a) Show that in lowest order in the fine-structure constant α , the differential cross section for the scattering of polarized electrons with momentum $\mathbf{p} \rightarrow \mathbf{p}' = \mathbf{p} + \mathbf{q}$ is

$$\frac{d\sigma}{d\Omega} = \frac{4Z^2\alpha^2 m^2 |\bar{u}_{s'}(\mathbf{p}') \gamma^0 u_s(\mathbf{p})|^2}{q^4} \quad (15.87)$$

- (b) Show that the nonrelativistic limit of this cross section is the familiar Rutherford scattering cross section.
 (c) Show that for the scattering of an unpolarized beam from the nucleus, the cross section for undetected final polarization is

$$\frac{d\sigma}{d\Omega} = \frac{Z^2\alpha^2(1 - \beta^2 \sin^2 \theta/2)}{4p^2\beta^2 \sin^4 \theta/2}, \quad (\beta = v). \quad (15.88)$$

7. Prove that spin is a good quantum number only in the rest frame of a particle or for motion along the spin quantization axis:

- (a) Examine solutions of the Dirac equation for a particle at rest and for one moving with $p_x = p_y = 0, p_z \neq 0$, and show that the spinors will be eigenstates of Σ_z .
 (b) Show and explain why the lower components are necessary to have these eigenstates.

8. Show that helicity ($\stackrel{\text{def}}{=} \boldsymbol{\sigma} \cdot \hat{\mathbf{p}}$) is a good quantum number for a free relativistic electron.

9. Insert the c 's and $\hbar$'s needed to make the plane-wave solutions dimensionally correct, and then check the dimensions of the orthogonality relation, the density, the current, and the projection operators.

CHAPTER 16

INTERACTIONS IN DIRAC THEORY

In this chapter we actually solve the Dirac equation when a potential acts. We start with the “simplifications” possible for a central potential that allow the equation to be “split” into two coupled, first-order differential equations. We then solve these equations for a potential approximating that in a hydrogen atom and find a solution that improves upon the Bohr result yet agrees with it in the nonrelativistic limit. We end the chapter by indicating some more general ways in which potentials couple into the Dirac equation and the type of physics they may contain. Many of the results of this chapter are used again in the next chapter where we describe scattering and the momentum-space Dirac equation.

16.1 Central-Force Problems

We wish to solve the Dirac equation exactly for an electron experiencing a central force (Bethe and Salpeter, 1977; Sakurai, 1967; Rose, 1961). We assume a general form for the Hamiltonian:

$$H = \alpha \cdot \mathbf{p} + \beta m + V(r). \quad (16.1)$$

From our study of the nonrelativistic limit, we know that this H contains the magnetic moment interaction of the electron, the spin–orbit force, relativistic kinematics, and of course a central force. However, the *constants of motion* may not be obvious, or even how to determine them. Fortunately we have conscientiously constructed a canonical Hamiltonian theory, and so we have the same rules as in Schrödinger theory. Because spin is included in H , we do not expect the orbital angular momentum $\mathbf{l}$ to be a good quantum number, and sure enough it does not commute with the Hamiltonian:

$$[H, l_1] = [\alpha \cdot \mathbf{p}, x_2 p_3 - x_3 p_2] \quad (16.2)$$

$$= \alpha_2 p_3 (p_2 x_2 - x_2 p_2) - \alpha_3 p_2 (p_3 x_3 - x_3 p_3) \quad (16.3)$$

$$= -i(\alpha_2 p_3 - \alpha_3 p_2) = -i(\boldsymbol{\alpha} \times \mathbf{p})_1, \quad (16.4)$$

$$\Rightarrow [H, \mathbf{l}] = -i\boldsymbol{\alpha} \times \mathbf{p}. \quad (16.5)$$

Likewise, the spin operator $\Sigma = (\sigma^{23}, \sigma^{32}, \sigma^{12})$ is not good:

$$[H, \Sigma_k] = 2i(\alpha \times \mathbf{p})_k \neq 0. \quad (16.6)$$

Exercise Verify (16.6). ◇

Constants of Motion

There are a number of constants which we shall find very useful in solving the hydrogen atom, some of which are different from the Schrödinger ones. The total angular momentum $\mathbf{j}$ is a constant of the motion because

$$[H, \mathbf{j}] = -i(\alpha \times \mathbf{p}) + i(\alpha \times \mathbf{p}) = 0. \quad (16.7)$$

Owing to the symmetry of the central-force problem, there is another operator, the Runge–Lenz or eccentricity vector,

$$\tilde{\kappa} = -\beta(\Sigma \cdot \mathbf{l} + 1) = -\beta(\Sigma \cdot \mathbf{j} - \frac{1}{2}), \quad (16.8)$$

with conserved eigenvalues (Thompson, 1994).

Exercise Show that $\tilde{\kappa}$ measures the alignment of the spin and angular momentum by showing that they are parallel or antiparallel for different signs of κ . ◇

Exercise Show that the eigenvalues of $\tilde{\kappa}$ and $\mathbf{j}$ are related by

$$\kappa^2 = j(j+1) + \frac{1}{4} \Rightarrow \kappa = \pm(j + \frac{1}{2}). \quad \diamond \quad (16.9)$$

Exercise Show that $\tilde{\kappa}$ is a good quantum number by showing

$$[H, \tilde{\kappa}] = 0. \quad \diamond \quad (16.10)$$

Form of Wave Function

In Chapter 9, *Spin Theory*, we found that the solution of the Schrödinger equation containing the interaction of a spin 0 with a spin $\frac{1}{2}$ particle separates into 1D radial equations if the wave function is an eigenstate of total angular momentum j and proportional to a spin-angle function $\mathcal{Y}_{jm}^{l=j \mp \frac{1}{2}}$. We generalize that result to include the extra degree of freedom associated with the negative-energy components of Dirac wave functions by assuming

$$\Psi_{jm}^{4 \times 1}(\mathbf{r}) = \begin{pmatrix} \psi_U^{jm} \\ \psi_L^{jm} \end{pmatrix}, \quad (16.11)$$

where the upper and lower components ψ_U and ψ_L contain Pauli spinors. Because a state of total angular momentum j and spin $\frac{1}{2}$ contains orbital angular momenta $l = j - \frac{1}{2}$ and $l' = j + \frac{1}{2}$, the solution is further delineated as

$$\Psi_{jm}^{4 \times 1}(\mathbf{r}) = \frac{1}{r} \begin{pmatrix} iG_{lj}^{(+)}(kr)\mathcal{Y}_{jm}^l(\sigma, \Omega) + iG_{l'j}^{(-)}(kr)\mathcal{Y}_{jm}^{l'}(\sigma, \Omega) \\ F_{lj}^{(+)}(kr)\mathcal{Y}_{jm}^l(\sigma, \Omega) + F_{l'j}^{(-)}(kr)\mathcal{Y}_{jm}^{l'}(\sigma, \Omega) \end{pmatrix}, \quad (16.12)$$

Table 16.1: Quantum Numbers of Radial Wave Functions

$\kappa = \pm(j + \frac{1}{2})$	$l = l_U$	$l' = l_L$
$j + \frac{1}{2} = l$	$j + \frac{1}{2}$	$j - \frac{1}{2}$
$-(j + \frac{1}{2}) = -(l + 1)$	$j - \frac{1}{2}$	$j + \frac{1}{2}$

where

$$(+) \Rightarrow j = l - \frac{1}{2} \quad (j = l + \frac{1}{2}), \quad (-) \Rightarrow l' = j + \frac{1}{2} \quad (j = l' - \frac{1}{2}). \quad (16.13)$$

The solution simplifies if we assume total parity is a good quantum number. Because the orbital angular momentum states have external parity $(-1)^l$, and $|l - l'| = 1$, the upper and lower parts must each contain only one (possibly different) l value. We therefore write the wave function as:

$$\Psi_{jm}^{4 \times 1}(\mathbf{r}) = \frac{1}{r} \begin{pmatrix} iG_{lj}^{(\pm)}(kr) \mathcal{Y}_{jm}^l(\sigma, \Omega) \\ F_{l'j}^{(\pm)}(kr) \mathcal{Y}_{jm}^{l'}(\sigma, \Omega) \end{pmatrix}, \quad (16.14)$$

where there is still the choice to use the $j = l \pm \frac{1}{2}$ solution. And now, because the internal space's separate parity operator is β (§ 14.5),

$$\mathcal{P} = \beta = \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix}, \quad (16.15)$$

the upper and lower parts have opposite *intrinsic parity*. This means that if the full-wave function has just one parity, the lower part must have an orbital angular momentum value l' differing by an odd integer from the upper part's l . With (16.13) permitting 0 or 1, we choose $l' = l - 1$. An elegant way to build this into our wave function is to use the identity

$$\mathcal{Y}_{jm}^{l=j-\frac{1}{2}} \propto \frac{\boldsymbol{\sigma} \cdot \mathbf{r}}{r} \mathcal{Y}_{jm}^{l=j+\frac{1}{2}}, \quad (16.16)$$

and to remember that because $\boldsymbol{\sigma} \cdot \mathbf{r}$ is a pseudoscalar, $\boldsymbol{\sigma} \cdot \mathbf{r} \mathcal{Y}_{jm}^l$ has the opposite parity eigenvalue from $\mathcal{Y}_{jm}^l$. The total wave function for the central-force problem is accordingly

$$\boxed{\Psi_{jm}^{4 \times 1}(\mathbf{r}) = \frac{1}{r} \begin{pmatrix} iG_{lj}^{(\pm)}(kr) \mathcal{Y}_{jm}^l(\sigma, \Omega) \\ F_{lj}^{(\pm)}(kr) \boldsymbol{\sigma} \cdot \hat{\mathbf{r}} \mathcal{Y}_{jm}^l(\sigma, \Omega) \end{pmatrix}}, \quad (16.17)$$

where l is the orbital angular momentum of the upper part (since l is not a good quantum number, having different l 's mix in the solution is quite acceptable). The assignment of quantum numbers for the different parts of the wave function are given in Table 16.1.

Coupled Radial Equations

In realistic two-body systems such as hydrogen, the bodies move about their center of momentum. In the Schrödinger two-body problem, their relative motion separates from

the motion of their CM, and we solve for an effective particle of reduced mass μ moving in an external potential. Because the Dirac equation is a single-particle equation (two-particle relativistic generalizations such as the those in Chapter 25, *Wave Equations from Field Theory*, do not separate), we want it to have the correct nonrelativistic limit, so we *choose* the mass in the Dirac equation as the reduced mass $\mu = m_e m_p / (m_p + m_e)$,¹ and the potential as the fourth component of a 4-vector,

$$V(r) = qA^0 = -e\phi. \quad (16.18)$$

The Dirac equation is thus

$$[\boldsymbol{\alpha} \cdot \mathbf{p} + \beta\mu] \Psi(\mathbf{r}, t) = \left[i \frac{\partial}{\partial t} - V(r) \right] \Psi(\mathbf{x}, t). \quad (16.19)$$

When we assume the upper and lower parts have the same time dependence,

$$\Psi(\mathbf{x}, t) = e^{-iEt} \Psi(\mathbf{x}), \quad (16.20)$$

we obtain:

$$\begin{pmatrix} 0 & \boldsymbol{\sigma} \cdot \mathbf{p} \\ \boldsymbol{\sigma} \cdot \mathbf{p} & 0 \end{pmatrix} \Psi + \mu \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \Psi + V\Psi = E\Psi, \quad (16.21)$$

where $\boldsymbol{\sigma}$ is a 2×2 Pauli matrix and $\mathbf{p}$ is the momentum operator $-i\nabla$. If we substitute the split, central-force solution (16.17) into (16.21), we obtain the two 2D equations:

$$\boldsymbol{\sigma} \cdot \mathbf{p} \frac{F_{lj}}{r} \boldsymbol{\sigma} \cdot \hat{\mathbf{r}} \mathcal{Y}_{jm}^l + \mu i \frac{G_{lj}}{r} \mathcal{Y}_{jm}^l + V i \frac{G_{lj}}{r} \mathcal{Y}_{jm}^l = E i \frac{G_{lj}}{r} \mathcal{Y}_l, \quad (16.22)$$

$$\boldsymbol{\sigma} \cdot \mathbf{p} i \frac{G_{lj}}{r} \mathcal{Y}_{jm}^l - \mu \frac{F_{lj}}{r} \boldsymbol{\sigma} \cdot \hat{\mathbf{r}} \mathcal{Y}_{jm}^l - V \frac{F_{lj}}{r} \boldsymbol{\sigma} \cdot \hat{\mathbf{r}} \mathcal{Y}_{jm}^l = E \frac{F_{lj}}{r} \boldsymbol{\sigma} \cdot \hat{\mathbf{r}} \mathcal{Y}_{jm}^l. \quad (16.23)$$

We simplify (16.23) by finding the proportionality constant between $\boldsymbol{\sigma} \cdot \mathbf{p}$ and $\boldsymbol{\sigma} \cdot \mathbf{r}$, in which case all the $\boldsymbol{\sigma} \cdot \mathbf{r}$'s can be factored out and canceled:

$$\boldsymbol{\sigma} \cdot \mathbf{p} \frac{G}{r} \mathcal{Y}_{jm}^l \equiv \frac{\boldsymbol{\sigma} \cdot \mathbf{r}}{r^2} [\boldsymbol{\sigma} \cdot \mathbf{r} \boldsymbol{\sigma} \cdot \mathbf{p}] \frac{G}{r} \mathcal{Y}_{jm}^l = \frac{\boldsymbol{\sigma} \cdot \mathbf{r}}{r^2} \left[\frac{r}{i} \frac{\partial}{\partial r} + i\boldsymbol{\sigma} \cdot \mathbf{l} \right] \frac{G}{r} \mathcal{Y}_{jm}^l, \quad (16.24)$$

where we obtained the second line via the familiar identity

$$\boldsymbol{\sigma} \cdot \mathbf{A} \boldsymbol{\sigma} \cdot \mathbf{B} = \mathbf{A} \cdot \mathbf{B} + i\boldsymbol{\sigma} \cdot (\mathbf{A} \times \mathbf{B}). \quad (16.25)$$

We reduce (16.23) further by noting that $l = j \mp \frac{1}{2}$ for fixed j , yet since the constant of the motion κ (16.8) has the values $\kappa = \pm(j + \frac{1}{2})$ (see Table 16.1), we must have

$$\boldsymbol{\sigma} \cdot \mathbf{l} = \left(j - \frac{1}{2} \right) = -(1 + \kappa). \quad (16.26)$$

Finally, we have the proportionality between $\boldsymbol{\sigma} \cdot \mathbf{p}$ and $\boldsymbol{\sigma} \cdot \mathbf{r}$:

$$\boldsymbol{\sigma} \cdot \mathbf{p} \frac{G}{r} \mathcal{Y}_{jm}^l = \left[\frac{d(G/r)}{dr} - i(1 + \kappa) \frac{G}{r^2} \right] \boldsymbol{\sigma} \cdot \hat{\mathbf{r}} \mathcal{Y}_{jm}^l. \quad (16.27)$$

¹For scattering with large kinetic energies, it would be better still to use $\mu = E_e E_p / (E_e + E_p)$ since this produces the correct relativistic flux factors.

Now $\sigma \cdot \mathbf{r} \mathcal{Y}_{jm}^l$ is factored out and canceled, leaving

$$\frac{dG_{lj}(kr)}{dr} + \frac{\kappa}{r} G_{lj}(kr) = [E + \mu - V] F_{lj}(kr). \quad (16.28)$$

The other radial equation (16.22) is reduced to a simple differential equation by systematically eliminating all Pauli operators:²

$$\sigma \cdot \mathbf{p} \frac{F}{r} \sigma \cdot \hat{\mathbf{r}} \mathcal{Y}_{jm}^l = \sigma \cdot \mathbf{p} \sigma \cdot \hat{\mathbf{r}} \frac{F}{r} \mathcal{Y}_{jm}^l = \sigma \cdot \mathbf{p} \sigma \cdot \mathbf{r} \frac{F}{r^2} \mathcal{Y}_{jm}^l \quad (16.29)$$

$$= [\mathbf{p} \cdot \mathbf{r} + i\sigma \cdot \mathbf{p} \times \mathbf{r}] \frac{F}{r^2} \mathcal{Y}_{jm}^l. \quad (16.30)$$

Yet from simple analysis we know:

$$\mathbf{p} \cdot \mathbf{r} = \mathbf{r} \cdot \mathbf{p} - i\nabla \cdot \mathbf{r} = -ir\partial_r - 3i, \quad \mathbf{p} \times \mathbf{r} = -i\mathbf{l}, \quad (16.31)$$

$$\Rightarrow \sigma \cdot \mathbf{p} \frac{F}{r} \sigma \cdot \hat{\mathbf{r}} \mathcal{Y}_{jm}^l = [-ir\partial_r - 3i - i\sigma \cdot \mathbf{l}] \frac{F}{r^2} \mathcal{Y}_{jm}^l \quad (16.32)$$

$$= -\frac{i}{r} \left[\frac{\partial F}{\partial r} - \kappa \frac{F}{r} \right] \mathcal{Y}_{jm}^l. \quad (16.33)$$

The two radial equations are (at last):

$$-\frac{dF_{lj}(kr)}{dr} + \frac{\kappa}{r} F_{lj}(kr) = [E - \mu - V] G_{lj}(kr), \quad (16.34)$$

$$\frac{dG_{lj}(kr)}{dr} + \frac{\kappa}{r} G_{lj}(kr) = [E + \mu - V] F_{lj}(kr). \quad (16.35)$$

The $E - \mu$ in (16.34) is the familiar binding energy $-E_B$ (negative for bound states) and $V(r)$ the timelike component of a 4-vector potential. These equations are highly symmetrical, with the main difference being that G (which is “large” in the nonrelativistic limit) is multiplied by the small number $(E - \mu - V)$, while F (small) is multiplied by a large number $(E + \mu - V)$; this is as it should be to make this an equality.

16.2 Hydrogen Atom

In a hydrogen atom, $V(r)$ is the Coulomb potential between an electron and a proton.³ The potential is the fourth component (not “scalar”) potential,

$$V(r) = -e\phi = \frac{-Ze^2}{r} = \frac{-Z\alpha}{r}, \quad (16.36)$$

with α the fine structure constant. Equations (16.34)–(16.35) are solved with the conventional techniques used for the nonrelativistic hydrogen atom; we outline them in what follows.

²We thank V. A. Madsen for this clear reduction.

³Scattering solutions for the Dirac equation are in Chapter 17.

First we make the equations *dimensionless* by a change of variables:

$$\frac{dG(\rho)}{d\rho} + \frac{\kappa}{\rho}G - \left(\sqrt{\frac{\epsilon_1}{\epsilon_2}} + Z\frac{\alpha}{\rho} \right) F = 0, \quad (16.37)$$

$$\frac{dF(\rho)}{d\rho} - \frac{\kappa}{\rho}F - \left(\sqrt{\frac{\epsilon_2}{\epsilon_1}} - Z\frac{\alpha}{\rho} \right) G = 0, \quad (16.38)$$

$$\rho = \sqrt{\epsilon_1 \epsilon_2} \tau, \quad \epsilon_{1,2} = \mu \pm E. \quad (16.39)$$

We deduce the asymptotic forms of these equations by letting $\rho \rightarrow \infty$:

$$\frac{dG}{d\rho} - \sqrt{\frac{\epsilon_1}{\epsilon_2}} F = 0, \quad \frac{dF}{d\rho} - \sqrt{\frac{\epsilon_2}{\epsilon_1}} G = 0. \quad (16.40)$$

Taking the derivative of (16.40) with respect to ρ and substituting it back into (16.37) yields:

$$\frac{d^2G}{d\rho^2} - \sqrt{\frac{\epsilon_1}{\epsilon_2}} \frac{dF}{d\rho} = 0, \quad (16.41)$$

$$\Rightarrow \frac{d^2G}{d\rho^2} - \sqrt{\frac{\epsilon_1}{\epsilon_2}} \sqrt{\frac{\epsilon_2}{\epsilon_1}} G = 0, \quad (16.42)$$

$$\Rightarrow \frac{d^2G}{d\rho^2} = G, \quad G \simeq e^{-\rho}. \quad (16.43)$$

Likewise, $F \simeq e^{-\rho}$. For $\rho \rightarrow 0$, these equations have the limiting forms:

$$\frac{dG(\rho)}{d\rho} + \frac{\kappa}{\rho}G - Z\frac{\alpha}{\rho}F = 0, \quad (16.44)$$

$$\frac{dF(\rho)}{d\rho} - \frac{\kappa}{\rho}F - Z\frac{\alpha}{\rho}G = 0. \quad (16.45)$$

Substitution of the same power for both F and G ,

$$F \simeq a_0 \rho^s, \quad G \simeq b_0 \rho^s, \quad (16.46)$$

determines the relation between a_0 and b_0 , and the two possible s values:

$$a_0 = \frac{-Z\alpha}{s - \kappa} b_0, \quad s = \pm \sqrt{\kappa^2 - (Z\alpha)^2}. \quad (16.47)$$

Generally, $Z\alpha$ is a small number whereas $\kappa^2 = (j + \frac{1}{2})^2 \geq 1$, so the two possible powers are $s \simeq \pm 1$.⁴ We demand the integrated probability be finite,

$$\int d^3x \Psi^\dagger \Psi = 1, \quad (16.48)$$

which implies Ψ is less singular than $r^{-\frac{1}{2}}$ at the origin. Yet since $\Psi \propto \{F, G\}/r$, only the positive s is allowed, so

$$s = \sqrt{\kappa^2 - (Z\alpha)^2}. \quad (16.49)$$

⁴Exceptions occur. For example, $Z\alpha > 1$ is possible in the interaction of two heavy ions, in which case imaginary s in (16.49) is also possible. This is a situation like that found in Klein's paradox in which the Coulomb potential has become strong enough to produce electron-positron pairs.

Note that because $\kappa = 1$ is possible, the wave function of the Dirac hydrogen atom diverges at the origin, even though the integrated probability remains finite.

We now assume the upper and lower parts are power series multiplied by the $\rho = 0$ and $\rho = \infty$ behaviors:

$$G = e^{-\rho} \rho^s \sum_{m=0}^{\infty} b_m \rho^m, \quad F = e^{-\rho} \rho^s \sum_{m=0}^{\infty} a_m \rho^m. \quad (16.50)$$

Substitution determines the recursion relations among the power series coefficients:

$$(s + m - \kappa)a_m - a_{m-1} + Z\alpha b_m - \sqrt{\epsilon_2/\epsilon_1} b_{m-1} = 0, \quad (16.51)$$

$$(s + m + \kappa)b_m - b_{m-1} - Z\alpha a_m - \sqrt{\epsilon_1/\epsilon_2} a_{m-1} = 0. \quad (16.52)$$

Exercise Derive (16.51)–(16.52). $\diamond$

A full solution would start with an arbitrary value of b_0 [to be determined eventually by the normalization (16.48)], a determination of a_0 from (16.47), and then use of the recursion relations (16.51)–(16.52) to determine all terms in the series needed for the desired precision.

While there are situations in which the infinite series must actually be summed, for the hydrogen bound state we would run into a problem because the large- m limit of the recurrence relations,

$$a_m \simeq \frac{a_{m-1}}{m}, \quad b_m \simeq \frac{b_{m-1}}{m}, \quad (16.53)$$

is that of the exponential $e^{+\rho}$. Accordingly, if the power series continued out to $m = \infty$, the total solution would diverge as $\rho \rightarrow \infty$ and consequently would not be normalizable. Yet because we want the bound-state solution and because bound states must be normalizable, both series must terminate at some value, say $m = n' + 1$ (we assume both F and G terminate at the same m). This requires

$$a_{n'+1} = b_{n'+1} = 0 \Rightarrow a_{n'} = -\sqrt{\frac{\epsilon_2}{\epsilon_1}} b_{n'}, \quad (m = n' + 1). \quad (16.54)$$

Eigenenergy

If we combine (16.54) with the recursion relations for $m = n'$, we obtain a single equation involving the unknown n' :

$$2(s + n') + Z\alpha \frac{\epsilon_2 - \epsilon_1}{\sqrt{\epsilon_1 \epsilon_2}} = 0. \quad (16.55)$$

Exercise Derive (16.55). (*Hint:* Multiply one of the recursion relations by a single factor permitting the $a_{n'-1}$ and $b_{n'-1}$ terms to cancel). $\diamond$

Equation (16.55) is the quantum condition; n' is an integer, s is a constant, (16.49), which depends on the angular momentum, and the ϵ depends on the energy via (16.39). If we

substitute for all these constants, we obtain

$$2 \left(\sqrt{\kappa^2 - (Z\alpha)^2} + n' \right) + Z\alpha \frac{\mu - E - \mu - E}{\sqrt{(\mu - E)(\mu + E)}} = 0, \quad (16.56)$$

which we solve for the total energy

$$E_{n',j} = \mu \left[1 + \left(\frac{Z\alpha}{n' + \sqrt{(j + \frac{1}{2})^2 - (Z\alpha)^2}} \right)^2 \right]^{-1/2}. \quad (16.57)$$

We see that in order for (16.57) to yield Bohr's result with n as the *principal quantum number*, we need assign

$$n = n' + (j + \frac{1}{2}) = n' + |\kappa|. \quad (16.58)$$

The energy then takes the form

$$E_{n,j} = \mu \left[1 + \left(\frac{Z\alpha}{n - (j + \frac{1}{2}) + \sqrt{(j + \frac{1}{2})^2 - (Z\alpha)^2}} \right)^2 \right]^{-1/2}. \quad (16.59)$$

The relations of these quantum numbers to the spectroscopic ones and to the energy are given in Table 16.2.

Exercise Show that the nonrelativistic limit of (16.59) is

$$E_{n,j} \simeq \mu \left\{ 1 - \frac{Z^2\alpha^2}{2n^2} \left[1 + \frac{Z^2\alpha^2}{n} \left(\frac{1}{j + \frac{1}{2}} - \frac{3}{4n} \right) \right] + \dots \right\}. \quad \diamond \quad (16.60)$$

Curiously, this expression for the Dirac hydrogen energy is the same as the Klein-Gordon expression with l replaced by j . As is clear in the expansion, from the level diagram in Figure 16.1, and from Table 16.2, the energy depends on the principal quantum number n and the total angular momentum $j = l \pm \frac{1}{2}$, but not l . The fine-structure splitting is included properly (higher j less bound). Furthermore, the degeneracy of the Schrödinger equation for different l 's is removed, while the degeneracy of levels with same principal shell (n) and different l 's remain; for example, the $2S_{1/2}$ and $2P_{1/2}$ levels are degenerate.⁵

As is evident from Figure 16.1, all the lines in the experimental hydrogen spectrum are closely spaced doublets; the splitting is smaller than the fine structure and is called *hyperfine structure*. The doubling is due to the dipole-dipole interaction between the magnetic moments of the electron and proton. The small value of the splitting arises from the small value of the proton's magnetic moment (nuclear magnetons e/m_p with $m_p \simeq 2000m_e$ are small). Whereas the fine-structure splitting is 11,000 MHz (69×10^{-7} eV), the hyperfine splitting is only 1420 MHz (9×10^{-7} eV), a small value relative to the

⁵As discussed in Chapter 21, *Applications of Nonrelativistic QED*, this degeneracy is removed by the coupling to the radiation field known as the Lamb shift.

Table 16.2: Quantum Numbers for Hydrogen Energy Levels

nL_j	n	κ	n'	j	L	$E_{n\,2j}$
$1S_{1/2}$	1	-1	0	$1/2$	0	E_{11}
$2S_{1/2}$	2	-1	1	$1/2$	0	E_{21}
$2P_{1/2}$	2	+1	1	$1/2$	1	E_{21}
$2P_{3/2}$	2	-2	0	$3/2$	1	E_{23}
$3S_{1/2}$	3	-1	2	$1/2$	0	E_{31}
$3P_{1/2}$	3	+1	2	$1/2$	1	E_{31}
$3P_{3/2}$	3	-2	1	$3/2$	1	E_{33}
$3D_{3/2}$	3	+2	1	$3/2$	2	E_{33}

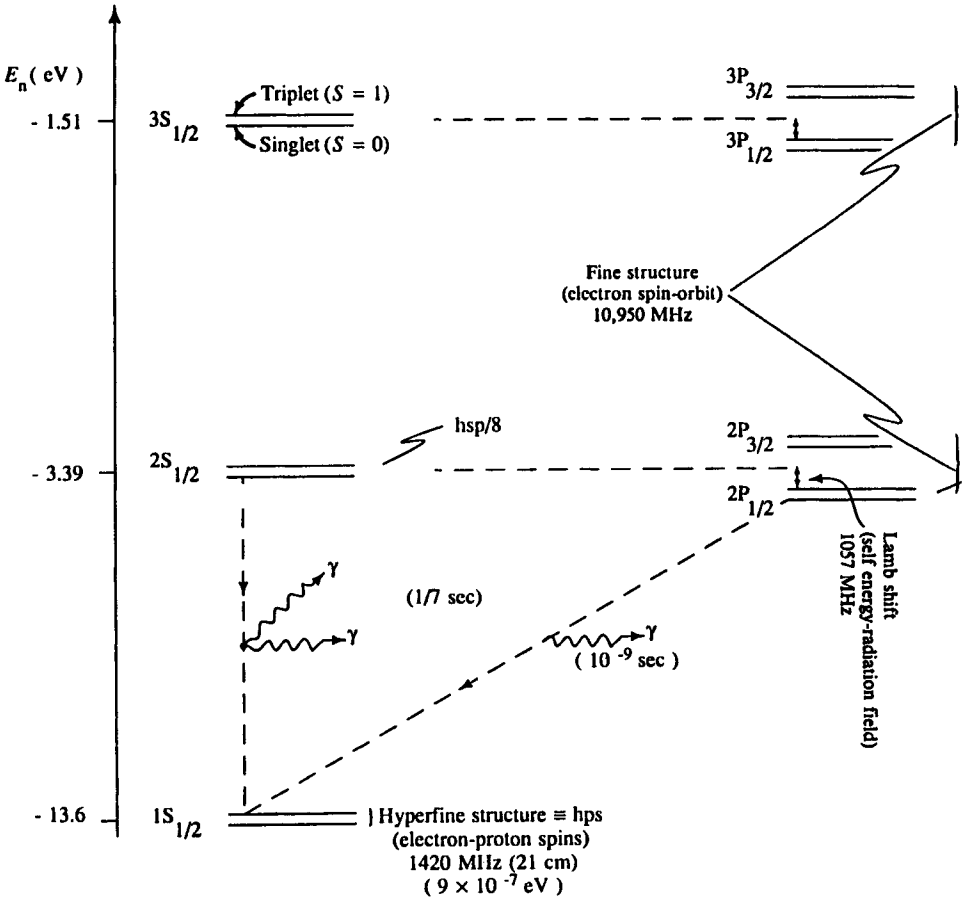


Figure 16.1: The energy levels of atomic hydrogen (not to scale). They are labeled by the quantum numbers of the large component. (Adapted from Bjorken and Drell, 1964).

principal splitting of $13.6 \text{ eV}/n^2$, yet important in applications (1420 MHz corresponds to the 21-cm hydrogen line of key importance to radio astronomy). Because the hyperfine interaction is a magnetic dipole–dipole interaction, in lowest order it is proportional to $\mathbf{S}_e \cdot \mathbf{S}_p$ and thus has two values depending on whether the total ep spin is 0 or 1. Because the triplet state with spins aligned would have the magnetic moments of the oppositely charged particles in opposition ($\uparrow\downarrow$), this is a repulsive configuration and is the less-bound one.

16.3 General Force Problem $\odot$

So far we have shown how to solve the Dirac equation for potentials, such as the Coulomb potential, that transform like the time component of a 4-vector. The same minimal coupling principle (15.46) could also have been used to include the spacelike parts **A**. More general (but still local) potential couplings produce a Dirac equation with the Hamiltonian form:⁶

$$i \frac{\partial \Psi(\mathbf{r}, t)}{\partial t} = (\boldsymbol{\alpha} \cdot \nabla / i + \beta [\mu + V^S(\mathbf{r}) + \gamma^5 V^{PS}(\mathbf{r}) + \gamma^\mu V_\mu^V(\mathbf{r}) + \gamma^\mu \gamma^5 V_\mu^A(\mathbf{r}) + \sigma^{\mu\nu} V_{\mu\nu}^T(\mathbf{r})]) \Psi(\mathbf{r}, t). \quad (16.61)$$

Here the superscripts (S, V, PS, A, T) indicates the transformation nature of the potential (scalar, vector, pseudoscalar, axial vector, and tensor), as discussed in § 14.5. We are already familiar with the vector potential from the electromagnetic interaction, and in Chapter 23, *The Breit–Pauli and Meson-Exchange Interactions*, we shall see that these other potentials may arise when a boson is exchanged between two particles. Physically, scalar and pseudoscalar potentials couple to mass, vector and axial vector potentials to 4-momenta, and tensor potentials to magnetic moments. Because an interaction including all these couplings would not conserve parity, we make our presentation more realistic and simpler by choosing to eliminate the pseudo couplings:

$$i \frac{\partial \Psi(\mathbf{r}, t)}{\partial t} = (\boldsymbol{\alpha} \cdot \nabla / i + \beta [\mu + V^S(\mathbf{r}) + \gamma^\mu V_\mu^V(\mathbf{r}) + \sigma^{\mu\nu} V_{\mu\nu}^T(\mathbf{r})]) \Psi(\mathbf{r}, t). \quad (16.62)$$

Equivalent Schrödinger Potential

While in some sense the simplest thing to do when faced with this complicated equation is to just solve it, we illuminate some of its physics by converting it to a Schrödinger-like equation (analogous to our study of the Pauli equation). For spherically symmetric potentials (which is not the same as central),

$$V^S(\mathbf{r}) = V^S(r), \quad (16.63)$$

$$\gamma^\mu V_\mu^V(\mathbf{r}) = \gamma^0 V_0^V(r) - \boldsymbol{\gamma} \cdot \hat{\mathbf{r}} V_r^V(r), \quad (16.64)$$

$$\sigma^{\mu\nu} V_{\mu\nu}^T(\mathbf{r}) = -\gamma^0 \boldsymbol{\gamma} \cdot \hat{\mathbf{r}} V_r^T(r) = \beta i \boldsymbol{\alpha} \cdot \hat{\mathbf{r}} \beta V_r^T(r), \quad (16.65)$$

where the subscript r indicates the radial part of the spacelike components and $\boldsymbol{\gamma} \cdot \hat{\mathbf{r}}$ is the component of the γ matrix in the radial direction. The stationary-state Dirac equation

⁶Further discussion of solutions to the Dirac equation with these couplings and their phenomenological effects is found in Clark et al. (1985).

(16.62) is then

$$[\alpha \cdot \nabla / i + \beta(\mu + V^S) - (E - V_0^V - q\phi) - \beta\gamma \cdot \hat{\mathbf{r}} V_r^V + i\alpha \cdot \hat{\mathbf{r}} \beta V^T] \Psi(\mathbf{r}) = 0, \quad (16.66)$$

where the potentials are functions of $\mathbf{r}$ and the Coulomb potential $q\phi$ is included for completeness.

Exercise Show that if the split-wave function (16.12) is inserted into (16.66), the upper and lower Pauli spinors are related by

$$\psi_L(\mathbf{r}) = \frac{-i\sigma \cdot \nabla - \sigma \cdot \hat{\mathbf{r}} V_r^V + i\sigma \cdot \hat{\mathbf{r}} V^T}{\Phi(\mathbf{r})} \psi_U(\mathbf{r}), \quad (16.67)$$

$$\Phi(\mathbf{r}) = \mu + V^S + E - V_0^V - q\phi. \quad \diamond \quad (16.68)$$

The second-order differential equation for ψ_U obtained by substituting ψ_L back into the coupled Dirac equations (15.50) and (15.51) is complicated:

$$[\nabla^2 + \{(E - V_0^V - q\phi)^2 - (m + V^S)^2 - (V_r^V)^2 - T^2 + (iV_r^V + T)(\Phi' - 2/r)\} + \{V_{so}\sigma \cdot \mathbf{l} + V_{dar}\}] \psi_U(\mathbf{r}) = 0, \quad (16.69)$$

$$\frac{1}{\Phi} \frac{\partial \Phi}{\partial r} \stackrel{\text{def}}{=} \Phi'. \quad (16.70)$$

The part in the first braces (which resembles the Klein–Gordon equation) has contributions to the central potential from all the components of the 4-vector potential, from the Coulomb potential,⁷ from the scalar potential, and from a new tensor potential,

$$T(\mathbf{r}) \equiv V^T(\mathbf{r}) + V_{anm}(\mathbf{r}). \quad (16.71)$$

We see that T is the sum of the input tensor potential V^T plus an induced one V_{anm} arising from the interaction of the anomalous magnetic moment (15.66) of the projectile with the Coulomb field of the target,

$$V_{anm}(\mathbf{r}) = -\frac{q\kappa}{2m} \frac{\partial \phi(\mathbf{r})}{\partial r}. \quad (16.72)$$

Here κ is the anomalous magnetic moment of the projectile (κ is 1.79 for protons, -1.91 for neutrons, 1.00116 for electrons, and 1.00117 for muons). The second brace in (16.69) contains a spin–orbit potential,

$$V_{so}(\mathbf{r}) = \frac{\Phi}{r} - 2\frac{T}{r}, \quad (16.73)$$

which has contributions from the derivatives of the scalar and zeroth component potentials plus a direct tensor contribution. The second brace in (16.69) also contains the uniquely

⁷We assume the target is charged but produces no electromagnetic vector potential $\mathbf{A}$. The interaction of the beam fermion's magnetic moment with the magnetic field arising from the motion of the target is included properly.

relativistic Darwin potential [see (15.72)], which here is related to the Laplacian of the central potential,

$$V_{\text{dar}} = \frac{1}{r} \left(\mathbf{r} \cdot \frac{\nabla}{i} \right) (V_r^V - iT) - \left(i\Phi' - 2 \frac{V_r^V}{r} \right) \left(\mathbf{r} \cdot \frac{\nabla}{i} \right). \quad (16.74)$$

As discussed in § 15.5, this contact interaction is important at very short distances and consequently affects mainly S-waves.

Because (16.69) contains the first derivative of the wave function (the ∇ 's in V_{dar}), it is not strictly the same as a Schrödinger equation. Yet the references show how to eliminate these derivative terms by transforming to a new wave function and new potentials. The basic result still holds: the equivalent nonrelativistic central and spin-orbit potentials are energy dependent and a mixture of more fundamental (Dirac) potentials and their derivatives. Because of this mixture, the shape (geometry) of the effective central and spin-orbit potentials is expected to be quite different from each other. It may then be simpler to understand the basic coupling in the Dirac equation (16.62) than the equivalent Schrödinger equation.

16.4 Problems

- Set up the equations needed to solve for the lowest-energy bound state of a Dirac electron in a 1D square well of depth V_0 and range R .
 - For a depth and range appropriate to the deuteron (80 MeV and 1.2 fm), compare the binding energy predicted by the Schrödinger equation and the Dirac equation.
 - Determine the large and small components of the wave function for a Dirac deuteron and plot them to scale.
 - Under what conditions will the small component be as large as the large one?
 - Would any of the problems associated with the Klein paradox occur when V_0 becomes comparable with or larger than $2m$?
- A nucleon ($m = 938$ MeV) bound in a nucleus, initially has the wave function

$$\psi(\mathbf{r}, t = 0) = \frac{e^{-r^2/2R^2} u_1^{(+)}(0)}{(\pi R^2)^{3/4}}, \quad R = 2.5 \text{ fm}. \quad (16.75)$$

- Estimate the probability for a negative-energy nucleon to be in the nucleus.
 - Estimate the most likely speed of the negative-energy nucleon.
 - Estimate the most likely speed of the positive-energy nucleon.
 - What is time-dependence of the wave packet describing this nucleon?
- Show that the wave function and probability density of the Dirac hydrogen atom diverges at the origin even though the integrated probability remains finite.
 - Construct the wave functions for the first S, P, D, and F states of Dirac hydrogen.
 - Give all the quantum numbers that occur for each of these states.

- (c) Compare the quantum numbers and wave functions to the equivalent ones of the Schrödinger hydrogen.
5. What would be the properties of the solution of the Dirac equation for an atom with $Z\alpha > 140$?
6. Examine the electron–proton problem for a total energy greater than m .
- Show that the power series for the hydrogen atom's upper spinor G_{lj} no longer terminates.
 - Show that the Dirac equation is no longer an eigenvalue problem, that is, solutions exist for all positive kinetic energies.
7. Show that for continuum energies ($E > m$), the hydrogen atom's G_{lj} satisfies a differential equation with the same asymptotic behavior as the nonrelativistic solution:
- $$G_{lj} \sim \sin(kr - l\pi/2 + \sigma_l - \eta \ln 2kr). \quad (16.76)$$
8. Consider a simple model for the dipole–dipole interaction between the magnetic moments of the electron and proton in hydrogen (the hyperfine interaction).
- Prove that this interaction is proportional to $\mathbf{s}_e \cdot \mathbf{s}_p$.
 - Assuming the interaction will be treated as a perturbation, compare the value of its matrix element for total ep spin of 0 and 1.
 - Show that because the triplet state with spins aligned has the magnetic moments of the oppositely charged particles in opposition ($\uparrow\downarrow$), this is a repulsive configuration and therefore less bound.
 - The experimental hyperfine splitting is only 1420 MHz (9×10^{-7} eV), in comparison to the fine structure splitting of 11,000 MHz (69×10^{-7} eV). Use actual values of physical constants to explain the difference and numerical values of these splittings.

CHAPTER 17

SCATTERING AND DIRAC INTEGRAL EQUATIONS

In this chapter we place the Dirac equation into a scattering theory analogous to the one developed for the Schrödinger equation and reduce the theory to abstract operator relations. The transition and potential matrices, Green's function operators, partial-wave analyses, momentum-space techniques, handling of spin, and definitions of scattering observables should seem like old friends. We have encountered some of this in Chapter 15, *Components of Dirac Wave Functions*, and now look more closely at the momentum-space representation, and the role of the negative-energy degrees of freedom.

17.1 Distorted and Plane Waves

Consider again Figure 15.1, the energy regimes available to a fermion interacting with a potential. We now want to study solutions to the Dirac wave functions for particles in the continuum, that is, for energy $E_k = \sqrt{k^2 + m^2} > 0$. For stationary states, we write these distorted waves as

$$\Psi(\mathbf{x}, t) = \begin{pmatrix} \psi_1 \\ \psi_2 \\ \psi_3 \\ \psi_4 \end{pmatrix} = e^{-iE_k t} \begin{pmatrix} \psi_U(\mathbf{x}) \\ \psi_L(\mathbf{x}) \end{pmatrix}. \quad (17.1)$$

Here we have split the 4×1 Ψ into upper and lower (large and small) 2×1 parts, as done to derive the Pauli equation or to solve the hydrogen atom.¹ When (17.1) is substituted into a Dirac equation containing a potential that couples like the zeroth component of a 4-vector, there results the 4×1 differential equations (16.21), or in the *split notation* (15.50)–(15.51),

$$[E_k - m - V(\mathbf{r})] \psi_U(\mathbf{r}) = \boldsymbol{\sigma} \cdot \frac{\nabla}{i} \psi_L(\mathbf{r}), \quad (17.2)$$

¹Recall that the terms “large” and “small” refer to the relative size of these terms in the nonrelativistic limit; since in the scattering of electrons and protons, the kinetic energy of the projectile often exceeds its rest mass, the lower components of the wave function can get large.

$$[E_k + m - V(r)] \psi_L(r) = \boldsymbol{\sigma} \cdot \frac{\nabla}{i} \psi_U(r). \quad (17.3)$$

Because both the momentum operator ∇/i and the spin operator $\boldsymbol{\sigma}$ appear in these equations, they are simultaneously, simultaneous differential and matrix equations.

Asymptotic States

When the fermion is far from the center of force, V vanishes and the asymptotic wave function (the part that determines all scattering observables) must be a solution of the $V = 0$ forms of (17.2)–(17.3):

$$[E_k - m] \psi_U = \boldsymbol{\sigma} \cdot \frac{\nabla}{i} \psi_L, \quad (r \rightarrow \infty), \quad (17.4)$$

$$[E_k + m] \psi_L = \boldsymbol{\sigma} \cdot \frac{\nabla}{i} \psi_U, \quad (r \rightarrow \infty). \quad (17.5)$$

While it is true that plane waves are solutions to these equations, what is important here is the relation of upper to lower components:

$$\psi_L = \frac{\boldsymbol{\sigma} \cdot \mathbf{k}}{E_k + m} \psi_U. \quad (17.6)$$

The relation (17.6) implies that in the asymptotic region, the lower part of the wave function ψ_L is completely determined by knowledge of the upper part. It also means that ψ_L is smaller than ψ_U by a factor of $O(v)$, and that ψ_L has a spin structure different from that of ψ_U .

Because the asymptotic upper part determines the lower part of the wave function, all scattering observables are determined from just ψ_U . The lower components of the wave function are present, of course, yet their contributions just change the normalizations,

$$\psi_U^\dagger \psi_U + \psi_L^\dagger \psi_L = \left(1 + \frac{k^2}{(E_k + m)^2}\right) \psi_U^\dagger \psi_U = N \psi_U^\dagger \psi_U. \quad (17.7)$$

And, because we are describing a continuous state whose normalization is chosen for our convenience anyway (it just “drops out” when we calculate cross sections), the change is of no consequence.

Exercise Show that if Ψ satisfies the scattering boundary conditions of the incident plane wave and outgoing spherical wave shown in Figure 1.2,

$$\Psi \sim \Phi^{\text{in}} + \Psi^{\text{sc}}, \quad (17.8)$$

then the cross sections depend only upon the upper part of the asymptotic wave function; that is, show

$$\sigma \propto \frac{j_{\text{sc}}(\Psi)}{j_{\text{in}}(\Psi)} \propto \frac{N j_{\text{sc}}(\psi_U)}{N j_{\text{in}}(\psi_U)} = \frac{j_{\text{sc}}(\psi_U)}{j_{\text{in}}(\psi_U)}. \quad \diamond \quad (17.9)$$

Even though the scattering observables do not depend on the lower parts of the wave function, these parts enter when solving the Dirac equation in the inner region where a potential acts.

One possible set of solutions to the potential-free equations (17.4)–(17.5) is composed of the plane waves (15.16), which are complete but clearly do not satisfy the scattering boundary conditions (17.8).

Exercise Show that the plane waves split into the Pauli spinors:

$$\Phi_s^{(+)}(\mathbf{x}, t) = e^{i(\mathbf{k} \cdot \mathbf{x} - E_k t)} \begin{pmatrix} \phi_U \\ \phi_L \end{pmatrix} \quad (17.10)$$

$$\phi_{U\uparrow}(k) = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \phi_{L\uparrow}(k) = \begin{pmatrix} \frac{k_+}{E_k + m} \\ \frac{k_-}{E_k + m} \end{pmatrix}, \quad (17.11)$$

$$\phi_{U\downarrow}(k) = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad \phi_{L\downarrow}(k) = \begin{pmatrix} \frac{k_-}{E_k + m} \\ \frac{-k_+}{E_k + m} \end{pmatrix}, \quad (17.12)$$

where we have dropped the normalization factor and have used $\mathbf{k}$ to denote the electron's momentum (which agrees with our preceding notation for scattering). $\diamond$

Spin Scattering

When we extended our discussion of scattering to include spin in Chapter 9, *Spin Theory*, and Chapter 10, *Spin Phenomenology and Identical Particles*, we found that we could define wave functions and scattering amplitudes as in the spinless case as long as we specified the spin orientation of the incident and scattered particles. The treatment for relativistic scattering is analogous. For a beam with incident momentum $\mathbf{k}$ along the z axis and spin quantized along the z axis, ψ_U could initially be up or down:

$$\psi_{U\uparrow}^{\text{in}} \sim e^{i\mathbf{k} \cdot \mathbf{x}} |\uparrow\rangle, \quad \psi_{U\downarrow}^{\text{in}} \sim e^{i\mathbf{k} \cdot \mathbf{x}} |\downarrow\rangle. \quad (17.13)$$

The scattered electron's spinor is more complicated (see the Problems section). For an incident electron with spin up, the asymptotic upper part of the full distorted wave has the same form as the Schrödinger wave function in Chapter 9, *Spin Theory*,

$$\psi_U \sim e^{i\mathbf{k} \cdot \mathbf{x}} |\uparrow\rangle + \frac{e^{ikr}}{r} [f_{\uparrow\uparrow}(\theta, \phi) |\uparrow\rangle + f_{\downarrow\uparrow}(\theta, \phi) |\downarrow\rangle]. \quad (17.14)$$

In terms of individual components:

$$\psi_1(\mathbf{r}) \sim e^{ikz} + f_{\uparrow\uparrow}(\theta, \phi) \frac{e^{ikr}}{r}, \quad (17.15)$$

$$\psi_2(\mathbf{r}) \sim f_{\downarrow\uparrow}(\theta, \phi) \frac{e^{ikr}}{r}, \quad (17.16)$$

where we do not give ψ_3 and ψ_4 because they are calculable from ψ_1 and ψ_2 via (17.6). Because the meaning of wave function has not changed in Dirac theory, the differential cross sections and polarizations are also given by our expressions in Chapter 10, *Spin Phenomenology and Identical Particles*. For example, for unpolarized beam and undetected final polarization,

$$\frac{d\sigma}{d\Omega}(\theta, \phi) = |f_{\uparrow\uparrow}|^2 + |f_{\downarrow\uparrow}|^2. \quad (17.17)$$

The relation of $f_{\uparrow\uparrow}$ and $f_{\downarrow\uparrow}$ in terms of phase shifts, (9.62) and (9.65), remain unchanged.

17.2 Coulomb Scattering

In the last chapter we did the hard work needed to understand electron scattering from a Coulomb potential. There we found that the Dirac wave function for an eigenstate of total angular momentum j has the form (16.17)

$$\Psi_{jm}^{4 \times 1} = \frac{1}{r} \begin{pmatrix} iG_{lj}^{(+)}(kr) \mathcal{Y}_{jm}^{(l)}(\sigma\Omega) \\ F_{lj}^{(+)}(kr) \mathcal{Y}_{jm}^{(l)}(\sigma\Omega) \end{pmatrix}. \quad (17.18)$$

Exercise Determine the relation between the asymptotic value of F_{lj} and G_{lj} . $\diamond$

In our solution for the hydrogen atom, we found that the upper wave function G_{lj} behaves very much like the nonrelativistic solution. It has the same asymptotic form and is a power series whose termination at power n' determines the bound-state energy. If the energy parameter is greater than m , the termination condition no longer obtains, the wave function oscillates as $r \rightarrow \infty$, and we have a scattering wave function. As shown in the problems, G_{lj} has the same asymptotic behavior as found for the nonrelativistic Coulomb wave (4.42),

$$G_{lj} \sim \sin(kr - l\pi/2 + \sigma_l - \eta \ln 2kr). \quad (17.19)$$

(Note: The Coulomb F of Part I is the present G). Consequently, the techniques used in Chapter 4, *Scattering Applications: Lengths, Resonances, Coulomb*, to handle the Coulomb force are also applicable with the Dirac equation.

It is also straightforward to develop a Born series for the Dirac equation (as we did in part in the Problems of Chapter 15, *Components of Dirac Wave Functions*) and apply it to electron scattering from a nucleus of charge Ze . For an unpolarized electron with velocity β , momentum k , and undetected final polarization, the Born approximation yields

$$\frac{d\sigma}{d\Omega} = \frac{e^4 Z^2}{[2k\beta \sin^2(\frac{\theta}{2})]^2} \left[1 - (\beta \sin \frac{\theta}{2})^2 + \pi Z\alpha\beta \sin \frac{\theta}{2} (1 - \sin \frac{\theta}{2}) + O(\alpha^2) \right]. \quad (17.20)$$

The fraction out front is the nonrelativistic (and classical) Rutherford cross section generalized to relativistic kinematics, the $(\beta \sin \frac{\theta}{2})^2$ term arises from the electron's spin (magnetic moment scattering), and the next term arises from the distortion of the wave function.

17.3 Momentum-Space Equation

In Chapter 15, *Components of Dirac Wave Functions*, we studied the stationary-state, Hamiltonian form of the Dirac equation,

$$E\Psi(\mathbf{r}) = [\boldsymbol{\alpha} \cdot \mathbf{p} + \beta m + V(\mathbf{r})] \Psi(\mathbf{r}, t), \quad (17.21)$$

where $\mathbf{p}$ is the differential operator ∇/i , and the potential couples into the energy (zeroth component of a 4-vector). If the potential $V(\mathbf{r})$ were nonlocal, (17.21) would become the integro-differential equation

$$E\Psi(\mathbf{r}) = (\boldsymbol{\alpha} \cdot \mathbf{p} + \beta m)\Psi(\mathbf{r}) + \int d^3r' V(\mathbf{r}, \mathbf{r}')\Psi(\mathbf{r}'). \quad (17.22)$$

An equation of this type is best solved in momentum space, where it has the form:²

$$(\boldsymbol{\alpha} \cdot \mathbf{p} + \beta m) \Psi(\mathbf{p}) + \int d^3 p' V(\mathbf{p}, \mathbf{p}') \Psi(\mathbf{p}') = E \Psi(\mathbf{p}). \quad (17.23)$$

In (17.23) the momentum $\mathbf{p}$ is no longer a differential operator and $V(\mathbf{p}, \mathbf{p}')$ is in the momentum representation. As discussed in Chapter 6, *The Transition and Potential Matrices*, local potentials in momentum space are functions of just the momentum transfer $\mathbf{q} = \mathbf{p}' - \mathbf{p}$:

$$V(\mathbf{r}', \mathbf{r}) = (2\pi)^3 \delta(\mathbf{r} - \mathbf{r}') v(\mathbf{r}) \Rightarrow V^{\text{local}}(\mathbf{p}, \mathbf{p}') = v(\mathbf{q}), \quad (17.24)$$

where $v(\mathbf{q})$ is the Fourier transform of $v(\mathbf{r})$ [e.g., $v(\mathbf{q}) = 4\pi Ze^2/q^2$ for the Coulomb potential].

Exercise Show that for local potentials

$$(\boldsymbol{\alpha} \cdot \mathbf{p} + \beta m) \Psi(\mathbf{p}) + \int d^3 q v(\mathbf{q}) \Psi(\mathbf{p} - \mathbf{q}) = E \Psi(\mathbf{p}). \quad \diamond \quad (17.25)$$

Exercise Verify that the plane waves of Chapter 15, *Components of Dirac Wave Functions*, are $v(\mathbf{q}) = 0$ solutions of (17.25). $\diamond$

Partial Waves

It is easiest to solve (17.23) after decomposing Ψ and $v(\mathbf{q})$ in the partial-wave basis. The decompositions are just like the ones in coordinate space, only now with the spin-angle functions $\mathcal{Y}_{jm}^{l=j \pm \frac{1}{2}}$ referring to the direction of $\mathbf{p}$:

$$\Psi_{jm}(\mathbf{p}) = \frac{1}{p} \begin{pmatrix} iG_{lj}^{(\pm)}(p) \mathcal{Y}_{jm}^{l=j+1/2}(\sigma\Omega) \\ F_{lj}^{(\pm)}(p) \mathcal{Y}_{jm}^{l=j-1/2}(\sigma\Omega) \end{pmatrix}. \quad (17.26)$$

As worked out in the problems, for $j = l + \frac{1}{2}$, the Dirac equation becomes the coupled 1D integral equations:

$$\int_0^\infty dp' p'^2 v_l(p, p') G_{lj}^{(\pm)}(p') - p F_{lj}^{(\pm)}(p) = (E - m) G_{lj}^{(\pm)}(p), \quad (17.27)$$

$$\int_0^\infty dp' p'^2 v_{l+1}(p, p') F_{lj}^{(\pm)}(p') - p G_{lj}^{(\pm)}(p) = (E + m) F_{lj}^{(\pm)}(p), \quad (17.28)$$

where v_l is the potential in the partial-wave representation. The techniques used to solve these equations are discussed in the next chapter.

Internal Negative-Energy States

There is nothing unphysical about a solution of the Dirac equation for an electron containing negative-energy parts. This occurs, generally, whenever a potential acts.

²This also follows directly from the formal equation (17.51).

Exercise Show that if we reverse the sign of the potential and of the momentum operator $\mathbf{p}$, and interchange ψ_U and ψ_L , the Dirac equation describes a wave function of a positive-energy particle of the same mass as an electron but of opposite charge. $\diamond$

We see a symmetry at this level. The amplitude for electron scattering elastically from some field equals that for positron scattering from the reversed field. It is natural to try to take this a step further and ask if our theory predicts processes in which an antiparticle is actually produced (such as pair creation). While it is possible to manipulate the theory to predict such processes, it does not make good sense because we have postulated a theory for only one particle. (As we will see in Part III, *Quantum Fields*, a field theory properly describes processes in which the number of particles changes.) Nevertheless, negative-energy electrons do enter into our 1-body theory as virtual states, and we now build that into our formalism by developing a momentum-space representation that exhibits the positive- and negative-energy components of our states.

We start with the $\mathbf{r}$ representation of momentum eigenstates,

$$\Phi_{ps}^{(\pm)}(\mathbf{x}, t) = e^{\mp i p^\mu x_\mu} u_s^{(\pm)}(p), \quad (17.29)$$

or in a handy version of the split notation,

$$\Phi_{ps}^{(+)}(\mathbf{x}) = \frac{N e^{i \mathbf{p} \cdot \mathbf{x}}}{(2\pi)^{3/2}} \begin{pmatrix} 1_{2 \times 2} \\ \Gamma(\mathbf{p})_{2 \times 2} \end{pmatrix} \chi_s^{2 \times 1}(0), \quad (17.30)$$

$$\Phi_{ps}^{(-)}(\mathbf{x}) = \frac{N e^{i \mathbf{p} \cdot \mathbf{x}}}{(2\pi)^{3/2}} \begin{pmatrix} -\Gamma(\mathbf{p})_{2 \times 2} \\ 1_{2 \times 2} \end{pmatrix} \chi_s^{2 \times 1}(0), \quad (17.31)$$

$$N = \sqrt{\frac{E_p + m}{2E_p}}. \quad (17.32)$$

Here $\Gamma(\mathbf{p})$ is the 2×2 matrix,

$$\Gamma(\mathbf{p})_{2 \times 2} = \frac{\boldsymbol{\sigma} \cdot \mathbf{p}}{E_p + m}, \quad (17.33)$$

and χ_s is an ordinary, momentum-independent 2×1 Pauli spinor. The 4×2 matrices multiplying the χ 's yield 4×1 Dirac spinors. The terms multiplying the exponentials in (17.29) are Fourier expansion coefficients, and hence are proportional to the momentum-space representation of a momentum eigenstate,

$$\Phi_{ps}^{(\pm)}(\mathbf{p}') \propto \delta^3(\mathbf{p} - \mathbf{p}') u_s^{(\pm)}(p). \quad (17.34)$$

Exercise Verify that the momentum state $\Phi_{\mathbf{k}}(\mathbf{p})$ is the sum of positive- and negative-energy parts:

$$\Phi_{\mathbf{k}}(\mathbf{p}) = \Phi_{\mathbf{k}}^{(+)}(\mathbf{p}) + \Phi_{\mathbf{k}}^{(-)}(\mathbf{p}), \quad (17.35)$$

$$\Phi_{\mathbf{k}}^{(+)}(\mathbf{p}) = \delta(\mathbf{p} - \mathbf{k}) \begin{pmatrix} \chi^+(0) \\ \Gamma(\mathbf{p}) \chi^+(0) \end{pmatrix}, \quad (17.36)$$

$$\Phi_{\mathbf{k}}^{(-)}(\mathbf{p}) = \delta(\mathbf{p} - \mathbf{k}) \begin{pmatrix} -\Gamma(\mathbf{p}) \chi^-(0) \\ \chi^-(0) \end{pmatrix},$$

with $\chi^\pm(0)$ the 2×1 Pauli spinors $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$. $\diamond$

Just as the $u(\mathbf{p})$'s of momentum states split into upper and lower Pauli spinors (which for a particle at rest are identified with positive- and negative-energy spin states), so we split a general wave function,

$$\Psi_{4 \times 1} = \begin{pmatrix} \psi_U(\mathbf{p}) \\ \psi_L(\mathbf{p}) \end{pmatrix}, \quad (17.37)$$

where $\psi_U(\mathbf{p})$ and $\psi_L(\mathbf{p})$ are 2×1 Pauli spinors. With this Ψ , the $\mathbf{p}$ -space Dirac equation for a local, timelike potential splits into the coupled integral equations

$$(E - m)\psi_U(\mathbf{p}) = \boldsymbol{\sigma} \cdot \mathbf{p}\psi_L(\mathbf{p}) - \int d^3q v(\mathbf{q})\psi_U(\mathbf{p} - \mathbf{q}), \quad (17.38)$$

$$(E + m)\psi_L(\mathbf{p}) = \boldsymbol{\sigma} \cdot \mathbf{p}\psi_U(\mathbf{p}) - \int d^3q v(\mathbf{q})\psi_L(\mathbf{p} - \mathbf{q}). \quad (17.39)$$

In analogy with (17.35), we represent a general state as a sum of positive- and negative-energy parts,

$$\Psi(\mathbf{P}) = \Psi^{(+)}(\mathbf{p}) + \Psi^{(-)}(\mathbf{p}), \quad (17.40)$$

$$\Psi^{(+)}(\mathbf{p}) = \begin{pmatrix} \chi^+(\mathbf{p}) \\ \Gamma(\mathbf{p})\chi^+(\mathbf{p}) \end{pmatrix}, \quad (17.41)$$

$$\Psi^{(-)}(\mathbf{p}) = \begin{pmatrix} -\Gamma(\mathbf{p})\chi^-(\mathbf{p}) \\ \chi^-(\mathbf{p}) \end{pmatrix}, \quad (17.42)$$

with $\chi^\pm(\mathbf{p})$ a Pauli spinor with a momentum dependence that describes the probability distribution within Ψ . For example, if Ψ contains a plane wave (as scattering waves do), $\chi(\mathbf{p})$ will have a delta function embedded in it.

Because the ratios of upper to lower components in (17.41)–(17.42) are the same as in plane waves, it is the spinor functions $\chi^\pm(\mathbf{p})$ that describe the distortion. Normally we express a 4×1 wave function in terms of the unit basis vectors $u_s^{(\pm)}(0)$, (14.22)–(14.23), which describe free positive- and negative-energy particles *at rest*. The resulting upper and lower spinors are consequently eigenstates of the matrix β . Now when we use the plane-wave spinors $u_s^{(\pm)}(\mathbf{p})$ as basis vectors, we obtain 2×1 upper and lower spinors that are eigenstates of the free Hamiltonian H_0 .

Exercise Show that for a general $\Psi(\mathbf{p})$,

$$\Psi(\mathbf{p})_{4 \times 1} = \begin{pmatrix} 1_{2 \times 2} \\ \Gamma(\mathbf{p})_{2 \times 2} \end{pmatrix} \chi_+^{2 \times 1}(\mathbf{p}) + \begin{pmatrix} -\Gamma(\mathbf{p})_{2 \times 2} \\ 1_{2 \times 2} \end{pmatrix} \chi_-^{2 \times 1}(\mathbf{p}), \quad (17.43)$$

where Γ is given by (17.33). $\diamond$

The expansion (17.43) separates a solution of the Dirac equation with the full Hamiltonian H into a 4×1 part that is an eigenstate of the free Hamiltonian H_0 with positive

energy and a second 4×1 part which is an eigenstate of H_0 with negative energy:

$$H_0 \begin{pmatrix} 1 \\ +\Gamma(\mathbf{p}) \end{pmatrix} \chi_+^{2 \times 1} = +E_p \begin{pmatrix} 1 \\ +\Gamma(\mathbf{p}) \end{pmatrix} \chi_+^{2 \times 1}, \quad (17.44)$$

$$H_0 \begin{pmatrix} -\Gamma(\mathbf{p}) \\ 1 \end{pmatrix} \chi_-^{2 \times 1} = -E_p \begin{pmatrix} -\Gamma(\mathbf{p}) \\ 1 \end{pmatrix} \chi_-^{2 \times 1}. \quad (17.45)$$

Consequently, we can consider

$$\hat{\epsilon}_+ = \begin{pmatrix} 1_{2 \times 2} \\ \Gamma(\mathbf{p})_{2 \times 2} \end{pmatrix}, \quad \hat{\epsilon}_- = \begin{pmatrix} -\Gamma(\mathbf{p})_{2 \times 2} \\ 1_{2 \times 2} \end{pmatrix} \quad (17.46)$$

as momentum-dependent “basis vectors”, and project any Ψ onto them:

$$\Psi(\mathbf{p})_{4 \times 1} = \begin{pmatrix} \chi_+(\mathbf{p}) \\ \chi_-(\mathbf{p}) \end{pmatrix}_{+-}, \quad (17.47)$$

where the $+-$ subscript indicates the energy basis.

Exercise Verify that using ψ_U and ψ_L as basis vectors leads to eigenvectors of β (determine the eigenvalues), while using $\chi^{(\pm)}$ leads to eigenvectors of H_0 . $\diamond$

Because Ψ does *not* describe a free particle, (17.43) does not mean that an arbitrary (and possibly weak) potential has created physical antiparticles. Rather, because every solution of the Dirac equation has four components, there is always some projection onto the free-particle, negative-energy states. Yet because the full wave function describes an interacting particle, these negative-energy states are not eigenstates of the full Hamiltonian (this is analogous to off-energy-shell scattering in scattering theory). This positive- and negative-energy basis in which we will formalize Dirac theory is elucidated by Bethe and Salpeter (1977) in their momentum-space derivation of the Pauli equation, and is related to the *Foldy–Wouthuysen* transformation theory described in modern notation in Bjorken and Drell (1964).

17.4 Integral Dirac Equations

We now develop a formal version of Dirac theory which blends the split of Dirac wave functions into positive- and negative-energy parts (17.43) with the operator formalism of Part I, *Scattering and Integral Quantum Mechanics*. We examine basis states, Green's functions, and Dirac forms of the Lippmann–Schwinger (LS) equation.

Basis States

Because scattering theory is formulated in terms of momentum *eigenfunctions* [that is, (17.30)–(17.32)], we choose as *rays* in our Hilbert space the direct product of a Pauli spinor with the (momentum $\mathbf{k}$, spin s , sign of energy b) basis kets used in (17.43):

$$|\mathbf{k} s b\rangle \stackrel{\text{def}}{=} |\mathbf{k} b\rangle |\chi_s\rangle. \quad (17.48)$$

Because these basis are noninteracting, there is no momentum dependence in the χ 's. To construct the formal theory, we use Hermitian adjoints to transform kets into the dual-space bras, and choose the normalization of (17.48) such that the orthogonality and completeness relations are

$$\langle \mathbf{k} s b | \mathbf{k}' s' b' \rangle = \delta_{s s'} \delta_{b b'} \delta(\mathbf{k} - \mathbf{k}'), \quad (17.49)$$

$$\sum_{b=\pm, s=\uparrow, \downarrow} \int d^3 k |\mathbf{k} s b\rangle \langle \mathbf{k} s b| = \tilde{1}. \quad (17.50)$$

Note that this differs from the orthogonality and completeness relations (15.21) and (15.23), which used Dirac adjoints ($\bar{u}$, not $u^\dagger$).

Green's Function

The stationary-state Dirac equation in operator form is

$$(\not{p} - m) |\Psi\rangle = V |\Psi\rangle, \quad (17.51)$$

$$[E\gamma^0 - \mathbf{p} \cdot \boldsymbol{\gamma} - m] |\Psi\rangle = V |\Psi\rangle, \quad (17.52)$$

where we keep the notation simple by taking a local and scalar potential (one that adds into the mass). The Green's function operator is the inverse of the operator on the LHS of (17.51) (see § 5.1 and § 7.2 for review of these techniques):

$$G_D^{(+)}(E) = (E\gamma^0 - \mathbf{p} \cdot \boldsymbol{\gamma} - m + i\epsilon\gamma^0)^{-1} \equiv \frac{1}{E\not{\mathbf{p}} - \mathbf{p} \cdot \boldsymbol{\gamma} - m + i\epsilon\gamma^0}, \quad (17.53)$$

where to avoid the singularity and to ensure outgoing scattered waves, we have added a small positive imaginary part to the energy. As it stands, $G_D^{(+)}$ looks simple enough, yet with matrices and operators in the denominator its content and method of application are not transparent.

Exercise Show that the Dirac Green's function also can be expressed as

$$G_D = \frac{1}{\not{p} - m} = \frac{\not{p} + m}{p^2 - m^2} = \frac{\not{p} + m}{E^2 - p^2 - m^2 + i\epsilon}. \quad \diamond \quad (17.54)$$

The denominator in (17.54) looks like $(E^{(+)} - E_p)(E^{(+)} + E_p)$, which is proportional to the nonrelativistic G 's denominator. The numerator is proportional to the projection operator Λ^+ (15.25). Yet because G_D has poles at $E = \pm E_p$, it contains two different modes of electron propagation. To see this, we rewrite G_D as the sum of two terms, one with a positive-energy pole and one with a negative-energy pole:

$$(\not{p} - m)^{-1} = \int d^3 k \left(\frac{|\mathbf{k}+\rangle \langle \mathbf{k}+|}{E_p - E_k} + \frac{|\mathbf{k}-\rangle \langle \mathbf{k}-|}{E_p + E_k} \right). \quad (17.55)$$

The contribution to G_0 from each pole produces positive-energy or negative-energy scattered waves. Yet if our initial system has $E_p > 0$, then only the first terms in (17.55) has an accessible pole, and there will be physical scattered waves of positive energy that

travel beyond the range of the potential, but only virtual decaying waves of negative energy within the potential's range. Adding $i\epsilon$ to the energy does not change this; it only ensures that the scattered waves are outgoing. Further discussion of the propagator viewpoint is found in the original work of Feynman (1948A, 1948B, 1949, 1962A, 1962B), Volume I of Bjorken and Drell (1964), and Feynman and Hibbs (1965).

Wave Function

The abstract form of the Dirac equation is

$$|\Psi_{\mathbf{k}}\rangle = G_D(E_k^{(+)})V|\Psi_{\mathbf{k}}\rangle, \quad (17.56)$$

where the propagator G_D depends parametrically on the energy, and where we are building in outgoing-wave boundary conditions (the plus). Scattering solutions exist for any energy $E_k = \sqrt{k^2 + m^2}$ in the continuum, as do solutions $|\Phi_{\mathbf{k}}\rangle$ to the homogeneous Dirac equation,

$$(\not{k} - m)|\Phi_{\mathbf{k}}\rangle = 0. \quad (17.57)$$

The most general solution to (17.51) has the familiar incident plane wave plus scattered wave form,

$$|\Psi_{\mathbf{k}}\rangle = |\Phi_{\mathbf{k}}\rangle + G_D^{(+)}(E_k)V|\Psi_{\mathbf{k}}\rangle, \quad (17.58)$$

$$G_D^{(+)}(E_k) = \frac{1}{E_k^{(+)}\gamma^0 - \mathbf{k} \cdot \boldsymbol{\gamma} - m} = \frac{1}{\not{k}^{(+)} - m}. \quad (17.59)$$

Integral Equation for T

We again define the T matrix as the operator that has the same effect acting on a free positive-energy state $|\Phi_{\mathbf{k}}\rangle \equiv |\mathbf{k}\rangle$, as the potential V has acting on a distorted state $|\Psi_{\mathbf{k}}\rangle$:

$$T_E|\Phi_{\mathbf{k}}\rangle = V|\Psi_{\mathbf{k}}\rangle. \quad (17.60)$$

We should also include the spin and sign of energy as designators on these states; we leave the spin off until needed, and because we scatter only positive-energy projectiles, the plane-wave in (17.60) is always of positive energy. If we replace $V\Psi$ by $T\Phi$, the LS equation for $|\Psi\rangle$ (17.58) becomes

$$|\Psi_{\mathbf{k}}\rangle = |\Phi_{\mathbf{k}}\rangle + G_D^{(+)}(E_k)T_E|\Phi_{\mathbf{k}}\rangle. \quad (17.61)$$

If we multiply all term in (17.59) by V and again use (17.60), we obtain the LS equation for T :

$$V|\Psi_{\mathbf{k}}\rangle = V|\Phi_{\mathbf{k}}\rangle + V \frac{1}{\not{k}^{(+)} - m} T|\Phi_{\mathbf{p}}\rangle, \quad (17.62)$$

$$\Rightarrow T_E = V + V \frac{1}{\not{k} - m + i\epsilon\gamma^0} T_E, \quad (17.63)$$

where T and V are abstract operators in a 4D spin-energy space.

Energy Decomposition

By taking matrix elements of T and V between explicit energy states,

$$T_E^{b'b}(\mathbf{k}', \mathbf{k})_{2 \times 1} = \langle \mathbf{k}' b' | T_{4 \times 1} | \mathbf{k} b \rangle, \quad (17.64)$$

$$V^{b'b}(\mathbf{k}', \mathbf{k})_{2 \times 1} = \langle \mathbf{k}' b' | V_{4 \times 1} | \mathbf{k} b \rangle, \quad (17.65)$$

and replacing $G_D^{(+)}(E)$ by its positive- and negative-energy expansions, we convert the 4D abstract equation (17.63) into a set of integral equations in Pauli spin space (which can then be solved with the techniques of Chapter 9, *Spin Theory*, and Chapter 10, *Spin Phenomenology and Identical Particles*).

Exercise Show that the preceding steps lead to the coupled integral equations relating 2×2 matrix functions:

$$\begin{aligned} T_E^{++}(\mathbf{k}', \mathbf{k}) &= V^{++}(\mathbf{k}', \mathbf{k}) + \int d^3p V^{++}(\mathbf{k}', \mathbf{p}) \frac{1}{E^{(+)} - E_p} T_E^{++}(\mathbf{p}, \mathbf{k}) \\ &\quad + \int d^3p V^{+-}(\mathbf{k}', \mathbf{p}) \frac{1}{E^{(+)} + E_p} T_E^{-+}(\mathbf{p}, \mathbf{k}), \end{aligned} \quad (17.66)$$

$$\begin{aligned} T_E^{-+}(\mathbf{k}', \mathbf{k}) &= V^{-+}(\mathbf{k}', \mathbf{k}) + \int d^3p V^{-+}(\mathbf{k}', \mathbf{p}) \frac{1}{E^{(+)} - E_p} T_E^{++}(\mathbf{p}, \mathbf{k}) \\ &\quad + \int d^3p V^{--}(\mathbf{k}', \mathbf{p}) \frac{1}{E^{(+)} + E_p} T_E^{-+}(\mathbf{p}, \mathbf{k}). \end{aligned} \quad \diamond (17.67)$$

Exercise Explain why these are still matrix equations. $\diamond$

Equations (17.66)–(17.67) extend those in Chapter 7, *Formal Quantum Mechanics*, to include negative-energy basis states: Amplitude T^{++} is for transitions from an incident positive-energy momentum state into an outgoing positive-energy scattered wave. Amplitude T^{-+} is for transitions from an incident positive-energy momentum state into a virtual negative-energy state; because there are transitions between the positive- and virtual negative-energy states, the equations for T^{++} and T^{-+} are coupled. The potential V^{++} induces transitions of a positive-energy electron into another positive-energy electron that either propagates in a virtual states with $1/(E^{(+)} - E_p)$, or radiates out directly as a scattered wave (the Born term). The potential V^{-+} , in contrast, induces transitions of a positive-energy electron into a negative-energy one that can only propagate as a virtual state with $1/(E^{(+)} + E_p)$. The negative-energy electrons are scattered into other negative-energy electrons by V^{--} or back into positive-energy ones by V^{+-} .

To illustrate that the negative-energy states are just virtual, we use the completeness relation (17.50) to expand the wave function,

$$|\Psi\rangle = \int d^3p \left[\psi^{(+)}(\mathbf{p}) |\mathbf{p}+\rangle + \psi^{(-)}(\mathbf{p}) |\mathbf{p}-\rangle \right]. \quad (17.68)$$

We combine this with the LS equation for Ψ (17.61) to obtain equations for the positive- and negative-energy momentum-space wave functions,

$$\psi_{\mathbf{k}}^{(+)}(\mathbf{p}) = \delta(\mathbf{p} - \mathbf{k}) + \frac{1}{E_{\mathbf{k}}^{(+)} - E_p} T_{E_{\mathbf{k}}}^{++}(\mathbf{p}, \mathbf{k}), \quad (17.69)$$

$$\psi_{\mathbf{k}}^{(-)}(\mathbf{p}) = \frac{1}{E_{\mathbf{k}}^{(+)} + E_{\mathbf{p}}} T_{E_{\mathbf{k}}}^{-+}(\mathbf{p}, \mathbf{k}). \quad (17.70)$$

Exercise Verify that the prescribed steps lead to (17.69) and (17.70). $\diamond$

We see that $\psi^{(-)}$ contains no plane wave, which would be represented by a delta function as in (17.69), and no outgoing scattered wave, which would require a pole for positive energies.

17.5 Problems

1. An electron is scattered in the (θ, ϕ) direction (Figure 9.3).
 - (a) If spin is quantized along the z axis, show that the upper parts of the electron's spinors have the asymptotic forms

$$\psi_{U\uparrow}(r \rightarrow \infty) \propto \begin{pmatrix} \cos \theta/2 \\ -\sin \theta/2 e^{-i\phi} \end{pmatrix}, \quad \psi_{U\downarrow}(r \rightarrow \infty) \propto \begin{pmatrix} -\sin \theta/2 e^{i\phi} \\ \cos \theta/2 \end{pmatrix}. \quad (17.71)$$

- (b) Show that the scattered electron is *not* an eigenstate of spin but is an eigenstate of the helicity operator $\boldsymbol{\sigma} \cdot \mathbf{k}$.
2. Determine the relation between the scalar functions $b^{(\pm)}(p)$ in the plane wave expansion (15.34) and the two-component $\chi^{(\pm)}(p)$'s in the expansion of the general solution $\Psi(\mathbf{p})$ (17.43).
 3. What is the coordinate-space form $G_D(\mathbf{x}', \mathbf{x})$ of the Dirac propagator? (An integral expression is okay.)
 4. Show that the equation satisfied by the Dirac propagator $G_D(\mathbf{x}', \mathbf{x})$ has a 4D unit source (delta function) at the origin.
 5. Show that the nonrelativistic limit of $G_D(\mathbf{x}', \mathbf{x})$ is the Schrödinger result.
 6. Show that for an unpolarized electron with velocity β and momentum $\mathbf{k}$ scattering from a nucleus of charge Ze (with some screening for convergence), the Born approximation yields

$$\frac{d\sigma}{d\Omega} = \frac{e^4 Z^2}{[2k\beta \sin^2(\frac{1}{2}\theta)]^2} \left[1 - (\beta \sin \frac{1}{2}\theta)^2 + \pi Z\alpha\beta \sin \frac{1}{2}\theta (1 - \sin \frac{1}{2}\theta) + O(\alpha^2) \right]. \quad (17.72)$$

7. Derive the momentum-space Dirac equation (17.23) from the LS, operator form.
8. Study the Dirac equation with a zeroth-component, nonlocal, separable potential

$$V(\mathbf{r}', \mathbf{r}) = \frac{-1}{m} v(r)v(r'). \quad (17.73)$$

- (a) Show that only S waves are affected by this potential.

- (b) Determine the integral equations for the Dirac wave functions.
- (c) Determine the integral equations for T^{++} and T^{+-} .
- (d) Derive the solutions for Ψ and T in the Born approximation.
- (e) Go for the full solution!

9. The probability of finding some electron along the z axis is given by

$$P(z) = P_0 \sin kz. \quad (17.74)$$

- (a) What are the four components of its wave function in the standard basis?
 - (b) What are its upper and lower spinors in terms of eigenstates of β ?
 - (c) What are its upper and lower spinors in terms of eigenstates of H_0 ?
10. An electron moves along the positive z axis with speed $v = 0.6$, spin up, and 10% uncertainty in momentum.
- (a) What are the four components of its wave function?
 - (b) What are its upper and lower spinors in terms of eigenstates of β ?
 - (c) What are its upper and lower spinors in terms of eigenstates of H_0 ?
11. Develop a formal version of Dirac theory starting with the Hamiltonian form of the Dirac equation (in § 17.3 we did this starting with the covariant Dirac equation). How do the Green's functions differ?
12. Show that if Ψ satisfies the scattering boundary conditions of incident plane wave and outgoing spherical wave shown in Figure 1.2,

$$\Psi \sim \Phi_{\text{in}} + \Psi_{\text{sc}}, \quad (17.75)$$

then the cross sections depend only upon the upper part of the asymptotic wave function, that is, show that

$$\sigma \propto \frac{j_{\text{sc}}(\Psi)}{j_{\text{in}}(\Psi)} \propto \frac{N j_{\text{sc}}(\psi_U)}{N j_{\text{in}}(\psi_U)} = \frac{j_{\text{sc}}(\psi_U)}{j_{\text{in}}(\psi_U)}. \quad (17.76)$$

13. Deduce that in momentum space, the Dirac equation for $j = l \pm \frac{1}{2}$ are the coupled 1D integral equations

$$\int_0^\infty dp' p'^2 v_l(p, p') G_{lj}^{(\pm)}(p') - p F_{lj}^{(\pm)}(p) = (E - m) G_{lj}^{(\pm)}(p), \quad (17.77)$$

$$\int_0^\infty dp' p'^2 v_{l+1}(p, p') F_{lj}^{(\pm)}(p') - p G_{lj}^{(\pm)}(p) = (E + m) F_{lj}^{(\pm)}(p), \quad (17.78)$$

where v_l is the potential in the partial-wave representation.

14. It is also possible to develop a formal version of Dirac theory closer to our work in Part I, *Scattering and Integral Quantum Mechanics*, by starting with the Hamiltonian form of the Dirac equation.

- (a) Show that right-multiplying G_D by γ^0 produces a Green's function with the following properties:

$$G_D \gamma^0 = \frac{1}{\not{p} - m} \gamma^0 = \frac{1}{E - \boldsymbol{\alpha} \cdot \mathbf{p} - \gamma^0 m} = \frac{1}{E - H_0} \quad (17.79)$$

$$= \int d^3k \left(\frac{|\mathbf{k}+\rangle \langle \mathbf{k}+|}{E - E_k} + \frac{|\mathbf{k}-\rangle \langle \mathbf{k}-|}{E + E_k} \right). \quad (17.80)$$

- (b) Show that this $G\gamma^0$ is the inverse operator for the Hamiltonian form of the Dirac equation.

CHAPTER 18

SOLVING EVEN RELATIVISTIC INTEGRAL EQUATIONS ☉

The power and accessibility of high-speed computers has changed our view as to what theories and equations in physics are soluble. In particular, it is now easy to numerically evaluate integrals and to solve integral equations, and in this chapter we discuss the techniques and the resulting matrix equations. Although these solutions may “only” be numerical, they are in some sense still “exact,” and they free us from the consideration of only simple potentials. We thus convert the momentum-space techniques we have previously developed from theoretical to practical tools.

18.1 Singular Integrals

In Chapter 5, *Green’s Functions: Integral Quantum Mechanics*, we used Cauchy’s theorem to evaluate a singular integral. Another approach is to use the theorem (5.30) to convert these singular integrals to a principal-value part plus an on-shell term:

$$\int_{-\infty}^{+\infty} dk \frac{f(k)}{k - k_0 \pm i\epsilon} = \mathcal{P} \int_{-\infty}^{+\infty} dk \frac{f(k)}{k - k_0} \mp f(k_0). \quad (18.1)$$

Because the explicit principal-part prescription for integrating through a singularity,

$$\mathcal{P} \int_{-\infty}^{+\infty} dk \frac{f(k)}{k - k_0} = \lim_{\epsilon \rightarrow 0} \left(\int_{-\infty}^{k_0 - \epsilon} \dots + \int_{k_0 + \epsilon}^{\infty} \dots \right), \quad (18.2)$$

is awkward on a computer, we base our numerical technique on the identity

$$\mathcal{P} \int_{-\infty}^{\infty} \frac{dk}{k - k_0} = 0. \quad (18.3)$$

Equation (18.3) means that on both sides of the singular point k_0 in Figure 18.1, the curve of $1/(k - k_0)$ has equal areas under it. A change of variable $k \rightarrow -k$ permits (18.3) to be

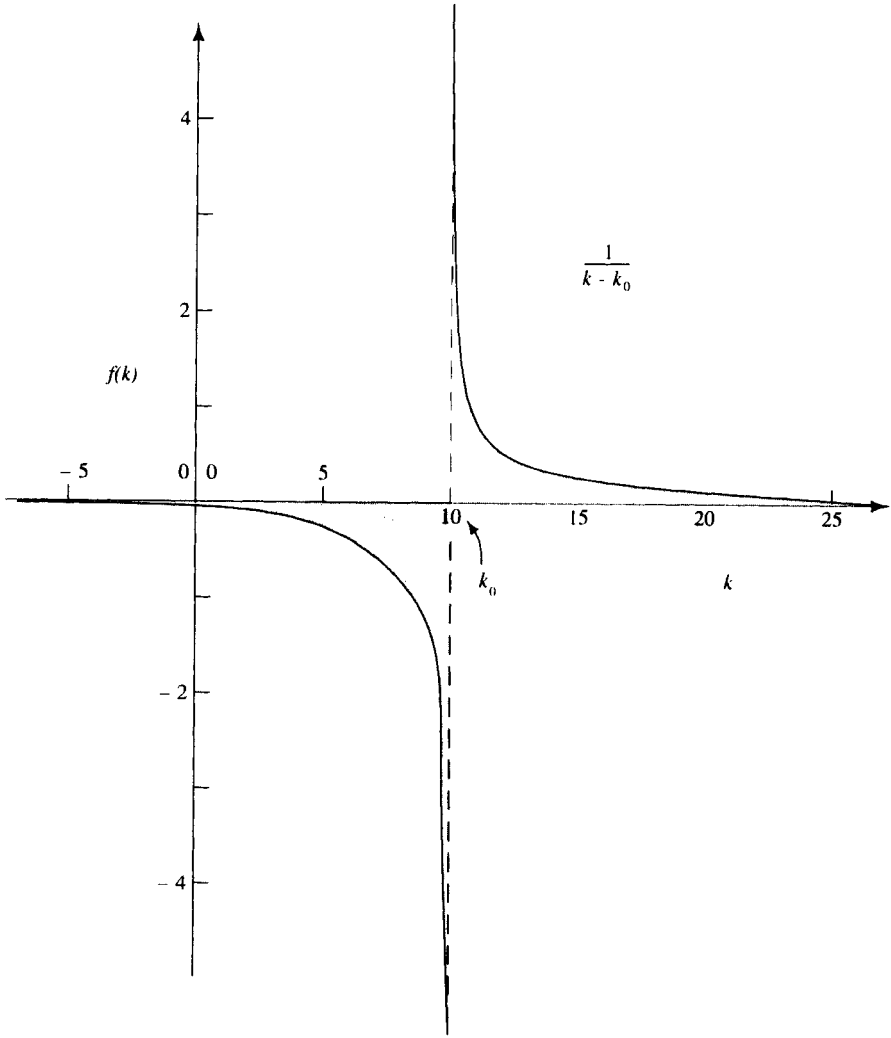


Figure 18.1: The function $1/(k - k_0)$ versus k . The total area under the curve is zero.

written as

$$\mathcal{P} \int_{-\infty}^{\infty} \frac{dk}{k - k_0} = -\mathcal{P} \int_0^{\infty} \frac{dk}{k + k_0} + \mathcal{P} \int_0^{\infty} \frac{dk}{k - k_0}, \quad (18.4)$$

$$\Rightarrow \mathcal{P} \int_0^{\infty} \frac{dk}{k^2 - k_0^2} = 0. \quad (18.5)$$

Although the new integrand (Figure 18.2) is distorted, there are still equal areas on either side of the pole.

We apply (18.5) by observing that the principal-part exclusion of the singular point's contribution (18.2) is equivalent to the subtraction of a zero integral,

$$\boxed{\mathcal{P} \int_0^{\infty} dk \frac{f(k)}{k^2 - k_0^2} = \int_0^{\infty} dk \frac{[f(k) - f(k_0)]}{k^2 - k_0^2}.} \quad (18.6)$$

The integrand on the RHS is nonsingular, and can therefore be evaluated numerically.

Exercise Show that the $k = k_0$ value of the integrand equals $[df(k)/dk]/2k$. $\diamond$

18.2 Numerical Integration

We approximate an integral as a sum over N Gaussian quadrature points $\{k_j; j = 1, N\}$, each weighted by w_j :¹

$$\int_{-1}^1 dk f(k) \simeq \sum_{j=1}^N w_j f(k_j). \quad (18.7)$$

The points k_j and weights w_j for integration from -1 to 1 are widely available (Abramowitz and Stegun, 1964). For other intervals they are obtained from these by scaling, for example,

$$\int_A^B dp f(p) \simeq \sum_{j=1}^N w'_j f(p_j) \quad (18.8)$$

$$p_j = \frac{1}{2}(B + A) + \frac{1}{2}(B - A)k_j, \quad w'_j = \frac{1}{2}(B - A)w_j. \quad (18.9)$$

In the equations to follow we have the interval $(A, B) = (0, \infty)$. Integration points and weights for this range can be obtained by the scaling:

$$p_j = \frac{1 + k_j}{1 - k_j} C, \quad w'_j = \frac{2C}{(1 - k_j)^2} w_j, \quad (18.10)$$

where C is a parameter which sets the scale for the point distribution (it is the midpoint).

¹If $f(k)$ is a polynomial of degree $2N - 1$, Gaussian quadrature is exact if the k_j 's are the zeroes of the Legendre polynomials. Modifications of the method exist for $f(k)$ containing exponential factors, square roots, powers of k , and so on. Although the goodness of the approximation depends on the details of the function f , many physical applications typically use 32 or fewer grid points for seven or more places of precision.

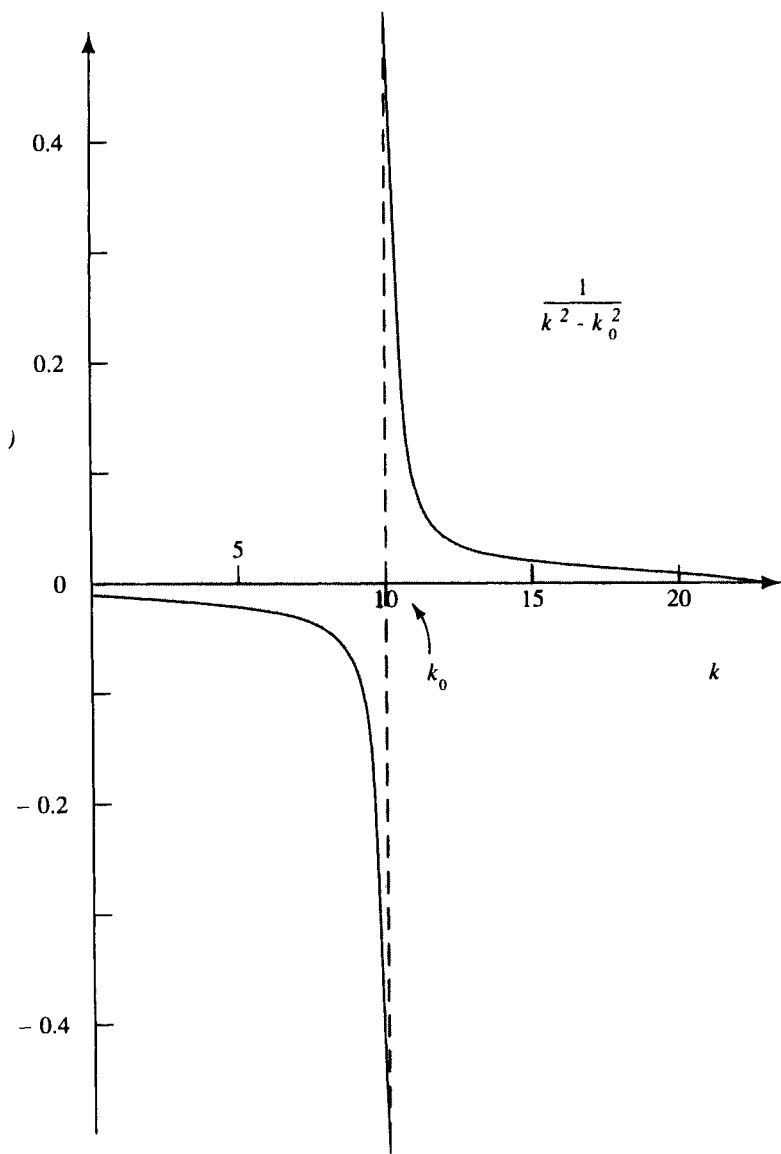


Figure 18.2: The function $1/(k^2 - k_0^2)$ versus k . The total area under the curve is zero.

18.3 Reduction of LS Equation to Linear Equations

To solve the partial-wave Lippmann–Schwinger equation,

$$T_l(k', k; E) = V_l(k', k) + \frac{2}{\pi} \int_0^\infty dp \frac{p^2 V_l(k', p) T_l(p, k; E)}{E - E_p + i\epsilon}, \quad (18.11)$$

we could break the integral into principal-value and delta function parts and keep track of each (we will in soon). Alternatively, we keep the analysis simpler (and real) by only including the principal part of the integral,

$$R_l(k', k; E) = V_l(k', k) + \frac{2}{\pi} \mathcal{P} \int_0^\infty dp \frac{p^2 V_l(k', p) R_l(p, k; E)}{E - E_p}, \quad (18.12)$$

that is, solving for the R matrix. As we know from Chapter 8, *The Angular Momentum Basis*, the solutions of (18.11) and (18.12) are related because both determine the same scattering phase shift:

$$R_l(k_0, k_0; E_{k_0}) = -\frac{\tan \delta_l}{\rho_T}, \quad \rho_T = 2\mu k_0, \quad (18.13)$$

$$T_l(k_0, k_0; E_{k_0}) = -\frac{e^{i\delta_l} \sin \delta_l}{\rho_T}, \quad \mu = \frac{m_T m_P}{m_T + m_P}. \quad (18.14)$$

To solve (18.12), we rewrite (Haftel and Tabakin, 1970) the principal-part prescription in a form which can be computed:

$$R_l(k', k) = V_l(k', k) + \frac{2}{\pi} \int_0^\infty dp \frac{p^2 V_l(k', p) R_l(p, k) - k_0^2 V_l(k', k_0) R_l(k_0, k)}{(k_0^2 - p^2)/2\mu}. \quad (18.15)$$

We convert the integral equation into linear equations by approximating the integral as a sum over N Gaussian quadrature points $\{k_j; j = 1, N\}$, each weighted by w_j :

$$\begin{aligned} R(k, k_0) = & V(k, k_0) + \frac{2}{\pi} \sum_{j=1}^N \frac{k_j^2 V(k, k_j) R(k_j, k_0) w_j}{(k_0^2 - k_j^2)/2\mu} \\ & - \frac{2}{\pi} \left[\sum_{m=1}^N \frac{w_m}{(k_0^2 - k_m^2)/2\mu} \right] k_0^2 V(k, k_0) R(k_0, k_0). \end{aligned} \quad (18.16)$$

Note that the last term in (18.16) implements the principal-part prescription and is needed to cancel the singular parts of the first term. [Even though the term in brackets looks like an approximation to the zero integral (18.5), it cannot be discarded because it cancels off the numerical errors arising from the (otherwise) singular integrand.]

Equation (18.16) contains the $N + 1$ unknowns, $R(k_j, k_0)$ for $j = 1, N$ and $R(k_0, k_0)$. To solve it for the on-shell $R(k_0, k_0)$ requires solving it for the half-off-shell $R(k_j, k_0)$ for all k_j . This is done by evaluating the equation for k on a *grid*.²

$$k_i = \begin{cases} k_j, & j = 1, N \text{ (quadrature points)}, \\ k_0, & i = 0 \text{ (on-energy-shell point)}. \end{cases} \quad (18.17)$$

²The same method is used to solve for the fully off-shell R or T matrix. In that case $[R]$ would be a square matrix rather than a vector.

There are now $N + 1$ unknowns $R(k_i, k_0) \equiv R_i$, and $N + 1$ linear equations for them:

$$R_i = V_i + \frac{2}{\pi} \sum_{j=1}^N \frac{k_j^2 V_{ij} R_j w_j}{(k_0^2 - k_j^2)/2\mu} - \frac{2}{\pi} \left(\sum_{m=1}^N \frac{w_m}{(k_0^2 - k_m^2)/2\mu} \right) k_0^2 V_{ii} R_0. \quad (18.18)$$

To assist computation, we express these equations as *matrix equations* by combining the energy denominators and weights into a single denominator function:

$$D_i = \begin{cases} +\frac{2}{\pi} \frac{w_i k_i^2}{E(k_i) - E_{k_0}}, & \text{for } i = 1, N, \\ -\frac{2}{\pi} 2\mu k_0^2 \sum_{j=1}^N \frac{w_j}{k_j^2 - k_0^2}, & \text{for } i = N + 1. \end{cases} \quad (18.19)$$

This reduces (18.18) to

$$R_i + \sum_{j=0}^N V_{ij} D_j R_j = V_i, \quad (18.20)$$

or, in matrix form,

$$[F]_{N+1 \times N+1} [R]_{N+1} = [V]_{N+1}, \quad (18.21)$$

$$F_{ij} \stackrel{\text{def}}{=} \delta_{ij} + D_j V_{ij} \quad (\text{no implied sum}), \quad (18.22)$$

where $[R]$ and $[V]$ are column vectors and $[F]$ is a $(N + 1) \times (N + 1)$ matrix. The solution to (18.21) is

$$[R] = [F]^{-1} [V], \quad (18.23)$$

where $[F]^{-1}$ is the inverse of the wave matrix $[F]$ (18.22) (or equivalently the matrix representation $[F^{-1}]$ of the inverse operator F^{-1}). Because the inversion of (even complex) matrices is a standard subroutine on scientific computers (18.23) represents a *direct solution* for the half-off-shell R -matrix elements which via (18.13) yields the phase shifts. (*Note:* Although this is an elegant solution, it is numerically more efficient to solve for R without explicit inversion of F .)

Solution for T Matrix

The numerical solution of (18.11) for the T matrix follows the same steps used to solve (18.12) for the R matrix. We use the identity (18.6) to convert the $i\epsilon$ integral into a principal-value part and an on-shell term:

$$\begin{aligned} T_l(k', k; E) &= V_l(k', k) + \frac{2}{\pi} \mathcal{P} \int_0^\infty dp \frac{p^2 V_l(k', p) T_l(p, k; E)}{E - E_p} \\ &\quad - 2i\mu(k_0) k_0 V_l(k', k_0) T_l(k_0, k; E), \end{aligned} \quad (18.24)$$

where μ is the reduced mass (18.14). Now the last term is incorporated into the numerical analysis by adding an imaginary term to the on-shell Green's function (18.19):

$$D_{N+1} = -\frac{2}{\pi} 2\mu k_0^2 \sum_{j=1}^N \frac{w_j}{k_j^2 - k_0^2} - 2\mu i k_0. \quad (18.25)$$

The solution proceeds as before, only now with complex matrices.

Relation to Formal Theory and Bound States

The relation between the abstract and matrix forms of the dynamical equations is the same for both the R and T matrices. Comparison of (18.20)–(18.23), our matrix approximation to the LS equation, with the operator form of the LS equation [(7.65) and (7.71), respectively],

$$T = V + V \frac{1}{E - H_0 + i\epsilon} T, \quad (18.26)$$

$$T = \left(1 - V \frac{1}{E - H_0 + i\epsilon} \right)^{-1} V = \Omega^{(+)} V, \quad (18.27)$$

shows that we have matrix representations of the abstract operators T , V , and Ω , with $[F]^{-1}$ the matrix representing the wave operator $\Omega^{(+)}$ (7.36). Yet because $\Omega^{(+)}$ produces the distorted wave from a plane wave,

$$|\psi\rangle = \Omega^{(+)} |\phi\rangle, \quad (18.28)$$

the solution for R or T also gives us the wave function:

$$u_l(kr) = N_0 \sum_{i=1}^N j_l(k_i r) F^{-1}(k_i, k_0), \quad (18.29)$$

where N_0 is a normalization constant.³

Other formal equations can also be represented and solved. For example, the bound-state connection (7.75),

$$\det(1 - GV) = 0, \quad (18.30)$$

becomes the matrix relation

$$\det[F] = 0, \quad (18.31)$$

which can be solved even for complex matrices and coupled channels (see footnote). Of course, (18.31) is just one way of looking at the problem; when there are true bound states, we also could solve the eigenvalue problem:

$$H\psi = E\psi, \quad (18.32)$$

$$\int d^3p \langle \mathbf{k} | H | \mathbf{p} \rangle \psi(\mathbf{p}) = E\psi(\mathbf{k}). \quad (18.33)$$

In the partial-wave $|klm\rangle$ basis of Chapter 8, (and, for central potentials), (18.33) becomes

$$\frac{k^2}{2\mu} \psi_l(k) + \frac{2}{\pi} \int_0^\infty dp p^2 V_l(k, p) \psi_l(p) = E\psi_l(k). \quad (18.34)$$

By evaluating H and ψ on a grid,

$$H_{ij} = \frac{k_i^2}{2\mu} \delta_{ij} + \frac{2}{\pi} V_{ij} k_j^2 w_j, \quad \psi_i = \psi(k_i), \quad \text{for } i, j = \{1, N\}, \quad (18.35)$$

we obtain the corresponding matrix *eigenvalue* problem:

$$[H]_{N \times N} [\psi]_N = E [\psi]_N. \quad (18.36)$$

Note that because the energy is the eigenvalue of this equation, there is no *on-shell* grid point.

³Details regarding the numerical solution and the conversion to outgoing wave boundary conditions are found in the Haftel and Tabakin (1970) and Landau (1982). The extensions for bound states and coupled channels are in Kwon and Tabakin (1978) and Landau (1983).

18.4 Relativistic Generalizations

Relativistic Schrödinger Equation

As described in Chapter 13, *Relativistic Wave Equations for Spinless Particles*, we generalize the Schrödinger equation by using relativistic expressions for the energy of the projectile and target:

$$E_p = E_P(p) + E_T(p) = \sqrt{p^2 + m_P^2} + \sqrt{p^2 + m_T^2}. \quad (18.37)$$

Consequently, the singular integrals over Green's functions in the LS equation (18.11) become

$$\int_0^\infty \frac{dp f(p)}{E_P(k_0) + E_T(k_0) - E_P(p) - E_T(p) + i\epsilon} = \quad (18.38)$$

$$\begin{aligned} & \mathcal{P} \int_0^\infty \frac{dp f(p)}{E_P(k_0) + E_T(k_0) - E_P(p) - E_T(p)} - \frac{i}{k_0} \pi \mu(k_0) f(k_0), \\ \mu(k_0) &= \frac{E_P(k_0) E_T(k_0)}{E_P(k_0) + E_T(k_0)}. \end{aligned} \quad (18.39)$$

Here $\mu(k_0)$ is the relativistic generalization of the reduced mass and arises from converting the p integration in (18.38) to an integration over energy:

$$\frac{1}{E - E' + i\epsilon} = \frac{\mathcal{P}}{E - E'} - i\pi\delta(E - E'), \quad (18.40)$$

$$p dp = \mu(p) dE_p. \quad (18.41)$$

Exercise Verify (18.41) for relativistic *and* nonrelativistic E_p . ◊

The previous numerical techniques for implementing the principal-value prescription apply even for relativistic energies. Because the singularity still occurs at $p = k_0$, we again subtract a function that equals the integrand at the singular point and has zero integral:

$$\begin{aligned} & \mathcal{P} \int_0^\infty \frac{dp f(p)}{E_P(p) + E_T(p) - E_T(k_0) - E_P(k_0)} = \\ & \int_0^\infty dp \left[\frac{f(p)}{E_P(p) + E_T(p) - E_T(k_0) - E_P(k_0)} - \frac{f(k_0)}{(p^2 - k_0^2)/2\mu(k_0)} \right]. \end{aligned} \quad (18.42)$$

Exercise Show that (18.42) follows from the limit

$$\lim_{p \rightarrow k_0} \frac{p^2 - k_0^2}{2\mu(k_0)} = E_p - E_{k_0}, \quad [E_p \stackrel{\text{def}}{=} E_P(p) + E_T(p)]. \quad (18.43) \quad \diamond$$

The integral form of the relativistic Schrödinger equation now becomes

$$R_l(k', k_0) = V_l(k', k_0) + \frac{2}{\pi} \int_0^\infty dp \left[\frac{p^2 V_l(k', p) R_l(p, k)}{E_{k_0} - E_p} - \frac{k_0^2 V_l(k', k_0) R_l(k_0, k_0)}{(k_0^2 - p^2)/2\mu(k_0)} \right], \quad (18.44)$$

where the subtracted term makes the integrand nonsingular but has zero integral.

Bound States

The bound-state energies for the relativistic Schrödinger equation are determined from either (18.31) or (18.32) just by using a relativistic definition of the energy:

$$\langle \mathbf{p} | H | \mathbf{p}' \rangle = \sqrt{p^2 + m^2} \delta(\mathbf{p} - \mathbf{p}') + \langle \mathbf{p} | V | \mathbf{p}' \rangle. \quad (18.45)$$

The nonrelativistic $p^2/2\mu$ term gets replaced by a radical, and the zero point of energy changes.

Klein–Gordon Equation

The form of the Klein–Gordon equation with interactions depends on whether the interaction is a scalar, the timelike component of a 4-vector, or the spacelike component of a 4-vector. As an example, we consider a local potential that behaves timelike. The momentum-space Klein–Gordon equation is then

$$[E - V]^2 \psi(p) = [p^2 + \mu^2] \psi(p), \quad (18.46)$$

where the reduced mass μ is chosen to agree with the nonrelativistic limit of the potential. One way to solve (18.46) is to cast it into the form of an eigenvalue problem,

$$[p^2 + 2EV - V^2] \psi(p) = \epsilon \psi(p), \quad (18.47)$$

$$\epsilon = (E - \mu)(E + \mu), \quad (18.48)$$

and then solve it for the “eigenvalue” ϵ . Yet because the effective Hamiltonian (term in brackets) depends on the energy E and thus on the eigenvalue, the equation must be solved iteratively; the old energy is used in the bracket when solving for a new energy, then the new energy is used in the bracket for yet a newer energy, until the change in energy is insignificant. In practice, this technique converges rapidly, with the solution given in line 3 of Table 18.1 requiring three or four iterations.

Dirac Equation

In Chapter 16, *Interactions in Dirac Theory*, we decomposed the Dirac equation into the coupled radial equations in coordinate space, (16.34) and (16.35). In momentum space, these equations are represented as

$$\int_0^\infty dp p^2 V_l(k, p) G_{lj}(p) - k F_{lj}(k) = (E - \mu) G_{lj}(k), \quad (18.49)$$

$$\int_0^\infty dp p^2 V_{l+1}(k, p) F_{lj}(p) - k G_{lj}(k) = (E + \mu) F_{lj}(k). \quad (18.50)$$

For bound states, these equations constitute an eigenvalue problem of the type we solved analytically for the hydrogen atom. If the equations are solved on a grid, they become the matrix eigenvalue problem (18.36), where the matrices are enlarged to accommodate the coupling of the upper and lower components:

$$\begin{pmatrix} [V_{ij} k_j^2 w_j]_{N \times N} & [k_i \delta_{ij}]_{N \times N} \\ [k_i \delta_{ij}]_{N \times N} & [2\mu \delta_{ij} - V_{ij} k_j^2 w_j]_{N \times N} \end{pmatrix} \begin{pmatrix} [G_{ij}]_{N \times 1} \\ [F_{ij}]_{N \times 1} \end{pmatrix} = (\mu - E) \begin{pmatrix} [G_{ij}]_{N \times 1} \\ [F_{ij}]_{N \times 1} \end{pmatrix}. \quad (18.51)$$

Here the Hamiltonian and wave functions have become supermatrices constructed from $N \times N$ and $N \times 1$ submatrices. While the larger matrices may be more work for the computer, conceptually it is the same problem.

18.5 Coulomb-Like Forces in Momentum Space

The theory and equations of quantum mechanics are represented equally well in coordinate or momentum space. Bound states problems, which by definition deal with normalizable wave functions, can actually be solved equally well in either space, while scattering problems, which in the time-independent Schrödinger theory deal with non-normalizable states, are more challenging in momentum space. This challenge arises, in part, because boundary conditions are more naturally imposed in coordinate space, and, in part, because non-normalizable states contain singularities in momentum space and, accordingly, have no Fourier transforms (Hernandez and Mondragon, 1984). In spite of the difficulties, momentum-space calculations are important because momentum space is where one derives the nonlocal potentials of many-body and field theories, and because there are fewer approximations needed in momentum space to handle them.

The Coulomb problem in momentum space has actually been “solved” a number of times, possibly starting with Fock’s study of the hydrogen atom (Fock, 1935), yet no one numerical approach appears to provide the requisite precision for all applications. The real “problem” is that the Coulomb potential arising from a finite-size target,

$$V_c(\mathbf{k}', \mathbf{k}) = -\frac{Ze^2}{2\pi^2 q^2} \rho(q), \quad q = |\mathbf{k} - \mathbf{k}'|, \quad (18.52)$$

has a $1/q^2$ singularity arising from the infinite range of the Coulomb potential. (The form factor $\rho(q)$ accounts for the finite size of the target’s charge distribution and makes the potential well-behaved at large q , but not at $q = 0$.) The singularity in (18.52) must be regularized before a numerical solution is implemented, and in this section we describe one method used for bound states and a different one used for scattering.

Bound States in p -Space with Coulomb Forces

For bound-state problems we wish to solve a momentum-space equation such as the Schrödinger equation,

$$\frac{k^2}{2\mu} \psi_l(k) + \frac{2}{\pi} \int_0^\infty dp p^2 V_l(k, p) \psi_l(p) = E \psi_l(k), \quad (18.53)$$

where V_l contains the Coulomb plus a short-range potential (in $\mathbf{r}$ -space). We consider the standard attractive Coulomb potential for a point charge,

$$\langle \mathbf{r} | V^c | \mathbf{r}' \rangle = -\frac{Ze^2}{r} \delta(\mathbf{r} - \mathbf{r}'). \quad (18.54)$$

It has the momentum-space forms:

$$\langle \mathbf{k} | V^c | \mathbf{p} \rangle = -\frac{Ze^2}{2\pi^2} \frac{1}{|\mathbf{k} - \mathbf{p}|^2}, \quad (18.55)$$

$$V_l^c(k, p) = -\frac{Ze^2}{2} \int_{-1}^1 \frac{dx P_l(x)}{k^2 + p^2 - 2kpx} = -\frac{Ze^2}{2kp} Q_l(z_{kp}), \quad (18.56)$$

$$z_{kp} \stackrel{\text{def}}{=} \frac{p^2 + k^2}{2kp}. \quad (18.57)$$

Here Q_l is the Legendre function of the second kind, for example,

$$Q_0(z) = \frac{1}{2} \ln \left| \frac{z+1}{z-1} \right| = \ln \left| \frac{k+p}{k-p} \right|. \quad (18.58)$$

The logarithmic singularity of the Coulomb potential (18.57) at $p = p'$ makes numerical solution of the Schrödinger equation (18.53) difficult. This singularity is removed (Kwon and Tabakin, 1978; Landau, 1983) by subtracting a term from the integrand in (18.53), which makes it nonsingular, and then adding in a correction term $S_l(k)$ to account for the nonvanishing of the integral of the subtracted term. For the pure Coulomb problem, the integral Schrödinger equation is then

$$E\psi_l(k) = \frac{k^2}{2\mu} \psi_l(k) + \frac{2}{\pi} \int_0^\infty dp V_l(k, p) \left[\psi_l(p)p^2 - \frac{\psi_l(k)k^2}{P_l(z_{kp})} \right] + \frac{2}{\pi} \psi_l(k)k^2 S_l(k), \quad (18.59)$$

$$S_l(k) = \int_0^\infty dp \frac{V_l^c(k, p)}{P_l(z_{kp})} = -\frac{Ze^2}{2k} \int_0^\infty dp \frac{Q_l(z_{kp})}{pP_l(z_{kp})}. \quad (18.60)$$

Amazingly, the integral $S_l(k)$ can be evaluated analytically, for example,

$$S_l(p) = -Z \frac{\pi e^2}{2p} \left(\frac{\pi}{2} - I_l \right), \quad I_{0,1,2,3,\dots} = 0, 1.0, 1.224745, 1.322855, \dots \quad (18.61)$$

Because the term in brackets in (18.59) vanishes for $k = p$, the integral is nonsingular and can be evaluated numerically. For k on the grid $k = k_m$, $m = 1, N$, there results:

$$\begin{aligned} \frac{k_m^2}{2\mu} \psi_l(k_m) + \frac{2}{\pi} \sum_{n \neq m}^N w_n V_l(k_m, k_n) \psi_l(k_n) k_n^2 \\ + \frac{2}{\pi} \left(\sum_{n \neq m}^N \frac{-V_l^c(k_m, k_n)}{P_l(z_{mn})} + S_l(k_m) \right) k_m^2 \psi_l(k_m) = E \psi_l(p). \end{aligned} \quad (18.62)$$

Table 18.1: Coulomb Binding Energies (eV) Calculated in Momentum Space

Atomic State	Theory	Exact	$N=20$	$N=40$	$N=60$
2P (μ^- – ^{40}Ca)	DE	280,912	282,120	280,801	280,905
3D (K^- – ^{32}S)	SE	367,866	36,7842	367,866	367,866
1S (π^- – ^{16}O)	KGE	236,653	23,6805	236,674	236,659
1S (K^- – p)	SE	8,613	8,623	8,613	8,613
2S (K^- – p)	SE	2,153	2,192	2,154	2,153

Abbreviations: DE, Dirac equation; SE, Schrödinger equation; KGE, Klein–Gordon equation.
(From Kwon and Tabakin, 1978; Landau, 1983.)

Note that the diagonal ($n = m$) part of the Coulomb potential makes no contribution to the sum in (18.62). By defining a pseudopotential term along the diagonal,

$$V_l^c(k_m, k_m) \equiv \frac{1}{w_m} \left(S_l(k_m) - \sum_{n \neq m}^N \frac{V_l^c(k_m, k_n)}{P_l(z_{mn})} \right), \quad (18.63)$$

equation (18.62) reduces to the matrix eigenvalue problem:

$$\left[\frac{k_m^2}{2\mu} \delta_{mn} + \frac{2}{\pi} V_l(k_m, k_n) w_n k_n^2 \right]_{N \times N} [\psi]_{N \times 1} = E [\psi]_{N \times 1}. \quad (18.64)$$

This method of solution is general and is also used with relativistic equations. For example, it has been used to solve for bound states arising from the combined Coulomb plus nuclear force (exotic hydrogen atoms with additional strong interactions). In Table 18.1 we show a comparison of numerical and analytic results for a Coulomb potential in both relativistic and nonrelativistic wave equations. We see that the integral equations can be solved for five-place precision by using 60 integration points (we can see that here, but once the strong interaction is included there will be no analytic answer with which to compare).

Momentum Space Scattering with Coulomb Forces

We have just solved the *bound-state* problem with the momentum-space Coulomb potential (18.52) by subtracting a term from the potential which makes its integral finite, and then adding in a correction for the integral of the subtracted term. In this section we solve the *scattering problem* in momentum space for a Coulomb potential acting in concert with a short-ranged, or nuclear, potential. This procedure, deduced by Vincent and Phatak (1974), effectively gives the Coulomb potential a finite range by cutting it off beyond some radius R_{cut} , as shown in Figure 18.3. After the problem is solved, the resulting wave function and T matrix are corrected for the cutoff.

Consider scattering from the combination of a short-ranged nonlocal nuclear potential $V_n(\mathbf{r}', \mathbf{r})$ and the infinite-ranged Coulomb potential $V_c(\mathbf{r})$. Because the nuclear potential

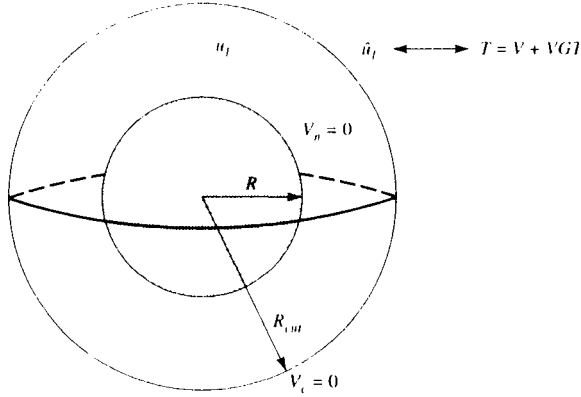


Figure 18.3: The VP procedure's partition of coordinate space into a region $r > R$ in which the nuclear potential vanishes, and a region in which the Coulomb potential is set equal to zero. The wave function in the outer region is denoted by $\hat{u}_l(k_0 r)$ and that in the intermediate region by $u_l(\eta, k_0 r)$.

V_n is nonlocal, the preferred method to obtain the scattering amplitude is to solve the Lippmann–Schwinger equation,

$$T_j(k', k) = V_j(k', k) + \frac{2}{\pi} \int_0^\infty p^2 dp \frac{V_j(k', p) T_j(p, k)}{E_{k_0} + i\epsilon - E_p}. \quad (18.65)$$

Here $V_{j=l\pm}(k', k)$ is the partial-wave matrix element of the momentum-space potential $V(\mathbf{k}', \mathbf{k})$, l is the orbital angular momentum, and $j = l \pm \frac{1}{2} \stackrel{\text{def}}{=} l \pm$ is the total angular momentum.⁴

Equation (18.65) is valid as long as the coordinate-space potential is of finite range, which in practice means that at some radius the potential is small enough to be ignored without significantly changing the predicted scattering observables. We indicate by the shaded area in Figure 18.3 the region in which the nuclear potential V_n acts, and the range for the nuclear potential by R . The coordinate-space Coulomb potential does not vanish rapidly enough to be considered as having a finite range, and although its strength may be weaker than the nuclear potential, it cannot be included with the nuclear potential in (18.65). The Vincent-Phatak procedure sets the coordinate-space Coulomb potential to zero (cuts it off) for all radii r greater than some fixed value R_{cut} :

$$V_c^{\text{cut}}(r) = V_c(r) \theta(R_{\text{cut}} - r). \quad (18.66)$$

The coordinate-space regions are illustrated in Figure 18.3 where we assume that R_{cut} is larger than the range R of the nuclear potential. The truncated Coulomb potential (18.66) has the momentum-space transform,

$$V_c^{\text{cut}}(\mathbf{k}', \mathbf{k}) = \frac{Z_P Z_T e^2}{2\pi^2 q^2} [\rho(q) - \cos(q R_{\text{cut}})], \quad q = |\mathbf{k} - \mathbf{k}'|. \quad (18.67)$$

⁴For a spin 0 projectile scattering from a spin 0 or spin $\frac{1}{2}$ target, there is no coupling of different channels and we need to solve only (18.65) with the Vincent–Phatak procedure. The extra complications of two spin $\frac{1}{2}$ particles can be found in Mefford (1994).

Since this has the $q \rightarrow 0$ limit of $Z_P Z_T e^2 R_{\text{cut}}^2 / (6\pi^2)$, we see that the $q = 0$ singularity of (18.55) has indeed been removed. Because the cutoff Coulomb potential is of finite range in coordinate space and without singularities in momentum space, its partial-wave decomposition can be added to that of the nuclear potential,

$$V_{l\pm}(k', k) = V_{n,l\pm}(k', k) + V_{c,l}^{\text{cut}}(k', k). \quad (18.68)$$

When inserted in the Lippmann–Schwinger equation (18.65), this combined potential produces a well-defined solution.

The solutions $T_{l\pm}(k', k)$ of the Lippmann–Schwinger equation can readily be transformed into coordinate-space wave functions for all values of r , as discussed in Chapter 6, *Transition and Potential Matrices*, and Chapter 8, *The Angular Momentum Basis*. Alternatively, just the on-shell element $T_{l\pm}(k_0, k_0)$ can be used to obtain the wave function any place outside of the shaded region in Figure 18.3. In the “outer” region, $r > R_{\text{cut}}$, both the nuclear and cutoff Coulomb potentials vanish. Consequently, the (unnormalized) wave function there is expressed as a linear combination of the regular plus either irregular or outgoing solutions of the potential-free Schrödinger equation [$F_l(k_0 r)$ plus either $G_l(k_0 r)$ or $H_l^{(+)}(k_0 r)$]:

$$\hat{u}_{j=l\pm 1/2}(k_0 r) = \begin{cases} e^{i\delta_{l\pm}} [\sin \delta_{l\pm} G_l(k_0 r) + \cos \delta_{l\pm} F_l(k_0 r)], & \text{for } r > R_{\text{cut}}, \\ F_l(k_0 r) + \hat{T}_{l\pm}(k_0) H_l^{(+)}(k_0 r), & \text{for } r > R_{\text{cut}}, \\ \sin(k_0 r - l\pi/2 + \delta_{l\pm}), & \text{for } r \rightarrow \infty. \end{cases} \quad (18.69)$$

In (18.69) we have used two equivalent forms for the free wave functions as well as their asymptotic limit. The reduced T matrix element $\hat{T}_{l\pm}(k_0)$ in (18.69) is related to a “preliminary” phase shift $\delta_{l\pm}$ by:

$$\hat{T}_{l\pm}(k_0) = e^{i\delta_{l\pm}(k_0)} \sin \delta_{l\pm}(k_0), \quad (18.70)$$

and to the solution $T_{l\pm}(k', k)$ of the Lippmann–Schwinger equation (18.65) by:

$$\hat{T}_{l\pm}(k_0) = -\rho_E T_{l\pm}(k_0, k_0), \quad \rho_E = 2k_0 \frac{E_P(k_0) E_T(k_0)}{E_P(k_0) + E_T(k_0)}. \quad (18.71)$$

To describe physical scattering observables we need a wave function which incorporates the full extent of the Coulomb force (or at least one with a cutoff of atomic dimensions, which is essentially at infinity in Figure 18.3). This, in turn, requires that the preliminary phase shifts δ_j be corrected for the artificial cutoff. The heart of the VP procedure is the observation that while there is no nuclear potential acting in the intermediate region between R and R_{cut} of Figure 18.3, there is the Coulomb potential there and that means the wave function in the intermediate region must be a linear combination of the regular and irregular Coulomb waves introduced in Chapter 4, *Scattering Applications: Lengths, Resonances, Coulomb*:

$$u_{l\pm}(\eta, k_0 r) = \begin{cases} F_l(\eta, k_0 r) + \hat{T}_{l\pm}^c(k_0) H_l^{(+)}(\eta, k_0 r), & \text{for } R \leq r \leq R_{\text{cut}}, \\ \sin[k_0 r - l\pi/2 + \delta_{l\pm}^c + \sigma_l - \eta \ln(2k_0 r)], & \text{for } r(\leq R_{\text{cut}}) \rightarrow \infty. \end{cases} \quad (18.72)$$

Here $\eta = Z_P Z_T e^2 / v$ is the Sommerfeld parameter, $F_l(\eta, k r)$ is the regular Coulomb function, and $H_l^{(+)} = G_l + iF_l$ is the outgoing Coulomb function.

The Coulomb-modified T matrix,

$$\hat{T}_{l\pm}^c(k_0) \stackrel{\text{def}}{=} e^{i\delta_{l\pm}^c} \sin \delta_{l\pm}^c, \quad (18.73)$$

is unknown, and our present purpose is to determine it, or equivalently the Coulomb-modified phase shift $\delta_{l\pm}^c$. This is done by requiring the logarithmic derivative of the intermediate region's wave function $u_{l\pm}(\eta, k_0 r)$ to match that of the exterior wave function $\hat{u}_{l\pm}(k_0 r)$ at $r = R_{\text{cut}}$:

$$\left. \frac{u'_{l\pm}(\eta, k_0 r)}{u_{l\pm}(\eta, k_0 r)} \right|_{r=R_{\text{cut}}-} = \left. \frac{\hat{u}'_{l\pm}(k_0 r)}{\hat{u}_{l\pm}(k_0 r)} \right|_{r=R_{\text{cut}}+}. \quad (18.74)$$

While r is not large enough (i.e. of atomic dimensions) for the $r \rightarrow \infty$ forms in (18.69) and (18.72) to be valid, in the intermediate region we can match the linear combination of free and Coulomb waves:

$$\hat{T}_{l\pm}^c(k) = \frac{\hat{T}_{l\pm}(k)[F_l(\eta, kr), H_l^{(+)}(kr)] + [F_l(\eta, kr), F_l(kr)]}{[F_l(kr), H_l^{(+)}(\eta, kr)] + \hat{T}_{l\pm}(k)[H_l^{(+)}(kr), H_l^{(+)}(\eta, kr)]}, \quad (18.75)$$

where the brackets indicate Wronskians evaluated at $r = R_{\text{cut}}$.⁵

As we expand the intermediate region by taking $R_{\text{cut}} \rightarrow \infty$, the intermediate region's wave function $u_{l\pm}(\eta, k_0 r)$ becomes the final physical wave function from which we can extract the experimental scattering observables. Consequently, we can now use the standard expression for the scattering amplitude describing scattering from a short-range potential in the presence of the Coulomb potential.

It is illuminating to note that if instead of matching, we had set the phase of the asymptotic limit of the intermediate-region wave function $u_{l\pm}(\eta, k_0 r)$ (18.72) equal to that of the asymptotic limit of the exterior wave function $\hat{u}_{j=l\pm\frac{1}{2}}(k_0 r)$ (18.69), we would have obtained

$$\delta_{l\pm} \sim \delta_{l\pm}^c + \sigma_l - \eta \ln(2kR_{\text{cut}}). \quad (18.76)$$

The $\ln(2kR_{\text{cut}})$ term, which arises from the specific distortion of wave functions caused by the point Coulomb force, is treated in the $R_{\text{cut}} \rightarrow \infty$ limit, just as we did in § 4.3. Accordingly, for all but the most forward of scatterings, the standard expansion of the scattering amplitude can be used with

$$\delta_{l\pm} \simeq \delta_{l\pm}^c + \sigma_l. \quad (18.77)$$

Substitution of (18.77) into the usual partial-wave expansion of the scattering amplitude, and some rearrangement, leads to the final expression for the (spin-nonflip) amplitude for scattering:

$$f(\theta) = f_{\text{pt}}^c(\theta) + f^{\text{nc}}(\theta), \quad (18.78)$$

$$f_{\text{pt}}^c(\theta) = -\frac{\eta}{2k_0 \sin^2(\theta/2)} \exp\{2i[\sigma_0 - \eta \ln \sin(\theta/2)]\}, \quad (18.79)$$

$$f^{\text{nc}}(\theta) = \frac{1}{2ik_0} \sum_{l=0}^{\infty} (2l+1) e^{2i\sigma_l} (e^{2i\delta_l^c} - 1) P_l(\cos \theta) \quad (18.80)$$

$$= \frac{1}{k_0} \sum_{l=0}^{\infty} (2l+1) e^{2i\sigma_l} \hat{T}_{l\pm}^c P_l(\cos \theta), \quad (18.81)$$

⁵Since this Wronskian is a functional of the solutions from two different Schrödinger equations, it is *not* independent of r .

where f_{pt}^c is the scattering amplitude for a point Coulomb potential, and f^{nc} is the amplitude for nuclear scattering in the presence of the Coulomb force. Note, that since the Coulomb-modified phase shift $\delta_{l\pm}^c$ is defined in (18.72) relative to Coulomb waves which are already shifted by the point Coulomb phase σ_l , the amplitude f^{nc} also includes the correction to point Coulomb scattering arising from the finite extent of the charge distribution.

18.6 Problems

1. Evaluate numerically the scattering from the delta shell potential examined in the problems of Chapter 8, *The Angular Momentum Basis*:

$$V(r) = \frac{\lambda}{2m} \delta(r - b). \quad (18.82)$$

Consider $\lambda b = \pi/2$ and $b = 2\pi$ and solve for the phase shift δ_1 by using the matrix form of the LS equation with four or eight grid points. Explore either of the following approaches.

- (a) Keep evaluating the terms in the Born (Neumann) series until you have two places of accuracy.
 - (b) Solve the entire problem by matrix inversion.
2. Compare the numerical and analytic evaluations of the singular integral J in the Problems section of Chapter 5, *Green's Functions: Integral Quantum Mechanics*.
 3. Consider the usual attractive Coulomb potential:

$$\langle \mathbf{r} | V^c | \mathbf{r}' \rangle = -Z \frac{e^2}{r} \delta(\mathbf{r} - \mathbf{r}'). \quad (18.83)$$

Show that it has the momentum-space forms:

$$\langle \mathbf{p} | V^c | \mathbf{p}' \rangle = -Z \frac{e^2}{2\pi^2} \frac{1}{|\mathbf{p} - \mathbf{p}'|^2}, \quad (18.84)$$

$$V_l^c(p, p') = -Z \frac{e^2}{2} \int_{-1}^1 \frac{dx P_l(\cos \theta)}{p^2 + p'^2 - 2pp' \cos \theta} \quad (18.85)$$

$$= -\frac{Ze^2}{2pp'} Q_l \left(\frac{p'^2 + p^2}{2pp'} \right), \quad (18.86)$$

where Q_l is the Legendre function of the second kind.

PART III

QUANTUM FIELDS

CHAPTER 19

SECOND QUANTIZATION

*The Naming of Cats is a difficult matter,
It isn't just one of your holiday games;
You may think at first I'm as mad as a hatter
When I tell you, a cat must have THREE DIFFERENT NAMES.*

—T. S. Eliot

We have found that theories combining relativity and quantum mechanics predict that strong potentials or short-distance confinement causes particles to be created or destroyed, in spite of our assumption of only one particle being present. To remove this inconsistency we need the formalism of second quantization. It is able to describe varying numbers of particles and ensure the proper symmetry (statistics) of the many-particle wave function. As a formalism, second quantization does not provide any new physical laws or replace any wave equations. It is just a consistent mathematical framework. After specifying the formalism in this chapter, we go on to apply it to creation and destruction of photons, electrons, positrons, fermion pairs, mesons, phonons, and hadrons. Our approach is deliberately the “old fashioned” perturbation theory exemplified by Heitler (1954). For a difficult subject such as field theory, we feel this makes the best connection with conventional quantum mechanics and develops best the needed physical intuition.

19.1 Occupation Number Space

The two basic postulates of second quantization are:

1. States of interacting particles can be expanded using non-interacting particle states as the basis.
2. The variables used to describe one-particle systems are capable of describing many-particle systems; that is, there is no basic change in the nature of a particle caused by the presence of other particles.

It is possible to imagine difficulties with these assumptions: If the elementary particles get polarized by their interactions, we are incorrect in assuming they are elementary; if the

vacuum is really a collective (that is, interacting) state of many particles, there may be no such thing as a noninteracting basis.

Occupation number or *Fock space* is an abstract mathematical space in which complete sets of states are enumerated by indicating the number of particles with a particular eigenvalue for a particular operator. The operator may be any of the familiar ones such as energy, 3-momentum, the harmonic-oscillator quantum number n , or any of the unfamiliar ones. The state vector for occupation number space has the form

$$\Psi \equiv |\Psi\rangle = |n_1, n_2, \dots, n_i, \dots\rangle \equiv \Psi(n_1, n_2, \dots, n_i, \dots), \quad (19.1)$$

where n_i is the number of particles in each state (slot), and the total number N of particles is the sum

$$N = \sum_{i=0}^{\infty} n_i. \quad (19.2)$$

This state vector Ψ has the properties:

- It is analogous to (but not equal to) to the wave function of the one-particle theory.
- It is a complete description of the N -body system.
- It is not a function of any particular set of coordinates.
- It is part of a set which forms a complete orthogonal basis, that is,

$$\langle \Psi | \Psi' \rangle = \langle n_1, n_2, \dots, n_i, \dots | n'_1, n'_2, \dots, n'_i, \dots \rangle = \prod_{i=1}^{\infty} \delta_{n_i, n'_i}. \quad (19.3)$$

The *vacuum state* $|0\rangle$ (which should be distinguished from its eigenvalues 0) is the state which has zero particles in every slot,¹

$$|0\rangle = |0, 0, \dots, 0, \dots\rangle \neq 0. \quad (19.4)$$

Construction of States

We develop here the second-quantized formalism to describe both fermion and boson states with the proper symmetry. For fermions, n_i must equal 0 or 1. If we place more than 1 fermion in a slot, the state fails to exist,

$$|n \geq 2\rangle_F = 0 |0\rangle = 0, \quad (19.5)$$

where the zero on the RHS does *not* mean we have the vacuum. For bosons, n_i can be any number.

The basic building element in Fock space is the *creation* operator $a_i^\dagger$ that creates one additional particle in position i . Its adjoint a_i is a *destruction* or *annihilation* operator that removes one particle from position i . Mathematically,

$$a_i |n_1, n_2, \dots, n_i, n_{i+1}, \dots\rangle = \pm \sqrt{n_i} |\dots, n_i - 1, n_{i+1}, \dots\rangle, \quad (19.6)$$

$$a_i^\dagger |n_1, n_2, \dots, n_i, n_{i+1}, \dots\rangle = \pm \sqrt{n_i + 1} |\dots, n_i + 1, n_{i+1}, \dots\rangle, \quad (19.7)$$

¹Yet (19.4) is only tentative. As we have already seen with the Dirac sea, the vacuum will not remain empty for long.

where for bosons only the positive sign occurs, whereas for fermions the sign depends on the specifics of the state. In either case, the $\sqrt{n_i}$ factor ensures that destroying a particle from an empty slot gives zero.

The Bose or Fermi statistics of the particles are built into the theory by using these definitions and by specifying the commutation and anticommutation relations of the creation and destruction operators. We start with a separate treatment for bosons and fermions and then see that both follow the same rules if appropriate commutation and anticommutation relations are substituted.

- *Boson* creation and destruction operators satisfy the *commutation relation*

$$[a_i, a_i^\dagger] = 1. \quad (19.8)$$

Here both operators act on particles in slot i , that is, (19.8) means

$$[a_i a_i^\dagger - a_i^\dagger a_i] |n_i\rangle = \left[(\sqrt{n_i+1})^2 - (\sqrt{n_i})^2 \right] |n_i\rangle = |n_i\rangle, \quad (19.9)$$

where we do not indicate the (unaffected) numbers of particles in the other slots.

- *Fermion* operators satisfy a similar relation involving the *anticommutator*

$$\{a_i, a_i^\dagger\} \equiv [a_i, a_i^\dagger]_+ = 1. \quad (19.10)$$

This means

$$\{a_i, a_i^\dagger\} |n_i\rangle = (a_i a_i^\dagger + a_i^\dagger a_i) |n_i\rangle = |n_i\rangle, \quad (19.11)$$

which follows from $aa^\dagger$ giving 0 or 1 while $a^\dagger a$ gives 1 or 0.

Because $a_i^\dagger$ is the adjoint of a_i [$a_i^\dagger = (a_i)^\dagger$], a and $a_i^\dagger$ reverse roles when acting to the left; $a_i^\dagger$ removes particles from a bra and a_i adds particles to a bra:

$$\langle n_i | a_i^\dagger = \sqrt{n_i} \langle n_i - 1 |, \quad \langle n_i | a_i = \sqrt{n_i + 1} \langle n_i + 1 |. \quad (19.12)$$

Proof

$$\langle n | a_i^\dagger | m \rangle = \sqrt{m+1} \langle n | m+1 \rangle = \sqrt{m+1} \delta_{n,m+1}. \quad (19.13)$$

Let $m = n - 1$:

$$\langle n | a_i^\dagger | m \rangle = \sqrt{n} \langle n-1 | m \rangle = (a_i | n \rangle)^\dagger | m \rangle. \quad (19.14)$$

Exercise Prove the second part of (19.12). $\diamond$

The operator

$$N_i \equiv \tilde{N}_i = a_i^\dagger a_i \quad (19.15)$$

counts the number of particles in position i and leaves the state unchanged:²

$$N_i | \cdots n_i \cdots \rangle = n_i | \cdots n_i \cdots \rangle. \quad (19.16)$$

²This is a funny way to count, annihilate whatever is in the slot and, if the slot still remains, refill it. The Three Stooges would approve, no doubt.

Exercise Prove (19.16) for both bosons and fermions and show that the total number operator

$$\tilde{N} = \sum_i a_i^\dagger a_i, \quad (19.17)$$

$$\text{where } \tilde{N}\psi = N\psi, \quad (19.18)$$

and N is the total number of particles in all slots. $\diamond$

Manipulating particles in different slots is also part of this formalism, although it means we must watch some more indices. If we take the positive (negative) sign for bosons (fermions), the basic relation for different slots is:

$$a_i^\dagger a_j^\dagger = \pm a_j^\dagger a_i^\dagger. \quad (19.19)$$

For bosons this means

$$[a_i^\dagger, a_j^\dagger] = [a_i, a_j] = 0. \quad (19.20)$$

That is, we may rearrange the particles at will because the order does not matter. Likewise, the original commutation relation (19.8) extends to different slots:

$$[a_i, a_j^\dagger] = \delta_{ij}. \quad (19.21)$$

To prove (19.20) and (19.21), we need to specify how the many-particle state vector Ψ is constructed. For bosons we use the convention:

$$|n_1, n_2, \dots\rangle = \dots \frac{(a_2^\dagger)^{n_2}}{\sqrt{n_2!}} \frac{(a_1^\dagger)^{n_1}}{\sqrt{n_1!}} |0, 0, \dots\rangle. \quad (19.22)$$

Exercise Verify that the normalization of the state (19.22) agrees with (19.3). $\diamond$

For fermions, the slot change operation (19.19) can be written as

$$\{a_i^\dagger, a_j^\dagger\} = \{a_i, a_j\} = 0. \quad (19.23)$$

This means the order of operators matters since for $i \neq j$ the state vector acquires a minus sign upon interchanging two identical fermions. If $i = j$, we may neither create nor destroy two fermions in the same slot because

$$(a_i^\dagger)^2 \Psi = (a_i)^2 \Psi = 0 \Psi = 0. \quad (19.24)$$

For fermions, we also use the definition (19.22) for the many-body wave function, that is, *we fill the first place first and the last place last*. Yet now the $n_i!$ denominators are superfluous (but benign) since for fermions $0! = 1! = 1$.

Example Is the two-particle fermion state $|1_1, 1_2\rangle$ properly symmetrized, that is, does

$$|1_1, 1_2\rangle = -|1_2, 1_1\rangle? \quad (19.25)$$

The answer is “yes” since

$$|1_1, 1_2\rangle = a_2^\dagger a_1^\dagger |0\rangle, \quad (19.26)$$

$$|1_2, 1_1\rangle = a_1^\dagger a_2^\dagger |0\rangle = -a_2^\dagger a_1^\dagger |0\rangle = -|1_1, 1_2\rangle. \quad \diamond \quad (19.27)$$

Once we have stacked the particles into slots with the proper symmetry via the construction procedure (19.22), we usually must go to the trouble of rearranging them in order to remove one. And since the last in is not necessarily the first out, this may give some unexpected results. The next exercise is an example.

Exercise Prove

$$a_1 |1, 1\rangle = -|0, 1\rangle, \quad a_1^\dagger |0, 1\rangle = -|1, 1\rangle, \quad \text{yet } a_2 |1, 1\rangle = |1, 0\rangle. \quad \diamond (19.28)$$

Apparently, because we have filled the first slot first, if we want to preserve the symmetry, we need to empty the first slot last.

19.2 Second Quantized Operators

The operators which act on Fock-space states are constructed from creation and destruction operators, as already seen for the number operator. Other *1-body operators* are variants of the number operator in that they count the amount of some variable contributed by the particles in each slot. For example, the kinetic energy and Hamiltonian operators are:³

$$T = \sum_i \frac{p_i^2}{2m} a_i^\dagger a_i, \quad H = \sum_i a_i^\dagger a_i E_i, \quad (19.29)$$

where p_i is the momentum in state i . Note that while (19.29) gives the Hamiltonian operator, the total energy,

$$E = \sum_i E_i n_i, \quad (19.30)$$

is *not* an operator in Fock space.

Exercise Check that these definitions of operators makes sense by evaluating them between Fock-space states. $\diamond$

Actually these two operators are especially simple because they are diagonal, that is, the creation and annihilation operators refer to the same slot. More general operators (such as the potential interaction V we shall use in field theory and to describe fermion pairing) may refer to different slots,

$$V = \sum_{i,j} \langle i|v|j\rangle a_j^\dagger a_i, \quad (19.31)$$

or even different slots for more than one particle [just wait until you cope with interaction Hamiltonians like (20.28)].

³We generally use no special symbol to indicate operators in Fock space, with the exception being the number operator N , which we sometimes call $\tilde{N}$ for clarity.

Field Operators

We now specialize the index i to represent momentum eigenstates in a box (Appendix A, *Natural Units and Plane Waves*):

$$i \equiv \mathbf{p}_i = \frac{2\pi}{L}(i_x, i_y, i_z). \quad (19.32)$$

Thus $a_i = a_{\mathbf{p}_i}$ destroys a free particle of momentum $\mathbf{p}$ and spin s (we generally will leave off the spin index unless needed).

It is natural to ask if there is an operator which is the coordinate space analog of $a_{\mathbf{p}}$, that is, an operator which *destroys* a particle at position $\mathbf{r}$. There is, it is called the *field operator* $\Psi(\mathbf{r})$, and its mathematical definition is identical to that obtained by considering $\Psi(\mathbf{r})$ as “ $a_{\mathbf{r}}$ ”:

$$\Psi(\mathbf{r}) \leftrightarrow a_{\mathbf{r}}, \quad (19.33)$$

$$\Psi(\mathbf{r}) = \sum_{\mathbf{p}} \langle \mathbf{r} | \mathbf{p} \rangle \langle \mathbf{p} | a \rangle. \quad (19.34)$$

This leads to the standard form for the field operator:

$$\Psi(\mathbf{r}) = \sum_{\mathbf{p}} \frac{e^{i\mathbf{p} \cdot \mathbf{r}}}{\sqrt{V}} a_{\mathbf{p}}. \quad (19.35)$$

While the definition (19.35) is correct, the identification (19.33) of $\Psi(\mathbf{r})$ with $a_{\mathbf{r}}$ is just pedagogical because $\mathbf{r}$ is not a quantized variable. Yet this is what the “second” in second quantization stands for; the field operator $\Psi(\mathbf{r})$ is the analog of the wave function only now quantized. Indeed, (19.35) looks like the expansion of the wave function in plane waves, with $a_{\mathbf{p}}$ as the expansion coefficients.

The adjoint $\Psi^\dagger(\mathbf{r})$ of the field operator Ψ *creates* a particle at point $\mathbf{r}$, that is,

$$|\mathbf{r}\rangle = \Psi^\dagger(\mathbf{r}) |0\rangle. \quad (19.36)$$

Proof

$$\Psi^\dagger(\mathbf{r}) |0\rangle = \sum_{\mathbf{p}} \frac{e^{-i\mathbf{p} \cdot \mathbf{r}}}{\sqrt{V}} a_{\mathbf{p}}^\dagger |0\rangle \quad (19.37)$$

$$= \sum_{\mathbf{p}} \frac{e^{-i\mathbf{p} \cdot \mathbf{r}}}{\sqrt{V}} |\mathbf{p}\rangle = \sum_{\mathbf{p}} |\mathbf{p}\rangle \langle \mathbf{p} | \mathbf{r} \rangle = |\mathbf{r}\rangle. \quad (19.38)$$

Exercise Confirm that $\langle \mathbf{r}' | \mathbf{r} \rangle = \delta(\mathbf{r}' - \mathbf{r})$ for $|\mathbf{r}\rangle$ defined by (19.36). $\diamond$

The field operator for bosons and its adjoint satisfy the commutation relations:

$$[\Psi(\mathbf{r}), \Psi(\mathbf{r}')] = 0, \quad [\Psi^\dagger(\mathbf{r}), \Psi^\dagger(\mathbf{r}')] = 0, \quad [\Psi(\mathbf{r}), \Psi^\dagger(\mathbf{r}')] = \delta(\mathbf{r} - \mathbf{r}'). \quad (19.39)$$

In this Fock-space shell game, there is no interference if you destroy or create two bosons at any point in space, although creating one and then destroying it at the same point is not equivalent to destroying it and then creating it.

Because the requirement of wave function antisymmetry for *fermions* produces the possibility of interference over all space, fermions fields do not satisfy commutation relations. They do, however, satisfy the analogous anticommutation relations,

$$\{\Psi(\mathbf{r}), \Psi(\mathbf{r}')\} = 0, \quad \{\Psi^\dagger(\mathbf{r}), \Psi^\dagger(\mathbf{r}')\} = 0, \quad \{\Psi(\mathbf{r}), \Psi^\dagger(\mathbf{r}')\} = \delta(\mathbf{r} - \mathbf{r}'). \quad (19.40)$$

Exercise Prove (19.39) and (19.40) by direct substitution of the expansions for the field operator and its adjoint. $\diamond$

Dynamical Operators

There are two ways in which dynamical variables are expressed in terms of fields. The direct way is to take the expression in terms of creation and destruction operators, such as (19.35), and substitute for the a 's in terms of the fields. This uses the inverse relation to the field expansion:

$$a_{\mathbf{p}}^\dagger = \int d^3r \frac{e^{i\mathbf{p}\cdot\mathbf{r}}}{\sqrt{V}} \Psi^\dagger(\mathbf{r}), \quad a_{\mathbf{p}} = \int d^3r \frac{e^{-i\mathbf{p}\cdot\mathbf{r}}}{\sqrt{V}} \Psi(\mathbf{r}). \quad (19.41)$$

The other way is to take a (familiar) expression using wave functions and substitute the field operator for the wave function. After trying the first method a few times, the second method becomes more obvious.

For example, consider the number operator with the first method:

$$N \equiv \tilde{N} = \sum_{\mathbf{p}} a_{\mathbf{p}}^\dagger a_{\mathbf{p}} = \int d^3r d^3r' \sum_{\mathbf{p}} \frac{e^{i\mathbf{p}\cdot\mathbf{r}}}{\sqrt{V}} \frac{e^{-i\mathbf{p}\cdot\mathbf{r}'}}{\sqrt{V}} \Psi^\dagger(\mathbf{r}) \Psi(\mathbf{r}'). \quad (19.42)$$

We replace the sum by its equivalent integral (A.19):

$$\frac{1}{V} \sum_{\mathbf{p}} \rightarrow \int \frac{d^3p}{(2\pi)^3}, \quad (19.43)$$

and obtain the “familiar:”

$$\tilde{N} = \int d^3r \Psi^\dagger(\mathbf{r}) \Psi(\mathbf{r}). \quad (19.44)$$

Likewise, the density operator is

$$\rho = \Psi^\dagger(\mathbf{r}) \Psi(\mathbf{r}). \quad (19.45)$$

We see again that what was a continuous wave function in Schrödinger theory is now quantized into a field operator (the “second” quantization), and that these expressions for N and ρ could have been deduced from the familiar wave function expressions. Some other operators are more obvious after the fact, for example, the nonrelativistic kinetic energy operator (19.29),

$$T = \sum_i \frac{p_i^2}{2m} a_i^\dagger a_i = - \int d^3r \frac{\nabla \Psi^\dagger \cdot \nabla \Psi}{2m}, \quad (19.46)$$

and the current operator,

$$\mathbf{j}(\mathbf{r}) = \frac{1}{2im} \left[\Psi^\dagger \nabla \Psi - (\nabla \Psi^\dagger) \Psi \right]. \quad (19.47)$$

Exercise Prove (19.46) and (19.47), in one case by substituting for a_i and in the other case by substituting for Ψ . $\diamond$

Calculations with second-quantized operators almost always come down to evaluating commutators. In doing so it is helpful to remember two basic theorems:

$$[A, BC] = B[A, C] + [A, B]C, \quad (19.48)$$

$$[AB, C] = A[B, C] + [A, C]B. \quad (19.49)$$

The number operator is a particularly pleasant one to work with because of the commutation relations it has with the creation and destruction operators.

Exercise Prove that for either bosons or fermions:

$$\begin{aligned} [N_{\mathbf{k}}, a_{\mathbf{k}'}^\dagger] &= a_{\mathbf{k}}^\dagger \delta_{\mathbf{k}\mathbf{k}'}, & [N_{\mathbf{k}}, a_{\mathbf{k}'}] &= -a_{\mathbf{k}} \delta_{\mathbf{k}\mathbf{k}'}, \\ [N_{\mathbf{k}}, (a_{\mathbf{k}}^\dagger)^m] &= m(a_{\mathbf{k}}^\dagger)^m, & [N_{\mathbf{k}}, N_{\mathbf{k}'}] &= 0. \end{aligned} \quad (19.50)$$

The sense of these expressions should become (somewhat?) clearer as we use them in the applications to follow. Yet it is good to remember that these operators are all evaluated between Fock-space states, which themselves depend on the creation and destruction operators. For this reason, field-theory problems are nonlinear and difficult to solve exactly; in this book we use the simplest perturbation theory and refer the reader to Feynman and Hibbs (1965), Sakurai (1967), Bjorken and Drell (1964, 1965), Landau and Lifshitz (1971), Roman (1969), and Schweber (1959) for the fuller solutions and covariant perturbation theory.

19.3 Time Dependence

Schrödinger theory is inherently time dependent. Because second quantization is just a means of applying Schrödinger theory to a variable number of particles, we must include time dependence into second quantization. There are various ways to do this and each is acceptable as long as it yields the correct value for experimental observables. To be specific, consider the expectation value of an arbitrary operator O :

$$\langle O \rangle = \langle \phi(t) | O(t) | \phi(t) \rangle. \quad (19.51)$$

Here ϕ is the wave function (not field operator) that, along with O , may be a function of time. In the *Schrödinger picture*, the wave function has a time dependence generated by the full Hamiltonian:

$$i \frac{\partial}{\partial t} |\phi(t)\rangle = H |\phi(t)\rangle \quad (19.52)$$

$$\Rightarrow |\phi(t)\rangle = e^{-iHt} |\phi(0)\rangle, \quad \langle \phi(t) | = \langle \phi(0) | e^{iHt}. \quad (19.53)$$

If the operator O has no explicit time dependence, that is, no time-dependent external fields, the expectation value (19.51) takes the form:

$$\langle O \rangle = \langle \phi(0) | e^{iHt} O e^{-iHt} | \phi(0) \rangle. \quad (19.54)$$

In the *Heisenberg picture*, we have the same experimental observables but ascribe to the operators the time dependence generated by the Hamiltonian; that is, we take

$$O(t) = e^{iHt} O e^{-iHt}. \quad (19.55)$$

The wave functions are no longer time-dependent so the matrix element is

$$\langle O \rangle = \langle \phi(0) | O(t) | \phi(0) \rangle = \langle \phi(0) | e^{iHt} O e^{-iHt} | \phi(0) \rangle, \quad (19.56)$$

which is the same final result as in the Schrödinger picture. Because the operators now have an implicit time dependence, they satisfy a time-dependent Schrödinger equation,

$$i \frac{\partial O}{\partial t} = i [iH e^{iHt} O e^{-iHt} + e^{iHt} O (-iH) e^{-iHt}], \quad (19.57)$$

$$i \frac{\partial O}{\partial t} = [O(t), H]. \quad (19.58)$$

The Heisenberg equation of motion for operators is seen to be very similar to the Poisson-bracket equation of classical mechanics (see, for example, Goldstein, 1977).

Heisenberg Picture in Fock Space

We assume our elementary creation and destruction operators are time dependent, that is, we assume the Heisenberg picture with the a 's satisfying (19.58):

$$\begin{aligned} i \frac{\partial a_n(t)}{\partial t} &= [a_n, H] = \left[a_n, \sum_m a_m^\dagger a_m E_m \right] = (a_n a_n^\dagger - a_n^\dagger a_n) a_n E_n \\ &= E_n a_n. \end{aligned} \quad (19.59)$$

The annihilation operators thus have the familiar time dependence

$$a_n(t) = a_n(0) e^{-iE_n t}. \quad (19.60)$$

It follows from this that the field operator has the time dependence

$$\Psi(\mathbf{r}, t) = \sum_{\mathbf{p}} \frac{e^{i\mathbf{p} \cdot \mathbf{r}}}{\sqrt{V}} a_{\mathbf{p}}(t) = \sum_{\mathbf{p}} \frac{e^{-i(E_{\mathbf{p}} t - \mathbf{p} \cdot \mathbf{r})}}{\sqrt{V}} a_{\mathbf{p}}(0). \quad (19.61)$$

This (again) looks like a plane-wave expansion even though the a 's are operators.

It is not so obvious what dynamical equation the field operator $\Psi(\mathbf{r}, t)$ satisfies. Because it is an operator in Fock space and because we are in the Heisenberg picture, it must satisfy the Heisenberg equation of motion,

$$i \frac{\partial \Psi(\mathbf{r}, t)}{\partial t} = [\Psi(\mathbf{r}, t), H]. \quad (19.62)$$

Yet this actually is a nonlinear equation because the Fock-space Hamiltonian also depends on the field operator:

$$H = H[\Psi] = \int d^3\mathbf{r} \Psi^\dagger(\mathbf{r}, t) \left[\frac{-\nabla^2}{2m} + V(\mathbf{r}) \right] \Psi(\mathbf{r}, t). \quad (19.63)$$

In the Problems section you will prove that the field operator also satisfies the time-dependent Schrödinger equation,

$$i \frac{\partial \Psi(\mathbf{r}, t)}{\partial t} = \left[\frac{-\nabla^2}{2m} + V(\mathbf{r}) \right] \Psi(\mathbf{r}, t). \quad (19.64)$$

Likewise, when we quantize the Dirac field, the field operator satisfies the time-dependent Dirac equation.

19.4 Problems

1. Consider two identical fermions in coordinate space described by the basis vectors

$$|\mathbf{r}_1, \mathbf{r}_2\rangle = \Psi^\dagger(\mathbf{r}_2) \Psi^\dagger(\mathbf{r}_1) |0\rangle, \quad (19.65)$$

where the Ψ 's are field operators.

- (a) Calculate and interpret $\langle \mathbf{r}'_1, \mathbf{r}'_2 | \mathbf{r}_1, \mathbf{r}_2 \rangle$.
 - (b) Show that $\langle \mathbf{r}_1, \mathbf{r}_2 | \phi \rangle$ is the two-particle, symmetrized, coordinate-space representation of the state $|\phi\rangle$.
2. The configuration space operator used to describe identical particles at $\mathbf{r}'$ and $\mathbf{r}$ interacting via a *local* pairwise potential $V(\mathbf{r}' - \mathbf{r})$ is

$$V^{\text{op}} = \frac{1}{2} \int d^3\mathbf{r}' d^3\mathbf{r} V(\mathbf{r}' - \mathbf{r}) \Psi^\dagger(\mathbf{r}) \Psi^\dagger(\mathbf{r}') \Psi(\mathbf{r}) \Psi(\mathbf{r}'), \quad (19.66)$$

where $\Psi(\mathbf{r})$ is a field operator.

- (a) Verify the sense of this definition by calculating the matrix element

$$\langle \phi' | V^{\text{op}} | \phi \rangle, \quad (19.67)$$

and expressing your answer in terms of the usual wave function ϕ .

- (b) Express V^{op} in terms of the momentum-space creation and destruction operators $a_{\mathbf{p}}$ and $a_{\mathbf{p}}^\dagger$. Interpret your answer.
3. Prove that if the field operator Ψ satisfies the Heisenberg equation of motion,

$$i \frac{\partial \Psi(\mathbf{r})}{\partial t} = [\Psi(\mathbf{r}), H], \quad (19.68)$$

$$H = \int d^3\mathbf{r} \Psi(\mathbf{r})^\dagger \left[\frac{-\nabla^2}{2m} + V \right] \Psi(\mathbf{r}), \quad (19.69)$$

it will also satisfy the Schrödinger equation:

$$i \frac{\partial \Psi(\mathbf{r})}{\partial t} = \left[\frac{-\nabla^2}{2m} + V \right] \Psi(\mathbf{r}). \quad (19.70)$$

Why are these considered *nonlinear* equations for the fields?

4. Because the annihilation operator a changes the number states $|m\rangle$, the $|m\rangle$'s are not eigenfunctions of a .
- (a) Use the $|m\rangle$'s as a basis to construct a new state which is an eigenfunction of a with eigenvalue b .
 - (b) Is b real? (Explain)
 - (c) What is the average energy of such a state?
5. ☉ Consider the second-quantized Hamiltonian

$$H = \omega(a^\dagger a + b^\dagger b) + \epsilon(a^\dagger b^\dagger + ab), \quad (19.71)$$

in which a and b are destruction operators for two types of bosons and ϵ is a small number.

- (a) Suggest a physical situation which may be described by this H .
- (b) How does the Hamiltonian and the physical system change as the interaction between the a and b particles gets larger?
- (c) Solve for the energy eigenvalues of this Hamiltonian and indicate how they differ from the result for two independent particle groups.
- (d) What are the energy eigenstates for this Hamiltonian?

CHAPTER 20

QUANTIZED ELECTROMAGNETIC FIELDS

It is ironic that one of the first phenomena discussed in quantum mechanics, the emission of radiation, is one of the last to be treated properly. As discussed in Chapter 19, *Second Quantization*, the reason is that we need a theoretical framework (second quantization) in which it is permitted to create and destroy photons, yet this is Not what the Schrödinger equation gives us. Our first application of second quantization is to the theory of the electromagnetic (EM) field, *quantum electrodynamics* (QED). Our discussion and that of Mandl (1966), and Harris (1972) are introductory in nature. More thorough treatments found in the Bjorken and Drell (1964, 1965), Sakurai (1967), Gross (1993), Schweber (1959), the Feynman papers and books in the references, and Feynman and Hibbs (1965).

20.1 Classical Electromagnetic Fields: Review

In Gaussian (CGS) units, the classical equations for the electric and magnetic fields are the no-source Maxwell equations,

$$\begin{aligned}
 \nabla \cdot \mathbf{E} &= 0, & \nabla \times \mathbf{E} + \frac{\partial \mathbf{B}}{\partial t} &= 0, \\
 \nabla \cdot \mathbf{B} &= 0, & \nabla \times \mathbf{B} - \frac{\partial \mathbf{E}}{\partial t} &= 0.
 \end{aligned} \tag{20.1}$$

While these are considered the basic equations, it is the 4-vector potential $A^\mu = (\phi, \mathbf{A})$ which forms a *field* by satisfying the wave equation. Once A^μ is known, the $\mathbf{E}$ and $\mathbf{B}$ fields are also known because

$$\mathbf{E} = -\frac{\partial \mathbf{A}}{\partial t} - \nabla \phi, \quad \mathbf{B} = \nabla \times \mathbf{A}. \tag{20.2}$$

For simplicity in developing field theory, we work in the *radiation* or *transverse gauge*,

$$\nabla \cdot \mathbf{A}(\mathbf{x}, t) = 0, \tag{20.3}$$

and assume the static external field vanishes,

$$\phi(\mathbf{x}, t) = 0. \tag{20.4}$$

If there is a ϕ , we can eliminate it by a gauge transformation.¹ With these choices, the electric field $\mathbf{E}$ is parallel to the vector potential $\mathbf{A}$, and the magnetic field $\mathbf{B}$ is perpendicular to it:

$$\mathbf{E} = -\frac{\partial \mathbf{A}}{\partial t}, \quad \mathbf{B} = \nabla \times \mathbf{A}. \quad (20.5)$$

Exercise Show by direct substitution that the radiation gauge automatically satisfies three out of the four Maxwell equations (20.1), and that the fourth equation becomes the wave equation

$$\nabla^2 \mathbf{A}(\mathbf{x}, t) - \frac{\partial^2 \mathbf{A}(\mathbf{x}, t)}{\partial t^2} = 0. \quad \diamond \quad (20.6)$$

Because $\mathbf{A}$ transforms as a vector, the EM field is a *vector field* (in quantum mechanics this vector nature means the photon has spin 1).

When we specify the *polarization of light*, we actually give the direction of the $\mathbf{E}$ field, which in the radiation gauge is also the direction of $\mathbf{A}$. If we consider light traveling in the z direction, so that its wave vector is $\mathbf{k} = k\hat{\mathbf{z}}$, there are two directions possible for $\mathbf{E}$, both of which must be transverse to $\mathbf{k}$. It is conventional to use one of the two sets of polarization vectors:

$$\hat{\mathbf{e}}_x = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \hat{\mathbf{e}}_y = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad (20.7)$$

or

$$\hat{\mathbf{e}}_R \equiv \hat{\mathbf{e}}_+ = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \\ 0 \end{pmatrix}, \quad \hat{\mathbf{e}}_L \equiv \hat{\mathbf{e}}_- = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \\ 0 \end{pmatrix}. \quad (20.8)$$

The set $\hat{\mathbf{e}}_x$ and $\hat{\mathbf{e}}_y$ is clearly the choice to describe plane polarized waves, while $\hat{\mathbf{e}}_R$ and $\hat{\mathbf{e}}_L$ is the choice for waves having right- or left-circular polarizations. Equivalently, $\hat{\mathbf{e}}_+$ and $\hat{\mathbf{e}}_-$ is the choice for positive- and negative-helicity states. For either set, the vectors span the 2D space and are orthogonal to their mates:²

$$\hat{\mathbf{e}}_i^* \cdot \hat{\mathbf{e}}_j = \delta_{ij}. \quad (20.9)$$

If the classical EM field is confined to a box of volume V , the most general field $\mathbf{A}$ is a linear combination of plane-wave solutions to the wave equation, that is, the *Fourier series*,

$$\mathbf{A}(\mathbf{x}, t) = \sum_{\mathbf{k}} \sum_{i=1}^2 \sqrt{\frac{2\pi}{V\omega_k}} \left[\hat{\mathbf{e}}_{\mathbf{k}i} a_{\mathbf{k}i}(t) e^{i\mathbf{k}\cdot\mathbf{x}} + \hat{\mathbf{e}}_{\mathbf{k}i}^* a_{\mathbf{k}i}^*(t) e^{-i\mathbf{k}\cdot\mathbf{x}} \right]. \quad (20.10)$$

¹A divergentless $\mathbf{A}$ is also known as the *transversality condition*. It is shown (Sakurai, 1967, p. 304) that (20.3) and (20.4) can always be arranged. Sometimes the terms “Coulomb” or “radiation gauge” refer to the transversality condition on $\mathbf{A}$ with A_0 given by a nonzero, instantaneous Coulomb potential. The field theory is equivalent, because (to quote Sakurai’s paraphrase of Fermi) “we can forget about the dynamical degrees of freedom associated with A_0 or $\mathbf{E}_\perp$. In quantizing the Maxwell field it is sufficient to quantize just the transverse vector potential $\mathbf{A}_\perp$ provided we work in the radiation gauge.”

²Because in our gauge there is no polarization in the z direction, it is convenient to simply drop the z components. Nevertheless, this should not lead the reader to believe that the EM field is a two-component (spinor) field; it is a vector (three-component) field with one component chosen to be zero for convenience.

Here the polarization index i can be either (x, y) or (R, L) , $a_{\mathbf{k}i}$ is a (time-dependent) Fourier expansion coefficient, the square-root renormalization factor is chosen for later convenience, and the addition of the complex conjugate term ensures $\mathbf{A}$'s reality.

Exercise Take the expression for $\mathbf{A}(\mathbf{x}, t)$, substitute it into the wave equation (20.6), and show that this leads to an equation which when solved gives the time dependence of $a_{\mathbf{k}i}$:

$$a_{\mathbf{k}i}(t) = e^{-i\omega_k t} a_{\mathbf{k}i}(0), \quad \omega_k = k. \quad \diamond \quad (20.11)$$

Accordingly, the vector potential is,

$$\mathbf{A}(\mathbf{x}, t) = \sum_{\mathbf{k}, i} \sqrt{\frac{2\pi}{V\omega_k}} \left[\hat{\mathbf{e}}_{\mathbf{k}i} a_{\mathbf{k}i}(0) e^{i(\mathbf{k} \cdot \mathbf{x} - \omega_k t)} + \hat{\mathbf{e}}_{\mathbf{k}i}^* a_{\mathbf{k}i}^*(0) e^{-i(\mathbf{k} \cdot \mathbf{x} - \omega_k t)} \right]. \quad (20.12)$$

Equation (20.12) indicates that the EM field is represented as a collection of independent oscillators, each with its own wave vector $\mathbf{k}$, frequency ω_k , and polarization $\hat{\mathbf{e}}_{\mathbf{k}i}$.

20.2 Photon State Vector

It is an empirical fact that photons are bosons with zero rest mass and with an energy (frequency) and momentum (wave vector) related by

$$E = \hbar\omega = \omega = kc = k. \quad (20.13)$$

In addition, because there is an electric field $\mathbf{E}$ associated with a photon (that is, light), we demand three independent numbers to describe the state of the photon, that is, it must have a *vector* wave function (the vector aspect refers to the mixing of the components under rotations). By convention, we take the three components to be the electric field strengths:

$$|\psi_\gamma\rangle = N \begin{pmatrix} E_x \\ E_y \\ E_z \end{pmatrix}. \quad (20.14)$$

Because photons in the radiation gauge satisfy the transversality condition,

$$\mathbf{k} \cdot \hat{\mathbf{e}}_{\mathbf{k}} = 0, \quad (20.15)$$

one component of the photon wave function is identically zero. We display our laziness by not displaying the zero component and writing:

$$|\psi_\gamma\rangle = N \begin{pmatrix} E_x \\ E_y \end{pmatrix}, \quad \text{or} \quad |\psi_\gamma\rangle = N \begin{pmatrix} E_R \\ E_L \end{pmatrix}. \quad (20.16)$$

Even though we indicate only two components, the photon is still a spin-one particle with the right-handed circular polarization vector (20.8) corresponding to the z component of angular momentum $s_z = +\hbar \equiv 1$, and the left-handed circular polarization state corresponding to $s_z = -\hbar \equiv -1$. For spin-one particles with nonzero mass, or for nonfree

photons, all three of the components in (20.14) occur. In summary, we have assumed the photon wave function is proportional to the polarization vector of light.

To quantize these photons (Planck's hypothesis), we view the vector potential,

$$\mathbf{A}(\mathbf{x}) = \sum_{\mathbf{k}, i} \sqrt{\frac{2\pi}{V\omega_{\mathbf{k}}}} \hat{\mathbf{e}}_{\mathbf{k}i} \left[a_{\mathbf{k}i}(0) e^{i(\mathbf{k} \cdot \mathbf{x} - \omega_{\mathbf{k}} t)} + a_{\mathbf{k}i}^\dagger(0) e^{-i(\mathbf{k} \cdot \mathbf{x} - \omega_{\mathbf{k}} t)} \right], \quad (20.17)$$

as a three component (A_x, A_y, A_z) form of the field operator introduced in Chapter 19, *Second Quantization*.³ That is, $\mathbf{A}(\mathbf{x}, t)$ is an operator which destroys a photon at the point $(\mathbf{x}, t)$. The a 's in (20.17) are creation and annihilation operators which satisfy the familiar commutation relations, for example,

$$[a_{\mathbf{k}i}(t), a_{\mathbf{k}'i'}^\dagger(t)] = \delta_{\mathbf{k}\mathbf{k}'} \delta_{ii'}, \quad (20.18)$$

where the operators are acting at equal times. The EM field $\mathbf{A}$ is thus a collection of photons being created and destroyed, each with an energy $E_{\mathbf{k}} = \omega_{\mathbf{k}}$ (Planck's condition) and momentum $\mathbf{p} = \mathbf{k}$ (deBroglie's relation).⁴ In the Problems section we find that the use of these second-quantized fields in the classical relation for the energy of the field leads to Planck's Hamiltonian and his hypothesis for the energy in the *free* electromagnetic field:

$$H = \frac{1}{8\pi} \int d^3x (E^2 + B^2) \quad (20.19)$$

$$= \frac{1}{2} \sum_{\mathbf{k}, i} \omega_{\mathbf{k}} (a_{\mathbf{k}i}^\dagger a_{\mathbf{k}i} + a_{\mathbf{k}i} a_{\mathbf{k}i}^\dagger) = \sum_{\mathbf{k}, i} \omega_{\mathbf{k}} \left[a_{\mathbf{k}i}^\dagger a_{\mathbf{k}i} + \frac{1}{2} \right], \quad (20.20)$$

$$\Rightarrow E = \sum_{\mathbf{k}, i} N_{\mathbf{k}i} \omega_{\mathbf{k}}, \quad (20.21)$$

where $N_{\mathbf{k}i}$ is the number of photons of wave vector $\mathbf{k}$ and polarization $\hat{\mathbf{e}}_{\mathbf{k}i}$. In (20.21) we do not include the $\frac{1}{2}$ from (20.20) because it leads to an (unobservable) infinite, zero-point energy. Or in other words, the vacuum state with no photons has infinite energy, and all energies are measured relative to this.

20.3 Electron–Photon Interaction

The usefulness of this second-quantized theory is its ability to describe and predict natural phenomena. Because a measurement of a natural phenomenon requires interaction, we now apply second quantization to deduce the *electron–photon interaction Hamiltonian*.⁵ We deduce it from the unperturbed Hamiltonian for a *nonrelativistic* electron,

$$H_0^e = \frac{p^2}{2m} + V(\mathbf{x}), \quad (20.22)$$

³We assume real $\epsilon_{\mathbf{k}i}$ to keep the notation simpler.

⁴This is an incoherent photon field (the photon states have no phase relation to each other) with an indeterminate number of photons: as such, it does not have a classical limit. In § 21.2 we discuss a coherent state of the radiation field that has a classical limit.

⁵The Planck Hamiltonian (20.21) describes the free-photon field without interactions.

where $V(\mathbf{x})$ does not include the interaction with the radiation field. We include the EM field by changing the canonical momentum in (20.22) according to the minimal-coupling hypothesis (§ 13.5):⁶

$$\mathbf{p} \rightarrow \mathbf{p} - \frac{q}{c} \mathbf{A} = \mathbf{p} + \frac{e}{c} \mathbf{A} \Rightarrow H \rightarrow H + H^{int}. \quad (20.23)$$

This produces the interaction Hamiltonian

$$H^{int} = \frac{e^2}{2m} \mathbf{A}^2(\mathbf{x}) + \frac{e}{m} \mathbf{A} \cdot \mathbf{p}(\mathbf{x}) \stackrel{\text{def}}{=} H_\gamma + H_{2\gamma}. \quad (20.24)$$

It is conventional to break this Hamiltonian into a piece linear in $e\mathbf{A}$,

$$H_\gamma = \frac{e}{m} \mathbf{A} \cdot \mathbf{p}, \quad (20.25)$$

and one quadratic in it,

$$H_{2\gamma} = \frac{e^2}{2m} \mathbf{A}^2, \quad (20.26)$$

where e is the magnitude of the electron's charge.

Interaction Hamiltonian

The second-quantized version of the electron-photon interaction is obtained by substituting our expression for the field operator $\mathbf{A}$ expressed as an expansion of creation and annihilation operators (20.17) into these expressions for H_γ and $H_{2\gamma}$:

$$H_\gamma = \frac{e}{m} \sum_{\mathbf{k}, i} \sqrt{\frac{2\pi}{V\omega_k}} \hat{\mathbf{e}}_{\mathbf{k}i} \cdot \mathbf{p} \left[a_{\mathbf{k}i}(t) e^{i\mathbf{k} \cdot \mathbf{x}} + a_{\mathbf{k}i}^\dagger(t) e^{-i\mathbf{k} \cdot \mathbf{x}} \right], \quad (20.27)$$

$$H_{2\gamma} = \frac{e^2}{2m} \sum_{\mathbf{k}, i, \mathbf{k}', i'} \sqrt{\frac{2\pi 2\pi}{V\omega_k V\omega_{k'}}} \hat{\mathbf{e}}_{\mathbf{k}'i'} \cdot \hat{\mathbf{e}}_{\mathbf{k}i} \left[a_{\mathbf{k}i} a_{\mathbf{k}'i'} e^{i(\mathbf{k}+\mathbf{k}') \cdot \mathbf{x}} \right. \\ \left. + a_{\mathbf{k}i}^\dagger a_{\mathbf{k}'i'}^\dagger e^{-i(\mathbf{k}+\mathbf{k}') \cdot \mathbf{x}} + a_{\mathbf{k}i} a_{\mathbf{k}'i'}^\dagger e^{i(\mathbf{k}-\mathbf{k}') \cdot \mathbf{x}} + a_{\mathbf{k}i}^\dagger a_{\mathbf{k}'i'} e^{i(\mathbf{k}'-\mathbf{k}) \cdot \mathbf{x}} \right]. \quad (20.28)$$

As before, e is the magnitude of electron's charge, and to keep the equations "simple" we have assumed the polarization vector $\hat{\mathbf{e}}$ is real.

Exercise Verify (20.27) and (20.28). ◇

⁶Because minimal coupling is for point particles only, finite-size effects must be included explicitly. The magnetic moment interaction of a nonrelativistic electron is included by explicitly adding on an additional $(e/2m)g\boldsymbol{\sigma} \cdot \nabla \times \mathbf{A}$ term. If a Dirac Hamiltonian is used in place of (20.22), this magnetic interaction is included automatically.

20.4 Following the Golden Rule

Because the EM interaction is weak (that is, $e^2 = \alpha \simeq 1/137$), perturbation theory works well with H_γ and $H_{2\gamma}$. Consequently, we use it to calculate the transition rate Γ_{fi} , that is, the number of transitions per unit time (inverse of the lifetime) from level f to i of the unperturbed Hamiltonian. This is the rate at which H^{int} induces transitions between states i and f and is given by *Fermi's golden rule*:⁷

$$\Gamma_{fi} = \frac{1}{\tau} = 2\pi |\mathcal{M}_{fi}|^2 \delta(E_f - E_i). \quad (20.29)$$

Here $\mathcal{M}_{fi}$ is proportional to the transition matrix (T -matrix) element familiar from Part I, *Scattering and Integral Quantum Mechanics*, and is given by a LS equation:⁸

$$\mathcal{M}_{fi} = \langle f | H^{\text{int}} | i \rangle + \sum_n \frac{\langle f | H^{\text{int}} | n \rangle \langle n | H^{\text{int}} | i \rangle}{E_i - E_n + i\epsilon} + \dots \quad (20.30)$$

In these equations, $|i\rangle$, $|f\rangle$, and $|n\rangle$ represent the initial, final, and intermediate states and H^{int} is the interaction (perturbing potential) which induces the transition. Because we take H^{int} to be an operator in Fock space, the states must also be in Fock-space, and then the matrix elements in (20.30) are simple numbers.

In the Born-approximation applications of Part I (Chapter 5), we kept only the first term in (20.30). Now we will examine processes for which this first term vanishes, or for which greater precision is required, and in these cases higher-order terms shall be kept. Because the higher-order terms describe processes involving intermediate *virtual states*, their energies may differ from E_i and E_f . It is also possible to have a singular energy denominator. In Part I we indicated a method for handling that singularity by including an $i\epsilon$ in the energy; other prescriptions, such as those used in degenerate perturbation theory, are also possible. Yet all too often in practical calculations, the higher-order “corrections” are infinite and thereby require special treatment. We indicate one treatment in § 21.3 in the discussion of the electron’s self-energy.

Our statement of the golden rule (20.29) has an energy-conserving delta function originating from time-dependent perturbation theory. While a singular function may appear awkward, in actual calculations the rate is summed or integrated over final states, so the golden rule can also be written as

$$\Gamma_{fi} = 2\pi |\mathcal{M}_{fi}|^2 \delta(E_f - E_i) dN_f, \quad (20.31)$$

$$dN_f = \frac{V d^3 k_1}{(2\pi)^3} \frac{V d^3 k_2}{(2\pi)^3} \dots, \quad (20.32)$$

where we “count” over all the particles in the final state. The actual counting replaces the delta function by a density-of-states factor ρ_E :

$$\sum_f \Gamma_{fi} = \int dE_f \frac{dN_f}{dE} \Gamma_{fi} = \int dE_f \frac{dN_f}{dE} 2\pi |\mathcal{M}_{fi}|^2 \delta(E_f - E_i),$$

⁷Good descriptions of perturbation theory for our purposes are found in Sakurai (1967) and Merzbacher (1970), who also examine exponential decay of levels.

⁸We generalize the golden rule to use the full T matrix rather than just the interaction potential. Note, however, that because we do field-theory calculations in a box of volume V , the exact relation of the amplitudes of Parts I and II may not be apparent here. We work out the details in Chapter 23, *The Breit–Pauli and Meson-Exchange Interactions*.

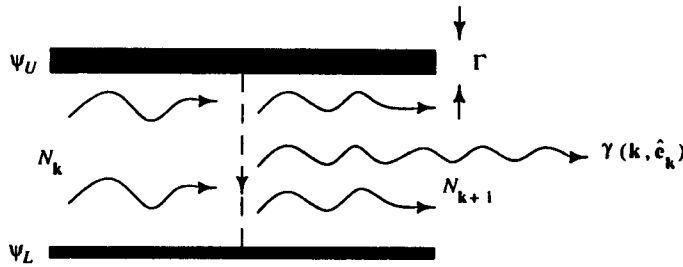


Figure 20.1: An upper level ψ_U decaying to a lower level ψ_L via single-photon emission. The width of ψ_U arises, at least in part, from this decay. The complete Fock-space state also contains photons as indicated by the wavy lines.

$$\Rightarrow \sum_f \Gamma_{fi} = 2\pi |\mathcal{M}_{fi}|^2 \rho_f, \quad \rho_f = \frac{dN_f}{dE}, \quad (20.33)$$

where ρ_f is the number of states per unit energy at energy E_f . We will evaluate ρ_f in a number of examples.

Decaying States

A quantum-mechanical state f decaying to state i may do so spontaneously at the rate Γ_{fi} , but does not do so instantaneously. For this reason there is uncertainty in the length of time for which level i exists and, correspondingly, uncertainty in the precise value of the energy of level i . This of course is just a statement of the uncertainty principle $\Delta E \Delta t \geq \hbar \equiv 1$. Often this is pictured as in Figure 20.1 with the initial level having a *width* Γ .

Exercise Show that the probability of an exponentially-decaying state has the time dependence:

$$|\psi|^2 \propto e^{-t/\tau} = e^{-\Gamma t}, \quad (20.34)$$

where the level width Γ equals the transition rate $d(|\psi|^2)/dt$ and τ is the mean lifetime of the state. $\diamond$

Once an electron couples to the radiation field, its state no longer corresponds to just one value of energy (it acquires a width) and is thus no longer an eigenstate of the unperturbed Hamiltonian (20.22). As discussed in § 4.2, these decaying states are sometimes described as eigenstates with complex energies.

Exercise Show that a stationary state with complex “eigenenergy” $E = E_R - i\Gamma/2$ corresponds to a state which decays exponentially with a lifetime $\tau = 1/\Gamma$. $\diamond$

These exponentially-decaying states, although no longer eigenstates of the unperturbed Hamiltonian, are eigenstates of the full Hamiltonian including the EM field. In practice, however, the full problem is rarely solved and the widths and shifts of levels caused by coupling to the radiation field is either ignored or calculated in perturbation theory.

20.5 Radiation from Two-Level System

As our first example, we calculate the rate for the system's transition from the upper to the lower energy level in Figure 20.1, with the accompanying emission of a photon of wave vector $\mathbf{k}$ and polarization $\hat{\mathbf{e}}_{\mathbf{k}i}$. The system may be an atom, a nucleus, a solid, or whatever, as long as we can describe it as having an upper-level wave function ψ_U and a lower-level wave function ψ_L (at present we ignore the system's recoil). In order to apply the Hamiltonian $H_\gamma + H_{2\gamma}$, we specify the initial and final states for the entire system with Dirac kets for the bound system and Fock-space states for the photon field

$$|i\rangle = |\psi_U\rangle |N_1, \dots, N_{\mathbf{k}i}, \dots\rangle, \quad |f\rangle = |\psi_L\rangle |N_1, \dots, N_{\mathbf{k}i} + 1, \dots\rangle. \quad (20.35)$$

Here the subscripts on the N 's indicate the photon's momentum and polarization. Note that the final photon state differ from the initial one by having one more photon of $\mathbf{k}$ and $\hat{\mathbf{e}}_{\mathbf{k}i}$ present (we leave open the possibility of there already being photons around).

If there is a nonvanishing matrix element of the Hamiltonian between $|f\rangle$ and $|i\rangle$, then we may use the golden rule (20.29) to calculate the rate. In general, there is an infinite number of terms in (20.30) which contribute to the transition, and calculating them all can be exhaustive, if not never ending. Yet because each higher-order term introduces an additional power of $\alpha = e^2 \simeq 1/137$, the higher-order terms make progressively smaller contributions.

We examine only the lowest-order matrix element of H_γ between $|f\rangle$ and $|i\rangle$:

$$\begin{aligned} \langle f | H_\gamma | i \rangle &= \langle \psi_L | \langle \dots N_{\mathbf{k}i} + 1 | \frac{e}{m} \sum_{\mathbf{k}', i} \sqrt{\frac{2\pi}{V\omega_{\mathbf{k}'}}} \hat{\mathbf{e}}_{\mathbf{k}'i} \cdot \mathbf{p} \\ &\quad \times \left[a_{\mathbf{k}'i}(t) e^{i\mathbf{k}' \cdot \mathbf{x}} + a_{\mathbf{k}'i}^\dagger(t) e^{-i\mathbf{k}' \cdot \mathbf{x}} \right] | \dots N_{\mathbf{k}i} \rangle | \psi_U \rangle, \end{aligned} \quad (20.36)$$

where we assume real $\hat{\mathbf{e}}$ for simplicity. To evaluate (20.36), we pick out those terms which connect the initial Fock-space state $|\dots N_{\mathbf{k}i} \dots\rangle$ to the final one with one more photon $|\dots N_{\mathbf{k}i} + 1 \dots\rangle$. First we note that all matrix elements involving the destruction operator $a_{\mathbf{k}}$ must vanish because it changes $|i\rangle$ into a state with one fewer photons, which cannot then have any overlap with the state $|f\rangle$ which has one extra photon. That leaves the infinite sum of terms involving $a_{\mathbf{k}i}^\dagger$:

$$\langle f | H_\gamma | i \rangle = \langle \psi_L | \frac{e}{m} \sum_{\mathbf{k}, i} \sqrt{\frac{2\pi}{V\omega_{\mathbf{k}}}} \hat{\mathbf{e}}_{\mathbf{k}i} \cdot \mathbf{p} \langle N_{\mathbf{k}i} + 1 | a_{\mathbf{k}i}^\dagger(t) | N_{\mathbf{k}i} \rangle e^{-i\mathbf{k} \cdot \mathbf{x}} | \psi_U \rangle. \quad (20.37)$$

Now the only term in the sum which survives is the one in which the dummy variable $\mathbf{k}'$ equals the momentum of the created photon:

$$\begin{aligned} \langle f | H_\gamma | i \rangle &= \frac{e}{m} \sqrt{\frac{2\pi}{V\omega_{\mathbf{k}}}} \langle \psi_L | \hat{\mathbf{e}}_{\mathbf{k}i} \cdot \mathbf{p} e^{-i\mathbf{k} \cdot \mathbf{x}} | \psi_U \rangle \langle N_{\mathbf{k}i} + 1 | a_{\mathbf{k}i}^\dagger(t) | N_{\mathbf{k}i} \rangle \\ &= \frac{e}{m} \sqrt{\frac{2\pi}{V\omega_{\mathbf{k}}}} \langle \psi_L | \hat{\mathbf{e}}_{\mathbf{k}i} \cdot \mathbf{p} e^{-i\mathbf{k} \cdot \mathbf{x}} | \psi_U \rangle \sqrt{N_{\mathbf{k}i} + 1}. \end{aligned} \quad (20.38)$$

We see that the Fock-space operator automatically chooses the term which contributes and then gives us a $\sqrt{N_{\mathbf{k}i} + 1}$ factor. The remaining (nonphoton) matrix element between ψ_U and ψ_L is a conventional one.

As (20.38) stands now, there is no restriction on the photon energy ω_k . However, when we follow the golden rule (20.29), the delta function requires conservation of energy and this requires that

$$\omega_k = E_i - E_f. \quad (20.39)$$

The predicted transition rate is accordingly

$$\Gamma_{fi} = \left(\frac{e}{m}\right)^2 \frac{(2\pi)^2}{V\omega_k} (N_{\mathbf{k}i} + 1) \left| \langle \psi_L | \hat{\mathbf{e}}_{\mathbf{k}i} \cdot \mathbf{p} e^{-i\mathbf{k} \cdot \mathbf{x}} | \psi_U \rangle \right|^2 \delta(E_f + \omega - E_i). \quad (20.40)$$

Even before we evaluate the bound-state matrix element, some illuminating physics is evident. The transition rate depends on $N_{\mathbf{k}i} + 1$, the number of photons plus one. This means there is *stimulated emission* proportional to $N_{\mathbf{k}i}$, which means it is more likely for the atom to emit a photon if some other photons of the right energy are already around. This also means that there is *spontaneous emission* proportional to 1 which is not influenced by the presence of other photons. Stimulated emission makes intuitive sense because the photons carry the EM field, and it is (classically) the acceleration of the electron by this field (that is, background photons) which causes radiation. Spontaneous emission might then can be thought of as arising from a fluctuation in the vacuum producing virtual photons which stimulate the transition.

Dipole Approximation

If we know the bound system wave functions $\psi_U(\mathbf{r})$ and $\psi_L(\mathbf{r})$ [say, the eigenstates of the Hamiltonian (20.22)], the matrix element in (20.40) can be evaluated:

$$\langle \psi_L | \hat{\mathbf{e}}_{\mathbf{k}i} \cdot \mathbf{p} e^{-i\mathbf{k} \cdot \mathbf{x}} | \psi_U \rangle = \left(\int d^3x \psi_L^*(\mathbf{x}) \mathbf{p} e^{-i\mathbf{k} \cdot \mathbf{x}} \psi_U(\mathbf{x}) \right) \cdot \hat{\mathbf{e}}_{\mathbf{k}i}. \quad (20.41)$$

Here the wave functions are in the $\mathbf{x}$ -space representation, and so $\mathbf{p}$ is the momentum operator $-i\nabla$ while $\mathbf{k}$ is the momentum of the emitted photon (and is not an operator).

Exercise Show that the wavelength of a photon of visible light is 100–1000 times larger than the size of an atom. $\diamond$

Exercise Show that the wavelength of a gamma ray of 1 MeV energy is 100–1000 times larger than the size of a nucleus. $\diamond$

The matrix element (20.41) can be approximated and simplified after realizing that for many practical cases the wavelength of the emitted photon is much larger than the size of the bound system. That is, $1/k$ is much greater than the values of $\mathbf{x}$ which enter into (20.41), and so it is reasonable to expand the exponential:

$$e^{-i\mathbf{k} \cdot \mathbf{x}} = 1 - i\mathbf{k} \cdot \mathbf{x} - \frac{1}{2}(\mathbf{k} \cdot \mathbf{x})^2 + \dots \quad (20.42)$$

In this case the first term of (20.42) produces electric dipole (E1) transitions, the second term electric quadrupole (E2) and magnetic dipole (M1) transitions, and so forth. Each successive term is progressively smaller by a factor of ~ 100 – 1000 .

Exercise Show that $\frac{e}{m}\hat{\mathbf{e}} \cdot \mathbf{p}$ is ~ 100 – 1000 times larger than $\frac{eg}{2m}\boldsymbol{\sigma} \cdot (\mathbf{k} \times \hat{\mathbf{e}})$. $\diamond$

This last exercise implies that the magnetic interaction contained in (20.30) arising from the electron's orbital motion is ~ 100 – 1000 times larger than the magnetic interaction arising from the intrinsic magnetic dipole (i.e., spin) of the electron.

In the *dipole approximation* to (20.41), only the first term of the exponential's expansion is kept:

$$\langle \psi_L | \mathbf{p} e^{-i\mathbf{k} \cdot \mathbf{r}} | \psi_U \rangle \simeq \langle \psi_L | \mathbf{p} | \psi_U \rangle. \quad (20.43)$$

The “dipole” nature of $\langle \psi_L | \mathbf{p} | \psi_U \rangle$ becomes evident after realizing that the effect of $\mathbf{p}$ on a wave function is proportional to the effect of $[\mathbf{r}, H_0]$ on it (H_0 is the unperturbed electron's Hamiltonian operator):

$$\begin{aligned} \mathbf{p}\psi &\stackrel{?}{=} \frac{m}{i}[\mathbf{r}, H]\psi = \frac{m}{i}(\mathbf{r}H\psi - H\mathbf{r}\psi) = im \left(\mathbf{r}i\frac{\partial\psi}{\partial t} - i\frac{\partial\mathbf{r}}{\partial t}\psi - i\mathbf{r}\frac{\partial\psi}{\partial t} \right) \\ &= m\frac{\partial\mathbf{r}}{\partial t}\psi = m\frac{\mathbf{p}}{m}\psi = \mathbf{p}\psi, \end{aligned} \quad (20.44)$$

where we assume there are no explicit $\mathbf{p}$ -dependent forces. We thus write the bound-system matrix element as

$$\langle \psi_L | \mathbf{p} | \psi_U \rangle = -im \langle \psi_L | \mathbf{r}H - H\mathbf{r} | \psi_U \rangle = -im[E_i - E_f] \langle \psi_L | \mathbf{r} | \psi_U \rangle. \quad (20.45)$$

In order to understand the role of moments in radiative transitions, it is helpful to look at a classical harmonic oscillator. If the electron is oscillating in simple harmonic motion, its classical acceleration (the cause of radiation) is proportional to the electron's displacement from equilibrium:

$$\mathbf{a} = -\omega^2 \mathbf{x}, \quad (20.46)$$

where ω is the oscillator's angular frequency. In the quantum regime, (20.46) is true for expectation values, as shown by invoking Newton's law $\mathbf{a} = (d\mathbf{p}/dt)/m$:

$$\begin{aligned} \langle \psi_L | \mathbf{a} | \psi_U \rangle &= \frac{1}{m} \frac{d}{dt} \langle \psi_L | \mathbf{p} | \psi_U \rangle = -i(E_i - E_f) \left\langle \psi_L \left| \frac{d\mathbf{r}}{dt} \right| \psi_U \right\rangle \\ &= -i(E_i - E_f)(-i) \langle \psi_L | [\mathbf{r}, H] | \psi_U \rangle = -(E_i - E_f)^2 \langle \psi_L | \mathbf{r} | \psi_U \rangle. \end{aligned} \quad (20.47)$$

Hence, if the dipole matrix element for the transition exists, a matrix element of acceleration exists, and we can say an acceleration has caused the radiation.

If the matrix element of $\mathbf{r}$ between states does not vanish, we have a non-vanishing dipole moment existing during the transition. Because $\mathbf{r}$ is proportional to the Legendre polynomial $Y_l^m(\hat{\mathbf{r}})$, it is a vector operator, and there are certain values of the quantum numbers for Ψ_L and Ψ_U for which the matrix element must vanish. In particular (see the Problems section),

$$\langle \psi_L | \mathbf{r} | \psi_U \rangle = \begin{cases} 0, & \text{if } \psi_L \text{ and } \psi_U \text{ are spherically symmetric,} \\ \text{not zero,} & \text{when } \Delta l = \pm 1, \Delta m = \pm 1, 0. \end{cases} \quad (20.48)$$

Equation (20.48) is still valid even if the entire exponential in (20.42) is kept.

Transition Rate

The golden rule (20.29) proclaims that the number of transitions per unit time from state $|i\rangle$ to $|f\rangle$ is

$$\Gamma_{fi} = \sum_{\mathbf{k}} \left(\frac{2\pi e}{m} \right)^2 \frac{N_{\mathbf{k}+1}}{V} \frac{\left| \langle \psi_L | \hat{\mathbf{e}}_{\mathbf{k}i} \cdot \mathbf{p} e^{-i\mathbf{k} \cdot \mathbf{x}} | \psi_U \rangle \right|^2}{\omega_{\mathbf{k}}} \delta(E_f + \omega_{\mathbf{k}} - E_i). \quad (20.49)$$

Here $\mathbf{k}$ and $\hat{\mathbf{e}}_{\mathbf{k}i}$ are the wave vector and polarization of the emitted photon, and we have added in a summation over final photon momenta to account for the finite energy acceptance of our photon detector. As derived in Appendix A, *Natural Units and Plane Waves*, we change the normalization of the photon's plane wave states from that of a finite box to continuum states with the replacement

$$\frac{1}{V} \sum_{\mathbf{k}} \rightarrow \int \frac{d^3\mathbf{k}}{(2\pi)^3} = \int \frac{k^2 dk d\Omega_{\mathbf{k}}}{(2\pi)^3}. \quad (20.50)$$

The integration over $d\mathbf{k}$ removes the delta function and constrains $\mathbf{k}$ to conserve energy,

$$k = \omega_{\mathbf{k}} = E_i - E_f. \quad (20.51)$$

This leaves the angular integration:

$$\begin{aligned} \Gamma_{nm} &= \frac{e^2(N_{\mathbf{k}i} + 1)k}{2\pi m^2} \int d\Omega_{\mathbf{k}} \left| \langle \psi_L | \hat{\mathbf{e}}_{\mathbf{k}i} \cdot \mathbf{p} e^{-i\mathbf{k} \cdot \mathbf{x}} | \psi_U \rangle \right|^2 \\ &= \frac{e^2(N_{\mathbf{k}i} + 1)\omega^3 m^2}{2\pi m^2} \int d\Omega_{\mathbf{k}} \left| \langle \psi_L | \mathbf{r} | \psi_U \rangle \cdot \hat{\mathbf{e}}_{\mathbf{k}i} \right|^2. \end{aligned} \quad (20.52)$$

The transition rate (20.52) depends on the polarization of the emitted photon $\hat{\mathbf{e}}_{\mathbf{k}i}$, an experimental observable. If this polarization is not observed, we take the rate to be the sum over the rates (probabilities) for both possible polarizations. This *polarization sum* is straightforward if we use a diagram such as Figure 20.2 (actually the diagram is somewhat symbolic because it assumes the dipole matrix element $\langle \psi_L | \mathbf{r} | \psi_U \rangle$ is real). This matrix element $\langle \mathbf{r} \rangle$, which we place along the z axis, imposes a preferred direction to space. We eliminate any azimuthal ϕ dependence by choosing the photon momentum $\mathbf{k}$ to lie in the yz plane with a polar angle θ . With this geometry one of the photon's polarization vectors $\hat{\mathbf{e}}_{\mathbf{k}1}$ makes an angle $\pi/2 - \theta$ with $\langle \mathbf{r} \rangle$ and the other is orthogonal to it:

$$\langle \mathbf{r} \rangle \cdot \hat{\mathbf{e}}_{\mathbf{k}1} = \langle \mathbf{r} \rangle \sin \theta, \quad \langle \mathbf{r} \rangle \cdot \hat{\mathbf{e}}_{\mathbf{k}2} = 0. \quad (20.53)$$

The rate is therefore zero if the photon has polarization $\hat{\mathbf{e}}_{\mathbf{k}2}$, but finite for polarization $\hat{\mathbf{e}}_{\mathbf{k}1}$. The rate summed over polarizations is

$$\Gamma_{fi} = e^2 \omega^3 (N_{\mathbf{k}} + 1) \int_{-1}^{+1} d(\cos \theta) 2\pi \sin^2 \theta |\langle \psi_L | \mathbf{r} | \psi_U \rangle|^2. \quad (20.54)$$

Although we have picked a geometry to make life easy on us, the sum of squares is the same for all geometries. The angular integration now gives

$$\Gamma_{fi} = \frac{1}{\tau_{fi}} = \frac{4}{3} (N_{\mathbf{k}} + 1) e^2 \omega^3 |\langle \psi_L | \mathbf{r} | \psi_U \rangle|^2. \quad (20.55)$$

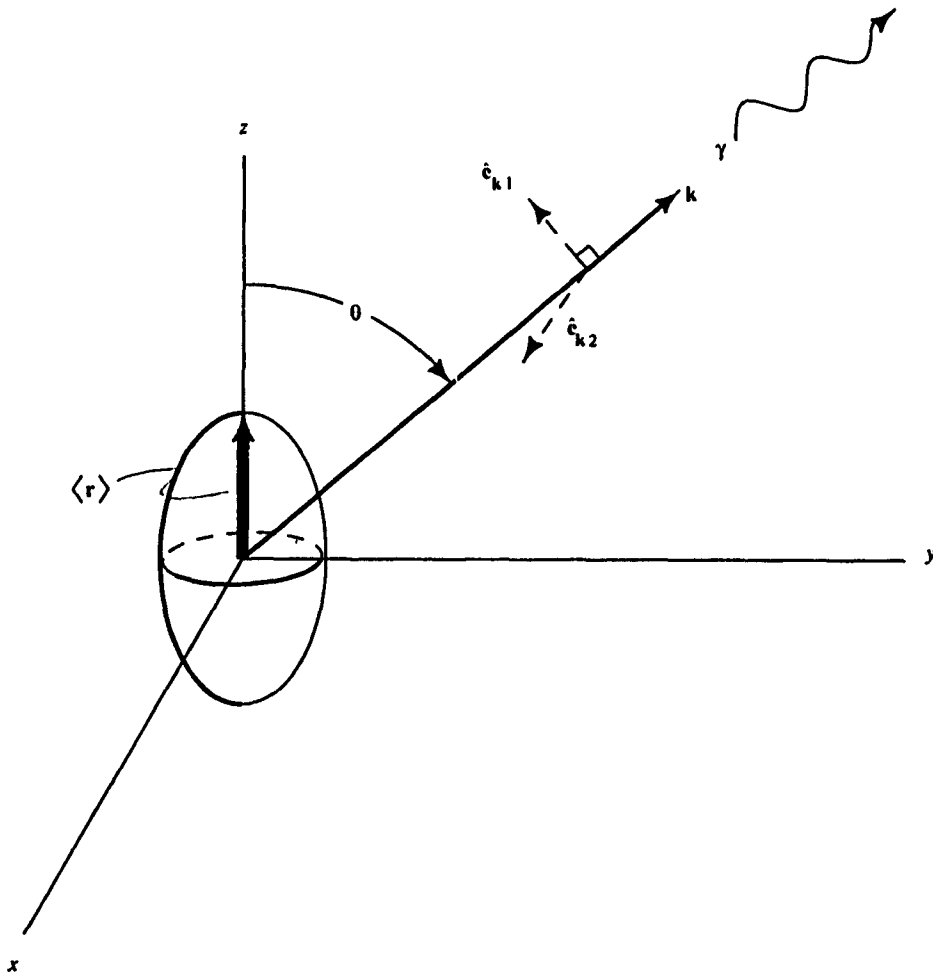


Figure 20.2: A photon emitted from a bound system having its dipole moment along the z axis. The photon's momentum $\mathbf{k}$ is used to define the yz plane, and the two possible polarization vectors of the photon are indicated.

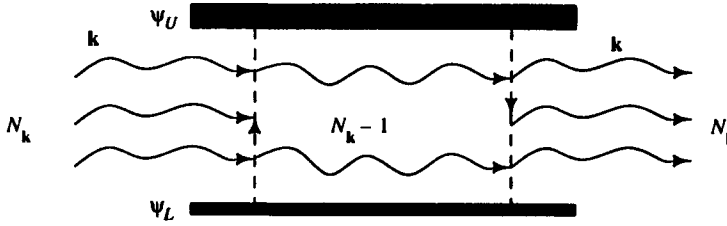


Figure 20.3: The emission and absorption of photons in a two-level system leading to Planck's distribution law. There are $N_{\mathbf{k}}$ photons exciting the absorption from ψ_L to ψ_U and, accordingly, $N_{\mathbf{k}} - 1$ photons exciting the emission from ψ_U to ψ_L .

The relation (20.55) gives the approximate rate for emission of radiation accompanying a transition from bound-state level U to L . As examined in the Problems section, for atomic hydrogen the dipole matrix element is evaluated with the hydrogenic wave functions and determines the $2P \rightarrow 1S$ lifetime as:

$$\tau(2p \rightarrow 1s) = 1.6 \times 10^{-9} \text{sec.} \quad (20.56)$$

Similar calculations for electric quadrupole (E2) or magnetic dipole (M1) transitions employ higher-order terms in the exponential series (20.42) and yield:

$$\tau(E2, M1) = O(10^{-3}) \text{sec.} \quad (20.57)$$

Note, longer lifetimes corresponds to *less* probable transitions. The lifetimes we calculate here are for the lowest-order transitions, that is, those proportional to the fine structure constant $\alpha = e^2$ and arising from the first term in the Born series (20.30). Higher-order terms in H add higher-order-of- α corrections to the rate and may be important for transitions which are forbidden in lowest order. For example, the $2S \rightarrow 1S$ transition in hydrogen occurs with the emission of two photons (see the Problems section) and thus has an unusually long lifetime:

$$\tau(2s \rightarrow 1s) \simeq \frac{1}{7} \text{sec.} \quad (20.58)$$

Light Absorption, Planck's Law

*He, to keep the warm side inside,
Put the cold side, skin-side outside;
He, to keep the cold side outside,
Put the warm side, fur-side, inside;
That's why he put the cold side outside,
Why he put the warm side inside,
Why he turned them inside outside.*

—Anonymous

We now return to the problem which made quantum mechanics famous, Planck's law. That is, we want to calculate the spectral distribution of radiation (photons) in thermal equilibrium with radiating atoms. We take a field-theory point of view of the two-level system represented in Figure 20.3, in which photons are being continuously created and destroyed:

$$\psi_U \leftrightarrow \psi_L + \gamma. \quad (20.59)$$

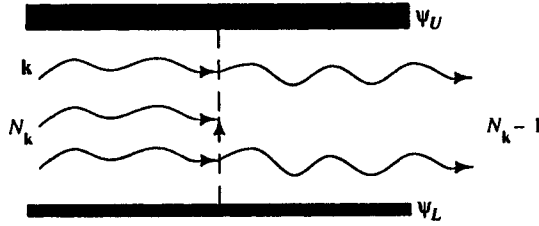


Figure 20.4: The state ψ_L being excited to ψ_U via single-photon absorption.

Absorption of Photons

We have not yet spoken about the absorption of photons, Figure 20.4, which is very much like the left side of Figure 20.3. For absorption, the initial and final Fock states are:

$$|i\rangle = |\psi_L\rangle |N_{\mathbf{k}}\rangle, \quad |f\rangle = |\psi_U\rangle |N_{\mathbf{k}} - 1\rangle, \quad (20.60)$$

that is, the final state has one fewer photon than the initial state. Consequently, the destruction of a photon from the initial state yields the final state with a factor $\sqrt{N_{\mathbf{k}}}$:

$$a_{\mathbf{k}} |i\rangle = |\psi_L\rangle |N_{\mathbf{k}} - 1\rangle \sqrt{N_{\mathbf{k}}}. \quad (20.61)$$

Accordingly, we can calculate the absorption of radiation with the same steps used to determine the rate for emission (20.40).

Exercise Show that

$$\frac{\Gamma_{LU}^{\text{ems}}}{\Gamma_{UL}^{\text{abs}}} = \frac{(N_{\mathbf{k}_i} + 1) \left| \langle \psi_L | \mathbf{p} \cdot \hat{\mathbf{e}}_{\mathbf{k}_i} e^{-i\mathbf{k} \cdot \mathbf{x}} | \psi_U \rangle \right|^2}{N_{\mathbf{k}_i} \left| \langle \psi_U | \mathbf{p} \cdot \hat{\mathbf{e}}_{\mathbf{k}_i} e^{i\mathbf{k} \cdot \mathbf{x}} | \psi_L \rangle \right|^2}. \quad \diamond \quad (20.62)$$

Yet because we know the interaction Hamiltonian and because it is Hermitian, the ratio of moduli of matrix elements in (20.62) is one because

$$\langle \psi_L | \mathbf{p} \cdot \hat{\mathbf{e}}_{\mathbf{k}_i} e^{-i\mathbf{k} \cdot \mathbf{x}} | \psi_U \rangle \equiv \langle \psi_U | \mathbf{p} \cdot \hat{\mathbf{e}}_{\mathbf{k}_i} e^{i\mathbf{k} \cdot \mathbf{x}} | \psi_L \rangle^*. \quad (20.63)$$

The ratio (20.62) is thus

$$\frac{\Gamma_{LU}^{\text{ems}}}{\Gamma_{UL}^{\text{abs}}} = \frac{N_{\mathbf{k}_i} + 1}{N_{\mathbf{k}_i}}. \quad (20.64)$$

We now return to our cavity-model problem in Figure 20.3 and postulate large numbers N_U and N_L of particles in the upper and lower levels in equilibrium with each other and with $N_{\mathbf{k}_i}$ photons on the right and left (but only $N_{\mathbf{k}_i} - 1$ in the middle). Because we have equilibrium, the emission and absorption rates equal each other:

$$N_U \Gamma^{\text{ems}} = N_L \Gamma^{\text{abs}}, \quad (20.65)$$

$$\frac{N_L}{N_U} = \frac{\Gamma^{\text{ems}}}{\Gamma^{\text{abs}}} = \frac{N_{\mathbf{k}_i} + 1}{N_{\mathbf{k}_i}}. \quad (20.66)$$

We relate this ratio to the temperature of the system via the Boltzmann distribution law of statistical mechanics, which states that the probability of finding a system with energy E is proportional to $\exp(-E/k_b T)$, where k is Boltzmann's constant. For our case:

$$\frac{N_L}{N_U} = \frac{e^{-E_L/k_b T}}{e^{-E_U/k_b T}} = e^{\omega_k/k_b T}, \quad \omega_k = E_U - E_L. \quad (20.67)$$

Exercise Show that substitution of (20.67) into (20.66) gives the number of photons in equilibrium with a two-level system at temperature T :

$$N_k(T) = \frac{1}{e^{\omega_k/k_b T} - 1}. \quad \diamond \quad (20.68)$$

The number of photons in the state $|k_i\rangle$ can vary with temperature, with large numbers of photons occurring for a hot, dense medium. If the photons are actually in a cavity, enough energy levels are available to the atoms for us to consider ω to be continuous. In this case the number of photon states in a box of volume V is (Appendix A, *Natural Units and Plane Waves*):

$$dN_k = 2 \frac{d^3 k}{(2\pi)^3} V, \quad (20.69)$$

the factor of 2 out front accounting for the two polarization directions. The total energy in the frequency range from ω to $\omega + d\omega$ is thus the product of the number of photons in the state with frequency ω (20.68), times the number of states available in the cavity for this angular frequency (20.69):

$$dE = \frac{\omega}{e^{\omega/k_b T} - 1} \frac{2d^3 k V}{(2\pi)^3} = \frac{\omega}{e^{\omega/k_b T} - 1} \frac{8\pi k^2 d\omega V}{(2\pi)^3}. \quad (20.70)$$

The spectral distribution per unit volume per unit frequency is Planck's radiation law:

$$\frac{dE}{V d\nu} = \frac{8\pi\nu^3}{e^{2\pi\nu/k_b T} - 1} \equiv \frac{8\pi\hbar(\nu/c)^3}{e^{2\pi\hbar\nu/k_b T} - 1}. \quad (20.71)$$

We see that the basic dynamics of Planck's law arises from the annihilation and creation operators giving the ratio (20.67). The Hermiticity of the Hamiltonian and the number of particles in a box are in some sense just kinematic or symmetry conditions.

20.6 Problems

1. Review questions:

- What is the effect of the field operator $\Psi(\mathbf{r})$ when it operates on a state?
- Indicate why *minimal electromagnetic coupling* is so important in QED.
- How are the electron-photon interaction Hamiltonians H_γ and $H_{2\gamma}$ deduced?
- Describe what is meant by the "radiation" and "Coulomb" gauges.
- What is Planck's hypothesis about the radiation field?

2. An electron is in a time-dependent, external electromagnetic field.

(a) Show that the gauge transformation

$$\mathbf{A} \rightarrow \mathbf{A} + \nabla\chi(\mathbf{r}, t), \quad \phi \rightarrow \phi - \frac{\partial\chi(\mathbf{r}, t)}{\partial t} \quad (20.72)$$

induces only a phase change on the quantum-mechanical wave function:

$$\psi(\mathbf{r}) \rightarrow e^{i\chi(\mathbf{r}, t)}\psi(\mathbf{r}). \quad (20.73)$$

(b) Why is the procedure $\mathbf{p} \rightarrow \mathbf{p} - e\mathbf{A}$ called *gauge invariant* in classical physics?

3. Use the dipole approximation.

(a) Show that the lifetime for spontaneous decay between the $2P$ and $1S$ levels in atomic hydrogen is $\tau = 1/\Gamma = 1.6 \times 10^{-9}$ sec.

(b) Verify that the $\Delta m = 1$ rate equals the $\Delta m = 0$ rate.

4. Consider a three-level system in which level 1 (the highest) decays exponentially with lifetime τ_1 , level 2 decays with lifetime τ_2 , and level 3 (the lowest) is stable.

(a) Show that τ_1 is the mean (average) time it takes for a particle in level 1 to decay to level 2.

(b) If at time $t = 0$ there are $N_1(0)$ atoms in state 1 and no atoms in states 2 and 3, derive an expression for $N_3(t)$, the number of atoms in state 3 as a function of time.

5. As an example of the preceding problem, consider the $3D \rightarrow 1P$ transition in hydrogen.

(a) Can this transition occur within the confines of the dipole approximation (Explain).

(b) When the lifetime for the $3D \rightarrow 2P$ transition is calculated in dipole approximation (much like the $2P \rightarrow 1S$ calculation), we obtain $\tau_2 = 1.56 \times 10^{-8}$ seconds. Derive an approximate expression for the mean lifetime for the $3D \rightarrow 1S$ transition in hydrogen. (This requires little calculation.)

6. Consider the radiative processes responsible for the $2S \rightarrow 1S$ transition in a (non-relativistic) hydrogen atom.

(a) Prove that the transition cannot occur via single-photon emission.

(b) Estimate (say, within a factor of 10) the lifetime for this transition if it occurs via the magnetic (spin) interaction of the electron.

(c) Determine the matrix element appropriate for two-photon emission transitions (both H_γ and $H_{2\gamma}$ contribute).

(d) Show that in dipole approximation the two photons are preferentially emitted in the same or opposite directions and that the probability of observing two photons with an angle between them of θ is proportional to $1 + \cos^2 \theta$. Assume that the polarizations are not observed.

(e) Estimate the order of magnitude for the lifetime of this metastable $2S$ state.

7. Consider the various $P_{1/2} \rightarrow S_{1/2}$ and $P_{3/2} \rightarrow S_{1/2}$ transitions for a one-electron atom in a weak magnetic field.
- Determine the relative shifts of the lines from the zero-field case. (*Hint:* Use the correct g factor, the dipole selection rules, and a clear energy-level diagram.)
 - Determine the angular distribution of each line (in dipole approximation).
 - Determine the relative line intensities, assuming equal initial-state populations. (You can leave your answers in terms of ratios of reduced matrix elements.)
 - Determine the polarization of the light emitted in the preferred direction for each line.
8. In describing the electromagnetic radiation field in terms of quanta (photons), the vector potential operator $\mathbf{A}$ is expanded as a type of Fourier series.
- Write down the expansion of $\mathbf{A}$ and *briefly* explain the meaning of each term.
 - What is the electric field operator $\mathbf{E}$ in this representation?
 - What is the magnetic field operator $\mathbf{B}$ in this representation?
 - Show that the expectation value of $\mathbf{E}$ must vanish for an occupation number (Fock) space state with a fixed number of photons N , that is,

$$\langle N | \mathbf{E} | N \rangle = 0. \quad (20.74)$$

This shows that a fixed number of photons is not compatible with the picture of a classical EM field with its definite value of $\mathbf{E}$.

- To further illustrate the incompatibility of a state with a simultaneous fixed number of photons and well-defined electric field, show that the number operator $N = a^\dagger a$ and the electric field operator $\mathbf{E}$ do not commute.
9. Show that the use of second-quantized fields in the classical relation for the energy of the electromagnetic field leads to Planck's hypothesis for the energy in the EM field [(20.20 and (20.21)]:

$$H = \frac{1}{8\pi} \int d^3x (E^2 + B^2) \quad \Rightarrow \quad E = \sum_{\mathbf{k}, i} N_{\mathbf{k}, i} \omega_{\mathbf{k}}. \quad (20.75)$$

10. Derive the dipole selection rules (20.48):

$$\langle \psi_L | \mathbf{r} | \psi_U \rangle = \begin{cases} 0, & \text{if } \psi_L \text{ and } \psi_U \text{ are spherically symmetric,} \\ \text{not zero,} & \text{when } \Delta l = \pm 1, \Delta m = \pm 1, 0. \end{cases} \quad (20.76)$$

Show that the rules are still valid even if the dipole approximation is not made.

CHAPTER 21

APPLICATIONS OF NONRELATIVISTIC QUANTUM ELECTRODYNAMICS

Whereas the basic Hamiltonians of QED are fairly simple when expressed as fields, they appear rather complicated when the fields are expanded in annihilation and destruction operators. Likewise, the calculation of a transition rate or a cross section may involve just one or two powers of the Hamiltonian yet, when expanded, the accounting may appear overwhelming. In this chapter we encounter a few more calculations with the nonrelativistic theory to get a better feel for what goes into a QED calculation and to learn some of the short cuts. In particular, we shall see how each part of a Feynman diagram such as Figure 21.4 corresponds to definite steps in an otherwise lengthy calculation. While attempting to teach all the rules for constructing and calculating directly from these diagrams is beyond the scope of this book, we use graphs as a pedagogic path to the rules in texts such as Gross (1993), Bjorken and Drell (1964, 1965), Schweber (1959), and Halzen and Martin (1984).

*It was a beauty that I saw,
So pure, so perfect, as the frame
Of all the universe was lame
To that one figure, could I draw
Or give least line of it a law.*

—Ben Jonson

21.1 Light Scattering from Electrons

Classical: Rayleigh, Thompson, and Compton

Classically, when light shines on the bound electron shown schematically in Figure 21.1, the incident electric field causes the electron to vibrate. Since the vibration is an acceleration, this causes the emission of radiation from the electron. That radiation is emitted at the same frequency as the incident radiation and is viewed as light elastically “scattered” by the electron as illustrated in Figure 21.2. If ω_0 is the characteristic (resonant) frequency of the bound system and the incident radiation has a frequency ω , the classical cross section

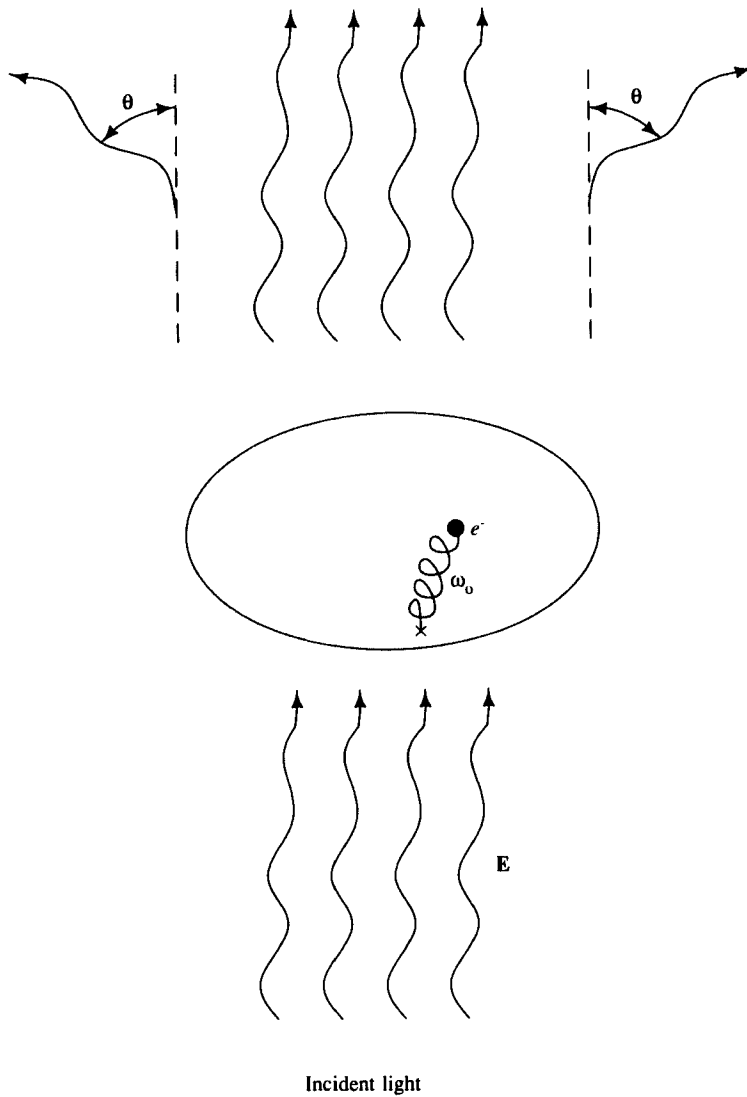


Figure 21.1: The classical scattering of an incident light beam **E** from an electron e^- bound to a force center.

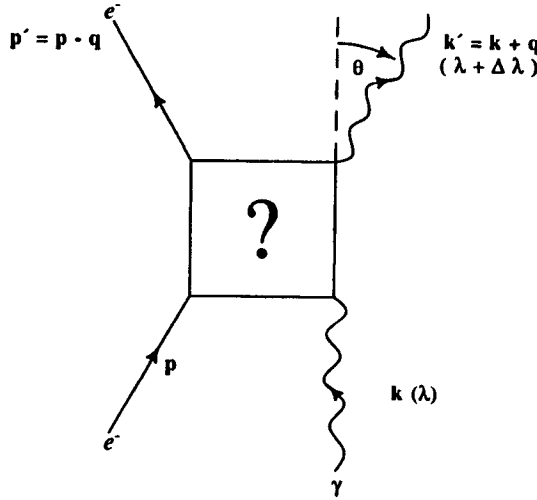


Figure 21.2: Kinematics for Compton scattering of a photon of momentum $\mathbf{k}$ from an electron of momentum $\mathbf{p}$. The box represents all possible reaction mechanisms.

(given, for example, by Jackson, 1975) is

$$\sigma = \frac{8\pi r_e^2}{3} \frac{\omega^4}{(\omega_0^2 - \omega^2)^2}. \quad (21.1)$$

Here r_e is a constant with the dimension of length known as the *classical radius of the electron*,

$$r_e = \frac{e^2}{m} \equiv \frac{[e^2/(\hbar c)](\hbar c)}{mc^2} \simeq \frac{[1/137](197 \text{ MeV fm})}{0.51 \text{ MeV}} \simeq 2.82 \text{ fm}. \quad (21.2)$$

In spite of its dimensions, r_e is not the size of anything physical. Its magnitude of nearly 3 fm is very big (the size of light nuclei), whereas measurements of the actual electron's size are consistent with zero (a point particle). Yet r_e is a useful quantity because it is the product of the Compton wavelength of the electron (the scale at which relativity enters),

$$\lambda_c = \frac{1}{m} \equiv \frac{\hbar c}{mc^2} \simeq 386 \text{ fm}, \quad (21.3)$$

with the fine-structure constant $\alpha = e^2$,

$$r_e = \alpha \lambda_c. \quad (21.4)$$

There are two limits of the electron-light scattering cross section (21.1) that appear frequently. The first, *Rayleigh scattering*, is the low-energy limit,

$$\sigma \rightarrow \frac{8}{3} \pi r_e^2 \frac{\omega^4}{\omega_0^4}, \quad (\omega \ll \omega_0). \quad (21.5)$$

The rapid increase in scattering of the larger ω 's explains why the sky is blue. The second limit, *Thompson scattering*, is the high-energy limit,

$$\sigma \rightarrow \frac{8}{3} \pi r_e^2, \quad (\omega \gg \omega_0), \quad (21.6)$$

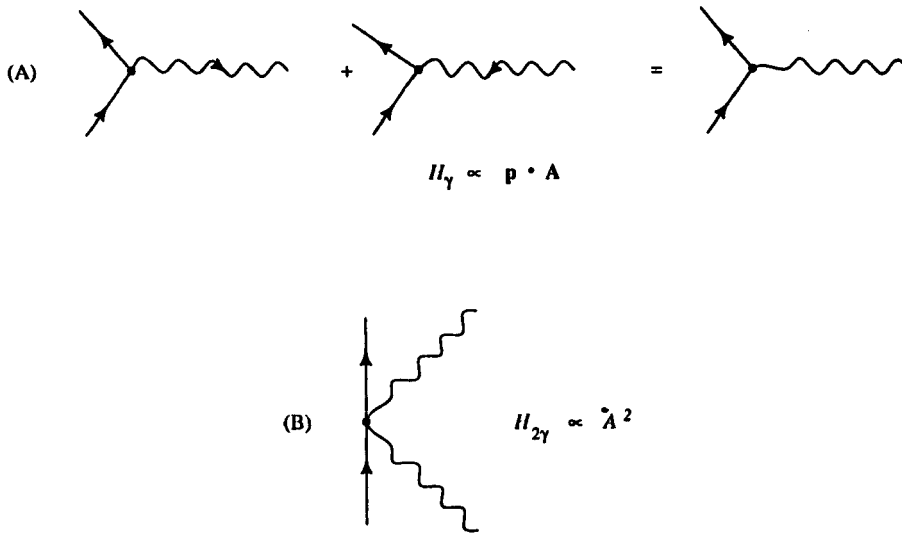


Figure 21.3: The nonrelativistic electron–photon Hamiltonian. (A) 1-photon coupling via $H_{1\gamma}$; (B) 2-photon coupling via $H_{2\gamma}$.

which clearly ignores the binding of the electron. While dropping names, we remind the reader that *Compton scattering* refers to the scattering of very high energy light (X-rays or gamma rays) from electrons in which the electron recoils with enough energy for the quantum (particle) nature of the photon to become evident. This is the kinematical effect illustrated in Figure 21.2 that leads to the famous relation:

$$\Delta\lambda = \lambda_c(1 - \cos\theta). \quad (21.7)$$

Exercise Show that (21.7) follows from setting the recoil energy of the electron equal to the energy loss of the photon. $\diamond$

The constant on the RHS of (21.7) is the electron’s Compton wavelength (21.3), which we see from its derivation has nothing to do with the size of the electron.

Quantum Photon–Electron Scattering

After all the scattering formulations of Part I, *Scattering and Integral Quantum Mechanics*, you would think we could just take one of those powerful equations, choose the appropriate potential, and come up with the cross section for photon–electron scattering. But it is to no avail; we developed scattering theory for situations in which particles are neither created nor destroyed, and so we must use field theory.¹

Consider Thompson scattering in which the photon’s energy ω is large enough so that we can ignore the binding of the electron, yet much smaller than the rest energy of the electron, so we can use nonrelativistic kinematics for the electron. The process is pictured in Figure 21.2. Our problem is to figure out what goes on inside the box.

¹Indeed, Sakurai (1967) obtains many QED results by describing photon emission and absorption with time-dependent potentials.

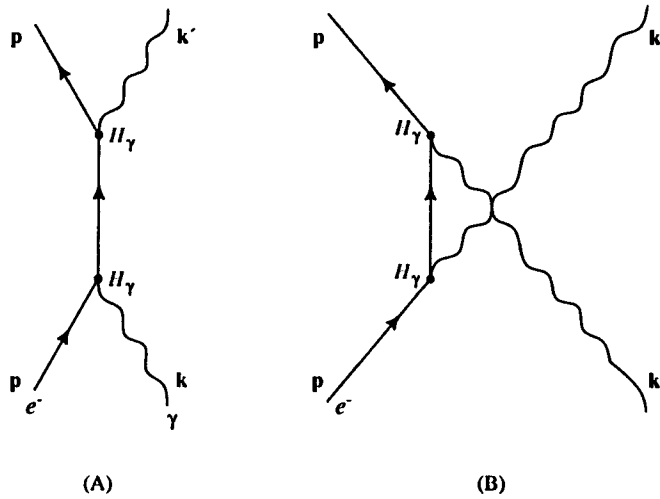


Figure 21.4: Two momentum-space Feynman diagrams contributing to Compton scattering in second order. (A) The initial photon is absorbed before the final photon is emitted; (B) the final photon is emitted before the initial one is absorbed.

Because in field theory what goes on inside the box is some convolution of the absorption and emission of photons, our problem is not much different from emission and absorption of radiation studied in § 20.5. We start (as usual) with the Hamiltonian describing the coupling of electrons with charge $-e$ and mass m to photons (20.24):

$$H = H_\gamma + H_{2\gamma} = \frac{-e}{m} \mathbf{A} \cdot \mathbf{p}_e + \frac{e^2}{2m} A^2, \quad (21.8)$$

where H_γ is a sum over photon creation and annihilation operators $a_{\mathbf{k}}$ and $a_{\mathbf{k}}^\dagger$ (20.27), and $H_{2\gamma}$ is a sum over their squares aa , $aa^\dagger$, $a^\dagger a$, and $a^\dagger a^\dagger$, (20.28). The individual pieces of H_γ are pictured in Figure 21.3A, and those of $H_{2\gamma}$ in Figure 21.3B. It is seen that by itself, H_γ does not conserve 4-momentum and so cannot directly connect one free electron to another; if present, it must act more than once within the box of Figure 21.2. In contrast, $H_{2\gamma}$ does permit a direct connection between the initial and final states.

For photon scattering, H_γ does not have a matrix element between $|i\rangle$ and $|f\rangle$ because H_γ does not contain a term which annihilates the initial photon and create the final one (this would take a term like $a_{\mathbf{k}'}^\dagger a_{\mathbf{k}}$). However, there is such a connection possible in second order:

$$M_{fi} = 0 + \sum_n \frac{\langle f | H_\gamma | n \rangle \langle n | H_\gamma | i \rangle}{E - E_n} + \dots \quad (21.9)$$

The two possible second-order contributions are diagrammed in Figure 21.4. They differ inasmuch as Figure 21.4A has no photons present in the intermediate state, whereas Figure 21.4B has two. Although by no means obvious, these two diagrams tend to cancel each other (and do so perfectly in the classical limit); we leave their evaluation to the Problems section, state their value in (21.33), and actually evaluate them in the next chapter for relativistic electrons.

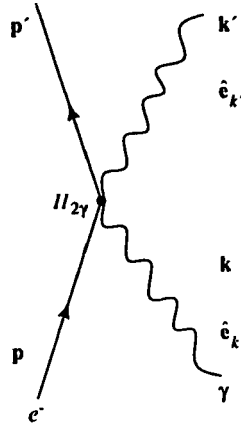


Figure 21.5: The seagull graph, the lowest-order diagram contributing to Compton scattering via the 2-photon Hamiltonian $H_{2\gamma}$.

Each of the diagrams in Figure 21.4 are second order in H_γ and therefore of order e^2 ; that is, there is a factor of e introduced each time a photon is created or destroyed at a vertex. The second power of e also occurs in lowest order with $H_{2\gamma}$ and, (Figure 21.5) is rather unusual in that it directly connects the two particles in the initial and final states.² We describe these states in Fock space for the photons and momentum space for electrons:

$$|i\rangle = |p\rangle_e |N_{\mathbf{k}_i}\rangle_\gamma, \quad |f\rangle = |p'\rangle_e |N_{\mathbf{k}'_i}\rangle_\gamma. \quad (21.10)$$

Although unusual, this “seagull graph” produces the classical limit. The $H_{2\gamma}$ of (20.28) is the sum over four distinct operators: one which creates two photons, one which destroys two photons, and two which create a photon while destroying a different one. It is clearly the latter two,

$$H_{2\gamma} = \frac{e^2}{2m} \sum_{\mathbf{k}, i} \sum_{\mathbf{k}', i'} \sqrt{\frac{2\pi}{V\omega_{\mathbf{k}}}} \sqrt{\frac{2\pi}{V\omega_{\mathbf{k}'}}} \times \hat{\mathbf{e}}_{\mathbf{k}'_i'} \cdot \hat{\mathbf{e}}_{\mathbf{k}_i} \left[a_{\mathbf{k}_i} a_{\mathbf{k}'_i'}^\dagger e^{i(\mathbf{k}-\mathbf{k}')\cdot\mathbf{x}} + a_{\mathbf{k}_i}^\dagger a_{\mathbf{k}'_i'} e^{i(\mathbf{k}'-\mathbf{k})\cdot\mathbf{x}} + \dots \right], \quad (21.11)$$

which contribute to the process described in Figure 21.5. To apply the golden rule (20.29) in lowest order, we need the matrix element of $H_{2\gamma}$ between the initial and final states (21.10).

Exercise Show that the matrix elements of the sums over $a^\dagger a$ and $aa^\dagger$ both give the same result [there are only contributions when $(\mathbf{k}, \mathbf{k}')$ equals $(\mathbf{k}_i, \mathbf{k}_f)$ or $(\mathbf{k}_f, \mathbf{k}_i)$]:

$$\begin{aligned} \langle f | H_{2\gamma} | i \rangle &= 2 \frac{e^2}{2m} \sqrt{\frac{2\pi}{V\omega_{\mathbf{k}_i}}} \sqrt{\frac{2\pi}{V\omega_{\mathbf{k}_f}}} \\ &\times \hat{\mathbf{e}}_{\mathbf{k}'_i'} \cdot \hat{\mathbf{e}}_{\mathbf{k}_i} \langle \mathbf{p}_f | e^{i(\mathbf{k}_i-\mathbf{k}_f)\cdot\mathbf{x}} | \mathbf{p}_i \rangle \sqrt{N_i(N_f+1)}, \end{aligned} \quad (21.12)$$

²It is interesting to observe that because $H_{2\gamma}$ does not occur in relativistic QED, there are no four-point functions there, that is, no four fields acting at the same point.

where the factor of 2 comes from the two possibilities for $(\mathbf{k}, \mathbf{k}')$, and N_i (N_f) is the number of initial (final) photons present. $\diamond$

Calculating Cross Sections

To calculate a cross section as defined in (2.10), we take the golden rule (20.29),³

$$\Gamma_{fi} = 2\pi |\mathcal{M}_{fi}|^2 \delta(E_f - E_i), \quad (21.13)$$

which determines the number of scatterings (transitions) per unit time, and divide it by the incident beam flux. The beam flux is the number of photons incident on the target per unit time per unit area of target, that is, the number density of the beam times the beam velocity v . Putting the rate and flux together gives the differential cross section:

$$d\sigma = \frac{\Gamma_{fi}}{(N_i/V)} = \frac{\Gamma_{fi} V}{N_i}. \quad (21.14)$$

For our calculation of the seagull graph, this is:

$$\begin{aligned} d\sigma = & \frac{V}{N_i} 2^2 2\pi \left(\frac{e^2}{2m} \right)^2 \left(\sqrt{\frac{2\pi}{V\omega_k}} \sqrt{\frac{2\pi}{V\omega_{k'}}} \right)^2 \\ & \times |\hat{\mathbf{e}}_{\mathbf{k}',f} \cdot \hat{\mathbf{e}}_{\mathbf{k},i}|^2 \left| \langle \mathbf{p}' | e^{i(\mathbf{k}-\mathbf{k}') \cdot \mathbf{x}} | \mathbf{p} \rangle \right|^2 N_i (N_f + 1) \delta(E_f - E_i). \end{aligned} \quad (21.15)$$

We call (21.15) a “differential cross section,” yet it may be more differential than those measured in realistic experiments which employ instruments with finite angular and energy resolution. A more appropriate differential cross section is then a sum over the range of final photon momenta actually measured:

$$\begin{aligned} d\sigma = & \sum_{\mathbf{k}'} \frac{V}{N_f} 2^2 2\pi \left(\frac{e^2}{2m} \right)^2 \left(\sqrt{\frac{2\pi}{V\omega_k}} \sqrt{\frac{2\pi}{V\omega_{k'}}} \right)^2 |\hat{\mathbf{e}}_{\mathbf{k}',f} \cdot \hat{\mathbf{e}}_{\mathbf{k},i}|^2 \\ & \times \left| \langle \mathbf{p}' | e^{i(\mathbf{k}-\mathbf{k}') \cdot \mathbf{x}} | \mathbf{p} \rangle \right|^2 N_i (N_f + 1) \delta(E_f - E_i). \end{aligned} \quad (21.16)$$

Here we do not sum over the initial and final photon polarizations since they are variables which an experimenter has the option of measuring.

The matrix element of the exponential in (21.16) gives the momentum conservation,

$$\langle \mathbf{p}' | e^{i(\mathbf{k}-\mathbf{k}') \cdot \mathbf{x}} | \mathbf{p} \rangle = \delta_{\mathbf{p}+\mathbf{k}, -\mathbf{p}'+\mathbf{k}'}, \quad (21.17)$$

already assumed in Figure 21.2 and in (21.16). For our box normalization (see Appendix A, *Natural Units and Plane Waves*) a Kronecker delta function occurs in (21.17). It equals 1 when momentum is conserved and still equals 1 when squared in (21.16) (if we had gone to continuum normalizations, we would get a Dirac delta function and its square is harder

³In Chapter 23, *The Breit–Pauli and Meson-Exchange Interactions*, we discuss the calculation of cross sections for 1-body and 2-body reactions, and relate them to the equivalent potential theory calculation.

to handle).⁴ To reach the classical limit, we set $N_f = 1$ so that there are no final-state photons before the scattering (else we get “stimulated” emission), and consider photons of high, but not too high, energy such that their energy loss upon scattering is negligible while still having $\omega \ll m_e$. This kinematic constraint is reasonable for photons of lower energy (less “hard”) than X-rays. Specifically, since the Compton relation (21.7) tells us that

$$\Delta\lambda = \frac{1}{m_e}(1 - \cos\theta) = (0.004 \text{ \AA})(1 - \cos\theta), \quad (21.19)$$

the fractional energy loss by the incident photon is

$$\frac{\Delta E}{E} = \frac{\Delta\omega}{\omega} = \frac{\Delta\lambda}{\lambda} \simeq \frac{0.004 \text{ \AA}}{4000 \text{ \AA}} \simeq 0. \quad (21.20)$$

Therefore we set $\omega_f \simeq \omega_i = \omega$ and $k' \simeq k$. Because the photons are detected in a “large” box, we replace the sum over final photon momenta with the integral (A.19):

$$\sum_k \rightarrow V \int \frac{d^3k}{(2\pi)^3}. \quad (21.21)$$

Yet because the direction of $\mathbf{k}$ is something an experiment may measure, we take a step back and do not integrate over $\Omega_{\mathbf{k}}$:

$$d\sigma = \frac{V 2\pi r_e^2 (2\pi)^2}{N_f 4\omega^2 V^2} 4 \left| \hat{\mathbf{e}}_{\mathbf{k}'f} \cdot \hat{\mathbf{e}}_{\mathbf{k}i} \right|^2 N_f V d\Omega_{\mathbf{k}} \int \frac{dk k^2}{(2\pi)^3} \delta(E_f - E_i), \quad (21.22)$$

where we have combined some of the constants into the classical radius of the electron r_e , (21.2). The integral is evaluated with the help of the energy delta function $\delta(E_f - E_i)$, which, if we ignore the recoil energy of the electron, we can approximate with the momentum delta function $\delta(k_f - k_i)$:

$$d\Omega_{\mathbf{k}} \int_0^\infty \frac{dk k^2}{(2\pi)^3} \delta(k_f - k_i) = \frac{\Omega_{\mathbf{k}} k^2 d}{(2\pi)^3}. \quad (21.23)$$

Note that in (21.23) we have not integrated over $d\Omega_{\mathbf{k}}$, the differential solid angle of the emitted photon, because this is a variable which the experimenter can control. The (Thompson) cross section arising from the seagull graph Figure 21.5 is thus independent of photon energy (as long as $\omega \gg \omega_0$),

$$d\sigma = r_e^2 \left| \hat{\mathbf{e}}_{\mathbf{k}'f} \cdot \hat{\mathbf{e}}_{\mathbf{k}i} \right|^2 \delta_{\mathbf{k}'\mathbf{k}} d\Omega_{\mathbf{k}}. \quad (21.24)$$

Division by $d\Omega$ gives us the conventional differential cross section (per unit solid angle):

$$\boxed{\frac{d\sigma}{d\Omega}(\theta, \phi) = r_e^2 \left| \hat{\mathbf{e}}_{\mathbf{k}'f} \cdot \hat{\mathbf{e}}_{\mathbf{k}i} \right|^2 \delta_{\mathbf{k}'\mathbf{k}},} \quad (21.25)$$

where the delta function reminds us that this is the result for elastic scattering. Equation (21.25) is the same as the classical result for Thomson scattering including the dependence on photon polarization.

⁴In this case one of the delta functions gets integrated over in summing over final states while the other one gets removed by convention when we evaluate matrix elements in a two-body Hilbert space, e.g.,

$$\langle \mathbf{p}_1, \mathbf{p}_2 | T | \mathbf{p}'_1, \mathbf{p}'_2 \rangle = \delta(\mathbf{p}'_1 + \mathbf{p}'_2 - \mathbf{p}_1 - \mathbf{p}_2) \delta(E_f - E_i) T(\mathbf{k}' \leftarrow \mathbf{k}). \quad (21.18)$$

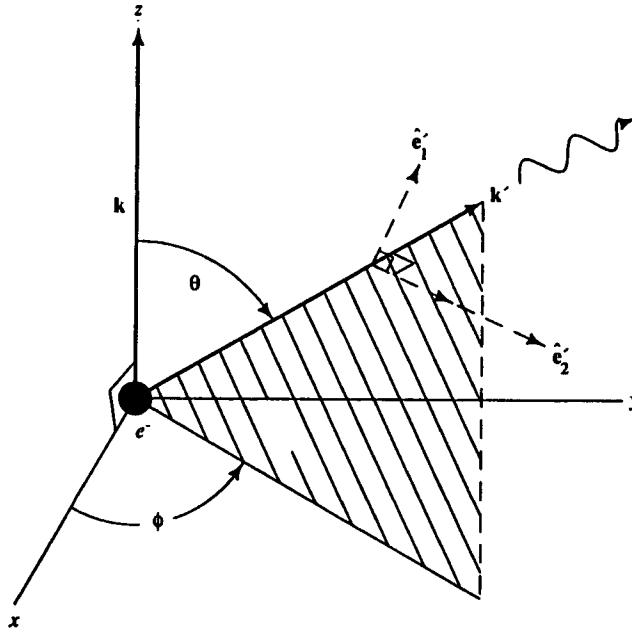


Figure 21.6: Explicit geometry of the photon's momenta and polarization vectors in Compton scattering. Primes denote final states.

Photon Polarization Sum

If the experimenter does not observe the polarization of the scattered photon $\hat{\mathbf{e}}_{\mathbf{k}'f}$, then he or she has effectively summed the cross section over the two orientations of $\hat{\mathbf{e}}_{\mathbf{k}'f}$, and we must do the same theoretically to predict that cross section. Likewise, if the initial photon beam is unpolarized, $\langle \hat{\mathbf{e}}_{\mathbf{k}i} \rangle = 0$, the theoretical cross section is an average over the two orientations for $\hat{\mathbf{e}}_{\mathbf{k}i}$ (that is, half the sum of the cross sections for both $\hat{\mathbf{e}}$'s).⁵

To carry out these summations, it is helpful to view a diagram such as Figure 21.6 where we assume real $\hat{\mathbf{e}}$'s and orient the coordinate system to simplify the calculation. The initial photon's direction $\hat{\mathbf{k}}$ is chosen as the z axis, and its polarization $\hat{\mathbf{e}}_{\mathbf{k}i}$ is chosen along the x axis:

$$\hat{\mathbf{e}}_{\mathbf{k}i} = (1, 0, 0). \quad (21.26)$$

The scattered photon's momentum $\mathbf{k}'$ is then described with the polar and azimuthal angles (θ, ϕ) . In addition, scattered photon's spin is described by having its polarization either normal to ("out of") or "in" the scattered plane:

$$\begin{aligned} \hat{\mathbf{e}}_{\mathbf{k}'1} &= (\sin \phi, -\cos \phi, 0), & \text{out of plane,} \\ \hat{\mathbf{e}}_{\mathbf{k}'2} &= (\cos \theta \cos \phi, \cos \theta \sin \phi, -\sin \theta), & \text{in plane.} \end{aligned} \quad (21.27)$$

The cross section (21.25) for the two possible final polarizations is thus

$$\frac{d\sigma}{d\Omega}(\theta, \phi) = r_e^2 \left| \hat{\mathbf{e}}_{\mathbf{k}'f} \cdot \hat{\mathbf{e}}_{\mathbf{k}i} \right|^2 \delta_{\mathbf{k}', \mathbf{k}} = r_e^2 \begin{cases} \sin^2 \phi, & \text{out of plane,} \\ \cos^2 \theta \cos^2 \phi, & \text{in plane.} \end{cases} \quad (21.28)$$

⁵See Chapter 9, *Spin Theory*, for a discussion of the meaning of unpolarized beams.

There are four special cases of this formula:

1. If the initial beam is polarized along $\hat{e}_x$, (21.28) gives the cross sections for elastic scattering into the two final states.
2. If the initial beam is unpolarized, the scattering cross section is the average over all beam polarization directions [the two terms in (21.29)],

$$\left. \frac{d\sigma}{d\Omega} \right|_{\text{un}} = \frac{1}{2\pi} \int_0^{2\pi} d\phi \frac{d\sigma}{d\Omega}(\theta, \phi) = \frac{1}{2} r_e^2 \begin{cases} 1, & \text{out of plane,} \\ \cos^2 \theta, & \text{in plane.} \end{cases} \quad (21.29)$$

These equations show that the scattering of unpolarized light by unpolarized electrons polarizes the light. The degree of polarization changes with the scattering angle; for example, at $\theta = \pi$ the scattered beam will be completely polarized and “out” of the scattering plane because scattering to the “in” polarization vanishes.

3. If the initial beam is polarized but the final photon polarization is not observed, the cross section is the sum of cross sections for scattering to the two final states,

$$\left. \frac{d\sigma}{d\Omega} \right|_{\text{un} \leftarrow P} = \left. \frac{d\sigma}{d\Omega} \right|_{\text{in} \leftarrow P} + \left. \frac{d\sigma}{d\Omega} \right|_{\text{out} \leftarrow P} = r_e^2 (\sin^2 \phi + \cos^2 \theta \cos^2 \phi). \quad (21.30)$$

As also observed for the scattering of spin 0 on $\frac{1}{2}$, the forces are obviously not axial symmetric because there is some ϕ dependence.

4. If the beam is unpolarized and the final polarization is not observed, we integrate (21.30) over ϕ and sum over all spin directions in the initial beam,

$$\left. \frac{d\sigma}{d\Omega} \right|_{\text{un} \leftarrow \text{un}} = \frac{1}{2} r_e^2 (1 + \cos^2 \theta). \quad (21.31)$$

As illustrated in Figure 21.7, this cross section (radiation pattern) is equally peaked in the forward and backward directions.

5. Finally we obtain the *total cross section* for the scattering of unpolarized photons with undetected final photon polarization by integrating (21.31) over $d\Omega$:

$$\sigma_t = \int d\Omega \frac{d\sigma}{d\Omega} = \frac{8}{3} \pi r_e^2 \simeq 665 \text{ mb}. \quad (21.32)$$

We thus see that on this submicroscopic level, the classical scattering of light arises from the seagull graph and has a cross section of hundreds of millibarns.

Rayleigh Scattering

If we evaluate the second-order graphs of Figure 21.4 we would obtain (Sakurai, 1967) the more general and complete *Kramers–Heisenberg* relation:

$$\left. \frac{d\sigma}{d\Omega} \right|_{f \leftarrow i} = r_e^2 \frac{\omega'}{\omega} \left| \delta_{\mathbf{k}'\mathbf{k}} \hat{\mathbf{e}}_{\mathbf{k}'f} \cdot \hat{\mathbf{e}}_{\mathbf{k}i} - \frac{1}{m} \sum_n \left(\frac{\langle L | \mathbf{p} \cdot \hat{\mathbf{e}}_f | n \rangle \langle n | \mathbf{p} \cdot \hat{\mathbf{e}}_i | U \rangle}{E_n - E_U - \omega} + \frac{\langle L | \mathbf{p} \cdot \hat{\mathbf{e}}_i | n \rangle \langle n | \mathbf{p} \cdot \hat{\mathbf{e}}_f | U \rangle}{E_n - E_U + \omega} \right) \right|^2. \quad (21.33)$$

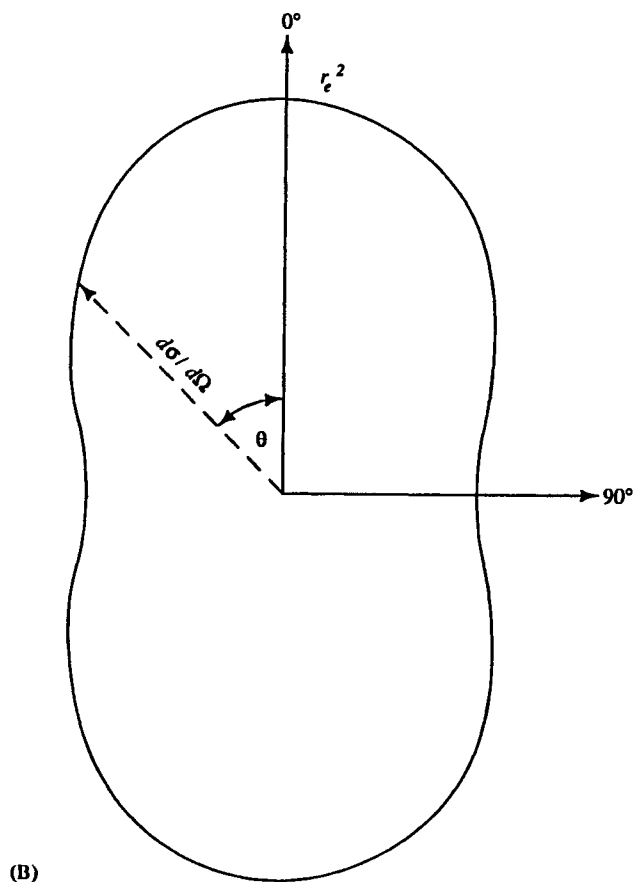
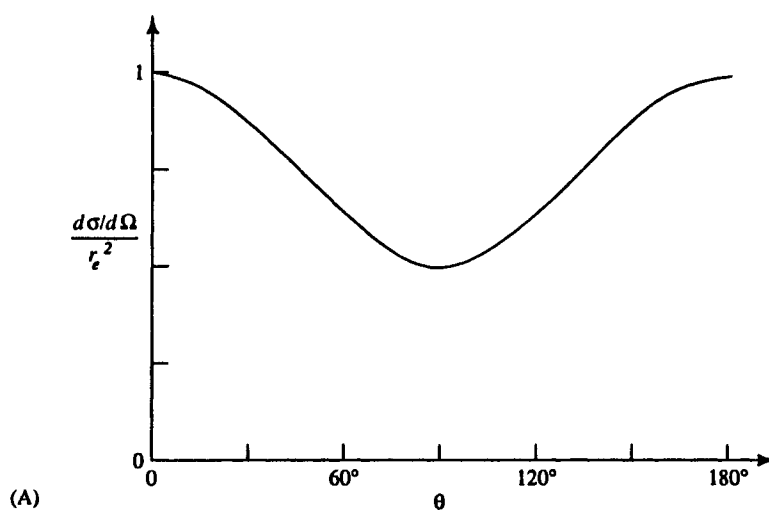


Figure 21.7: The differential cross section for Compton scattering. (A) As a function of scattering angle; (B) as a polar plot.

In (21.33), U and L refer to the upper and lower atomic states of the electron, and the sum is over virtual atomic intermediate states n .

Exercise Show that the extra terms in (21.33) cancel in the classical limit. $\diamond$

Rayleigh scattering is the limit of (21.33) for $\omega \ll E_n - E_U \stackrel{\text{def}}{=} \omega_{nU}$. In this limit the three terms combine to yield

$$\frac{d\sigma}{d\Omega} \simeq \left(\frac{r_e}{m}\right)^2 \left| \sum_n \frac{\omega^4}{\omega_{nU}^3} [\langle U | \mathbf{p} \cdot \hat{\mathbf{e}}'_f | n \rangle \langle n | \mathbf{p} \cdot \hat{\mathbf{e}}_i | U \rangle + i |\mathbf{p} \cdot \hat{\mathbf{e}} | n \rangle \langle n | \mathbf{p} \cdot \hat{\mathbf{e}}'_f | U \rangle] \right|^2. \quad (21.34)$$

Again, the ω^4 dependence of (21.34) makes the sky blue.

21.2 Coherent States of the Radiation Field: Tutorial

A quantum electromagnetic field with photons being emitted and absorbed on electrons appears quite different from a classical electromagnetic field propagating through space via waves. In part this is the wave-particle duality, yet in part it reflects a fundamental difference at a microscopic level. In the present section we lead the reader through a tutorial which examines the correspondence principle for the electromagnetic field and derives the type of second-quantized state that models lasers.

Nonclassical Aspects of Fock States

We have built the electromagnetic radiation field from photons by converting the Fourier series for the vector potential $\mathbf{A}$ into a field operator.

1. What is the electric field operator $\mathbf{E}$ in the second-quantized representation?
2. What is the magnetic field operator $\mathbf{B}$ in this representation?
3. Show that the expectation value of $\mathbf{E}$ must vanish for an occupation number (Fock space) state with a fixed number N of photons, that is,

$$\langle N | \mathbf{E} | N \rangle = 0. \quad (21.35)$$

This shows that states with a fixed number of photons are not compatible with the classical picture of an electromagnetic field having a definite value for the electric field $\mathbf{E}$.

4. Further show the incompatibility of a state with both a fixed number of photons and a well-defined electric field by showing that the number operator $\tilde{N}$ and the electric field operator $\mathbf{E}$ do *not* commute

$$[\mathbf{E}, \tilde{N}] \neq 0. \quad (21.36)$$

5. Evaluate the commutator $[\mathbf{B}(\mathbf{r}, t), \mathbf{E}(\mathbf{r}', t')]$ and interpret your result in terms of experimental measurements of electric and magnetic fields.

Construction of Coherent States

We now wish to define a new type of state in occupation number space which approximates a classical field when the number of photons becomes very large. These states are useful in understanding the classical limit and in the describing the quantum mechanics of lasers. We start by constructing a state $|\Phi_b\rangle$, that is, 'a linear combination of Fock-space states:

$$|\Phi_b\rangle = \sum_{n=0}^{\infty} c_n |n\rangle, \quad (21.37)$$

where the state $|n\rangle$ has n photons all of momentum $\mathbf{k}$.

1. Determine the result of the annihilation operator $a_{\mathbf{k}}$ acting on $|\Phi_b\rangle$.
2. If $|\Phi_b\rangle$ is an *eigenstate* of $a_{\mathbf{k}}$, its action would be a multiplication of the state by an eigenvalue,

$$a_{\mathbf{k}} |\Phi_b\rangle = b |\Phi_b\rangle, \quad (21.38)$$

where b is a complex number. Deduce the relation among the expansion coefficients c_n in (21.37) such that (21.38) is true. *Hint:* While c_0 is arbitrary, show that the other c 's are determined by:

$$c_n = \frac{b}{\sqrt{n}} c_{n-1}. \quad (21.39)$$

3. Show that the eigenstate can be written:

$$|\Phi_b\rangle = c_0 \sum_{n=0}^{\infty} \frac{b^n}{\sqrt{n!}} |n\rangle, \quad (21.40)$$

where, again, each of the n photons in $|n\rangle$ has momentum $\mathbf{k}$.

4. Show that the normalization condition,

$$\langle \Phi_b | \Phi_b \rangle = 1, \quad (21.41)$$

determines the overall normalization constant c_0 to be

$$|c_0| = e^{-|b|^2/2}. \quad (21.42)$$

5. Show that $a_{\mathbf{k}}$ and $a_{\mathbf{k}}^\dagger$ have the reasonable expectation values

$$\langle \Phi_b | a_{\mathbf{k}} | \Phi_b \rangle = b, \quad \langle \Phi_b | a_{\mathbf{k}}^\dagger | \Phi_b \rangle = b^*. \quad (21.43)$$

6. Show that there is a catch here, that is, the nonhermiticity of $a_{\mathbf{k}}$ also means that

$$a_{\mathbf{k}}^\dagger |\Phi_b\rangle \neq b^* |\Phi_b\rangle. \quad (21.44)$$

7. Show that the number operator $\tilde{N}$ has the expectation value

$$\langle \tilde{N} \rangle = bb^*. \quad (21.45)$$

8. Show that, in spite of (21.45), $|\Phi_b\rangle$ is *not* an eigenstate of $\tilde{N}$. (Hint: Show that $\langle N^\dagger \rangle \neq \langle N \rangle$, and use this to deduce the result.)
9. Determine the expectation value of the electric field operator $\mathbf{E}$ in the state $|\Phi_b\rangle$ and use this to show that $|b|$ is proportional to the magnitude of the field and that the phase of b determines the phase of the electric field. This shows that b is proportional to the amplitude of the classical, electric field.
10. Show that while this coherent state is not an eigenstate of total energy, the expectation value of the energy is proportional to the average value of the number of photons and to the field intensity, that is,

$$\langle \Phi_b | \omega_{\mathbf{k}} a_{\mathbf{k}}^\dagger a_{\mathbf{k}} | \Phi_b \rangle = \omega_{\mathbf{k}} |b|^2 = \omega_{\mathbf{k}} \langle \tilde{N} \rangle. \quad (21.46)$$

11. The relative uncertainty of the number of photons in the state $|\Phi_b\rangle$ has the conventional definition

$$\Delta N = \langle \Phi | (\tilde{N} - \langle \tilde{N} \rangle)^2 | \Phi \rangle^{1/2}. \quad (21.47)$$

Use the above relations to show that the relative uncertainty is

$$\frac{\Delta N}{\langle N \rangle} = \frac{1}{\langle N \rangle^{1/2}}. \quad (21.48)$$

Thus, even though we do not have an eigenstate of $\tilde{N}$, the relative uncertainty in the number of photons becomes very small when there is a very large number of photons. In this way we approach a classical field with a definite value of $\mathbf{E}$ and energy.

12. More specifically, show that

$$\langle \Phi_b | E^2 | \Phi_b \rangle - |\langle \Phi_b | \mathbf{E} | \Phi_b \rangle|^2 = \frac{2\pi\omega}{V}. \quad (21.49)$$

13. Comment on this relation in the limits $\hbar \rightarrow 0$ and $V \rightarrow 0$.

21.3 Self-Energies and Their Handling

*At the round earth's imagined corners, blow
Your trumpets, angels; and arise, arise
From death, you numberless infinities*

—John Donne

While there are many more problems in nonrelativistic quantum electrodynamics left to study, in the remainder of this chapter we look at how the theory copes with some problems of its own. In particular, we study the interaction of an electron with the field that it itself has generated, its so-called *self energy*. Although the relativistic field theory of Chapter 22, *Interaction of Photons with Quantized Matter, QED*, is more accurate, the basic physical effects are present (and easier to understand) with the nonrelativistic theory.

We have already encountered two examples of infinite energies in our theory: the filled sea of negative-energy electrons in Dirac's picture, (Figures 15.1 and 21.8A), and the

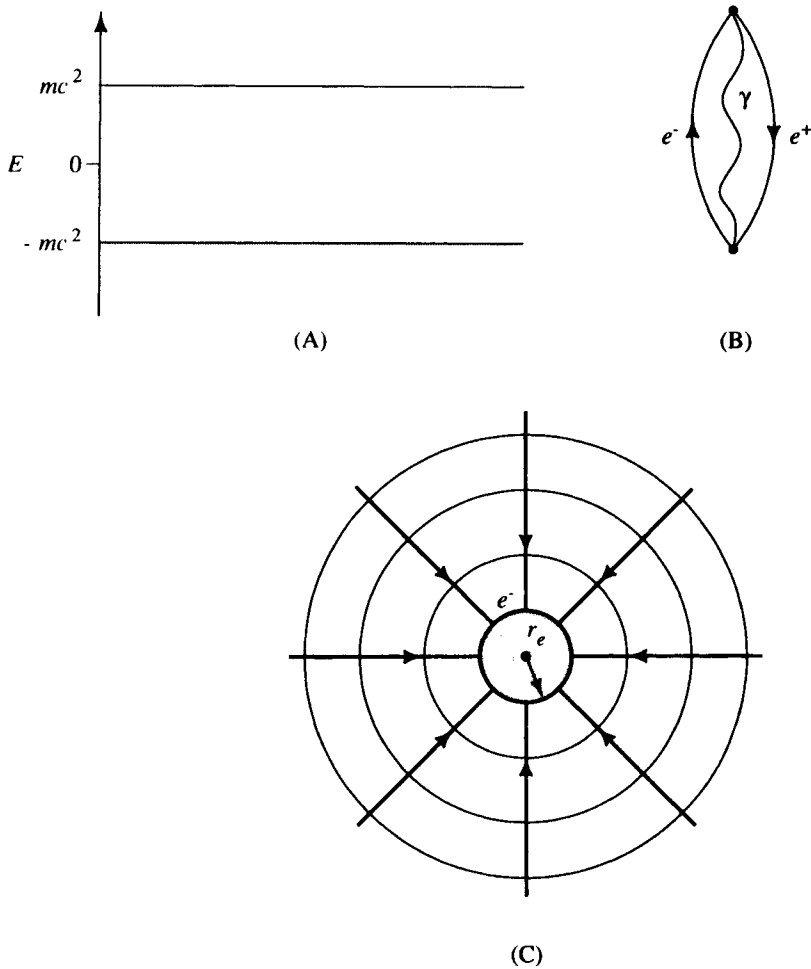


Figure 21.8: Three sources of the infinite energy of an electron. (A) the Dirac sea; (B) a vacuum fluctuation; (C) the electron's interaction with its own field (self-energy).

radiation field $E_\gamma = \langle 0 | \sum_{\mathbf{k}} (a_{\mathbf{k}}^\dagger a_{\mathbf{k}} + \frac{1}{2}) \omega_{\mathbf{k}} | 0 \rangle$. Likewise, the second-quantized Dirac sea has fluctuations in which an electron-positron pair self-creates and then self-destructs, as diagrammed in Figure 21.8B. The energy associated with this process is:

$$\langle 0 | \Delta E | 0 \rangle \propto e^2 \sum_{s,s'} \int d^3p d^3k \frac{|\bar{u}_{s'}(p+k) \gamma \cdot \hat{\epsilon}_{\mathbf{k}} u_s(p)|^2}{0 - E_{\mathbf{p}+\mathbf{k}} - E_p - k} = \infty, \quad (21.50)$$

where the zero of the denominator leads to the singularity. Although it is not comforting to have these infinities around, they are not viewed as too bothersome as long as they have no connections with the external world and thus no influence on measurable quantities; they are just part of what we call “zero energy.” There are, however, some infinities in our theory which are bothersome.

Classical Self-Energy

Even before quantum mechanics, Abraham and Lorentz (Lorentz, 1915) used the simple model of Figure 21.8C to consider the interaction of an electron with its own field. Here the electron is modeled as a sphere of radius r_e with only surface charge, so its self-energy is

$$E = eV(r_e) = \frac{e^2}{r_e}. \quad (21.51)$$

In an argument reminiscent of Mach’s principle,⁶ Abraham and Lorentz set the self-energy of the electron equal to its rest energy and hence introduced the classical value of the electron’s radius:

$$e^2 r_e = m \Rightarrow r_e = \frac{e^2}{m} = 2.82 \text{ fm}. \quad (21.52)$$

Of course this argument has flaws; (21.52) implies that as the electron’s size approaches zero (which we now think is true), the electron’s mass must approach infinity (which we know is not true).

Quantum Self-Energy

In the quantum regime the self-energy is the interaction of the electron with the photons which exists due to the presence of this electron, such as diagrammed in Figure 21.9. Because there is no way in nature to “turn off” the electromagnetic interaction, the cloud of virtual photons around the electron pictured here is an integral part of what we have come to know as an electron, and there is no direct way to separate off these photons while the electron is free. Because of the interaction in Figure 21.9A (which looks like a short-circuited or virtual version of Thompson scattering, Figure 21.5), a free electron has an energy which differs slightly from the $p^2/2m$ we usually assume. We calculate (in natural units) the energy of these fluctuations by taking the matrix element of $H_{2\gamma}$ between initial and final electron states of momentum $\mathbf{p}$ and empty photon states:

$$\Delta E^{(A)} = \langle f | H_{2\gamma} | i \rangle = \frac{e^2}{2m} \langle \mathbf{p}, 0 | A^2 | \mathbf{p}, 0 \rangle \quad (21.53)$$

⁶The inertial properties of an object are determined by the energy momentum throughout all space (Einstein et al., 1952.)

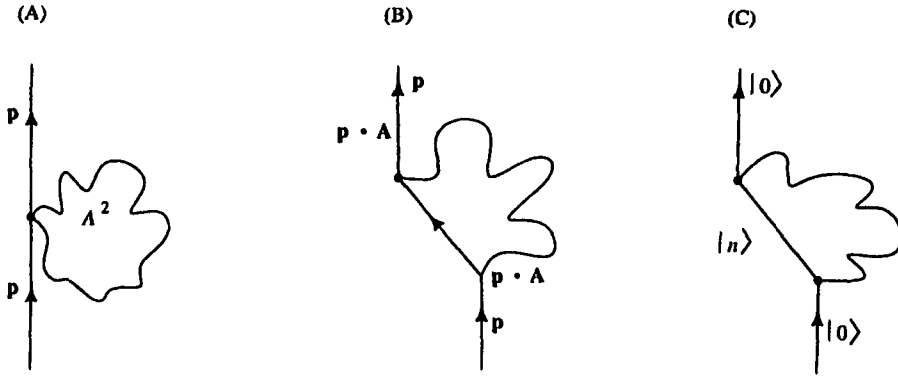


Figure 21.9: Infinite-energy diagrams in nonrelativistic QED. (A) arises from $H_{2\gamma}$; (B) arises from H_γ acting twice on a free electron; (C) arises from H_γ acting twice on a bound electron.

$$= \frac{e^2}{2m} \sum_{\mathbf{k}, i} \frac{2\pi}{V k} \hat{\mathbf{e}}_{\mathbf{k}i} \cdot \hat{\mathbf{e}}_{\mathbf{k}i} = \frac{2e^2}{\pi m} \int_0^\infty dk k = \infty.$$

In the last line we have replaced the sum over the small box's $\mathbf{k}$ by an integral over the large box's $d^3k/(2\pi)^3$ (see Appendix A, *Natural Units and Plane Waves*). Although (21.53) is infinite, it is independent of the electron's momentum $\mathbf{p}$, therefore it is independent of the state of the electron, therefore it is the same for all electrons, therefore it only shifts the zero point of energy, and, consequently, it is not interesting!

Mass Renormalization, Free Electron

The energy shift caused by the virtual emission and absorption of radiation pictured in Figure 21.9B is more interesting because it will not be the same for all electrons. To see this, we write down its matrix element in analogy with (20.38) for the emission and absorption from a bound level [the (B) superscript]:

$$\Delta E^{(B)} \simeq \frac{e^2}{m^2} \sum_{\mathbf{k}, i} \frac{\langle 1\mathbf{p}, 0_\gamma | \mathbf{p} \cdot \mathbf{A} | 1\mathbf{p}-\mathbf{k}, 1\mathbf{k} \rangle \langle 1\mathbf{p}-\mathbf{k}, 1\mathbf{k} | \mathbf{p} \cdot \mathbf{A} | 1\mathbf{p}, 0_\gamma \rangle}{E_p - E_{\mathbf{p}-\mathbf{k}} - k}. \quad (21.54)$$

Exercise Verify (21.54) by enumerating the initial, final, and intermediate states and then using the golden rule. $\diamond$

We evaluate (21.54) in the dipole approximation (§ 20.5), which is basically a low- k approximation. Although it is logical to believe that the calculation of infinite energies requires a high-energy theory, these infinities arise from $E_{\mathbf{p}-\mathbf{k}} \simeq E_p$, and so are essentially low-energy effects. The relativistic theory only serves to give a “more accurate” value for infinity! The good agreement with experiment we will find justifies our approach.

We substitute into (21.54) the expansion (20.17) for the field operator $\mathbf{A}$ and approximate the electron's energy $E_{\mathbf{p}-\mathbf{k}}$ by E_p (the increased magnitude of the energy denominator

at high k tends to make the high- k part less important). This produces:

$$\Delta E^{(B)} \simeq \frac{e^2}{m^2} \frac{2\pi^2}{V} \sum_{\mathbf{k}, i} \frac{1}{k} \frac{|\mathbf{p} \cdot \hat{\mathbf{e}}_{\mathbf{k}i}|^2}{-k}. \quad (21.55)$$

Exercise Show that if θ is the angle between $\mathbf{p}$ and $\mathbf{k}$, the polarization sum in (21.55) is

$$\sum_i |\mathbf{p} \cdot \hat{\mathbf{e}}_{\mathbf{k}i}|^2 = p^2 \sin^2 \theta. \quad \diamond \quad (21.56)$$

Exercise Show that conversion of the sum over $\mathbf{k}$ to the familiar integral

$$\frac{1}{V} \sum_{\mathbf{k}} \Rightarrow \int \frac{d^3 k}{(2\pi)^3} = \int \frac{dk d(\cos \theta) 2\pi k^2}{(2\pi)^3} \quad (21.57)$$

leads to

$$\Delta E^{(B)} \simeq \frac{-2e^2}{3\pi m^2} p^2 \int_0^\infty dk = \infty. \quad \diamond \quad (21.58)$$

As written, (21.58) is infinite like (21.53), yet it is also proportional to the square of the electron's momentum. The p^2 dependence (to which we shall return) means that the shift is state dependent; the infinity presumably means that there is something fundamentally wrong in the basic structure of our theory, possibly as $r \rightarrow 0$ (maybe spacetime is somewhat granular or stringy). Under these circumstances, it is reasonable to eliminate the very large k (small $\lambda = 2\pi/k$) contributions to the integral by replacing the upper, infinite limit by the large, but finite, number $k_{\max} \simeq m$ [which is on the order of the inverse Compton wavelength $\lambda_C^{-1} = m$]:

$$\Delta E^{(B)} \simeq \frac{-2e^2}{3\pi m^2} p^2 \int_0^{k_{\max}} dk = \mathcal{K} p^2, \quad \mathcal{K} = \frac{-2e^2}{3\pi m^2} k_{\max}. \quad (21.59)$$

Equation (21.59) means that electrons have a negative (downwards) shift to their energy which increases with increasing p . Yet because the physical electron is “dressed” with a cloud of virtual photons, such as in Figure 21.9B, the e is really a *quasiparticle* with its kinetic energy already including the shift. Accordingly, the energy observed is the bare energy for a bound electron plus the shift:

$$E_{\text{obs}} = E_{\text{bare}} + \Delta E^{(B)}, \quad (21.60)$$

$$\frac{p^2}{2m_{\text{obs}}} = \frac{p^2}{2m_{\text{bare}}} + \Delta E^{(B)}. \quad (21.61)$$

This means the observed mass of an electron is an “effective” mass dressed by photons from the vacuum.

Exercise Solve (21.61) for the observed free and bare electron masses, and show that for $k_{\max} \simeq m_e$, the two masses differ by $\sim 1/3\%$. $\diamond$

Mass Renormalization, Bound Electron (Lamb Shift)

In Schrödinger's theory of the hydrogen atom, energy levels with the same principal quantum number n and different l values were degenerate. In Dirac's theory, the degeneracy for different l 's is removed, but not between levels with the same total angular momentum j (for example, the $2P_{1/2}$ and $2S_{1/2}$ states are still degenerate). Before the relativistic calculations of self-energies were complete, Bethe argued that because the energy shift (21.59) depends on p^2 , it is different for different levels and thereby removes the j degeneracy. In essence, while the shift may be infinite for each of two levels, it is a different infinity for each, with the difference in infinities equal to a finite and observable number. Although these infinities may not seem like the natural world, the estimated splitting (Bethe and Salpeter, 1977) of the $2P_{1/2}$ and $2S_{1/2}$ levels is in close agreement with the value $\sim 4 \cdot 10^{-6}$ eV observed first by Lamb and Retherford (1947).

Bethe's Estimate

To outline Bethe's estimate, we recalculate the energy shift $\Delta E^{(B)}$ for a bound state $|n_0\rangle$ of hydrogen in Figure 21.9C instead of the free electron in Figure 21.9B. The calculation of the shift is analogous to (21.54)–(21.58):

$$\Delta E^{(B)} = \frac{2e^2}{3\pi m^2} \sum_{n=0}^{\infty} \int_0^{k_{\max}} dk k \frac{\langle n_0 | \mathbf{p} | n \rangle \cdot \langle n | \mathbf{p} | n_0 \rangle}{E_0 - E_n - k} \quad (21.62)$$

$$= \frac{2e^2}{3\pi m^2} \sum_{n=0}^{\infty} \left(|\langle n_0 | \mathbf{p} | n \rangle|^2 \int_0^{k_{\max}} \frac{dk k}{E_0 - E_n - k} \right). \quad (21.63)$$

While this is an infinite (unobservable) shift if we set $k_{\max} = \infty$, because we have already used the infinite $p^2/2m_{\text{obs}}$ (21.61) in calculating the unperturbed energy levels, the observable shift should be calculated as the difference,

$$\Delta E_{\text{obs}}^{(B)} = \Delta E^{(B)} - \left\langle n_0 \left| \left(\frac{\mathbf{p}^2}{2m_{\text{obs}}} - \frac{\mathbf{p}^2}{2m} \right) \right| n_0 \right\rangle \quad (21.64)$$

$$= \Delta E^{(B)} - \mathcal{K} \langle n_0 | \mathbf{p}^2 | n_0 \rangle. \quad (21.65)$$

We now substitute our previous expression (21.59) for $\mathcal{K}$ and utilize the completeness relation to write

$$|\langle n_0 | \mathbf{p}^2 | n_0 \rangle|^2 = \sum_n \langle n_0 | \mathbf{p} | n \rangle \cdot \langle n | \mathbf{p} | n_0 \rangle = \sum_n |\langle n_0 | \mathbf{p} | n \rangle|^2. \quad (21.66)$$

Exercise Show that these substitutions lead to the observable energy shift:

$$\Delta E_{\text{obs}}^{(0)} = \frac{2e^2}{3\pi m^2} \int_0^{k_{\max}} dk \left[\sum_n \left(\frac{k |\langle n_0 | \mathbf{p} | n \rangle|^2}{E_0 - E_n - k} + |\langle n_0 | \mathbf{p} | n \rangle|^2 \right) \right]. \quad \diamond \quad (21.67)$$

If we now combine the fractions in (21.67) and evaluate the simple integral, we obtain

$$\Delta E_{\text{obs}}^{(0)} = \frac{2e^2}{3\pi m^2} \sum_n \left[|\langle n_0 | \mathbf{p} | n \rangle|^2 (E_0 - E_n) \ln \frac{k_{\max}}{|E_0 - E_n|} \right], \quad (21.68)$$

where the sum is over all states of hydrogen, and $|n_0\rangle$ is the state whose shift is being evaluated. We see that the answer still diverges for $k_{\max} \rightarrow \infty$, only now logarithmically (which makes it much less sensitive to the exact value of $k_{\max}$). To obtain a numerical approximation to the expression in (21.68), we remove the logarithm term from the sum and replace it by its value averaged over all n :

$$\Delta E_{\text{obs}}^{(0)} = \frac{2e^2}{3\pi m^2} \sum_n \left[|\langle 0|\mathbf{p}|n\rangle|^2 (E_0 - E_n) \right] \ln \frac{k_{\max}}{\langle |E_0 - E_n| \rangle}. \quad (21.69)$$

The sum now depends solely on the bound-state wave functions and has been evaluated (Bethe and Salpeter, 1977; Sakurai, 1967) as:

$$\sum_{n=0}^{\infty} \left[|\langle 0|\mathbf{p}|n\rangle|^2 (E_0 - E_n) \right] = -\frac{1}{2} \int d^3x |\psi_0(\mathbf{x})|^2 \nabla^2 V_{\text{cent}}, \quad (21.70)$$

where V_{cent} is the central potential which binds the system. For hydrogen, $\nabla^2 V_{\text{cent}} = 4\pi e^2 \delta(\mathbf{r})$, so

$$\sum_{n=0}^{\infty} \left[|\langle 0|\mathbf{p}|n\rangle|^2 (E_0 - E_n) \right] = -2\pi e^2 |\psi_0(0)|^2 = \begin{cases} \frac{-2e^2}{n_0^3 a_B^3}, & S \text{ states,} \\ 0, & \text{otherwise.} \end{cases} \quad (21.71)$$

Here a_B is the Bohr radius and only S states are shifted since only they are nonzero at the origin. In summary, if we have an S state in hydrogen with principal quantum number n_0 , the electron interacts with its own Coulomb field, and this shifts its energy by

$$\Delta E_{\text{obs}}^{(0)} = \frac{8}{3\pi} (e^2)^3 \frac{1}{2a_B n_0^3} \ln \frac{k_{\max}}{\langle |E_0 - E_n| \rangle}. \quad (21.72)$$

We recognize the term in parenthesis as the cube of the fine-structure constant α^3 , the next factor as the Rydberg = 13.6 eV, and $\langle E_n - E_0 \rangle$ as the excitation energy of the virtual state averaged over the entire hydrogen spectrum. To obtain an estimate of the splitting of the $2S_{1/2}$ - $2P_{1/2}$ levels of Figure 16.1, we substitute $k_{\max} = m$, $\langle E_n - E_0 \rangle \simeq 17.8 \text{ Ry}$, $\ln(m/17.8\text{Ry}) = 7.65$, and so obtain:

$$\Delta E \equiv \frac{\Delta E}{\hbar} = 1040 \text{ megacycles (MH)}. \quad (21.73)$$

The experimental value of the shift⁷ measured by Lamb and Retherford (1947) is $1057.8 \pm 0.1 \text{ MH}$. The value predicted from the relativistic theory (which has the infinities removed) is $1057.7 \pm 0.2 \text{ MH}$. Clearly this small shift of about $4 \times 10^{-6} \text{ eV}$ is basically a nonrelativistic effect. It is also a very good test of our ability to understand electrodynamics at very short distances. In fact, the agreement of the relativistic theory with experiment at this level leads to the belief that we have found an awkward but workable way to renormalize the infinities out of the theory and obtain experimental predictions in the process. [Actually, tremendous progress has been made in the renormalization of the entire electroweak interaction (Guidry, 1991).]

⁷ Actually the interaction with the radiation field both shifts and broadens atomic levels.

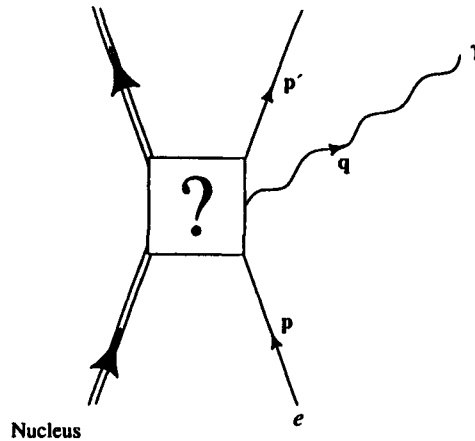


Figure 21.10: Electron *bremsstrahlung* in the field of a heavy nucleus (double line). The question mark represents all possible reaction mechanisms.

21.4 Problems

1. Include the magnetic interaction into our field theory Hamiltonian for electrons and photons:
 - (a) What new Hamiltonian would you add to H_γ and $H_{2\gamma}$? (Express your answer in terms of creation and destruction operators.)
 - (b) Estimate the ratio of magnetic to electric transition probabilities (assuming all “else” but the H ’s is the same).
 - (c) Given that the size of an orbit in the hydrogen atom is approximately the Bohr radius, evaluate the ratio of magnetic to electric transition strengths.
2. Consider the scattering of a low-energy, polarized photon from a neutron (the neutron has no net charge, spin $\frac{1}{2}$, a magnetic moment, and a very large mass).
 - (a) What are the initial, final, and two lowest-order (in e) intermediate states for this scattering. Give your answer in terms of momentum states for the neutron and Fock-space states for the photons.
 - (b) Photon–neutron scattering can occur via the magnetic interaction of the preceding problem. Evaluate the matrix elements of this interaction for the initial, final and intermediate states just enumerated. Make sure to show how momentum conservation is present in each matrix element.
 - (c) Without a detailed calculation, try to deduce for which photon polarizations $d\sigma/d\Omega = 0$.
3. Consider the radiation of a photon of momentum $\mathbf{q}$ from an electron of momentum $\mathbf{p}$ in the presence of the heavy nucleus pictured in Figure 21.10.
 - (a) Why is it that a free electron can’t radiate a photon?
 - (b) What must be the relations between $\mathbf{p}$, $\mathbf{p}'$, and $\mathbf{q}$?

- (c) What are the two lowest-order graphs that can appear within the box in the figure?
 - (d) What are the matrix elements of the two lowest-order graphs (for nonrelativistic electrons)?
 - (e) Is there any spin flip here in the nonrelativistic limit?
 - (f) Use your expressions to deduce that the matrix elements should vanish when the photon and electron are both emitted in the initial electron's direction.
 - (g) Deduce that the differential cross section for this process is the Rutherford cross section times a probability-of-emission factor.
4. Without referring to the text, describe a real hydrogen atom by giving
 - (a) its good quantum numbers,
 - (b) the degeneracy removed by mass renormalization,
 - (c) the approximate energies for,
 - i. level spacings,
 - ii. fine structure,
 - iii. hyperfine structure,
 - iv. Lamb shift.
 5. Construct the coherent state of the radiation field as outlined in § 21.2.
 6. Prove that negative-energy electron states are needed to obtain the classical limit of electron–photon scattering. Discuss the meaning of this.
 7. Verify the evaluation of the sum in (21.69).
 8. Deduce how the divergence properties of the integrals in the electron's mass renormalization would change for relativistic kinematics.
 9. Prove that the two diagrams in Figures 21.4A and 21.4B for Thompson scattering tend to cancel each other (and do so perfectly in the classical limit).

CHAPTER 22

INTERACTION OF PHOTONS WITH QUANTIZED MATTER, QED

In this chapter we introduce the last ingredient in our development of quantum electrodynamics (QED), namely, the possibility of electrons being created and destroyed (as were photons in Chapter 20, *Quantized Electromagnetic Fields*). In the § 22.1 we use the second-quantization formalism to describe electrons in the presence of the negative-energy sea, and see why it is mandatory to include positron degrees of freedom when describing electrons (and vice versa). In § 22.2 we generalize the electromagnetic Hamiltonian to include quantized photons and electrons, and in § 22.3, we step through a calculation of Compton scattering. We take the view that in field theory, as in learning a language, an operational knowledge should precede the formal rules of grammar. Accordingly, our treatment is direct with little emphasis on mathematical proof or completeness.

22.1 Quantized Holes and Particles

The plane-wave solutions to the Dirac equation were studied in Chapter 15, *Components of Dirac Wave Functions*, and are enumerated in Appendix B, *Dirac Notation and Representations*. You will recall, the solutions have a spacetime dependence given by an exponential, and an internal space dependence given by a 4D Dirac spinor:

$$\psi_s^{(\pm)}(x) = e^{\mp i p^\mu x_\mu} u_s^{(\pm)}(p). \quad (22.1)$$

As it stands here, $\psi_s^{(\pm)}$ is a wave function (not field operator) describing four possible states: $e^{-i p^\mu x_\mu} u_\uparrow^{(+)}(p)$ describes an electron with momentum $\mathbf{p}$, energy E_p , and spin up; $e^{-i p^\mu x_\mu} u_\downarrow^{(+)}(p)$ describes an electron with $(\mathbf{p}, E_p, \downarrow)$; $e^{+i p^\mu x_\mu} u_\uparrow^{(-)}(p)$ describes an electron with $(-\mathbf{p}, -E_p, \downarrow)$; $e^{+i p^\mu x_\mu} u_\downarrow^{(-)}(p)$ describes an electron with $(-\mathbf{p}, -E_p, \uparrow)$.¹ Because $\sqrt{m/V E_p} \psi_s^{(b)}(x)$ is normalized to 1, a convenient expression for the most general solution

¹ Recall we said this reversal of spin and momentum will be useful for the hole interpretation. It is now about to be useful.

(wave packet) Ψ of the Dirac equation is

$$\Psi(\mathbf{x}, t) = \sum_{\mathbf{p}, s} \sqrt{\frac{m}{V E_p}} \left[e^{i\mathbf{p} \cdot \mathbf{x}} u_s^{(+)}(\mathbf{p}) b_{\mathbf{p}s}^{(+)} + e^{-i\mathbf{p} \cdot \mathbf{x}} u_s^{(-)}(\mathbf{p}) b_{\mathbf{p}s}^{(-)} \right], \quad (22.2)$$

where the b 's are expansion coefficients.

New physics enters by second-quantizing this wave packet into the spin- $\frac{1}{2}$ *field operator* that destroys an electron at point $(\mathbf{x}, t)$. The expansion follows the general prescriptions from Chapter 19, *Second Quantization*, of replacing the Fourier coefficients $b^{(+)}$ and $b^{(-)}$ in (22.2) by creation and destruction operators (with no change in symbol):

$$\tilde{\Psi}(\mathbf{x}, t) = \sum_{\mathbf{p}, s} \sqrt{\frac{m}{V E_p}} \left[e^{i\mathbf{p} \cdot \mathbf{x}} u_s^{(+)}(\mathbf{p}) b_{\mathbf{p}s}^{(+)} + e^{-i\mathbf{p} \cdot \mathbf{x}} u_s^{(-)}(\mathbf{p}) b_{\mathbf{p}s}^{(-)} \right]. \quad (22.3)$$

The adjoint field operator is thus

$$\tilde{\Psi}^\dagger(\mathbf{x}, t) = \sum_{\mathbf{p}, s} \sqrt{\frac{m}{V E_p}} \left[e^{-i\mathbf{p} \cdot \mathbf{x}} u_s^{(+)}(\mathbf{p})^\dagger b_{\mathbf{p}s}^{(+)\dagger} + e^{i\mathbf{p} \cdot \mathbf{x}} u_s^{(-)}(\mathbf{p})^\dagger b_{\mathbf{p}s}^{(-)\dagger} \right], \quad (22.4)$$

where (for a while) we use the tilde to indicate an operator in Fock space. Here $b_{\mathbf{p}s}^{(+)}$ destroys an electron of momentum $\mathbf{p}$, spin s , and energy E_p ; $b_{\mathbf{p}s}^{(-)}$ destroys an electron of momentum $-\mathbf{p}$, spin $-s$, and energy $-E_p$, while the adjoints create electrons with these same properties.

Hole Picture

In the hole picture of Dirac, Figure 15.1, the vacuum is viewed as a filled sea of negative-energy electrons. A physical positron with $(e, \mathbf{p}, s, E_p)$ is identified as a hole in the sea, that is, the absence of an electron with $-(e, \mathbf{p}, s, E_p)$. In terms of positive- and negative-energy electrons, the second-quantized, free electron Hamiltonian (19.29) becomes

$$H_e = \sum_{\mathbf{p}, s} E_p \left[b_{\mathbf{p}s}^{(+)\dagger} b_{\mathbf{p}s}^{(+)} - b_{\mathbf{p}s}^{(-)\dagger} b_{\mathbf{p}s}^{(-)} \right] \quad (22.5)$$

$$= \sum_{\mathbf{p}, s} E_p \left[\tilde{N}_{\mathbf{p}s}^{(+)} - \tilde{N}_{\mathbf{p}s}^{(-)} \right], \quad (22.6)$$

where we explicitly separate positive- and negative-energy terms and always keep $E_p > 0$. Likewise, the charge operator is

$$Q = -e \sum_{\mathbf{p}, s} \left[\tilde{N}_{\mathbf{p}s}^{(+)} + \tilde{N}_{\mathbf{p}s}^{(-)} \right]. \quad (22.7)$$

Exercise Show that (22.6) and (22.7) are also obtained from the second-quantization expressions,

$$\tilde{H}_e = \int d^3x \tilde{\Psi}^\dagger (H_{\text{Dirac}}) \tilde{\Psi}, \quad (22.8)$$

$$\tilde{Q} = -e \int d^3x \tilde{\Psi}^\dagger \tilde{\Psi}, \quad (22.9)$$

where H_{dirac} is the Hamiltonian (22.5), and $\tilde{\Psi}$ is the field operator. $\diamond$

Because the vacuum is an infinite sea of negative-energy electrons, the energy and charge operators for the vacuum,

$$\tilde{H}_{e0} = \sum_{\mathbf{p},s} (-E_p) \tilde{N}_{\mathbf{p}s}^{(-)}, \quad (22.10)$$

$$\tilde{Q}_0 = -e \sum_{\mathbf{p},s} \tilde{N}_{\mathbf{p}s}^{(-)}, \quad (22.11)$$

have infinite eigenvalues:

$$\langle 0 | \tilde{H}_{e0} | 0 \rangle = -\infty, \quad \langle 0 | \tilde{Q}_0 | 0 \rangle = -\infty. \quad (22.12)$$

Dirac's hole picture is incorporated into this formalism by calibrating the energy and charge operators so that only fluctuations from the (infinite) vacuum are observables:

$$\tilde{H}'_e = \tilde{H}_e - \tilde{H}_{e0} = \sum_{\mathbf{p},s} E_p \left[\tilde{N}_{\mathbf{p}s}^{(+)} - \tilde{N}_{\mathbf{p}s}^{(-)} + 1 \right], \quad (22.13)$$

$$\Rightarrow \tilde{H}'_e = \sum_{\mathbf{p},s} E_p \left[\tilde{N}_{\mathbf{p}s}^{(+)} + (1 - \tilde{N}_{\mathbf{p}s}^{(-)}) \right], \quad (22.14)$$

$$Q = -e \sum_{\mathbf{p},s} \left[\tilde{N}_{\mathbf{p}s}^{(+)} - (1 - \tilde{N}_{\mathbf{p}s}^{(-)}) \right]. \quad (22.15)$$

These equations express the hole interpretation: The sea contributes to H and Q only when a negative-energy electron is *missing*, that is, when $N^{(-)} = 0$.

The Positron Relation

We incorporate the hole picture into field theory by reviewing the practical advice given by Eliot in Chapter 19, *Second Quantization*, and renaming the operator $b^{(-)}$ which destroyed an electron with $-(\mathbf{p}, E_p, s)$, as a new operator $d^\dagger$ which creates a *positron* with $(\mathbf{p}, E_p, s)$:

$$d_{\mathbf{p}s}^\dagger \equiv b_{\mathbf{p}s}^{(-)}. \quad (22.16)$$

Destroying a negative-energy particle (and thereby creating a hole in the sea) is now equivalent to creating an antiparticle. Consequently, rather than consider the entire negative-energy sea, we only treat the holes and the positive-energy electrons, and we treat the holes as positrons.

Exercise Show that the renaming (22.16) leads to positron operators satisfying the anti-commutation relations (ACR):

$$\{d_{\mathbf{p}s}^\dagger, d_{\mathbf{p}'s'}\} = \delta_{ss'} \delta_{\mathbf{p}\mathbf{p}'}, \quad \{d_{\mathbf{p}s}, b_{\mathbf{p}'s'}\} = 0, \quad (0 \text{ for other mixed ACR's too}). \quad \diamond \quad (22.17)$$

Accordingly, we now have positive-energy electron and positive-energy positron number

operators:

$$\tilde{N}_{ps}^{(\epsilon^+)} = 1 - \tilde{N}_{ps}^{(\epsilon^-)} = 1 - b_{ps}^{(-)\dagger} b_{ps}^{(-)} = b_{ps}^{(-)} b_{ps}^{(-)\dagger} \quad (22.18)$$

$$= d_{ps}^\dagger d_{ps}. \quad (22.19)$$

Exercise Verify that (22.19) is correct only if the b 's satisfy anticommutation relations $\diamond$.

Because $|0\rangle$ contains no positive-energy electrons or positrons, with these new number operators the vacuum finally appears empty,

$$\tilde{N}_{ps}^{(\epsilon^+)} |0\rangle = \tilde{N}_{ps}^{(\epsilon^-)} |0\rangle = 0. \quad (22.20)$$

It is also useful to replace the negative-energy spinors $u^{(-)}(p)$ by the positron spinors $v(p)$'s introduced in § 15.2. These spinors have the properties (all with argument p)

$$\begin{aligned} (\not{p} - m)u_s &= 0, & \bar{u}_s(\not{p} - m) &= 0, \\ (\not{p} + m)v_s &= 0, & \bar{v}_s(\not{p} + m) &= 0, \\ \bar{u}_s u_{s'} &= -\bar{v}_s v_{s'} = \delta_{ss'}, & u_s^\dagger u_s &= v_s^\dagger v_s = \frac{E_p}{m} \delta_{ss'}. \end{aligned} \quad (22.21)$$

The field operator (22.4) which creates an electron at point $\mathbf{x}$,

$$\tilde{\Psi}^\dagger(\mathbf{x}, t) = \sum_{p,s} \sqrt{\frac{m}{VE_p}} [e^{-ip \cdot x} u_s^\dagger(p) b_{ps}^\dagger + e^{ip \cdot x} v_s^\dagger(p) d_{ps}], \quad (22.22)$$

is seen to also destroy a positron. The field operator which destroys an electron at point $\mathbf{x}$,

$$\tilde{\Psi}(\mathbf{x}, t) = \sum_{p,s} \sqrt{\frac{m}{VE_p}} [e^{ip \cdot x} u_s(p) b_{ps} + e^{-ip \cdot x} v_s(p) d_{ps}^\dagger], \quad (22.23)$$

is seen to also create a positron.

Free-Electron Hamiltonian

The effect of this renaming of electron holes as positrons is evident in the free-electron Hamiltonian,

$$\tilde{H}_e = \sum_{p,s} E_p [\tilde{N}_{ps}(e^-) + \tilde{N}_{ps}(e^+)] = \sum_{p,s} E_p [b_{ps}^\dagger b_{ps} + d_{ps}^\dagger d_{ps}], \quad (22.24)$$

which is now the sum over physical electrons and positrons. We want to test if this deduction is consistent with the “standard procedure” introduced in Chapter 19, *Second Quantization*, for deriving the second-quantized Hamiltonian operator (taking the matrix element of a nonquantized H between field operators). We evaluate the matrix element of the standard Dirac Hamiltonian,

$$H_D = \boldsymbol{\alpha} \cdot \frac{\nabla}{i} + \beta m, \quad (22.25)$$

between the electron field operator and its adjoint:

$$\tilde{H}_e = \int d^3x \tilde{\Psi}^\dagger(\mathbf{x}, t) \left[\boldsymbol{\alpha} \cdot \frac{\nabla}{i} + \beta m \right] \tilde{\Psi}(\mathbf{x}, t). \quad (22.26)$$

Because the Dirac wave functions are eigenstates of H_D ,

$$H_D \psi^{(\pm)} = \pm E_P \psi^{(\pm)}, \quad (22.27)$$

the action of H_D on $\tilde{\Psi}$ is easy to determine.

Exercise Verify by explicit substitution and integration that

$$\tilde{H}_e = \sum_{p,s} E_p [b_{ps}^\dagger b_{ps} - d_{ps} d_{ps}^\dagger]. \quad \diamond \quad (22.28)$$

This expression for $\tilde{H}_e$ is almost (22.24). We make them alike by substituting the anticommutation relation:

$$d_{ps} d_{ps}^\dagger = 1 - d_{ps}^\dagger d_{ps} \Rightarrow \tilde{H}_e = \sum_{p,s} E_p [b_{ps}^\dagger b_{ps} + d_{ps}^\dagger d_{ps}] - \sum_{p,s} E_{ps}. \quad (22.29)$$

The first sum is the desired result while the second sum is the (unobservable) infinite energy of the sea. We have encountered infinities of this sort in the last chapter where we found that they may be removed by *renormalization* procedures which leave small, but measurable, consequences. While we must leave further discussions to the references, we remind the reader that a proper handling of them is crucial to ensure the mathematical rigor of QED.

22.2 Interaction Hamiltonian

*Gather ye rosebuds while ye may,
Old Time is still a-flying:
And this same flower that smiles to-day
To-morrow will be dying.*

...

*That age is best which is the first,
When youth and blood are warmer;
But being spent, the worse, and worst
Times still succeed the former.*

—Robert Herrick

In Chapter 20, *Quantized Electromagnetic Fields*, we described quantized electromagnetic radiation via the field operator which creates a photon at spacetime point x :

$$\mathbf{A} = \sum_{\mathbf{k}, i} \sqrt{\frac{2\pi}{V\omega_{\mathbf{k}}}} \hat{\mathbf{e}}_{\mathbf{k}i} \left[a_{\mathbf{k}i} e^{i\mathbf{k}\cdot\mathbf{x}} + a_{\mathbf{k}i}^\dagger e^{-i\mathbf{k}\cdot\mathbf{x}} \right]. \quad (22.30)$$

When we used the postulate of minimal electromagnetic coupling, we deduced that the Hamiltonian density operator describing the interaction of photons with nonquantized electrons had the form:

$$H_{e\gamma} = \frac{e}{m} \mathbf{A} \cdot \mathbf{p}, \quad (22.31)$$

where the charge $q = -e$. After substitution for $\mathbf{A}$ we have

$$H_{e\gamma} = \frac{e}{m} \sum_{\mathbf{k}, i} \sqrt{\frac{2\pi}{V\omega_k}} \left[a_{\mathbf{k}i} e^{i\mathbf{k}\cdot\mathbf{x}} + a_{\mathbf{k}i}^\dagger e^{-i\mathbf{k}\cdot\mathbf{x}} \right] \hat{\mathbf{e}}_{\mathbf{k}i} \cdot \mathbf{p}, \quad (22.32)$$

where $\mathbf{k}$ and $\mathbf{p}$ refer to the momenta of the photon and electron, and the time dependence is in the exponential. To obtain the Fock-space Hamiltonian density for photons *and quantized electrons*, we follow the canon:

$$\tilde{H}_{e\gamma} = \int d^3x \tilde{\Psi}^\dagger(x) \left[\frac{e}{m} \mathbf{A} \cdot \mathbf{p} \right] \tilde{\Psi}(x), \quad (22.33)$$

[a canon we already used in (22.26) as a check with the free-electron Hamiltonian]. The resulting Dirac–Fock Hamiltonian has eight terms (two from $\tilde{\Psi}^\dagger$ times two from $\tilde{\Psi}$ times two from $\mathbf{A}$):

$$\begin{aligned} \tilde{H}_{e\gamma} = & \frac{e}{c} \sum_{p, s, s'} \sum_{\mathbf{k}i} \sqrt{\frac{2\pi}{V\omega_k}} \left[u_{s'}^\dagger(p+k) \alpha \cdot \hat{\mathbf{e}}_{\mathbf{k}i} u_s(p) b_{p+k, s'}^\dagger b_{p, s} a_{\mathbf{k}i} \sqrt{\frac{m}{E_{p+k}}} \sqrt{\frac{m}{E_p}} \right. \\ & + u_{s'}^\dagger(-k-p) \alpha \cdot \hat{\mathbf{e}}_{\mathbf{k}i} v_s(p) b_{\mathbf{k}-\mathbf{p}, s'}^\dagger d_{p, s}^\dagger a_{\mathbf{k}i} \sqrt{\frac{m}{E_{\mathbf{k}-\mathbf{p}}}} \sqrt{\frac{m}{E_p}} \\ & + u_{s'}^\dagger(k-p) \alpha \cdot \hat{\mathbf{e}}_{\mathbf{k}i} v_s(p) b_{-\mathbf{k}-\mathbf{p}, s'}^\dagger d_{p, s}^\dagger a_{\mathbf{k}i} \sqrt{\frac{m}{E_{\mathbf{k}+\mathbf{p}}}} \sqrt{\frac{m}{E_p}} \\ & \left. + v_{s'}^\dagger(p-k) \alpha \cdot \hat{\mathbf{e}}_{\mathbf{k}i} v_s(p) d_{p-k, s'}^\dagger d_{p, s}^\dagger a_{\mathbf{k}i} \sqrt{\frac{m}{E_{p-k}}} \sqrt{\frac{m}{E_p}} \right] \\ & + \text{Hermitian adjoint}, \end{aligned} \quad (22.34)$$

where we do not repeat the Hermitian adjoint terms, and where (p, s, s', b, d) refer to the fermions and $(\mathbf{k}, i, a)$ to the photons.

A way to make sense of such a long expression is to remember that we must still use it with the golden rule to obtain a physical transition amplitude. Hence $\tilde{H}_{e\gamma}$ will always be evaluated between Fock-space states which describe specific initial and final states. To make sense of the individual pieces in $\tilde{H}_{e\gamma}$, in Figure 22.1 we graph the operators in each piece. We imagine time running *upwards* in these graphs, and so positrons (with the opposite sign for the time dependence) are viewed as electrons running *backwards in time* (see Chapter 15, *Components of Dirac Wave Functions*). These pieces of diagrams represent the most elementary electron–photon interactions: They describe the absorption and creation of photons on an individual electron (Figure 22.1A), on a positron (Figure 22.1B), and the creation or destruction of an e^+e^- pair by a photon (Figure 22.1C). While these individual pieces do not describe processes which could occur in free space (they do not conserve energy), all physical processes are built up from combinations of them. For example, in Figure 22.2 we show how electron–electron scattering is built up from photon exchange; (A) is lowest-order direct, (B) is lowest-order exchange, (C) is higher-order, direct, and (D) is higher-order exchange. In Figure 22.3 we show how still other physical processes are obtained by putting these elementary vertices together. (Compton scattering, Figures 22.4 and 22.5, is calculated in the next section.)

Returning now to Figure 22.1, we note that the $b^\dagger d^\dagger a^\dagger$ and bda pieces of (D) are exceptional, they are fluctuations of the vacuum in which a photon plus an electron–positron pair are spontaneously created from nothing or annihilate into nothing (in other

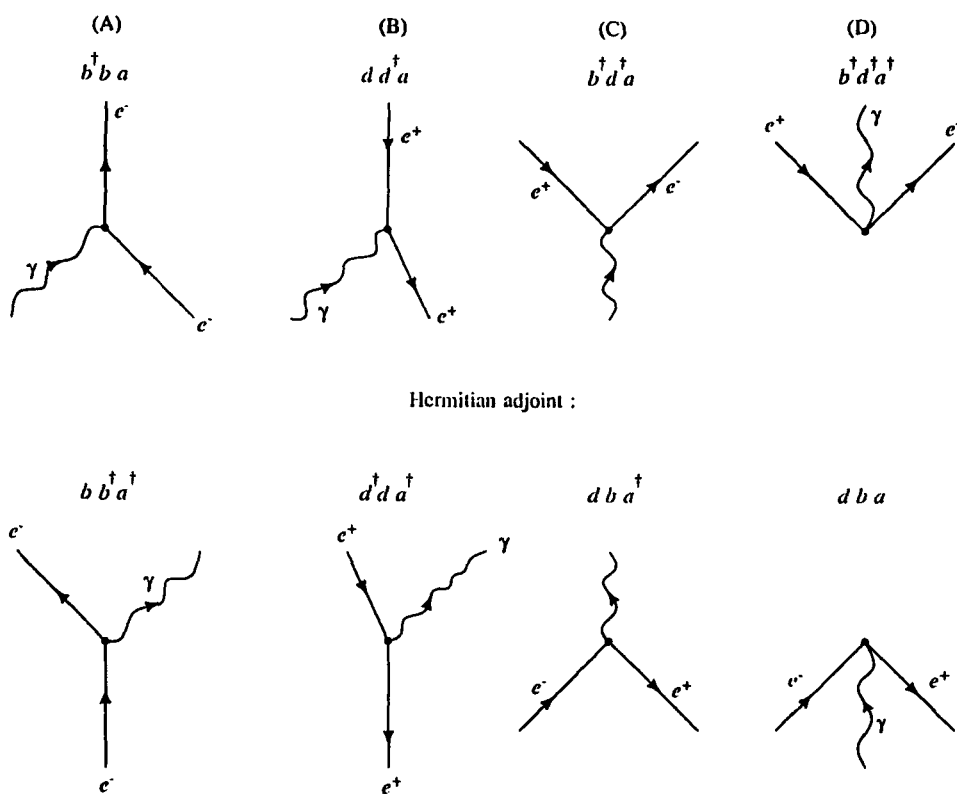


Figure 22.1: (Top) The coupling of the electron and photon fields; (Bottom) the coupling of the adjoint fields.

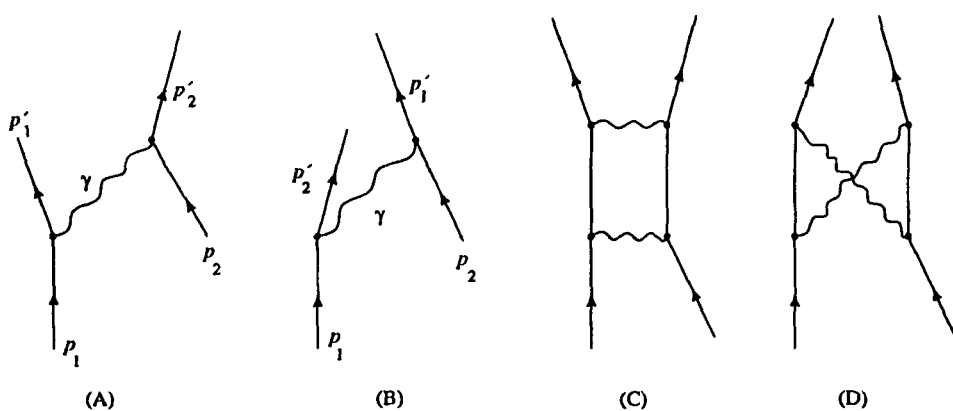


Figure 22.2: (A) Direct electron–electron scattering in second order; (B) exchange second-order ee scattering; (C, D) direct and exchange fourth-order ee scattering.

words, an electron jumps out of, or into the negative-energy sea). These pieces join together to form *vacuum diagrams* such as Figure 22.3. Because this process has no connection to measurable external particles, it has no physical consequences (which is fortunate because it is infinite).

Returning now to the Hamiltonian (22.33) and examining the spinors which about the creation and destruction operators, we observe the following rules:

- $u_s^\dagger(p)$ occurs when there is an electron with (p, s) in the final state.
- $v_s(p)$ occurs when there is a positron with (p, s) in the final state.
- $u_s(p)$ occurs when there is an electron with (p, s) in the initial state.
- $v_s^\dagger(p)$ occurs when there is a positron with (p, s) in the initial state.
- $\sqrt{m/E_p}$ occurs when each fermion of momentum p enters.
- $\sqrt{2\pi/V\omega_k}$ occurs when each photon of momentum k enters.
- An e factor occurs each time an electron and photon couple.

In summary, the Hamiltonian (22.34) is equivalent to the simple expression (22.33). Because $\tilde{H}_{e\gamma}$ is linear in $\mathbf{A}$, each term in the QED Hamiltonian contains a single photon creation or destruction operator; because $\tilde{H}_{e\gamma}$ is quadratic in electron fields, each term also contains two fermion operators. And since in (22.32) we integrated over all space, the Hamiltonian (22.34) is in *momentum space* with 3-momentum conserved at each vertex. Consequently, once we draw a Feynman diagram in momentum space, such as Figure 22.5, there no longer is a time arrow present [to remove the time dependence covariantly, we should integrate (22.33) over time as well as space].²

Because the individual terms in the Hamiltonian (the pieces in Figure 22.1) do not conserve energy, intermediate states in the physical processes shown in Figures 22.2 and 22.3 are virtual. This is standard for the intermediate states in perturbation theory. Yet other conservation laws such as charge and angular momentum still hold for these virtual states; charge, since the Hamiltonian pieces conserve it, and angular momentum, since the Hamiltonian is rotationally invariant.

Longitudinal and Timelike Photons

Before applying $\tilde{H}_{e\gamma}$ we generalize the basic coupling (22.33) to 4-vector form. In doing this we remind the reader of the footnote in § 20.1 which indicates that we have quantized the Maxwell field by quantizing just the transverse vector potential $\mathbf{A}_\perp$. An observer in a reference frame moving relative to the frame in which we have quantized the photon field will see a Lorentz-transformed polarization vector $\hat{\mathbf{e}}_{\mathbf{k}_i}$. Because a Lorentz transformation of a 4-vector mixes the transverse components with the longitudinal and timelike ones, the

²Although we shall generalize this theory to incorporate the 4D (spacetime) aspects of $\mathbf{A}$ and $\mathbf{p}$, that is not full covariance. For example, we only have 3-momentum conservation at each vertex. Because we use the same expression for the energy of virtual particles as for free particles (for example, $E^2 - p^2 = m^2$), this energy is not conserved at each vertex and we have off-energy-shell scattering. In the covariant reformulation, 4-momentum is conserved at each vertex, yet the energy is not given by the same expression as for a free particle, and we have “off-mass-shell scattering”, that is, $E^2 - p^2 \neq m^2$. The two formulations are equivalent in that they yield the same predictions.

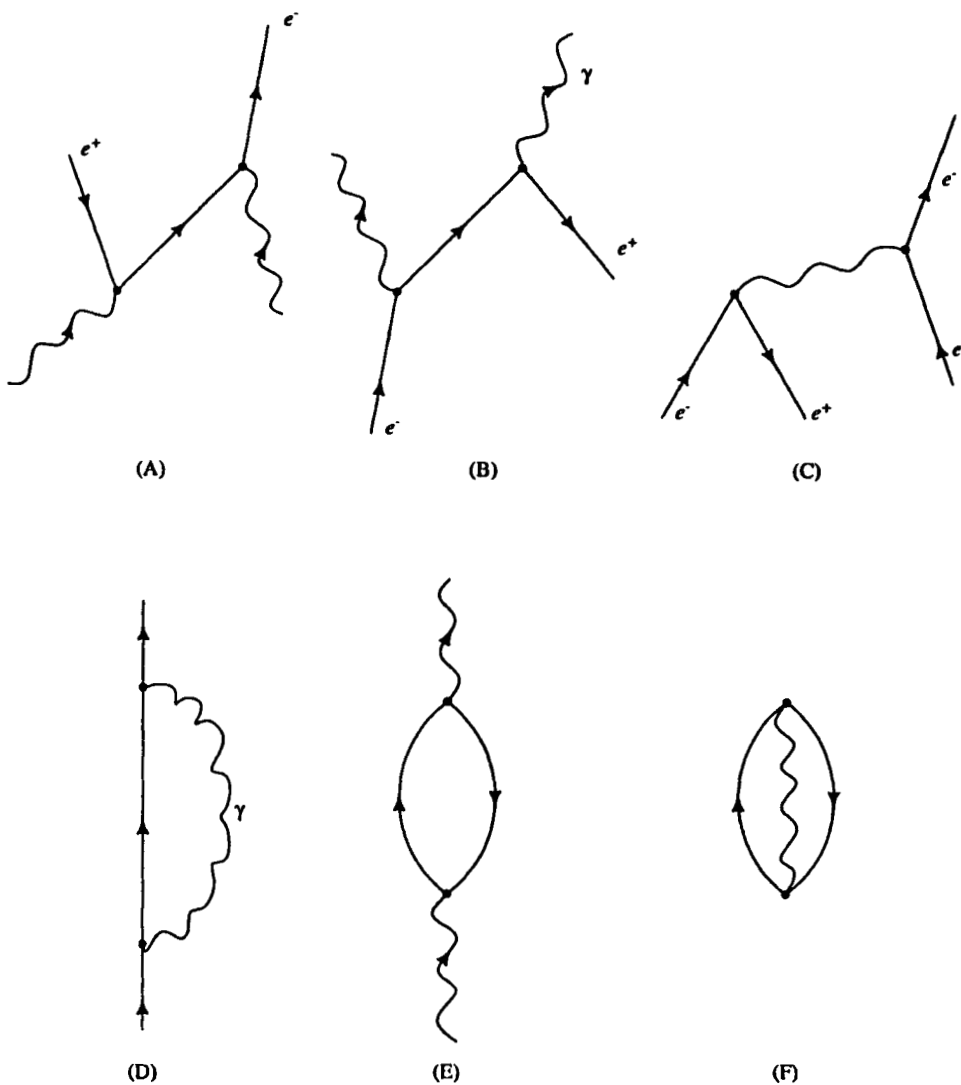


Figure 22.3: (A) Pair creation by two photons; (B) pair annihilation into two photons; (C) pair annihilation in the field of an electron; (D) virtual photon emission and absorption by an electron; (E) virtual pair creation and destruction by a photon; (F) a vacuum fluctuation.

moving observer will see a field with all polarizations possible. If a colleague at rest has informed the observer that only $\hat{\epsilon}_1$ and $\hat{\epsilon}_2$ are needed for radiation, after some thought the observer might conclude that the timelike part of $\hat{\epsilon}$ arises from the timelike part of $\mathbf{A}$, that is, from the static Coulomb field,

$$A_0 = \phi = -Z \frac{e}{r}. \quad (22.35)$$

Likewise, because this Coulomb field is not radiating, the observer might also deduce that longitudinal photons (with unit polarization vector $\hat{\epsilon}_3$ parallel to $\mathbf{k}$) arise from nonradiating (Coulomb) fields.

The existence of timelike photons leads to the generalization of the electron–photon interaction Hamiltonian (22.31),

$$H_{e\gamma} = \frac{e}{m} \mathbf{A} \cdot \mathbf{p} \rightarrow \frac{e}{m} A^\mu p_\mu. \quad (22.36)$$

Note, it is still just $\mathbf{A}_\perp$ which is dynamic and quantized. To see how this generalization affects the Fock-space Hamiltonian (22.34), we rewrite the spinor matrix element

$$u^\dagger \boldsymbol{\alpha} \cdot \hat{\epsilon} u = u^\dagger \gamma^0 \gamma^0 \boldsymbol{\alpha} \cdot \hat{\epsilon} u = \bar{u} \boldsymbol{\gamma} \cdot \hat{\epsilon} u, \quad (22.37)$$

where we have used the relations $\gamma_0 \gamma_0 = 1$ and $\bar{u} = u^\dagger \gamma_0$ from our study of Dirac theory. The elementary coupling now generalizes to the scalar product of two 4-vectors, $\bar{u} \gamma^\mu u \epsilon_\mu$, and the Hamiltonian density to

$$\tilde{H}_{e\gamma} = \int d^3x \bar{\Psi}(x) \left[\frac{e}{m} A^\mu p_\mu \right] \Psi(x) \quad (22.38)$$

$$\begin{aligned} &= \frac{e}{c} \sum_{p,s,s',k} \sqrt{\frac{2\pi}{V\omega_k}} \left[\bar{u}_{s'}(p+k) \gamma^\mu \epsilon_\mu u_s(p) b_{p+k,s'}^\dagger b_{p,s} a_{k,i} \sqrt{\frac{m}{E_{p+k}}} \sqrt{\frac{m}{E_p}} \right. \\ &\quad + \bar{v}_{s'}(-k-p) \gamma^\mu \epsilon_\mu v_s(p) b_{-k-p,s'}^\dagger d_{p,s}^\dagger a_{k,i} \sqrt{\frac{m}{E_{-k-p}}} \sqrt{\frac{m}{E_p}} \\ &\quad + \bar{u}_{s'}(k-p) \gamma^\mu \epsilon_\mu v_s(p) b_{k-p,s'}^\dagger d_{p,s}^\dagger a_{k,i} \sqrt{\frac{m}{E_{k-p}}} \sqrt{\frac{m}{E_p}} \\ &\quad \left. + \bar{v}_{s'}(p-k) \gamma^\mu \epsilon_\mu v_s(p) d_{p-k,s'}^\dagger a_{k,i} \sqrt{\frac{m}{E_{p-k}}} \sqrt{\frac{m}{E_p}} \right] \\ &\quad + \text{Hermitian adjoint}, \end{aligned} \quad (22.39)$$

where we leave the tilde off the $\tilde{\Psi}$ because it would interfere with the overbar. Both forms of the Hamiltonian (22.38) and (22.39) are useful. For example, we use (22.38) in the Problems section to calculate Moller scattering (electron scattering from a fixed Coulomb potential), and (22.39) for Compton scattering.

22.3 Relativistic Compton Scattering

In the last chapter we used field theory to calculate the cross section for scattering of photons from electrons when the electrons were nonrelativistic and nonquantized. In this section we study the same process with the relativistic quantized-electron theory, and see

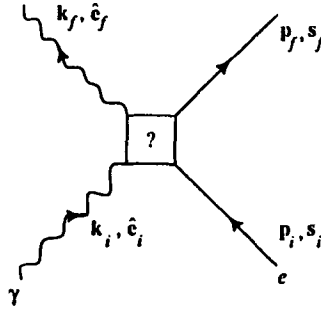


Figure 22.4: Electron–photon (Compton) scattering.

how the new theory gives the old answers and more. In outlining this calculation we also indicate how the Feynman diagrams are useful in leading us through the calculations, and how each part of the diagram gets associated with a term in the calculation. After a little practice it is possible to draw diagrams and do the calculation without explicitly writing down the Hamiltonian. Although we do not develop the rules here, after going through the next section the reader should be closer to accepting the standard recipes found in Bjorken and Drell (1964), Schweber (1959), and Halzen and Martin (1984).

Set-Up

We visualize Compton scattering as the process in Figure 22.4 in which a photon scatters elastically from an electron. In a general reference frame, the momenta of all particles differ, even for elastic scattering. Because each term in the Hamiltonian (22.40) conserves 3-momentum, so does the total process:

$$\mathbf{k}_i + \mathbf{p}_i = \mathbf{k}_f + \mathbf{p}_f. \quad (22.40)$$

The initial and final Fock-space states contain one photon and one electron, each with a definite momentum and spin:

$$|i\rangle = |1_{\mathbf{p}_i, s_i}\rangle_e |1_{\mathbf{k}_i, \epsilon_i}\rangle_\gamma, \quad |f\rangle = |1_{\mathbf{p}_f, s_f}\rangle_e |1_{\mathbf{k}_f, \epsilon_f}\rangle_\gamma. \quad (22.41)$$

To figure out what goes in the box in Figure 22.4, we recall that we calculate the rate for this process by means of the golden rule,

$$\Gamma_{fi} = 2\pi |\mathcal{M}_{fi}|^2 \delta(E_f - E_i), \quad (22.42)$$

$$\mathcal{M}_{fi} = \langle f | H_{e\gamma} | i \rangle + \sum_n \frac{\langle f | H_{e\gamma} | n \rangle \langle n | H_{e\gamma} | i \rangle}{E_i - E_n} + \dots, \quad (22.43)$$

where $H_{e\gamma}$ is represented by the momentum-space pieces in Figure 22.1. While there is an infinite number of terms (graphs) from (22.43) which may go inside the box, we calculate those of lowest order in $H_{e\gamma}$. Each higher-order one is smaller by a power of $e^2 = \alpha \simeq 1/137$, and therefore the lowest-order terms should be a good first approximation. Because the relativistic Hamiltonian is first order in $\mathbf{A}$, there are no lower-order terms in H which take a photon directly to another photon, so the lowest-order possibilities are the two second-order terms given in Figure 22.5.

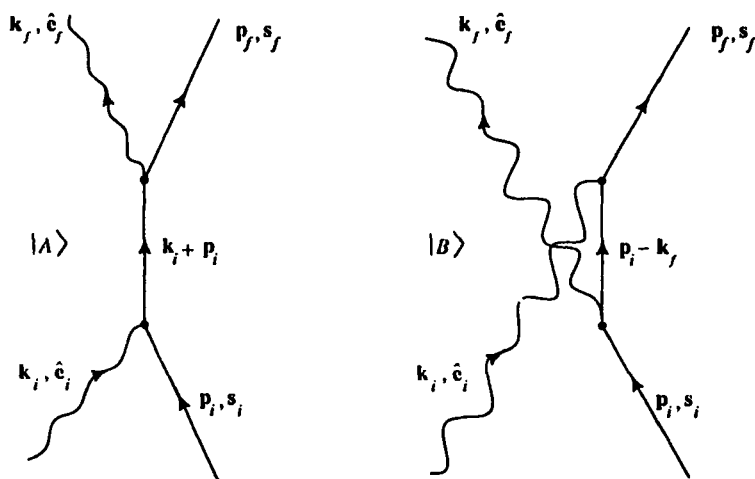


Figure 22.5: Direct and exchange Compton scattering in second order.

Exercise Explain why there was an A^2 term in the nonrelativistic theory directly connecting an $e\gamma$ to an $e\gamma$, but none in the relativistic theory. $\diamond$

Exercise Show that the matrix elements for Compton scattering which contain virtual e^+e^- pairs are of higher order in α than those in Figure 22.5. $\diamond$

The processes in Figures 22.5A and B are clearly different because in A there is an intermediate state $n = |A\rangle$ with no photons present, while in B the state $|n\rangle = |B\rangle$ has two photons present:

$$|A\rangle = |1_{k_i+p_i, s'}\rangle_e |0\rangle_\gamma, \quad |B\rangle = |1_{p_i-k_f, s'}\rangle_e |1_{k_i, \epsilon_i}, 1_{k_f, \epsilon_f}\rangle_\gamma. \quad (22.44)$$

Exercise Verify that the use of $H_{e\gamma}$ at each vertex in Figure 22.5 leads to the internal momenta as labeled and to overall momentum conservation (22.40). $\diamond$

Inasmuch as the intermediate-state spinors have four components, we are including intermediate states with both positive- and negative-energies components, even though we do not explicitly include the positron states with $v(p)$'s. In fact, unless these negative-energy components are included, we would not have a complete set of states for the electrons and we would not even obtain the correct nonrelativistic limit.

Alternatively, if we were to do this calculation in spacetime rather than in momentum space, that is, using the direct form (22.38),

$$\tilde{H}_{e\gamma} = e \int d^3x \bar{\Psi}^\dagger(x) [A(x) \cdot \alpha] \bar{\Psi}(x), \quad (22.45)$$

we would have to consider the different *time orderings* of the intermediate state shown in Figure 22.6. In this spacetime view, it looks as if there are virtual positrons in the intermediate state because the electrons are “running backward” in time. However, the two spacetime diagrams of Figure 22.6 sum to give the single $\mathbf{p}$ -space graph of Figure 22.5A in

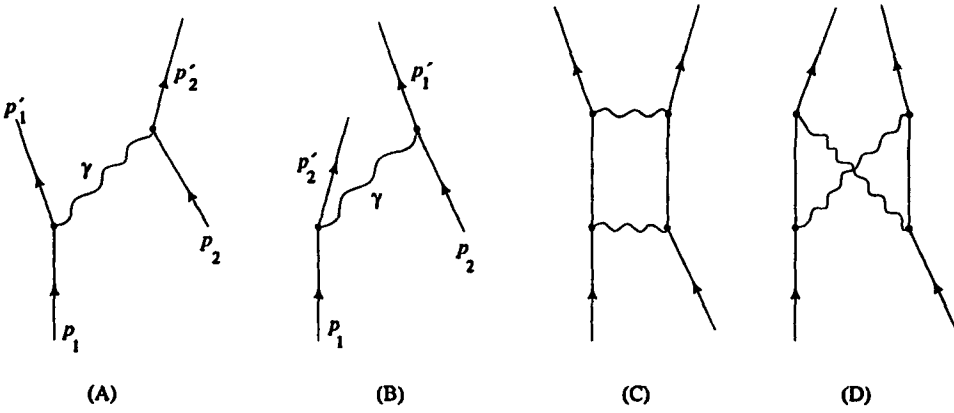


Figure 22.6: The two spacetime graphs included in one of the momentum-space graphs in Figure 22.5.

which the negative-energy states are included via the lower components of the spinors. For an amplitude of this order in e evaluated in momentum space, there are no virtual positrons.

Second-Order Matrix Elements

The virtual, intermediate states $|n\rangle$ which contribute to the sum in (22.43) are just the states $|A\rangle$ and $|B\rangle$ of Figure 22.5A and B. The four, nonvanishing matrix elements are hence:

$$\langle A | \hat{H}_{e\gamma} | i \rangle = e \sqrt{\frac{2\pi}{V\omega(k_i)} \frac{m}{E_{p_i+k_i}} \frac{m}{E_{p_i}}} [\bar{u}_{s'}(k_i + p_i) \not{\epsilon} u_{si}(p_i)], \quad (22.46)$$

$$\langle f | \hat{H}_{e\gamma} | A \rangle = e \sqrt{\frac{2\pi}{V\omega(k_f)} \frac{m}{E_{p_f}} \frac{m}{E_{p_i+k_i}}} [\bar{u}_{sf}(p_f) \not{\epsilon} u_{s'}(k_i + p_i)], \quad (22.47)$$

$$\langle B | \hat{H}_{e\gamma} | i \rangle = e \sqrt{\frac{2\pi}{V\omega(k_f)} \frac{m}{E_{p_i-k_f}} \frac{m}{E_{p_i}}} [\bar{u}_{s'}(p_i - k_f) \not{\epsilon} u_{si}(p_i)], \quad (22.48)$$

$$\langle f | \hat{H}_{e\gamma} | B \rangle = e \sqrt{\frac{2\pi}{V\omega(k_i)} \frac{m}{E_{p_i-k_f}} \frac{m}{E_{p_f}}} [\bar{u}_{sf}(p_f) \not{\epsilon} u_{s'}(p_i - k_f)], \quad (22.49)$$

where $\not{\epsilon}$ is the Dirac shorthand for the 4-vector dot product $\gamma \cdot \epsilon$.

Exercise Verify that (22.46)–(22.49) are the only nonvanishing elements. $\diamond$

If we look at (22.46)–(22.49) and Figure 22.5, we see that it is possible to associate each part of the figure with a factor in the matrix element:

- $e\gamma \cdot \epsilon$ for each electron–photon vertex,
- $\sqrt{2\pi/V\omega_k}$ each time a photon of momentum k enters (created or destroyed),
- $\sqrt{m/E_p}$ each time an electron of momentum p enters,

- $\bar{u}(p)$ for an electron in the final state, and
- $u(p)$ for an electron in the initial state.

We put these pieces together to form the second-order perturbation-theory matrix elements:

$$\mathcal{M}_A = \left[e^2 \sqrt{\frac{2\pi}{V\omega_{k_i}} \frac{2\pi}{V\omega_{k_f}} \frac{m}{E_{p_i}} \frac{m}{E_{p_f}}} \right] \left[\bar{u}_{s_f}(p_f) \not{\epsilon} \sum_{s'} \{ u_{s'}(k_i + p_i) \} \right] \left[\bar{u}_{s'}(k_i + p_i) \right] \not{\epsilon} u_{s_i}(p_i) \frac{m}{E_{p_i+k_i}} \frac{1}{E_{p_i+k_i} - E_{p_i+k_i}}, \quad (22.50)$$

$$\mathcal{M}_B = \left[e^2 \sqrt{\frac{2\pi}{V\omega_{k_i}} \frac{2\pi}{V\omega_{k_f}} \frac{m}{E_{p_i}} \frac{m}{E_{p_f}}} \right] \left[\bar{u}_{s_f}(p_f) \not{\epsilon} \sum_{s'} \{ u_{s'}(p_i - k_f) \} \right] \left[\bar{u}_{s'}(p_i - k_f) \right] \not{\epsilon} u_{s_i}(p_i) \frac{m}{E_{p_i-k_f}} \frac{1}{E_{p_i-k_f} - E_{p_i-k_f} - k_i - k_f}, \quad (22.51)$$

where the sum is over the spin s' of the virtual electron in the intermediate state.

Extracting Cross Sections

The preceding matrix elements are products of 4×1 , 1×4 , and 4×4 matrices. While products of this sort can be evaluated explicitly, they occur often enough that a number of tricks have been developed to help evaluate them. If we look at the sums over spinors in (22.50) and (22.51), we notice that the terms in braces are the electron projection operators introduced in § 15.2. The sums are thus replaced by analytic expression involving the dot products of momenta and gamma matrices:

$$\mathcal{M}_A = \{ \cdots \} \bar{u}_{s_f}(p_f) \left[\not{\epsilon} \frac{\not{k} + \not{p} + m}{2m} \not{\epsilon} \frac{m/E_{p_i+k_i}}{E_{p_i+k_i} - E_{k_i+p_i}} \right] u_{s_i}(p_i), \quad (22.52)$$

$$\mathcal{M}_B = \{ \cdots \} \bar{u}_{s_f}(p_f) \left[\not{\epsilon} \frac{\not{p} - \not{k} + m}{2m} \not{\epsilon} \frac{m/E_{p_i-k_f}}{E_{p_i-k_f} - E_{k_f+p_i}} \right] u_{s_i}(p_i). \quad (22.53)$$

Equations (22.52) and (22.53) are the matrix elements needed for electron–photon scattering. In the most general case they would be evaluated for initial electron spin s_i , final spin s_f , initial photon polarization ϵ_i , and final polarization ϵ_f . While varying each of these spins is an experimental possibility, the most common (and simplest) experiments would have a target with unpolarized electrons and a final state in which the spin polarization of the electron is unobserved, exactly along the lines we have discussed in Chapter 9, *Spin Theory*, and Chapter 10, *Spin Phenomenology and Identical Particles*. For this simple case, the transition rate is

$$\Gamma_{fi} = \frac{2}{\pi} \delta(E_f - E_i) \frac{1}{2} \sum_{s_f, s_i} |\mathcal{M}_A + \mathcal{M}_B|^2, \quad (22.54)$$

where we have averaged over initial states (the $\frac{1}{2}$) and summed over final states. Because there is no experimental means to determine if a particular scattering event has proceeded through state A or B in Figure 22.5A, the two processes add *coherently*, and the amplitudes

are added before squaring to form a probability. As discussed in § 22.1, the cross section for this process is proportional to the transition rate

$$\sigma = \frac{\Gamma_{fi}}{\text{flux}} = \frac{\Gamma_{fi}}{v_{\text{rel}}/V}. \quad (22.55)$$

The actual evaluation of the sums in (22.54) is a lengthy process in general. It can be evaluated directly by substitution of some representation of the γ matrices, computed algebraically, or reduced further. For example, it is possible to take the square, regroup terms and replace some sums by projection operators. The sums then are written as the trace of a product of vectors dotted into gamma matrices, which in turn are simplified using theorems. While the basic physics should be clear, these steps are too tedious for us to repeat here so we just give the result:

$$\frac{d\sigma}{d\Omega} = \frac{1}{4} \left[\frac{e^2}{m} \right]^2 \left[\frac{k_f}{k_i} \right]^2 \left[\frac{k_f}{k_i} + \frac{k_i}{k_f} + 4(\epsilon_f \cdot \epsilon_i)^2 - 2 \right]. \quad (22.56)$$

This is the *Klein–Nishina formula* and includes the effects of the recoil and negative-energy states of the electron, but is correct only to order α^2 because higher order graphs were not included.

Exercise Verify that (22.56) reduces to the nonrelativistic expression for Thompson scattering (21.25) if we take $k_i \simeq k_f$. $\diamond$

Many QED calculations are similar to this one.³ When enough higher-order terms are included, the resulting expressions are found to be in excellent agreement with the most precise experiments, so good in fact that QED is considered to be the most accurate of all theories. To help develop some physical feel of what physics is included in these kinds of calculations, in the next chapter we examine some potentials derived from field theory.

22.4 Problems

1. Use field theory to calculate the scattering of a Dirac electron from a very heavy nucleus (that is, a fixed external potential).

- (a) Show that in lowest order, the differential cross section for polarized electrons is

$$\frac{d\sigma}{d\Omega} = \frac{4Z^2\alpha^2m^2 |\bar{u}_s(p')\gamma^0 u_s(p)|^2}{q^4}, \quad (22.57)$$

where q is the momentum transfer and α the fine structure constant.

- (b) Show that the nonrelativistic limit of part (a) is the Rutherford scattering cross section.

³Their path is similar but not identical because modern calculations use the 4D generalization of the golden rule known as covariant perturbation theory in which the matrix elements are invariants, the density of states is invariant, and only a flux factor depends on the particular reference frame. It is for this form of the theory that the Feynman rules are usually listed.

2. Now explicitly evaluate the preceding expression:

- (a) Show that the spinors describing electrons with momentum making an angle θ with the z axis, and with positive and negative *helicity* are:

$$u_+(p) = N \begin{pmatrix} \cos \theta/2 \\ \sin \theta/2 \\ \frac{p \cos \theta/2}{E_p + m} \\ \frac{p \sin \theta/2}{E_p + m} \end{pmatrix}, \quad u_-(p) = N \begin{pmatrix} -\sin \theta/2 \\ \cos \theta/2 \\ \frac{p \sin \theta/2}{E_p + m} \\ \frac{-p \cos \theta/2}{E_p + m} \end{pmatrix}. \quad (22.58)$$

- (b) Evaluate the cross section for the scattering of a spin up electron into a positive helicity state.
 (c) Evaluate the cross section for the scattering of a spin up electron into a negative-helicity state.
 (d) Show that for the scattering of an unpolarized beam from the nucleus, the differential cross section for undetected final polarization is

$$\frac{d\sigma}{d\Omega} = \frac{Z^2 \alpha^2 [1 - \beta^2 \sin^2(\theta/2)]}{4p^2 \beta^2 \sin^4(\theta/2)}, \quad (22.59)$$

where $\beta = v/c = v$. (*Hint:* You can evaluate this explicitly without any “tricks” by summing over the helicity states.)

- (e) Compare this answer with the Rutherford cross section and discuss the differences.

3. Consider the matrix element $\mathcal{M}_{fi}$ for electron–electron scattering which is of lowest order in α (it is discussed in § 22.2).

- (a) Fill in the steps left out in the text and work out $\mathcal{M}_{fi}$ from first principles. You may assume nonrelativistic, spinless electrons in the static limit.
 (b) Show that these approximations yield the Rutherford scattering cross section. (This does mean that the Rutherford cross section is only approximate).

4. Try answering the following questions without resort to the text.

- (a) Indicate the two lowest-order field theory diagrams which can lead to photon–photon scattering.
 (b) Discuss why you would expect differences and similarities in electron–electron and positron–electron scattering.

5. A new interaction Hamiltonian,

$$H' = \lambda \sum_{\mu, \nu} \gamma^0 \sigma^{\mu\nu} (\partial_\mu A_\nu - \partial_\nu A_\mu), \quad (22.60)$$

is added in to the familiar Dirac Hamiltonian. Evaluate the nonrelativistic limit of the expectation value of this interaction and interpret your results. (You can use limits deduced previously.)

6. The Coulomb potential is the 4th component of a 4-vector. Its simplest coupling to a fermion field is obtained by forming another 4th component from the fermion field, and then coupling the two together as if they are part of the scalar product of two 4-vectors:

$$\tilde{H} \propto A_0 \bar{\Psi} \gamma_0 \Psi. \quad (22.61)$$

For this problem assume the Coulomb potential is scalar.

- (a) How would the Coulomb field now couple to fermion fields?
- (b) Show that the Rutherford cross section is now obtained for low-energy scattering, but not for high energies. (This contradicts experiment, as it should.)

CHAPTER 23

THE BREIT–PAULI AND BOSON-EXCHANGE INTERACTIONS

In this chapter we derive potentials based on the quantum field theory descriptions of the electromagnetic and nuclear interactions. These potentials are useful in giving an intuitive understanding of the physics and often have wider applications than does a field theory, for example, to many body-problems. Specifically, we show that the exchange of a photon between two electrons is the origin of the Coulomb force, and much more. We also show that the exchange of bosons with mass explains many of the properties of the strong force between nucleons. Not shown—but related—is the exchange of gluons leading to the color force between quarks.

*Mourn not! The terminology's misleading:
Decay is virtual particle breeding.
And seething, though mindless, can serve noble ends:
The clone stuff, exchanged, is a bond between friends.*

*To be or not? The choice seems clear enough.
Hamlet vacillated—so does this stuff.*

—Frank Wilczek

23.1 Relating Field Theory and Potential Amplitudes

To find the potential that is equivalent to a given field-theory diagram, we compare the Born approximation of potential theory (Chapter 5, *Green's Functions: Integral Quantum Mechanics*) with the Born approximation of field theory (the golden rule). We do the comparison in center-of-momentum frame (it is there that the two-body potential problem separates), and assume equal-mass particles to keep the equations simpler. For elastic scattering from a potential, the differential cross section is related to the scattering amplitude by

$$\frac{d\sigma}{d\Omega} = |f_B|^2. \quad (23.1)$$

Here the Born scattering amplitude is related to the (local) potential via

$$f_B = \frac{\mu}{2\pi} \int d^3x e^{-i\mathbf{p}' \cdot \mathbf{x}} V(\mathbf{x}) e^{i\mathbf{p} \cdot \mathbf{x}}, \quad (23.2)$$

with μ the reduced mass:

$$\mu_E = \frac{E_1 E_2}{E_1 + E_2} \rightarrow \frac{m_1 m_2}{m_1 + m_2} \quad (\text{nonrelativistic}). \quad (23.3)$$

In field theory we use the golden rule to determine the rates via:

$$\Gamma_{fi} = \frac{2}{\pi} |\mathcal{M}_{fi}|^2 \delta(E_f - E_i), \quad (23.4)$$

and convert rates to cross sections by dividing out the flux (density times relative velocity) of the incident beam:

$$d\sigma = \frac{\Gamma_{fi}}{\text{flux}} = \frac{\Gamma_{fi}}{(N/V)v_{\text{rel}}} = \frac{V\Gamma_{fi}}{p_i/E_i} = V \frac{E_i}{p_i} \sum_f 2\pi |\mathcal{M}_{fi}|^2 \delta(E_f - E_i). \quad (23.5)$$

To convert (23.5) to a specific differential cross section, we sum (integrate) over the final states accepted by the scattered-particle detector. As discussed in § 21.1, this usually removes the delta function. Because these phase-space manipulations to obtain observables are general, yet often difficult for students, we work them out for one- and two-body interactions.

One-Body Interactions

When a single particle is scattered from a fixed potential, we take the asymptotic final state as a plane wave, that is, an eigenstate of the momentum $\mathbf{p}$. The summation over available states is then equivalent to the integration:

$$\sum_f \rightarrow V \int \frac{d^3p}{(2\pi)^3}, \quad (23.6)$$

$$\begin{aligned} \Rightarrow d\sigma &= V \int \frac{d^3p}{(2\pi)^3} V \frac{E_i}{p_i} 2\pi |\mathcal{M}_{fi}|^2 \delta(E_f - E_i) \\ &= V \int \frac{p^2 dp d\Omega_p}{(2\pi)^3} V \frac{E_i}{p_i} 2\pi |\mathcal{M}_{fi}|^2 \delta(E_f - E_i). \end{aligned} \quad (23.7)$$

The details of this integration depend on the specific experiment, for example, whether the direction and energy loss of the scattered particle is observed. To implement the delta function in (23.7), we convert the p integration to one over total energy:

$$dp = \frac{dp}{dE_f} dE_f = \frac{E_f}{p} dE_f. \quad (23.8)$$

We recognize the ratio E_f/p as the velocity for either a relativistic or nonrelativistic particle. Equation (23.7) now becomes

$$d\sigma = \frac{V^2 E^2}{(2\pi)^2} d\Omega_p |\mathcal{M}_{fi}|^2, \quad (23.9)$$

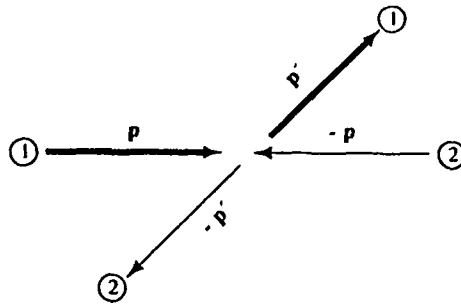


Figure 23.1: Kinematics for elastic scattering in the CM.

where we label the energy as $E \equiv E_f \equiv E_i$. Equation (23.9) for the cross section $d\sigma$ includes an integration over the solid angle Ω_p into which the particle is scattered. Although this cross section is still differential in that particles with only one energy are observed, it is what in Chapter 2, *Currents and Cross Sections*, we called the angle-integrated or total cross section. The usual differential cross section $d\sigma/d\Omega$ is for scattering into some solid angle, and is obtained by *not* performing the $d\Omega$ part of the integration in (23.9):

$$\frac{d\sigma}{d\Omega} = \frac{V^2 E^2}{(2\pi)^2} |\mathcal{M}_{fi}|^2. \quad (23.10)$$

Two-Body Interactions

When we studied the Schrödinger equation, we eliminated the overall motion of the center of mass and solved for the motion of an effective particle in an external potential. To start with a field theory process and deduce the equivalent potential, it is simplest to stay in the CM system, Figure 23.1. For two identical particles, we have twice the number of states available and therefore the sum over final states in (23.6) gets multiplied by 2,

$$\sum_f \rightarrow 2V \int \frac{d^3p}{(2\pi)^3}. \quad (23.11)$$

There is not an independent phase space associated with each particle because the particles' momenta are equal and opposite. The total energy is now

$$E_f = E_1(p) + E_2(p). \quad (23.12)$$

Exercise Show that for relativistic two-particle phase space,

$$dp = \frac{\mu_E dE}{p}, \quad (23.13)$$

$$\Rightarrow \frac{d\sigma}{d\Omega} = \frac{2V^2 E \mu_E}{(2\pi)^2} |\mathcal{M}_{fi}|^2, \quad (23.14)$$

where μ_E is the relativistic reduced mass (23.3). $\diamond$

Because the target and projectile masses are equal, the reduced mass μ_B equals $E/2$, and so $d\sigma/d\Omega$ (23.14) is identical to (23.10). We thus proceed with both cases simultaneously by setting the expressions for the cross sections in potential and field theories [(23.1) and (23.10)] equal:

$$V(q) = \int d^3x e^{-i\mathbf{q}\cdot\mathbf{x}} V(\mathbf{x}) = 2V\mathcal{M}_{fi}, \quad (23.15)$$

where we have assumed the potential is local. The coordinate-space potential is obtained by multiplying both sides by $\exp(i\mathbf{q}\cdot\mathbf{x})$ and integrating over all q (momentum transfers) to invert (23.15) to:

$$V(\mathbf{x}) = \int \frac{d^3q}{(2\pi)^3} 2V\mathcal{M}_{fi} e^{i\mathbf{q}\cdot\mathbf{x}}. \quad (23.16)$$

23.2 The Electron-Electron Interaction

In the Problems section of Chapter 22, *Interaction of Photons with Quantized Matter, QED*, we used field theory to calculate an electron scattering from an instantaneous Coulomb potential by setting A_0 (the zeroth component of the vector potential) proportional to $1/r$. We found that to lowest order in e^2 , the exchange of these timelike photons yields classical Rutherford scattering. In Figure 23.2 we see the related problem of electron–electron (Moller) scattering in the CM frame with two electrons interacting via the exchange of a *covariant photon*. The covariance of the photon means that its polarization vector has independent transverse and longitudinal space parts as well as a timelike component. Realizing that this is the simplest electron–electron interaction possible in field theory, we may suspect it must be equivalent to the Coulomb potential. To check out our suspicion, we look at the direct scattering, Figure 23.2A. In nonrelativistic quantum mechanics, exchange scattering, Figure 23.2B, is included by imposing the identical-particle symmetry on the wave function (and consequently scattering amplitude) at the end of the calculation (as we did in § 10.4).

Because the photon exchange in Figure 23.2 is second order in e (there are two vertices), it corresponds to the second-order matrix element:

$$\mathcal{M}_{ee} = \sum_n \frac{\langle f | \tilde{H}_{e\gamma} | n \rangle \langle n | \tilde{H}_{e\gamma} | i \rangle}{E - E_n}, \quad (23.17)$$

where n is an intermediate state and $\tilde{H}_{e\gamma}$ is the electron–photon interaction Hamiltonian. The energy E is that of the two (initial or final) electrons:

$$E = E_1(p) + E_2(p) = 2E_p. \quad (23.18)$$

The energy E_n of the intermediate state is that of one final electron, one initial electron, and the exchanged photon:

$$E_n = E_1(p - q) + E_2(p) + \omega_q = E + \omega_q, \quad (23.19)$$

where we use the elastic scattering condition $E_{p'} = E_p$.

We first see what potential results for *massive photons*, that is, by taking

$$\omega_q = \sqrt{m_\gamma^2 + q^2}. \quad (23.20)$$

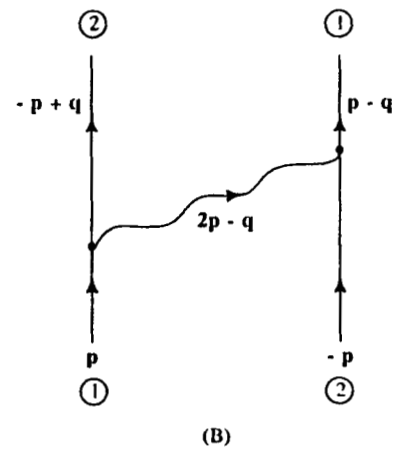
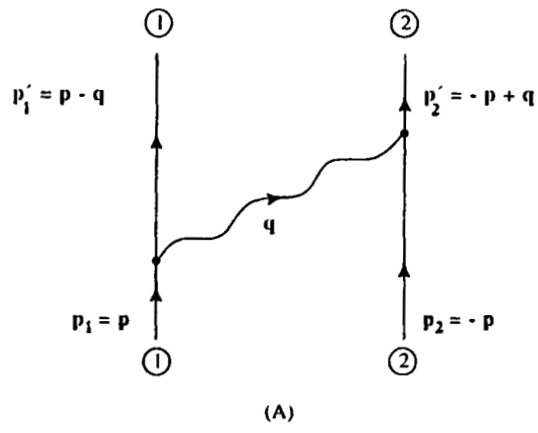


Figure 23.2: (A) Direct and (B) exchange graphs contributing to the lowest-order electron–electron interaction in the CM.

We assume an arbitrary polarization vector ϵ^μ for the exchanged “photon”, which means we are actually calculating the potential arising from the *exchange* of any spin 1 (vector) boson. In this way we see what aspects of the electron–electron potential are sensitive to the photon’s mass, and latter use this calculation to determine the nuclear potential.

Exercise Verify that using (22.39) for $\tilde{H}_{e\gamma}$ determines the amplitude equivalent to Figure 23.2A to be:

$$\mathcal{M}_{ee} = \sqrt{\frac{2\pi}{\omega_q} \frac{2\pi}{\omega_q} \frac{m}{E_{p_1}} \frac{m}{E_{p'_1}} \frac{m}{E_{p_2}} \frac{m}{E_{p'_2}}} e^2 \frac{[\bar{u}_1(p'_1) \gamma_\mu \epsilon^\mu u_1(p_1)][\bar{u}_2(p'_2) \gamma_\nu \epsilon^\nu u_2(p_2)]}{\omega_q}, \quad (23.21)$$

where the factors in the square-root arise from the normalizations for boson and fermion states, and the same polarization vector ϵ appears in both brackets since they refer to the same exchanged photon. $\diamond$

The Coulomb Interaction: An Approximation

We start by finding the potential equivalent to Figure 23.2A in the limit in which the electrons are at rest (static limit) and then look at the corrections for small electron velocities.¹ For static electrons:

$$E \rightarrow m, \quad \sqrt{\frac{m}{E}} \rightarrow 1, \quad \gamma_\mu \epsilon^\mu \rightarrow \gamma_0 \epsilon^0, \quad (23.22)$$

$$\bar{u}(p') \gamma^0 u(p) \equiv u^\dagger(p') u(p), \quad (23.23)$$

$$\chi_{s'}^\dagger \left[1 + \frac{\mathbf{p} \cdot \mathbf{p}'}{(2m)^2} + \frac{i\sigma_1 \times (\mathbf{p}' \times \mathbf{p})}{(2m)^2} \right] \chi_s \rightarrow \delta_{s's}, \quad (23.24)$$

$$\Rightarrow \mathcal{M}_{ee} \rightarrow e^2 \frac{2\pi}{V} \frac{\delta_{s'_1 s_1} \delta_{s'_2 s_2}}{m_\gamma^2 + q^2}, \quad (23.25)$$

where the $\omega_q^2 = m_\gamma^2 + q^2$ in the denominator of (23.25) arises from the exchanged photon (one ω_q from the normalization, another from the energy-denominator).

Exercise Show that only timelike photons contribute in the static limit. $\diamond$

The potential equivalent to (23.25) is obtained via the relation we deduced between potential and field theory amplitudes, (23.16):

$$\begin{aligned} V(\mathbf{x}) &= \int \frac{d^3q}{(2\pi)^3} 2V \mathcal{M}_{ee} e^{i\mathbf{q} \cdot \mathbf{x}} = \frac{2e^2 \delta_{s'_1 s_1} \delta_{s'_2 s_2}}{(2\pi)^2} \int \frac{d^3q}{m_\gamma^2 + q^2} e^{i\mathbf{q} \cdot \mathbf{x}} \\ &= e^2 \delta_{s'_1 s_1} \delta_{s'_2 s_2} \frac{e^{-m_\gamma r}}{r}. \end{aligned} \quad (23.26)$$

¹This procedure is similar to the one used in § 15.5 to obtain the Pauli equation as the nonrelativistic limit of the Dirac equation. In this section we are deducing a potential for the Schrödinger equation. In Chapter 25, *Wave Equations from Field Theory*, we make the related deduction of some new forms for the wave equation.

Exercise Verify that the 3D integration over the photon's momentum transfer leads to (23.26). $\diamond$

We see that if the mass of the photon m_γ is exactly zero, we obtain Coulomb's inverse-square law as the static limit of the exchange of one timelike photon. If the exchanged particle were spin one but with mass, the equivalent potential would fall off exponentially with a range $R = 1/m \equiv \hbar/mc$. Consequently, observing a deviation from Gauss's law (which derives from the potential being proportional to $1/r$), say by observing the nonvanishing of an electric field within a hollow conductor, is a way of measuring the photon's mass.

The Breit Interaction: An Improvement

When the electrons are not static, but not necessarily relativistic, we obtain contributions to the ee potential from the exchange of spacelike photons as well as timelike ones. We know from our study in Chapter 15, *Components of Dirac Wave Functions*, of the Pauli equation and the Gordon decomposition that a moving Dirac electron creates a convection current, a dipole current, and a spin-orbit interaction. And because we have Dirac spinors in the one-photon exchange matrix element (23.21), we expect these physical effects to be present in the electron-electron potential. These corrections to the Coulomb potential between two electrons, deduced by Breit (1929) and Breit et al. (1936–1939) are called the *Breit interaction*, and the generalization of the Schrödinger equation which includes these interactions is called the *Breit equation* (Bethe and Salpeter, 1977).

The new terms of the Breit interaction arise from the spacelike part of (23.21):

$$\mathcal{M}_{ee} = \sum_{i=1}^2 (2\pi) e^2 \frac{[\bar{u}_1(p'_1) \gamma_1 \cdot \hat{\epsilon}_q u_1(p_1)] [\bar{u}_2(p'_2) \gamma_2 \cdot \hat{\epsilon}_q u_2(p_2)]}{q^2} \quad (23.27)$$

where we sum only the transverse polarizations of the exchanged photon.² For simplicity we have set $m/E = 1$ and thus ignore some velocity dependences, some recoil effects, and some nonlocalities which the potential should contain. The summation in (23.27) is evaluated by use of the identity:

$$\sum_{i=1}^2 (\gamma_i \cdot \hat{\epsilon}_q) (\gamma_i \cdot \hat{\epsilon}_q) = \gamma_1 \cdot \gamma_2 - \frac{(\gamma_1 \cdot \mathbf{q})(\gamma_2 \cdot \mathbf{q})}{q^2}, \quad (23.28)$$

where $\mathbf{q}$ is the photon's momentum,

$$\mathbf{q} = \mathbf{p}_1 - \mathbf{p}'_1 = \mathbf{p}'_2 - \mathbf{p}_2. \quad (23.29)$$

The second term in (23.28) vanishes when evaluated between spinors which satisfy the Dirac equation because

$$\bar{u}_1(p'_1) (\gamma_1 \cdot \mathbf{q}) u_1(p_1) = \bar{u}_1(p'_1) (\gamma_1 \cdot [\mathbf{p}_1 - \mathbf{p}'_1]) u_1(p_1) = 0, \quad (23.30)$$

$$\bar{u}_1(p'_1) (\gamma_0 [E_1 - E'_1]) u_1(p_1) = 0. \quad (23.31)$$

²We do not include the timelike and related longitudinal parts since they produce the previously derived Coulomb interaction.

The non-Coulomb part of the one-photon exchange matrix element is thus

$$\mathcal{M}_{ee} = 2\pi e^2 \frac{[\bar{u}_1(\mathbf{p}'_1)\boldsymbol{\gamma}u_1(\mathbf{p}_1)] \cdot [\bar{u}_2(\mathbf{p}'_2)\boldsymbol{\gamma}u_2(\mathbf{p}_2)]}{q^2}. \quad (23.32)$$

Working out all the details for the equivalent potential and wave equation is tedious, and we refer the reader to Bethe and Salpeter (1977) and Sakurai (1967). The physics arising from the exchange of a transverse photon is, however, evident. Each bracket in (23.32) is the Dirac equation's expression for the electron's current, which we know from the Gordon decomposition (§ 15.4) contains charge and dipole pieces:

$$\bar{u}(\mathbf{p}')\boldsymbol{\gamma}^\mu u(\mathbf{p}) = \bar{u}(\mathbf{p}') \left[\frac{(\mathbf{p} + \mathbf{p}')^\mu}{2m} + \frac{i\sigma^{\mu\nu}(\mathbf{p}' - \mathbf{p})_\nu}{2m} \right] u(\mathbf{p}). \quad (23.33)$$

We see from this relation that the electron's spin couples to the momentum transfer, so there is no spin interaction for static electrons. The matrix element $\mathcal{M}_{ee}$ (23.32) consequently contains the interactions of the charge “c” plus dipole “d” currents of one electron, with the charge plus dipole currents of the other:

$$\mathcal{M}_{ee} = \mathcal{M}_{cc} + \mathcal{M}_{cd} + \mathcal{M}_{dc} + \mathcal{M}_{dd}. \quad (23.34)$$

The dipole–dipole interaction $\mathcal{M}_{dd}$ produces the $(\boldsymbol{\sigma}_1 \times \mathbf{q}) \cdot (\boldsymbol{\sigma}_2 \times \mathbf{q})$ term visualized in Figure 23.3. It is a magnetic spin–spin interaction similar to the hyperfine interaction in hydrogen, and is thus a “tensor force” between electrons. We leave it to the Problems section to show that $\mathcal{M}_{dd}$ is equivalent to the potential

$$V_{dd} = \mu^2 \left(-\frac{8\pi}{3} \boldsymbol{\sigma}_1 \cdot \boldsymbol{\sigma}_2 \delta(\mathbf{r}) - \frac{S_{12}}{r^3} \right). \quad (23.35)$$

This potential contains a contact term at the origin plus a long-range *tensor force* containing the tensor operator S_{12} :

$$S_{12} = \frac{3(\boldsymbol{\sigma}_1 \cdot \mathbf{r})(\boldsymbol{\sigma}_2 \cdot \mathbf{r})}{r^2} - \boldsymbol{\sigma}_1 \cdot \boldsymbol{\sigma}_2. \quad (23.36)$$

Note that the $1/r^3$ dependence in the tensor part of (23.35) is characteristic of two dipoles interacting ($V = -\boldsymbol{\mu}_1 \cdot \mathbf{B}_2$ with $\mathbf{B}_2$ the magnetic field from dipole 2).

The charge–dipole interaction $\mathcal{M}_{cd} + \mathcal{M}_{dc}$ produces a $(\boldsymbol{\sigma}_1 \times \mathbf{q}) \cdot (\mathbf{p}_2 + \mathbf{p}'_2)$ term which is equivalent to the spin–orbit potential

$$V_{cd} = \frac{-3}{4m^2} \frac{1}{r} \frac{d}{dr} \left(\frac{e}{r} \right)^2 \mathbf{l} \cdot (\boldsymbol{\sigma}_1 + \boldsymbol{\sigma}_2). \quad (23.37)$$

This spin–orbit potential is similar to the *ep* one in hydrogen but three times as large.

Exercise Show that a spin–orbit force with the same strength as in hydrogen is contained in the exchange of timelike photons. That is, show that

$$\bar{u}_1 \gamma_0 u_1 \simeq \chi_1^\dagger \left[1 + \frac{\mathbf{p}_1 \cdot \mathbf{p}'_1}{(2m)^2} + \frac{i\boldsymbol{\sigma}_1 \times (\mathbf{p}'_1 \times \mathbf{p}_1)}{(2m)^2} \right] \chi_1. \quad \diamond \quad (23.38)$$

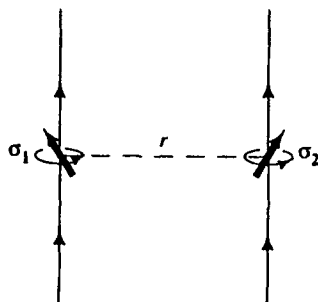


Figure 23.3: The dipole–dipole interaction of two electrons separated by a distance r . This is part of the interaction arising from the one-photon exchange in Figure 23.2.

An amazing realization about the terms in the Breit interaction (23.34) is that they are all included in the one-photon-exchange graph Figure 23.2A. Higher-order graphs produce an even more accurate (and complicated) electron–electron interaction. Other higher-order graphs are automatically included when the potential is iterated as part of the solution of wave equation.

23.3 One-Boson-Exchange Potentials

We have seen that the exchange of a photon is the origin of the electric and magnetic forces between electrons. Other forces in nature also arise from boson exchange; mass-zero bosons produce infinite-range forces (like gravity) while massive boson exchange produces a finite-range force (like the weak and strong interactions). Because the second-quantization rules of Chapter 19, *Second Quantization*, are general, the actual quantization of other spin-one (vector) fields Φ^μ follows the same formalism used for the photon field A^μ , but with no restriction to transverse polarizations (spins). A spin-zero (scalar) field Φ is also analogous to the photon field (both are bosons) but with no degrees of freedom associated with the internal polarization vectors.

Meson Origin of the Nuclear Force

We have seen that the exchange of a particle of mass m leads to a potential proportional to $\exp(-mr)/r$. In 1935, Yukawa proposed that the strong force between nucleons, which has a range $R \simeq 1.5$ fm, is caused by the exchange of bosons with mass $m = 1/R \equiv \hbar c/R \simeq 197 \text{ MeV fm}/1.5 \text{ fm} \simeq 130$ MeV. Shortly after his prediction, short-lived particles with masses 110–140 MeV were found in cosmic rays. One of them, the pi meson, is the major contributor to the long-range part of the nuclear force. We now study the nuclear force as an example of how a new interaction is built from fields. In Chapter 28, *Phonons*, we outline a similar construction for the lattice–phonon interaction in solids and for the weak force in nuclei.

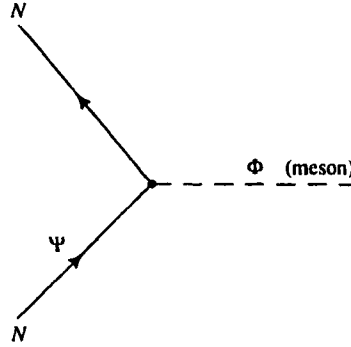


Figure 23.4: The elementary coupling of a boson field Φ (dashed) to a fermion field Ψ (solid).

Coupling of Fields

In QED we used the minimal electromagnetic coupling to obtain the $e\gamma$ interaction Hamiltonian:

$$\tilde{H}_{e\gamma} = -e (\bar{\Psi} \gamma_\mu \Psi) A^\mu. \quad (23.39)$$

In terms of Lorentz covariance, this interaction is just the scalar product of the the 4-vector electromagnetic field A^μ with the electron 4-current $\bar{\Psi} \gamma_\mu \Psi$. Because nucleons (neutrons and protons) are spin- $\frac{1}{2}$ fermions, they too are described by spinor fields $\bar{\Psi}$ and Ψ , with their couplings to the various meson fields generating the nuclear interaction. The different spins, parities, and masses of the mesons (Particle Data Group, 1994), produce potentials with different spin, angular, and functional forms. To construct these potentials, we look back at Table 14.3 and construct combinations of fields which transform as scalars (S), pseudoscalars (PS), vectors (V), pseudovectors (PV), and tensors (T):³

$$\begin{array}{ll} \text{S: } \bar{\Psi} \Psi & \text{PS: } \bar{\Psi} \gamma_5 \Psi \\ \text{V: } \bar{\Psi} \gamma_\mu \Psi & \text{PV: } \bar{\Psi} \gamma_\mu \gamma_5 \Psi \\ \text{T: } \bar{\Psi} \sigma_{\mu\nu} \Psi, & \end{array} \quad (23.40)$$

where γ_5 is the pseudoscalar under parity,

$$P^{-1} \gamma_5 P = \gamma_0 \gamma_5 \gamma_0 = -\gamma_5. \quad (23.41)$$

We next introduce boson fields Φ and Φ^μ that contains meson annihilation and creation. For example, scalar mesons are described by the Φ in Figure 23.4 (with no arrow associated with it), or mathematically by

$$\Phi(x) = \sum_k \sqrt{\frac{2\pi}{V\omega_k}} \left[a_k e^{-ik \cdot x} + a_k^\dagger e^{ik \cdot x} \right]. \quad (23.42)$$

A pseudoscalar meson is described by the field Φ^P having the same form as (23.42) but acquiring -1 under reflection P . Vector mesons, being spin-1 bosons, are described by a

³The pseudoscalar and pseudovector bosons acquire a -1 under the parity transformation in addition to that arising from angular momentum. For example, the pion is pseudoscalar.

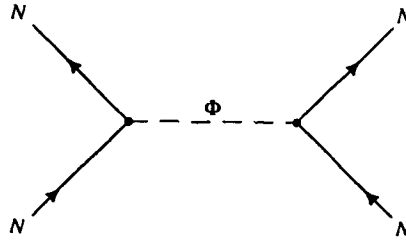


Figure 23.5: The nucleon–nucleon potential arising from meson exchange.

field Φ_μ which has the same form as the photon field A^μ (20.9), except ω_k now includes the meson's mass, and the polarization vector $\hat{\epsilon}$ is not restricted to be transverse.

The most general Hamiltonian describing the vertex in Figure 23.4 is the sum of Lorentz-invariant pieces. Each piece is constructed from scalar products of Dirac space operators formed out of fermion and boson fields:

$$\tilde{H} = \begin{cases} G\bar{\Psi}\Psi\Phi, & \text{for S-S,} \\ iG(\bar{\Psi}\gamma_5\Psi)\Phi^P, & \text{for PS-PS,} \\ i\sqrt{4\pi}g_v(\bar{\Psi}\gamma_\mu\Psi)\Phi^\mu, & \text{for V-V,} \\ \frac{i}{\mu}F(\bar{\Psi}\gamma_5\gamma_\mu\Psi)\frac{\partial\Phi}{\partial x_\mu}, & \text{for PV-PV,} \\ \frac{f}{4M}(\bar{\Psi}\sigma^{\mu\nu}\Psi)(\partial_\nu\Phi_\mu - \partial_\mu\Phi_\nu), & \text{for T-T,} \end{cases} \quad (23.43)$$

where the constants (charges) in front of each $\tilde{H}$ are common ones found in the literature. It is significant that once we accept nature's creation of mesons with different spins and parities, we know that (23.43) must be their interaction.

To see if these interactions explain the strong potential between nucleons, we take field-theory diagrams such as Figure 23.5, evaluate the matrix element $\mathcal{M}_{fi}$ via second-order perturbation theory, and then use (23.16) to deduce the equivalent potential. One can then predict the nuclear force with the known meson masses and coupling constants or try to reproduce it by varying the masses and coupling constants and see if the fitted values agree with the tabulated ones. In either case, the analytic forms of the potential agree with experiment, whereas the detailed numbers are close, but not in precise agreement. The inaccuracy may be because of the nonconvergence of perturbation theory for the strong interaction or because of the need to include quarks as the fundamental structure of the mesons and nucleons at short distances.

We have already gone through some of the steps needed to deduce the potential in the last section. For example, in looking at the electron–electron potential we deduced that the

exchange of a scalar meson gives

$$\mathcal{M}_{fi}^S = G^2 \frac{2\pi}{V} \left(\sqrt{\frac{m}{E}} \right)^4 \frac{(\bar{u}_1' u_1)(\bar{u}_2' u_2)}{-\omega_q^2}. \quad (23.44)$$

This has the static nucleon limit:

$$\mathcal{M}_{fi}^S \rightarrow -G^2 \frac{2\pi}{V} \frac{\delta_{s_1' s_1} \delta_{s_2' s_2}}{q^2 + \mu^2}, \quad (23.45)$$

with μ the mass of the exchanged particle and the minus sign originating in the energy denominator. The Born term $\mathcal{M}_{fi}^S$ Fourier transforms into the attractive potential,

$$V^S(\mathbf{r}) = -G^2 \frac{e^{-\mu r}}{r} \delta_{s_1' s_1} \delta_{s_2' s_2}, \quad (23.46)$$

which has range $1/\mu$.

The potentials derived here are local and velocity independent (in the static nucleon limit). Bethe and Salpeter (1977) and Brown and Jackson (1976) show how to make them more accurate (and much more complicated) by including more kinematic factors. Yet if it is important to include these dependences, it is probably best to solve for and use the deduced potentials with the momentum-space techniques developed in Part I, *Scattering and Integral Quantum Mechanics*.

Pseudoscalar Exchange

When a pseudoscalar particle such as a pi meson is exchanged, an extra factor of γ_5 gets introduced at each vertex in Figure 23.5, so

$$\mathcal{M}_{fi}^{PS} = G^2 \frac{2\pi}{V} \sqrt{\frac{m}{E}}^4 \frac{(\bar{u}_1' \gamma_5 u_1)(\bar{u}_2' \gamma_5 u_2)}{-\omega_q^2}. \quad (23.47)$$

In the static nucleon limit

$$\mathcal{M}_{fi}^{PS} \rightarrow -G^2 \frac{2\pi}{V} \frac{\boldsymbol{\sigma}_1 \cdot \mathbf{q} \boldsymbol{\sigma}_2 \cdot \mathbf{q}}{q^2 + \mu^2}, \quad (23.48)$$

that is, the spins of fermions couple to the boson's momentum, as in the Breit interaction. If we ignore the delta function at the origin (the finite nucleon size makes point interactions unphysical), we obtain:

$$V^{PS}(\mathbf{r}) \propto -G^2 \int \frac{d^3 q e^{i\mathbf{q} \cdot \mathbf{r}} \boldsymbol{\sigma}_1 \cdot \mathbf{q} \boldsymbol{\sigma}_2 \cdot \mathbf{q}}{q^2 + \mu^2} \quad (23.49)$$

$$\begin{aligned} &= G^2 \int \frac{d^3 q \boldsymbol{\sigma}_1 \cdot \nabla \boldsymbol{\sigma}_2 \cdot \nabla e^{i\mathbf{q} \cdot \mathbf{r}}}{q^2 + \mu^2} \\ &= G^2 \boldsymbol{\sigma}_1 \cdot \nabla \boldsymbol{\sigma}_2 \cdot \nabla \int \frac{d^3 q e^{i\mathbf{q} \cdot \mathbf{r}}}{q^2 + \mu^2} \\ &= G^2 \boldsymbol{\sigma}_1 \cdot \nabla \boldsymbol{\sigma}_2 \cdot \nabla \frac{\pi}{2} \frac{4\pi}{r} e^{-\mu r}. \end{aligned} \quad (23.50)$$

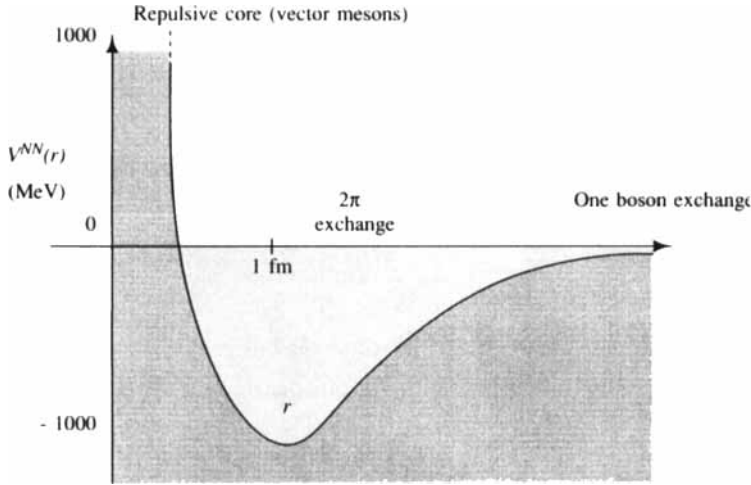


Figure 23.6: A schematic representation of the nucleon–nucleon potential showing regions in which different particle exchanges dominate. Within the core, quark and gluon exchanges should be important.

If we perform those derivatives for $r \neq 0$ we obtain

$$V^{PS}(\mathbf{r}) = \frac{G^2}{4\pi} \left(\frac{\mu}{2m} \right)^2 \left[\frac{1}{3}(\boldsymbol{\sigma}_1 \cdot \boldsymbol{\sigma}_2) + \left(\frac{1}{(\mu r)^2} + \frac{1}{\mu r} + \frac{1}{3} \right) S_{12} \right] \frac{e^{-\mu r}}{r}, \quad (23.51)$$

where S_{12} is the tensor operator (23.36). We see that the exchange of a pseudoscalar boson produces a potential similar to the magnetic interaction between two dipoles. Yet since a meson is being exchanged, the potential has a finite range, and because G is much larger than e (on some appropriate scale), the nuclear potential is much stronger. The spin–spin and tensor forces are responsible for strong spin effects in nuclei, with the noncentral tensor force also causing the mixing of the S and D states in the deuteron.

In summary, the nuclear force has central, spin–spin, spin–orbit and tensor parts:

$$V = V_c + V_\sigma \boldsymbol{\sigma}_1 \cdot \boldsymbol{\sigma}_2 + V_{ls} \mathbf{l} \cdot \mathbf{s} + V_t S_{12}, \quad (23.52)$$

which arise from the exchange of mesons with finite mass. While no one meson explains all aspects of the nuclear force, each seems to serve a purpose in generating some aspect. For example, as illustrated in Figure 23.6, vector mesons are needed to generate the spin–orbit force and the repulsive hard core which keeps nuclei from collapsing. The exchanges of multiple mesons (or possibly quarks) generate the short-range behavior.

23.4 Problems

1. Consider the interaction between two fermions arising from the exchange of a pseudo-scalar boson with mass μ .

- (a) Indicate what must be the simplest Hamiltonian, and show that the lowest-order matrix element for the fermion–fermion interaction is:

$$\mathcal{M}_{fi} \propto \frac{G^2(\boldsymbol{\sigma}_1 \cdot \mathbf{q})(\boldsymbol{\sigma}_2 \cdot \mathbf{q})}{\mu^2 + q^2}, \quad (23.53)$$

where $\mathbf{q}$ is the momentum transferred by the boson.

- (b) Show that in the static fermion limit, this matrix element is equivalent to the potential

$$V(\mathbf{r}) \propto G^2(\boldsymbol{\sigma}_1 \cdot \nabla)(\boldsymbol{\sigma}_2 \cdot \nabla) \frac{e^{-\mu r}}{r}. \quad (23.54)$$

- (c) Show that the potential of part b can be expressed as spin–spin and tensor forces

$$V(\mathbf{r}) \propto G^2 \mu^2 \left[\frac{\boldsymbol{\sigma}_1 \cdot \boldsymbol{\sigma}_2}{3} + \left(\frac{1}{(\mu r)^2} + \frac{1}{\mu r} + \frac{1}{3} \right) S_{12} \right] \frac{e^{-\mu r}}{r}. \quad (23.55)$$

2. Consider the interaction between two fermions and a pseudoscalar boson

$$\tilde{H}^{int} = iG\bar{\Psi}(x)\gamma_5\Psi(x)\phi(x). \quad (23.56)$$

- (a) Calculate the amplitude (*not* cross section) for
- fermion–fermion scattering
 - boson–fermion scattering
 - fermion–antifermion scattering (for example, $p + \bar{p} \rightarrow \pi + \pi$)
- (b) Deduce the analytic relation (known as crossing symmetry) among these amplitudes.
3. Try answering the following questions without resort to the text.
- Why does the γ^5 appear in the pion–nucleon strong Hamiltonian?
 - Is the nucleon–nucleon potential arising from single pion exchange local or nonlocal at high energies?
 - Explain the origin of the dipole–dipole part of the Breit interaction.
 - Why should not ladder graphs be included in the derivation of the Coulomb potential?
 - Is field theory more or less “correct” than quantum mechanics?
 - What general form would you expect for the neutron–neutron electromagnetic interaction?
 - Can the exchange of a scalar particle lead to a spin-dependent force?
 - It is found that the force between two nucleons becomes strongly repulsive at a separation of ~ 0.2 fm. If this is due to meson exchange, what is the heaviest mass of the exchanged boson?

- (i) How would you detect contributions from mesons which have heavier masses than pions?
 - (j) Explain why vector-meson exchange leads to a spin–orbit force.
4. Give an explanation of why the spin–orbit force is of opposite sign in the hydrogen atom and in the deuteron.
 5. How would the properties of the potentials deduced in this chapter change if higher-order field theory diagrams were included?
 6. How would the properties of the potentials deduced in this chapter change if higher-order kinematics were included?
 7. Postulate reasonable couplings which can be used as the interaction Hamiltonians for the interactions of
 - (a) scalar bosons,
 - (b) a pi meson with a photon,
 - (c) a fermion with a scalar boson.

CHAPTER 24

WEAK FIELDS

In this chapter we show how the techniques in Part III, *Quantum Fields*, are applicable to the weak force. This a beautiful theory, full of symmetry and simplicity, and this beauty only increases as the latest advances are included. There are many excellent books on the subject, and we recommend Gross (1993) and Guidry (1991) for extensions to the electroweak model, and deShalit and Feshback (1974) for details.

24.1 The Weak Force

In order of decreasing strength, the four fundamental forces of nature are strong, electromagnetic, weak, and gravity (the electromagnetic and weak force actually derive from the unified electroweak force). We have already examined the electromagnetic and strong forces, and now look at the weak force as an example of how the elements of Dirac theory and field theory are used to construct a basic Hamiltonian of nature. (Quantum gravity requiring spin-2 gravitons is beyond the scope of this book.)

Historical Puzzle

Beta decay is a natural process in which nuclei transmute and in the process emit a spectrum of electrons. It has been studied from the earliest days of quantum mechanics and field theory, and still has high pedagogical value. In beta decay, a neutron spontaneously converts into a proton, electron, and antineutrino:

$$n \rightarrow p + e^- + \bar{\nu}_e. \quad (24.1)$$

Whereas it took an experimental *tour de force* to actually detect the antineutrino in (24.1), its existence was theoretically inferred because an entire spectrum of electron energies would not be observed if there were only two particles in the final state.

Exercise Prove that if a neutron decays into two particles they will have definite energies, but if the decay is into three particles there will be a spectrum of energies. $\diamond$

Calling the neutrino emitted in beta decay an “anti”-neutrino is more than semantic. Once the electron is called a particle and assigned a *lepton number* of $+1$, lepton number conservation requires the other lepton present, the $\bar{\nu}$, to have lepton number -1 , that is, to be an antiparticle. Physically, the neutrino and antineutrino represent the particles in the solution of the mass-zero Dirac equation of § 15.6. There we found the two solutions to have opposite helicity $\sigma \cdot \hat{p}$. For beta decay this means the antiparticle $\bar{\nu}$ is right-handed and the particle ν is left-handed.

Form of the Weak Hamiltonian

Given no experimental knowledge of the weak Hamiltonian and no classical limit (both in contrast to QED), we postulate the most general form for the Hamiltonian and then call on experiment to pare it down. (Because parity is not a good symmetry for the weak interaction, we cannot even use *that* to restrict the form.) If we look at the decay (24.1), we see a neutron being annihilated, and a proton, electron and antineutrino being created.

Exercise Prove that since the n , p , and e^- all have spin $\frac{1}{2}$, angular momentum conservation requires the spin of the $\bar{\nu}$ to be half-integer. $\diamond$

Following the original path of Fermi (1932, 1934), the simplest thing we can say is that at one point in spacetime, these four fermion fields couple as indicated in Figure 24.1A. At this point the neutron is annihilated via the field operator Ψ_n , the proton created by $\bar{\Psi}_p$, the electron created by $\bar{\Psi}_e$, and the antineutrino created by $\bar{\Psi}_\nu$ (recall fermion field operators destroy particles and create antiparticles). “Four-point” functions, such as that in Figure 24.1A, are unlikely in microscopic physics (as likely as getting two couples to arrive at a rendezvous at the same time). A more correct view is that of Figure 24.1B in which a very massive *intermediate vector boson* W is exchanged between the nucleon couple and the lepton couple (the couples can then be at two different rendezvous points). Compared to Figure 24.1A, the matrix element of Figure 24.1B contains an additional $(m_W^2 + q^2)^{-1}$ factor from the propagating intermediate boson (q is the momentum transferred from the n to the p). Yet because the mass of the boson is so large ($m_W \equiv m_W c^2 \simeq 80$ GeV), the distance separating the couples is small enough to be considered a single point ($1/m_W \equiv \hbar c/m_W c^2 \leq 3 \times 10^{-3}$ fm!). The point-interaction model is valid for decay processes in which the energies are on the order of the neutron–proton mass difference (≤ 1 MeV). If the leptons are accelerated to energies comparable to m_W , the finite range becomes visible. By providing a small τ (or equivalently, large k) cutoff, the W ’s propagator also keeps the theory from diverging at high energy.

To create or destroy four particles at one point, the interaction Hamiltonian described by Figure 24.1A must be linear in the field of each particle. We build it by coupling covariants constructed from the nucleon fields with the corresponding ones constructed from the lepton fields:

Figure 24.1: Feynman diagrams for the beta decay of the neutron. In (A) four fields act at one point; in (B) an intermediate W boson connects the nucleons to the leptons.

Covariant	Nucleon	Lepton
S	$\bar{\psi}_p \psi_n$	$\bar{\psi}_e \psi_\nu$
V	$\bar{\psi}_p \gamma^\mu \psi_n$	$\bar{\psi}_e \gamma^\mu \psi_\nu$
PS	$\bar{\psi}_p \gamma_5 \psi_n$	$\bar{\psi}_e \gamma_5 \psi_\nu$
A	$\bar{\psi}_p \gamma^\mu \gamma_5 \psi_n$	$\bar{\psi}_e \gamma^\mu \gamma_5 \psi_\nu$
T	$\bar{\psi}_p \gamma_5 \psi_n$	$\bar{\psi}_e \sigma^{\mu\nu} \psi_\nu$

$$\begin{aligned} \bar{H} &= \bar{H}_F + H_{GT}, \\ \bar{H}_F &= C_S (\bar{\psi}_p \psi_n) (\bar{\psi}_e \psi_\nu) + C_{PS} (\bar{\psi}_p \gamma_5 \psi_n) (\bar{\psi}_e \gamma_5 \psi_\nu) \\ &\quad + C_V (\bar{\psi}_p \gamma^\mu \psi_n) (\bar{\psi}_e \gamma_\mu \psi_\nu), \\ H_{GT} &= C_A (\bar{\psi}_p \gamma_5 \psi_n) (\bar{\psi}_e \gamma_5 \psi_\nu) + C_T (\bar{\psi}_p \sigma^{\mu\nu} \psi_n) (\bar{\psi}_e \sigma_{\mu\nu} \psi_\nu). \end{aligned} \tag{24.2}$$

For historical and physical reasons, the Hamiltonian in (24.2) is separated into a piece responsible for *Fermi transitions* and another one responsible for *Gamow–Teller transitions*. We shall see that Fermi transitions are the simplest and only connect states in which there is no change in total angular momentum $\Delta j = 0$, whereas Gamow–Teller transitions are more complicated and can have $\Delta j = \pm 1, 0$ ($0 \rightarrow 0$ forbidden).

Nonrelativistic Nucleon

When the matrix element of the weak Hamiltonian between two states exists, the weak interaction induces transitions between the states. Because the neutron–proton mass difference is approximately 1 MeV and the weak interaction connects these two states of the nucleon, this 1 MeV is the energy released in the beta decay of a free neutron.

Exercise Prove that a 1 MeV nucleon is nonrelativistic, a 1 MeV electron is highly relativistic, and a neutrino with any energy is relativistic.

◇

Consequently, we simplify the Hamiltonian (24.2) by taking the nonrelativistic limit for

the nucleons, in which case:

$$\begin{aligned}\bar{\Psi}_p \gamma_5 \Psi_n &\rightarrow 0, & \bar{\Psi}_p \Psi_n &\rightarrow \psi^\dagger \psi, \\ \bar{\Psi}_p \sigma_{ij} \Psi_n &\rightarrow \psi^\dagger \sigma_k \psi, & \bar{\Psi}_p \gamma_\mu \gamma_5 \Psi_n &\rightarrow (0, i\psi^\dagger \sigma \psi), \\ \bar{\Psi}_p \gamma_\mu \Psi_n &\rightarrow (\psi^\dagger \psi, 0).\end{aligned}\quad (24.3)$$

Here Ψ denotes a 4-component Dirac spinor field and ψ denotes a 2-component Pauli spinor field. The matrix element of the Hamiltonian between nuclear states is then

$$\langle f | \hat{H} | i \rangle = \mathcal{M}_{fi} \quad (24.4)$$

$$\begin{aligned}&= u_p^\dagger u_n [C_S \bar{u}_e(p_e) v_{\bar{\nu}}(p_{\bar{\nu}}) + C_V \bar{u}_e(p_e) \gamma_0 v_{\bar{\nu}}(p_{\bar{\nu}})] \quad (\text{F}) \\ &+ u_p^\dagger \sigma u_n \cdot [C_A \bar{u}_e(p_e) \gamma_5 \gamma v_{\bar{\nu}}(p_{\bar{\nu}}) + C_T \bar{u}_e(p_e) \sigma v_{\bar{\nu}}(p_{\bar{\nu}})] \quad (\text{GT}),\end{aligned}\quad (24.5)$$

where we separate the Fermi (F) and Gamow–Teller (GT) parts. Note that because we have taken matrix elements, the spinors in (24.5) are wave functions not fields. We see from (24.5) that Fermi transitions are independent of the nucleon spin, whereas Gamow–Teller transitions are spin dependent; consequently, GT transitions have $\Delta j = \pm 1, 0$, while Fermi transitions have the electron and neutrino carrying off no net angular momentum, $\Delta j = 0$. The coupling constants or weak charges C_i in (24.5) must be determined from experiment, and like the electric charges for particles, some are zero.

Parity Violation

Lee and Yang (1956) were the first to point out (at least for some time) that no experiment had shown parity to be a good symmetry for the weak interactions. For our construction project, this means that the scalar covariant of one couple can couple to the pseudoscalar covariant of the other couple ($S \times PS$), and likewise for vector and axial vector ($V \times A$):

$$\langle f | \hat{H} | i \rangle = u_p^\dagger u_n [C_S \bar{u}_e(p_e) [1 + \alpha_S \gamma_5] v_{\bar{\nu}}(p_{\bar{\nu}}) + C_V \bar{u}_e [1 + \alpha_V \gamma_5] \gamma_0 v_{\bar{\nu}}] \quad (\text{F}) \quad (24.6)$$

$$+ u_p^\dagger \sigma u_n \cdot (C_A \bar{u}_e [1 + \alpha_A \gamma_5] \gamma_5 \gamma v_{\bar{\nu}} + C_T \bar{u}_e [1 + \alpha_T \gamma_5] \sigma v_{\bar{\nu}}) \quad (\text{GT}), \quad (24.7)$$

where for clarity we leave off most momentum labels. If any of the α 's in the brackets were nonzero, the Hamiltonian would differ in a reflected world. This means some characteristic of the weak decay (such as a correlation between some particle's spin and momentum) would be nonzero and therefore violate parity. By studying nuclear decays and observing correlations of this sort, it is found that the violation is maximal and symmetric: All the α 's in (24.7) are -1 , and so:

$$\langle f | \hat{H} | i \rangle = (u_p^\dagger u_n) \bar{u}_e [1 - \gamma_5] (C_S + C_V \gamma_0) v_{\bar{\nu}} \quad (\text{F}) \quad (24.8)$$

$$+ (u_p^\dagger \sigma u_n) \cdot \bar{u}_e [1 - \gamma_5] (C_T \sigma + C_A \gamma) v_{\bar{\nu}} \quad (\text{GT}). \quad (24.9)$$

The C 's are determined by further experiments to be

$$C_T \simeq 0, \quad C_S \simeq 0, \quad \frac{C_A}{C_V} \simeq 1.2, \quad (24.10)$$

which means that the strength of Fermi transitions is 1.2 times that of the Gamow–Teller ones, and that there are no scalar and tensor terms.

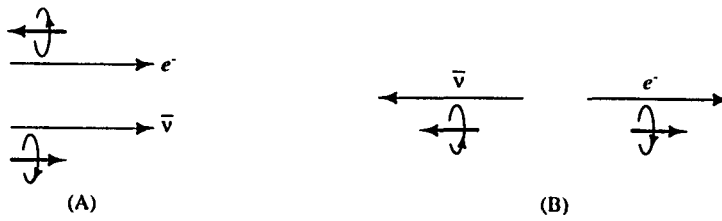


Figure 24.2: The orientation of the leptons' spins in beta decay for two possible directions of the antineutrino's momentum.

Exercise Show that the values in (24.10) reduce the full Hamiltonian to

$$\langle f | \tilde{H} | i \rangle = g[u_p^\dagger \gamma_\mu (1 - 1.2\gamma_5) u_n][\bar{u}_e \gamma^\mu (1 - \gamma_5) v_{-\nu}]. \quad \diamond \quad (24.11)$$

Neutrino Helicity

The $(1 - \gamma_5)$ factor in the weak Hamiltonian (24.11) requires neutrinos to be in helicity eigenstates. To determine the eigenvalues, we recall our examination of the Dirac equation for mass-zero particles in § 15.6 where we found right-handed and left-handed spinors for which

$$\boldsymbol{\sigma} \cdot \hat{\mathbf{p}} u_{\text{rh}} = +u_{\text{rh}}, \quad \boldsymbol{\sigma} \cdot \hat{\mathbf{p}} u_{\text{lh}} = -u_{\text{lh}}. \quad (24.12)$$

Yet we also know that $(1 - \gamma_5)$ is a *helicity projection operator* for these spinors, that is,

$$(1 - \gamma_5)u_{\text{rh}} = 0, \quad (1 - \gamma_5)u_{\text{lh}} = 2u_{\text{lh}}. \quad (24.13)$$

This means the $(1 - \gamma_5)$ acting on the neutrino spinor in the Hamiltonian (24.11) removes all the right-handed ones without disturbing the left-handed ones. Neutrinos emitted in beta decay are thus left-handed. The hole interpretation of Dirac solutions tells us that an antineutrino (the absence of a neutrino) is right-handed.

A consequence of the form of the Hamiltonian is that the antineutrino emitted in beta decay is completely polarized, as indicated in Figure 24.2. This implies that the electron emitted in beta decay must also be polarized. Thus even without observing the massless neutrino, its helicity was deduced by measuring the polarization of the electron.¹ In fact, the sheer existence of polarization for the electron emitted in a natural decay is proof that the neutrino was in a helicity eigenstate and consequently evidence of parity violation (the reflected world is different).

We show two final-state configurations in Figure 24.2 corresponding to decays in which the angular momenta of the neutron and proton remain unchanged. The electron and $\bar{\nu}$ are emitted with their momenta opposite in Figure 24.2B. Because the antineutrino is always right-handed and angular momentum is conserved, the electron must be left-handed while the one in B must be right-handed. Although we do not prove it here (see the Problems section), it is a characteristic of the weak-interaction Hamiltonian (24.11) that

¹ It was evident from our analysis in Chapter 15, *Components of Dirac Wave Functions*, that mass-zero solutions of the Dirac equation violate parity. While there is widely held confidence that the mass of the neutrino is small, it is hard to verify that it is exactly zero. Even if the neutrino is found to have mass, parity violation remains as an experimental fact; the algebra of the theory would, however, be a little more complicated.

the $(1 - \gamma_5)$ factor which causes the massless antineutrino to be right-handed also increases the likelihood that the light-mass electron be left-handed. Consequently, the decay in Figure 24.2 which produces the right-handed electron is suppressed, that is, there is an *anticorrelation* between the electron's momentum and spin. Extensive discussions of these correlations are found in the references.

Beta Decay Spectrum

In our deduction of the weak-interaction Hamiltonian (24.11), we indicated that experiments on radioactive decay determine the constants. As an example of what is measured and how the theory is tested, we calculate the rate for the beta decay of the neutron,

$$n \rightarrow p + e^- + \bar{\nu}_e. \quad (24.14)$$

For simplicity we keep only the scalar part of the Fermi interaction, yet since the decay is possible for both free and bound neutrons, we keep both those possibilities.

We want to calculate the matrix element of the scalar part of the Fermi Hamiltonian (24.8) as illustrated in Figure 24.1A:

$$\tilde{H}_F = g(\bar{\Psi}_p \Psi_n)(\bar{\Psi}_e \Psi_\nu). \quad (24.15)$$

There is one neutron in the initial state, and one proton, one electron, and one antineutrino in the final state. In Fock space the i and f states are

$$|i\rangle = |1_{\phi_i, n}\rangle, \quad |f\rangle = |1_{\phi_f, p}\rangle |1_e\rangle |1_{\bar{\nu}}\rangle, \quad (24.16)$$

where we use $\phi_{n,p}$ to denote the bound-state nucleon wave functions, and we assume momentum eigenstates (plane waves) for the final leptons.²

Exercise Show that for nonrelativistic nucleons, the matrix element of (24.15) is

$$\mathcal{M}_{fi} = \langle pe^- \bar{\nu} | \tilde{H}_F | n \rangle = \frac{g}{V} \int d^3x \phi_{f,p}^*(\mathbf{x}) \phi_{i,n}(\mathbf{x}) e^{-i(\mathbf{p}_e + \mathbf{p}_{\bar{\nu}}) \cdot \mathbf{x}}, \quad (24.17)$$

where the ϕ 's are wave functions for nucleons bound in a nucleus, or plane waves if the neutron is free. $\diamond$

Note that because the weak interaction shown in Figure 24.1A has all four fields acting at one point, in the language of Part I, *Scattering and Integral Quantum Mechanics*, this is a zero-range force. Accordingly, all the wave functions in (24.17) are evaluated at the same point $\mathbf{x}$. Likewise, $\mathcal{M}_{fi}$ is an *inelastic form factor*, that is, a Fourier transform of the overlap of the initial and final wave functions, as discussed in § 5.6. Even though the electron has a kinetic energy of about 1 MeV, it has a relatively low momentum (on the nuclear fm^{-1} scale for momenta):

$$p_e \simeq 1 \text{ MeV} \Rightarrow \frac{p_e}{\hbar} \simeq \frac{1 \text{ MeV}}{\hbar c} \simeq \frac{1 \text{ MeV}}{197 \text{ MeV fm}} \simeq \frac{1}{200} \text{ fm}^{-1}. \quad (24.18)$$

²It is possible to include the distortion of the electron's wave functions caused by the Coulomb field by using the Coulomb waves $F_l(kr, \eta)$ and $G_l(kr, \eta)$ of Chapter 4, *Scattering Applications: Lengths, Resonances, Coulomb*, in place of plane waves.

Because the electron and neutrino have similar momenta, all the momenta in the exponential in (24.17) are small compared with $\mathbf{x}$, and so we ignore the exponential factor (much as in the dipole approximation in Chapter 20, *Quantized Electromagnetic Fields*):

$$\mathcal{M}_{fi} \simeq \frac{g}{V} \int d^3\mathbf{x} \phi_{f,p}^*(\mathbf{x}) \phi_{i,n}(\mathbf{x}). \quad (24.19)$$

In this limit $\mathcal{M}_{fi}$ is just the overlap of the wave functions for the n and p .

Transition Rate

To calculate the decay rate or lifetime τ for beta decay, we apply the golden rule (20.29),

$$\Gamma_{fi} = \frac{1}{\tau} = 2\pi |\mathcal{M}_{fi}|^2 \delta(E_f - E_i). \quad (24.20)$$

Because the e and $\bar{\nu}$ are in the continuum, we sum over all states available to them:

$$\Gamma_{fi} = \sum_{\mathbf{p}_e, \mathbf{p}_{\bar{\nu}}} 2\pi |\mathcal{M}_{fi}|^2 \delta(E_f - E_i) = 2\pi \int \frac{d^3\mathbf{p}_e}{(2\pi)^3} V \frac{d^3\mathbf{p}_{\bar{\nu}}}{(2\pi)^3} V |\mathcal{M}_{fi}|^2 \delta(E_f - E_i). \quad (24.21)$$

One of the integrations in (24.21) is performed with the help of the delta function. To do it we write the delta function's argument in terms of $T_{\max}$, the maximum kinetic energy of the emitted electron:

$$\begin{aligned} E_f - E_i &= E_p + m_e + T_e + E_{\bar{\nu}} - E_n = T_e + E_{\bar{\nu}} - T_{\max}, \\ T_{\max} &= E_n - E_p - m_e, \quad E_{\bar{\nu}} = p_{\bar{\nu}} = E_{\max} - T_e. \end{aligned} \quad (24.22)$$

Here we have ignored the binding energy of the neutron and proton, and have assumed the neutrino mass is much less than the electron mass so $E_{\bar{\nu}} \simeq p_{\bar{\nu}}$.

Exercise Show that for a free neutron $T_{\max} \simeq 0.8$ MeV. Use this to calculate the maximum momentum of the emitted electron. (*Hint:* It is relativistic.) $\diamond$

Exercise Show that if the electron is emitted with maximum kinetic energy of about $\simeq 0.8$ MeV, the maximum kinetic energy of the recoiling proton is approximately 0.0008 MeV (we thus ignore the energy carried off by the recoiling proton). $\diamond$

The integration in (24.21) now gives

$$\begin{aligned} \Gamma_{fi} &= \frac{1}{\tau} = \frac{\pi V^2 g^2 |\langle \phi_f | \phi_i \rangle|^2}{(2\pi)^6 V^2} \int_0^\infty dp_e 4\pi p_e^2 4\pi p_{\bar{\nu}}^2 dp_{\bar{\nu}} \delta(T_e + E_{\bar{\nu}} - T_{\max}) \\ &= \frac{g^2 |\langle \phi_f | \phi_i \rangle|^2}{2\pi^3 V^2} \int_0^{p_{\max}} dp_e p_e^2 (T_{\max} - T_e)^2, \end{aligned} \quad (24.23)$$

where the matrix element is assumed independent of nuclear, electron, and neutrino spin. We leave it to the Problems section to finish the integration, obtain the decay rate, and thus deduce that the strength of the weak coupling constant g is approximately 10^{-7} that of the strong coupling constant; the force responsible for beta decay is in fact weak!

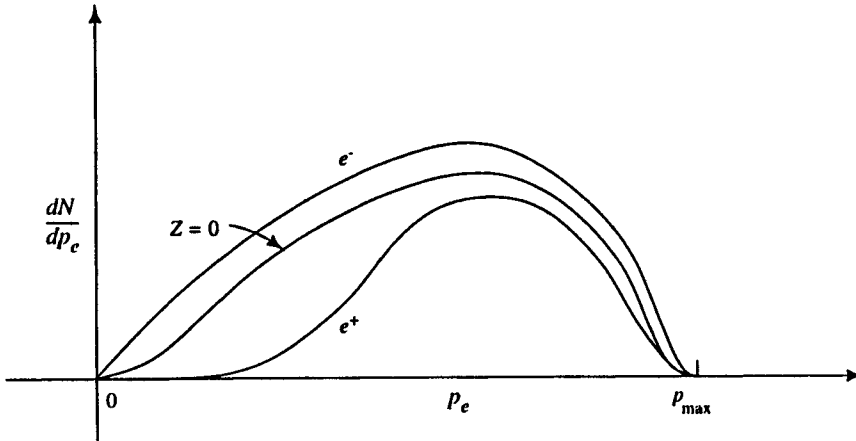


Figure 24.3: A possible momenta spectra for the electrons emitted in beta decay. In the middle curve the Coulomb force does not act, while the other two curves contain an attractive (e^-) and repulsive (e^+) final-state interaction with the residual proton. (Adapted from Blatt and Weisskopf, 1953).

A measurement of the lifetime for a neutron in ϕ_f to decay to a proton in ϕ_i provides us with just one number, a number which is sensitive to normalization factors, Coulomb distortions, and couplings. To obtain more and different information, decay experiments often measure the spectrum, that is, the number of emitted electrons per unit of time, per unit of electron momentum. To obtain a spectrum from (24.23), we suspend the final counting over electron momentum and write:

$$\frac{dN}{dt dp_e} = \frac{g^2 |\langle \phi_f | \phi_i \rangle|^2}{2\pi^3 V^2} p_e^2 (T_{\max} - T_e)^2 \theta(T_{\max} - T_e). \quad (24.24)$$

As seen in Figure 24.3, the spectrum rises from zero for electrons of zero momentum, and continues out to the maximum allowed energy consistent with energy conservation (24.22), at which point the spectrum has dropped to zero.

A good way to test the functional dependence of our predicted spectrum (24.24) (which is really just 3-body phase space with one of the bodies having zero mass) is to plot $\sqrt{[dN/(dt dp_e)]}/p_e^2$ versus T_e . If our prediction is correct, this *Kurie plot* should be a straight line for Fermi transitions (Gamow–Teller matrix elements will not be constant). As we see in Figure 24.4, this simple theory provides an accurate description of the spectrum shape, and confirms our assumption that the mass of the neutrino is essentially zero. Because the kinematics causes the spectrum to vanish, it is possible to deduce the neutrino mass by carefully examining the shape of the *endpoint* of the spectrum.³ The deduced values $m_\nu \leq 46$ eV is controversial, in part because of the simplicity of the theory.

³See Particle Data Group (1994) and its updates for the latest mass values.

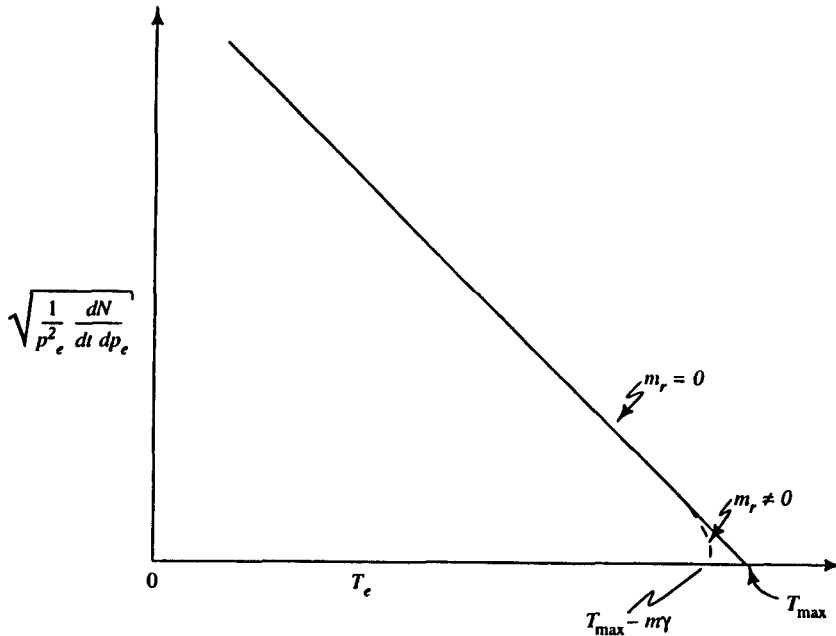


Figure 24.4: A Kurie or Fermi plot of the nondistorted electron spectrum for the beta decay of Figure 24.3. The dashed curve results when the neutrino is given a finite mass.

24.2 Problems

1. Why was the existence of a neutrino predicted before its discovery?
2. The pion decays via the weak interaction into a muon and neutrino:

$$\pi^+ \rightarrow \mu^+ + \nu_\mu. \quad (24.25)$$

(The muon μ is a particle with all the same properties as an electron e , only it is heavier.)

- (a) If the pion is at rest when it decays, what is the approximate momentum of the μ and ν ?
 - (b) Explain in what direction you would expect the neutrino's spin to point.
 - (c) For your answer to the preceding spin direction, in what direction would you expect the muon's spin to point if all final-state particles have zero orbital angular momentum?
3. What is the simplest interaction Hamiltonian which describes the weak process:

$$\mu^- \rightarrow e^- + \nu_\mu + \bar{\nu}_e. \quad (24.26)$$

4. Consider the decay of the free neutron

$$n \rightarrow p + e^- + \bar{\nu}_e. \quad (24.27)$$

How would you observe the violation of parity in this process?

5. The free neutron decays with a half life of ~ 10.6 minutes. Use this observation to estimate (within a factor of ~ 10) the strength g of the weak Hamiltonian. (You can use the fact that the neutron and proton have radii $\simeq 0.75$ fm.)
6. Prove that the $(1 - \gamma_5)$ factor which causes the antineutrino to be right-handed, increases the likelihood that the light-mass electron is left-handed.

CHAPTER 25

WAVE EQUATIONS FROM FIELD THEORY ☉

*If this be error and upon me proved,
I never writ, nor no man ever loved.*

—William Shakespeare

The Klein–Gordon and Dirac equations are fine for a free particle or for a particle interacting with an external potential. If, however, the potential arises from the interaction with another particle which is free to move in spacetime, we do not know what potential or wave equation is needed to satisfy Einstein's two postulates of relativity (Chapter 13, *Relativistic Wave Equations for Spinless Particles*). In contrast, the field-theory amplitudes and diagrams we used in Chapter 23, *The Breit–Pauli and Meson-Exchange Interactions*, to describe the particle–particle interaction via the exchange of a boson, do satisfy the postulates of relativity, but only for the particular order of perturbation theory for which we have calculated the amplitude.

In this chapter we outline how quantum field theory and relativistic potential theory are used to deduce relativistic wave equations for the interaction of two particles. This is a relativistic field theory problem since the particles interact via the creation and destruction of field quanta which take a finite time to propagate through spacetime.

25.1 Bethe–Salpeter Equation

Bethe and Salpeter (1952, 1957) have derived a covariant, quantum-mechanical wave equation describing the relativistic interaction of two particles. In order to be covariant and include the physics of an interacting field propagating between the particles with a speed less than c , the variables in this equation are the four components of the 4-momentum (or 4-coordinate) of each particle. This means there is a new variable, the relative energy (or relative time), which is an extension of the relative 3-momentum or 3-coordinate) used in the nonrelativistic 2-body problem. Although much progress has been made in solving the equations with this new variable, they are still difficult to solve and interpret.

The original derivation of the Bethe–Salpeter equation was rather intuitive, but nevertheless, the same equation obtains from more formal field theory arguments. Here we

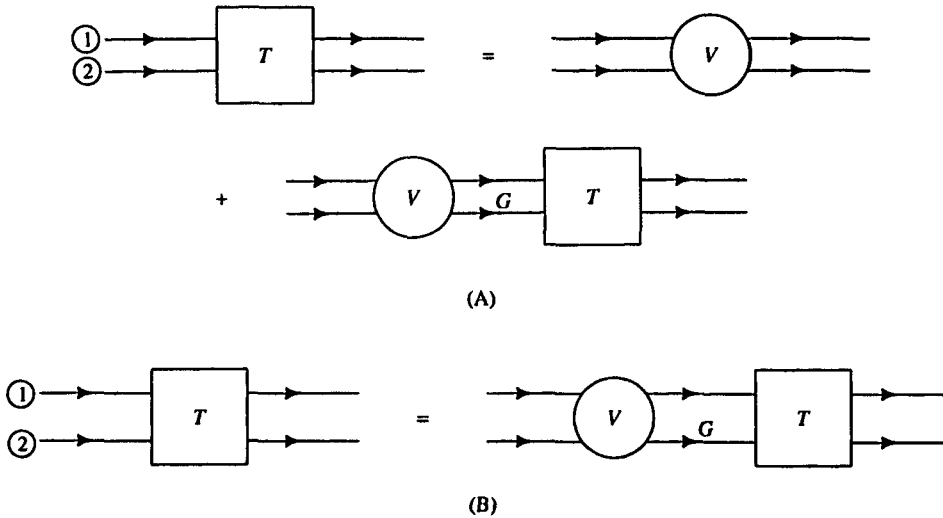


Figure 25.1: The Lippmann–Schwinger equation for (A) scattering and (B) bound states. The box is the transition matrix T , the circle is the interaction potential V , and the internal lines is the Green's function G .

outline its derivation and emphasize how it draws together the techniques of formal quantum mechanics (Part I, *Scattering and Integral Quantum Mechanics*) with those of Part II, *Relativistic Quantum Mechanics*, and Part III, *Quantum Fields*. To keep the analysis simple and emphasize the new physics, we consider only spinless particles; the case with spin yields a more complicated 4D version of the Breit equation (which we are happy to leave to the references).

Deduction

In formal quantum mechanics the Lippmann–Schwinger equation with scattering boundary conditions is

$$T = V + VGT \quad (\text{scattering}). \quad (25.1)$$

This is represented diagrammatically in Figure 25.1A where T is the transition matrix and V the interaction potential. As discussed in § 7.5, the Lippmann–Schwinger equation for bound states,

$$T = VGT \quad (\text{bound states}), \quad (25.2)$$

results by leaving off the homogeneous term in (25.1). This eigenvalue equation is represented diagrammatically in Figure 25.1B.

The Bethe–Salpeter equation has the same operator equations and diagrams as (25.1) and (25.2), but uses the full, 4D spacetime dependence of the potentials and propagators.¹ Additionally, the potential V is considered elementary in Schrödinger and Dirac theories, but the V 's of the Bethe–Salpeter equation are matrix elements (Feynman diagrams) from quantum field theory, explicitly, the 1-boson exchange diagrams, Figure 23.2 or Figure 23.5.

¹The relativistic Schrödinger equation of Chapter 18, *Solving Even Relativistic Integral Equations*, uses relativistic kinematics in G , but not relativistic dynamics.

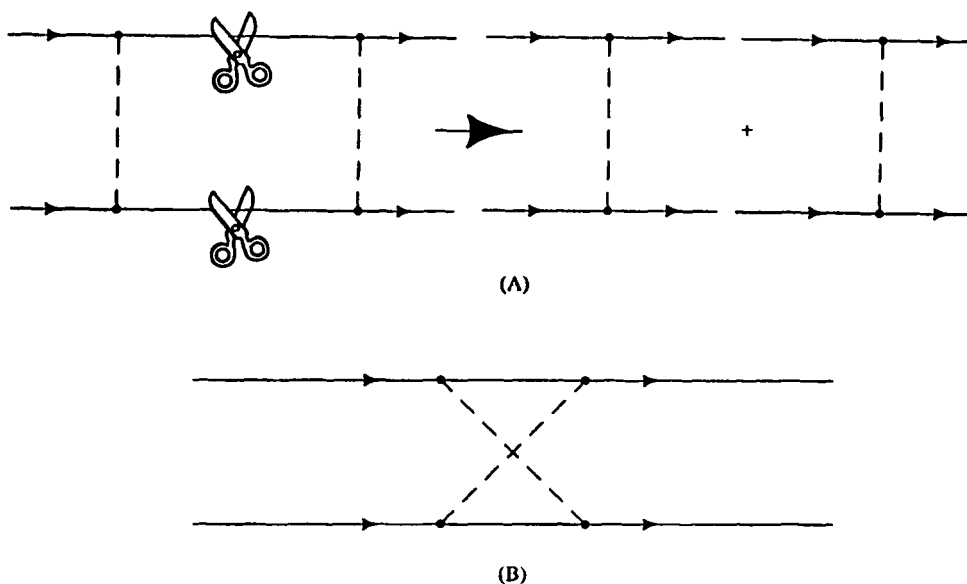


Figure 25.2: (A) The reduction of a reducible graph into two irreducible one by cutting internal lines; (B) an irreducible graph which cannot be cut and divided into two simpler ones (twisting not allowed).

Yet there is the restrictions that V cannot include ladder diagrams such as Figure 22.2C, because they will get generated automatically when V is iterated in (25.1). To ensure that there is no “double counting,” we require V to include all Feynman diagrams which cannot be reduced into two *disconnected diagrams* by cutting two internal lines (we ignore complications arising from self-energy corrections). For example, Figure 25.2A is reducible, while 25.2B is not (B is thus part of the potential).

If the potential arose from single-boson exchange, use of it in (25.1) would generate the series in Figure 25.3, that is, a “ladder” graph. Because the electromagnetic interaction and the long-range part of the nucleon–nucleon interaction are described well by one-boson-exchange, in these cases the ladder approximation may be good.

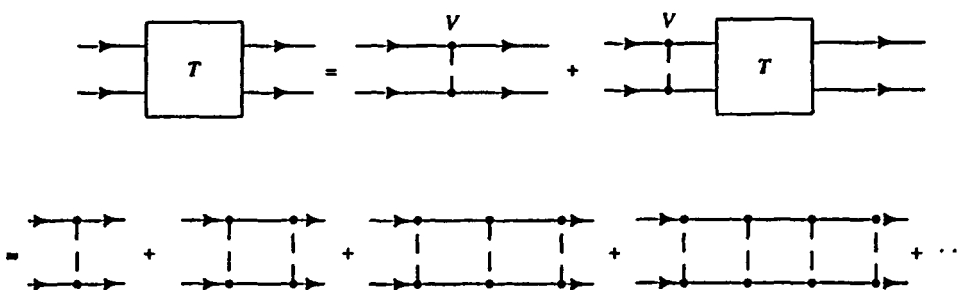


Figure 25.3: The Bethe–Salpeter equation in the ladder approximation with the potential generated by boson exchange (dashed line). The upper diagram is the exact equation, the lower diagram its Born series.

In Coordinate Space

We know from our study of formal quantum mechanics in Part I, *Scattering and Integral Quantum Mechanics*, that the distorted (full) wave function satisfies the Lippmann–Schwinger equation:

$$\psi = \phi + GV\psi. \quad (25.3)$$

This is equivalent to Figure 25.1A. Likewise, the bound-state wave function satisfies the homogeneous equation:

$$\psi = GV\psi, \quad (25.4)$$

and this is equivalent to Figure 25.1B. The Bethe–Salpeter representation of (25.4) in spacetime is

$$\psi(x_1, x_2) = \int d^4x'_1 d^4x'_2 d^4x''_1 d^4x''_2 G(x_1, x_2; x'_1, x'_2) V(x'_1, x'_2; x''_1, x''_2) \psi(x''_1, x''_2). \quad (25.5)$$

Here G is the Green's function for two free particles, and because free particles move independently, is the product of two single-particle G 's:

$$G(x_1, x_2; x'_1, x'_2) = G_1(x_1 - x'_1) G_2(x_2 - x'_2). \quad (25.6)$$

The use of 4-vectors in these G 's is a generalization to spacetime.² As introduced in Chapter 5, *Green's Functions and Integral Quantum Mechanics*, the one-particle Green's function G_i is the “unit impulse response function,” in this case, the solution of the Klein–Gordon equation for a point source:

$$(p_1^2 - m_1^2) G_1(x_1 - x'_1) = \delta^{(4)}(x_1 - x'_1), \quad (25.7)$$

$$(p_2^2 - m_2^2) G_2(x_2 - x'_2) = \delta^{(4)}(x_2 - x'_2), \quad (25.8)$$

$$p_j = (i\partial_t, -i\nabla_j). \quad (25.9)$$

Exercise Show that in momentum space, the Klein–Gordon, single-particle G has the representation

$$G(x) = \int \frac{d^4p}{(2\pi)^4} \frac{e^{-ip \cdot x}}{p^2 - m^2 + i\epsilon}. \quad \diamond \quad (25.10)$$

The generalization (25.6) is transparent. Each particle propagates with its own time coordinate and its own momentum. So (like the railroads) the interacting particles each move with different internal times yet occasionally interact at junctions.

Exercise Show that having the differential operators act on both side of (25.5) (as we did for the Schrödinger equation) leads to the doubly differential equation:

$$(p_1^2 - m_1^2)(p_2^2 - m_2^2)\psi(x_1, x_2) = V(x_1 - x_2)\psi(x_1, x_2). \quad \diamond \quad (25.11)$$

Equation (25.11) may look simple, yet it is a partial differential equation in eight variables including the time coordinates for each particle. Considerable progress has occurred in

²The Bethe–Salpeter equation for particles with spin replaces each particle's Klein–Gordon G with the Dirac G studied in Chapter 17, *Scattering and Dirac Integral Equations*.

solving this equation (Bethe and Salpeter, 1952, 1977; Roman, 1969; Schweber, 1959; Landau and Lifshitz, 1971; Brown and Jackson 1976), yet problems still remain in regard to normalizations, nonrelativistic limits, uniqueness, and especially the choice of relative times for the particles.

In Momentum Space

We return to the Lippmann–Schwinger equation (25.1) and consider it as the operator representation of the set of diagrams in Figure 25.1A. In the ladder approximation (Figure 25.3), the potential is generated by the exchange of a boson of mass μ and (as seen in the last chapter) has the covariant form

$$V(p, q) = \frac{g^2}{(p - q)^2 + \mu^2 + i\epsilon}. \quad (25.12)$$

Here p and q are the 4-momenta of the interacting particles and g is the coupling constant. The 4-momentum representation of the Bethe–Salpeter equation (25.1) is the integral equation,

$$T(p, q) = V(p, q) + i \int \frac{d^4 k}{(2\pi)^4} V(p, k) G(P, k) T(k, q). \quad (25.13)$$

The normalization here is such that

$$S_{fi} = \delta_{fi} + (2\pi)^4 i \delta^4(P_f - P_i) T_{fi}, \quad (25.14)$$

$$SS^\dagger = S^\dagger S = 1, \quad \langle p | p' \rangle = (2\pi)^4 \delta^4(p - p'). \quad (25.15)$$

The variables in (25.13) are natural generalizations of those in Chapter 1, *Scattering*, and are used as labels in Figure 25.4. Here P is the total 4-momentum of the system (a constant), while p , k , and q denote the relative 4-momentum in the initial, intermediate, and final states:

$$P = p_1 + p_2, \quad p = \frac{1}{2}(p_1 - p_2). \quad (25.16)$$

The Mandelstam variable s is the square of the total 4-momentum,

$$s = P^2 = (E_1 + E_2)^2 - (\mathbf{p}_1 + \mathbf{p}_2)^2, \quad (25.17)$$

and is conserved during interactions. Note that because this is a relativistic theory in which 4-momentum is conserved at each step, the total (mass plus kinetic) energy of the particles is conserved even during interactions, yet their masses, $m^2 \stackrel{\text{def}}{=} E^2 - p^2$, vary. This is off-mass-shell scattering, in contradistinction to the off-energy-shell scattering of the nonrelativistic theory in which the mass never changes but kinetic energy is not conserved.

Exercise Show that the energies E_i and E'_i for particle $i = 1, 2$ in Figure 25.4 are given in terms of the relative energies p_0 and k_0 and total energy $\sqrt{s}$ by

$$\begin{aligned} E_1 &= \sqrt{s}/2 + p_0, & E'_1 &= \sqrt{s}/2 + k_0, \\ E_2 &= \sqrt{s}/2 - p_0, & E'_2 &= \sqrt{s}/2 - k_0. \end{aligned} \quad \diamond \quad (25.18)$$

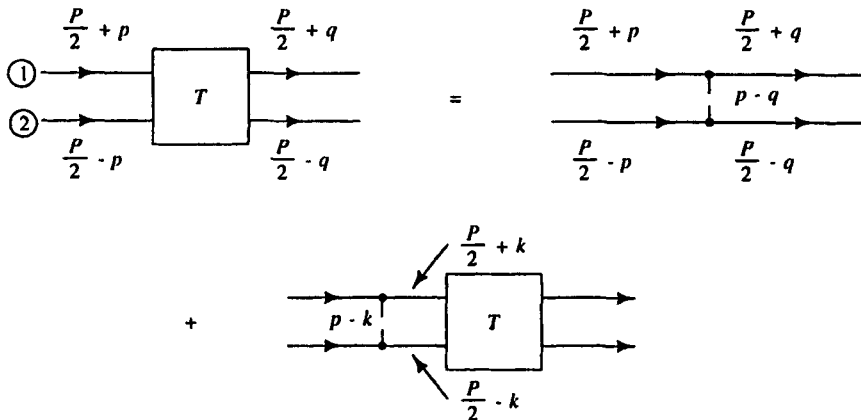


Figure 25.4: Relative momenta, p and q , and total 4-momenta P entering into the Bethe-Salpeter equation (25.13).

As follows from the definitions of the Green's function in coordinate space, (25.6)–(25.10), the momentum-space Green's function is the product of two single-particle G 's:

$$G(P, k) = \frac{1}{[(\frac{1}{2}P + k)^2 - m_1^2 + i\epsilon][(\frac{1}{2}P - k)^2 - m_2^2 + i\epsilon]}. \quad (25.19)$$

Because each individual G has an on-shell pole, the product (25.19) has a more complicated analytic structure. In some sense, the Green's function gives the Bethe-Salpeter equation its relativistic character, as we will see shortly.

Exercise Show that the 4D single-particle Green's function $G_i = 1/[p_i^2 + m^2]$ agrees with nonrelativistic form in the appropriate limit. $\diamond$

Properties

All the different forms of the Bethe-Salpeter equations (25.5), (25.11), and (25.13) are covariant and have “good” analytical properties (that is, typical of a wave equation). Consequently, they produce scattering amplitudes which are analytic, covariant, and contain the sum of an infinite number of diagrams. Because in practice there is no general way to sum all two-particle irreducible diagrams (those that should be in V), there may be a concern that the terms left out of V are still important. The hope is that these higher-order terms, which involve larger masses being exchanged than the lower-order terms, have their main influence at correspondingly small r . If we thus exclude very-small- r phenomena (possibly by including the finite size of hadrons to provide a natural small- r cutoff), we should have an improvement on nonrelativistic quantum mechanics and on single-particle relativistic quantum mechanics. Of course, if small- r effects are important, then the quark degrees of freedom should also be.

Aside from the difficulties of understanding the meaning of the relative time for two particles, a more mundane problem with the Bethe-Salpeter equation is the difficulty in solving it, even after partial-wave decomposition (it decomposes into two-dimensional equations with both relative energy and momentum as variables). In addition, there are

theoretical concerns with the Bethe–Salpeter equation containing multiparticle states such as those in Figure 25.3, because these interfere with analyticity and may be unphysical for a professed two-particle equation. In the next section we show the equation which results from eliminating the relative-time coordinate.

25.2 Blankenbecler–Sugar Equation

To help understand the content of the Bethe–Salpeter equation and obtain an equation which is more readily solved, we convert it into a 3D integral equation. First we limit ourselves to the CM where the variables in Figure 25.4 assume the simple form

$$P = \sqrt{s}, \quad E_1 = \sqrt{p_1^2 + m_1^2}, \quad E_2 = \sqrt{p_2^2 + m_2^2}. \quad (25.20)$$

Here P is the total momentum, p_1 and p_2 are (on-shell) relative momenta, and k_i is the (off-shell) intermediate momenta. Since the intermediate-state particles in Figure 25.4 are off the mass shell (they are interacting), we do not know how the energy and momentum are related, that is, we do not have free-particle kinematics,

$$E_i(k) \neq \sqrt{k_i^2 + m_i^2}. \quad (25.21)$$

With the variables (25.20), we reduce the Bethe–Salpeter equation (25.13) to three dimensions by utilizing the poles of the Green's function (25.19) to evaluate the integral.³ We observe that because the two-particle Green's function $G(P, k)$ is the product of two single-particle ones, it contains the product of the two on-shell delta functions, and they can be expressed as

$$\delta\left[\left(\frac{P}{2} + k\right)^2 - m_1^2\right]\delta\left[\left(\frac{P}{2} - k\right)^2 - m_2^2\right] = \frac{\delta(k_0 + \frac{\sqrt{s}}{2} - E_1)\delta(\sqrt{s} - E_1 - E_2)}{4E_1E_2}. \quad (25.22)$$

This leads to

$$\begin{aligned} G(P, k) &= \frac{\pi}{E_1E_2}\delta(k_0 - \frac{E_1}{2} + \frac{E_2}{2})\frac{E_1 + E_2}{(E_1 + E_2)^2 - s} \\ &= \frac{\pi}{E_1E_2}\delta(k_0 - \frac{E_1}{2} + \frac{E_2}{2})\left[\frac{1}{(E_1 + E_2) - \sqrt{s}} + \frac{1}{(E_1 + E_2) + \sqrt{s}}\right]. \end{aligned} \quad (25.23)$$

This expression for G is illuminating. It separates the singularities into a Schrödinger-like pole and another one at negative energies (arising from antiparticle degrees of freedom).

Exercise Show that the substitution of (25.23) into the Bethe–Salpeter equation (25.13) leads to the *Blankenbecler–Sugar equation*:

$$\begin{aligned} T(p, q) &= V(p, q) + \int \frac{d^3k}{(2\pi)^3} V(p, k) \left(\frac{1}{E_1 + E_2 - \sqrt{s}} + \frac{1}{\sqrt{s} + E_1 + E_2} \right) \frac{T(k, q)}{2E_1E_2} \\ &= V(p, q) + \int \frac{d^3k}{(2\pi)^3} V(p, k) \frac{E_1 + E_2}{[(E_1 + E_2)^2 - s]} \frac{T(k, q)}{2E_1E_2}. \quad \diamond \end{aligned} \quad (25.24)$$

³Unfortunately, the procedure is not unique and there are numerous ways to express G and, subsequently, numerous 3D covariant equations.

An alternate and illuminating approach to reducing the Bethe–Salpeter equation to three dimensions, is to choose k_0 and p_0 , the time components of the relative momentum, so that the 2-particle interaction potential $V(p, k)$ is independent of them. This makes it clear that retardation effects are being ignored. Because V is then independent of relative momenta, the k_0 integration is over only G and can be done easily. To be more specific, we write the one-boson-exchange potential (25.10) in terms of spacetime variables,

$$V(p, k) = \frac{g^2}{(p_0 - k_0)^2 - (\mathbf{p} - \mathbf{k})^2 + \mu^2 + i\epsilon}. \quad (25.25)$$

With this form it is clear that setting the relative energies equal, $p_0 = k_0$, makes V independent of any relative energy.

Setting the relative energies of the two interacting particles equal appears reasonable if both particles are of equal mass and are in the CM ($\mathbf{P} = 0$). With these symmetries it follows from the definition of relative 4-momentum (25.18) that $p_0 = k_0 = 0$. Whereas it is clear that for this highly symmetrical situation that the relative energy never changes and so the particles have no relative time difference, the dependence of the potential on relative time for unequal masses and other reference frames is not obvious. Yet even for unequal masses, we can see the approximation in obtaining the Blankenbecler–Sugar equation. Solving (25.18) for the relative momenta, we obtain:

$$p_0 = E_1 - \frac{\sqrt{s}}{2} = E_1 - \frac{E_1 + E_2}{2} = \frac{E_1 - E_2}{2}, \quad (25.26)$$

$$k_0 = E'_1 - \frac{\sqrt{s}}{2} = E'_1 - \frac{E'_1 + E'_2}{2} = \frac{E'_1 - E'_2}{2}, \quad (25.27)$$

$$\Rightarrow p_0 - k_0 = \frac{1}{2}[(E_1 - E'_1) - (E_2 - E'_2)]. \quad (25.28)$$

If we ignore retardation and propagations effects here, that is, the time it takes for the boson in Figure 25.4 to travel from particle 1 to 2, then $p_0 = k_0$ and the energy lost by particle 1 equals to the energy gained by 2. So again, ignoring finite propagation speed leads to $V(p, k)$ being independent of the relative energy k_0 .

Once the potential V is made independent of k_0 , the time integration in (25.13) is converted to a contour integration over just G :

$$i \int \frac{d^4 k}{(2\pi)^4} V(p, k) G(P, k) T(k, q) = i \int \frac{d^3 k}{(2\pi)^3} V(p, k) T(k, q) \int \frac{dk_0}{2\pi} G(P, k). \quad (25.29)$$

The relative-energy integral is evaluated using the expression (25.19) for G :

$$\int \frac{dk_0}{2\pi} G(P, k) = i \left[\frac{-1}{\sqrt{s} - E_1 - E_2} + \frac{1}{\sqrt{s} + E_1 + E_2} \right] \frac{1}{4E_1 E_2}, \quad (25.30)$$

and the Bethe–Salpeter equation becomes the Blankenbecler–Sugar equation (25.24).

Examination of the Blankenbecler–Sugar equation reveals it to be a relativistic generalization of the Schrödinger equation appropriate for two particles interacting via the exchange of a boson. While not unique, it is covariant, it conserves unitarity, and it is only three-dimensional (so the numerical techniques of Chapter 18 work for it). The Blankenbecler–Sugar equations does not, however, contain all the physics present in the full field theory treatment of two interacting particles, in particular, it has problems with unitarity and the negative-energy degrees of freedom (crossing). Nevertheless, it is used quite often, particularly in few-body physics.

25.3 Problems

1. Consider the 4-momentum-space potential arising from the exchange of a massive boson,

$$V(q) = \frac{g^2}{q^\mu q_\mu - \mu^2}. \quad (25.31)$$

Show how this potential exhibits the finite propagation time for fields.

2. Deduce a solution to the Bethe–Salpeter equation for noninteracting particles.
3. Deduce the spacetime form for the Bethe–Salpeter Green's function. How does this G exhibit finite signal speed?
4. Deduce how the T matrix and scattering amplitude are related in the Blankenbecler–Sugar equation. Show that our previous nonrelativistic expression can be generalized by using total energies in place of masses in the flux and phase-space factors.

PART IV

MANY-BODY THEORY

CHAPTER 26

MANY-BODY PROBLEMS

26.1 General Ideas

We have developed techniques to solve with style the Schrödinger equation for bound and scattering states. Although this may build confidence in our powers, we should keep in mind that we could solve the 2-particle problem only because we could reduce it to a problem in which an effective particle interacts with an effective external potential. Whereas it is routine to go beyond that and solve 3-body problems, even here the solutions are usually numerical and for restricted classes of potentials. For more than three interacting particles, exact solutions are usually not attempted, and instead approximate (yet powerful) many-body techniques are used. We discuss some of these techniques in this part of the book. The discussion of many-body theories is meant as an introduction to the subject and is deliberately less rigorous than our discussion of scattering theory. The aim is to make more specialized discussions more understandable and to show the applicability and unity of our theoretical framework. Further discussion can be found in Messiah (1961), Bethe and Jackiw (1968), Baym (1969), Fetter and Walecka (1971), Weissbluth (1980), deShalit and Feshbach (1974), Preston and Bhaduri (1975), and Koltun and Eisenberg (1988).

Given a “snapshot” of a many-body system at one instant in time which looks like Figure 26.1, we need to figure out a way to solve for the wave function of particle i . To be specific, we take the particles to be N electrons bound within an atom, and the particle–particle interaction to be the Coulomb potential (the theory is also useful for fermions in solids, atoms, or nuclei). Even though each electron–electron interaction $v(e, e)$ and electron–nucleus interaction $\Phi(e, Z)$ is a Coulomb potential, the potential felt by the electron i at this instant,

$$V_i(\mathbf{r}_1, \mathbf{r}_2, \mathbf{r}_3, \dots) = \sum_{j(\neq i)=1}^N v(e_i, e_j) + \Phi(e_i, Z) \quad (26.1)$$

$$= \sum_{j(\neq i)=1}^N \frac{e^2}{|\mathbf{r}_i - \mathbf{r}_j|} - Z \frac{e^2}{r_i}, \quad (26.2)$$

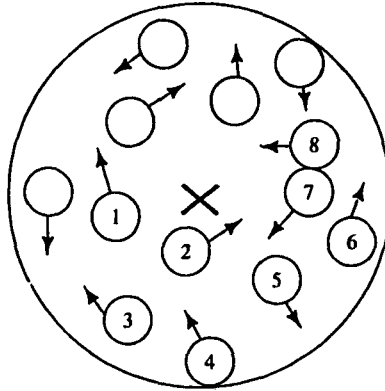


Figure 26.1: A bound many-body system. The X marks the center of momentum which, in a proper theory, does not move as the particles interact.

is neither $1/r_i$ nor spherical. Because electron i 's potential depends on the position of all other particles, it is not truly possible to solve for the motion of one electron without solving for the motion of all, that is, solving the many-body Schrödinger equation

$$\left[\sum_i^N K_i + V_i(\mathbf{r}_1, \mathbf{r}_2, \mathbf{r}_3, \dots) \right] \Psi(\mathbf{r}_1, \mathbf{r}_2, \mathbf{r}_3, \dots) = E\Psi(\mathbf{r}_1, \mathbf{r}_2, \mathbf{r}_3, \dots). \quad (26.3)$$

Some of the art of many-body theory is developing approximate procedures that work for all electrons simultaneously. Indeed, because all of the electrons in the atom are identical fermions, the exact wave function must be totally antisymmetric under the interchanged coordinates of any two, and so the proper solution for any one electron must be proper for all.

Exercise Show that if the potential (26.2) were the sum of single-particle operators, the solution of the Schrödinger equation would separate into the product of single-particle functions. $\diamond$

By applying the preceding exercise, we see that a potential such as (26.2), that is not the sum of single-particle operators, produces a wave function that cannot be written as the product of single-particle wave functions. Accordingly, attempts to do so can only be approximations.

26.2 Hartree Approximation

An intuitive and useful approximation to the many-fermion problem was suggested by Hartree in 1928. The motivation behind this approach arises in the observation that because there are bound states in nature in which electrons are in eigenstates, we should be able to describe a single electron i in state α with a single-particle, bound-state wave function $\phi_\alpha(\mathbf{r}_i) \equiv \phi_\alpha(\mathbf{r})$. Correspondingly, there must be an effective, single-particle

potential $V_a(\mathbf{r})$ whose use in the 1-body Schrödinger equation,

$$\left[\frac{-\nabla^2}{2m} + V_a(\mathbf{r}) \right] \phi_a(\mathbf{r}) = \epsilon_a \phi_a(\mathbf{r}), \quad (a = 1, N), \quad (26.4)$$

exactly reproduces $\phi_a(\mathbf{r})$.¹ Because $V_a(\mathbf{r})$ must contain effects due to the other electrons which, in turn, are influenced by i , this potential must somehow be determined simultaneously with the solution for all other $\phi_b(j)$'s. Hartree hypothesized that the single-particle potential is approximately:²

$$V_a(\mathbf{r}) \simeq \int d^3r' \frac{e^2}{|\mathbf{r} - \mathbf{r}'|} \sum_{b \neq a}^N |\phi_b(\mathbf{r}')|^2 - Z \frac{e^2}{r}, \quad (26.5)$$

where we take $\mathbf{r}$ as the coordinate for particle i and $\mathbf{r}'$ as the (dummy) coordinate for the other single-particle wave functions. Relation (26.5) states that the single-particle potential for electron i in state a is the average over the coordinates of all the other particles, with the probability of finding electron j in state b at $\mathbf{r}_j$ given by $|\phi_b(\mathbf{r}_j)|^2$.

Often in applications of many-body theories, some numerical complications are avoided by using the *central-field approximation*:

$$V_a(\mathbf{r}) \simeq V_a(r) = \frac{1}{4\pi} \int d\Omega_r V_a(\mathbf{r}), \quad (26.6)$$

that is, using an angle-averaged potential. Because we know from atomic theory that filled shells are spherically symmetric, approximation (26.6) has a good theoretical basis (see, e.g., Condon & Shortly, 1951).

It is now straight forward to solve the Schrödinger equation (26.4) with the partial-wave techniques developed in the earlier chapters. Because all of the ϕ_b 's are needed to calculate V_a , (26.5) and (26.6) are N integro-differential equations. Analytic solutions to them is usually impossible for realistic cases, but they can be solved numerically via *self-consistent iteration*. First we assume a convenient basis, for example, the hydrogen solutions:

$$\phi_a(\mathbf{r}) = \frac{R_{nl}(r)}{r} Y_{lm}(\Omega). \quad (26.7)$$

In Hartree's approximation, the wave function Ψ for the complete N -electron system is assumed to be a simple product of these single-particle states with no two electrons in the same state:³

$$\Psi(1, 2, \dots, N) = \phi_a(1) \phi_b(2) \cdots \phi_{A_N}(N), \quad (26.8)$$

where the subscripts on ϕ denote the N different states needed for N fermions. Relations (26.5) and (26.6) are now used to generate the effective potential V_a arising from this Ψ , and the N differential equations (26.4) are then solved with V_a . The improved wave functions are then used to determine an improved V_a , which, in turn, determines improved wave functions, and so forth. When the wave functions and potentials become self-consistent, that is, to some level of precision the two stop changing, the approximation scheme is considered convergent and successful.

¹ This is the same view taken for the optical potential in nuclear reactions and for the shell model for atomic and nuclear structure.

² It also follows more rigorously from a variation principle.

³ The Hartree method can also be thought of as finding the "best" product wave function (in a variational sense) for a many-body system. Yet as noted in § 26.1, since the particles interact, the product form cannot be exact.

26.3 Correlations and Determinantal Wave Functions

When all the fermions in a many-body system are identical, the wave function must be antisymmetric under the interchange of the coordinates of any two. Because this requirement imposes a relationship among the coordinates of all the particles in the system, it means these coordinates must be *correlated*. (Recall from § 10.4 that the odds favor the two fermions being in a spin-symmetric state, in which case the exclusion principle requires them to be in a space-antisymmetric state, that is, anticorrelated.) These *Pauli exclusion principle*, or *exchange* correlations tend to be repulsive and to have a range comparable to the size of the bound system, while *dynamical correlations* [those arising from particle-particle forces not included in $V_a(\mathbf{r})$] usually are short ranged and can be attractive or repulsive. (For nucleons in nuclei as well as electrons in superconductors, there are dynamic short-range attractive correlations and long-range repulsions—though the origins of the interactions are quite unrelated.)

Hartree's simple product wave function is improved by including Pauli correlations. This is accomplished by using a Slater determinant for the many-fermion wave function:

$$\Psi = \frac{\det}{\sqrt{N!}} \begin{pmatrix} \phi_a(1) & \phi_a(2) & \cdots & \phi_a(N) \\ \phi_b(1) & \phi_b(2) & \cdots & \phi_b(N) \\ \vdots & \vdots & \ddots & \vdots \\ \phi_{A_N}(1) & \phi_{A_N}(2) & \cdots & \phi_{A_N}(N) \end{pmatrix} \quad (26.9)$$

$$= \frac{1}{\sqrt{N!}} \sum_P (-1)^P \prod_{i=1}^N \phi_{PA_i}(i) = \frac{1}{\sqrt{N!}} \sum_P (-1)^P \prod_{i=1}^N \phi_{A_i}(Pi). \quad (26.10)$$

Here the subscripts on the ϕ 's (the rows) indicate orbitals (which can contain spin wave functions), the arguments (the columns) indicate particles, and the sum is over all possible permutations P of the orbitals (particles), with the sign depending on whether the permutation is odd or even.⁴

Exercise Confirm that all three relations (26.9)–(26.10) predict the determinantal wave function for two fermions to be

$$\Psi(1, 2) = \frac{\det}{\sqrt{2}} \begin{pmatrix} \phi_a(1) & \phi_a(2) \\ \phi_b(1) & \phi_b(2) \end{pmatrix} = \frac{\phi_a(1)\phi_b(2) - \phi_b(1)\phi_a(2)}{\sqrt{2}}. \quad \diamond \quad (26.11)$$

Exercise Show that the theorems for determinants lead to a wave function (26.9) that is antisymmetric under the interchange of the coordinates or orbitals of any two particles. $\diamond$

Because the potentials differ at each stage of a self-consistent calculation, the ϕ_a 's are not guaranteed to be orthogonal to each other. This can be a problem. It is corrected by adding orthogonality as a constraint on the wave functions, which means forming linear combinations of the ϕ_a 's that are orthogonal. Because the theorems of linear algebra tell us that we can add a constant times one row of a determinant to another row without changing

⁴In high-precision applications, the number of orbitals used may be larger than the number of particles present, i.e., partial filling of orbitals is possible. We ignore such complications here.

the value of the determinant, the constraint does not change the value of the total wave function Ψ , the binding energy of the system, or the form of the subsequent Hartree–Fock equations.

Correlation Function

If two particles in a system are correlated, the many-body wave function differs from the product of single-particle functions. For example, the Slater determinant (26.9) containing exclusion or exchange correlations is not a simple product. Let us call $\rho(\mathbf{r}_1, \mathbf{r}_2)$ the joint probability for finding one particle at $\mathbf{r}_1$ and another at $\mathbf{r}_2$. Ideally, this is obtained from the many-body wave function by integrating out all coordinates from the total probability:

$$\rho(\mathbf{r}_1, \mathbf{r}_2) = \int d^3r_3 d^3r_4 \cdots d^3r_N |\Psi(\mathbf{r}_1, \mathbf{r}_2, \mathbf{r}_3 \cdots \mathbf{r}_N)|^2 \delta\left(\frac{1}{N} \sum_i \mathbf{r}_i - \mathbf{R}_{\text{CM}}\right), \quad (26.12)$$

where the delta function is included to eliminate the overall motion of the CM (that is, to keep $\mathbf{X}$ in Figure 26.1 stationary as the particles move), and where we leave out any spin degrees of freedom.⁵

Exercise Calculate $\rho(1, 2)$ for the two-fermion wave function (26.11). ◇

For determinantal wave functions, the two-body density has a particularly transparent form. If the fermions are in any two states a and b , then

$$\rho(\mathbf{r}_1, \mathbf{r}_2) = \sum_{a,b} \rho(a, b) \quad (26.13)$$

$$= \sum_{a,b} |\phi_a(1)|^2 |\phi_b(2)|^2 - \text{Re} \sum_{a,b} \phi_a(1) \phi_b(2) \phi_a^*(2) \phi_b^*(1). \quad (26.14)$$

The first term in (26.14) is clearly the product of single particle densities, while the second is the correlation term arising from antisymmetrization.

For other systems, the correlation term is sometimes separated from the product of densities:

$$\rho(\mathbf{r}_1, \mathbf{r}_2) = \rho(\mathbf{r}_1) \rho(\mathbf{r}_2) [1 - C(\mathbf{r}_1, \mathbf{r}_2)]. \quad (26.15)$$

Here C is the *2-body correlation function* and is zero for no correlations, 1 for complete anticorrelations, and -1 for complete (attractive) correlations.

In the problems at the end of the chapter the student is asked to determine this correlation function for two cases. The first is for a gas of fermions in an infinite box. In this case the ϕ 's are plane waves, and C is:

$$C(\mathbf{r}_1, \mathbf{r}_2) = C(\mathbf{r}_1 - \mathbf{r}_2) = \left(3 \frac{j_1(k_f r)}{k_f r}\right)^2, \quad j_1(x) = \frac{\sin x - x \cos x}{x^2}, \quad (26.16)$$

⁵In the calculation of the states of few-body systems, spurious states associated with overall motion of the CM can contaminate the spectrum unless the CM is constrained to remain fixed. This is the reason for the delta function in (26.12). As the system gets heavier and heavier, the contamination become less significant.

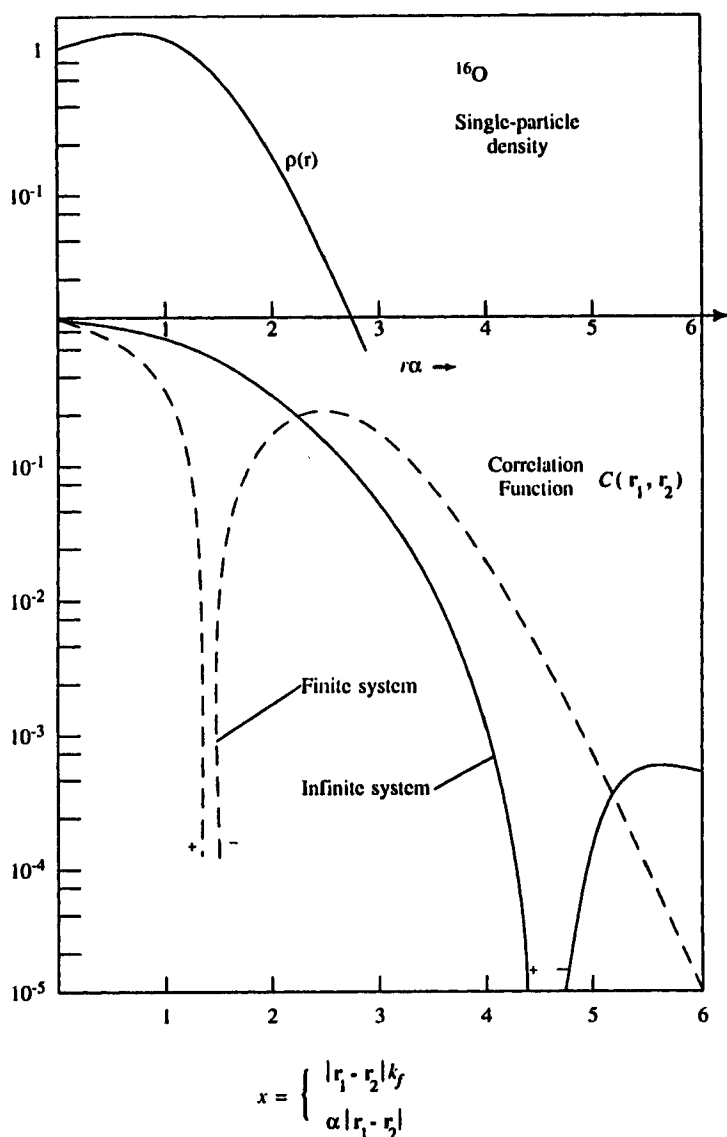


Figure 26.2: (Top): The single-particle density for 16 bound fermions; (Bottom): The exclusion-principle correlation function within a finite (dashed curve) and infinite (solid curve) system of fermions. *Note:* The logarithmic scale means the oscillations occur for small C and $\rho(r)$.

with $\tau = |\mathbf{r}_1 - \mathbf{r}_2|$ being the interparticle distance and k_f the *Fermi momentum* of the electron gas.⁶ As shown by the solid curve in Figure 26.2, this correlation function is mainly positive because the exclusion principle is an effective repulsion between fermions. It does oscillate around zero as the particle–particle distances increase, yet this is in the region where C has dropped three orders of magnitude. Accordingly, the fermions are kept apart for short distances, while for distances beyond several k_f^{-1} 's there is little effect. Consequently, particle–particle collisions at short distances are rare in fermion matter, and because the strongest part of the interaction is often at short distances, the effective force between the fermions is much less than in free space. Thus it is often a good approximation to treat bound electrons or nucleons as experiencing only a relatively weak central potential. For example, if a nucleon in ^{16}O felt the cumulative sum of 16 nucleon–nucleon potentials, it would feel a potential of ~ 1400 MeV; yet empirically it is known that the effective potential felt by a nucleon in the middle of a nucleus is only ~ 40 – 50 MeV deep.

The preceding properties of the correlation function also hold for finite systems. The second correlation function to be worked out in the Problems section (and shown by the dashed curve in Figure 26.2) is for a system of 16 fermions filling the first S and first P states of a 3D harmonic oscillator. If the CM of the two fermions is kept at the origin ($\mathbf{R} \equiv 0$), the correlation function is

$$C(\mathbf{r}_1, \mathbf{r}_2) = \left(1 - \frac{1}{2}\beta^2\tau^2\right)^2 e^{-\beta^2\tau^2/2}, \quad (26.17)$$

where again $\tau = |\mathbf{r}_1 - \mathbf{r}_2|$. As we see in Figure 26.2, even in finite systems the close approach of the two fermions is severely reduced [$C(0) = 1$]; yet this anticorrelation becomes small for distances on the order of the size $\sim \beta^{-1}$ of the bound system.

26.4 Hartree–Fock Equations

While it is intuitively obvious that the use of a determinantal wave function should improve Hartree theory (it puts in more quantum mechanics), the justification of Hartree–Fock theory follows from the Rayleigh–Ritz *variational principle* (Merzbacher, 1970, Gottfried, 1966) which states that the correct ground-state wave function minimizes the ground state energy E . The ground state energy is calculated as the expectation value of the many-body Hamiltonian between the many-body, ground-state wave function, and its variation vanishes:

$$\delta E = \delta \langle \Psi^* | H | \Psi \rangle = \delta \int d\tau \Psi^* H \Psi = 0. \quad (26.18)$$

Here $d\tau$ is the many-body volume element, Ψ is the ground-state wave function, and H is the exact many-body Hamiltonian,

$$H = \sum_{i=1}^N K_i + \sum_{i=1}^N \Phi(\mathbf{e}_i, Z) + \sum_{i < j}^N v(\mathbf{e}_i, \mathbf{e}_j). \quad (26.19)$$

The variation is on the wave function, $\Psi \rightarrow \Psi + \delta\Psi$, until a minimum in the ground-state energy is reached. For determinantal wave functions, the variation is over the individual single-particle states $\phi_a(\mathbf{r}_i)$, with the subsidiary constraint that the orbitals remain

⁶We review plane waves in Appendix A, *Natural Units and Plane Waves*, and further develop the Fermi gas model in the next chapter.

orthonormal:

$$\int d^3r \phi_a^*(\mathbf{r}) \phi_b(\mathbf{r}) = \delta_{a,b}. \quad (26.20)$$

If spin is included, a delta function arising from the spin wave function also appears in (26.20).

Two approaches are used to obtain explicit equations from (26.18). One (described by Baym, 1969) employs the second-quantization techniques discussed in Chapter 19, *Second Quantization*, to keep track of the symmetry of the many-body wave function. Once the notation is mastered, this is probably the simplest approach. The other approach (described by Bethe and Jackiw, 1968 and outlined in what follows) evaluates (26.18) algebraically.

To determine the consequences of the variational principle (26.18), we must evaluate the integral. First we note some theorems regarding the matrix element,

$$\langle \Psi | F | \Phi \rangle = \sum_P (-1)^P \prod_{i=1}^N \psi_i^*(i) F \phi_{P a_i}(i), \quad (26.21)$$

of a general operator F between determinantal wave functions.

1. If F is identically 1,

$$\langle \Psi | 1 | \Phi \rangle = 0, 1, \quad (26.22)$$

the 1 occurring for those permutations which make the ϕ 's and the ψ 's equal.

2. If F equals the sum of single-particle operators,

$$\left\langle \Psi \left| \sum_{j=1}^N f_j \right| \Phi \right\rangle = \begin{cases} 0, & \text{if } \psi_i \neq \phi_i \text{ for more than one } i, \\ \langle \psi_j | f | \phi_j \rangle, & \text{if } \psi_i = \phi_i \text{ for all } i \text{ except } i = j, \\ \sum_i \langle \phi_i | f | \phi_i \rangle, & \text{if } \psi_i = \phi_i \text{ for all } i. \end{cases} \quad (26.23)$$

3. If F equals the sum of two-particle operators,

$$\left\langle \Psi \left| \sum_{i < j}^N v_{ij} \right| \Phi \right\rangle = \begin{cases} 0, & \psi_i \neq \phi_i, \text{ more than 2 } i\text{'s}, \\ \sum_{i < j}^N [\langle ij | v | ij \rangle - \langle ij | v | ji \rangle], & \psi_i = \phi_i, \text{ all } i, \\ \sum_{i \neq j}^N [\langle ij | v | ij \rangle - \langle ij | v | ji \rangle], & \psi_j = \phi_i, \text{ except } i = j, \\ [\langle ij | v | ij \rangle - \langle ij | v | ji \rangle], & \psi_k = \phi_k, \text{ all } k \neq i, j. \end{cases} \quad (26.24)$$

The $\langle ij | v | ij \rangle$ term in (26.24) is called the *direct* integral while the $\langle ij | v | ji \rangle$ term is the *exchange* integral.

Exercise Verify (26.24) for two electrons. $\diamond$

If we return to the variational matrix element $\langle \Psi | H | \Psi \rangle$ of (26.18) and (26.19), we see that the kinetic energy K and the electron–nucleus potential Φ are single-particle operators (we treat the nucleus as fixed), so their evaluation is straightforward:

$$\langle K + \Phi(e, Z) \rangle = \sum_{a=1}^N \int d^3r \left(-\frac{|\nabla \phi_a(\mathbf{r})|^2}{2m} - Z \frac{e^2}{r} |\phi_a(\mathbf{r})|^2 \right), \quad (26.25)$$

that is, the sum of single-particle kinetic energies and Coulomb potential energies.

Exercise Show that the vanishing of the wave function at infinity means the kinetic energy term in (26.25) can be rewritten as

$$\int d^3r \frac{-|\nabla \phi_a(\mathbf{r})|^2}{2m} = \int d^3r \left(\phi_a^*(\mathbf{r}) \frac{-\nabla^2}{2m} \phi_a(\mathbf{r}) \right). \quad \diamond \quad (26.26)$$

The matrix element of the electron–electron interaction v is the sum of 2-particle operators, so according to (26.24),

$$\begin{aligned} \left\langle \sum_{i < j} v(e_i, e_j) \right\rangle &= \sum_{a < b} \left[\int d^3r d^3r' \frac{e^2}{|\mathbf{r} - \mathbf{r}'|} |\phi_a(\mathbf{r})|^2 |\phi_b(\mathbf{r}')|^2 \right. \\ &\quad \left. - \delta_{a,b} \int d^3r d^3r' \frac{e^2}{|\mathbf{r} - \mathbf{r}'|} \phi_a^*(\mathbf{r}) \phi_a(\mathbf{r}') \phi_b^*(\mathbf{r}') \phi_b(\mathbf{r}) \right], \end{aligned} \quad (26.27)$$

where the Kronecker delta function restricts contributions of the exchange integral to states in which the electrons are in the same spin state (their space wave function is then antisymmetric). The exchange potential, the one entering with a minus sign, is seen to lower the calculated binding energy relative to the Hartree values.

Putting these pieces together gives the total energy of the many-fermion system as a *functional* of the single-particle wave functions:

$$E[\Psi] = \langle H \rangle = \sum_i \langle \Phi_i + K_i \rangle + \sum_{i < j} \langle v_{ij} \rangle. \quad (26.28)$$

The easiest way to vary each ϕ_a subject to the constraint (26.20) that it remains orthogonal to the other ϕ_a 's, is to add the constraint to the energy with Lagrange multipliers λ_{ab} , that is, to require

$$\delta \left(E[\Psi] - \sum_{a < b} \lambda_{ab} [\langle a | b \rangle - \delta_{a,b}] \right) = 0. \quad (26.29)$$

Because the energy must be real, we must have $\lambda_{ab} = \lambda_{ab}^*$, and because the wave functions can be complex, we vary $\text{Re}\phi_a$ and $\text{Im}\phi_a$ (or ϕ_a and ϕ_a^*) independently. We now take the variation of the matrix element of the Hamiltonian,

$$\delta \int d^3r d^3r' \phi_a^*(\mathbf{r}) \phi_b^*(\mathbf{r}') H \phi_a(\mathbf{r}) \phi_b(\mathbf{r}') = 0, \quad (26.30)$$

and move the variation inside the integral,

$$\int d^3r d^3r' \delta [\phi_i^*(\mathbf{r}) \phi_j^*(\mathbf{r}') H \phi_i(\mathbf{r}) \phi_j(\mathbf{r}')] = 0. \quad (26.31)$$

This implies that the integrand must vanish, and leads to a set of integro-differential equations for the single-particle wave functions known as the *Hartree–Fock equations*:

$$\begin{aligned} \sum_{b=1}^N \lambda_{ab} \phi_b(\mathbf{r}) &= \left(-\frac{\nabla^2}{2m} - Z \frac{e^2}{r} \right) \phi_a(\mathbf{r}) \\ &\quad + \int d^3r' \phi_b^*(\mathbf{r}') \frac{e^2}{|\mathbf{r} - \mathbf{r}'|} \sum_{b=1}^N [\phi_b(\mathbf{r}') \phi_a(\mathbf{r}) - \phi_b(\mathbf{r}) \phi_a(\mathbf{r}') \delta_{a,b}]. \end{aligned} \quad (26.32)$$

The interpretation of the Lagrange multipliers (or eigenvalues) λ_{ab} is not obvious. Because the matrix $[\lambda_{ab}]$ is Hermitian, it is possible to make a unitarity transformation on the wave functions (basis vectors),

$$\phi_a^{(\text{new})} = \sum_b c_{ab} \phi_b^{(\text{old})}, \quad (26.33)$$

to a new set for which the eigenvalue matrix λ will be diagonal:

$$\lambda_{ab}^{(\text{old})} \rightarrow \delta_{a,b} \epsilon_a^{(\text{new})}. \quad (26.34)$$

With these new ϕ 's, the RHS of (26.32) becomes $\epsilon_a \phi_a$ and (26.32) becomes an eigenvalue problem. Thus we are led to believe that the ϵ_a 's are physical energies. By evaluating the expectation value of the Hamiltonian, (26.28) becomes:

$$E[\Psi] = \langle H \rangle = \sum_i^N \epsilon_i - \sum_{i < j}^N [\langle ij | v(e_i, e_j) | ij \rangle - \langle ij | v(e_i, e_j) | ji \rangle]. \quad (26.35)$$

We therefore see that ϵ_i is the energy needed to remove the i^{th} electron from the atom, provided the wave functions for the other electrons do not change. Yet because all the electrons interact with each other to form the central potential V , the wave function will change if one electron is removed, and so ϵ_i is not a true ionization energy.

The integral in (26.32) is quite fascinating. First, it involves the joint probability (*correlation*) of finding electrons in states a and b at points $\mathbf{r}$ and $\mathbf{r}'$. Second, it indicates that a fermion in a many-fermion system feels an effective single-particle potential that is both *spin dependent* and *nonlocal* (depends on both $\mathbf{r}$ and $\mathbf{r}'$). The nonlocal nature of Hartree–Fock theory is made clearer by writing (26.32) as an equation for just one wave function ϕ_a influenced by effective potentials V and W :

Exercise Show that (26.32) can be written as

$$\epsilon_a \phi_a(\mathbf{r}) = T_a \phi_a(\mathbf{r}) + V(\mathbf{r}) \phi_a(\mathbf{r}) + \int d^3 r' W(\mathbf{r}, \mathbf{r}') \phi_a(\mathbf{r}'), \quad (26.36)$$

$$W(\mathbf{r}, \mathbf{r}') = -\frac{e^2 \sum_b^N \phi_b^*(\mathbf{r}') \phi_b(\mathbf{r}) \delta_{s_a, s_b}}{|\mathbf{r} - \mathbf{r}'|} = v(\mathbf{r} - \mathbf{r}') \delta_{s_a, s_b} \rho(\mathbf{r}', \mathbf{r}), \quad (26.37)$$

$$V(\mathbf{r}) = \Phi(\mathbf{r}) + U(\mathbf{r}), \quad (26.38)$$

$$U(\mathbf{r}) = -e^2 \sum_{b=1}^N \int d^3 r' \frac{|\phi_b(\mathbf{r}')|^2}{|\mathbf{r} - \mathbf{r}'|}. \quad \diamond \quad (26.39)$$

We see from these equations that if the many-fermion wave function is approximated by a Slater determinant, the “best” potential from a variational point of view is the average (including exchange) potential generated by the other $(A - 1)$ fermions.⁷ If the interaction is not the Coulomb interaction, we only need to replace $e^2/|\mathbf{r} - \mathbf{r}'|$ in (26.37) by the appropriate $v(\mathbf{r} - \mathbf{r}')$. Note, because a determinantal wave function makes all particles and orbitals equivalent, the self-consistent potential is the same for all occupied states.

⁷The part of the interaction left over is often called the *residual* interaction and, if our approximations are good, it should be small enough to be handled with perturbation theory.

Table 26.1: Removal Energies of Ag^+ (in Ry $\simeq 13.6$ eV). (From Bethe and Jackiw, 1968.)

Level	Experiment	Hartree–Fock	Thomas–Fermi–Dirac
1S	1879.7	1828.	1805.
2S	282.0	270.	268.
2P	253.7	251.	245.
3S	53.4	52.2	—
3P	44.8	44.3	—
3d	27.6	29.8	29.6
4S	7.3	8.5	8.0
4P	5.35	5.82	—
4D	1.57	1.69	—

By recalling the origin of (26.37) we see that the nonlocality of the potential W arises from the correlated nature of the electron's wave function, that is, the exchange potential. Determining $\phi_a(\mathbf{r})$ thus requires knowing the wave functions of the fermions in all other orbits as well as all other $\mathbf{r}'$. This is reasonable because for an interacting system the fermions throughout the system respond if a fermion at one point is modified, and so the effective potential is nonlocal. An approximation technique to remove this nonlocality is to assume the range of nonlocality is small, so that $\phi_a(\mathbf{r}') \simeq \phi_a(\mathbf{r})$ inside the integral in (26.36), and then to remove ϕ_a to outside the integral. This leaves an approximate equivalent local potential:

$$\begin{aligned} \int d^3r' W(\mathbf{r}, \mathbf{r}') \phi_a(\mathbf{r}) &\approx W^{\text{eq}}(\mathbf{r}) \phi_a(\mathbf{r}), \\ W^{\text{eq}}(\mathbf{r}) &\simeq \int d^3r' W(\mathbf{r}, \mathbf{r}'). \end{aligned} \quad (26.40)$$

Equations (26.36)–(26.39) are solved with the self-consistent techniques and approximations discussed for the Hartree equations (if we ignore antisymmetry, Hartree–Fock theory reduces to the Hartree approximation). Some results are given in Table 26.4 (see also Table 27.3) where we see agreement at the several percentage level is common. Further discussions are in the references and Chapter 27, *Statistical Help with Many-Body Problems*.

26.5 Hartree–Fock for a Fermi Gas: Tutorial

While most Hartree–Fock applications require numerical evaluations, analytic solutions are possible if we put the electrons in a box. While this may be only a crude approximation for the valence electrons in a metal, it is educational; we present it here as a tutorial (based on Bethe and Jackiw, 1968). The reader can review the Fermi gas model of electrons in a box in Appendix A, *Natural Units and Plane Waves*, or jump ahead to the description of Thomas–Fermi theory in § 27.1 where the model is also described.

Consider a gas of N electrons confined to a box of volume $\mathcal{V}$. The electrons repel each other via the Coulomb interaction and are simultaneously attracted to the positive (lattice) charge density ρ_0 by the background potential

$$V_B(\mathbf{r}) = -e^2 \int d\mathbf{r}' \frac{\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|}, \quad \rho(\mathbf{r}') = \rho_0 = \frac{eN}{\mathcal{V}}. \quad (26.41)$$

Let there be N^+ electrons with spin up and N^- electrons with spin down. Each is described by a wave function,

$$\psi_i^\pm = u_i^\pm(\mathbf{r})\chi^\pm, \quad (26.42)$$

where u_i is the space part for orbital i , χ is the spin part, and where the $\pm$ indicate the spin.

1. Write down the Hartree–Fock equations for $u_i^\pm(\mathbf{r})$.
2. Show that the HF equations are solved by plane waves:

$$u_i^\pm(\mathbf{r}) = \frac{e^{i\mathbf{k}_i \cdot \mathbf{r}}}{\sqrt{\mathcal{V}}}. \quad (26.43)$$

3. Write down expressions for the direct and exchange potentials for spin $s = \pm$:

$$V(\mathbf{r}) = e^2 \int d\mathbf{r}' \sum_{i,s=\pm} \frac{|u_i^s(\mathbf{r}')|^2}{|\mathbf{r} - \mathbf{r}'|}, \quad (26.44)$$

$$U^s(\mathbf{r}_1, \mathbf{r}_2) = \frac{-e^2 \rho^s(\mathbf{r}_1, \mathbf{r}_2)}{|\mathbf{r}_1 - \mathbf{r}_2|}, \quad (26.45)$$

$$\rho^s(\mathbf{r}_1, \mathbf{r}_2) = \sum_i [u_i^s(\mathbf{r}_2)]^* u_i^s(\mathbf{r}_1). \quad (26.46)$$

4. Perform these sums using a Fermi gas model in which the spin up and down electrons fill separate Fermi spheres with respective Fermi momenta $k_f^\pm$. Show that:

$$V(\mathbf{r}) = e^2 \int d\mathbf{r}' \sum_{s=\pm} \frac{\rho^s(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|}, \quad (26.47)$$

$$\rho^s(\mathbf{r}) = \frac{N^s}{\mathcal{V}} = \frac{(k_f^s)^3}{6\pi^2}, \quad (26.48)$$

$$U^s(\mathbf{r}_1, \mathbf{r}_2) = \frac{-e^2 \rho^s(\mathbf{r}_1, \mathbf{r}_2)}{r_{12}} \quad (26.49)$$

$$= \frac{-e^2 \left(\sin k_f^s r_{12} - k_f^s r_{12} \cos k_f^s r_{12} \right)}{2\pi^2 r_{12}^4}, \quad (26.50)$$

$$\int d\mathbf{r}_2 U^s(\mathbf{r}_1, \mathbf{r}_2) u_i^s(\mathbf{r}_2) = -\frac{2}{\pi} e^2 k_f^s F(\eta_i^s) u_i^s(\mathbf{r}_1), \quad (26.51)$$

$$F(\eta) = \frac{1}{2} + \frac{1 - \eta^2}{4\eta} \ln \frac{1 + \eta}{1 - \eta}, \quad \eta_i^s = \frac{k_i}{k_f^s}. \quad (26.52)$$

5. Show that in (26.41), the direct potential V cancels the background potential V_B .

6. Show that for orbital i , the Hartree–Fock energy is

$$\epsilon_i^s = \frac{k_i^2}{2m} - \frac{2e^2 k_f^4}{\pi} F(\eta_i^s). \quad (26.53)$$

7. Show that the expectation value of the Hamiltonian for this Hartree–Fock model of a Fermi gas is

$$\langle H \rangle = \frac{V}{(2\pi)^3} \left(\frac{2\pi}{5m} \left[(k_f^+)^5 + (k_f^-)^5 \right] - e^2 \left[(k_f^+)^4 + (k_f^-)^4 \right] \right). \quad (26.54)$$

26.6 Problems

1. The pair correlation function C was defined in terms of two densities as

$$\rho(\mathbf{r}_1, \mathbf{r}_2) = \rho(\mathbf{r}_1)\rho(\mathbf{r}_2)[1 - C(\mathbf{r}_1, \mathbf{r}_2)]. \quad (26.55)$$

Show that for a uniform gas of spinless electrons,

$$C(\mathbf{r}_1, \mathbf{r}_2) = \left(\frac{3j_1(k_f r)}{k_f r} \right)^3, \quad (26.56)$$

where $r = |\mathbf{r}_1 - \mathbf{r}_2|$ is the interparticle separation and k_f the Fermi momentum.

2. Consider a finite system of 16 fermions filling the first S and first P states of a 3D harmonic oscillator. If the CM of the two fermions is kept at the origin ($\mathbf{R} \equiv 0$), show that the correlation function is

$$C(\mathbf{r}_1, \mathbf{r}_2) = [1 - \frac{1}{2}\beta^2 r^2]^2 \exp(-\beta^2 r^2/2), \quad (26.57)$$

where $r = |\mathbf{r}_1 - \mathbf{r}_2|$. This function is plotted in Figure 26.2.

3. Use first-order perturbation theory to calculate the ground-state energy of an atom with nuclear charge Ze and two electrons. As a zero-order approximation, assume each electron moves in the field of an effective charge Z' .

- (a) Evaluate the following (using $e^2/2a_0 = 13.6\text{eV} = 1\text{ Ry}$ as the unit of energy):

- The first-order ground-state energy.
- The (exact) lowest-order continuum energy.
- The ionization energy.

- (b) Determine the numerical values of the preceding energies for $Z = 1, 2$ and 3 and the two models $Z' = Z$ and $Z' = Z - 5/16$. Compare your answers to the experimental ionization energies:

H^- : 0.055 Ry, He: 1.807 Ry, and Li^+ : 5.560 Ry.

4. The wave function for the two electrons in a helium atom may depend on all the electron's quantum numbers.

- (a) Deduce the form of the product wave function for the helium ground state.

- (b) Deduce the form of the product wave function for an excited state of helium in which one electron is in a P state.
5. Consider the 3D potential
- $$V = \frac{1}{2}m\omega^2 r^2. \quad (26.58)$$
- (a) Assuming that you do not know the exact solution, use the wave function parameter α as a variational parameter to obtain the variational limits for the energy of a particle in this potential for the following:
- The lowest S state using $\psi = e^{-\alpha r}$.
 - The lowest P state using $\psi = \sin \theta e^{-\alpha r} e^{i\phi}$.
 - The lowest P state using $\psi = r \cos \theta e^{-\alpha r}$.
- (b) Compare with the exact energies in each case and discuss why part ii gives a poor result.
6. To include the nuclear motion in the helium atom, it is convenient to change variables to

$$\rho_1 = \mathbf{r}_1 - \mathbf{r}_N, \quad \rho_2 = \mathbf{r}_2 - \mathbf{r}_N, \quad \mathbf{R} = \frac{m_N \mathbf{r}_N + m_e \mathbf{r}_1 + m_e \mathbf{r}_2}{m_N + 2m_e}, \quad (26.59)$$

where N refers to the nucleus.

- (a) Carry out this change of variables for the Schrödinger equation and show that the corrections to the infinite-nuclear-mass case are of three types:
- motion of the atom as a whole,
 - use of a reduced mass in place of an electron mass, and
 - addition of a correlation term $H' = \mathbf{p}_1 \cdot \mathbf{p}_2 / m_N$.
- (b) Calculate the size of each of these corrections for the variational wave function

$$\psi = e^{-\alpha(\rho_1 + \rho_2)}, \quad (26.60)$$

where α is determined for the infinite-mass case by the minimization

$$\delta \frac{\langle \psi | H | \psi \rangle}{\langle \psi | \psi \rangle} = 0. \quad (26.61)$$

7. We want a solution of the Schrödinger equation

$$H\psi = E\psi \quad (26.62)$$

Rather than solve this directly, derive the equivalent equation if the wave function is expanded in a complete set ϕ_a of known functions:

$$\psi = \sum_a^N c_a \phi_a. \quad (26.63)$$

- (a) Under what conditions will this ψ be exact?
- (b) What variation will determine the equivalent Schrödinger equation and maintain the wave function normalization?

- (c) Introduce a Lagrange multiplier ϵ to incorporate the normalization constraint into the variation.
- (d) Show that the variation leads to the equations

$$\sum_b \langle b | H | a \rangle c_b = \epsilon \sum_b \langle b | a \rangle c_b. \quad (26.64)$$

- (e) Interpret the meaning of each term in this equation especially the Lagrange multiplier ϵ .
- (f) Show that these linear equations constitute an eigenvalue problem, and that the eigenvalues must satisfy the matrix condition

$$\det[\langle b | H | a \rangle - \epsilon \langle b | a \rangle] = 0. \quad (26.65)$$

CHAPTER 27

STATISTICAL HELP WITH MANY-BODY PROBLEMS

Solving Hartree–Fock equations becomes a significant undertaking when the number of fermions becomes large (the tutorial in the last chapter is an exception). When the number is approximately 100, as it may be for molecules and solids, less accurate (and less quantum-mechanical) theories are frequently used either as a starting point for self-consistent calculations or in place of them. In the statistical theories discussed in this chapter, Fermi–Dirac statistics and semiclassical arguments are used to deduce a closed-form expression for the effective, single-particle potential. These models, which extend the theory of a homogeneous, fermion gas to inhomogeneous fermion gases, are used for atoms, molecules, solids, and nuclei. Because these are models and not exact theories, they have difficulties explaining some physics such as the finite size of atoms, shell structure, the existence of negative ions, and exchange effects, but they do permit us to understand and calculate some basic properties of many-fermion systems.¹

We start by highlighting some aspects of *Thomas–Fermi theory* and go on to its modern generalization, *density functional theory*. We then examine the *Bethe–Goldstone* (or *Brueckner*) *approach* in which the Lippmann–Schwinger (LS) formalism we have developed in the first part of this book gets implanted into a Fermi sea. This last application is an excellent summary, using ideas from scattering theory, Hartree–Fock models, and quantum statistics.

27.1 Thomas–Fermi Theory

We assume the number of interacting electrons (or more generally fermions) is large enough for a statistical view to be valid. To relate the potential $V(\mathbf{r})$ which an electron feels at point $\mathbf{r}$ to the density of this and other electrons at that point, $\rho(\mathbf{r})$, we apply Poisson's

¹Some of the difficulties are removed by extensions to the theory and approximations. We refer the reader to the insightful discussion in Bethe and Jackiw (1968) and the more modern ones in March (1983) and Dreizler and de Providencia (1985).

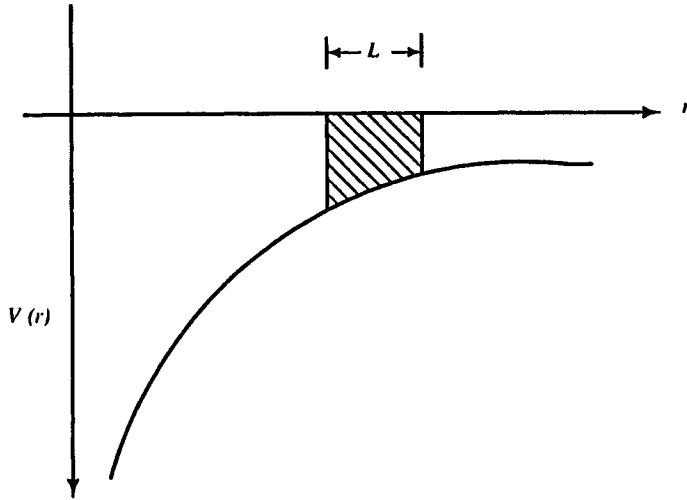


Figure 27.1: The potential V and a region of space in which V is assumed constant in Thomas–Fermi theory.

equation from classical electrostatics:

$$\nabla^2 \left[\frac{V(r)}{e} \right] = -4\pi e \rho(r). \quad (27.1)$$

In Thomas–Fermi theory, ρ is taken to be the fundamental variable (wave functions are not calculated) with V ultimately determined by solving (27.1). In contrast to the Hartree–Fock potential, which varies as do the wave functions of the electrons, we assume the potential in the small, hatched region of space in Figure 27.1 is constant. Next we imagine the hatched region to be a box (spherical shell) with a side L larger than the characteristic wavelength for an electron in the atom, yet smaller than the atom’s size:

$$\lambda_e < L < R_{\text{atom}}. \quad (27.2)$$

Because V is constant within each box, the wave function is approximately a plane wave there:

$$\phi_a \simeq e^{i\mathbf{k} \cdot \mathbf{r}}, \quad (27.3)$$

yet with $\mathbf{k}$ varying from box to box. The number of particles N in the box (and correspondingly the volume of the box) should be large enough for the kinetic energy to be much larger than the potential, yet the box must be small enough for the potential to be approximately constant. The number N is related to the box size and the Fermi wavenumber k_f by

$$N = 2 \left(\frac{L}{2\pi} \right)^3 \int_0^{k_f} d^3k = \frac{L^3}{3\pi^2} k_f^3(r). \quad (27.4)$$

For an electron gas, the Fermi momentum $k_f \equiv \hbar k_f$ is related to the local density by

$$k_f^3(r) = \frac{3\pi^2 N}{L^3} = 3\pi^2 \rho(r), \quad (27.5)$$

$$\Rightarrow k_f(r) = [3\pi^2 \rho(r)]^{1/3}, \quad \rho = \frac{k_f^3}{3\pi^2}. \quad (27.6)$$

If we view an atom as a uniform sphere of radius r_s , the Fermi momentum and energy are

$$k_f = \frac{2.81 \text{ \AA}^{-1}}{r_s/a_B}, \quad \epsilon_f = \frac{k_f^2}{2m} = \frac{30.1 \text{ eV}}{(r_s/a_B)^2}, \quad (27.7)$$

where a_B is the Bohr radius $\hbar^2/me^2 \equiv 1/me^2$. For electrons in atoms, ϵ_f is about 10 eV or less; for electrons in a metal, ϵ_f is about 5 eV.

For nucleon matter (a homogeneous gas of neutrons and protons intermixed), there are twice as many states for each momentum value of a nucleon, and accordingly,

$$N = \frac{2L^3 k_f^3}{3\pi^2} \Rightarrow k_f^3(r) = \frac{3}{2}\pi^2 \rho(r), \quad \text{or,} \quad (27.8)$$

$$k_f^3(r) = \frac{2k_f^3}{3\pi^2} \quad (\text{nucleon matter}). \quad (27.9)$$

From the observed density in the central region of nuclei, $\rho(0) \simeq 1.72 \times 10^{38}$ particles/cc $= 0.17$ nucleons/fm³, the Fermi momentum and energy for nucleons are:

$$k_f = 1.36 \text{ fm}^{-1} \equiv \hbar k_f = 268 \text{ MeV}/c, \quad \epsilon_f = 38 \text{ MeV}. \quad (27.10)$$

The momentum-space density within the box of Figure 27.1 is uniform, the probability of finding an electron with momentum in the interval $p \rightarrow p + dp$ is

$$d\mathcal{P}(p + dp) = \begin{cases} 3p^2 dp / k_f^3(r), & \text{for } p < k_f, \\ 0, & \text{for } p > k_f. \end{cases} \quad (27.11)$$

The average kinetic energy of an electron or nucleon in our box is accordingly:

$$\langle K \rangle = \int_0^{k_f} d\mathcal{P} \frac{p^2}{2m} = \frac{3}{5} \epsilon_f, \quad (27.12)$$

which is 30% of the maximum electron energy ϵ_f . The average kinetic energy density of an electron gas t is this energy times the local density:

$$t \equiv \langle K \rangle \rho = \frac{3}{5} \frac{k_f^2}{2m} \rho = \frac{3(3\pi^2)^{2/3}}{10m} \rho^{5/3}. \quad (27.13)$$

Exercise Prove (27.13) ◇

For an electron in the hatched area of Figure 27.1, $k_f(r)^2/2m$ is the maximum kinetic energy possible, and is related to the local density by (27.6). If we denote the chemical potential (the classical total energy of the fastest electron) by ξ , then

$$\xi = V(r) + \frac{k_f^2(r)}{2m}. \quad (27.14)$$

Because ξ is a constant throughout the atom, this again says the Fermi momentum varies with the potential, or equivalently, the local density varies with the potential as:

$$\rho(r) = \frac{k_f^3(r)}{3\pi^2} = \frac{(2m[\xi - V(r)])^{3/2}}{3\pi^2}. \quad (27.15)$$

The relation (27.15) is key to Thomas–Fermi theory.² Whereas we have deduced it intuitively, it also derives from a variational principle.³

The Thomas–Fermi equation for the potential V is Poisson's equation (27.1) applied to the density (27.15):

$$\nabla^2 V(r) = -4\pi e^2 \rho(r) \quad (27.16)$$

$$\frac{1}{r} \frac{\partial^2}{\partial r^2} (rV), = \frac{-4\pi e^2 (2m[\xi - V(r)])^{3/2}}{3\pi^2}. \quad (27.17)$$

Exercise Show that the change of variable and function,

$$x = br, \quad b = \left(\frac{4}{3\pi} \right)^{2/3} 2me^2 Z^{1/3}, \quad \Phi(x) = \frac{r[\xi - V(r)]}{Ze^2}, \quad (27.18)$$

leads to the *Thomas–Fermi equation* for the atomic potential:

$$\boxed{\frac{d^2 \Phi(x)}{dx^2} = x \left(\frac{\Phi(x)}{x} \right)^{3/2}. \quad \diamond} \quad (27.19)$$

Solutions

Although general analytic solutions to the Thomas–Fermi equation have not been found, the numerical solutions exhibit intriguing features, some of which are shown in Figure 27.2. First, the Z independence of (27.18) means there is a universal shape for the effective potential Φ in *all* atoms. Second, the dynamical equation (27.19) is universal and rather simple and thus we expect its solution also to be. Third, when solving this classical equation, the atomic potential Φ is set equal to zero in classically forbidden regions (negative $\xi - V$, i.e., negative kinetic energy). Consequently, if with increasing x , Φ intercepts the x axis from above, it vanishes for all larger x (the positive ion graph in Figure 27.2).

Exercise Show that the nuclear charge at the origin of the atom leads to the boundary condition

$$\Phi(0) = 1. \quad \diamond \quad (27.20)$$

Exercise Show that the neutrality of an atom leads to the boundary condition

$$\Phi(\infty) = 0. \quad \diamond \quad (27.21)$$

A consequence of the boundary conditions (27.20) and (27.21) is the absence of solutions to the Thomas–Fermi equation for neutral atoms of finite size, and for negative ions.

²Since ρ is used in a classical Poisson's equation, this relation is often simplified by making a gauge transformation and replacing $\xi - V$ with just $-V$.

³Bethe and Jackiw (1968); March (1983).

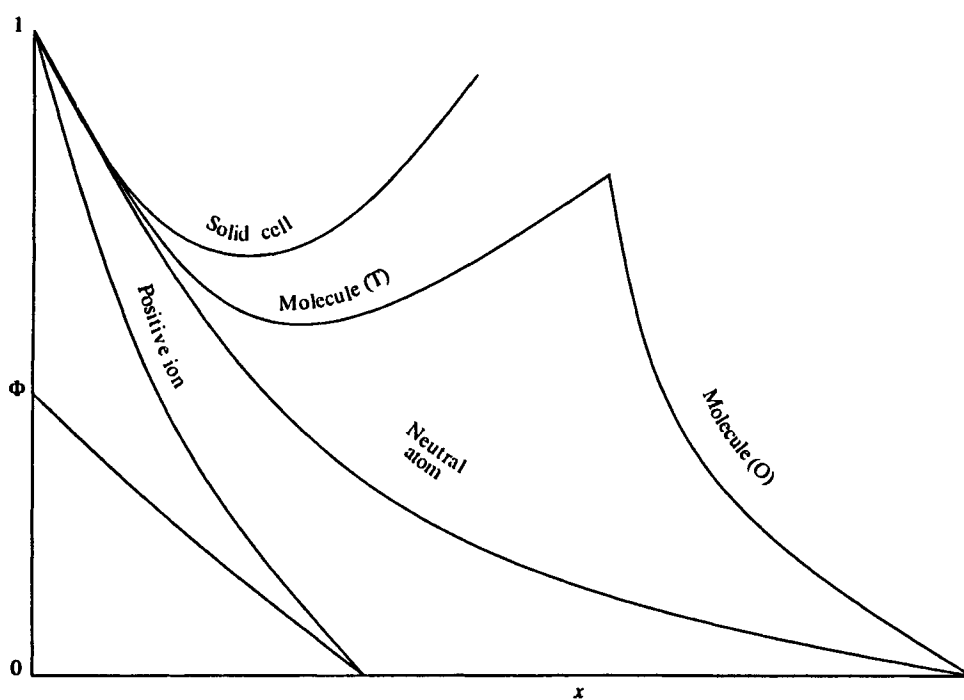


Figure 27.2: The potential function Φ for the Thomas–Fermi equation. The different curves are for positive ion, neutral atom, tetrahedral molecule, octahedral molecule, and solid cell. Note the discontinuity in Φ when it equals zero and for different molecular types. (From March, 1983).

Thus in Figure 27.2, the initial slope of Φ may not be adjusted, and the only acceptable atomic solutions are those that are asymptotically tangent to the x axis. Solutions with less slope intercept the x axis and then vanish; solutions with more slope diverge and are not normalizable (and thus not bound). Furthermore, solutions for positive ions are seen to be discontinuous!

The asymptotic limit for Φ ,

$$\Phi \sim \frac{144}{x^3}, \quad (27.22)$$

is a solution of (27.19), yet is not complete because it does not satisfy the boundary condition at the origin (27.20). This form of solution, with x the scaling parameter given by (27.18), implies that the functional form of the outer densities of all atoms is the same. Likewise, the radii of all atoms are proportional to $Z^{-1/3}$.

Exercise Verify (27.22). ◇

Also obtained from the solutions of (27.19) is the small- r behavior:

$$V(r \simeq 0) \rightarrow \frac{-Z}{r} e^2 + \frac{1.8Z^{4/3}}{a_B} e^2, \quad (27.23)$$

where a_B is the Bohr radius. This universality helps explain why hydrogen-like wave functions are so useful even for complicated atoms.

Exercise Verify (27.23). ◇

In general, Thomas–Fermi theory is most valid for high densities where the semiclassical approximation and the neglect of exchange become better approximations. It becomes exact for very large atomic numbers (Lieb, 1976), and gives good agreement with atomic densities when the densities are large (Politzer, 1980).

27.2 Thomas–Fermi–Dirac Equation

The Thomas–Fermi equation (27.19) incorporates quantum mechanics by using plane waves in a box to determine the electrons' density. Yet, as discussed in Chapter 26, *Many-Body Problems*, the exclusion principle leads to the particles in a box being correlated, even if they are not interacting with each other! Including this correlation (exchange) arising from antisymmetrization changes the Thomas–Fermi equation into the Thomas–Fermi–Dirac equation. Because the new equation still contains the assumption that the kinetic-energy density is proportional to the five-thirds power of the number density, this is not yet density-functional theory.

We have found that for a free-fermion gas, the correlation function due to exchange is (26.16):

$$C(\mathbf{r}_1, \mathbf{r}_2) = C(\mathbf{r}_1 - \mathbf{r}_2) = \left(3 \frac{j_1(k_f r)}{k_f r} \right)^2. \quad (27.24)$$

This effectively places a “hole” around each electron in an atom so that other electrons of the same spin are repelled. If the Coulomb force among electrons is also included, there would

be an additional repulsive correlation even for electrons with opposite spins. A simple way to include the effect of this hole is to include it in the Thomas–Fermi Hamiltonian and then use the variational principle to derive a modified Thomas–Fermi equation. For an individual box with one electron, the self-energy of this hole in the two-particle density $\rho(\mathbf{r}_1, \mathbf{r}_2)$, is called the *exchange energy*, and is

$$\langle E_x \rangle = \frac{2}{\rho} \int d^3 r_2 \frac{e\rho(\mathbf{r}_1, \mathbf{r}_2) e\rho(\mathbf{r}_1, \mathbf{r}_2)}{r_{12}} = \frac{-3e^2}{2\pi} k_f. \quad (27.25)$$

Exercise Verify (27.25) by explicit substitution of the two-body density for a uniform electron gas. $\diamond$

The total energy contains the kinetic energy term (27.13) and the potential interactions,

$$\langle V \rangle = \int d^3 r \rho \left[-Z \frac{e^2}{r} + \frac{1}{2} \int d^3 r_2 \frac{e^2}{r_{12}} \rho(r_2) - \frac{3}{4} e^2 \left(\frac{3\rho}{\pi} \right)^{1/3} \right], \quad (27.26)$$

where we have substituted (27.6) for k_f . The terms in (27.26) correspond to different interactions. The first is the interaction with the atomic nucleus, the second among all electrons, and the third with the hole (there are factors of $\frac{1}{2}$ in the last two terms to avoid double counting).

The modification of the Thomas–Fermi equation is obtained by requiring the energy of the system to be stationary under variations of the density ρ . This determines the relation

$$y = \frac{\sqrt{2}}{\pi} \left(\sqrt{\Psi} + \frac{1}{\pi\sqrt{2}} \right), \quad (27.27)$$

$$y = a_B \left(\frac{3\rho}{\pi} \right)^{1/3} = \frac{a_B k_f}{\pi}, \quad \Psi = \frac{1}{2\pi^2} - \frac{a_B V(r)}{e^2}, \quad (27.28)$$

where we have changed to a new variable and function. Equations (27.27)–(27.28) are similar to the Thomas–Fermi relation (27.15) in which ρ varies basically like $V^{3/2}$. If (27.27) is used in Poisson’s equation and converted to dimensionless form, we obtain the *Thomas–Fermi–Dirac equation*,

$$\boxed{\frac{d^2 \Psi(x)}{dx^2} = x \left(\sqrt{\frac{\Psi}{x}} + \beta \right)^3}, \quad (27.29)$$

where $\beta = \sqrt{t/2a_B Z \pi^2}$. Because the β term introduces an explicit Z dependence, (27.29) is not as universal as the Thomas–Fermi equation (which it equals as $Z \rightarrow \infty$). While there are problems in solving this equation, it is often possible to work around them and obtain quite respectable results. This is illustrated in Table 26.4 of the last chapter where the results of the Thomas–Fermi–Dirac equation are compared with those of the more correct Hartree–Fock theory and to experiment. The Thomas–Fermi–Dirac results are seen to provide agreement with experimental measurements comparable to that of Hartree–Fock, but without the need to solve multiple, coupled Schrödinger equations self-consistently.

27.3 Density Functional Theory

In recent times, aspects of Hartree–Fock and Thomas–Fermi theory have been combined to produce *density functional theory* (DFT). As in Thomas–Fermi theory, the basic dynamical quantity is the single-particle density, yet whereas Thomas–Fermi theory is for a locally homogeneous gas of fermions, DFT is for an inhomogeneous gas. Density functional theory differs from Hartree–Fock theory in its use of densities rather than wave functions and in its inclusion of *both* exchange and dynamical correlations. As formulated, Hartree–Fock theory treats exchange exactly but leaves out correlations (except for those arising from the exchange). On the other hand, DFT includes both effects, but neither exactly. In some sense these two theories compliment each other; the DFT equations contain more approximations and produce less information (densities rather than wave functions). On the other hand, DFT contains more physical effects and is solved for systems with larger number of particles than Hartree–Fock theory can handle.

In the simplest form of DFT, and in atomic units, the energy of the system is viewed as a functional of the density $\rho(\mathbf{r})$ with the constraint,⁴

$$\int d^3\mathbf{r} \rho(\mathbf{r}) = N. \quad (27.30)$$

We assume the energy separates as

$$E[\rho] = K[\rho] + \int d^3\mathbf{r} \Phi(\mathbf{r})\rho(\mathbf{r}) + \frac{1}{2} \int d^3\mathbf{r} d^3\mathbf{r}' \frac{\rho(\mathbf{r})\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} + E_{xc}[\rho], \quad (27.31)$$

that is, a kinetic energy for the electrons K , an interaction Φ of the electrons with the nucleus, an electron–electron potential interaction ($\rho\rho$), and the *exchange-correlation energy* E_{xc} (which includes correlations). If we demand that the energy E be an extremum for variations of the density,

$$\delta E[\rho] = 0, \quad (27.32)$$

the DFT dynamical equations result. The equations are equivalent to the single-particle equations:

$$\left[-\frac{1}{2}\nabla^2 + V_{\text{eff}}(\mathbf{r})\right] \psi_a(\mathbf{r}) = \epsilon_a \psi_a(\mathbf{r}), \quad (27.33)$$

$$\rho(\mathbf{r}) = \sum_{a=1}^N |\psi_a(\mathbf{r})|^2, \quad (27.34)$$

where the sum is over the N lowest occupied eigenstates. The physics is in the potential,

$$\boxed{V_{\text{eff}}(\mathbf{r}) = V(\mathbf{r}) + U_{xc}(\mathbf{r})}, \quad (27.35)$$

where V is the total, classical potential energy (nuclear centers plus electron–electron),

$$V(\mathbf{r}) = \Phi(\mathbf{r}) + \int d^3\mathbf{r}' \frac{\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|}. \quad (27.36)$$

⁴Kohn and Vashista (1983).

The new ingredient here is the *exchange correlation potential* U_{xc} ,

$$U_{xc}(\mathbf{r}) \stackrel{\text{def}}{=} \frac{\delta E_{xc}[\rho(\mathbf{r})]}{\delta \rho(\mathbf{r})}, \quad (27.37)$$

which is seen to be a functional derivative of the (still to be determined) exchange correlation energy.

Equations (27.33)–(27.37) must be solved self-consistently for the density ρ while treating the kinetic and potential terms as functionals of ρ . The exchange correlation energy functional E_{xc} is expressed in terms of another (unknown) functional g ,

$$E_{xc}[\rho] = \int d^3r \{g[\rho(\mathbf{r})] - K[\rho(\mathbf{r})]\}, \quad (27.38)$$

where g contains kinetic, exchange, and correlation energy parts, each of which are functionals of ρ . If g were known exactly for the atom, molecule, or solid under study, DFT could be an exact theory for the single-particle density, at least if the dynamical quantities were all proportional to ρ .

In practice, g is determined approximately for a number of special cases. For example, if the density is slightly inhomogeneous,

$$\rho(\mathbf{r}) \simeq \rho_0 + \Delta\rho(\mathbf{r}), \quad (27.39)$$

g is given by a power series in the inhomogeneity:

$$g[\rho] = g[\rho_0] + \int d^3r' \mathcal{K}(|\mathbf{r} - \mathbf{r}'|) \Delta\rho(\mathbf{r}) \Delta\rho(\mathbf{r}') + O(\Delta\rho^3). \quad (27.40)$$

The Kernel $\mathcal{K}$, in turn, is expanded as a Fourier series in the momentum transfer dependence of the dielectric constant ϵ of a homogeneous electron gas:

$$\mathcal{K}(\mathbf{r} - \mathbf{r}') = \frac{1}{L^3} \sum_{\mathbf{q}} \frac{2\pi e^{-i\mathbf{q} \cdot (\mathbf{r} - \mathbf{r}')}}{q^2[\epsilon(q) - 1]}. \quad (27.41)$$

At this stage, models such as the Fermi gas are used to approximate $\epsilon(q)$. This yields the following kinetic, exchange, and correlation parts of g in terms of the Fermi momentum (Kohn and Vashista, 1983):

$$g_{\text{kin}}[\rho] = \frac{3k_f^2\rho}{10}, \quad g_{\text{exh}}[\rho] = -0.239k_f\rho, \quad g_{\text{cor}}[\rho] = \frac{k_f\rho}{4.34 + 17.7k_f}. \quad (27.42)$$

In Table 27.3 we compare some DFT results with experiment. We see agreement around the 1% level, which is somewhat better than found in Table 26.4 for Hartree–Fock and Thomas–Fermi theories. Similar comparisons occur for atoms, molecules and solids. In Table 27.4 we compare some eigenvalues determined by both DFT and Hartree–Fock to experiment. For these cases, Hartree–Fock and DFT provide comparable agreement.

27.4 Bethe–Goldstone Equation

In order to describe the nuclear interior, many-body theory has been combined with the techniques of scattering theory to produce models such as the Bethe–Goldstone theory, the

Table 27.1: Atomic Energies (in Ry). (From March, 1983).

State	Experiment	DFT
H ⁻ , total	14.4	14.4
He, total	79.0	77.8
He, 1S	5.807	5.651
Li ⁺ , total	198.1	195.2
Ne, 1S	257.880	256.349
Ar, 1S S	1055.21	1051.709

$\mathcal{G}$ matrix approach, and the linked-cluster Rayleigh–Schrödinger expansion. We discuss the essence of them here. For fuller discussions the reader might wish to consult Koltun and Eisenberg (1988), Haftel and Tabakin (1970), Preston and Bhaduri (1975), deShalit and Feshback (1974), and Bethe and Goldstone (1957).

One of the problems encountered in early attempts to apply Hartree–Fock theory to nuclei was that the repulsion of the core of the nucleon–nucleon potential made conventional expansion techniques fail. This problem was solved in two ways. First, by incorporating the exclusion principle and other correlations, the close approach of two nucleons is suppressed and the effect of the short-range part of the potential is greatly reduced. Second, by replacing the actual potential operator by a nonsingular, pseudo potential $\mathcal{G}$ (like the T of scattering theory), the mathematical behavior of the equation is improved.

The Bethe–Goldstone equation combines both these ideas by calculating the nucleon–nucleon (NN) $\mathcal{G}$ matrix in the presence of other nucleons, and then uses this $\mathcal{G}$ matrix to calculate nuclear properties. For instant, consider two nucleons within a bound system of A identical nucleons. We assume the nucleons interact only as *independent pairs* (no 3-, 4-, $\dots$, A -body interactions), so the effect of the background nucleons arises solely from the exclusion principle. The LS equation (7.22) for the wave function of particle i ,

$$|\psi_k^{(+)}(i)\rangle = |\phi_k(i)\rangle + \frac{1}{E_k - H_0} v |\psi_k^{(+)}(i)\rangle, \quad (27.43)$$

is now generalized to two particles k and l within a nucleus:

$$|\Psi_{kl}(1, 2)\rangle = |\phi_k \phi_l\rangle + \frac{Q_{kl}}{E_{kl} - H_0} v |\Psi_{kl}\rangle. \quad (27.44)$$

Here v is the NN potential, H_0 is the unperturbed Hamiltonian,

$$H_0 = K + U = \sum_i^A (K_i + U_i), \quad (27.45)$$

and $\phi_k \phi_l$ is the product of the single-particle wave functions used to describe the ground state of the nucleus. The single-particle energies ϵ_k are the eigenvalues of the equation:

$$H_0 |\phi_k\rangle = (K_k + U_k) |\phi_k\rangle = \epsilon_k |\phi_k\rangle, \quad (27.46)$$

and so

$$H_0 |\phi_k \phi_l\rangle = (\epsilon_k + \epsilon_l) |\phi_k \phi_l\rangle. \quad (27.47)$$

Table 27.2: Orbital Eigenvalues (in Ry). (From Dreizler and da Providência, 1985).

State	Experiment	DFT	HF
Be, 1S	9.1	8.5	9.5
Be, 2S	0.68	0.68	0.62
Ne, 1S	64.	61.	66.
Ne, 2S	3.6	3.3	3.9
Ne, 2P	1.6	1.6	1.7

The new ingredient in (27.44) is the *exclusion-principle projection operator* Q that restricts intermediate states to those in which at least one orbital is not filled. Explicitly, with the ϕ_i 's as basis, this *Pauli operator* is

$$Q_{kl} = \sum_{ab \neq kl} |ab\rangle \langle ab| + \sum_{a \neq kl} |ka\rangle \langle ka| + \sum_{b \neq kl} |bl\rangle \langle bl|, \quad (27.48)$$

where $|ab\rangle$ is shorthand for $|\phi_a \phi_b\rangle$.

Exercise Explain why, in the independent pair approximation, (27.48) simplifies to

$$Q_{kl} = \sum_{ab} |ab\rangle \langle ab|. \quad \diamond \quad (27.49)$$

A key difference in the LS and Bethe–Goldstone equations (27.43) and (27.44) is the energy denominator or Green's function. In scattering, the Green's function singularity is moved away from the physical domain by adding a small positive imaginary part $+i\epsilon$ to the energy; this also builds in outgoing-wave boundary conditions. Within a nucleus, the projection operator Q_{kl} eliminates the possibility of reaching the singular point. This means there is no scattered wave, and that for large separations of the particles the distorted wave Ψ “heals” back to plane waves:

$$\Psi_{kl}(1, 2) \sim \phi_k(1)\phi_l(2). \quad (27.50)$$

In our study of scattering theory we found it convenient to express the dynamics in terms of the transition matrix (or pseudopotential) T in (??) and (??):

$$T_E(\mathbf{k}', \mathbf{k}) = v(\mathbf{k}', \mathbf{k}) + \int d^3p v(\mathbf{k}', \mathbf{p}) \frac{1}{E - E_p + i\epsilon} T_E(\mathbf{p}, \mathbf{k}), \quad (27.51)$$

$$T_E = v + v \frac{1}{E - H_0} T_E, \quad (27.52)$$

rather than the wave function ψ . Likewise, it is now convenient to have an operator G which acts like a T in the nuclear medium:⁵

$$G = v + v \frac{Q_{kl}}{E_{kl} - H_0} G. \quad (27.53)$$

⁵Not to be confused with the Green's function G_B .

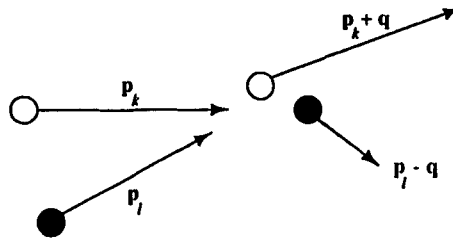


Figure 27.3: Momenta for a scattering event in fermion matter. Unless the momentum transfer q equals zero, the final momentum must be greater than the Fermi momentum k_f .

The $\mathcal{G}$ matrix is finite for singular v 's, much the way we found T to be.

Both the Ψ and $\mathcal{G}$ forms of the Bethe–Goldstone equation are solved with techniques similar to those developed earlier in this book. In coordinate space this involves integro-differential equations (or approximations to make the equations local); in momentum space this involves integral equations (which reduce to quadrature with the techniques in Chapter 18, *Solving Even Relativistic Integral Equations*). Specifically, to convert (27.53) to an integral equation, we look at its representation in the momenta basis $|\mathbf{p}_1, \mathbf{p}_2\rangle$ and employ the symmetry of the two body problem by transforming to relative and CM momentum κ and $\mathbf{P}$:

$$\kappa = \frac{1}{2}(\mathbf{p}_1 - \mathbf{p}_2), \quad \mathbf{P} = \frac{1}{2}(\mathbf{p}_1 + \mathbf{p}_2). \quad (27.54)$$

In this case the Bethe–Goldstone equation (27.53) has the representation

$$\langle \kappa', \mathbf{P} | \mathcal{G} | \kappa, \mathbf{P} \rangle = \langle \kappa' | v | \kappa \rangle + \int d^3 p \frac{\langle \kappa' | v | \mathbf{p} \rangle Q(\mathbf{p}) \langle \mathbf{p}, \mathbf{P} | \mathcal{G} | \kappa, \mathbf{P} \rangle}{E(\mathbf{P}, \kappa) - E(\mathbf{P}, \mathbf{p})}, \quad (27.55)$$

where the energy $E(\mathbf{P}, \kappa)$ is determined self-consistently and the Q function ensures that the exclusion principle is met.

Fermion Matter

To obtain simple forms for projection operator Q , we restrict ourselves to infinite matter. Translational invariance requires the total, two-nucleon momentum to be constant, as indicated in Figure 27.3 by one nucleon acquiring a momentum transfer of $\mathbf{q}$ and the other of $-\mathbf{q}$.

Exercise Show that with the independent-pair approximation, the exclusion principle requires *both* intermediate states a and b to have momentum greater than k_f . How would this change if we permitted more than pairwise interactions? $\diamond$

For the plane-wave states of nuclear matter, the Pauli operator Q has the property that it is zero if the final-state nucleons are below the Fermi sea, and one if they are above:

$$Q = \begin{cases} 1, & \text{if } p_a \text{ and } p_b > k_f, \\ 0, & \text{otherwise.} \end{cases} \quad (27.56)$$

Exercise Show that if the total momentum $\mathbf{P}$ is chosen as the z axis, Q takes the form

$$Q(P, \kappa, \cos \theta_{P\kappa}) = \begin{cases} 1, & \text{if } P^2 + \kappa^2 - 2P\kappa |\cos \theta| > k_f^2, \\ 0, & \text{otherwise.} \end{cases} \quad \diamond \quad (27.57)$$

In most calculations that actually solve the Bethe–Goldstone equation (27.55), the matrix elements of v , Q and $\mathcal{G}$ are expanded in spherical harmonics (as in Chapter 8, *The Angular Momentum Basis*); for example,

$$Q(\mathbf{P}, \kappa) = \sum_{l=0}^{\infty} Q_l(P, \kappa) P_l(\cos \theta). \quad (27.58)$$

To simplify the calculations and remove some of the severity of the infinite, nuclear matter assumption, often only an “angle-averaged Pauli operator” is used in which only the Q_0 term is kept.

Exercise Verify that

$$Q_0(P, \kappa) = \begin{cases} 1, & \text{if } \kappa < k_f + P, \\ \tilde{Q}_0, & \text{if } k_f + P > \kappa > \sqrt{k_f^2 - P^2}, \\ 0, & \text{if } \sqrt{k_f^2 - P^2} > \kappa, \end{cases} \quad (27.59)$$

$$\tilde{Q}_0 = \frac{(\kappa^2 + P^2 - k_f^2)}{2P\kappa}. \quad \diamond \quad (27.60)$$

Energy of Nuclear Matter

To obtain the total energy of the many-nucleon system, we write the total Hamiltonian as a sum over the single-particle kinetic energies and the two-body interactions:

$$H = \sum_i^N K_i + \frac{1}{2} \sum_{i(\neq j)=1}^N v(i, j). \quad (27.61)$$

Just as the effect of the transition operator T on the exact (distorted) wave of scattering was the same as the effect of T on the unperturbed plane wave:

$$V|\psi\rangle = T|\phi\rangle, \quad (27.62)$$

so the effect of $v(i, j)$ on the exact wave function for two nucleons within nuclear matter is the same as G on uncorrelated two-particle states:

$$v|\Psi_{kl}\rangle = \mathcal{G}|\phi_k\phi_l\rangle. \quad (27.63)$$

If we now calculate the expectation value of the full Hamiltonian H with the exact state Ψ we obtain

$$E = \langle \Psi | H | \Psi \rangle = \left\langle \Psi \left| \sum_i^N K_i + \frac{1}{2} \sum_{i,j=1}^N v(i, j) \right| \Psi \right\rangle, \quad (27.64)$$

$$= \sum_i^N \langle i | K | i \rangle + \frac{1}{2} \sum_{i,j=1}^N [\langle ij | \mathcal{G} | ij \rangle - \langle ij | \mathcal{G} | ji \rangle], \quad (27.65)$$

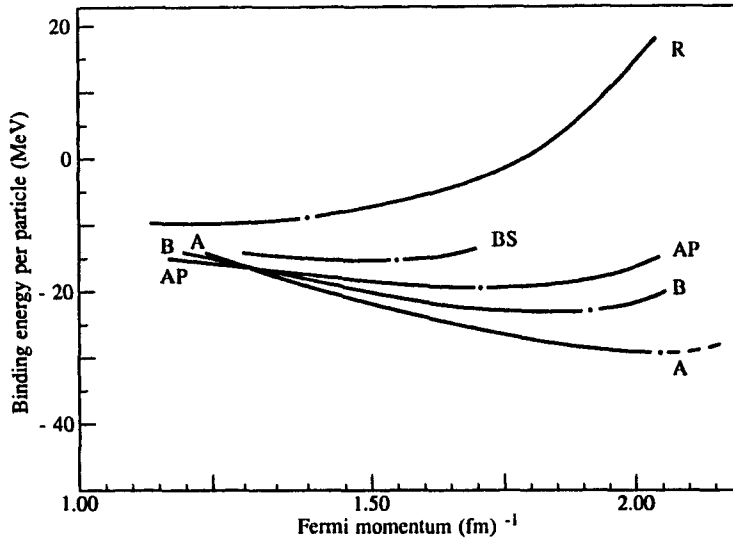


Figure 27.4: The total binding energy per nucleon as a function of the Fermi momentum k_f (equivalent to the dependence on the nuclear density). The different curves arise from different nucleon–nucleon potentials, with increased binding correlated with increased smoothness. (From Haftel and Tabakin, 1970).

where we have used theorem (26.24) regarding matrix elements of two-body operators between antisymmetrized states (which is how the exchange term enters), and the fact that both the exact and plane wave states are eigenstates of the kinetic energy operator. The term in (27.65) added to the kinetic energy describes the interaction of each nucleon with all others. By definition, its matrix element is the effective single-particle potential,

$$\langle i | U | i \rangle \equiv U_i \stackrel{\text{def}}{=} \sum_{j=1}^N [\langle ij | \mathcal{G} | ij \rangle - \langle ij | \mathcal{G} | ji \rangle]. \quad (27.66)$$

Equation (27.65) is essentially the same as the Hartree–Fock results of § 26.4 only now with $\mathcal{G}$ matrix elements rather than the actual potential. It is solved in much the same way, too. First, one chooses a convenient, that is, approximate, central potential U_i and solves the single-particle Schrödinger equation for $\{\phi_a\}$ and $\{\epsilon_a\}$ (27.46). Second, one uses these ϕ 's as basis states in solving the Bethe–Goldstone equation (27.44) or (27.55). [To do this, Q is determined via sums such as (27.49) or via approximate expressions such as (27.59).] The $\mathcal{G}$'s are then used to determine the potential U via (27.66), which in turn produces new and better wave functions ϕ_i 's, which are used to produce new and better $\mathcal{G}$'s, and so forth. The whole procedure is repeated until self-consistency is attained.

A prime prediction from these calculations is the binding energy per particle in nuclear matter. This is a particularly important quantity since it is (experimentally) approximately the same for all nuclei. In Figure 27.4 we show some results of these types of calculations. The binding energy of nuclear matter versus the Fermi momentum (or equivalently density) is plotted. The different curves corresponding to different nucleon–nucleon potentials. The minimum of each curve, the “saturation point,” has the maximum binding and thus

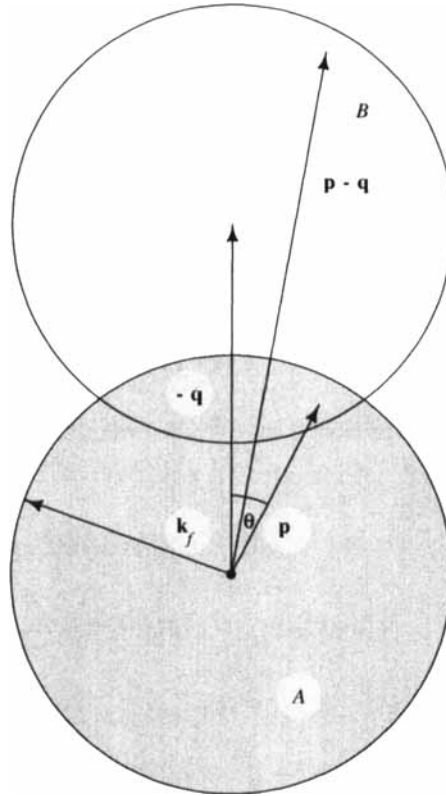


Figure 27.5: Two intersecting Fermi spheres A and B describing the scattering of a fermion with momentum $\mathbf{p}$ into one with $\mathbf{p} - \mathbf{q}$. The exclusion principle requires the final momentum $\mathbf{p} - \mathbf{q}$ to lie outside of the first sphere, while momentum conservation requires it to be within the second.

determines the equilibrium binding energy and density. The experimental result of -15.7 MeV/nucleon for the binding energy and 1.36 fm^{-1} for the Fermi momentum are close to the line of minima predicted with various nucleon–nucleon potentials. Presumably, other effects such as higher-order correlations, relativity, and meson and quark currents are needed to obtain even better agreement with experiment.

27.5 Problems

1. Relate the exchange potential of Thomas–Fermi–Dirac Theory to the two-body density function, and therefore to the electron–electron correlation function.
2. Use a variational principle to derive the Thomas–Fermi equation.
3. To understand the Pauli reduction factor Q , derive an explicit expression for it for the collision pictured in Figure 27.5 (a fermion of momentum $\mathbf{p}$ is scattered into one with $\mathbf{p}' = \mathbf{p} - \mathbf{q}$).

- (a) Show that the Pauli principle applied to this fermion is equivalent to taking an average of Q over all momentum $\mathbf{p} \equiv \mathbf{p}_k$ appropriate to a free Fermi gas:

$$Q_{\text{av}}(q) = \int d^3p \rho(p) \Theta(p'^2 - k_f^2). \quad (27.67)$$

- (b) Show that the values of $\mathbf{p} - \mathbf{q}$ allowed by the Pauli principle *and* momentum conservation lie inside sphere B and outside of sphere A (the clear region).
 (c) Show that the average Pauli factor Q is the ratio of volumes,

$$Q_{\text{av}} = \frac{4\pi k_f^3/3 - \text{excluded volume}}{4\pi k_f^3/3}, \quad (27.68)$$

where the spheres' overlap is the excluded volume.

- (d) Show that if the momentum transfer q is greater than $2k_f$, the spheres do *not* overlap, and so $Q_{\text{av}} = 1$.
 (e) Show that if $q < 2k_f$, the boundary of the excluded region is determined by

$$|\mathbf{q} + \mathbf{p}| = k_f, \quad \cos \theta_{qp} = \frac{k_f^2 - q^2 - p^2}{2qp}. \quad (27.69)$$

- (f) Show that the allowed volume $V = \pi(qk_f^2 - q^3/12)$, and so

$$Q_{\text{av}} = \begin{cases} \frac{3}{4} \left(q/k_f - q^3/(12k_f^3) \right), & \text{for } q \leq 2k_f, \\ 1, & \text{for } q > 2k_f. \end{cases} \quad (27.70)$$

CHAPTER 28

PHONONS

*Next, when I cast mine eyes and see
That brave vibration each way free,
—O how that glittering taketh me!*

—Robert Herrick

In this chapter we show how the techniques of Part III, *Quantum Fields*, are used to describe quantized lattice vibrations. Our discussion is only a sample of the richness in this field, and a student would benefit from reading specialized texts such as Taylor (1970), Callaway (1976), Ziman (1960, 1964), and Fetter and Walecka (1971).

28.1 Phonons

We have seen throughout our study of field theory that photons and mesons play dual roles. They are the virtual quanta whose exchanges are the origins of the electromagnetic and nuclear forces, as well as physical particles which can be formed into beams and used in laboratories. *Phonons* are the quanta of vibrational fields and have the same relation to sound as photons have to light; they do not, however, exist as real particles in free space.

In their simplest form, phonons arise in the model of a solid as a chain of inert ions connected by springs, symbolize in Figure 28.1. Because the number of degrees of freedom in a solid equals the number of atoms (a very, very large number), there are many *types* of phonons possible. In addition, as the intensity of the sound increases the numbers of phonons increases. Yet we know from the classical study of vibrations (Fetter and Walecka, 1980; Goldstein, 1977) that all possible vibrations can be expressed as the sum of *normal modes*, each normal mode containing vibrations with just one frequency. These normal modes are *collective excitations* involving a large fraction of the particles in the solid, yet are described with just one (collective) coordinate. Since atoms are heavy, their motion is essentially classical and we are able to generalize the classical analysis without much change.

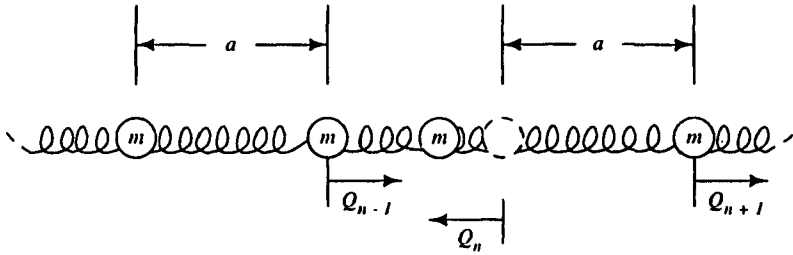


Figure 28.1: A linear chain of masses and springs. The equilibrium separation of the masses is a and the displacement from equilibrium of mass number n is Q_n .

Classical One-Dimensional Vibrations

Consider the infinite chain of atoms in Figure 28.1 connected by springs each with force constant K . Newton's second law of motion gives the classical equation of motion for mass m at position Q_n (measured relative to the mass's equilibrium position):

$$F = ma, \quad (28.1)$$

$$K[(Q_{n+1} - Q_n) - (Q_n - Q_{n-1})] = m \frac{d^2 Q_n}{dt^2}. \quad (28.2)$$

While springs may not appear realistic, (28.2) is equivalent to expanding the potential V for the entire chain as a power series in the displacements about the equilibrium position Q_n ,

$$V(Q_1, Q_2, \dots) = V(0, 0, \dots) + \frac{1}{2} \sum_{n,m} \left. \frac{\partial^2 V}{\partial Q_n \partial Q_m} \right|_0 Q_n Q_m + O(Q^3), \quad (28.3)$$

and assuming small Q_n so that the anharmonic (third- and higher-order) terms are negligible. Note, there is no linear term in (28.3) because the oscillations are about the equilibrium position where the forces $\partial V / \partial Q_n$ vanish. If we include only nearest-neighbor interactions and identify K as the second derivative of the potential, we obtain (28.2).

Because the chain is very long, only the phase of the solution to (28.2) changes as $n \rightarrow n + 1$. For normal modes of vibrations this means

$$Q_{n+1} = e^{i\phi} Q_n. \quad (28.4)$$

Yet because we must have invariance under translations of multiples of the link length a , and because we are looking for harmonic solutions, we guess

$$Q_n = C e^{i(k_n a - \omega_k t)}. \quad (28.5)$$

Here C is a normalization constant (or function in more complete analysis), k is the wavevector, and ω_k is the eigenfrequency associated with k .

Exercise Use (28.2) to show that the allowed frequencies for the chain are

$$\omega_n = \pm \omega_0 \sin\left(\frac{1}{2} k_n a\right), \quad \omega_0 = \sqrt{4 \frac{K}{m}}. \quad \diamond \quad (28.6)$$

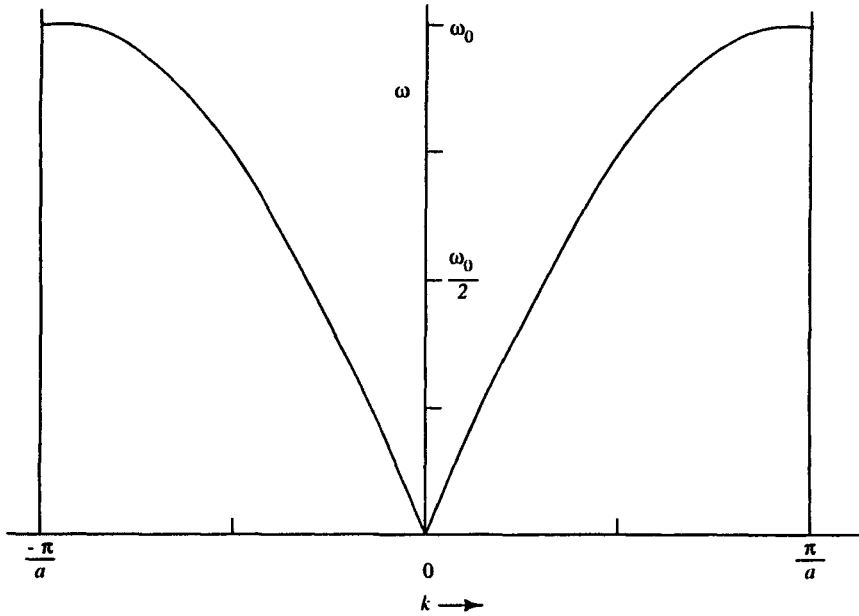


Figure 28.2: The dispersion curve for the linear chain of Figure 28.1.

The actual values of k_n are determined by imposing *periodic boundary conditions*¹ on (28.5), that is, requiring Q_n to match onto itself after N atoms,

$$Q_{n+N} = Q_n \Rightarrow k_n = \frac{2\pi}{Na} n \quad (n = 1, N). \quad (28.7)$$

The frequencies ω_k allowed for each k are given by (28.6) and graphed as the *dispersion curve* in Figure 28.2. We restrict the N allowed values of k to the range (first *Brillouin zone*):

$$-\frac{\pi}{a} \leq k \leq \frac{\pi}{a}. \quad (28.8)$$

Because N is very large, k is dense and uniform throughout this zone.

Quantized One-Dimensional Vibrations

We quantize the vibrations of the linear chain in four steps.

1. Write down the Hamiltonian for the chain.
2. Transform to generalized momenta and coordinates which describe the normal modes (H is then the sum of harmonic-oscillator Hamiltonians).
3. Quantize the vibrations by replacing the generalized coordinates by operators.
4. Use second quantization to describe the phonons.

¹ The details of the boundary conditions have very little effect on the deduced properties of phonons because the number of surface atoms is small relative to the total number of atoms. This is obviously not true for surface phonons.

The Hamiltonian (step 1) expressed in terms of the Cartesian coordinates $\{P_n, Q_n\}$ is

$$H = K + V \quad (28.9)$$

$$= \sum_n^N \frac{P_n^2}{2m} + \frac{1}{2} \sum_{n,m} Q_n Q_m \left. \frac{\partial^2 V}{\partial Q_i \partial Q_j} \right|_0, \quad (28.10)$$

where $P_n = m dQ_n/dt$ and where we leave off the equilibrium value of V because it just sets the zero of energy. Because we assume normal modes of vibration, the modes are uncoupled and thus no off-diagonal terms occur in (28.10). In step 2 we change from the microscopic coordinates $\{P_n, Q_n\}$ to generalized coordinates $\{p_i, q_i\}$ which diagonalize the force constant matrix $[\partial^2 V / \partial Q_i \partial Q_j]$. This converts:

$$\frac{1}{2} \sum_{n,m} Q_n Q_m \left. \frac{\partial^2 V}{\partial Q_n \partial Q_m} \right|_0 \Rightarrow \frac{1}{2} \sum_i q_i q_i^\dagger K_{ii}, \quad (28.11)$$

where the Hermitian adjoint $q_i^\dagger$ appears in (28.11) because the generalized coordinates may be complex. Fortunately and miraculously, when the kinetic energy is expressed in terms of the conjugate, canonical variable $p_i = -i\partial/\partial q_i$, those normal coordinates which make the potential matrix diagonal also maintain the diagonal form of the kinetic energy:²

$$\sum_n \frac{P_n^2}{2m} = \sum_i \frac{p_i p_i^\dagger}{2m}. \quad (28.12)$$

The Hamiltonian for the linear chain in normal coordinates is accordingly

$$H = \sum_i \left[\frac{p_i p_i^\dagger}{2m} + \frac{1}{2} K_{ii} q_i q_i^\dagger \right]. \quad (28.13)$$

Exercise Show (step 3) that the normal coordinates are canonical by showing that they satisfy the commutation relations

$$[q_i, p_j] = i\delta_{ij}. \quad \diamond \quad (28.14)$$

Exercise Show that the normal-mode frequencies are

$$\omega_i = \sqrt{\frac{K_{ii}}{m}}. \quad \diamond \quad (28.15)$$

For step 4, we observe that each of these oscillators (normal modes) has energy $N_i \omega_i$, where we identify N_i as the number of phonons in this mode—as if these phonons are particles. We thus define the phonon creation and destruction operators:

$$a_i^\dagger = \frac{1}{\sqrt{2m\omega_i}} (m\omega_i q_i^\dagger - ip_i), \quad (28.16)$$

$$a_i = \frac{1}{\sqrt{2m\omega_i}} (m\omega_i q_i + ip_i^\dagger). \quad (28.17)$$

²This is also true in the classical study of small vibrations.

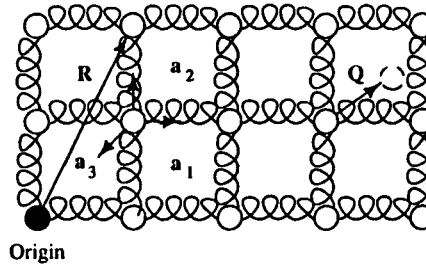


Figure 28.3: A 3D lattice of masses and springs. The vectors $(\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3)$ are elementary lattice vectors, $\mathbf{R}$ is the location of the equilibrium position of the mass relative to the origin (filled circle), and $\mathbf{Q}$ is the displacement of a mass from its equilibrium position.

Exercise Show that the phonon field is described by the second-quantized Hamiltonian,

$$H = \sum_i \omega_i (a_i^\dagger a_i + \frac{1}{2}), \quad (28.18)$$

and for N_i phonons with frequency ω_i , the energy of the chain is

$$E = \langle H \rangle = \sum_i (N_i + \frac{1}{2}) \omega_i. \quad \diamond \quad (28.19)$$

As shown in the Problems section, the generalized coordinates are expressed in terms of the creation and destruction operators as

$$q_i = \frac{1}{\sqrt{2m\omega_i}} (a_i^\dagger + a_i), \quad (28.20)$$

$$p_i = i\sqrt{\frac{m\omega_i}{2}} (a_i^\dagger - a_{-i}). \quad \diamond \quad (28.21)$$

Three-Dimensional Phonons

We extend the linear-chain model to one for a 3D solid by assuming that the actual lattice forces may be modeled as in Figure 28.3 by interconnected springs with appropriate force constants.³ Because the atoms in our model solid occupy all of space, the bookkeeping is more complicated than for a linear chain, but not the basic physics. The location of any atom is determined by the vector

$$\mathbf{R} = n_1 \mathbf{a}_1 + n_2 \mathbf{a}_2 + n_3 \mathbf{a}_3, \quad (28.22)$$

where $\{a_i\}$ are the primitive translation vectors and $\{n_i\} \equiv \mathbf{R}$ are three integers. The displacement from equilibrium of the atom at $\mathbf{R}$ is given by the vector $\mathbf{Q}_R$ (or $\mathbf{Q}_R^I$,

³The values of the force constants or more realistic potentials can be found using the many-body techniques of Chapter 26 and Chapter 27.

with $I = 1, 3$ for components). The Hamiltonian for the 3D lattice in the harmonic approximation has a simple structure, as before, but now with four sums:

$$H = \sum_{R,I}^{N,3} \frac{(P_R^I)^2}{2m} + \frac{1}{2} \sum_{R,R',I,J}^{N,N,3,3} K_{RR'}^{IJ} Q_R^I Q_{R'}^J, \quad (28.23)$$

where $[K]$ is a matrix formed from the second derivatives of the potential [as in (28.11)].

The normal modes of the 3D solid are also described by coordinates which make both the potential and kinetic energy matrices diagonal. The diagonalization is handled with matrix techniques familiar from classical vibrations and produces the Hamiltonian

$$H = \sum_{R,e}^{N,3} \left(\frac{p_R^e p_R^e}{2m} + \frac{1}{2} K_R^e q_R^e q_R^e \right). \quad (28.24)$$

Here the R sum runs over atoms, q_R are generalized coordinates equivalent to the atomic displacement, and the e sum runs over the basis vectors ($\hat{e}_1, \hat{e}_2, \hat{e}_3$) which diagonalize $[K]$. The directions $\hat{e}_i$ may differ for different wave vectors $\mathbf{k}$, yet for each $\mathbf{k}$ they are the directions in which the crystal vibrates collectively.⁴

We quantize 3D vibrations by replacing the second quantization's slot index i by the phonon's wave vector and polarization ($\mathbf{k}, e$). The creation and destruction operators are then

$$a_{\mathbf{k}e}^\dagger = \frac{1}{\sqrt{2m\omega_{\mathbf{k}e}}} (m\omega_{\mathbf{k}e} q_{\mathbf{k}}^e - ip_{\mathbf{k}}^e), \quad (28.25)$$

$$a_{\mathbf{k}e} = \frac{1}{\sqrt{2m\omega_{\mathbf{k}e}}} (m\omega_{\mathbf{k}e} q_{\mathbf{k}}^e + ip_{\mathbf{k}}^e). \quad (28.26)$$

Phonons of wave vector⁵ $\mathbf{k}$ have a frequency $\omega_{\mathbf{k}e}$ given in terms of the diagonal elements of the force constant matrix by

$$\omega_{\mathbf{k}e} = \sqrt{\frac{K_R^e}{M}}. \quad (28.27)$$

When these relations are substituted into (28.23), we obtain the 3D second-quantized phonon Hamiltonian

$$H_p = \sum_{\mathbf{k}e} \omega_{\mathbf{k}e} \left(a_{\mathbf{k}e}^\dagger a_{\mathbf{k}e} + \frac{1}{2} \right). \quad (28.28)$$

We are ready now to do field theory.

⁴Unless the wave is traveling along some symmetry axis of the crystal, the phonon's polarization is not pure but rather some combination of longitudinal and transverse.

⁵The vector $\mathbf{k}$ is now in 3D momentum space yet with a complication. Since for a crystal there is also translational invariance, in momentum space this means $\mathbf{k} + \mathbf{K}$ is equivalent to $\mathbf{k}$ for certain vectors $\mathbf{K}$ of the reciprocal lattice. Accordingly, momentum may not always appear to be conserved.

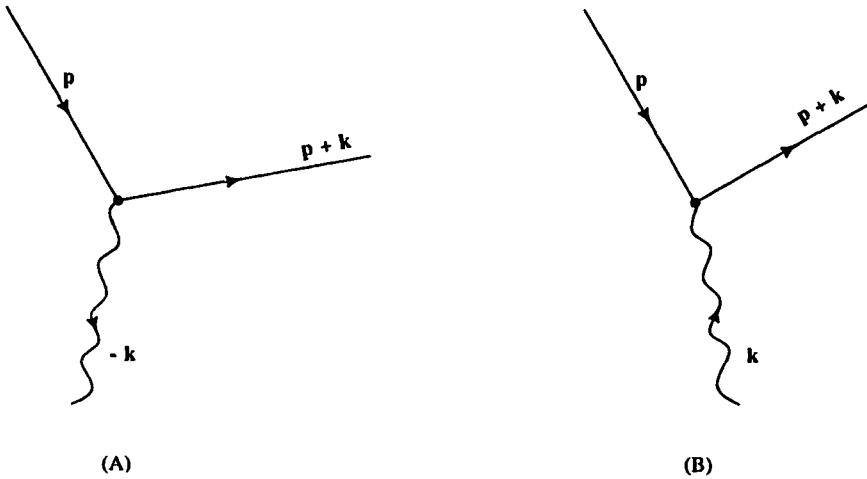


Figure 28.4: Feynman diagrams for (A) phonon creation by an electron and (B) photon absorption by an electron.

28.2 Electron-Phonon Interactions

The phonon Hamiltonian (28.19) or (28.28) by itself is useful in understanding energy transfer processes in solids and indeed is the basis of the Debye model of heat capacities.⁶ To show the similarities of photon and phonon interactions, we deduce how phonons couple to electrons by imagining electrons moving independently (and rapidly) through the lattice in Figure 28.3. If we ignore the zero-point energies, the unperturbed second-quantized Hamiltonian for electrons and phonons is

$$H_0 = \sum_{\mathbf{p}} E_{\mathbf{p}} b_{\mathbf{p}}^{\dagger} b_{\mathbf{p}} + \sum_{\mathbf{k}, \epsilon} \omega_{\mathbf{k}\epsilon} a_{\mathbf{k}\epsilon}^{\dagger} a_{\mathbf{k}\epsilon}, \quad (28.29)$$

where the first term “counts” the number $b_{\mathbf{p}}^{\dagger} b_{\mathbf{p}}$ of electrons with momentum $\mathbf{p}$ (we ignore the electron’s spin) and the second counts the number of phonons with wave vector $\mathbf{k}$ and polarization $\hat{\epsilon}$.

The electron-phonon interaction arises when an electron causes an ion to move slightly and thereby sets the crystal vibrating. If an electron of momentum $\mathbf{p}$ is scattered into momentum $\mathbf{p}'$ by the ion potential V (Figure 28.4), we describe this with the second-quantized interaction Hamiltonian,

$$H_I = \sum_{\mathbf{p}', \mathbf{p}, \mathbf{R}} \langle \mathbf{p}' | V(\mathbf{r} - [\mathbf{R} + \mathbf{Q}_R]) | \mathbf{p} \rangle b_{\mathbf{p}'}^{\dagger} b_{\mathbf{p}}, \quad (28.30)$$

where $\mathbf{r}$ is the electron’s position and $\mathbf{R} + \mathbf{Q}$ is the ion’s.

⁶A quantum classic, described by Kittel (1971) and Taylor (1970).

Exercise Use the completeness relation to express the matrix element in (28.30) in momentum space:

$$H_I = \sum_{\mathbf{p}', \mathbf{p}, \mathbf{R}} e^{i(\mathbf{p}-\mathbf{p}') \cdot (\mathbf{R} + \mathbf{Q}_R)} V(\mathbf{p}' - \mathbf{p}) b_{\mathbf{p}}^\dagger b_{\mathbf{p}}, \quad (28.31)$$

where $V(\mathbf{p}' - \mathbf{p})$ is the Fourier transform of $V(\mathbf{r})$. $\diamond$

Because the ion displacement $\mathbf{Q}_R$ is assumed small, the exponential in (28.31) may be expanded (as we did in the electromagnetic dipole approximation):

$$e^{i(\mathbf{p}-\mathbf{p}') \cdot \mathbf{Q}_R} \simeq 1 + i(\mathbf{p} - \mathbf{p}') \cdot \mathbf{Q}_R + \dots \quad (28.32)$$

We ignore the 1 in (28.32) because it is independent of the ion's displacement and thus does not couple electrons to phonons. To obtain the coupling to normal modes (phonons), we express $\mathbf{Q}_R$ in (28.31) in terms of the collective variable $q_{\mathbf{k}}$:

$$\mathbf{Q}_R = \sqrt{\frac{1}{N}} \sum_{\mathbf{k}} e^{i\mathbf{k} \cdot \mathbf{R}} q_{\mathbf{k}}. \quad (28.33)$$

Here we have made use of the result proven in the Problems section that the linear relation between collective and microscopic variables is also a sum over wave vectors. The part of interaction Hamiltonian (28.31) dependent upon the lattice displacement is thus

$$H_I \simeq \frac{i}{\sqrt{N}} \sum_{\mathbf{p}', \mathbf{p}} (\mathbf{p} - \mathbf{p}') \cdot q_{\mathbf{p}' - \mathbf{p}} V_{\mathbf{p}' - \mathbf{p}} b_{\mathbf{p}}^\dagger b_{\mathbf{p}}. \quad (28.34)$$

We obtain the Fröhlich Hamiltonian, quantized in both electron and phonon spaces, from (28.34) by expressing the coordinate q in terms of the creation and annihilation operators (28.20):

$$H_I = \sum_{\mathbf{p}', \mathbf{p}} i \sqrt{\frac{N}{2m\omega_{\mathbf{k}}}} |\mathbf{p} - \mathbf{p}'| V_{\mathbf{p}' - \mathbf{p}} (a_{-\mathbf{k}}^\dagger + a_{\mathbf{k}}) b_{\mathbf{p}}^\dagger b_{\mathbf{p}}, \quad (28.35)$$

where $\mathbf{k} = \pm(\mathbf{p}' - \mathbf{p})$ because the phonon is created or destroyed by the electron's momentum transfer. The phonon creation and annihilation processes described by this Hamiltonian are pictured in Figure 28.4. The interaction and the calculational rules are analogous to those of photons.

28.3 Problems

1. Let $\mathbf{R}_i$ and $\mathbf{R}_j$ be the position of two ions in a 3D lattice. Show that the lattice potential V depends on $\mathbf{R}_i$ and $\mathbf{R}_j$ only through the combination $\mathbf{R}_i - \mathbf{R}_j$.
2. Show that the collective variables for the normal modes of a linear chain satisfy the canonical commutation relations (28.14):

$$[q_i, p_j] = i\delta_{ij}. \quad (28.36)$$

Why does this permit the classical expressions to be quantized?

3. Show that for the linear chain, the normal-mode frequencies are given by (28.15):

$$\omega_i = \sqrt{K_{ii}/m}. \quad (28.37)$$

4. Show that the generalized coordinates for the linear chain can be expressed in terms of the creation and destruction operators as (28.20),

$$q_i = \frac{a_i^\dagger + a_i}{\sqrt{2m\omega_i}}, \quad p_i = i\sqrt{m\omega_i/2}(a_i^\dagger - a_i). \quad (28.38)$$

5. Show that the generalized coordinates for a 3D lattice satisfy the commutation relations

$$[(q_{\mathbf{R}}^e)^\dagger, p_{\mathbf{R}'}^{e'}] = i\delta(\mathbf{R} - \mathbf{R}')\delta_{ee'}, \quad (28.39)$$

$$[(q_{\mathbf{R}}^e)^\dagger, q_{\mathbf{R}'}^{e'}] = [(p_{\mathbf{R}}^e)^\dagger, p_{\mathbf{R}'}^{e'}] = [q_{\mathbf{R}}^e, p_{\mathbf{R}'}^{e'}] = 0. \quad (28.40)$$

6. Prove that phonons of wave vector $\mathbf{k}$ have a frequency (28.27):

$$\omega_{\mathbf{k}e} = \sqrt{K_q^e/m}. \quad (28.41)$$

7. Use the results of Chapter 26 and Chapter 27 on many-body theory to

- (a) estimate the force constant for a solid,
- (b) estimate the phonon frequency for a solid,
- (c) estimate the phonon speed in a solid.

8. Consider the symmetry in wave vector space (*reciprocal lattice space*) arising from the symmetry of a lattice.

- (a) Define what is meant by $(\hat{\mathbf{b}}_1, \hat{\mathbf{b}}_2, \hat{\mathbf{b}}_3)$, the basis vectors of reciprocal lattice space.
- (b) Show that if

$$\mathbf{K} = n_1\hat{\mathbf{b}}_1 + n_2\hat{\mathbf{b}}_2 + n_3\hat{\mathbf{b}}_3, \quad (28.42)$$

the vectors $\mathbf{k} + \mathbf{K}$ and $\mathbf{k}$ are equivalent, that is, they both correspond to phonons of the same frequency.

9. Show that the linear relation between the collective and microscopic variables, which must also satisfy translational invariance, is a sum over wave vectors:

$$Q_{\mathbf{R}} = \frac{1}{\sqrt{N}} \sum_{\mathbf{k}} e^{i\mathbf{k} \cdot \mathbf{R}} q_{\mathbf{k}}. \quad (28.43)$$

10. Fill in the step needed to prove (28.33).

CHAPTER 29

FERMION PAIRING

*After some minutes of just standing there saying nothing, he said,
"We did it last night."*

—J. Bardeen, as related to author by C. Slichter.

One of the most fascinating consequences of quantum mechanics is that in spite of the exclusion principle, fermions confined to a metal or a nucleus sometimes bind into stable pairs. These pairs are known as *Cooper pairs* and lead to the phenomenon of *superconductivity* in metals and pairing in nuclei. While the technical details of these phenomena can get quite complicated when real materials or nuclei are studied, in this chapter we use simple models to examine the basic physics of fermion pairing. We use the second quantization formalism and focus our study on how paired fermions sometimes behave like bosons and sometimes behave like fermions. A treatment of Cooper pairs without the second quantization formalism is given by Baym (1969). A more complete and realistic treatment of the Bardeen, Cooper and Schrieffer (BCS) theory of superconductivity is found in Lipkin (1973), Koltun and Eisenberg (1988), Fetter and Walecka (1971), and Schrieffer (1964).

29.1 Why Pairs?

In our discussion of many-body problems in Chapter 26 and Chapter 27, we saw that when a large number of fermions are confined to a region of space, they behave like a gas of independent particles experiencing some average, central potential. The crystal structure or presence of other fermions is secondary in nature. This is primarily a consequence of the *Pauli principle* which *blocks* many of the fermion–fermion interactions because the final states are occupied, and thereby permits the fermions to move as if they were free particles.

In some systems of bound fermions there exists an even more subtle consequence of the Pauli principle. The identical fermions near the Fermi surface may develop an effective attraction to each other which leads to bound pairs with integer spin. Because the number of these correlated pairs can be large, pairing can lead to a coherence throughout the whole

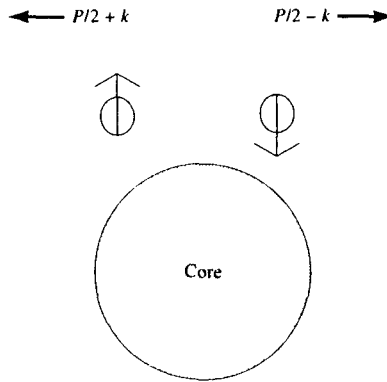


Figure 29.1: A correlated pair of interacting fermions with opposite spin, total momentum $\mathbf{P}$, and relative momentum $\mathbf{k}$. These fermions lie just outside the Fermi surface and are in the presence of a passive core of other identical or nonidentical particles.

material which is so strong that a macroscopic amount of energy is needed to break even one pair.

The type of pairing we envision is illustrated in Figure 29.1. We see fermions with opposite spin, total momentum $\mathbf{P}$, and relative momentum $\mathbf{k}$. In a superconductor's ground state, the total momentum $\mathbf{P}$ of each pair is zero; in nuclei, the angular momentum of each pair is zero.

In addition to understanding pairing because of its important applications, in this chapter we also try to shed some light on the quantum mechanical rule-of-thumb that an odd number of fermions behave like a fermion, while an even number of fermions behave like a boson. In our discussion of the particle concept in Chapter 1, *Scattering*, we indicated that if there is not enough energy to excite the internal degrees of freedom of a system, then the composite nature of the system will not be evident. In the present chapter we are more microscopic. We will show that if the density of the many body system is low, the bound pairs do not overlap much and their composite nature is not evident; they behave as bosons. When the density of the system is high, the fermion nature of the elementary constituents becomes more evident.

In developing a model of superconductivity, we must deal with a large and variable number of particles correlated through the Pauli principle. To avoid the horrendous accounting associated with explicitly antisymmetrizing many-body wave functions, and to handle pair creation and annihilation, we use the second-quantized formalism. To help with the algebraic manipulations we introduce *quasispin* operators. With them we need to perform only the familiar angular momentum algebra (the tedium of angular momenta being acceptable because it is woven into the fabric of quantum mechanics).

Origin of Pairing Attraction

In most of this chapter we will look at the quantum mechanics of pairing within a fermion gas. A realistic description of the attractive force responsible for pairing is left to the more technical discussions of the references. In nuclei, the attraction arises from the short range part of the nucleon–nucleon interaction which remains after the longer range (average

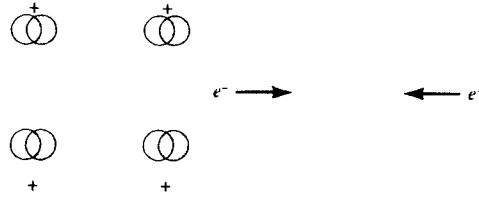


Figure 29.2: The attractive pairing force on the electron to the right arises from the displacement of the lattice toward the electron to the left.

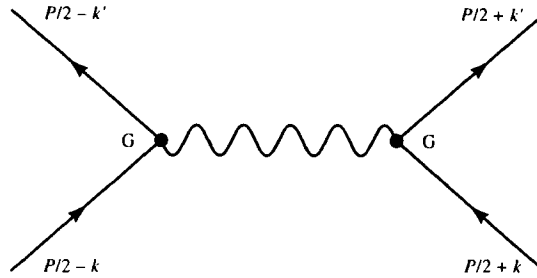


Figure 29.3: The exchange of a phonon between two electrons, which leads to the attractive pairing force.

Hartree–Fock) potential is removed. For this reason this nuclear interaction is often called the *residual interaction*.

In a superconductor the pairing force arises from an electron–lattice–electron interaction. A classical model of this interaction is given in Figure 29.2 and a quantum model in Figure 29.3. Normally an electron in a metal travels freely through a neutral background composed of a negatively-charged electron gas plus positively charged ions. In some materials the polarization of the lattice by one electron leads to a short-range attractive force on a different electron. Specifically, as the electron on the left in Figure 29.2 moves to the right, the positive ions are attracted to it and move ever-so-slightly to the right. While the ions eventually return to their equilibrium positions, they move slowly and if in the meantime there is another electron nearby on the right, it will see an excess positive charge to the left and be attracted to it.

Quantum mechanically, the pairing force arises from the exchange of a phonon between two electrons, as illustrated in Figure 29.3. Phonons are discussed in Chapter 28, *Phonons*, while the similar exchange of a scalar meson between nucleons is discussed in Chapter 23, *The Breit–Pauli and Meson-Exchange Interactions*. In fact, in equations (23.44)–(23.44) we evaluate the Feynman diagram corresponding to Figure 29.3 and find it to have the form:

$$\mathcal{M}_{fi} = G^2 \frac{2\pi}{V} \frac{\delta_{s_1 s'_1} \delta_{s_2 s'_2}}{[E(\mathbf{P}/2 - \mathbf{k}) - E(\mathbf{P}/2 - \mathbf{k}')]^2 - \omega_q^2}, \quad (29.1)$$

where G is the strength of the electron–phonon coupling. Since Cooper pairs occur for scattering near the Fermi surface, many of the final scattering states are blocked by the Pauli principle. Of course putting the electrons back into their original states is not blocked, and this corresponds to zero momentum transfer q for which the electron energies in (29.1) are

essentially equal and the matrix element is

$$\mathcal{M}_{fi} \simeq -G^2 \frac{2\pi}{V} \frac{\delta_{s_1 s'_1} \delta_{s_2 s'_2}}{\omega_0^2}. \quad (29.2)$$

As also occurs for scalar meson exchange, we have an attractive interaction. Further, we have already investigated constant T matrix elements in Chapter 6, *The Transition and Potential Matrices*, and found them to correspond to zero-range forces. Likewise, the pairing interaction (29.2) is a zero-range force. If we improve upon the approximations, we find that the pairing force is very short ranged and nonlocal.

As indicated in Figure 29.1, the two fermions which pair have opposite spins. This configuration gives greater binding than the parallel spin configuration because of the exchange interaction (discussed in connection with the Hartree–Fock equations in Chapter 26, *Many-Body Problems*). The short range pairing force is most effective when the fermions are close together in space, and this occurs for the space-symmetric S state (orbital angular momentum zero). But because the fermions are identical particles, they must have a totally antisymmetric wave function. This requires the spin state to be antisymmetric, that is, spin zero.

29.2 The Fermion Pair State

We represent the two particles pictured in Figure 29.1 with the ket $|A_{\mathbf{P}}\rangle$ (a ray in a two-particle Hilbert space). We use an uppercase A to indicate the composite nature of the pair and the uppercase $\mathbf{P}$ to indicate the total momentum of the pair.

Pair Wave Packet

The variables needed for a two-particle wave function have already been described in equations (1.17)–(1.20). The two-particle wave function is described by (1.25) which we rewrite as

$$\Psi_{\mathbf{P}}(\mathbf{r}_1, \mathbf{r}_2) = \langle \mathbf{R}, \mathbf{r} | A_{\mathbf{P}} \rangle \quad (29.3)$$

$$= \Phi_{\text{CM}}(\mathbf{R})\psi(\mathbf{r}) = e^{i\mathbf{P}\cdot\mathbf{R}}\psi(\mathbf{r}). \quad (29.4)$$

Here the relative and center-of-mass coordinates and momenta are

$$\mathbf{r} = \mathbf{r}_1 - \mathbf{r}_2, \quad \mathbf{R} = \frac{\mathbf{r}_1 + \mathbf{r}_2}{2}, \quad (29.5)$$

$$\mathbf{k} = \frac{\mathbf{k}_1 - \mathbf{k}_2}{2}, \quad \mathbf{P} = \mathbf{k}_1 + \mathbf{k}_2. \quad (29.6)$$

To keep things simple we ignore the spin variables.

A physical pair localized in space requires a wave packet containing a spectrum of momentum values. In § 1.1 we represented a wave packet as an integral over momentum states, as is appropriate in the continuum limit. However, similar to the quantization of photons, we work here with the simpler basis of plane waves in a box. In this case the expansions are sums over a finite number of discrete momenta with the volume of the box (or a quantity proportional to it) appearing explicitly in some of the equations. Details

are given in Appendix A, *Natural Units and Plane Waves*, as well as simple rules for transforming between “little” and “big” boxes.

We represent the relative wave function $\psi(\mathbf{r})$ of a localized pair as a wave packet built up from relative momentum states with Fourier components (spectral function) $F_{\mathbf{k}}$:

$$\Psi_{\mathbf{P}}(\mathbf{r}_1, \mathbf{r}_2) = \frac{e^{i\mathbf{P}\cdot\mathbf{R}}}{\sqrt{V}} \sum_{i=1}^{\Omega} F_{\mathbf{k}_i} \frac{e^{i\mathbf{k}_i\cdot\mathbf{r}}}{\sqrt{V}} \quad (29.7)$$

$$\equiv \frac{e^{i\mathbf{P}\cdot\mathbf{R}}}{\sqrt{V}} \sum_{\mathbf{k}} F_{\mathbf{k}} \frac{e^{i\mathbf{k}\cdot\mathbf{r}}}{\sqrt{V}}, \quad (29.8)$$

where in (29.7) we have left off the discrete subscript i on $\mathbf{k}$. The maximum value of the sum's index, Ω , is the number of single-particle levels which can exist in the box, and is proportional to the volume of the box. In practice, the number of single states Ω is a model-dependent constant which better be larger than the number of particles placed in the box if our description is to make any sense.

Since the plane-wave basis states are in a box they have a finite norm, and so we can also normalize the pair wave function to one:

$$\langle \Psi_{\mathbf{P}} | \Psi_{\mathbf{P}} \rangle = \iint d\mathbf{r} d\mathbf{R} \Psi_{\mathbf{P}}^*(\mathbf{r}, \mathbf{R}) \Psi_{\mathbf{P}}(\mathbf{r}, \mathbf{R}) = 1. \quad (29.9)$$

Exercise Substitute the pair wave packet expansion (29.7) into the normalization condition (29.9) and show that this leads to the normalization of the spectral function,

$$\sum_{i=1}^{\Omega} |F_{\mathbf{k}_i}|^2 = 1. \quad \diamond \quad (29.10)$$

Exercise Show that the spectral function is the Fourier transform of the relative wave function:

$$F_{\mathbf{k}} = \int d\mathbf{r} \frac{e^{-i\mathbf{k}\cdot\mathbf{r}}}{\sqrt{V}} \psi(\mathbf{r}). \quad \diamond \quad (29.11)$$

Pair State in Hilbert Space

We have expressed a normalized wave packet for a pair in terms of relative and CM coordinates. In developing the second-quantized theory, it is more convenient to have the wave packet expressed in terms of the individual particle coordinates. To do that we combine the exponentials in (29.7) and write

$$\mathbf{P} \cdot \mathbf{R} + \mathbf{k} \cdot \mathbf{r} = \mathbf{P} \cdot \frac{\mathbf{r}_1 + \mathbf{r}_2}{2} + \mathbf{k} \cdot (\mathbf{r}_1 - \mathbf{r}_2), \quad (29.12)$$

$$\Rightarrow \Psi_{\mathbf{P}}(\mathbf{r}_1, \mathbf{r}_2) = \sum_{\mathbf{k}} F_{\mathbf{k}} \frac{e^{i(\mathbf{P}/2 + \mathbf{k})\cdot\mathbf{r}_1}}{\sqrt{V}} \frac{e^{i(\mathbf{P}/2 - \mathbf{k})\cdot\mathbf{r}_2}}{\sqrt{V}}. \quad (29.13)$$

Let us now look at (29.13) in terms of the representation theory used in Appendix B, *Dirac Notation and Representations*, and Part I, *Scattering and Integral Quantum Mechanics*.

We recognize the LHS as the $\langle \mathbf{r}_1, \mathbf{r}_2 |$ representation of the pair wave packet, and the RHS as the product of the $\langle \mathbf{r}_1 |$ and $\langle \mathbf{r}_2 |$ representations of plane waves:

$$\langle \mathbf{r}_1, \mathbf{r}_2 | \Psi_{\mathbf{P}} \rangle = \sum_{\mathbf{k}}^{\Omega} F_{\mathbf{k}} \langle \mathbf{r}_1 | \mathbf{P}/2 + \mathbf{k} \rangle \langle \mathbf{r}_2 | \mathbf{P}/2 - \mathbf{k} \rangle. \quad (29.14)$$

Because the two particles are in different spaces, $\langle \mathbf{r}_1 |$ and $\langle \mathbf{r}_2 |$ are independent rays. Since the two-particle space $\langle \mathbf{r}_1, \mathbf{r}_2 |$ is just the direct product $\langle \mathbf{r}_1 | \otimes \langle \mathbf{r}_2 | \equiv \langle \mathbf{r}_1 | \langle \mathbf{r}_2 |$, we have the Hilbert-space representation of a localized pair state as

$$|\Psi_{\mathbf{P}}\rangle = \sum_{\mathbf{k}}^{\Omega} F_{\mathbf{k}} |\mathbf{P}/2 + \mathbf{k}\rangle |\mathbf{P}/2 - \mathbf{k}\rangle = \sum_{\mathbf{k}}^{\Omega} F_{\mathbf{k}} |\mathbf{P}/2 + \mathbf{k}, \mathbf{P}/2 - \mathbf{k}\rangle. \quad (29.15)$$

Pair State in Fock Space

Fock space is occupation number space in which particles can be created and annihilated. In the Fock-space formalism described in Chapter 19, *Second Quantization*, the two-particle ket on the RHS of (29.15) is created by starting with the vacuum state $|0\rangle$ and *first* placing an electron with momentum $\mathbf{P}/2 + \mathbf{k}$ in slot 1, and *then* placing an electron with momentum $\mathbf{P}/2 - \mathbf{k}$ in slot 2:

$$|\Psi_{\mathbf{P}}\rangle = \sum_{\mathbf{k}}^{\Omega} F_{\mathbf{k}} a_{\mathbf{P}/2+\mathbf{k}}^{\dagger} a_{\mathbf{P}/2-\mathbf{k}}^{\dagger} |0\rangle, \quad (29.16)$$

where $a_{\mathbf{p}}^{\dagger}$ is the creation operator for an electron of momentum $\mathbf{p}$.¹

The ket on the LHS of (29.16) represents one pair of fermions with total momentum $\mathbf{P}$ and with enough of a spread of internal momenta to produce some space localization. In the Fock space view, the pair can also be created directly from the vacuum by the action of the *pair* creation operator $A_{\mathbf{P}}^{\dagger}$:

$$|\Psi_{\mathbf{P}}\rangle \stackrel{\text{def}}{=} A_{\mathbf{P}}^{\dagger} |0\rangle \equiv |A_{\mathbf{P}}\rangle. \quad (29.17)$$

The relation of the composite Fock-space operator to the elementary Fock-space operators can be read off from (29.16) and (29.17) after stripping off the kets:

$$A_{\mathbf{P}}^{\dagger} = \sum_{\mathbf{k}}^{\Omega} F_{\mathbf{k}} a_{\mathbf{P}/2+\mathbf{k}}^{\dagger} a_{\mathbf{P}/2-\mathbf{k}}^{\dagger}, \quad (29.18)$$

$$A_{\mathbf{P}} = \sum_{\mathbf{k}}^{\Omega} F_{\mathbf{k}} a_{\mathbf{P}/2-\mathbf{k}} a_{\mathbf{P}/2+\mathbf{k}}, \quad (29.19)$$

where the pair annihilation operator (29.19) is the Hermitian adjoint of the creation operator (29.18), and where for simplicity we assume the spectral function $F_{\mathbf{k}}$ is real. Equation (29.18) says that to create one pair with momentum $\mathbf{P}$ and some space localization, you sum over Ω electrons with momentum $\mathbf{P}/2 + \mathbf{k}$ and Ω electrons with momentum $\mathbf{P}/2 - \mathbf{k}$,

¹ You will recall that the momentum label is actually discrete, since the electron states are in a box.

using a weighting factor (spectral function) $F_{\mathbf{k}}$. Different values for the spectral function produce different degrees of localization.

To extend the theory to include the spin degree of freedom for the electrons, we need only add a spin index to the subscript:

$$A_{\mathbf{P}}^{\uparrow} = \sum_{\mathbf{k}} F_{\mathbf{k}} a_{\mathbf{P}/2+\mathbf{k}\uparrow}^{\dagger} a_{\mathbf{P}/2-\mathbf{k}\downarrow}^{\dagger}, \quad (29.20)$$

$$A_{\mathbf{P}} = \sum_{\mathbf{k}} F_{\mathbf{k}} a_{\mathbf{P}/2-\mathbf{k}\downarrow} a_{\mathbf{P}/2+\mathbf{k}\uparrow}, \quad (29.21)$$

where we have built in the constraint that the pairs have zero total spin.

29.3 Are Pairs Elementary?

The pair state (29.16) really does not look very complicated and so it should have simple properties. Nevertheless, many body problems are inherently complicated, and while ours is a relatively simple one, we still need a fair amount of commutator algebra before the simple results follow. Therefore, we first determine the commutation relations of the pair creation and annihilation operators with the particle operators, and then the commutation relations of the pair operators with other pair operators and with some dynamical operators. After we have answered the question as to whether the pairs are elementary, we study a simple model of interacting pairs and of superconductivity.

The building blocks of any Fock-space theory are the commutation and anticommutation relations for bosons and fermions:

$$[a_i, a_j^{\dagger}] = \delta_{ij}, \quad [a_i, a_j] = [a_i^{\dagger}, a_j^{\dagger}] = 0 \quad (\text{bosons}), \quad (29.22)$$

$$\{a_i, a_j^{\dagger}\} = \delta_{ij}, \quad \{a_i, a_j\} = \{a_i^{\dagger}, a_j^{\dagger}\} = 0 \quad (\text{fermions}), \quad (29.23)$$

where we remind the reader that these relations have concrete meaning when acting on states containing particles. When a number of creation and destruction operators are juxtaposed, they can usually be simplified by use of the identities:

$$[A, BC] = B[A, C] + [A, B]C, \quad (29.24)$$

$$[AB, C] = A[B, C] + [A, C]B. \quad (29.25)$$

Particle–Pair Correlations

The nonvanishing of the commutators of a_i and $a_i^{\dagger}$ indicate the effect of the Pauli principle when we try to place two particles in the same state. This is the origin of the *Pauli correlations* in many particle wave functions. Analogously, we evaluate the commutator of a particle destruction and a pair creation operator (to see if particles and pairs are correlated):

$$\begin{aligned} [a_{\mathbf{P}\uparrow}, A_{\mathbf{P}}^{\dagger}] &= \sum_{\mathbf{k}} F_{\mathbf{k}} [a_{\mathbf{P}}, a_{\mathbf{P}/2+\mathbf{k}\uparrow}^{\dagger} a_{\mathbf{P}/2-\mathbf{k}\downarrow}^{\dagger}] = \sum_{\mathbf{k}} F_{\mathbf{k}} [a_{\mathbf{P}}, a_{\mathbf{P}/2+\mathbf{k}\uparrow}^{\dagger}] a_{\mathbf{P}/2-\mathbf{k}\downarrow}^{\dagger} \\ &= F_{\mathbf{P}-\mathbf{P}/2} a_{\mathbf{P}-\mathbf{P}\downarrow}^{\dagger}. \end{aligned} \quad (29.26)$$

Here we have suppressed the spin index in the intermediate steps and have used the fact that a state can contain only one fermion per slot.

Exercise Fill in the steps leading to (29.26). ◇

Similar manipulations produce the remaining particle–pair commutation relations:

$$[a_{\mathbf{p}\uparrow,\downarrow}^\dagger, A_{\mathbf{p}}] = [a_{\mathbf{p}\uparrow,\downarrow}, A_{\mathbf{p}}] = 0, \quad (29.27)$$

$$[a_{\mathbf{p}\downarrow}, A_{\mathbf{p}}^\dagger] = -F_{\mathbf{p}/2-\mathbf{p}} a_{\mathbf{p}-\mathbf{p}\uparrow}^\dagger, \quad (29.28)$$

$$[a_{\mathbf{p}\downarrow}^\dagger, A_{\mathbf{p}}] = F_{\mathbf{p}/2-\mathbf{p}} a_{\mathbf{p}-\mathbf{p}\uparrow}^\dagger, \quad (29.29)$$

$$[a_{\mathbf{p}\uparrow}^\dagger, A_{\mathbf{p}}] = -F_{\mathbf{p}-\mathbf{p}/2} a_{\mathbf{p}-\mathbf{p}\downarrow}. \quad (29.30)$$

Even though we have constructed the electron pair to have zero spin, it is not truly a boson or an elementary particle. (If the pair were an elementary boson, all of these pair-particle commutation relations would vanish.) However, if the Fourier component $F_{\mathbf{p}/2-\mathbf{p}}$ could be made smaller and smaller, the RHS's of the commutation relations would become smaller and smaller, and the pair would behave more and more like an elementary boson. This makes sense. We know from basic Fourier analysis that a high degree of localization is obtained by combining many small and fairly constant Fourier components. Many small $F_{\mathbf{k}}$'s lead to a highly localized pair which is unlikely to have much overlap with a single electron, and so the pair appear elementary.

On a more microscopic level we make the same argument by analyzing the wave packet normalization constraint (29.10). For the sum over a large number Ω of nearly equal components, we expect

$$\Omega |F_{\mathbf{k}}|^2 \simeq 1 \quad \Rightarrow \quad |F_{\mathbf{k}}| \simeq \frac{1}{\sqrt{\Omega}}. \quad (29.31)$$

Yet from the counting of states in a box (A.22), we know that the volume and number of states are proportional,

$$\Omega \propto V, \quad (29.32)$$

$$\Rightarrow |F_{\mathbf{k}}| \propto \frac{1}{\sqrt{V}}. \quad (29.33)$$

So again, if a pair and an electron are confined to a macroscopic box which has a volume much greater than the microscopic volume of the pair, we expect the pair to behave like an elementary boson.

Pair–Pair Correlations

We have just determined that if a system is dilute enough, an electron and a pair will behave as separate elementary particles, but as the probability of overlap increases, the composite nature of the pair becomes more evident. If the system contains only pairs, we might expect them to behave like identical bosons if their concentration is low, and somewhat like fermions if their concentration is high. To see if this is true, we examine the

commutation relations between pair operators. We define the state containing N_A pairs, each with momentum $\mathbf{P}$ as:

$$|N_A\rangle \stackrel{\text{def}}{=} (A_{\mathbf{P}}^\dagger)^{N_A} |0\rangle. \quad (29.34)$$

Exercise Use the electron–pair commutation relations to show that:

$$[a_{\mathbf{P}\uparrow}, (A_{\mathbf{P}}^\dagger)^{N_A}] = N_A F_{\mathbf{P}-\mathbf{P}/2} (A_{\mathbf{P}}^\dagger)^{N_A-1} a_{\mathbf{P}-\mathbf{P}\uparrow}^\dagger. \quad \diamond \quad (29.35)$$

As expected, (29.35) shows that an electron has some interference with an N_A –pair state, and that the resulting correlation (lack of commutivity) is proportional to the number of pairs present.

To see how two pairs interfere with each other, we evaluate

$$\begin{aligned} [A_{\mathbf{P}'}, A_{\mathbf{P}}^\dagger] &= \sum_{\mathbf{k}} F_{\mathbf{k}} \left(a_{\mathbf{P}'/2-\mathbf{k}\downarrow} [a_{\mathbf{P}'/2+\mathbf{k}\uparrow}, A_{\mathbf{P}}^\dagger] \right. \\ &\quad \left. + [a_{\mathbf{P}'/2-\mathbf{k}\downarrow}, A_{\mathbf{P}}^\dagger] a_{\mathbf{P}'/2+\mathbf{k}\uparrow} \right). \end{aligned} \quad (29.36)$$

The result is important, and so we rewrite it as

$$\begin{aligned} [A_{\mathbf{P}'}, A_{\mathbf{P}}^\dagger] &= \delta_{\mathbf{P}',\mathbf{P}} - \Delta_{\mathbf{P}',\mathbf{P}}, \\ \Delta_{\mathbf{P}',\mathbf{P}} &= \sum_{\mathbf{k}} F_{\mathbf{k}} \left(F_{\mathbf{P}'/2-\mathbf{P}/2+\mathbf{k}} a_{\mathbf{P}-\mathbf{P}'/2-\mathbf{k}\downarrow}^\dagger a_{\mathbf{P}'/2-\mathbf{k}\downarrow} \right. \\ &\quad \left. + F_{\mathbf{P}'/2-\mathbf{P}/2-\mathbf{k}} a_{\mathbf{P}-\mathbf{P}'/2+\mathbf{k}\uparrow}^\dagger a_{\mathbf{P}'/2+\mathbf{k}\uparrow} \right). \end{aligned} \quad (29.38)$$

We observe that $\Delta_{\mathbf{P}',\mathbf{P}}$ is a measure of the composite nature of the pairs. Since the sum in the definition of Δ is over the number of single-particle states, and since $|F_{\mathbf{k}}|^2$ is proportional to $1/V$, we observe that Δ is proportional to the density of pairs (the Problems section at the end of the chapter explores this more analytically). Again, this is as expected; if there is little chance for the pairs to overlap, they behave like elementary bosons; if there are pairs nearly everywhere in the box, their behavior is not like that of elementary particles.

29.4 Simple Model of Pairs

Our description of pairs is simple. We have assumed that each pair has momentum $\mathbf{P}$, is in a box of volume $V \propto \Omega$, and is described by the same wave packet (same $F_{\mathbf{k}}$'s). We now simplify the description further by assuming the spectral function $F_{\mathbf{k}}$ is a constant for all the states used to construct the wave packet:

$$F_{\mathbf{k},i} = \begin{cases} 1/\sqrt{\Omega}, & \text{for } i \leq \Omega, \\ 0, & \text{otherwise.} \end{cases} \quad (29.39)$$

Here Ω is the number of plane wave states we use in our expansions, and the value of $F_{\mathbf{k}}$ is chosen to satisfy the normalization condition (29.10). With this model, the pair creation and destruction operators are the sums over Ω elementary terms:

$$A_{\mathbf{P}}^{\dagger} = \frac{1}{\sqrt{\Omega}} \sum_{\mathbf{k}} a_{\mathbf{P}/2+\mathbf{k},\uparrow}^{\dagger} a_{\mathbf{P}/2-\mathbf{k},\downarrow}^{\dagger}, \quad (29.40)$$

$$A_{\mathbf{P}} = \frac{1}{\sqrt{\Omega}} \sum_{\mathbf{k}} a_{\mathbf{P}/2-\mathbf{k},\downarrow} a_{\mathbf{P}/2+\mathbf{k},\uparrow}. \quad (29.41)$$

It will be convenient for us to define an effective “pair-number operator” $\tilde{N}_A$ using the standard form for a number operator and treating $A_{\mathbf{P}}$ as if it were an elementary operator:

$$\tilde{N}_A \stackrel{\text{def}}{=} A_{\mathbf{P}}^{\dagger} A_{\mathbf{P}} \quad (29.42)$$

$$= \frac{1}{\Omega} \sum_{\mathbf{k}',\mathbf{k}} a_{\mathbf{P}/2+\mathbf{k},\uparrow}^{\dagger} a_{\mathbf{P}/2-\mathbf{k},\downarrow}^{\dagger} a_{\mathbf{P}/2+\mathbf{k}',\downarrow} a_{\mathbf{P}/2-\mathbf{k}',\uparrow}. \quad (29.43)$$

Only for a dilute system will $\tilde{N}_A$ acting on a state have as its eigenvalue the number of pairs; as the system gets more concentrated, $\tilde{N}_A$ acts more like an operator which annihilates two fermions and creates another two. With this number operator and with our simplifying model (29.39), the expression for $\Delta_{\mathbf{P}\mathbf{P}}$ (the measure of compositeness) (29.38), takes the particularly simple form:

$$\Delta_{\mathbf{P}\mathbf{P}} = \sum_{\mathbf{k}} \left(F_{\mathbf{k}}^2 \tilde{N}_{\mathbf{k}\downarrow} + F_{\mathbf{k}}^2 \tilde{N}_{\mathbf{k}\uparrow} \right) = \frac{1}{\Omega} \sum_{\mathbf{k}} \left(\tilde{N}_{\mathbf{k}\downarrow} + \tilde{N}_{\mathbf{k}\uparrow} \right) \quad (29.44)$$

$$= \frac{\tilde{N}_A}{\Omega}, \quad (29.45)$$

where in (29.45) we have replaced the sum over explicit number operators by the total number operator.

One nice thing about working with number operators is that they have nice commutation relations regardless of whether they refer to bosons or fermions:

$$\begin{aligned} [\tilde{N}_i, a_j^{\dagger}] &= a_i^{\dagger} \delta_{ij}, & [\tilde{N}_i, a_j] &= -a_i \delta_{ij}, \\ [\tilde{N}_i, (a_i^{\dagger})^m] &= m(a_i^{\dagger})^{m-1}, & [\tilde{N}_i, \tilde{N}_j] &= 0. \end{aligned} \quad (29.46)$$

We use these commutation relations and the associativity relations (29.24)–(29.25) to determine relations which will be useful in constructing the Hamiltonian. For example,

$$\begin{aligned} [\Delta_{\mathbf{P}\mathbf{P}}, A_{\mathbf{P}}^{\dagger}] &= \frac{1}{\Omega} \sum_{\mathbf{k}',\mathbf{k}} F_{\mathbf{k}'} (\tilde{N}_{\mathbf{k}'}, a_{\mathbf{P}/2+\mathbf{k}}^{\dagger} a_{\mathbf{P}/2-\mathbf{k}}^{\dagger}) \\ &= \frac{1}{\Omega} \sum_{\mathbf{k}',\mathbf{k}} F_{\mathbf{k}'} (a_{\mathbf{P}/2+\mathbf{k}}^{\dagger} [\tilde{N}_{\mathbf{k}'}, a_{\mathbf{P}/2-\mathbf{k}}^{\dagger}] + [\tilde{N}_{\mathbf{k}'}, a_{\mathbf{P}/2+\mathbf{k}}^{\dagger}] a_{\mathbf{P}/2-\mathbf{k}}^{\dagger}) \\ &= \frac{1}{\Omega} \sum_{\mathbf{k}} F_{\mathbf{k}} (a_{\mathbf{P}/2+\mathbf{k}}^{\dagger} a_{\mathbf{P}/2-\mathbf{k}}^{\dagger} + a_{\mathbf{P}/2+\mathbf{k}}^{\dagger} a_{\mathbf{P}/2-\mathbf{k}}^{\dagger}) \end{aligned} \quad (29.47)$$

$$[\Delta_{\mathbf{P}\mathbf{P}}, A_{\mathbf{P}}^\dagger] = \frac{2A_{\mathbf{P}}^\dagger}{\Omega}, \quad (29.48)$$

where for clarity we have left off the spin indices.

Exercise Follow similar steps to show that

$$[\Delta_{\mathbf{P}\mathbf{P}}, A_{\mathbf{P}}] = \frac{-2A_{\mathbf{P}}}{\Omega}. \quad \diamond \quad (29.49)$$

Quasispin

The tedium of working out commutation relations is relieved somewhat by switching to *quasispin* operators which have the same commutation relations as angular momenta. While not related to true angular momenta, their conservation reflects some symmetry of the Hamiltonian. The z component of quasispin is defined as

$$S_z \stackrel{\text{def}}{=} -\frac{\Omega}{2} (1 - \Delta_{\mathbf{P}\mathbf{P}}) = -\frac{\Omega}{2} \left(1 - \frac{\tilde{N}}{\Omega}\right) = \frac{\tilde{N} - \Omega}{2}, \quad (29.50)$$

and therefore equals half of the difference between the total number of electrons and the number of single-particle states. The ladder operators in quasispin space are defined as

$$S_+ \stackrel{\text{def}}{=} \sqrt{\Omega} A_{\mathbf{P}}^\dagger = \sqrt{\Omega} \sum_{\mathbf{k}} a_{\mathbf{P}/2+\mathbf{k}\uparrow}^\dagger a_{\mathbf{P}/2-\mathbf{k}\downarrow}, \quad (29.51)$$

$$S_- \stackrel{\text{def}}{=} \sqrt{\Omega} A_{\mathbf{P}} = \sqrt{\Omega} \sum_{\mathbf{k}} a_{\mathbf{P}/2-\mathbf{k}\downarrow} a_{\mathbf{P}/2+\mathbf{k}\uparrow}, \quad (29.52)$$

where we have used the model relation $F_{\mathbf{k}} \equiv 1/\sqrt{\Omega}$. These operators derive their names and usefulness from their angular-momentum-like properties:

$$\begin{aligned} [S_z, S_\pm] &= \pm S_\pm, & [S_+, S_-] &= 2S_z, \\ [S^2, S_\pm] &= 0, & [S^2, S_z] &= 0. \end{aligned} \quad (29.53)$$

As is true for other angular momenta, there is a total quasispin operator:

$$\tilde{S}^2 \equiv S^2 = \frac{1}{2}(S_+ S_- + S_- S_+) + S_z^2, \quad (29.54)$$

$$\tilde{S}^2 |S, M\rangle = S(S+1) |S, M\rangle. \quad (29.55)$$

The action of the ladder operators are:

$$S_\pm |S, S_z\rangle = \sqrt{S(S+1) - S_z(S_z \pm 1)} |S, S_z \pm 1\rangle. \quad (29.56)$$

Quasispin Eigenstates

The definition of the z component of quasispin (29.50) tells us that its eigenvalue equals half the difference between the total number of electrons and the number of single-particle states. So if we have a state with m pairs of electrons, the z component of its quasispin is

$$S_z |m\rangle \equiv S_z (A_{\mathbf{P}}^\dagger)^m = \frac{N - \Omega}{2} |m\rangle = \frac{2m - \Omega}{2} |m\rangle, \quad (29.57)$$

where we note that m pairs corresponds to $N = 2m$ electrons.

Some special care is needed when dealing with the vacuum in a many-particle theory. Even though the vacuum contains no electrons ($N = 0$), when S_z acts on it we do not get zero:

$$S_z |0\rangle = \frac{N - \Omega}{2} |0\rangle = -\frac{\Omega}{2} |0\rangle, \quad (29.58)$$

$$\Rightarrow |0\rangle = |S_0, M_0 = -\Omega/2\rangle, \quad (29.59)$$

where S_0 is the total quasispin of the vacuum. The reader may well be wondering “why, if the vacuum is by definition empty, doesn’t $\Omega |0\rangle = 0$?” The answer is that Ω is the number of single particle states we sum over in our expansion of a general state, and even if there are no particles present in the general state, we still make the expansion. As such Ω is an artifact of our model. The reader’s faith in the vacuum may be restored somewhat by noting that you cannot lower its quasispin,

$$S_- |0\rangle = 0, \quad (29.60)$$

which follows from S_- (29.52) being composed purely of electron annihilation operators.

A nonvanishing z component of quasispin for the vacuum implies that the vacuum must have a total quasispin S_0 which is greater than or equal to the magnitude of the z component.

Exercise Use angular momentum algebra to show that

$$\tilde{S}^2 |0\rangle = \frac{\Omega}{2} \left(1 + \frac{\Omega}{2}\right) |0\rangle, \Rightarrow S_0 = \frac{\Omega}{2}. \quad \diamond \quad (29.61)$$

This is consistent with a z component of $-\Omega/2$ for the vacuum.

Even though we cannot remove an electron from the vacuum or lower the vacuum’s quasispin, because S_+ (29.51) is composed of electron creation operators, we can raise the vacuum’s quasispin:

$$\begin{aligned} S_+ |0\rangle &= S_+ |S_0, M_0\rangle = \sqrt{S_0(S_0 + 1) - M_0(M_0 + 1)} |S_0, M_0 + 1\rangle \\ &= \sqrt{\Omega} \left| \frac{\Omega}{2}, 1 - \frac{\Omega}{2} \right\rangle. \end{aligned} \quad (29.62)$$

It is logical to suppose that as we create more pairs in our system, the quasispin of the system increases. To see if this is true, we have the total quasispin operator act on a state of m pairs:

$$\tilde{S}^2 |m\rangle = \tilde{S}^2 (A_{\mathbf{P}}^\dagger)^m |0\rangle. \quad (29.63)$$

To evaluate this, we note that

$$S_+ = \sqrt{\Omega} A_{\mathbf{P}}^\dagger, \quad \text{and} \quad [S_+, \tilde{S}^2] = 0, \quad (29.64)$$

$$\Rightarrow [A_{\mathbf{P}}^\dagger, \tilde{S}^2] = 0, \quad (29.65)$$

$$\Rightarrow \tilde{S}^2 (A_{\mathbf{P}}^\dagger)^m |0\rangle = (A_{\mathbf{P}}^\dagger)^m \tilde{S}^2 |0\rangle = \frac{\Omega}{2} \left(1 + \frac{\Omega}{2}\right) |m\rangle. \quad (29.66)$$

Apparently, a state with m bound pairs is an eigenstate of $\tilde{S}^2$ with the same total quasispin as the vacuum.

29.5 The Pair's Potential

We continue with our simple model of the pair spectral function $F_{\mathbf{k}} \equiv 1/\sqrt{\Omega}$, and now include a model for the potential shown in Figure 29.3. In the second-quantized formalism, this potential annihilates a localized pair of momentum $\mathbf{P}$ and opposed spins, and then creates another localized (wavepacket) pair with the same momentum and opposed spins. The operator which does this is just the effective, pair-number operator $\tilde{N}_A$ (29.42), and so we take the potential as proportional to it:

$$V = -E_B \tilde{N}_A \equiv -E_B A_{\mathbf{P}}^{\dagger} A_{\mathbf{P}} \quad (29.67)$$

$$= \frac{-E_B}{\Omega} \sum_{\mathbf{k}, \mathbf{k}'} a_{\mathbf{P}/2+\mathbf{k}\uparrow}^{\dagger} a_{\mathbf{P}/2-\mathbf{k}\downarrow}^{\dagger} a_{\mathbf{P}/2-\mathbf{k}'\downarrow} a_{\mathbf{P}/2+\mathbf{k}'\uparrow}. \quad (29.68)$$

Here E_B is a constant which measures the total binding energy of the pair.² It is an extensive quantity related to the two-particle interaction strength G (which appears at each vertex in Figure 29.3) via

$$E_B = \Omega G. \quad (29.69)$$

The potential (29.68) is convenient to work with because the pair state $|A_{\mathbf{P}}\rangle$ is an eigenfunction of it:

$$\begin{aligned} V |A_{\mathbf{P}}\rangle &= -E_B \tilde{N}_A |A_{\mathbf{P}}\rangle \\ &= -E_B \tilde{N}_A A_{\mathbf{P}}^{\dagger} |0\rangle = -E_B A_{\mathbf{P}}^{\dagger} A_{\mathbf{P}} A_{\mathbf{P}}^{\dagger} |0\rangle \\ &\equiv -E_B \left(A_{\mathbf{P}}^{\dagger} A_{\mathbf{P}} A_{\mathbf{P}}^{\dagger} - A_{\mathbf{P}}^{\dagger} A_{\mathbf{P}}^{\dagger} A_{\mathbf{P}} \right) |0\rangle \\ &= -E_B A_{\mathbf{P}}^{\dagger} [A_{\mathbf{P}}, A_{\mathbf{P}}^{\dagger}] = -E_B A_{\mathbf{P}}^{\dagger} \left(1 - \frac{\tilde{N}_A}{\Omega} \right) |0\rangle \\ &= -E_B |A_{\mathbf{P}}\rangle. \end{aligned} \quad (29.70)$$

Exercise Fill in the steps in the derivation of (29.70), that is, substitute for the pair commutator in terms of $\Delta_{\mathbf{P}\mathbf{P}}$. $\diamond$

A potential $V \propto \tilde{N}_A$ may appear somewhat odd. First, it may seem inconsistent for $|A_{\mathbf{P}}\rangle$ to be an eigenstate of V since we do not truly have a definite number of pairs except in the low density limit. But since $\tilde{N}_A$ is not a true number operator, it is alright for it to have eigenvalues; it is well to remember that these eigenvalues equal the number of pairs only in the low density limit. Second, the potential always binds two electrons into a single type of localized pair with the same quantum number $\mathbf{P}$ (total momentum in our case) and with the spins opposed. A physical system may well contain electrons which are unbound, or which are paired with other values for the momentum. The theory can accommodate this, but the potential would have to be more complicated.

²Normally the binding energy derives from both potential and kinetic terms. As part of the simple models used in this chapter, we ignore kinetic energy effects and refer to the eigenvalue of the potential as a “binding energy”.

Potential–Quasispin Relation

As we previously warned, working with a potential which is the product of four operators can be tedious, and so it is more convenient to express the potential in terms of quasispin operators. To accomplish this, we note that the quasispin raising operator is proportional to $A^\dagger$ (29.51), and that the lowering operator is proportional to A (29.52). The potential V is proportional to the product of $S_+ = \sqrt{\Omega} A_{\mathbf{P}}^\dagger$, and $S_- = \sqrt{\Omega} A_{\mathbf{P}}$:

$$V = -\frac{E_B}{\Omega} S_+ S_- . \quad (29.71)$$

Because we usually deal with eigenstates of S^2 and S_z , the work is made even easier by expressing the product $S_+ S_-$ in terms of S and S_z :

$$V = -E_B A_{\mathbf{P}}^\dagger A_{\mathbf{P}} = -\frac{E_B}{\Omega} (S^2 - S_z^2 - S_z) . \quad (29.72)$$

Exercise Perform the angular momentum algebra leading to (29.72), taking care to remember that the total spin operator S^2 has the eigenvalue $S(S+1)$. $\diamond$

29.6 Multiple Pairs

Now that we have done our quasispin homework we can extract the physics of pairing. The action of V (29.72) on the m pair state is

$$\begin{aligned} V |m\rangle &= -\frac{E_B}{\Omega} S_+ S_- (A_{\mathbf{P}}^\dagger)^m |0\rangle \\ &= \frac{E_B}{\Omega} \left[\frac{\Omega}{2} \left(1 + \frac{\Omega}{2} \right) - \left(\frac{2m - \Omega}{2} \right)^2 - \left(\frac{2m - \Omega}{2} \right) \right] |m\rangle \\ &= -m E_B \left(1 - \frac{m-1}{\Omega} \right) |m\rangle . \end{aligned} \quad (29.73)$$

The eigenvalue here is a binding energy and is seen to be m times the binding E_B for one pair times a correction term which decreases the binding. The decrease is due to the Pauli principle blocking transitions to occupied final states and thereby decreasing the phase space (and effectively the strength) otherwise available to the interaction.

Given two, independent, electron states, we must be able to construct two, independent, pair states. We have constructed one pair state which describes two bound electrons, and guess that there is another one in which the electrons interact but are *not bound*. Accordingly we define an operator $B^\dagger$ which creates a *nonbound* pair of electrons:

$$|B\rangle \stackrel{\text{def}}{=} B^\dagger |0\rangle . \quad (29.74)$$

Since the states are independent, we also demand that they be orthogonal,

$$\langle A_{\mathbf{P}} | B \rangle = 0 . \quad (29.75)$$

To determine the binding energy of $|B\rangle$, we observe that the quasispin operations carried on with $|A_{\mathbf{P}}\rangle$ can also be carried on with $|B\rangle$. For example:

$$S_z |B\rangle = \left(1 - \frac{\Omega}{2}\right) |B\rangle, \quad (29.76)$$

$$S_- |B\rangle = 0, \quad (29.77)$$

$$\tilde{S}^2 |B\rangle = \left(\frac{\Omega}{2} - 1\right) \left(\frac{\Omega}{2} - 1 + 1\right) |B\rangle \Rightarrow S_B = \frac{\Omega}{2} - 1. \quad (29.78)$$

These properties tell us that the action of the potential on the nonbound pair is simple:

$$V |B\rangle = -\frac{E_B}{\Omega} S_+ S_- |B\rangle = 0. \quad (29.79)$$

Therefore, $|B\rangle$ is a two-electron state with zero binding energy and is orthogonal to the bound state $|A\rangle$.

Mixed Bound and Nonbound States

We can also use the second-quantized formalism to build states with a combination of bound and nonbound pairs. For example, the state

$$|1, B\rangle \stackrel{\text{def}}{=} A_{\mathbf{P}}^\dagger |B\rangle \quad (29.80)$$

has one bound pair and one unbound pair of electrons (or equivalently, one pair of electrons plus two nonbound electrons). Conveniently enough, this simplest of mixed states is an eigenstate of the potential:

$$V |1, B\rangle = -\frac{E_B}{\Omega} S_+ S_- |1, B\rangle = -E_B \left(1 - \frac{2}{\Omega}\right) |1, B\rangle. \quad (29.81)$$

The binding energy of $|1, B\rangle$ is thus the product of E_B , the binding energy for a single pair, times a correction factor which shows twice the Pauli reduction as occurs for two bound pairs in (29.73). Clearly, unbound pairs block twice the phase space as do bound pairs.

The state with v nonbound electrons (*not* pairs),

$$|v\rangle \stackrel{\text{def}}{=} B^{v/2} |0\rangle, \quad (29.82)$$

is said to have *seniority*³ v . (Because the nonbound pairs are created before the bound pairs, as shown in (29.80), they have greater seniority.) The nonbound state is also an eigenstate of S_z at the bottom of the quasispin multiplet:

$$S_z |v\rangle = -\frac{\Omega - v}{2} |v\rangle, \quad S_- |v\rangle = 0. \quad (29.83)$$

The ket $|v\rangle$ has $S_v = (\Omega - v)/2$.

We now build states with m bound pairs and v unbound electrons as:

$$|m, v\rangle \stackrel{\text{def}}{=} (A_{\mathbf{P}})^m |v\rangle = (A_{\mathbf{P}})^m B^{v/2} |0\rangle. \quad (29.84)$$

³The symbol v for seniority was introduced by Racah, and comes from the Hebrew word for seniority, *vethek*.

Even this mixed state is an eigenstate of quasispin:

$$S^2 |m, v\rangle = \frac{\Omega - v}{2} \left(\frac{\Omega - v}{2} + 1 \right) |m, v\rangle, \quad (29.85)$$

$$S_z |m, v\rangle = \frac{2m + v - \Omega}{2} |m, v\rangle, \quad (29.86)$$

but with a different value of quasispin than the pure states. Consequently this mixed state with m bound pairs and v nonbound electrons is also an eigenstate of the potential:

$$V |m, v\rangle = E_{m,v} |m, v\rangle, \quad (29.87)$$

$$E_{m,v} = -mE_B \left(1 - \frac{m-1+v}{\Omega} \right). \quad (29.88)$$

We see here a binding energy m times the binding energy of one pair, reduced by a Pauli suppression factor which accounts for the pairs and nonbound electrons present.

29.7 Superconductivity

BCS theory models the ground state of a metallic superconductor as a state containing a large number m of bound pairs and no unbound electrons:

$$|\psi_{\text{BCS}}^{(0)}\rangle \simeq |m, 0\rangle = (A_0^\dagger)^m |0\rangle, \quad (29.89)$$

where we have given all the ground-state pairs zero total momentum. We assume the kinetic energies of the pairs and of the nonbound electrons are small compared with the potential energy so that all the binding derives from the potential. The binding energy of the ground state is according to (29.88):

$$E^{(0)} = E_{m,v=0} = -mE_B \left(1 - \frac{m-1}{\Omega} \right). \quad (29.90)$$

In normal conductors, current flow occurs when an electric field is applied and electrons are pulled through the Fermi surface into conduction bands. Because the energy levels of a normal conductor are very closely spaced, even a minuscule amount of energy is enough to move some electrons. Once in the conduction bands, the electrons behave as if they are in the continuum. In contrast, superconductors are characterized by a rather large energy gap between the ground state and the first excited state, and by the fact that once this gap energy is applied to the superconductor, a current exists for long periods of time with essentially zero resistance (and thus without further application of electric fields).

In BCS theory, the first excited state corresponds to the state $|m-1, s=2\rangle$ in which a single bound pair is broken up into two nonbound electrons. If we assume the ground and excited-state pairs have the same total momentum, the energy of this first excited state is

$$E^{(1)} = E_{m-1,v=2} = -(m-1)E_B \left(1 - \frac{m-2+2}{\Omega} \right). \quad (29.91)$$

The energy gap is the difference between the excited- and ground-state energies:

$$\begin{aligned} \Delta E &= E^{(1)} - E^{(0)} = -(m-1)E_B \left(1 - \frac{m-2+2}{\Omega} \right) + mE_B \left(1 - \frac{m-1}{\Omega} \right) \\ &= E_B \equiv \Omega G, \end{aligned} \quad (29.92)$$

where G is the elementary two-particle interaction strength shown in Figure 29.3.

Equation (29.92) appears very reasonable in stating that the gap energy equals the binding energy of a single, isolated bound pair. This is, however, an unexpected and important result. In a superconductor, all electrons at the Fermi surface form pairs with the same momentum $\mathbf{P}$, and so there is a very large overlap of the electron pairs. Consequently, the number of pairs m is a macroscopically large number, comparable to the number of atoms in the material. This means that m and the number of single-particle states Ω are comparable in size, and we would expect most of the phase space for the two nonbound electrons to be blocked by a Pauli factor $\simeq 1 - m/\Omega$. This would lead to a gap energy that is vanishingly small, say on the order of the potential strength G . Equation (29.92) tells us that the gap energy is Ω times larger, where Ω is a macroscopic number for a superconductor.

The very large gap energy arises from all the bound-electron pairs in a superconductor being correlated with each other; you cannot change one pair without affecting all others. This long-range correlation arises from the Pauli principle requirement that the many-electron wave function for the entire metal be antisymmetric under the interchange of any two electrons. (As you will recall, the short-range correlation which forms the pairs arises from the electron–lattice–electron interaction.) While the theory we have developed may not appear to impose this antisymmetry upon the many-electron wave function, the theory has imposed it implicitly through our use of the second-quantized formalism.

In more realistic models of superconductors, the momenta of the excited and ground-state pairs may differ, the momentum dependence of the pairing force may be included, and a nonconstant model for the spectral function $F_{\mathbf{k}}$ might be employed. This makes working out numbers more difficult, but does not change the essential physics of overlapping fermion pairs.

When the pairs in an excited state of a superconductor all have the same nonzero momentum, they carry a *pair current* that is quite different from the normal electronic current. Electronic currents obey Ohm's law and arise from a thermal equilibrium between electrons and lattice vibrations when an external electric field is applied to a conductor. The electrons can be described as taking a random walk through the metal during which they suffer a continual series of collisions. The electrons thereby acquire a statistical distribution of velocities, with the average or "drift" velocity and mean free path between collisions decreasing with increasing temperature. Pair currents, in contrast, have all the pairs moving with a common velocity. If the temperature is low enough, very few phonons or nonbound electrons have enough energy (on the order of the gap energy) to disturb the pairs, and so the pair current experiences very little resistance. The reason for the small number of collisions is that the pair current is collective and involves a macroscopic number of pairs, while the collisions are individual with a single phonon or nonbound electron. It is therefore very unlikely for a single phonon or nonbound electron at low temperature to have an energy comparable to the macroscopic gap energy (29.92).

29.8 Problems

1. Determine under what conditions the exchange of a scalar particle between two electrons leads to an attractive interaction.
2. The function $\Delta_{\mathbf{p},\mathbf{p}}$ defined by (29.38) measures the composite nature of the pairs

in a system. To make some sense of it, evaluate $\Delta_{\mathbf{P}\mathbf{P}}$ for $\mathbf{P}' = \mathbf{P}$.

- (a) Show that for real and symmetric F ,

$$\Delta_{\mathbf{P}\mathbf{P}} = \sum_{\mathbf{k}} F_{\mathbf{k}}^2 \left(\tilde{N}_{\mathbf{P}/2-\mathbf{k}\downarrow} + \tilde{N}_{\mathbf{P}/2+\mathbf{k}\uparrow} \right), \quad (29.93)$$

where $\tilde{N}$ is the number operator.

- (b) Use this sum to deduce that $\Delta_{\mathbf{P}\mathbf{P}}$ scales like the ratio of the number of pairs N_A to the number of electrons N_e :

$$\Delta_{\mathbf{P}\mathbf{P}} \sim \frac{N_A}{N_e}. \quad (29.94)$$

3. Prove (29.61).

APPENDIX A

NATURAL UNITS AND PLANE WAVES

A.1 Natural Units

It is common and useful to use natural units in derivations and problem solving. This serves to save time and make the equations more transparent by eliminating the physical constants which tend to clutter the equations. In contrast to the popular belief, once you have developed the knack of placing the constants back into the final answers, dimensional analysis is still possible. The basis of our natural units has Planck's constant $\hbar = 1$ and the speed of light $c = 1$.

To convert to natural units just take your formulas in conventional units and set $\hbar = 1$ and $c = 1$. The fine structure constant $\alpha = e^2/\hbar c \simeq 1/137.04$ becomes $\alpha = e^2 \simeq 1/137.04$. With these units, angular momenta are measured in $\hbar$'s, velocities in c 's, and masses as rest energies $mc^2 = m$ (for example the electron has $mc^2 = m = 0.511 \text{ MeV}$). Thus the Compton wavelength $\hbar/mc = 1/m$ is in inverse mass (energy) units. In summary:

$$\hbar = c = 1, \quad (\text{A.1})$$

$$\alpha = \frac{e^2}{\hbar c} = \frac{e^2}{4\pi\epsilon_0\hbar c} = e^2 \simeq \frac{1}{137.04}, \quad (\text{A.2})$$

$$\frac{\hbar}{mc} = \frac{1}{m} = \frac{1}{0.511 \text{ MeV}}. \quad (\text{A.3})$$

To convert from natural to conventional units, just insert the $\hbar$'s and c 's. In practice, the author finds it simplest to:

- Change mass m to mc^2 so it has energy units.
- Change e^2 to $\alpha = e^2/\hbar c$ so it is dimensionless.
- Insert $\hbar c$, which has the dimensions of energy $\times$ length, in either numerator or denominator to get the dimensions in the desired form (or close to it as described next).

- If needed, insert c , which has the dimension of length $\div$ time, in either numerator or denominator to get the dimensions correct (for example to convert length to inverse time).
- To obtain actual numbers for your answers, substitute the explicit values for the constants:

$$\begin{aligned} \hbar c &= 197.329 \text{ MeV fm} &= 1973.29 \text{ eV \AA}, \\ c &= 2.998 \times 10^8 \text{ m/sec}, \\ \hbar &= 6.582 \times 10^{-22} \text{ MeV sec} &= 1.055 \times 10^{-27} \text{ erg sec}. \end{aligned} \quad (\text{A.4})$$

For example, the Bohr radius of hydrogen (assuming an infinitely heavy proton) is $a_B = 1/me^2$ in natural units. In conventional units it is

$$a_B = \frac{1}{m_e e^2} = \frac{1}{m_e c^2} \frac{\hbar c}{e^2} = \frac{1}{0.511 \text{ MeV}} \times 137.04. \quad (\text{A.5})$$

Yet because we know a_B is a length, we now multiply by $\hbar c$ which cancels the energy unit in the denominator and inserts a length into the numerator:

$$a_B = \frac{137.036}{0.511 \text{ MeV}} \times (\hbar c = 197.33 \times 10^{-5} \text{ MeV \AA}) = 0.529 \text{ \AA} = 5.29 \times 10^{-9} \text{ cm}. \quad (\text{A.6})$$

A.2 Plane Waves in Little and Big Boxes

A plane wave is a mathematical abstraction, a solution to the wave equation which has constant phase along a 2D infinite plane. Although these may not be physically realizable, they are a convenient substitute for a wave packet of definite momentum and are the conventional basis for expanding the wave function of an interacting particle. The wave functions of quantum mechanics form a Hilbert space, that is, a linear vector space of infinite dimension.¹ Whereas the dynamical coordinates $\mathbf{r}$ and $\mathbf{p}$ of wave functions are continuous, the eigenvalues or parameters of these functions, such as the bound-state energies $E = -\kappa_i^2/2\mu$, are discrete. Any Hermitian Hamiltonian can be used to generate a complete, orthogonal set of wave functions. The free-particle Hamiltonian,

$$H_0 = \frac{\mathbf{p}^2}{2\mu} = -\frac{\nabla^2}{2\mu}, \quad (\text{A.7})$$

is particularly convenient because it generates the plane waves:

$$\begin{aligned} \tilde{\phi}_{\mathbf{k}}(\mathbf{r}) &= \mathbf{k} \phi_{\mathbf{k}}(\mathbf{r}), & k &= |\mathbf{k}|, \\ H_0 \phi_{\mathbf{k}}(\mathbf{r}) &= E_{\mathbf{k}} \phi_{\mathbf{k}}(\mathbf{r}), & E_{\mathbf{k}} &= k^2/2\mu, \\ \phi_{\mathbf{k}}(\mathbf{r}) &= N e^{i\mathbf{k} \cdot \mathbf{r}}, & N &= \begin{cases} (2\pi)^{-3/2} & \text{infinite domain,} \\ V^{-1/2} & \text{finite domain.} \end{cases} \end{aligned} \quad (\text{A.8})$$

For simplicity in developing the formalism (and a patina of mathematical rigor), it is useful to consider the plane waves as occupying a finite volume (a box). The box and the periodic boundary conditions we impose on the wave functions are just for convenience (scattered waves are certainly not periodic); eventually we go to the limit of an infinite domain.

¹Jackson (1962); Gottfried (1966).

Little Boxes

To determine the allowed eigenenergies, we place the plane waves (A.8) in a box of volume V with sides (L_x, L_y, L_z) , and demand that they satisfy periodic boundary conditions

$$\phi_{\mathbf{k}}(x + L_x, y + L_y, z + L_z) = \phi_{\mathbf{k}}(x, y, z), \quad (\text{A.9})$$

$$\Rightarrow (k_x L_x, k_y L_y, k_z L_z) = 2\pi(i_x, i_y, i_z). \quad (\text{A.10})$$

Here $(i_x, i_y, i_z) \equiv \mathbf{i}$ is a set of three positive or negative integers which determine the allowed, discrete wave vectors and thus energies:

$$\mathbf{k}_i = 2\pi\left(\frac{i_x}{L_x}, \frac{i_y}{L_y}, \frac{i_z}{L_z}\right), \quad E_i = \frac{k_i^2}{2\mu}. \quad (\text{A.11})$$

With these boundary conditions, the plane waves for different values of $\mathbf{i}$ and $\mathbf{j}$ are *orthogonal*. By choosing the normalization constant N we make the plane waves *orthonormal*:

$$\phi_{\mathbf{k}_i}(\mathbf{r}) \equiv \phi_i(\mathbf{r}) = \frac{e^{i\mathbf{k}_i \cdot \mathbf{r}}}{\sqrt{V}}, \quad (\text{A.12})$$

$$\Rightarrow \int d^3r \phi_i^*(\mathbf{r}) \phi_j(\mathbf{r}) = \delta_{ij}, \quad (\text{orthonormality}). \quad (\text{A.13})$$

Note that in the confined volume of the box, the variable $\mathbf{k}$ is discrete but the variable $\mathbf{r}$ is continuous (but limited). The discreteness of $\mathbf{k}_i$ leads to the *Kronecker delta function* in (A.13). Since the free Hamiltonian is Hermitian, plane waves form a *complete set* in which any solution of the Schrödinger equation $\psi(\mathbf{r})$ can be expanded:

$$\psi(\mathbf{r}) = \sum_i c_i \phi_i(\mathbf{r}). \quad (\text{A.14})$$

Orthonormality determines the c_i 's [multiply (A.14) by ϕ_i^* and integrate over $\mathbf{r}$]:

$$c_i = \int d^3r' \phi_i^*(\mathbf{r}') \psi(\mathbf{r}'). \quad (\text{A.15})$$

If we substitute this back into (A.14) and interchange the order of integration and summation, we obtain

$$\psi(\mathbf{r}) = \int d^3r' \left[\sum_i \phi_i^*(\mathbf{r}') \phi_i(\mathbf{r}) \right] \psi(\mathbf{r}'). \quad (\text{A.16})$$

Yet because (A.16) must be an identity, we identify the term in brackets as some kind of unit operator. This yields the *closure or completeness relation* for discrete states:

$$\sum_i \phi_i^*(\mathbf{r}') \phi_i(\mathbf{r}) = \delta(\mathbf{r}' - \mathbf{r}), \quad (\text{closure}). \quad (\text{A.17})$$

The Big Box

To obtain plane waves in an infinite domain, we let the box size approach infinity. In this limit of very large L and very large i , the index i is still an integer so $\Delta i \equiv 1$. The momenta

$\mathbf{k}_i$ in (A.11) remain finite but become continuous:

$$\frac{2\pi}{L_i} \Delta i \rightarrow d\mathbf{k}_i, \quad \Delta i_x \rightarrow \frac{L_x}{2\pi} dk_x, \quad (\text{A.18})$$

$$\sum \Delta i \rightarrow V \int \frac{d^3 k}{(2\pi)^3}. \quad (\text{A.19})$$

The relation (A.19) is the basis for the important result that the number of states in a volume V with momenta in the interval $\mathbf{k} \rightarrow \mathbf{k} + \Delta\mathbf{k}$ is:

$$dN = n_0 V \frac{d^3 \mathbf{k}}{(2\pi)^3}. \quad (\text{A.20})$$

Here n_0 is the number of states with the same momentum (for example, two for electrons with spins up and down in atoms, and four for nucleons with spins and isospins up and down in nuclei).

Equation (A.20) is often used to determine the Fermi momentum, $p_F = \hbar k_F = k_F$, for a gas of fermions confined to a box of volume V . If N electrons are placed in the box, they will progressively fill up all the levels until there are no particles left. The momentum at which all levels just get filled is k_F . Since N equals the momentum-space density times the momentum-space volume, we have

$$N = n_0 \frac{V}{(2\pi)^3} \int_0^{k_F} d^3 k = \frac{n_0 V}{6\pi^2} k_F^3(r), \quad (\text{A.21})$$

$$k_F^3 = \frac{6\pi^2 N/V}{n_0} = \frac{6\pi^2 \rho}{n_0}, \quad (\text{A.22})$$

$$k_F = \left(\frac{6\pi^2 \rho(r)}{n_0} \right)^{1/3}. \quad (\text{A.23})$$

Here $\rho = N/V$ is the number of particles per unit volume and, although the derivation assumes it to be a constant, it is sometimes taken to be a function of position r .

To generalize the closure relation (A.17) to a big box, we insert a $\Delta i \equiv 1$ into the sum in (A.17), and take the $L \rightarrow \infty$ limit:

$$\sum_i \Delta i \phi_i^*(\mathbf{r}') \phi_i(\mathbf{r}) = \delta(\mathbf{r}' - \mathbf{r}), \quad (\text{A.24})$$

$$\Rightarrow V \int \frac{d^3 k}{(2\pi)^3} \frac{e^{-i\mathbf{k} \cdot \mathbf{r}'}}{\sqrt{v}} \frac{e^{i\mathbf{k} \cdot \mathbf{r}}}{\sqrt{v}} = \delta(\mathbf{r}' - \mathbf{r}), \quad (\text{closure}). \quad (\text{A.25})$$

This gives the form for plane waves in an infinite domain:

$$\phi_i(\mathbf{r}) = \frac{e^{i\mathbf{k}_i \cdot \mathbf{r}}}{\sqrt{V}} \Rightarrow \phi_{\mathbf{k}}(\mathbf{r}) = \frac{e^{i\mathbf{k} \cdot \mathbf{r}}}{(2\pi)^{3/2}}. \quad (\text{A.26})$$

The orthogonality relation (A.13) for an infinite domain is now just the closure relation with a change of variable,

$$\delta_{ij} \rightarrow \int \frac{d^3 r}{(2\pi)^3} e^{-i\mathbf{k}' \cdot \mathbf{r}} e^{i\mathbf{k} \cdot \mathbf{r}} = \delta(\mathbf{k}' - \mathbf{k}) \quad (\text{orthogonality}). \quad (\text{A.27})$$

APPENDIX B

DIRAC NOTATION AND REPRESENTATIONS

This appendix is meant for reference. See elementary texts such as Merzbacher (1970) and Gottfried (1966) for a proper introduction, and Chapters 5–7 for applications.

B.1 Dirac Notation

A **state** is represented by the *ket* $|\psi\rangle$. This state is a ray in a linear vector space of infinite dimension, that is, an abstract vector in a Hilbert space.

A **dual or adjoint-space** state is represented by the *bra* $\langle\psi|$. The 1:1 correspondence between the spaces of kets and bras is shown by the adjoint operation:

$$\langle\psi| = |\psi\rangle^\dagger. \quad (\text{B.1})$$

The **scalar or inner product** of states $|\phi\rangle$ and $|\psi\rangle$ is given by the juxtaposed “bra-ket” = *braket*,

$$\langle\phi|\psi\rangle \equiv (\phi, \psi) = \langle\psi|\phi\rangle^*. \quad (\text{B.2})$$

General operators O or $\tilde{O}$ are objects which transform one state into another:

$$O|\psi\rangle = |\phi\rangle = |O\psi\rangle. \quad (\text{B.3})$$

Accordingly, $O|\psi\rangle$ is *not* proportional to $|\psi\rangle$ —although it may look that way.

An **operator** is formed by the juxtaposition $|\psi\rangle\langle\phi|$ of a ket and a bra. This is an operator and not a scalar product because it changes one ket into another.

A **complete set of states** is obtained as the eigenstates of any Hermitian operator H ,

$$H|\phi_a\rangle = a|\phi_a\rangle, \quad a = 1, \infty. \quad (\text{B.4})$$

A **basis** is formed by a complete set such as ϕ_a . Any state can be expanded as a sum of the ϕ_a ’s:

$$|\psi\rangle = \sum_a c_a |\phi_a\rangle, \quad c_a = \langle\phi_a|\psi\rangle. \quad (\text{B.5})$$

Here the sum is for discrete states and the integral is for continuum states. When integrating, there is a phase space factor, $\int \rightarrow \int d^3k / (2\pi)^3$. The quantity c_a is the probability amplitude for $|\psi\rangle$ to “contain” $|\phi_a\rangle$ or to be “at” a .

The **orthogonality relation** is

$$\langle \phi_a | \phi_{a'} \rangle = \begin{cases} \delta_{aa'}, & \text{discrete states,} \\ \delta(a - a') / \rho_a, & \text{continuum states.} \end{cases} \quad (\text{B.6})$$

In (B.6) the ρ_a is the *density-of-states factor*:

$$\rho_a \stackrel{\text{def}}{=} \frac{dN}{da} = \begin{cases} 1, & \text{for } \phi_a = \exp(i\mathbf{k} \cdot \mathbf{r}) / (2\pi)^{3/2}, \quad (\text{our choice}), \\ (2\pi)^{-3}, & \text{for } \phi_a = \exp(i\mathbf{k} \cdot \mathbf{r}), \quad (\text{others}). \end{cases} \quad (\text{B.7})$$

The **α representation** of a state is the expansion of that state:

$$|\psi\rangle = \sum_a c_a |\phi_a\rangle, \quad c_a = \langle \phi_a | \psi \rangle. \quad (\text{B.8})$$

The **completeness relation** follows from the preceding expansion,

$$|\psi\rangle = \sum_a c_a |\phi_a\rangle = \sum_a |\phi_a\rangle \langle \phi_a | \psi \rangle \quad (\text{B.9})$$

$$\Rightarrow \mathbf{I} = \sum_a |\phi_a\rangle \langle \phi_a|, \quad (\text{B.10})$$

where $\mathbf{I}$ is the *unit operator*.

The **matrix representation** of an operator O in the α representation is the bracket $\langle a' | O | a \rangle$. By changing basis we change the representation of an operator:

$$\langle b' | O | b \rangle = \sum_{a', a} \langle b' | a' \rangle \langle a' | O | a \rangle \langle a | b \rangle. \quad (\text{B.11})$$

The **complex conjugate** of a bracket can take different forms:

$$\langle \phi | O | \psi \rangle = \langle O^\dagger \phi | \psi \rangle = \langle \psi | O^\dagger \phi \rangle^* = \langle \psi | O^\dagger | \phi \rangle^*. \quad (\text{B.12})$$

The **wave function** in Dirac notation is

$$\psi(\mathbf{r}) \equiv \langle \mathbf{r} | \psi \rangle, \quad (\text{B.13})$$

which is just the probability amplitude for finding the state ψ at $\mathbf{r}$, that is, its projection onto the $\mathbf{r}$ basis (see too next section).

B.2 Explicit Representations

Examples of *explicit representations* include $|\mathbf{r}\rangle$, $|\mathbf{k}\rangle$, and $|klm\rangle$; that is, coordinate, momentum, and energy plus angular momentum space. These are developed in Chapters 5–8.

Coordinate Space

$$\langle \mathbf{r} | \psi \rangle \equiv \psi(\mathbf{r}) = \langle \psi | \mathbf{r} \rangle^* = \text{probability amplitude to be at } \mathbf{r} \quad (\text{B.14})$$

$$\langle \mathbf{r} | \mathbf{r}' \rangle = \delta(\mathbf{r} - \mathbf{r}') \quad (\text{B.15})$$

$$\int d^3r |\mathbf{r}\rangle \langle \mathbf{r}| = \tilde{1} \Rightarrow |\psi\rangle = \int d^3r |\mathbf{r}\rangle \langle \mathbf{r} | \psi \rangle = \int d^3r \psi(\mathbf{r}) |\mathbf{r}\rangle \quad (\text{B.16})$$

Momentum Space

$$|\phi_{\mathbf{k}}\rangle \equiv |\mathbf{k}\rangle = \text{plane wave ray} \quad (\text{B.17})$$

$$\langle \mathbf{r} | \phi_{\mathbf{k}} \rangle \equiv \langle \mathbf{r} | \mathbf{k} \rangle = \frac{e^{i\mathbf{k}\cdot\mathbf{r}}}{(2\pi)^{3/2}}, \quad \langle \mathbf{k} | \mathbf{r} \rangle = \frac{e^{-i\mathbf{k}\cdot\mathbf{r}}}{(2\pi)^{3/2}} \quad (\text{B.18})$$

$$\langle \mathbf{k} | \mathbf{k}' \rangle = \langle \mathbf{k} | \phi_{\mathbf{k}'} \rangle = \langle \phi_{\mathbf{k}} | \phi_{\mathbf{k}'} \rangle = \delta(\mathbf{k}' - \mathbf{k}) \quad (\text{B.19})$$

$$\tilde{1} = \int d^3k |\mathbf{k}\rangle \langle \mathbf{k}| \quad (\text{B.20})$$

$$\psi(\mathbf{k}) \equiv \langle \mathbf{k} | \psi \rangle = \text{probability amplitude to contain } \mathbf{k} \quad (\text{B.21})$$

Change of representations occur via insertion of completeness relations:

$$\psi(\mathbf{r}) = \langle \mathbf{r} | \psi \rangle = \int d^3k \langle \mathbf{r} | \mathbf{k} \rangle \langle \mathbf{k} | \psi \rangle = \int d^3k \langle \mathbf{r} | \mathbf{k} \rangle \psi(\mathbf{k}) \quad (\text{B.22})$$

$$= \int d^3k \frac{e^{i\mathbf{k}\cdot\mathbf{r}}}{(2\pi)^{3/2}} \psi(\mathbf{k}) \quad (\text{B.23})$$

$$\psi(\mathbf{k}) = \langle \mathbf{k} | \psi \rangle = \int d^3r \langle \mathbf{k} | \mathbf{r} \rangle \langle \mathbf{r} | \psi \rangle = \int d^3r \frac{e^{-i\mathbf{k}\cdot\mathbf{r}}}{(2\pi)^{3/2}} \psi(\mathbf{r}) \quad (\text{B.24})$$

$$\langle \mathbf{r} | klm \rangle \equiv \langle \mathbf{r} | \phi_{klm} \rangle = i^l \frac{F_l(kr)}{kr} Y_{lm}(\Omega_r) \quad (\text{B.25})$$

$$\langle \mathbf{r} | \psi_{klm} \rangle = i^l \frac{u_l(kr)}{kr} Y_{lm}(\Omega_r) \quad (\text{B.26})$$

$$\langle k'lm | \psi_k \rangle = \frac{(\pi/2)^{1/2} Y_{l,m}^*(\Omega_k)}{kk'} u_l(k'; E_k) \quad (\text{B.27})$$

K and *G* Operators (Nonrelativistic)

$$\langle \mathbf{r}' | K | \mathbf{r} \rangle = \delta(\mathbf{r} - \mathbf{r}') \frac{-\nabla_r^2}{2\mu} \quad (\text{kinetic energy}) \quad (\text{B.28})$$

$$\langle \mathbf{k}' | K | \mathbf{k} \rangle = \frac{\delta(\mathbf{k}' - \mathbf{k})}{2\mu} k^2 \quad (\text{B.29})$$

$$\langle \mathbf{r}' | G_E^{(+)} | \mathbf{r} \rangle = \frac{-m}{2\pi} \frac{e^{i\mathbf{k}|\mathbf{r}-\mathbf{r}'|}}{|\mathbf{r}-\mathbf{r}'|} \quad (\text{Green's function}) \quad (\text{B.30})$$

$$\langle \mathbf{k}' | G_E | \mathbf{k} \rangle = \langle \mathbf{k}' | \frac{1}{E - K} | \mathbf{k} \rangle = \frac{\delta(\mathbf{k} - \mathbf{k}')}{E - k^2/2\mu} \quad (\text{B.31})$$

$$\langle p|l m| G_E^{(+)} |p' l' m'\rangle = \frac{\pi}{2} \frac{\delta(p-p')}{p^2} \frac{\delta_{ll'} \delta_{mm'}}{E - E_p + i\epsilon} \quad (\text{B.32})$$

$$g_i^{(+)}(r, r'; E) = \frac{-2\mu}{k} F_i(kr_{<}) H_i^{(+)}(kr_{>}) \quad (\text{B.33})$$

Scattering Amplitude and T Matrix

$$T_E(\mathbf{k}', \mathbf{k}) = \langle \phi_{\mathbf{k}'} | T_E | \phi_{\mathbf{k}} \rangle = \langle \phi_{\mathbf{k}'} | V | \psi_{\mathbf{k}}^{(+)} \rangle \quad (\text{B.34})$$

$$f_E(\theta, \phi) = -4\pi^2 \mu T_E(\mathbf{k}', \mathbf{k}) \Big|_{\mathbf{k}'=\mathbf{k}} = -4\pi^2 \mu \langle \phi_{\mathbf{k}'} | V | \psi_{\mathbf{k}}^{(+)} \rangle \Big|_{\mathbf{k}'=\mathbf{k}} \quad (\text{B.35})$$

Energy and Angular Momentum Basis

Plane Wave

$$\langle \mathbf{r} | \mathbf{k} \rangle = \frac{e^{i\mathbf{k} \cdot \mathbf{r}}}{(2\pi)^{3/2}} = \sum_{l=0}^{\infty} i^l j_l(kr) \frac{(2l+1) P_l(\cos \theta)}{(2\pi)^{3/2}} \quad (\text{B.36})$$

$$\sum_{l,m} Y_{lm}^*(\Omega_k) Y_{lm}(\Omega_r) 4\pi = \sum_l (2l+1) P_l(\cos \theta_{kr}) \quad (\text{B.37})$$

Distorted Wave

$$\psi_{\mathbf{k}}^{(+)}(\mathbf{r}) = \langle \mathbf{r} | \psi(\mathbf{r}) \rangle = \sum_{l,m} i^l \frac{u_l(kr)}{kr} \frac{Y_{lm}^*(\Omega_k) Y_{lm}(\Omega_r) 4\pi}{(2\pi)^{3/2}} \quad (\text{B.38})$$

Angular Momentum and Energy Eigenstate

$$\phi_{klm}(\mathbf{r}) \equiv \langle \mathbf{r} | klm \rangle = i^l \frac{F_l(kr)}{kr} Y_{lm}(\Omega_r) \quad (\text{B.39})$$

Free Waves

$$F_l(kr) \equiv kr j_l(kr) = \begin{cases} \frac{(kr)^{l+1}}{1 \cdot 3 \cdot 5 \cdots (2l+1)}, & \text{when } r \rightarrow 0 \\ \sin(kr - l\pi/2), & \text{when } r \sim \infty \end{cases} \quad (\text{B.40})$$

$$G_l(kr) \equiv -kr n_l(kr) = \begin{cases} \frac{1 \cdot 3 \cdot 5 \cdots (2l-1)}{(kr)^l}, & \text{when } r \rightarrow 0 \\ \cos(kr - l\pi/2), & \text{when } r \sim \infty \end{cases} \quad (\text{B.41})$$

Momentum Ket Expansion

$$|\mathbf{k}\rangle = \sqrt{\frac{2}{\pi}} \sum_{l,m} Y_{lm}^*(\Omega_k) |klm\rangle \quad (\text{B.42})$$

Completeness Relation, Identity Operator

$$\bar{1} = \frac{2}{\pi} \oint_{l,m} dk k^2 |klm\rangle \langle klm| = \frac{2\mu}{\pi} \oint_{l,m} dE_k k |klm\rangle \langle klm| \quad (\text{B.43})$$

T and V Matrix Expansions, p Space

$$\langle \mathbf{k}' | \begin{pmatrix} V \\ T \end{pmatrix} | \mathbf{k} \rangle = \frac{2}{\pi} \sum_{l,m} \begin{pmatrix} V_l(k', k) \\ T_l(k', k) \end{pmatrix} Y_{lm}^*(\Omega_{k'}) Y_{lm}(\Omega_k) \quad (\text{B.44})$$

$$= \frac{1}{2\pi^2} \sum_{l,m} (2l+1) \begin{pmatrix} V_l(k', k) \\ T_l(k', k) \end{pmatrix} P_l(\cos \theta_{kk'}) \quad (\text{B.45})$$

Rotational Invariance

$$\langle k'l'm' | \begin{pmatrix} V \\ T \end{pmatrix} | klm \rangle = \delta_{ll'} \delta_{mm'} \begin{pmatrix} V_l(k', k) \\ T_l(k', k) \end{pmatrix} \quad (\text{B.46})$$

T and V Matrix Expansions, r Space

$$\langle \mathbf{r}' | \begin{pmatrix} V \\ T \end{pmatrix} | \mathbf{r} \rangle = \sum_{l,m} \begin{pmatrix} V_l(r', r) \\ T_l(r', r) \end{pmatrix} Y_{lm}^*(\Omega_{r'}) Y_{lm}(\Omega_r) \quad (\text{B.47})$$

Local Potential

$$V_l(r', r) = \frac{\delta(r - r')}{r^2} V(r) \quad (\text{all } l\text{'s}) \quad (\text{B.48})$$

Wave function Transform, Non-Local Potential

$$V_l(k', k) = \frac{1}{k'k} \int_0^\infty dr \int_0^\infty dr' r r' F_l(k'r') V_l(r', r) F_l(kr) \quad (\text{B.49})$$

Wave function Transform, Local Potential

$$V_l(k', k) = \frac{1}{k'k} \int_0^\infty dr F_l(k'r) V(r) F_l(kr) \quad (\text{B.50})$$

T matrix

$$T_l(k', k; E_k) = \frac{1}{k'k} \int_0^\infty dr \int_0^\infty dr' r r' F_l(k'r') V_l(r', r) u_l(kr) \quad (\text{B.51})$$

On-Energy-Shell Values

$$T_l(k, k; E_k) = -\frac{e^{2i\delta_l} - 1}{2i\rho_T} = \frac{R_l}{1 + i\rho_T R_l} \quad (\text{B.52})$$

$$R_l(k, k; E_k) = -\frac{\tan \delta_l(k)}{\rho_T}, \quad \rho_T = 2\mu k \quad (\text{B.53})$$

$$S_l(E[k]) = e^{2i\delta_l} \quad (\text{only defined on shell}) \quad (\text{B.54})$$

Scattering Amplitude

$$f_E(\theta, \phi) = -4\pi^2 \mu \langle \mathbf{k}' | T_E | \mathbf{k} \rangle |_{k'=k=k_0} \quad (\text{B.55})$$

$$= \sum_l (2l+1) \frac{e^{2i\delta_l} - 1}{2ik} P_l(\cos \theta_{kk'}) \quad (\text{B.56})$$

One-Dimensional Integral Equations

$$u_l(kr) = F_l(kr) + 2\mu \int_0^\infty dr' F_l(kr_{<}) H_l^{(+)}(kr_{>}) V(r') u_l(kr') \quad (\text{B.57})$$

$$u_l(k'; E_k) = \delta(k' - k) + \frac{2k' \int_0^\infty dp p V_l(k', p) u_l(p; E_k)}{\pi(E_k - E_{k'} + i\epsilon)} \quad (\text{B.58})$$

$$T_l(k', k; E) = V_l(k', k) + \frac{2}{\pi} \int_0^\infty dp p^2 \frac{V_l(k', p) T_l(p, k)}{E + i\epsilon - E_p} \quad (\text{B.59})$$

$$R_l(k', k) = V_l(k', k) + \frac{2}{\pi} \mathcal{P} \int_0^\infty dp p^2 \frac{V_l(k', p) R_l(p, k)}{E - E_p} \quad (\text{B.60})$$

APPENDIX C

FOUR-VECTORS AND LORENTZ TRANSFORMATIONS

Lorentz transformations relate the coordinates x^μ of a physical event in the inertial reference frame O to the coordinates x'^ν describing the *same event* in a different inertial reference-frame O' (Figure 14.1). We label events with the position vectors:

$$x^\mu = (t, x, y, z) \equiv (x^0, x^1, x^2, x^3), \quad (\text{C.1})$$

$$x'^\mu = (t', x', y', z') \equiv (x'^0, x'^1, x'^2, x'^3), \quad (\text{C.2})$$

where $t \equiv ct$ is called the timelike, zeroth, or fourth component of the position vector. Note that the 4D tensor indices are denoted by Greek letters $\mu, \nu, \dots$, which take on the values 0, 1, 2, 3 (in our notation there are no imaginary i 's in the metric and no difference between zeroth and fourth components). The Roman indices $i, j, k, \dots$, do not include the time component. The Lorentz transformation is the linear relation,

$$x'^\nu = \sum_{\mu=0}^3 a^\nu_{\mu} x^\mu, \quad (\text{C.3})$$

where the placement of the super and subscripts is important. *The summation convention* used with these four components is to sum over two repeated (i.e., identical) Greek indices if one is in the upper and the other is in the lower position. Accordingly, the Lorentz transformation (C.3) is also written as:

$$x'^\nu = a^\nu_{\mu} x^\mu. \quad (\text{C.4})$$

A *velocity boost* refers to the velocity acquired by a particle when viewing it in a different reference frame. If an observer in O sees O' moving with relative velocity u along the positive x axis, a particle at rest in O' gets boosted to u in O ; one at rest in O' gets boosted to $-u$ in O . The coordinates used by observer O are called x and are related to x' , those used by O' , by a velocity boost:

$$x^{0'} = \gamma_u (x^0 - \beta_u x^1), \quad (\text{C.5})$$

$$x^{1'} = \gamma_u (x^1 - \beta_u x^0), \quad (\text{C.6})$$

$$x^{2'} = x^2, \quad (\text{C.7})$$

$$x^{3'} = x^3, \quad (\text{C.8})$$

$$\gamma_u = \frac{1}{\sqrt{1 - \beta_u^2}}, \quad \beta_u = u \equiv \frac{u}{c}. \quad (\text{C.9})$$

The matrix $a^\nu{}_\mu$ of (C.4) is composed of the coefficients relating x' to x :

$$a^\nu{}_\mu = \begin{pmatrix} \gamma_u & -\gamma_u \beta_u & 0 & 0 \\ -\gamma_u \beta_u & \gamma_u & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \quad (\text{C.10})$$

Lorentz transformations in arbitrary directions can be generated as a combination of a rotation along one axis and a velocity transformation along one axis. Both velocity boosts and rotations are called Lorentz transformations and both are “proper,” that is, they have

$$\det[a^\nu{}_\mu] = 1. \quad (\text{C.11})$$

A *contravariant 4-vector* A^μ is a set of four numbers,

$$A^\mu = (A^0, A^1, A^2, A^3) \equiv (A^0, \mathbf{A}), \quad (\text{C.12})$$

which are mixed by a Lorentz transformation in exactly the same way as are the spacetime coordinates:

$$A'^\mu = a^\mu{}_\nu A^\nu. \quad (\text{C.13})$$

The *4-momentum*,

$$p^\mu = (E, \mathbf{p}), \quad (\text{C.14})$$

is a contravariant vector. A *covariant 4-vector* A_μ is the set

$$A_\mu = (A_0, A_1, A_2, A_3) \equiv (A^0, -A^1, -A^2, -A^3) \equiv (A^0, -\mathbf{A}), \quad (\text{C.15})$$

obtained by reversing the sign of the space components of the corresponding contravariant 4-vector.¹ Although we do not strictly need this “lower” type of 4-vector for the special relativity encountered in this book, it is useful for notational convenience and in general relativity. As indicated below, derivatives are covariant 4-vectors and they transform with a rule different from the covariant one (C.13). The contravariant rule is

$$A'_\mu = a_\mu{}^\nu A_\nu, \quad (\text{C.16})$$

$$a_\mu{}^\nu = \begin{pmatrix} \gamma_u & \gamma_u \beta_u & 0 & 0 \\ \gamma_u \beta_u & \gamma_u & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \quad (\text{C.17})$$

¹Remember, “co” is “low.”

The *derivative* with respect to a contravariant coordinate transforms as a covariant 4-vector. For example,

$$\partial_\mu = a_\mu{}^\nu \partial'_\nu, \quad (\text{C.18})$$

$$\partial_\mu = \frac{\partial}{\partial x^\mu} = \left(\frac{\partial}{\partial t}, \frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right) = \left(\frac{\partial}{\partial t}, +\nabla \right). \quad (\text{C.19})$$

On the other hand, the derivative with respect to a covariant coordinate,

$$\partial^\mu = \frac{\partial}{\partial x_\mu} = \left(\frac{\partial}{\partial t}, -\frac{\partial}{\partial x}, -\frac{\partial}{\partial y}, -\frac{\partial}{\partial z} \right) = \left(\frac{\partial}{\partial t}, -\nabla \right), \quad (\text{C.20})$$

transforms as a contravariant vector. The *4-momentum operators* p^μ and p_μ are proportional to the derivatives:

$$p^\mu = i\partial^\mu = i\frac{\partial}{\partial x_\mu} = i\left(\frac{\partial}{\partial t}, -\nabla \right), \quad (\text{C.21})$$

$$p_\mu = i\partial_\mu = i\frac{\partial}{\partial x^\mu} = i\left(\frac{\partial}{\partial t}, +\nabla \right). \quad (\text{C.22})$$

Raising and lowering of indices with the metric tensor $g_{\mu\nu}$ changes one type of vector to the other. For example,

$$\begin{aligned} x_\mu &= g_{\mu\nu} x^\nu, & x^\mu &= g^{\mu\nu} x_\nu, \\ g_{\mu\nu} &= \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} &\equiv g^{\mu\nu}. \end{aligned} \quad (\text{C.23})$$

If gravity were important, $g^{\mu\nu}$ would be distorted from this flat-space value.

The tensor indices are also interchangeable. For example, (C.17) follows from

$$a_\mu{}^\nu = g_{\mu\alpha} g^{\nu\beta} a^\alpha{}_\beta. \quad (\text{C.24})$$

The *dot* or *scalar product* of two 4-vectors A and B is

$$A \cdot B \equiv A_0 B_0 - \mathbf{A} \cdot \mathbf{B} = g_{\mu\nu} A^\mu B^\nu = A^\mu B_\mu = A_\mu B^\mu. \quad (\text{C.25})$$

The actual value of the scalar product is the same in all reference frames, that is, a scalar:²

$$A^\mu B_\mu = A'^\mu B'_\mu. \quad (\text{C.26})$$

This means the modulus of any 4-vector is also a scalar, for example,

$$p^\mu p_\mu = E_p^2 - p^2 = m^2. \quad (\text{C.27})$$

Since the 4-vector A' is related to A via the Lorentz transformation (C.13), the invariance of the scalar product leads to the constraint

$$a^\alpha{}_\gamma a^\beta{}_\delta g_{\alpha\beta} = g_{\gamma\delta}, \quad (\text{C.28})$$

or if $g_{\mu\nu}$ is used to lower or raise indices in place,

$$a^\alpha{}_\gamma a_\alpha{}^\beta = \delta_\gamma{}^\beta = \tilde{1}_{4 \times 4}. \quad (\text{C.29})$$

²The term “scalar” is also used to refer a state vector or particle with only one component, that is, spin 0. Calling the Coulomb potential a scalar, however, is a misnomer because it is the zeroth component of a 4-vector.

APPENDIX D

THE DIRAC EQUATION

Hamiltonian Form

$$i\frac{\partial\Psi}{\partial t}(\mathbf{x},t) = \mathbf{H}_0\Psi(\mathbf{x},t) \quad (\text{D.1})$$

$$\mathbf{H}_0 = \boldsymbol{\alpha} \cdot \mathbf{p} + \beta m \quad (\text{D.2})$$

$$= \frac{1}{i} \left(\alpha_1 \frac{\partial}{\partial x^1} + \alpha_2 \frac{\partial}{\partial x^2} + \alpha_3 \frac{\partial}{\partial x^3} \right) + \beta m \quad (\text{D.3})$$

Covariant Form

$$\left(i\gamma^0 \frac{\partial}{\partial t} - \boldsymbol{\gamma} \cdot \frac{\nabla}{i} \right) \Psi(\mathbf{x},t) = m\Psi(\mathbf{x},t) \quad (\text{D.4})$$

$$(\gamma^0 p_0 - \boldsymbol{\gamma} \cdot \mathbf{p})\Psi(\mathbf{x},t) = m\Psi(\mathbf{x},t) \quad (\text{D.5})$$

$$\gamma^\mu p_\mu \Psi(\mathbf{x},t) = m\Psi(\mathbf{x},t) \quad (\text{D.6})$$

Properties of α and β

$$\{\alpha_k, \alpha_j\} = 2\delta_{kj}, \quad \{\alpha_k, \beta\} = 0 \quad (\text{D.7})$$

$$\alpha_i^2 = \beta^2 = 1, \quad (\text{D.8})$$

$$\beta^\dagger = \beta, \quad \alpha^\dagger = \alpha \quad (\text{D.9})$$

Standard Matrices

$$\begin{aligned}
 \alpha_i &= \begin{pmatrix} 0 & \sigma_i \\ \sigma_i & 0 \end{pmatrix}, & \beta &= \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, & \gamma &= \beta\alpha, \\
 \gamma^i &= \begin{pmatrix} 0 & \sigma_i \\ -\sigma_i & 0 \end{pmatrix}, & \gamma^0 &= \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, & \gamma^5 &= \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \\
 \sigma^{ij} &= \begin{pmatrix} \sigma_k & 0 \\ 0 & \sigma_k \end{pmatrix}, & \sigma^{0i} &= i \begin{pmatrix} 0 & \sigma_i \\ \sigma_i & 0 \end{pmatrix}, & \sigma^{0i} &= i\alpha_i, \\
 \sigma_1 &= \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, & \sigma_2 &= \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, & \sigma_3 &= \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.
 \end{aligned} \quad (\text{D.10})$$

Properties of γ Matrices

$$\gamma^\mu \gamma^\nu + \gamma^\nu \gamma^\mu \stackrel{\text{def}}{=} \{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu} \quad (\text{D.11})$$

$$(\gamma^{i=1,2,3})^2 = -1, \quad (\gamma^0)^2 = 1 \quad (\text{D.12})$$

$$\gamma^\dagger = -\gamma, \quad \gamma_0^\dagger = \gamma_0 \quad (\text{D.13})$$

Bilinear Covariants

$$\bar{\Psi} \Gamma \Psi \quad (\text{D.14})$$

$\Gamma_{4 \times 4}$	Definition	Transforms	Number of
γ	$\beta \alpha$	Spacelike vector	3
γ_0	β	Timelike vector	1
$1 \equiv I$	Identity	Scalar	1
$\sigma^{\mu\nu}$	$\frac{i}{2} [\gamma^\mu, \gamma^\nu]$	Traceless tensor	6
γ_5	$i\gamma^0\gamma^1\gamma^2\gamma^3$	Pseudoscalar	1
$\gamma_5\gamma^\mu$	$\gamma_5\gamma^\mu$	Pseudovector	4

Current

$$j^\mu = (\rho, \mathbf{j}) = \bar{\Psi} \gamma^\mu \Psi \quad (\text{D.15})$$

$$\rho = \Psi^\dagger \Psi = |\psi_1|^2 + |\psi_2|^2 + |\psi_3|^2 + |\psi_4|^2 \quad (\text{D.16})$$

$$\mathbf{j} = \Psi^\dagger \boldsymbol{\alpha} \Psi \equiv \Psi^\dagger \beta \boldsymbol{\beta} \boldsymbol{\alpha} \Psi = \bar{\Psi} \boldsymbol{\gamma} \Psi \quad (\text{D.17})$$

Free-Particle Solutions

$$\Phi_s^{(\pm)}(x, t) = e^{\mp i p^\mu x_\mu} u_s^{(\pm)}(p); \quad (\text{D.18})$$

$$u_\uparrow^{(+)}(p) = N \begin{pmatrix} 1 \\ 0 \\ \frac{p_z}{E_p + m} \\ \frac{p_+}{E_p + m} \end{pmatrix}, \quad u_\downarrow^{(+)}(p) = N \begin{pmatrix} 0 \\ 1 \\ \frac{p_-}{E_p + m} \\ \frac{-p_z}{E_p + m} \end{pmatrix}, \quad (\text{D.19})$$

$$u_\downarrow^{(-)}(p) = N \begin{pmatrix} \frac{p_z}{E_p + m} \\ \frac{p_+}{E_p + m} \\ 1 \\ 0 \end{pmatrix}, \quad u_\uparrow^{(-)}(p) = N \begin{pmatrix} \frac{p_-}{E_p + m} \\ \frac{-p_z}{E_p + m} \\ 0 \\ 1 \end{pmatrix}. \quad (\text{D.20})$$

$$p_\pm = p_x \pm i p_y, \quad N = \sqrt{\frac{E_p + m}{2m}}. \quad (\text{D.21})$$

Free-Solution Properties

$$(\gamma^\mu p_\mu \mp m) u^{(\pm)}(p) \equiv (\not{p} \mp m) u^{(\pm)}(p) = 0 \quad (\text{D.22})$$

$$\bar{u}^{(\pm)}(p) (\not{p} \mp m) = 0 \quad (\text{D.23})$$

$$\bar{u}_s^{(b)}(p) u_{s'}^{(b')}(p) = b \delta_{bb'} \delta_{ss'}, \quad (\text{D.24})$$

$$\rho = \Phi^\dagger \Phi = u_s^{(b)}(p)^\dagger u_{s'}^{(b')}(p) = \frac{E_p}{m} \delta_{bb'} \delta_{ss'}, \quad (\text{D.25})$$

$$\mathbf{j} = \bar{u}_{s'}^{(b)}(p) \gamma u_s^{(b)}(p) = \frac{\mathbf{p}}{m} \delta_{ss'} \quad (\text{D.26})$$

$$\Lambda_+(p) = \sum_{s=1}^2 u_s(p) \bar{u}_s(p) = \frac{\not{p} + m}{2m} \quad (\text{D.27})$$

$$\Lambda_-(p) = \sum_{s=1}^2 v_s(p) \bar{v}_s(p) = \frac{-\not{p} + m}{2m} \quad (\text{D.28})$$

$$\Lambda_{\pm}^2 = \Lambda_{\pm}, \quad \Lambda_+ + \Lambda_- = I \quad (\text{D.29})$$

$$G_0 = \frac{1}{\not{p} - m} = \frac{\not{p} + m}{p^{\mu} p_{\mu} - m^2} \quad (\text{D.30})$$

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*When I am dead, I hope it may be said:
'His sins were scarlet, but his books were read.'*

—Hilaire Belloc

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